CHAPTER 4

Personality

IN THIS CHAPTER

4.1 ● What Personality Is: Definitions, Approaches and Early Type Theories: Define personality and its neighbouring terms, map the approaches and the six questions on which they differ, name the determinants and research methods, and describe the early type theories from the four humours to Sheldon's somatotypes.

4.2 ●● Freud's Psychoanalysis: Instincts, Structure and Psychosexual Stages: Trace how Freud built psychoanalysis from the study of hysteria, and use his methods, instincts, topographic and structural models, types of anxiety and psychosexual stages to explain personality, judging which of his ideas research supports.

4.3 ●●● Defence Mechanisms: How the Ego Protects Itself: Define defence mechanisms and their shared features, tell repression from its look-alikes, name the classic mechanisms from everyday vignettes, place them in Vaillant's four-level hierarchy, and match characteristic defences to particular disorders.

4.4 ●● Jung's Analytical Psychology and Adler's Individual Psychology: Explain how Jung's analytical psychology and Adler's individual psychology broke with Freud, and attribute each of their concepts (collective unconscious, archetypes, attitudes and functions, inferiority complex, fictional finalism, style of life, birth order) correctly.

4.5 ● Horney, Fromm, Sullivan and the Object-Relations Theorists: Explain how Horney, Fromm and Sullivan rooted personality in culture and relationships, and how ego psychology, object relations (Klein, Winnicott, Fairbairn, Mahler, Kernberg), Kohut's self psychology and Lacan revised Freud.

4.6 ● Traits I: Allport, Cattell and the Lexical Search: Explain the trait approach and the lexical hypothesis, Allport's personal dispositions, idiographic method, functional autonomy and proprium, and Cattell's surface and source traits, ergs and sentiments, L-, Q- and T-data and the 16PF.

4.7 ● Traits II: Eysenck, the Big Five and the Biology of Personality: Explain Eysenck's PEN model and its biology, the other biological temperament models, the Big Five with its facets and NEO inventories, HEXACO and the Dark Triad, what behaviour genetics shows about personality, and how the person-situation debate was resolved.

4.8 ● Humanistic and Existential Theories: Rogers, Maslow and May: Explain the humanistic image of the person through Rogers's self theory and Maslow's account of self-actualisation, and the existential view of Dasein, anxiety and meaning in Binswanger, May and Frankl.

4.9 • Behavioural, Social-Cognitive and Cognitive Theories: Explain personality through learning and cognition: Skinner's radical behaviourism, Dollard and Miller's early social learning, Bandura's reciprocal determinism, self-regulation and self-efficacy, Rotter's expectancy-value theory and locus of control, Mischel's CAPS, Kelly's personal constructs, Rep test and fixed-role therapy, and Witkin's and Kagan's cognitive styles.

4.10 • The Self, Positive Traits and Indian Approaches to Personality: Explain the classic theories of the self, separate self-concept from self-esteem, describe sensation seeking, explanatory style, well-being and character strengths, and identify the three gunas and the doshas of Indian approaches to personality.

Weightage. 202 of the 3180 questions mapped in this book (6.4%) test this chapter, 103 of them from real past papers. 1 of its 10 sections is High yield.

In 1920 a 22-year-old American named Gordon Allport was travelling home after a year teaching at Robert College in Istanbul. He stopped in Vienna to see one of his brothers and, on an impulse, sent a note to Sigmund Freud asking whether he might call. Freud, then in his sixties and already world famous, invited him over. When Allport entered the study, Freud said nothing at all. He simply sat and waited for the young visitor to explain why he had come.

The silence stretched until Allport, desperate for something to say, described an incident he had just seen on the tram. A small boy had seemed terrified of dirt: everything looked dirty to him, he changed his seat, and he told his mother not to let a dirty man sit beside him. Freud studied the neat, carefully groomed young man in front of him and asked, "Was that little boy you?"

Allport was shaken. He had told the story only to fill a silence; Freud had heard it as a disguised confession of his visitor's own unconscious fears. Allport denied for the rest of his life that he was the little boy, yet 47 years later he still wrote that his "single encounter with Freud was traumatic". The meeting convinced him that psychoanalysis dug too deep, and that psychology should first take people's conscious motives at face value. In 1922 he finished a Harvard dissertation on personality traits, the first research on traits in the United States, and went on to found the trait approach. Two great personality theorists had heard the same small story and seen entirely different things in it. Which of them was right is a question this whole chapter keeps asking.

4.1 What Personality Is: Definitions, Approaches and Early Type Theories

LOWER YIELD · 4 QS

You call a cousin moody and a friend a born leader, and you use those labels to predict what each will do next. Personality psychology does the same job with evidence. This section defines personality and its neighbouring terms, maps the approaches that fill this chapter, and ends with the oldest answers, which sorted people into types by body fluids, faces and physiques.

Persona and Allport's definition

The word personality comes from the Latin *persona*, the mask a Roman actor wore to show which character he was playing. The origin points to the public face we show others, and it raises the question every theory in this chapter must answer: is personality the mask, or what lies behind it?

The most quoted definition is that of **Gordon Willard Allport** (1897 to 1967), who weighed some 50 earlier definitions and offered his own in 1937: personality is "the dynamic organization within the individual of those psychophysical systems that determine his unique adjustments to his environment". In 1961 he replaced the last phrase with "characteristic behavior and thought". *Dynamic organisation* means personality grows, but in an organised way; *psychophysical* means mind and body work together; *determine* means it actively directs behaviour.

A working definition: **personality** is the relatively enduring, distinctive pattern of thoughts, feelings and behaviour that characterises a person, with internal aspects (motives, emotions) and external aspects (what others see), consistent across situations and over time. "Relatively" matters: a calm friend may still panic in an exam hall.

Temperament, character, trait and type

Temperament is the biologically based, early-appearing and relatively stable style of emotional reacting: how easily a person is aroused, how strongly they respond and what their usual mood is. Allport counted it, with physique and intelligence, as the inherited raw material of personality; infant temperament (Alexander Thomas and Stella Chess's easy, difficult and slow-to-warm-up babies) is in Chapter 6. **Character** is personality judged against moral standards; in Allport's phrase, "character is personality evaluated, and personality is character devaluated". A **trait** is a stable dimension along which people differ consistently, such as sociability; a **type** is a discrete category into which a person is placed, such as "melancholic". Table 4.1 adds three more terms.

Table 4.1 Personality and its neighbouring terms.

Term	Meaning	Example
Temperament	Biologically based, early-appearing emotional style	A baby who cries at every new face
Character	Personality evaluated morally; learned values	"She is honest and loyal"
Trait	Continuous dimension: more or less of it	7 out of 10 on sociability
Type	Discrete category: in or out	"He is an introvert"
Disposition	General tendency to act in a certain way; Allport's personal dispositions (Section 4.6)	A tendency to be generous
Habit	Specific learned, automatic response to a specific situation	Touching elders' feet on arrival
Attitude	Learned evaluation of a particular object, person or idea (Chapter 3)	Disliking fast food

Type versus trait is boxes versus a ruler: type theories sort people into a few categories; trait theories measure how much of each quality a person has, and find most people near the middle. Modern research favours traits; the early theories below are all type theories.

> EXAM TRAP
>
> "Biologically based", "inborn" or "visible from infancy" in a stem means **temperament**, not disposition (too loose a word), character (learned and moral) or trait (a descriptive dimension, not defined by biological origin). "Personality evaluated" is **character**.

What theories do: a map of the approaches

A **personality theory** is a set of related ideas that accounts for personality. A good one *describes* how people differ, *explains* why, *predicts* what a person will do and *generates research* that can test it. No theory does all four equally well, so the field is a family of approaches (Table 4.2, the map of this chapter).

Table 4.2 The approaches to personality and where they are taught.

Approach	Key names	Central idea	Where
Psychoanalytic	Freud	Unconscious conflict rooted in childhood	4.2, 4.3
Neo-psychoanalytic	Jung, Adler, Horney, Fromm, Sullivan	Social and cultural forces; less sex	4.4, 4.5
Ego psychology, object relations	Anna Freud, Melanie Klein	The ego's defences; early relationships	4.3, 4.5
Life-span	Erikson	Psychosocial stages across life	Chapter 6
Trait and type	Allport, Cattell, Eysenck, Costa and McCrae	Measurable stable dimensions	4.6, 4.7
Biological, evolutionary	Eysenck; twin researchers	Genes, brain, evolved tendencies	4.7
Humanistic, existential	Maslow, Rogers, May	Growth, choice, meaning	4.8
Behavioural	Skinner	Reinforcement shapes behaviour	4.9
Social-cognitive	Bandura, Rotter, Mischel	Learning plus thought, in situations	4.9
Cognitive	Kelly	Personal constructs	4.9
Limited-domain	Rotter, Zuckerman, Seligman	One facet at a time	4.9, 4.10
Indian	Samkhya, Ayurveda	Gunas and doshas	4.10

Six questions about human nature

Behind every theory lies an image of human nature. **Duane Schultz** and **Sydney Ellen Schultz** compare theorists on six questions: *free will versus determinism* (do we direct our lives?), *nature versus nurture* (inheritance or upbringing?), *past versus present* (is the adult fixed by childhood, the view called **historical determinism**?), *uniqueness versus universality*, *equilibrium versus growth* (do we only reduce tension, or strive to grow?) and *optimism versus pessimism* (are people basically good?).

Freud gave the darkest answers (Figure 4.1): behaviour determined by instincts and the first five years, life a struggle to reduce tension, people seen pessimistically; on nature-nurture and uniqueness he took a middle position. The humanists stand near the opposite pole: Carl Rogers saw people as free, shaped more by the present than by childhood, growing and basically good.

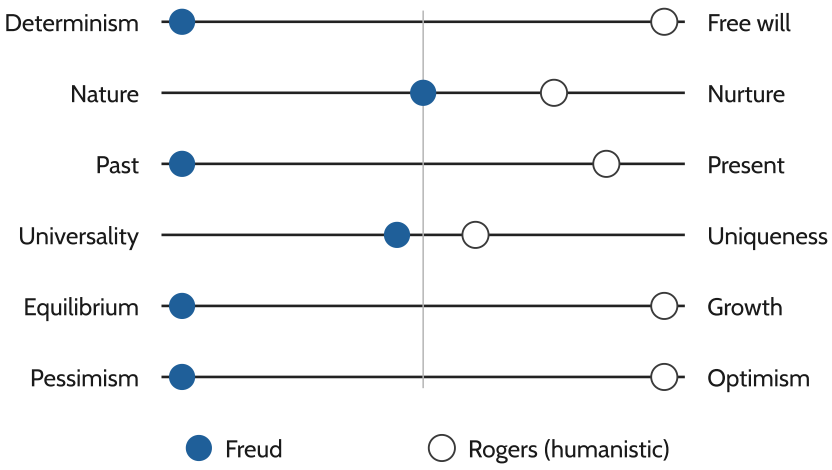


Figure 4.1 Schultz's six questions about human nature. Freud (filled) sits at the gloomy pole on four questions and near the middle on two; Rogers (open) sits near the opposite pole on most.

Determinants of personality

Heredity supplies temperament and more: twin studies (identical versus fraternal twins) and adoption studies (children compared with biological and adoptive parents) suggest genes account for up to about half the variation in traits (Section 4.7). **Environment** shapes the rest: the family (parenting style, parental models, and birth order, which Alfred Adler stressed; Section 4.4), culture (individualist and collectivist societies reward different traits) and social class (opportunities, stresses and values). Surprisingly, the environment siblings share matters less than each child's own experiences. The **situation** matters too: the student talkative in the canteen is silent in the viva. Walter Mischel's 1968 critique began a twenty-year person-situation debate (Section 4.9) that ended in **interactionism**: behaviour depends on the person, the situation and their interaction.

How personality is studied and assessed

The **clinical method** builds theory from the **case study**, a detailed history of one person from many sources; Freud, Jung and Adler built their theories from patients this way. It gives depth, but its data are subjective and open to the therapist's bias. The **experimental method** manipulates an independent variable and measures its effect on a dependent variable with all else held constant, so it alone shows cause; Skinner and Bandura worked this way, at the price of studying narrow slices of behaviour. The **correlational method** measures how strongly two variables go together, such as anxiety and exam marks, so it cannot prove cause. Its extension, **factor analysis**, finds the few factors underlying many intercorrelated measures; Raymond Cattell and Hans Eysenck built their trait theories with it.

The German philosopher **Wilhelm Windelband** (1848 to 1915) named a second divide in 1894. **Nomothetic** research (Greek *nomos*, law) seeks general laws from groups; **idiographic** research (Greek *idios*, one's own) studies the single individual in depth, the approach Allport championed (Section 4.6).

Assessment uses self-report inventories (MMPI-2, 16PF, NEO-PI-R, EPQ), projective techniques that ask for interpretations of ambiguous material (Rorschach inkblots, the Thematic

Apperception Test or TAT), interviews, behavioural observation and life records such as diaries and letters. Each is useful only if reliable (consistent) and valid (measures what it claims); the tests are in Chapter 2.

Humours and the four temperaments

The Greek physician **Hippocrates** (about 460 to 377 BC) taught that health depends on the balance of four bodily fluids, or **humours**: blood, yellow bile, black bile and phlegm. **Galen** (AD 129 to about 216), a Greek physician in Rome, is credited with turning this into a theory of temperament: the dominant humour sets a person's emotional style (Table 4.3). The chemistry is wrong, but the idea of a biological temperament was ahead of its time, and Hans Eysenck later fitted the four temperaments into the quadrants of his extraversion and neuroticism dimensions (Section 4.7).

Table 4.3 The four humours and temperaments, with Eysenck's later mapping.

Humour	Temperament	Character	Eysenck quadrant
Blood	Sanguine	Cheerful, hopeful, sociable	Stable extravert
Yellow bile (choler)	Choleric	Irritable, hot-tempered	Unstable extravert
Black bile	Melancholic	Sad, gloomy, anxious	Unstable introvert
Phlegm	Phlegmatic	Calm, slow, unexcitable	Stable introvert

Physiognomy and its look-alikes

Physiognomy judges character from facial features and outward appearance. The Greek *Physiognomonica*, attributed to **Aristotle** (384 to 322 BC) but probably by his followers, claimed that people who resemble certain animals share their characters. The Swiss pastor **Johann Kaspar Lavater** (1741 to 1801) made it a European fashion with his illustrated essays of the 1770s, which read each feature of the face as a sign of character. The Italian physician **Cesare Lombroso** (1835 to 1909) applied it to crime: in *Criminal Man* (1876) he claimed that "born criminals" carry **criminal stigmata**, such as a sloping forehead and large jaws, marks of a throwback to primitive humans. Later comparisons of prisoners with other citizens found no such type.

Phrenology, founded by **Franz Joseph Gall** (1758 to 1828) around 1800 and popularised under that name by **Johann Gaspar Spurzheim** (1776 to 1832), read the bumps of the skull, assuming each mental faculty has its own brain "organ" whose size pushes out the skull above it (Section 1.1): wrong about bumps, right that functions are localised. **Graphology** infers personality from handwriting style and shows no useful validity in controlled studies. Interpreting dreams for unconscious motives is psychoanalysis (Section 4.2).

> EXAM TRAP
>
> Face as a sign of character = **physiognomy**. Skull bumps and brain localisation of functions = **phrenology** (Gall). Handwriting = **graphology**. Dreams and unconscious motives = **psychoanalysis**. Body build = Kretschmer and Sheldon, below.

Body types: Kretschmer, Sheldon and later typologies ●●

The German psychiatrist **Ernst Kretschmer** (1888 to 1964) thought his patients' illnesses went with their build. In *Physique and Character* (1921) he described the **pyknic** type, short and round, with a **cyclothymic** temperament swinging between cheer and gloom and a liability to manic-depressive illness; the **asthenic** or leptosomic type, tall and thin, with a reserved, withdrawn **schizothymic** temperament and a liability to schizophrenia; the **athletic** type, muscular; and the **dysplastic** type, a disproportioned mix. One flaw: his manic-depressive patients were older, and people grow rounder with age.

The American physician **William Herbert Sheldon** (1898 to 1977) replaced boxes with dimensions. In *The Varieties of Human Physique* (1940) he rated thousands of standardised photographs of male college students on three components, named after the embryonic layers that form the dominant tissues: **endomorphy** (soft and round, digestive organs prominent), **mesomorphy** (muscle and bone) and **ectomorphy** (thin and fragile, a large skin surface for the body's mass). Each is rated from 1 to 7, so a **somatotype** is three numbers: 4-4-4 is the average build, 7-1-1 the extreme endomorph, 1-7-1 the extreme mesomorph and 1-1-7 the extreme ectomorph. In *The Varieties of Temperament* (1942) he matched each component to a temperament (Table 4.4).

Table 4.4 Sheldon's somatotypes and temperaments.

Physique	Build	Temperament	Described as
Endomorph	Round, soft	Viscerotonia	Relaxed, sociable, even-tempered; loves comfort, food and company
Mesomorph	Muscular, broad	Somatotonia	Assertive, energetic, noisy, callous; fond of physical activity and risk
Ectomorph	Thin, fragile	Cerebrotonia	Restrained, inhibited, private, self-conscious, introverted

KEY STUDY: SHELDON'S SOMATOTYPES (1940, 1942)

Does body build predict temperament? Sheldon rated young men on dozens of temperament traits and compared the ratings with their somatotypes, reporting correlations of about .8, far above almost anything in personality research. The flaw: he rated both physique and temperament himself, knowing each man's build, so his expectations and the stereotypes of the jolly fat man and the nervous thin one could create the correlation. Independent studies found only weak links. The theory is regarded as discredited.

EXAM TRAP

Sheldon's theory has been **discredited**: not supported by research, and not "accurate for thin people" or "for women only" (his subjects were men). Noisy, callous, fond of physical activity = **mesomorph**; relaxed and food-loving = endomorph; restrained and private = ectomorph. Pyknic and asthenic with psychosis = Kretschmer.

Type thinking outlived the body theories. The German philosopher **Eduard Spranger** (1882 to 1963) sorted people by dominant value into six types: theoretical (truth), economic (usefulness), aesthetic (beauty), social (love of people), political (power) and religious (unity); the Allport-Vernon-Lindzey *Study of Values* measures them (Section 4.6). Carl Jung's types are in Section 4.4; Type A and B (Meyer Friedman and Ray Rosenman), Type C and Johan Denollet's Type D in Section 1.11; the gunas and doshas in Section 4.10.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The word personality comes from the Latin _____, the mask of the Roman actor.
2. Allport (1937): the dynamic organisation within the individual of those _____ systems that determine his unique adjustments to his environment.
3. The biologically based, early-appearing style of emotional reacting is _____.
4. The view that adult personality is fixed in the early years is called _____ determinism.
5. The terms nomothetic and idiographic were introduced by _____.
6. In humoral theory, an excess of black bile produces the _____ temperament.
7. Judging character from facial features is _____; reading skull bumps is _____.
8. Kretschmer linked the round _____ build to manic-depressive illness, and Sheldon linked the mesomorph to _____.

Answers: (1) persona, (2) psychophysical, (3) temperament, (4) historical, (5) Wilhelm Windelband, (6) melancholic, (7) physiognomy; phrenology, (8) pyknic; somatotonia.

4.2 Freud's Psychoanalysis: Instincts, Structure and Psychosexual Stages

MEDIUM YIELD · 42 QS

You mean to thank a teacher and hear yourself use the name of one you disliked; you "forget" a dental appointment you were dreading. Most of us call these accidents. Sigmund Freud did not. To him every slip, dream and symptom had a hidden cause, and personality was an uneasy truce between inborn urges, the outside world and the morality we absorb as children. His system, **psychoanalysis**, was the first formal theory of personality and is still the best known: at once a theory of personality and a method of treating mental disorder. We follow it from hysteria to the verdict of research.

Freud's life and the birth of psychoanalysis ●●●

Sigmund Freud (1856 to 1939) was born in Freiberg, Moravia (now Příbor in the Czech Republic), and from the age of four lived in Vienna for almost 80 years. He qualified in medicine at Vienna in 1881 and began his career as a **neurologist**, not a psychiatrist or philosopher; "psychoanalyst" was a profession he had yet to invent. Neurologists were sent the patients nobody could explain, those with **hysteria**: genuine physical symptoms such as paralysis, blindness or loss of feeling (anaesthesia) in a limb, with no damage to the nerves. Freud came to see these as bodily reactions to emotional trauma, produced by painful memories kept out of awareness.

Three teachers pointed the way. In Paris (1885 to 1886) the neurologist **Jean-Martin Charcot** (1825 to 1893) showed him hysterical symptoms produced and removed by hypnosis, and hinted that their roots were sexual. At Nancy, which Freud visited in 1889, **Hippolyte Bernheim** (1840 to 1919) showed that hypnosis worked through suggestion. In Vienna the physician **Josef Breuer** (1842 to 1925) had found that a patient lost her symptoms when she talked through their origins under hypnosis. Breuer and Freud reported this in "On the Psychical

Mechanism of Hysterical Phenomena: Preliminary Communication" (1893), expanded into *Studies on Hysteria* (1895): hysterics, they wrote, suffer mainly from reminiscences.

KEY STUDY: ANNA O. AND THE TALKING CURE (BREUER AND FREUD, 1895)

Anna O., really **Bertha Pappenheim** (1859 to 1936), was a bright young Viennese woman who fell ill while nursing her sick father: blurred vision, weakness of her right arm and both legs, and trouble speaking. Breuer took one symptom at a time; under hypnosis she traced it back to the moment it began, relived the feeling, and the symptom faded. She called this the "talking cure". Breuer and Freud named the release of pent-up emotion **catharsis** (Greek for purification) and concluded that symptoms carry hidden meaning. Criticism: she was Breuer's patient, not Freud's, and she was not cured when treatment ended in 1882; she spent months in a Swiss sanatorium and took years to recover before becoming a pioneering social worker.

Freud went further than Breuer. By the mid-1890s he held that the buried trauma was always sexual: his **seduction theory** traced neurosis to sexual abuse in childhood, usually by an older male relative. In 1897 he abandoned it, deciding that many such memories were fantasies and that the wishes behind them mattered more than events. The same year he began a lifelong **self-analysis** through his own dreams, recovering childhood hostility to his father and attachment to his mother, the seed of the Oedipus complex. The major works followed (Table 4.5). He fled the Nazis to London in 1938 and died there in September 1939, after sixteen years of cancer of the mouth.

Table 4.5 Freud's path to psychoanalysis: the dates that are examined.

Year	Work or event	Why it matters
1893	"Preliminary Communication" (with Breuer)	Hysteria from traumatic memories
1895	*Studies on Hysteria* (with Breuer)	Anna O.; catharsis; the founding text
1900	*The Interpretation of Dreams*	Dreams as wish fulfilment; topographic model (appeared November 1899, dated 1900)
1901	*The Psychopathology of Everyday Life*	Parapraxes (slips)
1905	*Three Essays on the Theory of Sexuality*	Infantile sexuality; psychosexual stages
1909	Lectures at Clark University, Worcester, Massachusetts	Invited by G. Stanley Hall; American fame
1920	*Beyond the Pleasure Principle*	The death instinct
1923	*The Ego and the Id*	Structural model: id, ego, superego
1926	*Inhibitions, Symptoms and Anxiety*	Anxiety as the ego's danger signal
1930	*Civilization and Its Discontents*	Instinct versus civilisation; guilt

Freud's theory is **psychoanalytic**, not behaviouristic (Watson and Skinner: behaviour as learning), humanistic (Maslow and Rogers: growth and choice) or psychosocial (Erikson's social stages; Chapter 6). The broader family that grew from it is the **psychodynamic perspective**: many mental disorders can be traced to unconscious urges and to the struggle over their expression in the hidden depths of personality. Gestalt psychology studies perception as or-

ganised wholes (Chapter 1), and social learning theory explains behaviour by observation and reinforcement (Section 4.9).

CLINICAL LINK

Freud's hysteria survives as **conversion disorder**, called functional neurological symptom disorder in DSM-5-TR: real loss of movement or sensation that fits no neurological disease (Chapter 8).

EXAM TRAP

The 1893 paper and *Studies on Hysteria* are **Freud and Breuer**, never Freud with Charcot, Adler or Jung (Adler and Jung were later followers who broke away; Section 4.4). Freud began as a **neurologist**. To Freud, hysteria meant **physical symptoms produced by emotional trauma**; a phobia of horses is the case of Little Hans, and an unresolved Oedipus complex or oral behaviour are unrelated ideas. The father of psychoanalysis is Freud, not Bernheim, Adler or Jung.

Freud's methods: free association, dreams and slips ●●

Many patients could not be hypnotised, so Freud had them lie on a couch, with himself out of sight, and gave them one **fundamental rule**: say whatever comes to mind, however trivial, embarrassing or painful, without censoring or rearranging it. This is **free association**, a kind of daydreaming aloud, and it became the central technique of psychoanalysis. Because of psychic determinism (below), nothing in the chain was random; each idea led closer to the buried conflict.

The chain often stalls: the patient falls silent or changes the subject. This is **resistance**, the unconscious refusal to let painful material surface, which Freud took as a sign that the analysis was near the source. Patients also come to feel toward the analyst as they once felt toward their parents: **transference**. Both are worked through in therapy (Chapter 7).

Dream analysis was the second road in; Freud called the interpretation of dreams "the royal road to a knowledge of the unconscious". What you remember is the **manifest content**; its hidden meaning, a disguised wish, is the **latent content**. The dream-work that turns one into the other is taught in Section 1.5.

The third window was everyday mistakes. A **parapraxis** (plural parapraxes), popularly a **Freudian slip**, is a slip of the tongue or pen, a forgotten name or appointment, or a misplaced object that betrays an unconscious thought or wish. A student who writes to his professor "I look forward to missing your lecture" has, in Freud's eyes, said what he meant. Parapraxes are mistakes that reveal the unconscious; they need not be sexual. Their meaning is found as a symptom's is, through free association.

Freud reported his work as **case studies**, detailed histories of single patients: Anna O. (hysteria, Breuer's patient), Dora (hysteria, 1905), Little Hans (1909, a small boy's phobia of horses, read as Oedipal fear of his father), the Rat Man (obsessional neurosis, 1909), Schreber (paranoia, 1911, analysed only from his published memoir) and the Wolf Man (1918, a childhood dream of wolves). The method is rich but weak as evidence, as the evaluation shows.

EXAM TRAP

Lying on the couch saying whatever comes to mind is **free association**, not dreaming, narration or storytelling, and it is the central technique of **psychoanalysis**, not of behaviour, person-centred or cognitive therapy (Chapter 7). Freudian slips are analysed through **free association**: case studies are how Freud reported patients, behaviour analysis records observable acts and their consequences, and a sleep study records sleep stages, not meanings. A slip is not a **defence mechanism** (an unconscious strategy against anxiety; Section 4.3), a **cognitive distortion** (Aaron Beck's errors of thinking, such as overgeneralisation) or a **speech disorder** (such as stuttering).

Instincts and psychic energy ●●

What powers the mind? Freud's answer was the **instincts** (German *Triebe*, sometimes translated "drives"): the powerful innate forces that motivate all behaviour and set its direction. An instinct is the mental representation of a bodily need, the wish that hunger becomes in the mind. In "Instincts and Their Vicissitudes" (1915) Freud gave every instinct a **source** (the bodily need), an **aim** (to remove the excitation, that is, satisfaction), an **object** (whatever achieves the aim, the most changeable part) and an **impetus** or pressure (its strength). The aim is always tension reduction, so the model is homeostatic.

Freud grouped the instincts into two classes. The **life instincts**, or **Eros**, serve survival of the individual and the species: hunger, thirst and above all sex, which Freud defined broadly to include almost any bodily pleasure and affection. Their energy is the **libido**. In *Beyond the Pleasure Principle* (1920), shaken by the slaughter of the First World War, he added the **death instinct**, an unconscious drive toward decay and a return to the inanimate state, which his followers named **Thanatos**, a word Freud himself did not use in his writings. Turned inward it is self-destruction; turned outward it becomes the **aggressive drive** to destroy, conquer and kill, so wars and violence are, for Freud, evidence of the death instinct. Sex and aggression are what the id wants; the pleasure and reality principles (below) are the rules by which the id and the ego pursue them.

Libido can be attached to anything. **Cathexis** is the investment of psychic energy in the mental representation of an object, person or idea: if you adore a cricketer, your libido is cathected to him. An **anticathexis** is the opposing investment by which the ego or superego blocks an id cathexis; keeping a wish repressed costs energy. The cost matters because Freud saw the mind as a **closed energy system**, a fixed quantity of psychic energy that can be shifted but never added to, so whatever the ego spends holding back the id is lost to work and love. Beside it stands **psychic determinism**, the principle that every mental event has a cause, so no slip, dream or symptom is an accident.

EXAM TRAP

Freud's innate motivating forces are **instincts**; **needs** are Maslow's and Henry Murray's term, **drives** Clark Hull's, and the **id** is the reservoir that holds the instincts. Investing energy in an object is **cathexis**. Its neighbours: **libido** is the energy itself, **fixation** is libido left stuck at a childhood stage, **sublimation** channels it into socially valued work, **displacement** shifts an impulse to a safer target and **repression** keeps threatening material out of awareness (the last three are defences; Section 4.3).

The levels of the mind: the topographic model ●

Freud's first map, the **topographic model** (1900), sorts mental contents by how accessible they are. The **conscious** holds whatever you are aware of at this moment. The **preconscious** holds accessible memories, thoughts and perceptions that are not in awareness now but can be recalled at will: your mobile number, what you ate last night. The **unconscious**, by far the largest, holds instincts, repressed wishes and painful memories that cannot be recalled at will and show themselves only indirectly, in dreams, slips and symptoms. Freud likened the mind to an iceberg: the conscious is the small tip above the water, the preconscious lies just below the waterline, and the unconscious is the vast hidden mass, the real driving power of behaviour (Figure 4.2).

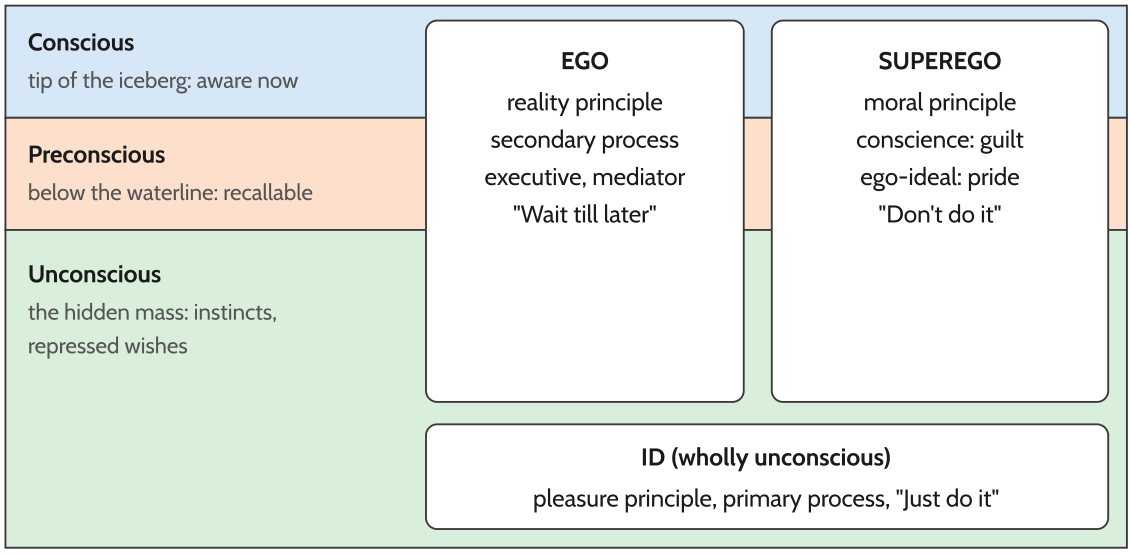


Figure 4.2 The two maps together. The shaded bands are the topographic levels of the iceberg; the boxes are the structures of 1923. The ego and superego reach into all three levels, while the id lies wholly below.

> EXAM TRAP
>
> Accessible memories recalled at will are **preconscious**, not conscious (in awareness now) or unconscious (not recallable at will). **Psychic determinism** is a principle, not a level of the mind.

Id, ego and superego: the structural model ●●●

In *The Ego and the Id* (1923) Freud replaced the three levels with three interacting structures, the **structural model**. The levels became properties of the structures (Figure 4.2).

The **id** is the reservoir of unorganised instinctual drives, present at birth, wholly unconscious and the source of all psychic energy. It operates on the **pleasure principle**: seek pleasure and avoid pain, now. Impulsive, irrational, selfish and amoral, it demands the immediate gratification of primitive needs and tolerates no delay ("Just do it"). Having no contact with reality, it can satisfy a need only by reflex or by **primary process thought**, forming a wish-fulfilling image of the object it lacks, as a hungry infant pictures the breast. This **wish fulfilment** also drives dreams.

Images feed no one, so from the id grows the **ego**, the rational part of personality, which works on the **reality principle**. Using **secondary process thought** (perception, memory, judgement and planning) and **reality testing**, it decides when, where and how the id's wishes can safely be met. It does not block the id; it postpones and redirects ("Wait till later"). Freud compared it to a rider guiding a stronger horse. It is called the **executive** of personality because it directs the energies supplied by the id and mediates between the id's desires, the superego and reality or social rules. The ego is partly conscious, partly preconscious and partly unconscious.

The **superego** is the moral arm, working on the moral principle. It forms at about five or six, when the Oedipus complex is resolved and the child internalises parental and social standards, so that parental control becomes self-control. It has two parts. The **conscience** is built from behaviours for which the child was punished and enforces them with guilt; the **ego-ideal** is built from good, correct behaviours for which the child was praised and rewards with pride. Like the id it admits no compromise: it seeks not to delay forbidden impulses but to stop them ("Don't do it"), and it strives for perfection. The ego thus serves three harsh masters: the id, reality and the superego (Table 4.6).

Table 4.6 The three structures of personality compared.

Feature	Id	Ego	Superego
Principle	Pleasure	Reality	Moral (perfection)
Thinking	Primary process; wish fulfilment	Secondary process; reality testing	Moral judgement
Awareness	Wholly unconscious	All three levels	All three levels, mostly unconscious
Appears	At birth	Infancy, out of the id	About 5 or 6, at Oedipal resolution
Role	Reservoir of instincts; all energy	Executive; mediator	Conscience (guilt) and ego-ideal (pride)
Motto	"Just do it"	"Wait till later"	"Don't do it"

EXAM TRAP

Gratification, impulsiveness and irrationality point to the **id**, and **only** the id seeks immediate gratification of primitive needs. The ego is the **executive**; "superordinate" and "subordinate" are not Freud's labels, and **congruence** is Carl Rogers's term for a match between self and experience (Section 4.8). The conscious and unconscious mind are levels, not principles. The **self** and the **libido** are not Freudian structures, and the superego's two parts are the **conscience and the ego-ideal**, not the id and the ego. "Do your own thing" and "Oh, sit on it" are not Freudian mottoes.

Anxiety: the ego's danger signal

Squeezed by three masters, the ego feels **anxiety**, fear without a clear object. In *Inhibitions, Symptoms and Anxiety* (1926) Freud made anxiety a **signal** that warns the ego of danger, with the overwhelming stimulation of birth as its prototype. He named three kinds by their source. **Reality anxiety** (objective anxiety) is fear of a real danger in the outside world, such as a speeding lorry; it is useful and fades when the danger passes. **Neurotic anxiety**, a conflict between id and ego, is the unconscious fear that id impulses will break through and

bring punishment. **Moral anxiety**, a conflict between id and superego, is fear of one's own conscience, felt as guilt or shame. When realistic action cannot remove the threat, the ego turns to unconscious defence mechanisms, the subject of Section 4.3.

The psychosexual stages ●●●

Freud believed adult personality was largely set by about the age of five, and his **psychosexual stages** (*Three Essays*, 1905) explain how. At each stage the libido centres on a different **erogenous zone**, a part of the body that gives pleasure, and each stage brings a conflict between the child's pleasure and the parents' demands. If a stage goes badly, part of the libido stays invested there: **fixation**. It results from **frustration** (needs unmet) or **overindulgence** (needs so richly met that the child will not move on), and it stamps a recognisable adult character. Under stress a person may show **regression**, a return to the behaviour of an earlier stage, usually the one where libido is fixated. Table 4.7 gives the sequence; the ages are approximate and differ between textbooks (some give oral birth to 1 year, anal 1 to 3, phallic 4 to 5 and latency 5 to puberty).

In the **oral stage** (birth to about 18 months) pleasure comes from the mouth, through sucking, biting and swallowing, and the central conflict is **feeding and weaning**. Its first mode, **oral-incorporative** (taking in), produces adults preoccupied with eating, drinking, smoking, kissing and nail-biting, who are dependent and gullible ("swallowing" whatever they are told). The later **oral-aggressive** or oral-sadistic mode, which comes with painful teething, leaves adults who are sarcastic ("biting" remarks), critical, hostile and argumentative. Freud linked alcohol dependence to oral fixation.

In the **anal stage** (about 18 months to 3 years) pleasure comes from holding back and expelling faeces, and **toilet training** is the first time society regulates an instinct. A child who defies it by soiling may become an **anal-expulsive** (anal-aggressive) adult: messy, wasteful, disorganised, even cruel. A child who resists by holding back may become **anal-retentive**: orderly, stingy, obstinate and compulsively neat.

The **phallic stage** (about 3 to 5 or 6 years) centres on the genitals and on incestuous fantasies. The boy faces the **Oedipus complex**, named after the king in Sophocles' play who unknowingly kills his father and marries his mother: he desires his mother and sees his father as a rival. Fearing that the father will cut off his penis (**castration anxiety**), he represses the desire and achieves **identification** with the same-sex parent, adopting his father's manner and standards; this gives him his sex role and creates the superego. The girl's first love is also her mother, but on finding she lacks a penis she feels **penis envy**, blames her mother and turns to her father; she resolves the conflict, less completely in Freud's view, by identifying with her mother. Freud called this the female Oedipus complex. **Carl Gustav Jung** (1875 to 1961) named it the **Electra complex** in 1913, after the Greek heroine who persuades her brother to kill their mother. Freud rejected the term, yet exams pair Electra with girls and Oedipus with boys.

In the **latency period** (about 6 to puberty) the sexual instinct is dormant, sublimated into school, sport, hobbies and friendships with children of the same sex; Freud did not count it as a true psychosexual stage, since no new zone appears. The **genital stage** (puberty onwards) reawakens sexuality, now directed to mature love. Its achievements are sex-role identity, adult social relationships and the capacity for love and work.

Table 4.7 Freud's psychosexual stages.

Stage and age	Erogenous zone	Central conflict	Adult fixation traits
Oral, birth to about 18 months	Mouth	Feeding and weaning	Incorporative: dependent, gullible, overeating, smoking, drinking, nail-biting. Aggressive: sarcastic, critical, hostile
Anal, about 18 months to 3 years	Anus	Toilet training	Retentive: orderly, stingy, obstinate. Expulsive: messy, wasteful, cruel
Phallic, about 3 to 5 or 6	Genitals	Oedipus complex; castration anxiety or penis envy; superego forms	Vain, brash, narcissistic
Latency, about 6 to puberty	None (dormant)	Sublimation into school, sport, same-sex friends	Not a fixation point
Genital, puberty onwards	Genitals (mature)	Mature love and work; sex-role identity; adult relationships	The healthy goal

EXAM TRAP

Smoking, drinking, overeating, nail-biting, sarcasm and criticalness all mean **oral** fixation, which comes from problems of **feeding and weaning**; toilet training is anal, Oedipal resolution phallic, and adolescent identity formation is Erik Erikson's psychosocial crisis (Chapter 6). Oral aggression is oral, not anal, and "genital identity confusion" is not a Freudian concept. Incestuous fantasies, the Oedipus complex and the emergence of the superego belong to the **phallic** stage; sublimation of the sex instinct is **latency**; sex-role identity and adult relationships are **genital**.

Evaluating Freud

Which parts of the theory survive testing? The strongest support is for an unconscious mind. In **subliminal perception** studies, stimuli flashed too briefly to be seen still shape thought: **Howard Shevrin** (1977) showed people a picture of a bee they could not report, and their free associations then included "sting" and "honey". **Implicit memory** (experience shaping performance without recollection) and **priming** point the same way. Some defences, such as repression, projection and displacement, have experimental support (Section 4.3), and **Michael Motley** (1987) found that men made anxious about an electric shock said "damn shock" for "sham dock", a laboratory Freudian slip. Reviewing some 2,500 studies, **Seymour Fisher** and **Roger Greenberg** (1977, 1996) judged the id, ego, superego, libido and death instinct untestable, found weak support for oral and anal character types and some Oedipal ideas, and found no support for the psychosexual stages or for dreams as disguised wishes.

The criticisms are heavy. Many concepts are **unfalsifiable**: if a man loves his mother it is Oedipal, and if he hates her he is defending against Oedipal love, so no result can count against the theory. The data came from Freud himself and a small, biased sample, a dozen or so published cases, mostly young, educated, well-off Viennese women. His notes were written from memory hours later, and patients' recollections were not corroborated. His account of women (penis envy, a weaker superego) is sexist, a charge led by Karen Horney (Section 4.5). His view of human nature is pessimistic and over-sexualised, and personality plainly goes on developing after five.

Still, Freud gave psychology its first comprehensive theory of personality, the ideas of unconscious motivation, defence and the weight of childhood, and the talking cure from which every psychotherapy descends. On the six questions about human nature (Section 4.1) Freud is a determinist (though analysis widens choice), takes a middle ground on nature and nurture (inborn id and stages, learned ego and superego), stresses the past (the first five years), sees both universality (shared instincts and stages) and uniqueness (the content of ego and superego), makes equilibrium (tension reduction) life's goal, and is a pessimist. In India, Girindrasekhar Bose, founder of the Indian Psychoanalytical Society (1922), corresponded with Freud and proposed his own theory of opposite wishes (Section 1.2).

> **EXAM TRAP**
>
> Research has begun to support the **unconscious mind**, not the existence of an id, ego and superego (untestable), the order of the psychosexual stages or a universal Oedipus complex.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Freud and _____ published *Studies on Hysteria* in 1895.
2. Saying whatever comes to mind on the couch, without censoring it, is _____.
3. The investment of psychic energy in an object, person or idea is _____.
4. Memories not in awareness but recallable at will lie in the _____.
5. The id works on the pleasure principle; the ego works on the _____ principle.
6. The two parts of the superego are the conscience and the _____.
7. Fear of one's own conscience, felt as guilt, is _____ anxiety.
8. Smoking, overeating, nail-biting and sarcasm suggest fixation at the _____ stage.

Answers: (1) Breuer, (2) free association, (3) cathexis, (4) preconscious, (5) reality, (6) ego-ideal, (7) moral, (8) oral.

4.3 Defence Mechanisms: How the Ego Protects Itself HIGH YIELD · 38 QS

Arjun has just failed his statistics paper. Within an hour he has told his friends that statistics is useless for a clinician anyway, and he has snapped at his younger brother over nothing. He is not lying, at least not knowingly. In the terms of Section 4.2, his ego is under threat: anxiety is signalling danger, and when the ego cannot remove a danger by realistic means it falls back on unconscious tricks that disguise it. This section defines those tricks, studies the most basic of them, repression, walks through the full catalogue, ranks them from psychotic to mature, and shows which ones mark particular disorders.

What defences are and what they share •

Sigmund Freud (1856 to 1939) wrote of defence from his first papers on hysteria in the 1890s, although for years he spoke almost only of repression. In *Inhibitions, Symptoms and Anxiety* (1926) he returned to the wider term: anxiety is a signal of danger to the ego, and defence is the name for every technique the ego uses to keep that danger out of awareness. Formally,

defence mechanisms are unconscious strategies the ego uses to protect itself against anxiety by denying or distorting reality. It was his daughter **Anna Freud** (1895 to 1982) who turned the idea into a catalogue. In *The Ego and the Mechanisms of Defence* (1936) she listed ten mechanisms (regression, repression, reaction formation, isolation, undoing, projection, introjection, turning against the self, reversal and sublimation), described others such as identification with the aggressor, and moved the ego to the centre of psychoanalysis. The standard list you learn today owes its full articulation to her.

The mechanisms differ in detail, but they share six features:

- They operate unconsciously. The person neither chooses a defence nor knows that one is working; if we knew we were lying to ourselves, the lie would no longer protect us.
- They deny or distort reality, either outer reality (a diagnosis, a loss) or inner reality (a wish, a feeling).
- They protect the ego and reduce anxiety. That is their whole function, and we usually use several at once.
- They have adaptive value in moderation. Freud held that everyone defends all the time and that we could not survive long without defences.
- They become pathological when they are rigid, overused or used where realistic action is needed; then they produce neurotic or even psychotic symptoms.
- They are not rational. A defence does not solve the problem; it changes how the problem is seen.

> **EXAM TRAP**
>
> Asked which statement does NOT describe defence mechanisms, the answer is "rational unconscious protective measures". "Having adaptive value", "pushing painful experiences out of consciousness" and "resulting in a distorted view of reality" are all true. The single word *rational* makes the option false.

Defence should not be confused with **coping**, the conscious, deliberate, flexible and reality-oriented effort to manage stress (problem-focused and emotion-focused coping are taught in Section 1.11). If Arjun draws up a revision timetable he is coping; if he decides the subject is useless he is defending. The line blurs only at the top of the maturity ladder (see below), where suppression and anticipation look much like good coping.

Repression, the primary defence ●●●

Freud called the theory of repression "the corner-stone on which the whole structure of psycho-analysis rests" (1914). In full, **repression** is the involuntary, unconscious blocking of threatening or anxiety-provoking memories, wishes and feelings from awareness, or the barring of them from ever reaching it. Freud separated two forms: **primal repression**, which keeps instinctual ideas from ever becoming conscious, and **repression proper** (after-pressure), which pushes out material that was once conscious and is now linked with the forbidden. Repression is the primary defence against anxiety: it is the most basic and most frequently used, the other mechanisms are often described as back-ups when it fails, and Freud saw it in all neurotic behaviour. It can act on memories, on perception (we may fail to see a disturbing event in front of us) and even on bodily function, as when a man represses his sex-

ual drive so strongly that he becomes impotent. What is repressed is not destroyed; it stays active in the unconscious and returns in disguise as dreams, slips and symptoms.

In memory terms, then, repression is an unconscious process that removes unpleasant memories from awareness. It is not a conscious effort to forget, not a strengthening of memory traces and not the deliberate recall of traumatic events. Several neighbouring terms are easily confused with it, and Table 4.8 sets them side by side.

Table 4.8 Repression and its look-alikes.

Term	Conscious?	What it is	Example
Repression	No, involuntary	Blocks threatening material from awareness	No memory of a childhood humiliation
Suppression	Yes, deliberate	Puts a worry aside to face later; mature in Vaillant's scheme	"I will think about the result after the exam"
Inhibition	No	Gives up an ego function (work, eating, sex) to avoid anxiety (Freud, 1926)	Cannot bring himself to start any work
Denial	No	Refuses to accept an outer reality that is perceived	"The biopsy report must be wrong"
Motivated forgetting	Can be either	The broad memory term: any forgetting driven by a motive (Chapter 1)	Includes both repression and suppression
Depression	Not a defence	A mood state and disorder (Chapter 8)	Persistent sadness, loss of interest

EXAM TRAP

Stems that say "involuntary", "unconscious" or "unable to gain access to conscious awareness" (Deepika cannot bring the threatening material into awareness) want **repression**. "Decides to put it out of mind" is **suppression**. Depression sounds similar but is a mood, not a mechanism, and "motivated forgetting" is the wider memory label, not the psychoanalytic defence.

The recovered memory controversy. From the late 1980s, adults in therapy began to "recover" memories of childhood sexual abuse, and some accused relatives were tried and imprisoned on that evidence. **Elizabeth Loftus** (born 1944) and others showed how easily suggestion, hypnosis and dream interpretation can implant vivid memories of events that never happened. Abuse is real and can go unrecalled for years, but Loftus, reviewing the evidence in 2006, concluded that there is little support for the idea that trauma is commonly banished from awareness and later reliably recovered. Repression remains a clinical concept rather than a proven memory mechanism (misinformation and false memory are taught in Chapter 1).

Repression, hysteria and dissociation. The concept was born in *Studies on Hysteria* (1895), written by Freud with **Josef Breuer** (1842 to 1925). Hysterical symptoms, they argued, came from painful memories pushed out of awareness that found expression through the body. For the same reason most psychodynamic theorists hold that dissociative symptoms (amnesia, fugue, multiple identities) are produced by repression: a whole segment of experience is walled off from consciousness. Conflict is what sets defence going, not the mechanism itself;

"ego states" describe the split-off parts rather than the process that splits them; and gender identity has no bearing on dissociation.

Five defences that redirect or disguise ●●●

Most questions on defences give a short story and ask you to name the mechanism. Five defences carry most of these stories. Three of them move an impulse somewhere, so the key question is where it goes; Figure 4.3 follows one impulse through all five.

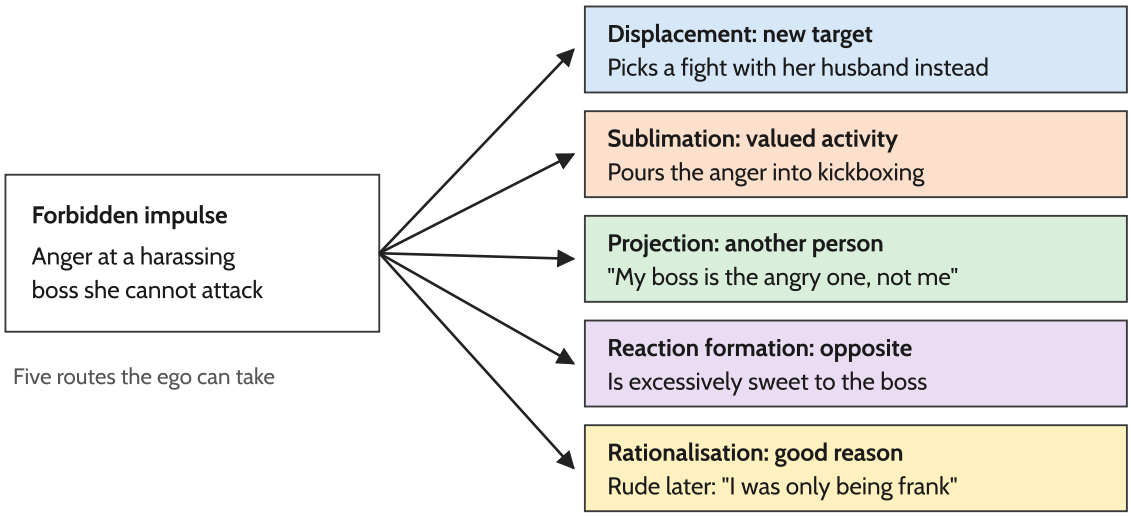


Figure 4.3 One impulse, five disguises. Notice what changes in each: the target (displacement), the activity (sublimation), the owner (projection), the direction (reaction formation) or the stated reason (rationalisation).

In **displacement** a feeling, usually anger, is redirected from a threatening target to a safer one that cannot hit back. The impulse stays the same; only its object changes. The woman angry with her husband shouts at her child; the teenager furious with a teacher goes home and shouts at his younger sibling; the man upset with his girlfriend yells at his dog; the woman harassed by her boss picks a fight with her husband. Because the substitute never discharges the tension as fully as the real target would, undischarged tension builds up. Displacement of fear, rather than anger, is the classic mechanism of phobias (see below).

In **projection** a person attributes his or her own unacceptable thoughts and impulses to someone else. The person says, in effect, "I don't hate him; he hates me." A husband who struggles with his own wish to stray accuses his faithful wife of flirting. The impulse is still expressed, but as somebody else's, which feels far less threatening.

In **sublimation**, sexual or aggressive energy is rechannelled into a socially valued activity. Roshan, who is full of anger and frustration, enrols in a kickboxing class; aggressive drives go into competitive sport, and frustrated sexual energy goes into painting or poetry. Displacement changes only the target, whereas sublimation transforms the impulse's aim into something society admires. Freud saw art, science and civilisation itself as built on sublimation and treated it as the healthiest defence; Vaillant later placed it among the mature ones.

The term **rationalisation**, introduced by Freud's follower **Ernest Jones** (1879 to 1958) in 1908, means a logical, socially acceptable reason invented after the act for behaviour that was really driven by unconscious or unacceptable motives. Clare complains about her duties, rarely finishes her tasks and is regularly late; when she is fired she says a co-worker badmouthed her. Arjun's "statistics is useless anyway" is the "sour grapes" form, after the fox in the fable who decides that the grapes he cannot reach are sour. Unlike a lie, the excuse is believed by the person who gives it.

In **reaction formation** the opposite of a threatening impulse is expressed: someone troubled by sexual longings becomes a crusader against pornography, and a child who resents a new baby smothers it with exaggerated affection. The giveaway is the excessive, rigid quality of the displayed feeling.

> EXAM TRAP
>
> Read the stem for its landing place: a **safer target** is displacement, a **socially valued activity** is sublimation, **another person's mind** is projection. Clare blames a co-worker, which tempts you towards projection or displacement, and she plays down her failings, which tempts you towards denial. But she does not deny the dismissal, and she moves no anger onto a safer target; she supplies an after-the-fact excuse that protects her self-esteem: rationalisation. Do not confuse rationalisation (an excuse) with intellectualisation (abstract thinking that keeps feeling away).

The wider catalogue

Beyond the big five lies a long list drawn from Freud, Anna Freud, **Alfred Adler** (1870 to 1937) and the object-relations school. Table 4.9 gives each one a definition and an everyday example; the defences found only in Vaillant's ladder are defined with it (below).

Table 4.9 The wider catalogue of defences.

Defence	What the person does	Everyday example
Denial	Refuses to acknowledge a painful outer reality	Parents keep a dead child's room unchanged
Regression	Retreats to the behaviour of an earlier, safer stage	A six-year-old wets the bed again after a sibling's birth
Fixation	Part of the libido stays tied to an earlier psychosexual stage (Section 4.2)	Lifelong oral habits such as nail-biting
Intellectualisation	Uses abstract, technical thinking to avoid feeling	Discusses her own cancer only as survival statistics
Isolation of affect	Keeps the thought but strips away its feeling	Describes a gruesome fantasy with no anxiety or disgust
Undoing	Performs an act that symbolically cancels a forbidden thought or act	Washes hands repeatedly after a hostile thought
Identification	Takes on another's qualities to feel safer or stronger	A boy copies his father's manner
Identification with the aggressor	Becomes like the person one fears (Anna Freud)	A child frightened at the dentist plays at being the dentist
Introjection	Takes another's values or qualities into the self; the reverse of projection	Parents' rules become one's own conscience

Defence	What the person does	Everyday example
Turning against the self	Redirects hostility from another person onto oneself	Bitter self-blame after a parent's death
Compensation and overcompensation	Makes up for a weakness by excelling elsewhere; or by over-striving in the very area of weakness (Adler, Section 4.4)	A frail boy becomes a chess champion; a stammerer becomes an orator
Conversion	Turns psychic conflict into a bodily symptom with symbolic meaning	An arm is paralysed in a man who wished to strike his father
Somatisation	Expresses distress as bodily complaints	Stomach aches every school morning
Dissociation	Drastically and temporarily alters memory, awareness or identity to escape distress	No memory of a terrifying accident
Splitting	Sees self or others as all good or all bad, never both	A doctor is "perfect" on Monday and "a monster" by Friday
Projective identification	Projects a disowned feeling into another, then behaves so that the other comes to feel it (Melanie Klein)	A patient provokes the therapist until the therapist feels the patient's rage
Acting out	Expresses an unconscious impulse in action so as not to feel it	A rejected teenager gets into fights instead of feeling hurt
Passive aggression	Expresses hostility indirectly, through passivity	Keeps "forgetting" a colleague's request

Several entries are worth separating. Three belong together because all appear in obsessional neurosis: **isolation of affect** keeps the idea in awareness but removes its emotion, **intellectualisation** goes further and buries the emotion under abstract analysis, and **undoing** adds an act that magically "cancels" the thought. Plain **identification** copies another's behaviour, and **identification with the aggressor**, Anna Freud's special case, copies the very person one fears; **introjection** goes deeper and swallows another's values whole, so that they feel like one's own. Note too that **regression**, a retreat to the behaviour of an earlier stage, is a general response to stress rather than the mark of any one disorder, and **dissociation**, though dramatic, is ranked neurotic rather than immature by Vaillant. In the same way, **conversion** (Freud's term from his work on hysteria) produces a single symbolic symptom such as paralysis or blindness, whereas somatisation is a general tendency to express distress through the body. Finally, **splitting**, described by **Melanie Klein** (1882 to 1960) in the infant's paranoid-schizoid position (Section 4.5), keeps good and bad images apart so that the bad cannot spoil the good; **projective identification**, also Klein's, adds the push that makes the other person actually carry the projected feeling. Adler's compensation is treated fully in Section 4.4.

Projection, introjection, empathy and denial apart

Projection has a broad sense as well as the narrow defensive one. Freud also used the word for any externalising of inner states, and in this broad sense projection is any distorting or selective effect of a person's inner world on his perception or conception of the outer world. This broad sense underlies the **projective hypothesis**: given an ambiguous stimulus, people read their own needs and conflicts into it. **Lawrence K. Frank** named such tests projective methods in 1939 (they are taught in Chapter 2).

The neighbouring terms run in other directions. Introjection moves something from outside to inside, the reverse of projection. By contrast, **empathy** is perceiving and sharing another person's feelings from that person's point of view; it is an accurate act of understanding, not a defence, and it distorts nothing. And **denial** blocks a particular, unwelcome fact of outer reality ("my son cannot be dead"); it does not colour perception in general.

> EXAM TRAP
>
> RML 2024 asked for "any distorting or selective effect of a person's inner world on his perception or conception of the outer world" and keyed **denial**. That wording is the broad definition of **projection**, the basis of the projective hypothesis, so the key is doubtful. Denial refuses a specific reality; introjection takes the world in; empathy reads others accurately. If the same stem appears, projection is the defensible answer.

Vaillant's hierarchy of maturity ●●●

Are some defences healthier than others? **George Eman Vaillant** (born 1934), an American psychiatrist, answered with decades of life-history data (see the Key study box) and arranged the defences in four levels of maturity (Table 4.10).

Table 4.10 Vaillant's four levels of defence (1977).

Level	Defences	Hallmark
I Psychotic	Delusional projection, psychotic denial, distortion	Reality testing lost; normal in dreams and early childhood
II Immature	Projection, schizoid fantasy, hypochondriasis, passive aggression, acting out	Normal in adolescence; in adults, common in personality disorders
III Neurotic (intermediate)	Intellectualisation, isolation, repression, displacement, reaction formation, dissociation	Common in healthy adults under stress; private, inner distortions
IV Mature	Altruism, anticipation, humour, sublimation, suppression	Integrate reality, feelings and relationships; most adaptive

Several names appear only in this ladder. At Level I, **delusional projection** hardens projection into a frank, usually persecutory delusion ("the neighbours are poisoning my water"); **psychotic denial** denies outer reality itself, as when a widow insists her husband is alive; **distortion** grossly reshapes reality to fit inner needs, as in a belief in one's own divine powers. At Level II, **schizoid fantasy** is retreat into private daydreams to avoid conflict and people, and **hypochondriasis** turns reproach towards others (after bereavement or loneliness) into complaints of illness. At Level IV, **altruism** gains satisfaction from constructive service to others, **humour** expresses painful feelings openly without hurting anyone, **anticipation** is realistic planning for future discomfort, and **suppression** is the conscious decision to postpone attention to a worry.

KEY STUDY: VAILLANT AND THE GRANT STUDY MEN (1977)

Question: do the defences a person habitually uses predict how well his life turns out? **Method:** the Grant Study of Adult Development has followed 268 Harvard men since 1938 through interviews, questionnaires and medical examinations. For *Adaptation to Life* (1977), judges rated the defences the men showed in crises over more than three decades. **Result:** the maturity of defences predicted adult adjustment: men who relied on mature defences had better marriages, careers, and physical and mental health, and defences tended to mature with age. **Criticism:** the sample was all male, mostly white and privileged, and the ratings rest on judges' inferences from life histories.

Some psychiatry textbooks extend the lists, calling Level I narcissistic and adding, for example, rationalisation and inhibition to the neurotic group and regression and somatisation to the immature group. The DSM-IV Defensive Functioning Scale (1994) later spread the defences over seven levels, from the high adaptive level down to defensive dysregulation, placing splitting and projective identification at the major image-distorting level.

EXAM TRAP

Repression, projection and reaction formation are **not** mature; only altruism, anticipation, humour, sublimation and suppression are. Splitting (primitive), sublimation (mature) and projection (immature) are **not** neurotic, so intellectualisation is the neurotic choice. **Dissociation is neurotic**, not immature. Denial, distortion and delusional projection together mark the psychotic level. No defence is "conscious" as a class; suppression is the one deliberate exception, and it sits in the mature group.

Defences typical of particular disorders ●●●

Psychoanalysts noticed that each neurosis seemed to favour its own defences, and Table 4.11 lists the classic pairings (the disorders themselves are taught in Chapter 8).

Table 4.11 Characteristic defences of particular disorders.

Disorder	Characteristic defences	Note
Obsessive-compulsive disorder	Isolation of affect, undoing, reaction formation, intellectualisation	Repression is the one not typical
Phobias	Displacement, with projection and avoidance	Little Hans (1909)
Hysteria, conversion and dissociative disorders	Repression, conversion, dissociation	*Studies on Hysteria* (1895)
Paranoia	Projection	Klein's splitting in the paranoid-schizoid position
Borderline personality disorder	Splitting, projective identification	Otto Kernberg
Depression	Introjection, turning against the self	Anger at a lost person turned inward
Mania	Denial	Denial of depressive pain
Schizophrenia	Psychotic-level defences	Delusional projection, distortion

KEY STUDY: LITTLE HANS (1909)

In *Analysis of a Phobia in a Five-Year-Old Boy* (1909), Freud described **Herbert Graf**, a Viennese boy who was terrified that a horse would bite him and would hardly go out into the street. The case was conducted mainly through the boy's father, a follower of Freud, who reported to him by letter. Freud concluded that Hans's Oedipal fear of his father had been displaced onto horses, whose blinkers and dark muzzles recalled the father's spectacles and moustache; a horse, unlike a father, can simply be avoided. **Criticism:** the evidence came from a sympathetic father asking leading questions, and **Joseph Wolpe** and **Stanley Rachman** (1960) argued that the phobia was conditioned after Hans saw a horse fall in the street.

The logic behind the table is worth knowing. In obsessional neurosis the disturbing idea stays fully conscious (that is what an obsession is), but its feeling is isolated, compulsions undo it, and excessive cleanliness or kindness stands in reaction to it; hysteria, by contrast, represses the idea itself. In phobia, a fear is displaced onto an object that can be avoided, as in Little Hans. In paranoia, the forbidden feeling is projected, so that hatred within becomes persecution from without. **Otto Kernberg** (born 1928) made splitting and projective identification the core of borderline personality organisation, which is why the person with borderline personality disorder swings between idealising and devaluing the same person; regression, undoing and sublimation are not characteristic. In depression, following Freud's *Mourning and Melancholia* (1917), the lost person is introjected and the anger meant for them is turned against the self.

EXAM TRAP

"Which defence is used in OCD: isolation, undoing, reaction formation, displacement?" Three options are true. When the key is **undoing** (RML 2024), take it: undoing is the defence most directly tied to compulsive rituals. When the stem asks which is NOT typical of OCD, the answer is **repression**. Asked where displacement is typically seen, answer **phobias** (IHBAS 2020), not hysteria, OCD or depression.

Development, research and clinical use

Defences have a developmental order. **Phebe Cramer**, coding children's Thematic Apperception Test stories (1987), found that denial dominates in early childhood, projection rises in later childhood, and identification takes over in adolescence, as Figure 4.4 shows. Each step demands more cognitive skill: denial needs only a refusal, projection needs a sense of other minds, identification needs a stable self that can borrow from others. A later study of pupils from Grade 2 to college found the same pattern: denial and projection fell with age while identification rose.

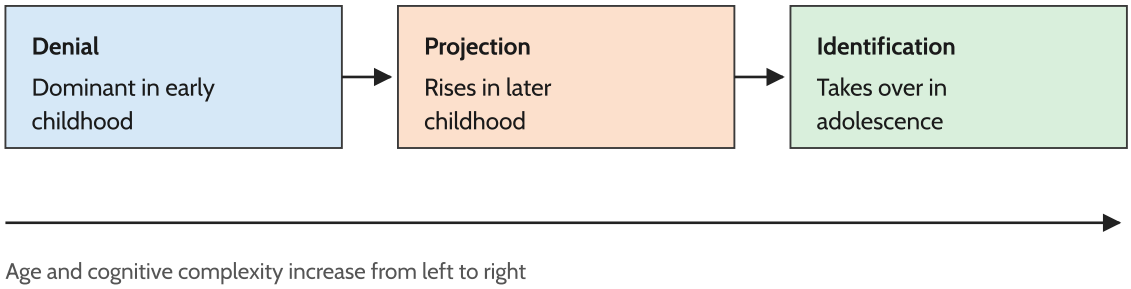


Figure 4.4 Cramer's developmental sequence. Simpler defences come first and give way to ones that need more cognitive skill.

Laboratory evidence for repression is debated. **Sam Glucksberg** and L. J. King (1967) paired some words with electric shock and found that people later forgot those words and related ones, which looks like repression. Others have found that habitual "repressors", people low in reported anxiety but high in defensiveness, recall fewer negative childhood memories. Yet **David S. Holmes** (1990), reviewing sixty years of studies, found no convincing controlled evidence that repression works as Freud described, and the recovered memory debate has kept the question open.

In everyday life defences are neither rare nor shameful: they soften losses, protect self-esteem and buy time. Clinically, they matter in two ways. The defences a person habitually uses help a clinician gauge the level of personality organisation, and in psychodynamic therapy defences show up as resistance, which the therapist gently interprets so that the patient can face what is being avoided and replace rigid, immature defences with mature ones (therapy is taught in Chapter 7).

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Anna Freud catalogued the defences in *The Ego and the Mechanisms of Defence*, published in _____.
2. The involuntary, unconscious blocking of threatening material from awareness is _____; its deliberate, conscious counterpart is _____.
3. A man upset with his girlfriend who yells at his dog is using _____.
4. Rechannelling aggression into kickboxing or competitive sport is _____.
5. Describing a gruesome fantasy without any anxiety or disgust shows _____ of affect.
6. In Vaillant's 1977 hierarchy, dissociation belongs to the _____ level, and altruism to the _____ level.
7. The primary defence associated with borderline personality disorder is _____.
8. Cramer's developmental sequence runs from denial to projection to _____.

Answers: (1) 1936, (2) repression; suppression, (3) displacement, (4) sublimation, (5) isolation, (6) neurotic; mature, (7) splitting, (8) identification.

4.4 Jung's Analytical Psychology and Adler's Individual Psychology

MEDIUM YIELD · 25 QS

In March 1907 a 31-year-old Swiss psychiatrist, Carl Jung, met Sigmund Freud in Vienna, and the two talked for almost thirteen hours. Six years later they were no longer speaking, and Alfred Adler had already walked out of Freud's circle. Each rebel kept part of Freud's framework but rebuilt the theory of personality at the point where he thought Freud had gone wrong.

The neo-Freudians and the break with Freud

The **neo-Freudians** began inside psychoanalysis, kept its belief in unconscious processes and in the weight of childhood, and then rebuilt the theory. The usual list is Carl Jung, Alfred Adler, Karen Horney, Erich Fromm, Harry Stack Sullivan and Erik Erikson (Horney, Fromm and Sullivan are in Section 4.5, Erikson in Chapter 6). Their changes ran one way: less sex and instinct; more social, cultural and conscious forces; and a person pulled toward goals, not only pushed by the past.

Adler left first. A member of Freud's circle from 1902 and president of the Vienna Psychoanalytic Society, he argued that inferiority and social striving, not sexuality, lay behind neurosis. In 1911 he resigned with 9 of the 23 members, and in 1912 he founded the Society for Individual Psychology. Jung was Freud's chosen successor. Freud called him his "crown prince", took him on the 1909 lecture trip to Clark University in Massachusetts, and saw him become first president of the International Psychoanalytic Association in 1910. But Jung would not accept that libido was purely sexual, and they parted in 1913 (Table 4.12).

Table 4.12 Freud, Jung and Adler compared.

Issue	Freud	Jung	Adler
School	Psychoanalysis	Analytical psychology	Individual psychology
Main motive	Sexual and aggressive instincts	General psychic energy	Striving for superiority; social interest
Unconscious	Personal; central	Personal and collective; even more central	Played down
Direction	Pushed by the past	Past and future	Pulled by future goals
Key period	First five years	Middle age	First four or five years
Left Freud	(founder)	1913	1911

> EXAM TRAP
>
> Lists of neo-Freudians hide outsiders: Hans Eysenck (traits, Section 4.7), Albert Bandura (social-cognitive, Section 4.9) and Carl Rogers (humanist, Section 4.8). Freud is the founder, not a neo-Freudian. Jung, Horney and Adler is the safe triple.

Jung's psyche: energy, ego and complexes

Carl Gustav Jung (1875 to 1961), a Swiss pastor's son, worked from 1900 at the Burghölzli hospital in Zurich under Eugen Bleuler. His **analytical psychology** begins with energy. Jung

widened libido from sexual energy to general psychic energy, the energy of all perceiving, thinking, feeling and wishing; sex is only one of its uses. It obeys three principles borrowed from physics. By the **principle of opposites**, conflict between polarities, such as love and hate, generates energy. By the **principle of equivalence**, energy withdrawn from one interest reappears in another of equal value, as when a retired cricketer's passion flows into coaching. By the **principle of entropy**, energy flows from stronger to weaker parts of the personality, tending toward a balance that is never reached.

The psyche has three levels (Figure 4.5). The ego is the conscious mind, the centre of awareness. Below it lies the **personal unconscious**, material once conscious but forgotten or suppressed, much like Freud's preconscious. There related memories and feelings gather into complexes. A **complex** is an emotionally charged cluster of ideas organised around a theme, such as a mother complex or a power complex; its owner rarely notices it, though others do. To detect complexes Jung developed, from 1904 with **Franz Riklin**, the **word association test**.

KEY STUDY: JUNG'S WORD ASSOCIATION TEST (1904 ONWARDS)

Question: can a hidden complex be detected objectively? **Method:** Jung read out 100 stimulus words (head, green, water, death), the patient answered each with the first word that came to mind, and Jung timed every reply and recorded physiological signs such as breathing and skin resistance. **Result:** certain words produced complex indicators: a long delay, no answer, repeating the stimulus word, the same reply to different words, a slip, or a bodily reaction. These marked the theme of a complex. **Legacy and criticism:** the test fed projective testing and lie detection, but its interpretation stays subjective. Francis Galton and Wilhelm Wundt had studied word association earlier; as a clinical tool for complexes it is Jung's.

EXAM TRAP

Word association test: Jung. Not Henry Murray (the Thematic Apperception Test, Chapter 2), not Freud (free association, talking freely with no stimulus words) and not Donald Winnicott (object relations: the transitional object and the good enough mother, Section 4.5).

The collective unconscious and the archetypes ●●●

Freud's unconscious was essentially personal. Jung held that the unconscious exists on two levels, the personal (individual) and the collective. The **collective unconscious** is the deepest layer of the psyche: inherited, universal, the same in every human being, the deposit of the experience of our human and pre-human ancestors. Jung accounted for it through evolution: the brain has the same structure in all human races, so it carries the same inherited predispositions. His evidence was that the same symbols (the dragon-slaying hero, the great flood, the sacred circle) recur in the myths of cultures that never met and in the dreams of patients who had never read those myths. We inherit readiness, not memories: not a fear of snakes, but the potential to fear them.

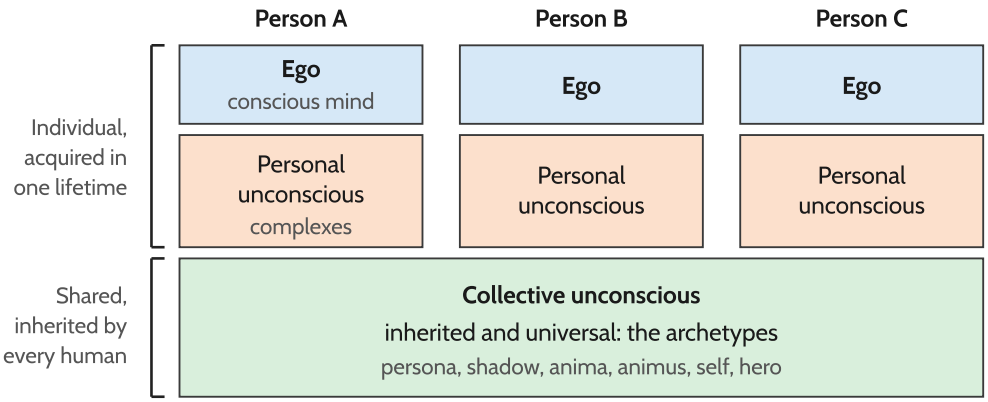


Figure 4.5 Jung's model of the psyche. Each person has an ego and a personal unconscious of his or her own, but all rest on the same collective unconscious, the inherited store of archetypes.

The contents of the collective unconscious are **archetypes**, which Jung also called primordial images: universal patterns that shape how we respond to life's recurring experiences, such as mother, birth and death (Table 4.13). The **persona** is the social mask; trouble starts when the ego mistakes it for the whole person. The **shadow**, the most powerful archetype, is the disowned animal side, the source of evil but also of vitality and creativity. The **anima** is the feminine side of a man and the **animus** the unconscious masculine side of a woman. The **self**, the archetype of wholeness and unity, is the centre of the whole psyche (the ego is only the centre of consciousness), symbolised by the **mandala**, a circle usually divided into four.

Table 4.13 Jung's main archetypes.

Archetype	What it is	Typical image
Persona	Social mask, public role	Uniform, stage mask
Shadow	Dark, animal, disowned side	Devil, villain
Anima	Feminine side of a man	Muse, idealised woman
Animus	Masculine side of a woman	Idealised man, rescuer
Self	Wholeness and unity	Mandala, circle
Hero	Triumph over great odds	Dragon slayer
Great mother	Nurturing source of life	Earth mother
Wise old man	Wisdom and guidance	Sage, guru
Trickster	Mischief that upsets order	Clown, cunning fox
Child	New beginnings, potential	Divine child

> **REMEMBER**
>
> *Anima* carries the feminine Latin ending -a (as in *femina*, woman): the woman within a man. *Animus* carries the masculine -us: the man within a woman.

> **EXAM TRAP**
>
> The collective unconscious is not repressed personal memories (the personal unconscious), not socially learned attitudes and not ruled by the reality principle (Freud's ego, Section 4.2). It is inherited, universal and archetypal, and it is Jung's, not Freud's, Adler's or Horney's. Asked for the two levels of the unconscious, answer collective and individual: anima and animus are archetypes, extraversion and introversion are attitudes, and intuitive and perceptive are type labels.

Attitudes, functions and the eight types

In *Psychological Types* (1921) Jung asked why people meet the same world so differently. His first answer was two attitudes, his own terms for the direction of psychic energy. In **extraversion** it flows outward, to people and things: the extravert is sociable and assertive. In **introversion** it flows inward, to one's own thoughts and feelings: the introvert is reserved and reflective. Everyone has both; one dominates consciousness, while the other sinks into the personal unconscious and still leaks out, as when a shy student is drawn to a loud friend.

His second answer was four functions in two opposed pairs (Figure 4.6). Thinking judges true or false; feeling judges pleasant or unpleasant. Because both evaluate, they are the **rational functions**. Sensing registers what is there, like a camera; intuiting grasps possibilities and hunches beyond the evidence. Because both perceive without evaluating, they are the **irrational functions** (also called nonrational). One function dominates; the other member of its pair, the **inferior function**, stays undeveloped and largely unconscious, so a strong thinker may be clumsy with feelings. Two attitudes times four functions give Jung's eight psychological types, from the extraverted thinking type (logical, rule-bound) to the introverted intuiting type (the visionary daydreamer).

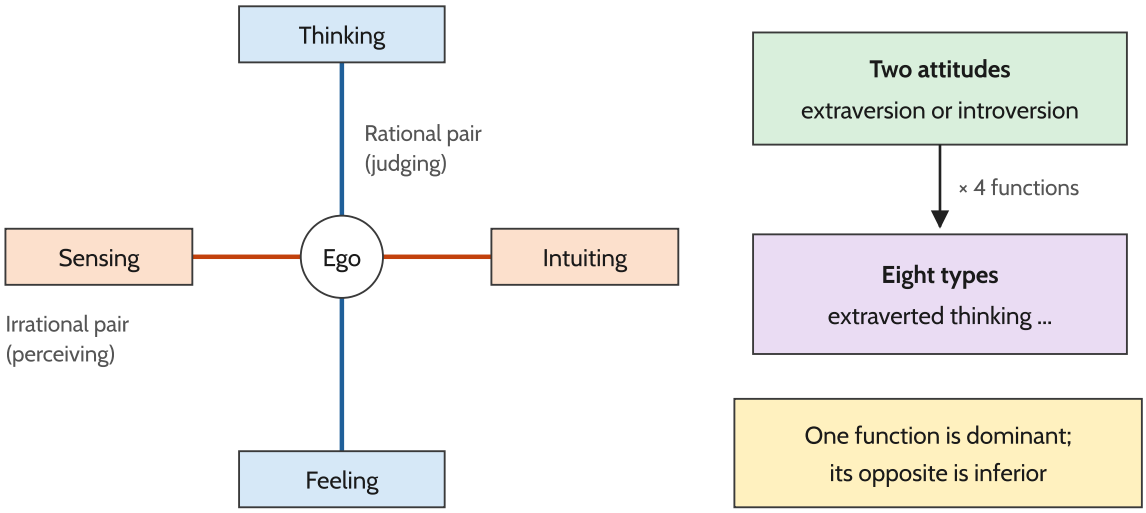


Figure 4.6 Jung's four functions form two opposed pairs: thinking and feeling judge (rational, vertical axis); sensing and intuiting perceive without judging (irrational, horizontal axis). Crossed with the two attitudes they give eight types.

Katharine Cook Briggs (1875 to 1968), who read Jung's book in 1923, and her daughter **Isabel Briggs Myers** (1897 to 1980), neither trained in psychology, turned the types into the **Myers-Briggs Type Indicator** (MBTI), whose first form appeared in 1943. It scores four di-

chotomies, extraversion-introversion (E-I), sensing-intuition (S-N), thinking-feeling (T-F) and judging-perceiving (J-P, their own addition), giving 16 types such as INTP. Hugely popular in workplaces, it is psychometrically weak: it forces continuous traits into either-or boxes, and its test-retest reliability is poor, with many people (in some studies about half) typed differently when retested a few weeks later. Researchers prefer trait inventories (Section 4.7; Chapter 2).

> EXAM TRAP
>
> Jung's two attitudes are extraversion and introversion; thinking, feeling, sensing and intuiting are functions; rational and irrational label the pairs of functions. The inventory based directly on Jung's types is the MBTI, not the MMPI (an empirically keyed clinical inventory), the 16PF (Raymond Cattell's factors, Section 4.6) or the NEO-PI-R (Paul Costa and Robert McCrae's Big Five, Section 4.7).

Individuation, method and evaluation of Jung

Freud thought personality was largely settled by five; Jung's crucial period was middle age, the time of his own breakdown after 1913. In the first half of life energy goes outward, to education, career and family, under consciousness and extraversion. Between about 35 and 40 many successful people feel empty. Jung saw this midlife crisis as natural, a signal to turn inward, from the material to the spiritual and from extraversion toward introversion. The goal of the second half of life is **individuation**, becoming a whole, undivided individual by integrating conscious and unconscious: dethrone the persona, accept the shadow, express the anima or animus, and transcend, so that the self, not the ego, becomes the centre. Such self-realisation is rarely complete. **Synchronicity**, a meaningful coincidence with no causal link, belongs to the same mystical side of Jung.

Jung assessed patients by word association, symptom analysis (free associations to the symptoms) and dream analysis. He studied series of dreams, saw them as prospective (preparing for the future) and compensatory (balancing a one-sided conscious attitude), and used **amplification**: circling back to one dream image and enlarging it with repeated associations and parallels from myth, instead of following free associations away from it. Critics call his evidence (myth, alchemy, the occult, his own visions) mystical, his writing obscure and the collective unconscious unfalsifiable. Yet introversion-extraversion (Section 4.7), complexes, word association, the midlife transition and individuation, a forerunner of Abraham Maslow's self-actualisation, all lasted.

Adler: the individual as a goal-seeking whole ●●

Alfred Adler (1870 to 1937), born near Vienna, was a sickly second son with rickets who nearly died of pneumonia at four and envied a healthy older brother. A physician, he joined Freud's circle as a colleague, never a pupil, and after the break founded **individual psychology**; he died on a lecture tour in Scotland. Individual here means indivisible: each person is a unique, unified whole, not an id, ego and superego at war, and the whole person moves toward a goal. The basic assumption of individual psychology is that human behaviour is purposeful and goal-directed: Freud explained behaviour by past causes, Adler by future goals that pull us forward. The goals are social. In place of Freud's sexual instinct, Adler held that people are motivated primarily by social urges, the need to belong and to count among oth-

ers. He gave the conscious more weight than the unconscious and saw people hopefully, as free to shape themselves.

> **EXAM TRAP**
>
> The basic assumption is purposeful, goal-directed behaviour. Optimism is a feature of Adler's view, not its basic assumption; "social beings understood through their relationships" describes his social emphasis but fits Sullivan's interpersonal theory better (Section 4.5); lifelong development with ego identity at the centre is Erik Erikson (Chapter 6). The motive is social urges: not conscious drives (drive is Freud's language), not neurotic needs (Karen Horney, Section 4.5) and not empathic understanding (a therapist condition of Carl Rogers, Section 4.8). Individual psychology is Adler's, not Horney's, Jung's or Erikson's.

Inferiority, compensation and striving for superiority ●●●

Adler's *Study of Organ Inferiority and Its Psychical Compensation* (1907) argued that a weak organ or body part drives the person to make up for it, sometimes overshooting into strength, as the Greek orator Demosthenes overcame a stutter. This idea of **organ inferiority** grew into a theory of all motivation. Every child starts small and dependent among powerful adults, so inferiority feelings are universal and normal, and they are the source of all human striving. The effort to overcome them is **compensation**; going beyond the deficit, as the frail boy who becomes an athlete does, is overcompensation (Figure 4.7).

When compensation fails, the feelings harden into an **inferiority complex**, Adler's term: a lasting lack of self-worth, with helplessness, hopelessness and avoidance of challenge. It has three childhood sources: organ inferiority, pampering (the spoilt child crumbles when school does not revolve around him) and neglect (the unwanted child learns that others are hostile). Children with organic defects or born into minority groups are therefore at risk. Overcompensation can instead produce a **superiority complex**, an exaggerated self-importance shown in boasting, vanity and belittling others.

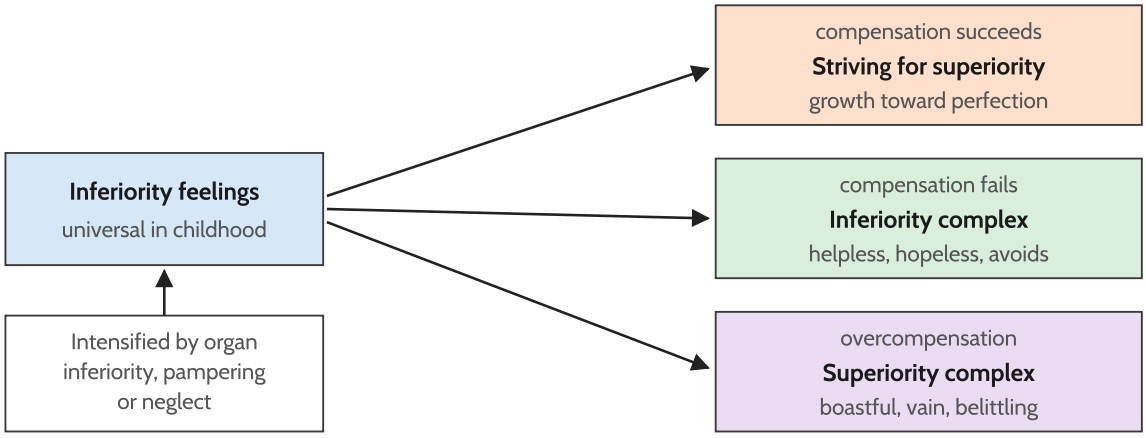


Figure 4.7 Adler's account of inferiority. Universal inferiority feelings drive all striving; successful compensation becomes striving for superiority, failure an inferiority complex, and overcompensation a superiority complex.

Neither complex is **striving for superiority**, the healthy, innate drive toward perfection (completion) that Adler called the fundamental fact of life: an effort to perfect oneself and

one's community, not to dominate others.

EXAM TRAP

The inferiority complex and the superiority drive are Adler's, not Jung's, Viktor Frankl's (logotherapy, Chapter 7) or Erikson's. Adler coined the term, not Franz Alexander (psychosomatic medicine), Gordon Allport (traits, Section 4.6) or Jung; obituaries wrongly credited Freud and Jung with it. Organ inferiority is Adler's too, not Sullivan's, Melanie Klein's (both Section 4.5) or Anna Freud's (Section 4.3). Overcompensation, lack of self-worth and hopelessness go with the complex; certainty in decision making is its opposite.

Fictional finalism, style of life and social interest

What goals pull us? Adler borrowed from the philosopher **Hans Vaihinger** (1852 to 1933), whose *The Philosophy of "As If"* (1911) argued that people live by useful fictions, unprovable ideas acted on as if they were true, such as "all people are created equal". Adler's **fictional finalism** is the idea that each person is guided by an imagined final goal, a subjective ideal of success or completeness; he preferred the terms subjective final goal or guiding self-ideal.

Each person pursues the goal through a unique **style of life**, the characteristic pattern of traits, habits and attitudes by which one strives for superiority. It grows from the child's compensations, birth order and parenting, and is largely fixed by four or five. Yet it is not fate: through the **creative power of the self** each child interprets heredity and environment and builds its own style.

The test of a healthy style is **social interest** (*Gemeinschaftsgefühl*, community feeling), an innate potential to cooperate with others for personal and social goals, which the mother must nurture. It is needed for the three life tasks of work, friendship and love, and its absence underlies neurosis and crime. Crossing social interest with degree of activity, Adler described four styles of life (Table 4.14).

Table 4.14 Adler's four styles of life.

Type	Social interest	Activity	Pattern
Ruling or dominant	Low	High	Dominates others; extreme forms turn delinquent
Getting or leaning	Low	Low	Expects to receive; dependent; the commonest type
Avoiding	Low	Lowest	Sidesteps problems to avoid failure
Socially useful	High	High	Cooperates and contributes; healthy

EXAM TRAP

Fictional finalism is an imagined goal that guides behaviour: not parental expectations, a repressed unconscious conflict or a biologically determined trait. In matching items, Jung with extraversion-introversion, Adler with the ruling dominant type and Albert Ellis with rational emotive therapy (Chapter 7) are correct pairs; Kohlberg with formal operations is wrong, because formal operations is Jean Piaget's stage and Lawrence Kohlberg's theory is of moral reasoning (Chapter 6).

Birth order, early recollections and Adler's legacy

For Adler, siblings raised by the same parents do not grow up in the same family, because each birth position is a different situation (Table 4.15). The first-born rules alone until a new baby dethrones it, and so knows power and wants it back; the second has a pacesetter ahead and, Adler said, behaves as if in a race. Large modern studies, however, find birth-order effects on personality traits weak or near zero.

Table 4.15 Adler's birth-order positions.

Position	Situation	Likely style
First-born	Dethroned by the next baby	Power-minded, past-oriented, conservative, responsible; may turn insecure
Second or middle	A pacesetter always ahead	Ambitious, competitive, rebellious, always in a race; may give up
Youngest	Never dethroned; family pet	Pampered; may stay spoilt and dependent or become highly ambitious
Only child	Never loses the centre	Matures early among adults; pampered; dislikes sharing the stage

> **EXAM TRAP**
>
> For Adler the middle (second) child is competitive and ambitious, racing to overtake the one ahead. "Overachieving" is too narrow (the second child may instead give up), "resentful of the older child's freedom" is not his description, and inferiority feelings are universal, not the mark of the middle child.

Adler's entrance gates to mental life were birth order, dreams and **early recollections**, a person's earliest memories which, true or imagined, reveal the style of life, because we remember what fits our present view of ourselves. Dreams, for him, rehearse solutions to present problems. His counselling methods are in Chapter 7. Critics call Adler commonsensical and unsystematic, but Abraham Maslow, Julian Rotter (social learning theory) and Albert Ellis (rational emotive therapy) built on his stress on goals and beliefs, and Karen Horney (Section 4.5) on his social emphasis.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Jung widened libido from sexual energy to general _____ energy.
2. Jung detected hidden complexes with the _____ test, timing each reply.
3. The deepest, inherited layer of the psyche, shared by all humanity, is the _____.
4. The feminine side of a man's psyche is the _____; the masculine side of a woman's is the _____.
5. Thinking and feeling are Jung's _____ functions; sensing and intuiting are his _____ functions.
6. Adler's 1907 book on _____ began his theory of compensation.
7. An imagined goal that guides behaviour is Adler's _____.
8. Adler said the _____ child behaves as if in a race.

Answers: (1) psychic, (2) word association, (3) collective unconscious, (4) anima; animus, (5) rational; irrational, (6) organ inferiority, (7) fictional finalism, (8) second (middle).

4.5 Horney, Fromm, Sullivan and the Object-Relations Theorists

LOWER YIELD · 10 QS

Freud's followers kept his couch but argued about what drives the person lying on it. Karen Horney, Erich Fromm and Harry Stack Sullivan moved the source of personality from instinct to culture and relationships. Others kept more of Freud and asked how the ego adapts, how loved people become figures inside the mind, how a cohesive self is built and how language shapes the unconscious. Jung and Adler are in Section 4.4.

Horney: basic anxiety and the neurotic trends ●●

Karen Horney (1885 to 1952), a Berlin-trained analyst who taught in the United States from 1932, argued in *The Neurotic Personality of Our Time* (1937) that personality is shaped primarily by social and cultural factors, above all the child's relationships with its parents, not by biological instincts, unconscious sexual conflicts or classical conditioning.

The helpless infant's overriding need is the childhood **need for safety and security**: to feel wanted and protected. What damages a child is **parental indifference**, a lack of genuine warmth shown through favouritism, unfair punishment, broken promises or ridicule. The child feels **basic hostility** towards such parents but must repress it, out of helplessness, fear, the wish to keep even counterfeit love, or guilt. The buried hostility returns as **basic anxiety**, in Horney's words an "insidiously increasing, all-pervading feeling of being lonely and helpless in a hostile world". It is the foundation on which every later neurosis is built.

EXAM TRAP

Pervasive loneliness and helplessness, the foundation of neurosis, is **basic anxiety**. Neurotic anxiety is Freud's fear that id impulses will break through (Section 4.2); existential anxiety is dread of death, freedom and meaninglessness (Section 4.8); "innate anxiety" is not a technical term. Instincts, sexual conflicts or conditioning are never Horney's answer.

To quiet basic anxiety the child seizes a strategy (winning affection, submitting, gaining power or withdrawing), and used rigidly a strategy hardens into a need. Horney listed ten **neurotic needs**, irrational defences against anxiety that become permanent parts of the personality: affection and approval, a dominant partner, power, exploitation, prestige, admiration, achievement, self-sufficiency and independence, perfection, and narrow limits to life. Everyone feels them at times; they are neurotic when pursued compulsively as the only route to safety. In *Our Inner Conflicts* (1945) she grouped them into three **neurotic trends** (Figure 4.8). The **compliant** person moves toward people, seeking love and leaning on others; the **aggressive** person moves against people, treating life as a struggle; the **detached** person moves away from people, keeping an emotional distance. A healthy person uses all three flexibly; the neurotic is locked into one, and the clash among them is the core of neurosis.

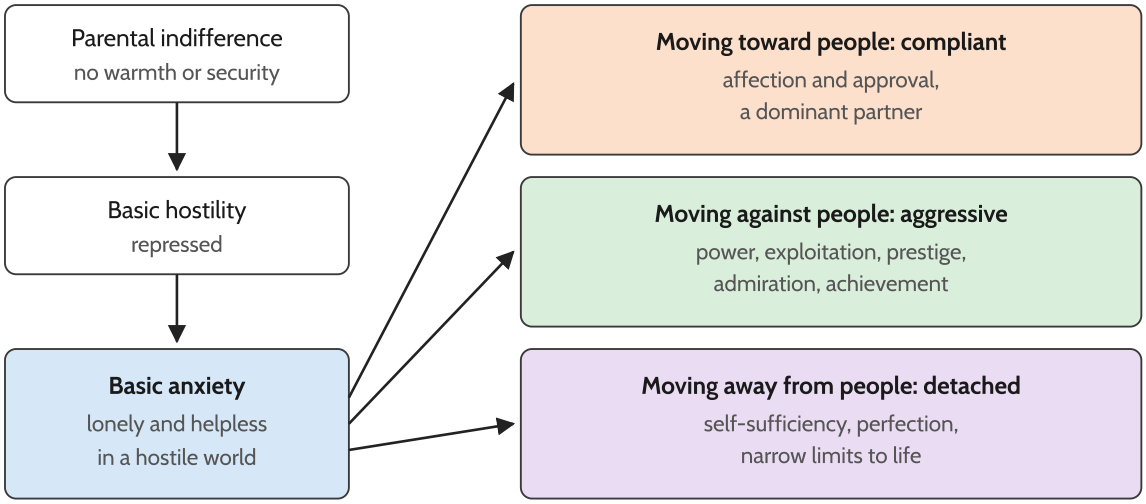


Figure 4.8 Horney's account of neurosis. Repressed hostility becomes basic anxiety, managed by one of three neurotic trends; the small text lists the neurotic needs in each.

INDIAN CONTEXT

For basic anxiety "in the Indian context" the key is **Sudhir Kakar** (born 1938), the psychoanalyst of Indian childhood and society (*The Inner World*, 1978; Section 1.2). He wins as the only psychoanalyst offered: Anand C. Paranjpe co-edited the *Handbook of Indian Psychology* (2008), Durganand Sinha (1922 to 1998) urged the indigenisation of psychology, and "Kamala Chaudhuri" is a garbled name.

Horney on the idealised self, women and "momism"

In *Neurosis and Human Growth* (1950) Horney turned to the self. A healthy self-image rests on a realistic appraisal of strengths and weaknesses. The neurotic, torn between incompatible trends, builds an **idealised self-image** of flawless virtue and tries to become it. The result is the **tyranny of the shoulds** ("I should be the perfect student, never tired, never angry"), demands no one can meet, which end in self-hatred.

Horney also challenged Freud's view of women. From 1922 she rejected Freud's penis envy as an idea drawn from neurotic women and seen through male eyes. Men, she countered, may feel **womb envy**, envy of women's power to bear children, which they sublimate into

achievement. In her **feminine psychology**, women's inferiority feelings have cultural causes, such as economic dependence and narrow roles, not anatomical ones. Her *Self-Analysis* (1942), drawn from her own lifelong practice, argued that people can ease many of their own neurotic conflicts without an analyst.

A fashion of the 1940s blamed mothers instead. **Momism** is the blaming of over-protective, idealised mothers for their sons' weakness. The writer **Philip Wylie** coined it in *Generation of Vipers* (1942), and the psychiatrist **Edward Strecker** took it up in *Their Mothers' Sons* (1946), blaming "moms" for the young men found unfit for military service in the Second World War.

> EXAM TRAP
>
> "Mom-ism" was coined by Wylie; when Wylie is not an option, the key is **Edward Strecker**, not Freud, Jung or Horney.

Fromm: escape from freedom and the existential needs

Erich Fromm (1900 to 1980), a German analyst trained also in sociology, called his approach humanistic psychoanalysis and studied whole societies. In *Escape from Freedom* (1941), written as fascism spread, he argued that modern people have won freedom from the old ties of church, clan and guild but pay for it with isolation. Many flee it through three **mechanisms of escape**: **authoritarianism** (submitting to a leader or dominating others), **destructiveness** (attacking the threatening world) and **automaton conformity** (becoming exactly what society expects). The healthy answer is positive freedom through spontaneous love and productive work.

In *The Sane Society* (1955) Fromm named five distinctly human **existential needs**, arising from our awareness of being separate from nature (Table 4.16).

Table 4.16 Fromm's five existential needs.

Need	What it is	Healthy versus unhealthy answer
Relatedness	Union with others	Productive love versus narcissism
Transcendence	Rising above passive existence	Creating versus destroying
Rootedness	Feeling at home in the world	Brotherliness versus clinging to mother and clan
Sense of identity	Knowing oneself as a separate being	Individuality versus herd conformity
Frame of orientation (consistent orientation)	A stable way of making sense of the world	Reason versus irrational belief

> EXAM TRAP
>
> Safety, love and belonging, esteem and self-actualisation are Maslow's (Section 1.10 compares the lists); competence, autonomy and relatedness are Deci and Ryan's self-determination theory; hunger, thirst and shelter are physiological needs.

Fromm's **character orientations** (*Man for Himself*, 1947) describe how people relate to the world. Four are nonproductive: the **receptive** person waits to be given all good things; the **exploitative** person takes them by force or cunning; the **hoarding** person holds on to possessions, feelings and thoughts; and the **marketing** person packages the self as a commodity for sale. The **productive** orientation is the healthy ideal: using one's own powers in love, reason and work.

Sullivan's interpersonal theory

The American psychiatrist **Harry Stack Sullivan** (1892 to 1949) went furthest. His **interpersonal theory** (*The Interpersonal Theory of Psychiatry*, published in 1953, after his death) holds that personality exists only in relations with other people, real or imagined: it is "the relatively enduring pattern of recurrent interpersonal situations which characterize a human life".

Anxiety, again, is the engine. It passes by empathy from the "mothering one" to the infant: a tense mother makes a tense baby. To avoid it the child builds a **self-system**, the organisation of experience that guards security, using **security operations** such as selective inattention, which ease anxiety but block learning. It forms **personifications**, images of self and others: the **good-me** (experiences met with tenderness), the **bad-me** (experiences met with the mother's anxiety) and the **not-me** (experiences so dreadful that they are dissociated, surfacing in nightmares and schizophrenia).

Experience matures through three modes. The **prototaxic** mode is the infant's undifferentiated flow of sensations. The **parataxic** mode links events that occur together as cause and effect, the prelogical thinking behind superstition. The **syntaxic** mode is logical and consensually validated through shared language. Sullivan's developmental epochs run from infancy and childhood through the juvenile era (the need for peers), preadolescence, and early and late adolescence to adulthood. Preadolescence brings the **chum**, a same-sex friend whose welfare matters as much as one's own: the first intimacy.

> **EXAM TRAP**
>
> A practice-mock key for "Sullivan's interpersonal theory" is "personality as an open system interacting with the environment". Accept it by elimination, but "open system" is Allport's phrase (*The Open System in Personality Theory*, 1960; Section 4.6). "Unconscious processes" is Freud, "observable behaviour" is behaviourism, and "social structures and institutions" is sociology, nearer Fromm. The eight stages from oral-sensory to maturity are Erikson's (Chapter 6), never Sullivan's.

Ego psychology: Anna Freud, Hartmann and Erikson

Ego psychology kept Freud's structural model but gave the ego more power. **Anna Freud** (1895 to 1982), Freud's youngest daughter, catalogued the defences in *The Ego and the Mechanisms of Defence* (1936; Section 4.3) and pioneered child analysis. **Heinz Hartmann** (1894 to 1970), the father of ego psychology, argued in *Ego Psychology and the Problem of Adaptation* (1939) that perception, memory, motor skill, language and thinking develop in a **conflict-free ego sphere**, not out of conflict with the id. These functions give the ego autonomy, and its task is adaptation to an "average expectable environment".

Erik Erikson (1902 to 1994), analysed by Anna Freud, carried the ego across the life span. His **epigenetic principle** holds that development unfolds by a ground plan, each part having its

own time of special ascendancy, as organs form in an embryo. His stages are psychosocial, not psychosexual: each is a crisis between the growing person and society's demands, not a shift of libido to a new body zone, and they continue into old age. The adolescent **identity crisis** is his term; the eight stages are in Chapter 6.

Object relations and Melanie Klein ●●

In psychoanalysis an **object** is the target of a drive or feeling: almost always a person, or the mental image of one, beginning with the mother. **Object relations** are the internalised mental representations of self and others (objects) and the relationships between them. A child does not just live with a mother; it carries a mother inside, coloured by love and hate, and later relationships are lived through these inner figures. The term does not mean the relationship among id, ego and superego, the use of defences, or the genital stage (Sections 4.2 and 4.3). The school's founder was **Melanie Klein** (1882 to 1960), who worked in Berlin and, from 1926, London.

Klein analysed very young children through her **play technique**, treating play with small toys as free association. At first, she concluded, the baby relates to **part objects**: the gratifying "good breast" and the frustrating "bad breast", felt as two different beings. In the **paranoid-schizoid position** of the first months, the infant protects the good by **splitting** it off from the bad, keeping love and hate apart, and projects its rage into the bad object, which then seems to persecute it. From about the middle of the first year, as the infant grasps that the loved and hated mother are one person, it enters the **depressive position**: it feels guilt over the harm its rage may have done and makes **reparation** through loving acts. Klein spoke of positions, not stages, because adults move between them all their lives (Table 4.17). In **projective identification** a person puts unwanted parts of the self into someone else and then treats that person as if they had them. **Envy** (*Envy and Gratitude*, 1957) is the infant's wish to spoil the good breast because it has what the infant lacks.

Table 4.17 Klein's two positions.

Feature	Paranoid-schizoid position	Depressive position
Begins	First months of life	About the middle of the first year
Object	Part objects: good breast, bad breast	Whole object: one mother, good and bad
Anxiety	Persecutory: fear of being attacked	Depressive: fear of having harmed the loved one
Main processes	Splitting, projection, projective identification	Guilt, mourning, reparation

> EXAM TRAP
>
> Klein's defence in paranoia is **splitting**: the paranoid-schizoid position rests on it, and projection works on what splitting has already separated. Projection is Freud's classic mechanism of paranoia (the Schreber case, 1911). Reaction formation (turning a feeling into its opposite) and suppression (conscious putting aside) are in Section 4.3.

Winnicott and the later object-relations theorists

The English paediatrician **Donald Winnicott** (1896 to 1971) held that the infant needs not a perfect mother but a **good-enough mother**, who meets its needs almost completely at first and then fails it in small, bearable ways, so that it learns to tolerate frustration. She provides

a **holding environment** of physical and emotional security. As the infant learns that the mother is separate, it adopts a **transitional object**, the teddy or blanket that is neither wholly "me" nor "not-me" and comforts it when she is away. Good-enough care lets a spontaneous **true self** develop; without it the infant builds a **false self**, a compliant shell that gives others what they expect.

Ronald Fairbairn (1889 to 1964) made the cleanest break with Freud: libido is object-seeking, not pleasure-seeking. **Margaret Mahler** (1897 to 1985) traced the infant's psychological birth through **separation-individuation**, from fusion with the mother to a separate self (Chapter 6). **Otto Kernberg** (born 1928) described **borderline personality organisation**: intense early frustration and aggression leave the person unable to unite good and bad images, so splitting persists into adulthood, identity is diffuse and others are all good or all bad, though reality testing is largely kept. **John Bowlby** (1907 to 1990), trained in this tradition, turned the infant's tie to its mother into attachment theory, an offshoot taught in Chapter 6.

Kohut's self psychology

Heinz Kohut (1913 to 1981) founded **self psychology**, which drew on object relations, in *The Analysis of the Self* (1971) and *The Restoration of the Self* (1977). Where Freud saw narcissism as love withdrawn from others, Kohut saw healthy narcissism as a normal line of development. The self grows through **selfobjects**, people experienced as part of the self because they perform functions the child cannot yet perform alone. Three selfobject needs must be met: **mirroring**, the gleam in the mother's eye that confirms the child's "Look at me!" and so sustains the **grandiose self**; **idealising**, merging with a calm, strong parent (the idealised parent imago); and **twinship** (alter ego), added later, feeling essentially alike with others. When parents respond well but imperfectly, each bearable failure leads the child to take over the selfobject's soothing function; this **transmuting internalisation** builds a cohesive self. When mirroring and idealising fail chronically, the self stays fragmented and enfeebled: empty and dependent on others to hold it together. Failed mirroring above all leaves the archaic grandiose self unintegrated, and it returns in adulthood as **narcissistic pathology**: grandiosity, a constant hunger for admiration and little empathy, as the person still seeks the mirroring that never came.

> EXAM TRAP
>
> Grandiosity, a need for constant admiration and a lack of empathy trace to failed parental **mirroring**. Failed idealising leaves a person searching for someone perfect to admire; twinship (alter ego) is the need to feel alike; "confrontation" is a therapist's technique, not a selfobject need.

> CLINICAL LINK
>
> A client says she is "no one", feels "empty inside", dreads abandonment and has intense, unstable relationships. Self psychology reads this as a fragmented self from failed parental mirroring and idealising. It is not oral fixation or an unresolved Oedipus complex with castration anxiety (Freud's psychosexual explanations, Section 4.2), nor guilt from a harsh superego, a neurotic conflict rather than a deficit of self. The borderline disorder itself is in Chapter 8.

Lacan, the "French Freud"

Jacques Lacan (1901 to 1981), the Paris analyst called the "French Freud", led a return to Freud, a close rereading of Freud's own texts against ego psychology's taming of them. In the **mirror stage**, between about 6 and 18 months, the infant delights in its unified mirror image; the ego is born from this identification with an outside image, and so is partly an illusion. Drawing on structural linguistics, Lacan taught that the unconscious is structured like a language. Experience moves in three registers: the **Imaginary** (images and identifications), the **Symbolic** (language, law and social order) and the **Real** (whatever resists being put into words).

EXAM TRAP

The other options are earlier French psychiatrists. Jean-Étienne Esquirol gave hallucination its modern meaning and described monomania. Jules Falret, with Charles Lasègue (1877), described folie à deux, a delusion shared by two people. Paul Briquet (1859) described the chronic, many-symptom hysteria called Briquet's syndrome, now somatisation (Chapter 8).

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Horney's all-pervading feeling of being lonely and helpless in a hostile world is _____ anxiety.
2. Horney's detached type moves _____ from people.
3. Fromm's escape by becoming exactly what society expects is _____ conformity.
4. Sullivan's mode that treats events occurring together as cause and effect is the _____ mode.
5. Hartmann's ego functions that develop free of conflict lie in the _____ ego sphere.
6. In Klein's _____ position the infant splits the good breast from the bad.
7. Winnicott's comforting teddy or blanket is a _____ object.
8. Kohut's people who perform functions the child's self cannot yet perform are _____.

Answers: (1) basic, (2) away, (3) automaton, (4) parataxic, (5) conflict-free, (6) paranoid-schizoid, (7) transitional, (8) selfobjects.

4.6 Traits I: Allport, Cattell and the Lexical Search

LOWER YIELD · 17 QS

Ask a friend to describe you and you will hear adjectives: friendly, stubborn, punctual, moody. Each one claims that you behave in a characteristic way across situations and across years. Trait theorists set out to make a science of this everyday habit. This section follows two of them, who began from the same dictionary and went opposite ways: Gordon Allport, who studied the unique individual, and Raymond Cattell, who used statistics to find the dimensions on which everyone differs. Eysenck and the Big Five follow in Section 4.7.

The trait approach and the lexical hypothesis

The **trait approach** describes personality through specific psychological attributes along which individuals differ in consistent and stable ways. A **trait** is one such attribute: an enduring dimension, such as sociability, on which everyone has a position. The approach assumes *stability* over time (the talkative child becomes a talkative adult), *consistency* across situations (sociable at a wedding and in the canteen alike) and *individual differences* (people sit at different points on the same dimension, so they can be compared and their behaviour predicted).

Three neighbouring ideas must be kept apart. A **type** sorts people into broad, discrete categories, such as Sheldon's body types (Section 4.1), and so describes personality through broad patterns; typing shades into stereotyping, a set of expected behaviours based on similarities among members of a group. **Situationism** holds that characteristics of the situation, not of the person, largely determine behaviour (Walter Mischel's challenge, Section 4.7). A state is a temporary condition: **Charles Spielberger** (1927 to 2013) separated state anxiety, a passing tension, from trait anxiety, a stable proneness to anxiety, and measured both with his State-Trait Anxiety Inventory (1970).

Where should one look for the traits that matter? The **lexical hypothesis** answers: in language. Differences between people that matter in daily life get noticed and named, so the important traits are already encoded as words, and the more important a trait, the more words a language has for it. In 1936 Gordon Allport and **Henry Odbert** searched an unabridged English dictionary and culled 17,953 words that distinguish one person's behaviour from another's. Only about 4,500 named stable traits (honest, sociable); the rest described temporary states (flustered), evaluations (admirable), or physical features and abilities. This list fed the whole factor-analytic tradition (Figure 4.9).

Such a list can be used in two ways. The **nomothetic approach** (Greek *nomos*, law) studies many people on the same dimensions to find general laws. The **idiographic approach** (*idios*, one's own) studies one person intensively to grasp his or her unique pattern. The terms are Wilhelm Windelband's (Section 4.1). Cattell went nomothetic; Allport preferred the idiographic path.

> **EXAM TRAP**
>
> The trait approach focuses on *specific psychological attributes along which individuals differ in consistent and stable ways*. "Broad patterns" describes the type approach; "expected behaviour based on similarities" describes stereotyping; "situational characteristics determine behaviour" is situationism, the critique of trait theory.

Allport: traits as real structures

Gordon Willard Allport (1897 to 1967) did more than anyone to make personality a recognised field. In 1920 he called on Sigmund Freud in Vienna and, to fill an awkward silence, described a small boy he had just seen on a tram who was terrified of dirt. Freud looked at his neat, meticulous visitor and asked, "Was that little boy you?" Allport decided that psychoanalysis digs too deep and that psychology should take conscious, present motives seriously. Back at Harvard, from 1924, he taught what was probably the first American college course on personality, and his *Personality: A Psychological Interpretation* (1937) established the field. It

defined personality as "the dynamic organization within the individual of those psychophysical systems that determine his unique adjustments to his environment" (Section 4.1).

For Allport a trait was not a convenient label but a real **neuropsychic structure**. It renders many stimuli functionally equivalent and guides consistent behaviour: a friendly man greets the stranger at the bus stop, the new colleague and the neighbour's guest with the same warmth. Traits therefore cause behaviour rather than merely summarising it. Allport distinguished **common traits**, shared in varying degrees by the members of a culture so that people can be compared on them, from individual traits, which he later renamed **personal dispositions** (*Pattern and Growth in Personality*, 1961): characteristics peculiar to one person.

Personal dispositions differ in how much of a life they govern, and Allport set them on three levels (Table 4.18). A **cardinal trait** is the most pervasive and powerful kind: a single ruling passion that dominates a whole life. Few people have one, and those who do often lend it a name: Machiavellian, narcissistic, Christlike. **Central traits** are the handful, five to ten, of outstanding traits that describe a person's behaviour, the words you would put in a reference letter. **Secondary traits** are the least important, displayed inconspicuously and inconsistently, such as a taste for filter coffee, and noticed only by close friends.

Table 4.18 Allport's three levels of personal dispositions.

Level	What it is	Who has it	Examples
Cardinal	The most pervasive and powerful trait; one ruling passion	Few people	Machiavellian, narcissistic, Christlike
Central	The handful (5 to 10) of outstanding traits that describe a person	Everyone	Honest, sociable, cynical: a reference letter
Secondary	The least important; shown inconspicuously and inconsistently	Everyone, many of them	Food or music preferences, passing attitudes

> EXAM TRAP
>
> Allport's levels are **cardinal, central and secondary**. "Primary, secondary, tertiary" is invented; "cognitive, emotional, physiological" are components of emotion; "id, ego, superego" is Freud's structure (Section 4.2). Match the definitions: most pervasive and powerful, cardinal; handful of outstanding traits, central; least important and inconsistent, secondary; shared by members of a culture, common traits.

The idiographic approach

Because every personality is unique, Allport argued, its science must study the individual case and not only group averages, which describe nobody in particular. He therefore preferred the idiographic approach to the nomothetic search for laws, with its factorial designs (several variables crossed over many subjects) and group statistics such as chi-square (Chapter 9). Later he called his approach **morphogenic**, the study of how traits are patterned within one person. His main method was the **personal-document technique**: analysing diaries, letters and autobiographies for the traits they reveal.

KEY STUDY: LETTERS FROM JENNY (1965)

Over twelve years a woman Allport called Jenny, the mother of his college roommate, wrote more than 300 letters to Allport and his wife. Thirty-six judges read them and listed nearly 200 trait names, which Allport reduced to eight clusters, among them quarrelsome-suspicious, self-centred and dramatic-intense. A later computer count of her words by one of his students found much the same eight. Criticism: the method rests on judges' interpretations, and one case is hard to generalise.

Allport was not purely idiographic. With **Philip Vernon** (1931), and later with **Gardner Lindzey**, he built the **Study of Values**, a nomothetic self-report test of the relative strength of six values taken from Eduard Spranger's types: theoretical, economic, aesthetic, social, political and religious. With Vernon he also showed (1933) that expressive behaviour, such as handwriting, gesture and voice, is consistent across tasks.

EXAM TRAP

Allport preferred the **idiographic** approach. The nomothetic approach, factorial designs and chi-square are the group methods he criticised. Nor did he use factor analysis: that is the hallmark of Cattell and Eysenck.

Functional autonomy, the proprium and maturity

Freud explained adult motives by tracing them to childhood. Allport's reply was **functional autonomy**: the motives of a mature adult become independent of the experiences in which they began, as a grown tree no longer needs its seed. A girl made to practise the veena to escape scolding may, as a woman, play for hours for its own sake. *Perseverative* functional autonomy covers low-level routines that persist without reward, such as addictions and daily rituals. *Propriate* functional autonomy covers motives bound up with one's values, self-image and lifestyle, and is the key to adult motivation.

"Propriate" comes from the **proprium**, Allport's word for the self. It develops through seven aspects: (1) the bodily self, (2) self-identity and (3) self-esteem, in about the first three years; (4) self-extension ("my house, my school") and (5) the self-image, actual and ideal, at about four to six; (6) the self as a rational coper, from six to twelve; and (7) propriate striving, the long-range goals of adolescence. The self returns in Section 4.10.

If childhood needs for affection and security are met, the result is the mature adult, marked by six criteria: (1) extension of the sense of self beyond oneself; (2) warm relating to others; (3) emotional security and self-acceptance; (4) realistic perception, skills and commitment to work; (5) self-objectification, meaning insight together with humour; and (6) a unifying philosophy of life. In 1960 Allport described personality as an **open system**, taking in energy and information and growing, not merely reducing tension. Critics note that functional autonomy says that motives become independent but not how, that the sharp break between child and adult personality is doubtful, and that the theory produced little research.

Cattell: surface and source traits ●●

Raymond Bernard Cattell (1905 to 1998) studied chemistry in London and then took a PhD in psychology in 1929 under Charles Spearman. Spearman had applied factor analysis to mental

abilities; Cattell resolved to apply it to personality. He moved to the United States in 1937 and worked mostly at the University of Illinois. His goal was prediction: "Personality is that which permits a prediction of what a person will do in a given situation" (1950).

Factor analysis examines the correlations among many measures and finds the few underlying factors that account for them: if people rated talkative are also rated sociable and cheerful, one factor may lie beneath all three. Cattell took Allport and Odbert's 4,500 stable-trait words, merged synonyms to leave 171, found that ratings on these formed about 35 clusters, and factor-analysed ratings, questionnaires and tests down to the 16 factors of his Sixteen Personality Factor Questionnaire, the 16PF (Figure 4.9).

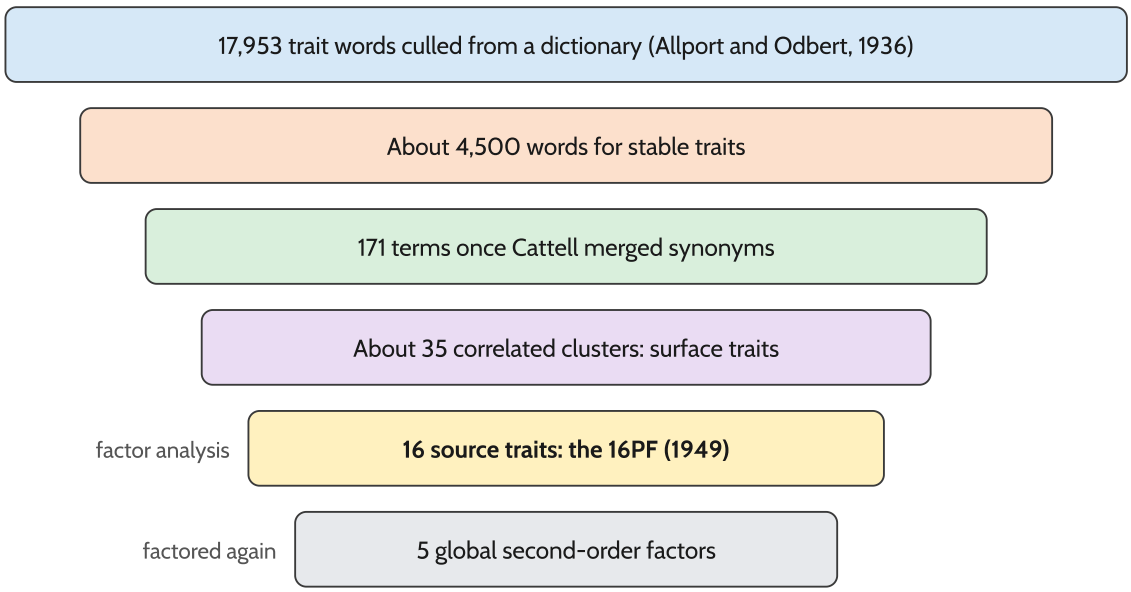


Figure 4.9 The lexical search, from dictionary to questionnaire. Each step keeps fewer, deeper units: the correlated surface clusters give way to the source traits beneath them.

The clusters and the factors give Cattell's most examined distinction. **Surface traits** are clusters of behaviours that correlate and appear together; anxiety, indecision and irrational fear, for example, form a surface trait one might call neuroticism. Because they arise from several sources, they are less stable and less important. **Source traits** are the stable, unitary factors beneath them, revealed by factor analysis: the building blocks of personality, as atoms are of matter. By origin, source traits are either **constitutional traits**, rooted in biology, or **environmental-mold traits**, learned from social and physical surroundings.

Cattell also separated common traits, which everyone has in some degree, from unique traits belonging to one or a few people (a passion for old coins), and divided traits by function into ability traits (how efficiently we pursue goals), temperament traits (emotional style) and **dynamic traits** (the motivating forces). Dynamic traits come in two kinds. **Ergs** are innate drives, such as hunger, sex, fear, curiosity, gregariousness and self-assertion. **Sentiments** are learned patterns of attitudes centred on an important object, such as career, religion, family or the self; the self-sentiment integrates the rest. A specific attitude ("I want to revise tonight") serves a sentiment, which in turn serves ergs, a chain Cattell called **subsidiation**. The web of such links is the **dynamic lattice** (Figure 4.10).

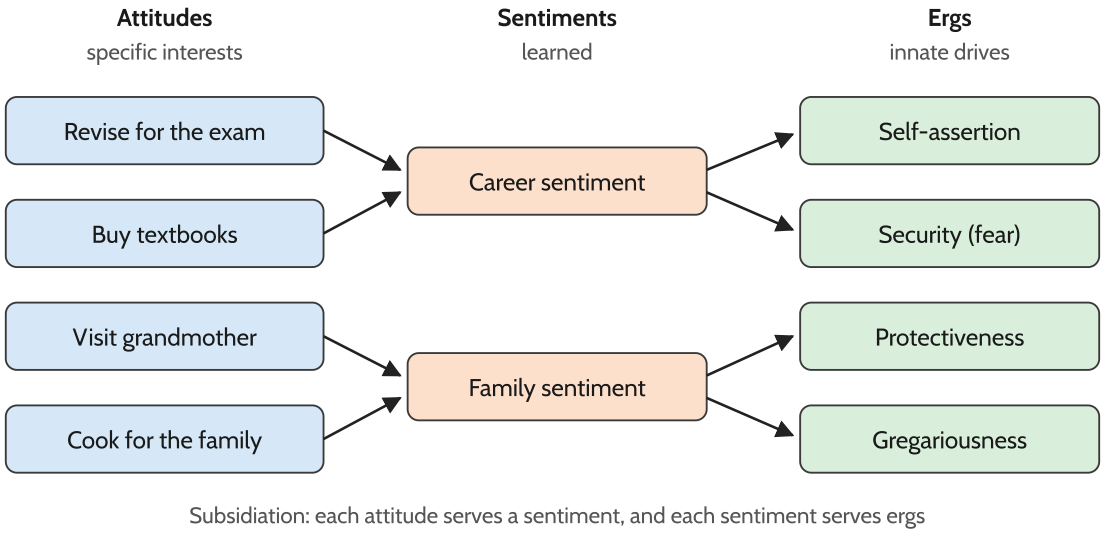


Figure 4.10 A simplified dynamic lattice. Read it from left to right: everyday attitudes are subsidiary to learned sentiments, which are subsidiary to innate ergs.

EXAM TRAP

Cattell's pair is **surface and source traits**. Cardinal and central traits are Allport's; openness and conscientiousness are Big Five factors (Section 4.7); id and ego are Freud's. An Amity 2023 item says Cattell divided "attitudes" into source and surface: the key is right, but the stem garbles traits as attitudes. Its option "egos and sentiments" garbles Cattell's **ergs and sentiments**, the two kinds of dynamic trait. "Traits and types" are two approaches; defence and coping mechanisms belong to Sections 4.3 and 1.11.

L-data, Q-data and T-data ●●

A factor found in only one kind of data might be an artefact of method, so Cattell required source traits to reappear across three kinds (Table 4.19). **L-data** (life-record data) capture everyday behaviour through records or the ratings of observers who know the person: school performance, absences from work, a criminal record. **Q-data** (questionnaire data) are self-report ratings of one's own characteristics, attributes and interests, limited by poor self-knowledge and by faking. **T-data** (objective test data) come from controlled test situations whose purpose is hidden, so responses are hard to fake. The four 16PF factors labelled Q1 to Q4 appeared only in questionnaire data, hence the Q.

Table 4.19 Cattell's three sources of personality data.

Data	Source	Examples
L-data: life record	Records and observers' ratings of everyday behaviour	School performance, job absences, criminal record
Q-data: questionnaire	Self-report ratings of one's own characteristics and interests	Self-report personality inventories
T-data: objective test	Behaviour in a controlled situation whose purpose is hidden	Reaction-time and performance tasks

EXAM TRAP

There are exactly three: L, Q and T. "P-data" (projective data) and "Z-data" do not exist; Cattell classed hard-to-fake tasks such as inkblots as T-data, and projective responses are never L-data. Q-data is not the Q-sort, in which a person sorts self-descriptive cards from "most like me" to "least like me" (Rogers, Section 4.8). Nor is it the **P-technique**, a research design that factor-analyses one person measured on many occasions (the usual R-technique correlates many people measured once). Self-report is Q-data; school and criminal records are L-data; the 16 source traits are results, not data.

The Sixteen Personality Factor Questionnaire ●●●

Cattell packaged his source traits in the **Sixteen Personality Factor Questionnaire** (16PF), first published in 1949; the fifth edition (1993) has 185 items for people aged 16 and over. It measures normal personality on 16 primary factors (Table 4.20), lettered roughly by their size in Cattell's early rating studies; D, J and K are missing because those factors did not reappear reliably in adult questionnaire data. Factor B is a brief reasoning scale, but the instrument measures personality, not intelligence. Factoring the 16 again yields five second-order **global factors**: extraversion, anxiety, tough-mindedness, independence and self-control. These approximate the Big Five (extraversion, neuroticism, low openness, low agreeableness and conscientiousness; Section 4.7). Sten-score norms, reliability and validity are in Chapter 2.

Table 4.20 The 16 primary factors of the 16PF (fifth edition names).

Factor	Name	Low scorer	High scorer
A	Warmth	Reserved, aloof	Warm, outgoing
B	Reasoning	Concrete thinking	Abstract thinking
C	Emotional stability	Easily upset	Calm, stable
E	Dominance	Deferential, docile	Dominant, assertive
F	Liveliness	Serious, restrained	Lively, enthusiastic
G	Rule-consciousness	Expedient	Dutiful, rule-bound
H	Social boldness	Shy, timid	Bold, venturesome
I	Sensitivity	Tough-minded	Tender-minded
L	Vigilance	Trusting	Suspicious
M	Abstractedness	Practical	Imaginative
N	Privateness	Forthright	Discreet, shrewd
O	Apprehension	Self-assured	Worried, self-doubting
Q1	Openness to change	Traditional	Experimenting
Q2	Self-reliance	Group-oriented	Self-sufficient
Q3	Perfectionism	Lax, tolerates disorder	Controlled, exacting
Q4	Tension	Relaxed	Tense, driven

EXAM TRAP

Any stem with 16 factors, 16 source traits or the 16PF points to **Raymond B. Cattell**. Eysenck also used factor analysis but found three dimensions (the Eysenck Personality Questionnaire, EPQ); Costa and McCrae built the NEO-PI-R for five factors; Allport built the Study of Values; Sheldon rated body types; Murray devised the Thematic Apperception Test (TAT); Freud and Rogers proposed no trait dimensions at all. The 16PF measures normal personality traits, not fluid, crystallised or general intelligence (Cattell's fluid and crystallised intelligence and Spearman's *g* are in Section 1.9), and it is not a personality-disorder scale, a Q-sort or a projective test.

Heredity, development and the wider trait family

Comparing twins and ordinary siblings reared together and apart, Cattell concluded that about one third of personality is genetically based and two thirds environmental, though intelligence (factor B) and boldness (factor H) looked about 80 percent hereditary. He proposed six life-span stages: infancy (birth to 6, the major formative period), childhood (6 to 14), adolescence (14 to 23), maturity (23 to 50), late maturity (50 to 65) and old age. His **specification equation** predicts a response in a given situation as the sum of the person's trait scores, each weighted by its relevance to that situation. The main criticism is that factor analysis is less objective than it looks: the researcher chooses the measures and names the factors, which may be why others could not replicate all 16.

Two contemporaries complete the picture (Table 4.21). **Henry Alexander Murray** (1893 to 1988) built his personology around motives rather than traits: internal needs (for achievement, affiliation, dominance and more) meet environmental press, the situational forces that help or hinder them, and a recurring combination of need and press is a thema (Section 1.10). **Joy Paul Guilford** (1897 to 1987), known for his model of intellect (Section 1.9), factor-analysed temperament into ten traits, such as general activity, ascendance, sociability and emotional stability, measured by the Guilford-Zimmerman Temperament Survey (1949).

Table 4.21 Who built which trait or motive model.

Theorist	Core idea	Signature measure	Where
Gordon Allport	Cardinal, central, secondary dispositions; idiographic study	Personal documents; Study of Values	4.6
Raymond B. Cattell	16 source traits by factor analysis; L, Q and T data	16PF (1949)	4.6
Henry Murray	Personology: needs, press and thema	TAT, with Christiana Morgan (1935)	1.10; Chapter 2
J. P. Guilford	Factor-analytic temperament traits	Guilford-Zimmerman Temperament Survey	4.6
Hans Eysenck	Three superfactors: E, N and P	EPQ	4.7
Costa and McCrae	Five-Factor Model	NEO-PI-R	4.7

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Allport and Odbert (1936) culled _____ trait words from an English dictionary.
2. A single ruling passion that dominates a whole life is a _____ trait.
3. Allport's preference for the intensive study of one person is the _____ approach.
4. Adult motives that have become independent of the experiences in which they began show functional _____.
5. Allport's term for the self is the _____.
6. Cattell's stable underlying factors, found by factor analysis, are _____ traits.
7. School performance and criminal records are _____-data.
8. Cattell's innate drives are _____; his learned attitude patterns are _____.

Answers: (1) 17,953, (2) cardinal, (3) idiographic, (4) autonomy, (5) proprium, (6) source, (7) L, (8) ergs, sentiments.

4.7 Traits II: Eysenck, the Big Five and the Biology of Personality

LOWER YIELD · 18 QS

Section 4.6 ended with Raymond Cattell's sixteen factors. Hans Eysenck thought three were enough and looked for them in the brain; Paul Costa and Robert McCrae settled on five, the answer that now dominates. This section follows both, then asks how much of a trait is inherited and whether traits predict what you will actually do tomorrow.

Eysenck's three superfactors ●●

Hans Jurgen Eysenck (1916 to 1997) left Berlin for London in 1934 and spent his career at the Maudsley Hospital and Institute of Psychiatry. Like Cattell he used factor analysis, but he distrusted purely statistical factors: a real dimension of personality must also show up in laboratory experiments and have a biological cause. Factor analysis joined to experiment and biology is his signature.

His **hierarchical model** has four levels (Figure 4.11). At the bottom are **specific responses**: today you chat with a stranger at the bus stop. Responses that recur become **habitual responses**: you chat at every bus stop. Correlated habits form **traits** (sociability, liveliness, activity), and correlated traits form a **type** or superfactor, here extraversion. An Eysenck type is a broad continuous dimension, not a box people fall into.

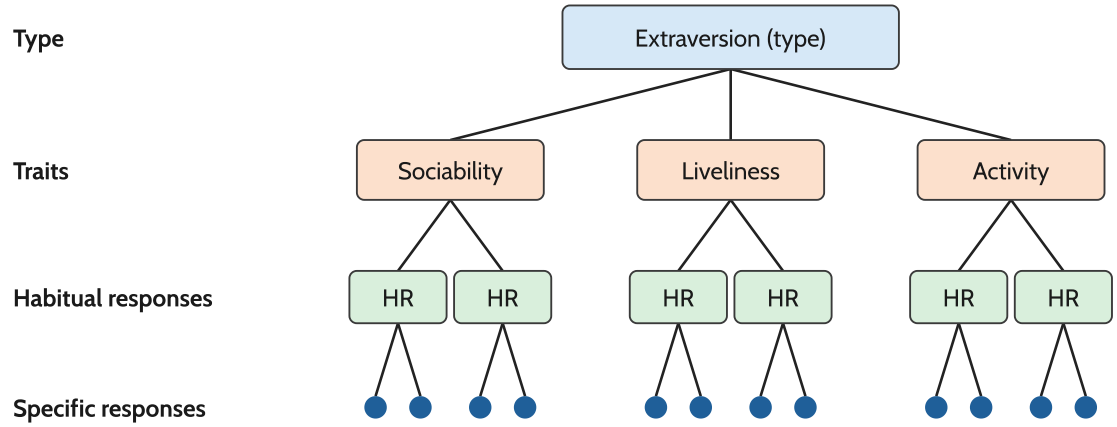


Figure 4.11 Eysenck's hierarchical model. Specific responses (dots) recur as habitual responses (HR), correlated habits form traits, and correlated traits form a type such as extraversion.

His early analyses (*Dimensions of Personality*, 1947) found two superfactors; a third was added in the 1970s, giving the **PEN model**:

- E, **extraversion** versus **introversion**. Extraverts are sociable, lively, active, talkative, assertive and sensation seeking; they enjoy parties and large gatherings and look for stimulation. Introverts are quiet, reserved and reflective; they prefer a few close friends, plan before acting and, in the popular phrase, gain energy from solitude while crowds drain them. Introversion is not unfriendliness, aggression or even shyness (Eysenck listed shyness under N).
- N, **neuroticism** versus emotional stability. High scorers are anxious, moody, tense, guilt-prone and emotionally reactive, with low self-esteem; low scorers are calm and even-tempered.
- P, **psychoticism** versus impulse control. High scorers are aggressive, cold, egocentric, impersonal, impulsive, antisocial and tough-minded, though sometimes creative. P is not psychosis; the scale is closer to psychopathy.

> EXAM TRAP
>
> Eysenck's three are **E, N and P**. "Extraversion, sensing-intuition, thinking-feeling" smuggles in the Myers-Briggs dichotomies from Carl Jung's types (Section 4.4); "psychoticism-conscientiousness" borrows a Big Five factor. His **earliest** model had only E and N: not openness and conscientiousness (Big Five), not inferiority and superiority striving (Alfred Adler, Section 4.4), not internal and external locus of control (Julian Rotter, Section 4.9). "Aggressive and dominant" never describes the introvert: dominance goes with E, aggression with P.

Arousal, the visceral brain and the four temperaments

Why should one student crave the roar of a cricket stadium while her roommate hides in the library? Eysenck's answer (*The Biological Basis of Personality*, 1967) was arousal. The **ascending reticular activating system** (ARAS) in the brainstem sets the level of **cortical arousal**. Introverts, he proposed, have a *higher* baseline arousal, so extra stimulation overloads them and they avoid it; extraverts sit below their comfortable level and seek stimulation to climb

up to it. The laboratory bore him out: given a few drops of lemon juice on the tongue, introverts salivate more than extraverts, and they are more sensitive to faint stimuli, tolerate pain less and condition faster. Later work confirmed these differences more firmly than the arousal explanation.

Neuroticism rests on the **visceral brain**, an older name for the limbic system (hippocampus, amygdala, cingulum, septum, hypothalamus), which drives the sympathetic nervous system; in high-N people this alarm fires at mild stress and settles slowly. For P he suggested male hormones, since men score higher; this biology is the weakest.

Because E and N are independent, crossing them gives four quadrants, and Eysenck pointed out that these reproduce the four temperaments of Galen (Section 4.1): the stable extravert is **sanguine**, the unstable extravert **choleric**, the unstable introvert **melancholic** and the stable introvert **phlegmatic** (Figure 4.12).

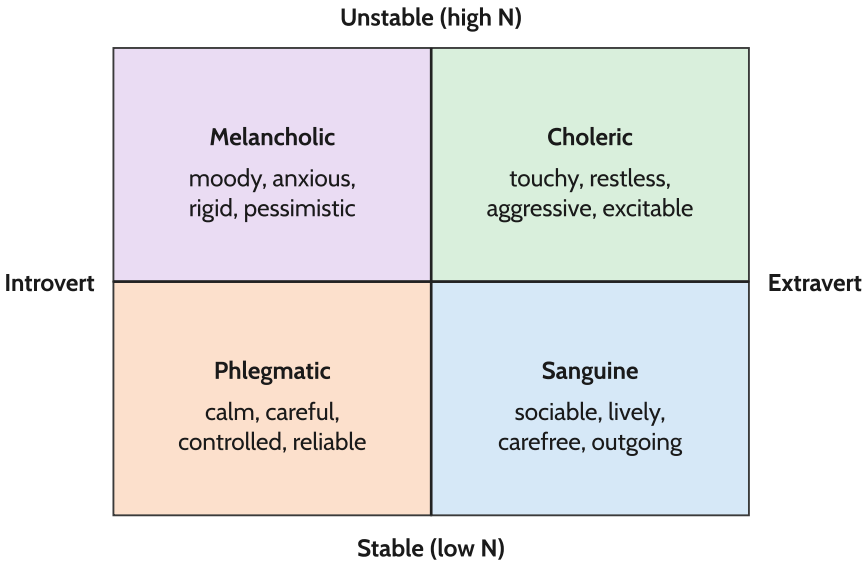


Figure 4.12 Eysenck's two independent dimensions crossed. Each quadrant matches one of Galen's temperaments; the trait words under each name are Eysenck's.

Eysenck's questionnaires, many built with his wife **Sybil B. G. Eysenck**, are the Maudsley Personality Inventory (MPI, 1959; E and N), the **Eysenck Personality Inventory** (EPI, 1964, or 1963 in some sources; E, N and a Lie scale), the **Eysenck Personality Questionnaire** (EPQ, 1975), which added P and kept the Lie scale (L) that catches faking good, and the EPQ-R (1985) with an improved P scale. Twin studies supported his view that the dimensions are largely inherited, E and N more than P, and all three appear in over 35 nations. In *Crime and Personality* (1964) he argued that offenders score high on E and N, adding P in later editions: extraverts condition poorly, so they are slow to acquire the conditioned anxiety that serves as conscience. The P link is the most consistent; the full theory has only modest support.

Other biological models of temperament

Eysenck opened the way to a biology of temperament, the inherited, early-appearing emotional style (Section 4.1) now seen as the biological core of personality. Table 4.22 sets out four successors. **Jeffrey Alan Gray** (1934 to 2004) rotated Eysenck's E and N axes in his **reinforce-**

ment sensitivity theory: anxiety runs from stable extravert to neurotic introvert, impulsivity from stable introvert to neurotic extravert. **C. Robert Cloninger** (1987, extended 1993) tied temperaments to neurotransmitters. **Jerome Kagan** (1929 to 2021) found that about one baby in five reacts to novelty with thrashing and crying, and these high-reactive infants tend to become shy, cautious children, showing **behavioural inhibition**. **Arnold Buss** and **Robert Plomin** (1984) proposed three heritable temperaments. Alexander Thomas and Stella Chess's infant categories are taught in Chapter 6.

Table 4.22 Biological models of temperament after Eysenck.

Model	Dimensions	Proposed basis
Gray: reinforcement sensitivity	behavioural inhibition system (BIS): anxiety, sensitivity to punishment; behavioural approach system (BAS, or activation system): impulsivity, sensitivity to reward	BIS: septo-hippocampal system; BAS: dopamine pathways
Cloninger: temperament and character	Temperament: novelty seeking, harm avoidance, reward dependence, persistence. Character: self-directedness, cooperativeness, self-transcendence	Dopamine, serotonin, noradrenaline for the first three; Temperament and Character Inventory
Kagan: behavioural inhibition	Inhibited (shy, withdrawn with the unfamiliar) versus uninhibited (bold, sociable) children	A highly reactive amygdala
Buss and Plomin: EAS	Emotionality, activity, sociability	Heritable; twin studies

The Big Five: discovery and the five factors ●●

If Cattell found sixteen factors and Eysenck three, who was right? Researchers who reanalysed trait ratings, following the lexical hypothesis of Section 4.6, kept arriving at five. **Donald Fiske** (1949) found five factors in ratings on Cattell's variables; **Ernest Tupes** and **Raymond Christal** (1961) found five recurrent factors in US Air Force ratings; **Warren Norman** (1963) replicated them; and **Lewis Goldberg** named them the **Big Five** (1981), "big" meaning broad, not important. Meanwhile **Paul Costa** (born 1942) and **Robert McCrae** (born 1949), at the US National Institutes of Health in Baltimore, reached the same five through questionnaires. The two roads met in the **Five-Factor Model** (FFM).

The five are remembered as **OCEAN** (or CANOE): **O**penness to experience, **C**onscientiousness, **E**xtraversion, **A**greeableness and **N**euroticism. Each is a continuous dimension with two poles, and most people fall near the middle. In the NEO-PI-R each factor, or domain, splits into six narrower **facets** (Table 4.23).

Table 4.23 The five factors, their poles and their six NEO-PI-R facets.

Factor	High scorer	Low scorer	Six facets
Openness (O)	Imaginative, curious, creative, unconventional	Conventional, practical, narrow interests	Fantasy, aesthetics, feelings, actions, ideas, values
Conscientiousness (C)	Organised, dependable, self-disciplined, dutiful, achievement-focused	Careless, disorganised, lax	Competence, order, dutifulness, achievement striving, self-discipline, deliberation
Extraversion (E)	Outgoing, sociable, talkative, assertive, energetic	Reserved, quiet, solitary	Warmth, gregariousness, assertiveness, activity, excitement-seeking, positive emotions

Factor	High scorer	Low scorer	Six facets
Agreeableness (A)	Trusting, kind, cooperative, softhearted	Suspicious, antagonistic, critical	Trust, straightforwardness, altruism, compliance, modesty, tender-mindedness
Neuroticism (N)	Anxious, moody, insecure, emotionally reactive	Calm, secure, even-tempered (emotional stability)	Anxiety, angry hostility, depression, self-consciousness, impulsiveness, vulnerability

To recognise a factor from a description, match it to the facets. Rohan is self-disciplined, achievement-focused and keen to do his duty: three facets of conscientiousness, so he scores high on C, not on E, A or N. Likewise "organised and dependable" is C, "outgoing and sociable" high E, "reserved and quiet" low E (introversion), and "imaginative and creative" O. Note that impulsiveness is a facet of N, and anxiety is a facet, not a factor.

Imposters and the NEO inventories ●●●

Table 4.24 lists the usual imposters and where each really belongs.

Table 4.24 Terms that are not Big Five factors.

Offered as a factor	What it really is
Psychoticism	Eysenck's third dimension (PEN); roughly low A plus low C
Intelligence	An ability, not a personality factor; it correlates with Openness
Intuition, sensing, thinking, feeling, judging, perceiving	Myers-Briggs Type Indicator (MBTI) dichotomies from Jung's types (Section 4.4)
Anxiety	One facet of Neuroticism
Sense of humour	An everyday trait word, not a factor
Honesty-humility	The sixth factor of HEXACO (below)
Sanguine, choleric, melancholic, phlegmatic	Galen's temperaments: the quadrants of Eysenck's E and N
Achievement, affiliation, power, intimacy, autonomy	Motives or needs, not traits (Section 1.10)

The standard measure is the NEO Personality Inventory. Costa and McCrae first measured only N, E and O, hence the name; the NEO-PI (1985) added A and C. The **NEO-PI-R** (1992) has 240 items, eight per facet, covering five domains with six facets each, 30 facets in all. The **NEO-FFI** (NEO Five-Factor Inventory) is a 60-item short form scoring domains only; the NEO-PI-3 (2005) simplified the wording for adolescents. The NEO belongs to the **trait approach**: it measures stable dispositions, not learned responses (behaviourist), thought processes (cognitive) or unconscious conflict (psychodynamic). Its psychometric properties are in Chapter 2.

> EXAM TRAP
>
> The NEO-PI-R rests on the **Five-Factor Model**: five domains, six facets each. It is not Cattell's 16 source traits (16PF, Section 4.6), Eysenck's three PEN dimensions (the EPQ), the MBTI's 16 types (Section 4.4), Henry Murray's needs and press (Section 4.6) or eleven personality-disorder scales (a clinical inventory, Chapter 2).

The Big Five across life and cultures

McCrae and Costa built the factors into **five-factor theory** (1996). The traits are **basic tendencies**: biologically based capacities that upbringing does not shape. What life shapes are **characteristic adaptations**: the habits, attitudes, skills, roles, relationships and self-concept through which a tendency is expressed in a particular culture. High E is a basic tendency; joining the college cultural committee in Mysuru is one adaptation of it.

The factors replicate in more than 50 nations, India among them, suggesting a common human structure. Critics note that most samples are urban and literate, that a remote Bolivian tribal group did not show the five clearly, and that lexical studies in some languages find six factors.

How stable are the factors? People keep their rank order, more so with age, but mean levels shift: from adolescence into middle age, conscientiousness and agreeableness rise while neuroticism falls, the **maturity principle**. Costa and McCrae borrowed **William James**'s 1890 phrase that by about 30 character is "set like plaster"; later work shows the plaster can still be chipped. Twin studies put the heritability of each factor at about 40 to 50 percent.

The factors also predict outcomes. **High conscientiousness and low neuroticism** go with better health and longer life. **Lewis Terman** began following about 1,500 bright Californian children in 1921; when **Howard Friedman** and colleagues (1993) traced their deaths, those rated conscientious and prudent as children had lived longer. Conscientious people exercise, wear seat belts, avoid smoking and follow medical advice. High N predicts distress and is the broadest personality risk factor for anxiety and depressive disorders (Chapter 8). Conscientiousness also predicted job performance in every occupational group in the meta-analysis of **Murray Barrick** and **Michael Mount** (1991).

> EXAM TRAP
>
> The health pair is **high C plus low N**. "Low agreeableness and low neuroticism" swaps in low A, which as hostility (the toxic core of the Type A pattern; Type A and Type D are in Section 1.11) predicts heart disease, not health. "Neuroticism exclusively" is backwards, since high N harms health, and "introversion exclusively" has no such link.

Beyond five: HEXACO and the Dark Triad

Not everyone accepts five. **Michael Ashton** and **Kibeom Lee** reanalysed trait words across many languages (from 2004) and found six factors, the **HEXACO** model: **H**onesty-Humility (sincere, fair, modest versus greedy, boastful, manipulative), **E**motionality (fearful, anxious, sentimental; close to N), e**X**traversion, **A**greeableness, **C**onscientiousness and **O**penness to experience. Honesty-Humility, the genuinely new factor, is low in offenders; the 100-item HEXACO-PI-R measures all six.

Delroy Paulhus and **Kevin Williams** (2002) described the **Dark Triad**, three overlapping, socially aversive but subclinical traits: **narcissism** (grandiosity, entitlement, hunger for admiration), **Machiavellianism** (cold, cynical, strategic manipulation) and **psychopathy** (callousness, impulsive thrill seeking, lack of remorse). All three share low agreeableness and low honesty-humility. The **Dark Tetrad** adds **everyday sadism**, enjoying others' suffering (Paulhus and colleagues, 2013).

Genes, environments and evolution

How can genes be separated from upbringing? **Behaviour genetics** uses two natural experiments. Identical (monozygotic) twins share all their genes and fraternal (dizygotic) twins about half, so greater similarity in identical pairs points to genes. In adoption studies, resemblance to the biological parents points to genes, resemblance to the adoptive parents to family environment. The **heritability** of a trait is the proportion of its variation *within a population* that is due to genetic differences; it says nothing about how "genetic" one person's trait is.

> **KEY STUDY: THE MINNESOTA STUDY OF TWINS REARED APART (FROM 1979)**
>
> Question: if identical twins grow up in different homes, how alike are they? Method: **Thomas Bouchard** and colleagues brought more than 100 sets of twins and triplets reared apart to the University of Minnesota for about a week of tests. Result (reported in 1990): identical twins reared apart were about as similar in personality as those reared together; heritability came to about 0.40 to 0.50 for personality and about 0.70 for IQ. Criticism: some pairs grew up in similar homes or had met before testing, and the sample was small and self-selected.

The same studies held a surprise: growing up in the same family adds almost nothing to personality similarity, and adoptive siblings reared together are scarcely more alike than strangers. Plomin and **Denise Daniels** (1987) asked why children in the same family are so different, and answered that what matters is the **non-shared environment**: different friends, teachers, illnesses, accidents and differential treatment by parents. The shared environment (the same home, parents, income and religion) has only a weak effect on adult personality.

Genes and environments are also entangled. **Sandra Scarr** and **Kathleen McCartney** (1983) described three kinds of **gene-environment correlation**. In the **passive** kind, parents give a child both genes and a matching home (musical parents, a house full of music); it is strongest in infancy. In the **evocative** kind, a child's inherited traits draw responses from others (a smiling baby is picked up more). In the **active** kind, or niche picking, people seek settings that suit them (the extravert joins the drama club); it grows with age. In **gene-environment interaction**, by contrast, a gene's effect depends on the environment: a vulnerable genotype may raise the risk of depression only after severe stress.

Why does variation persist at all? **David Buss** (born 1953) treats traits as **adaptive individual differences**: the Big Five summarise what our ancestors most needed to know about one another (who will lead, cooperate, work hard, panic or innovate), and different trait levels pay off in different environments. He is not Arnold Buss of the EAS model.

The person-situation debate

Traits assume consistency: an honest child should be honest in class, at play and at home. **Hugh Hartshorne** and **Mark May** tested this and found it largely missing (Key study below). Forty years later **Walter Mischel** (1930 to 2018) generalised the point in *Personality and Assessment* (1968). Trait measures, he found, correlated with behaviour in a given situation, and behaviour in one situation with behaviour in another, rarely above about 0.30, a ceiling he called the **personality coefficient**; it explains less than 10 percent of the variation in be-

haviour. His critique: **situational factors often override traits** in determining behaviour. So began the **person-situation debate**.

KEY STUDY: HARTSHORNE AND MAY'S HONESTY STUDIES (1928)

Question: is honesty a consistent trait? Method: in the Character Education Inquiry, thousands of American schoolchildren were given secret chances to cheat, lie or keep money in classroom, party, athletic and home settings. Result: honesty in one setting predicted honesty in another only weakly; a child who copied answers might still return a coin. Conclusion: honesty looked situation-specific. Criticism: single acts are unreliable samples; summed over many tests, the scores showed much more consistency.

Trait theorists replied in three ways. First, **aggregation**: **Seymour Epstein** (1979) showed that a single act is as unreliable as a single test item; averaged over many occasions, behaviour becomes far more stable, so traits predict most of the people much of the time. Second, a **strong situation** (a red traffic light, a funeral, an exam hall) has clear rules that make everyone behave alike, while a **weak situation** (a party, a free afternoon) lets traits show. Third, **interactionism** (**Norman Endler** and **David Magnusson**, 1976): behaviour depends on the person, the situation and their interaction. Mischel himself rebuilt consistency as stable if-then patterns in his cognitive-affective personality system (Section 4.9). Traits predict behaviour averaged across time and situations well, single acts poorly.

EXAM TRAP

Mischel's critique was that **situations often override traits**. He did not say that personality is entirely biological (closer to Eysenck), that traits outpredict situations (the opposite of his point), or that traits cannot be measured reliably: trait questionnaires were reliable; they simply predicted single behaviours poorly.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Eysenck's three dimensions are extraversion, neuroticism and _____.
2. Introverts have a higher baseline cortical arousal, regulated by the _____.
3. In Eysenck's scheme the unstable introvert corresponds to Galen's _____ temperament.
4. Cloninger linked harm avoidance to the neurotransmitter _____.
5. The label "Big Five" was coined by _____.
6. The NEO-PI-R has _____ items measuring 30 facets.
7. Plomin and Daniels showed that the _____ environment shapes personality more than the shared one.
8. Mischel's "personality coefficient" is a correlation of about _____.

Answers: (1) psychoticism, (2) ascending reticular activating system, (3) melancholic, (4) serotonin, (5) Lewis Goldberg, (6) 240, (7) non-shared, (8) 0.30.

4.8 Humanistic and Existential Theories: Rogers, Maslow and May

LOWER YIELD · 16 QS

Psychoanalysis pictured people driven by unconscious conflict; behaviourism pictured them shaped by reinforcement. By the 1950s a group of American psychologists objected that neither left room for what people experience most directly: choosing, hoping and trying to become someone. The humanists, Carl Rogers and Abraham Maslow, asked what a person could grow into. The existentialists, from Heidegger to Rollo May and Viktor Frankl, asked how a person faces freedom, anxiety and death.

The humanistic image of the person ●●

The **humanistic approach** to personality rests on four assumptions. People have **free will** and answer for their choices. Every person has an **innate tendency to grow** and develop, to reach his or her **maximum potential**. A person must be understood as a **whole**, not as a bundle of parts. What matters most is **conscious, present experience**, studied from the inside through the person's **phenomenological field**: the world as he or she perceives it, which may differ from the world as it is.

Table 4.25 sets this image beside its rivals. Humanism does not explain people by **unconscious conflict** (the psychodynamic view), **conditioned responses** (behaviourism) or information processing (the cognitive perspective), nor judge them against **statistical norms**, the "average is normal" yardstick of trait measurement and of one definition of abnormality (Chapter 8). Its founders were Abraham Maslow and Carl Rogers, with **Charlotte Buhler** (1893 to 1974), who studied the goals that organise the whole life course, and Rollo May, who brought in existential ideas. How it became psychology's "third force" is told in Section 1.2.

Table 4.25 Five images of the person.

Perspective	The person is	Behaviour explained by	Disorder seen as
Psychodynamic	Driven by hidden forces	Unconscious conflict, childhood	Repressed conflict
Behavioural	A product of learning	Conditioned responses, reinforcement	Faulty learning
Cognitive	An information processor	Thoughts, beliefs, schemas	Distorted thinking
Humanistic	Free, whole, growing	Conscious experience, the urge to grow	Blocked growth
Existential	Free, responsible, mortal	Choices made facing anxiety and death	Inauthentic, meaningless living

Disorder, on this view, is **blocked growth**: conditional love, unmet needs or a false self thwart the push to grow. Cure needs neither a reinforcement schedule nor the interpretation of buried conflict, but a relationship in which the therapist believes in the client's potential, offers positive regard and treats the client with respect (Chapter 7).

Critics reply that "actualisation" and "potential" are vague and nearly untestable, and that the focus on the individual's inner world misses the wider social and cultural contexts that shape experience and identity: the autonomous, self-made individual is a Western, individualist ideal.

EXAM TRAP

Maximum potential, an innate tendency to grow, free will and personal growth all point to **humanistic**. Maslow and Rogers propose the humanistic approach, not the existential (May, Frankl), cognitive or behavioural one. "Availability of reinforcement" is the behavioural view of disorder. The standard criticism is neglect of social and cultural context: "too much focus on cognition" fits the cognitive approach, "too much focus on past experience" fits psychoanalysis, and "neglecting personal experience" is the opposite of what humanists do.

Rogers: the self and its conditions of worth

Carl Ransom Rogers (1902 to 1987) built his theory from what clients told him about their own experience. Its single motive is the **actualising tendency**, the innate drive of every organism to maintain and enhance itself: it keeps the toddler walking despite falls, and later powers psychological growth. Once a self has formed, part of it becomes the **self-actualising tendency**, the push to actualise the self as perceived. From infancy, experience is judged by the **organismic valuing process**: what enhances the organism feels good and is sought, what hinders it feels bad and is avoided.

Each person lives in a private **phenomenal field**, also called the experiential world: all experience as the person perceives it, the only reality that counts for behaviour. Part of it is gradually set apart as "I" and "me": the **self-concept**, the organised picture of who one is. Beside it grows the **ideal self**, the self one would most like to be.

With the self comes a universal **need for positive regard** (warmth, love and acceptance from important others) and later a need for **positive self-regard**, approval from within. **Unconditional positive regard** values the child whatever he or she does: the act may be disapproved of, the child never is. **Conditional positive regard** is love with strings attached. The child then introjects (takes in as his or her own) the parents' standards as **conditions of worth**: "I am worthy only when I …". These work much like Freud's superego, and experience is now valued not by the organismic valuing process but by whether it earns approval.

The result is **incongruence**: a split between what the organism actually experiences and a self-concept burdened by conditions of worth, felt as tension, anxiety and confusion. Its opposite, **congruence**, is a close match between self-concept and experience. Threatening experience is kept out by **distortion** (admitting it in a changed form) or **denial** (keeping it out of awareness). Arjun, praised at home only when he tops the class in science, believes "I am the future doctor". When he finds he loves designing furniture and finds anatomy dull, he distorts it ("I am just tired") or denies it; tension builds and anxiety appears without a nameable cause. Figure 4.13 traces both routes.

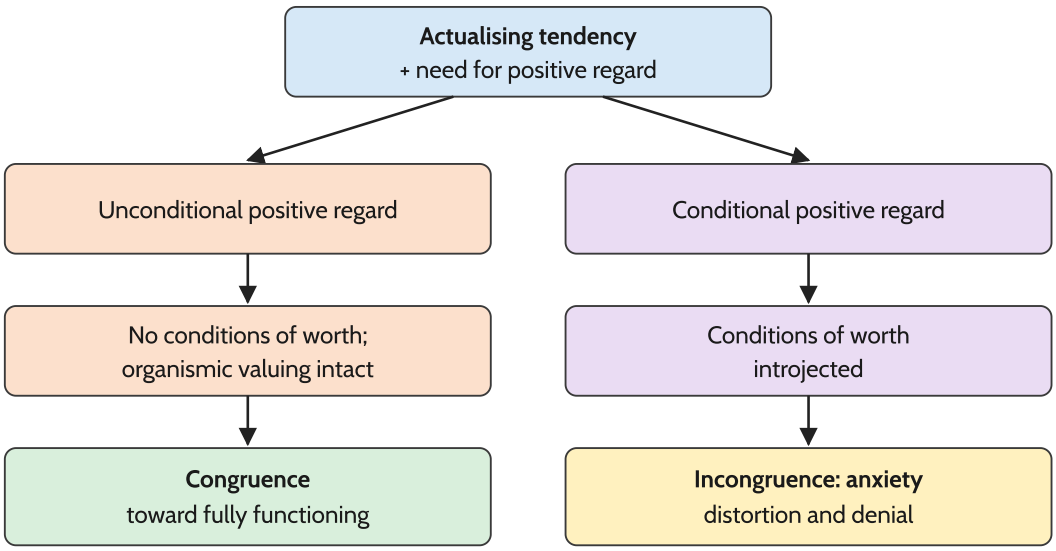


Figure 4.13 Two routes in Rogers's theory: the same need for positive regard leads to congruence when love is unconditional, and to incongruence and defence when it is conditional.

EXAM TRAP

"A split between the organismic valuing process and a self-concept burdened by conditions of worth, producing tension and confusion" is **incongruence**. Introjection is only how conditions of worth get in (and a defence mechanism, Section 4.3); conflict is a general term (id, ego and superego in Section 4.2, approach-avoidance in Section 1.10); the actualising tendency is the growth force that incongruence blocks.

The fully functioning person and the Q-sort

A child given unconditional positive regard stays congruent and moves toward what Rogers called the **fully functioning person**, "a direction, not a destination" (*On Becoming a Person*, 1961). Five features mark such a person: **openness to experience**, nothing distorted or denied; **existential living**, fully in each moment; **organismic trusting**, relying on one's own reactions; **experiential freedom**, a felt freedom to choose and responsibility for the choice; and **creativity**. Rogers stated the theory as **19 propositions** in *Client-Centered Therapy* (1951), beginning with the claim that each person lives at the centre of a continually changing world of experience and reacts to it as perceived. Therapy restores congruence; its core conditions are taught in Chapter 7.

Rogers also made the self measurable. The **Q-sort**, devised by the British psychologist **William Stephenson** (*The Study of Behavior*, 1953) and adopted by Rogers's group at Chicago, has the person sort self-descriptive statements ("I feel helpless", "I am emotionally mature") from "most like me" to "least like me" into a forced, roughly normal distribution: a few cards at each extreme, most in the middle. Because the statements are ranked against one another for the same person, the Q-sort captures within-person rather than between-person variation. Sorting once for the real self and once for the ideal self gives a self-ideal correlation, an index of congruence.

KEY STUDY: THE CASE OF MRS OAK (ROGERS, 1954)

Does successful therapy bring the perceived self closer to the ideal? Mrs Oak's self-ideal correlation rose from +.36 before client-centred therapy to +.79 a year after it, and in 25 clients studied by Rogers's associates **J. M. Butler** and **G. V. Haigh** (1954) the average rose from .01 before therapy to +.31 after it. Criticism: Q-sorts are self-reports, open to the wish to please the therapist, and a rising correlation can come from a lowered ideal as well as a changed self.

Research supports Rogers in part: conditionally loved children show unstable self-esteem, and large self-ideal gaps go with anxiety and depression. Critics note his reliance on conscious self-report and an actualising tendency assumed rather than shown; his lasting gifts were recorded therapy sessions as research data and the self-concept as a variable.

EXAM TRAP

The Q-sort was devised by Stephenson and used by Rogers to measure the self-concept and self-ideal congruence. It did not replace the MMPI (Chapter 2), and it is neither a test of infant reflexes nor a diagnostic test for schizophrenia.

Maslow: deficiency, being and the self-actualiser

Abraham Harold Maslow (1908 to 1970) asked what the healthiest people are like. His hierarchy of needs (physiological, safety, belongingness and love, esteem, self-actualisation, with lower needs stronger and prepotent) is taught in Section 1.10; here we follow what it implies for personality.

The four lower levels are **deficiency needs** (D-needs): a lack creates tension that drives us to fill it, and once filled they stop motivating. Behaviour driven by them is **D-motivation**. Once they are met, a person is moved differently. Maslow called this **metamotivation** (also B-motivation, from Being), the distinct motivation of self-actualisers: not making up a lack but growing and expressing oneself, even seeking tension. Its objects are being needs, the **metaneeds** or B-values, such as truth, goodness, beauty, justice, wholeness and meaningfulness. Frustrated metaneeds produce **metapathology**: cynicism, apathy and a sense of meaninglessness that the sufferer cannot trace to any lack (Table 4.26).

Table 4.26 Maslow's deficiency and being realms.

Feature	Deficiency (D)	Being (B)
Needs	Physiological, safety, belonging, esteem	Metaneeds (B-values): truth, beauty, justice
Motivation	D-motivation: fill a lack	Metamotivation: grow and express
Tension	Reduced	Sought and enjoyed
When thwarted	Illness, anxiety	Metapathology: cynicism, apathy
Typical of	Most people	Self-actualisers (1 percent or fewer)

Self-actualisation is the striving to fulfil one's potential, to become everything one is capable of becoming. The neurologist **Kurt Goldstein** (1878 to 1965) coined the term in *The Organism* (German 1934, English 1939); Maslow made it the summit of his hierarchy. Finding

almost none among students, he studied older people and historical figures, among them the later **Abraham Lincoln**, **Thomas Jefferson**, **Albert Einstein** and **Eleanor Roosevelt**, and estimated that 1 percent of people or fewer self-actualise. They showed efficient perception of reality; acceptance of self, others and nature; spontaneity; problem-centring (a mission outside themselves); a need for privacy; autonomy and resistance to enculturation; freshness of appreciation; peak experiences; social interest (Adler's term, Section 4.4); deep but few relationships; a democratic character; and creativity.

A **peak experience** is a brief, intense moment of ecstasy and awe in which the sense of self is transcended; late in life Maslow added the **plateau experience**, a calmer, longer-lasting and more voluntary serenity. Why do so few self-actualise? Besides unmet lower needs and poor upbringing, Maslow blamed the **Jonah complex**, the fear of one's own greatness, named after the prophet who ran from God's call. The **Personal Orientation Inventory** (POI) of **Everett Shostrom** (1964) later measured self-actualisation with 150 paired statements. Critics note that the sample was small and hand-picked by Maslow's own values, and that his terms are vague. The transpersonal "fourth force" he foresaw is in Section 1.2.

> EXAM TRAP
>
> The distinct motivation of self-actualisers is **metamotivation**; D-motivation drives everyone else, and "E-motivation" and "F-motivation" do not exist. "The striving for fulfilment of one's potential" is **self-actualisation**, not the self-concept (one's picture of oneself), the ideal self (the self one wishes to be) or self-efficacy (Bandura's belief that one can perform a task, Section 4.9).

Existential roots: Dasein and the three worlds ●●

Both schools reject determinism, but humanism assumes an inner nature, a seed of potential waiting to unfold, and existentialism denies even that. It grew from the Danish philosopher **Soren Kierkegaard** (1813 to 1855), who wrote of the anxiety of choosing oneself; the German **Martin Heidegger** (1889 to 1976), author of *Being and Time* (1927); and the French **Jean-Paul Sartre** (1905 to 1980), whose slogan was **existence precedes essence**: we exist first and then make ourselves by our choices. Nothing fixes the meaning of a life in advance, so each person must continually recreate himself or herself, and the meaning of existence keeps changing. This is existential psychology: not humanistic psychology, which assumes a potential to fulfil, nor experimental psychology, which seeks general laws by controlled experiment, nor philosophy alone, since it carries the philosophers' ideas into the study and treatment of persons.

Heidegger's key term was **Dasein** (German, "being there"): human existence as **being-in-the-world**. A person is not a mind looking out at a separate world but one unit with it, so Dasein expresses the basic unity of person and environment. Behaviourists, cognitivists and psychoanalysts all split person from world; existentialists refuse the split. The Swiss psychiatrists **Ludwig Binswanger** (1881 to 1966) and **Medard Boss** (1903 to 1990) applied Heidegger to patients as **Daseinsanalysis**, describing each patient's world rather than explaining symptoms.

Binswanger, and after him May, described three modes of being-in-the-world, lived at once (Figure 4.14). The **Umwelt** ("world around") is the biological and physical world: body, drives, sleep and death. The **Mitwelt** ("with-world") is the world of relationships with others. The

Eigenwelt ("own world") is one's relationship with oneself. Modern people, May argued, are alienated in all three.

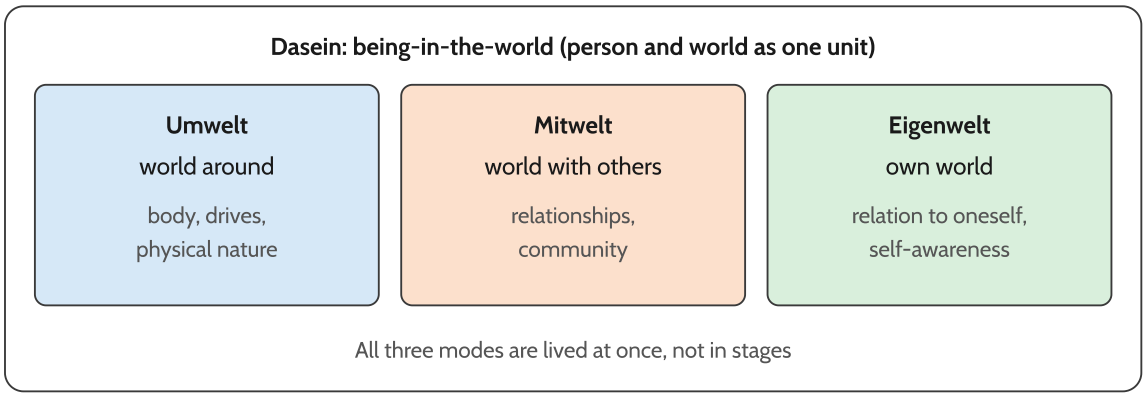


Figure 4.14 Dasein is the whole, the unity of person and world; Umwelt, Mitwelt and Eigenwelt are three sides of it that every person lives simultaneously.

> EXAM TRAP
>
> Dasein and being-in-the-world are **existential**, not behavioural, cognitive or psychoanalytic. Dasein is the unity of person and environment; Umwelt, Mitwelt and Eigenwelt are its three modes, so none of them is "the basic unity". The modes are not id, ego and superego, not conscious, preconscious and unconscious (both Section 4.2), and not Piaget's sensorimotor, preoperational and concrete operational stages (Chapter 6).

Rollo May: anxiety, the daimonic and love

Rollo May (1909 to 1994) brought existential psychology to America. After months in a tuberculosis sanatorium, unsure whether he would live, he wrote *The Meaning of Anxiety* (1950). For May anxiety is **ontological**, part of being itself: it arises whenever our existence, or a value we identify with, might be destroyed, and its root is awareness of **nonbeing**, death but also loss of self in conformity. **Normal anxiety** is proportionate to the threat, involves no repression and can be faced and used creatively; there is no freedom without it. **Neurotic anxiety** is disproportionate, involves repression and is handled by blocking off awareness.

In *Love and Will* (1969) May added three ideas. **Intentionality** is the structure that gives meaning to experience and lets a person decide about the future; it underlies will. The **daimonic** is any natural function that can take over the whole person, such as sex, anger or the craving for power: directed, it fuels creativity; unintegrated, it destroys. Love takes four forms: **sex** (biological release), **eros** (the longing to create and unite), **philia** (intimate, nonsexual friendship) and **agape** (selfless concern for another's welfare). In *Freedom and Destiny* (1981) he argued that freedom has meaning only against **destiny**, the givens of body, history and death; healthy people accept their destiny and use their freedom within it.

Frankl and the existential constructs

The Viennese psychiatrist **Viktor Emil Frankl** (1905 to 1997) survived Auschwitz and observed that prisoners who held on to a reason to live were the likelier to endure. In *Man's Search for Meaning* (German original 1946) he argued that the primary human motive is neither the will

to pleasure (Freud) nor the will to power (Adler) but the **will to meaning**. Frustrated, it leaves an **existential vacuum** of emptiness and boredom. If symptoms follow, the result is a **noogenic neurosis** (Greek *noos*, mind or spirit), arising from an existential problem such as loss of meaning or a clash of values rather than from conflict between drives; its anxiety is noogenic anxiety. Frankl's therapy, logotherapy, is taught in Chapter 7.

Across the existential writers five constructs recur: coming to terms with anxiety; confronting the universal fear of death; accepting freedom and the responsibility it brings; understanding our ultimate isolation, since no one can live or die for us; and facing meaninglessness by creating meaning. Irvin Yalom's four ultimate concerns restate the list for therapy (Chapter 7). Existentialists do not dig for unconscious fears and desires; that is the psychoanalytic aim. Critics find existential writing vague and hard to test.

> **EXAM TRAP**
>
> Noogenic neurosis and noogenic anxiety are Frankl's and belong to the **existential** school, not the humanistic, Gestalt (the school of perception in Section 1.2, or Perls's Gestalt therapy in Chapter 7) or psychodynamic school. "Uncovering our unconscious fears and desires" is psychoanalytic, the one option that is not an existential construct.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Rogers's term for the innate drive of every organism to maintain and enhance itself is the _____ tendency.
2. A split between experience and a self-concept burdened by conditions of worth is _____.
3. The Q-sort was devised by William _____.
4. Maslow called the distinct motivation of self-actualisers _____.
5. The fear of one's own greatness is Maslow's _____ complex.
6. Heidegger's term for being-in-the-world, the unity of person and environment, is _____.
7. The mode of being made up of relationships with others is the _____.
8. Frankl called a neurosis arising from loss of meaning a _____ neurosis.

Answers: (1) actualising, (2) incongruence, (3) Stephenson, (4) metamotivation, (5) Jonah, (6) Dasein, (7) Mitwelt, (8) noogenic.

4.9 Behavioural, Social-Cognitive and Cognitive Theories

LOWER YIELD · 16 QS

Why does one student freeze before every viva while his classmate, no cleverer, walks in calm? The theorists of this section ask what he has learned, what he expects and how he reads the situation. The story runs from Skinner, who denied any inner person, through the social learning theorists who put expectation and thought back in, to George Kelly, for whom personality is a way of construing the world.

Skinner: personality without a self

Burrhus Frederic Skinner (1904 to 1990) gave the most radical answer: personality is only a **collection of learned behaviour patterns**, each kept alive by its consequences. This is **radical behaviourism**: thoughts and feelings are behaviour obeying the same laws, and inner causes explain nothing ("he shouts because he is aggressive" is circular, since the only evidence of aggressiveness is the shouting).

Skinner distinguished two kinds of behaviour. **Respondent behaviour** is elicited by a specific stimulus: the knee-jerk to a tap, or salivation to Pavlov's bell after conditioning. **Operant behaviour** is emitted spontaneously or voluntarily and operates on the environment to change it: the rat presses a lever and food drops, the child cries and a parent comes. Most human behaviour is operant, so a personality is a history of reinforcement. Reinforcement and its schedules are taught in Section 1.6.

Shaping (successive approximation) builds a new response by reinforcing closer and closer approximations to it. **Superstitious behaviour** is behaviour strengthened by accidental reinforcement: pigeons fed every 15 seconds, whatever they did, kept repeating whatever they happened to be doing when food arrived, as a batsman clings to the bat that once scored a century. **Self-control**, for Skinner, is not willpower but control of the outside variables that control us, by four techniques: stimulus avoidance (studying away from the phone), self-administered satiation (eating sweets until sick of them), aversive stimulation (announcing a diet so that lapses bring ridicule) and self-reinforcement.

Walden Two (1948) is his novel of a utopian community run by positive reinforcement; *Beyond Freedom and Dignity* (1971) argued that freedom is an illusion and that cultures should design their contingencies deliberately. The approach is precise, testable and the source of behaviour modification and token economies; critics say it ignores cognition, heredity and inner experience and generalises too freely from rats and pigeons.

> EXAM TRAP
>
> "Behaviour emitted spontaneously or voluntarily that operates on the environment to change it" is **operant behaviour** (Skinner). The distractors are other theorists' terms: the **opposition principle** is Jung's (psychic energy arises from conflict between opposites; Section 4.4), **behaviour potential** is Rotter's (below) and **personal dispositions** are Allport's individual traits (Section 4.6).

Dollard and Miller: Freud in learning terms

Earlier, two Yale scholars had married psychoanalysis to learning theory. **John Dollard** (1900 to 1980) and **Neal Elgar Miller** (1909 to 2002) restated Freud's clinical insights in the drive-reduction terms of **Clark Leonard Hull** (1884 to 1952), in *Social Learning and Imitation* (1941) and *Personality and Psychotherapy* (1950): the **early social learning theory**. Learning needs four things ("want something, notice something, do something, get something"): a **drive**, a strong stimulus that impels action; a **cue**, which signals when, where and how to respond; a **response**; and **reinforcement**, the drive reduction that strengthens the cue-response link.

Their key idea was the **learned drive**. Fear is learned: Miller's rats, once shocked in a white box, learned to turn a wheel to escape it with no shock present, reinforced by fear reduction alone. Freud then translates easily. **Repression** is learned not-thinking: a forbidden thought

arouses fear, dropping it reduces fear, and the relief reinforces not-thinking. Neurotic conflict is competing approach and avoidance tendencies. Both grow stronger nearer the goal, but the avoidance gradient is steeper, so in an **approach-avoidance conflict** a person approaches a mixed goal from afar, retreats near it and hovers where the two gradients cross (the other conflict types are in Section 1.10). With Yale colleagues they also proposed the **frustration-aggression hypothesis** (1939): frustration always leads to some form of aggression, and aggression always presupposes frustration (Chapter 3). Table 4.27 shows the label each theorist carries.

Table 4.27 Who goes with which label.

Theorist	Label	Core ideas
B. F. Skinner	Radical behaviourism	Operant behaviour shaped by reinforcement
Dollard and Miller	Early social learning	Freud in Hull's terms: drive, cue, response, reinforcement
Bandura and Walters	Later social learning; social cognitive theory	Observational learning, reciprocal determinism, self-efficacy
Julian Rotter	Social learning (expectancy-value)	Behaviour potential, locus of control
Abraham Maslow	Self-actualisation (Section 4.8)	Humanistic hierarchy of needs

EXAM TRAP

Matching items pair Dollard and Miller with **early** social learning (Hull's drive theory) and Bandura and Walters with **later** social learning (observation and cognition); Skinner is **radical behaviourism**, Maslow **self-actualisation**.

Bandura: reciprocal determinism and self-regulation ●●

Albert Bandura (1925 to 2021) began where Dollard and Miller stopped. With **Richard Walters** he wrote *Social Learning and Personality Development* (1963): people learn much of what they do by watching models, without performing the act or being reinforced, which is **observational** or **vicarious learning** (its four processes and the Bobo doll studies are in Section 1.6). In *Social Foundations of Thought and Action* (1986) he renamed the approach **social cognitive theory**, stressing that thought, not reinforcement alone, guides action.

Its organising principle is **reciprocal determinism** (Figure 4.15): person (cognitive) factors such as beliefs and expectations, behaviour and the environment each influence the other two, in both directions. The master is not the environment (Skinner), the genes (biological theories) or the unconscious (psychodynamic theory). A student who believes she can master statistics (person) joins a study group (behaviour); its praise (environment) lifts her confidence, and she picks harder problems, changing the environment again.

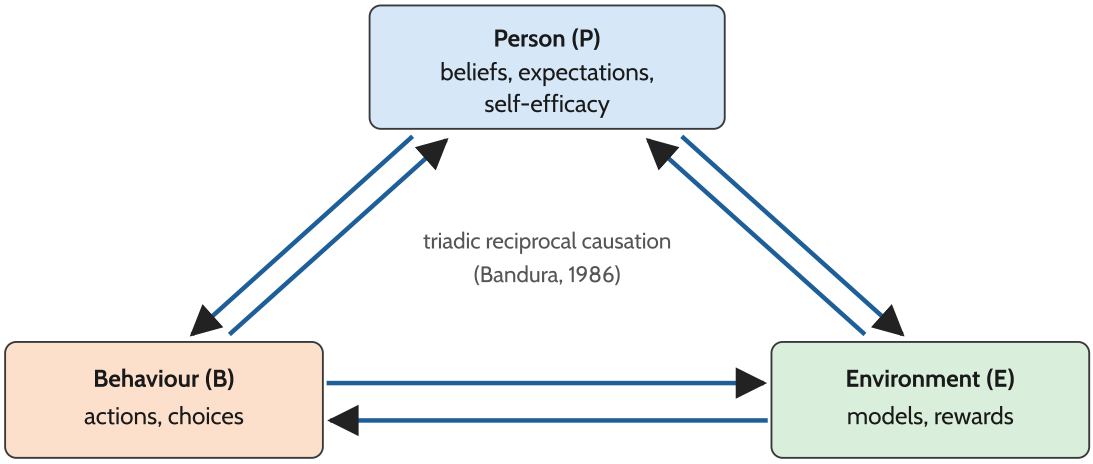


Figure 4.15 Reciprocal determinism. Each pair of factors is joined by arrows running both ways, so no single factor is the master cause.

EXAM TRAP

Reciprocal determinism is central to **social cognitive theory**, not psychodynamic, humanistic or biological theory, and it means two-way influence, not control by environment alone or genes alone, nor a denial of social reinforcement. Bandura's constructs are reciprocal determinism and observational (vicarious) learning, not **reciprocal inhibition** (Joseph Wolpe: a response incompatible with anxiety, such as relaxation, suppresses it; the basis of systematic desensitisation), **reciprocal altruism** (Robert Trivers: helping those who will help in return) or the invented "reciprocal interpretation" and "reciprocal dualism".

Bandura noticed that most behaviour occurs with no external reward or punishment in sight: nobody pays a novelist per page. Most behaviour must therefore stem from **self-regulation**. People adopt standards, largely from models such as parents and teachers, and run three processes: **self-observation** (watching their own performance), **judgement** (comparing it with the standard) and **self-response** (pride when they meet it, self-criticism when they fall short). Rewarding or punishing oneself this way is **self-reinforcement**. It is neither negative reinforcement (removal of an unpleasant stimulus, still an external consequence) nor Skinner's behavioural determinism. The machinery can also be switched off: **moral disengagement** silences self-condemnation during cruelty through moral justification, euphemistic labels, displaced responsibility and dehumanising the victim.

Self-efficacy: believing you can ●●

At the centre of Bandura's theory sits **self-efficacy**: a person's belief in his capability to organise and execute the actions needed to achieve a specific goal. Bandura introduced it in 1977 as a unifying theory of behavioural change, a cognitive-behavioural account of how people change in therapy (his concept, not Aaron Beck's, Albert Ellis's or Skinner's). It is the key motivational factor of social cognitive theory, ahead of external rewards, biological drives or goal difficulty (goal-setting theory, Section 1.10): people attempt what they believe they can do, aim higher and persist after failure.

Bandura separated two expectations (Figure 4.16). An **outcome expectancy** is the belief that a behaviour will produce an outcome ("steady revision gets a good rank"); an **efficacy ex-**

pectancy is the belief that one can perform it ("I can revise three hours a day"). Only the second is self-efficacy, and it is task-specific: confident at chess, helpless in a viva.

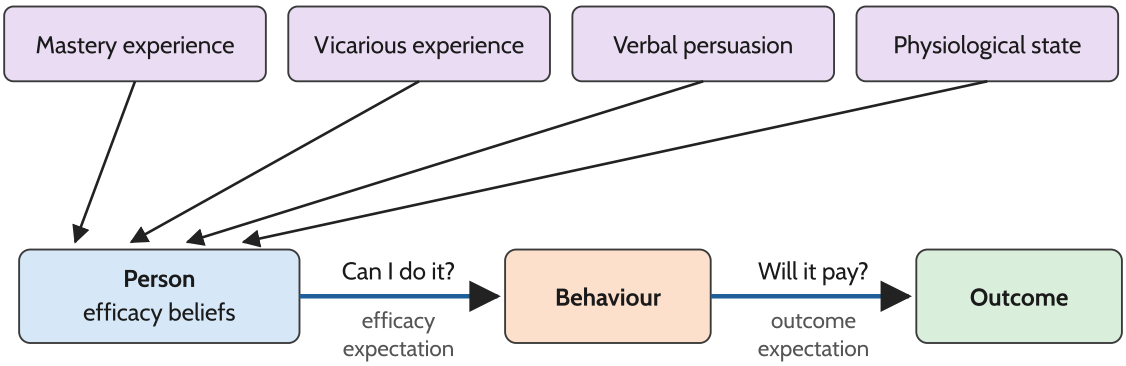


Figure 4.16 Efficacy versus outcome expectation. The four sources feed only the first, "Can I do it?".

Efficacy beliefs come from four sources, the first being the strongest:

1. **Performance (enactive) mastery**: one's own past successes; early failures weaken efficacy most.
2. **Vicarious experience**: seeing someone similar succeed ("if she can, so can I").
3. **Verbal persuasion**: being told "you can do it", which helps only if realistic.
4. **Physiological and emotional states**: calm is read as readiness, a pounding heart as inability.

KEY STUDY: MASTERING SNAKE PHOBIA (1969)

Bandura, Edward Blanchard and B. Ritter treated adults with intense snake phobia by a film of models approaching snakes, by systematic desensitisation, or by **participant modelling**: watching a live model handle a snake, then being guided, at first in gloves, to touch and hold it. Participant modelling, which gives direct mastery, worked best, removing the phobia in almost everyone. In later studies (1977) clients' efficacy beliefs after treatment predicted how close they would approach a snake, whichever treatment they had received. Critics reply that efficacy ratings may be predictions of behaviour rather than causes.

Collective efficacy is a group's shared belief that together it can succeed, as in a cricket team or a village self-help group. Bandura later framed the theory around **human agency**: people are agents who plan, anticipate, regulate and reflect on themselves, not puppets. Table 4.28 separates self-efficacy from its look-alikes.

Table 4.28 Self-efficacy and the concepts confused with it.

Concept	Theorist	What it is
Self-efficacy	Bandura	"I can do this task": a specific capability belief
Self-esteem, self-worth	Many	Global liking and acceptance of oneself (Section 4.10)
Locus of control	Rotter	Whether outcomes depend on me or on fate, chance, others
Inferiority complex	Adler	Exaggerated inadequacy behind striving for superiority (Section 4.4)

Concept	Theorist	What it is
Collective unconscious	Jung	Inherited layer of the psyche holding archetypes (Section 4.4)
Congruence	Rogers	Match between self-concept and experience (Section 4.8)

Rotter: expectancy, value and locus of control ●●

Julian Bernard Rotter (1916 to 2014) built his own **social learning theory** in *Social Learning and Clinical Psychology* (1954), asking which of several possible behaviours a person will perform in a situation. Four concepts answer:

- **Behaviour potential**: "the potentiality of any behaviour occurring in any given situation as calculated in relation to any single reinforcement or set of reinforcements".
- **Expectancy**: the subjective probability that the behaviour will bring that reinforcement.
- **Reinforcement value**: how much he prefers that reinforcement to others, expectancies being equal.
- **Psychological situation**: the situation as the person perceives it, which shapes both expectancy and value.

Behaviour potential is a function of expectancy and reinforcement value, BP = f(E & RV), often read as a product: a student who prizes a seat at NIMHANS but believes no study will win it will not study. There is no "punishment value" in the theory. Expectancies carried across many situations are **generalised expectancies**, which give personality its consistency; the best known are locus of control and interpersonal trust (belief that others' word can be relied on; Interpersonal Trust Scale, 1967).

Locus of control is a generalised expectancy about whether one's outcomes are controlled by internal efforts or external forces. A person with an **internal locus of control** believes outcomes depend on his own effort, ability and preparation, like the student who credits his exam result to his own work. One with an **external locus of control** credits luck, fate, chance, powerful others or an easy examiner. Rotter's **Internal-External (I-E) Scale** (1966) has 29 forced-choice pairs, 23 scored and 6 fillers; higher scores mean more externality. Internals, on average, achieve more, seek information, look after their health and cope actively; externals report more anxiety. Warm, consistent parents who praise achievement and encourage independence tend to raise internals. **Hanna Levenson** (1973) later split it into three separate dimensions: **internality**, **powerful others** and **chance** (her IPC scale).

> EXAM TRAP
>
> Locus of control has two poles, **internal** and **external**; there is no "neutral" or "situational" locus. It concerns **external forces versus internal efforts**, not cognitive schemas (Beck), genetic predisposition or social status. It is not self-efficacy: locus of control asks "do outcomes depend on what I do?", self-efficacy asks "can I do it?". Attribution theory, with its own locus dimension, is in Section 1.10.

Mischel's cognitive-affective personality system

Walter Mischel (1930 to 2018) is known for the person-situation critique of trait theory (Section 4.7): people behave less consistently across situations than traits predict. His positive answer, with **Yuichi Shoda**, was the **cognitive-affective personality system** (CAPS, 1995), a

stable network of five kinds of units: **encodings** (how one construes self, others and situations), **expectancies and beliefs**, **affects**, **goals and values**, and **competencies and self-regulatory plans**. Different situations activate different units, so what is stable is not an average level of behaviour but a **behavioural signature** of if-then relations: Ravi is aggressive *if* teased by classmates and meek *if* scolded by a teacher, every time. His delay of gratification research shows self-regulatory competencies at work (the marshmallow test is in Chapter 6).

Kelly: the person as scientist ●●

George Alexander Kelly (1905 to 1967) set out the most purely cognitive theory in *The Psychology of Personal Constructs* (1955). Ordinary people, he argued, behave like scientists: they form theories, predict events and revise the theories when predictions fail. The theories are **personal constructs**, bipolar ways of construing events, such as friendly versus hostile. A construct may be **permeable** (admitting new events, open to revision) or **impermeable** (rigid whatever the evidence, as in prejudice). Since no one is trapped in one construal, people can always re-construe their lives: **constructive alternativism**.

The theory has a **fundamental postulate**, "a person's processes are psychologically channelised by the ways in which he anticipates events", and eleven corollaries (Table 4.29). Since anticipation itself drives behaviour, Kelly needed no separate motivation theory, and he redefined emotions through constructs. **Anxiety** is recognising that events lie outside the range of one's constructs; **threat** is awareness of imminent comprehensive change in one's core constructs; **fear** is the same for an incidental change; **guilt** is perceiving oneself dislodged from one's core role; **hostility** is continued effort to extort evidence for a prediction that has already failed.

Table 4.29 Kelly's eleven corollaries.

Corollary	Meaning
Construction	We anticipate events by construing their replications
Individuality	People differ in how they construe events
Organisation	Constructs form a hierarchy, superordinate to subordinate
Dichotomy	Every construct is bipolar (honest versus dishonest)
Choice	We choose the pole that best extends our system
Range	Each construct fits a limited range of events
Experience	Constructs are revised as events unfold
Modulation	Revision is limited by the permeability of constructs
Fragmentation	A person may hold mutually inconsistent subsystems
Commonality	People who construe alike are psychologically alike
Sociality	To relate to someone we must construe his constructions

To find a person's constructs Kelly devised the **Role Construct Repertory Test** (Rep test). The client names people who fill role titles in his life (mother, father, a liked teacher, a disliked person), then takes them three at a time and says how two are alike and differ from the third; each answer yields a construct. Set out as a matrix of people by constructs, it becomes the **repertory grid**. To change constructs he used **fixed-role therapy**: from the client's self-

characterisation sketch, the therapist writes a fixed-role sketch of a new character with different constructs, and the client acts that character in daily life for about two weeks, discovering that new constructs work better. Kelly is praised for respecting the person's own view and criticised for neglecting emotion and development.

> EXAM TRAP
>
> Kelly is to the **Rep test** as Cattell is to the **16PF** (Section 4.6). The others: **California Psychological Inventory**, Harrison Gough; **Q-sort**, William Stephenson, used by Rogers; **MMPI** (Chapter 2). **Fixed-role therapy** is Kelly's; Aaron Beck developed cognitive therapy, Albert Ellis rational emotive behaviour therapy and Fritz Perls Gestalt therapy (Chapter 7).

Cognitive styles

A **cognitive style** is a habitual way of perceiving and processing information, distinct from ability. **Herman Witkin** (1916 to 1979) described **field dependence-independence**. In the **rod-and-frame test** a person in a dark room sets a luminous rod to vertical inside a tilted luminous frame: field-dependent people tilt the rod towards the frame, field-independent people trust their bodily sense of upright and also find shapes hidden in complex designs faster on the **Embedded Figures Test**. Field-independent people are more analytic, field-dependent people more global and socially attentive. **Jerome Kagan** (1929 to 2021) described **reflection-impulsivity**: on the Matching Familiar Figures Test, where a child picks which of several near-identical pictures matches a standard, impulsive children answer fast with many errors, reflective children slowly and accurately.

> SCAN YOUR BRAIN
>
> Fill in the blanks. The answers are below; cover them first.
>
> 1. Emitted behaviour that operates on the environment is _____ behaviour.
> 2. Dollard and Miller described learning by drive, cue, response and _____.
> 3. Bandura's principle that person, behaviour and environment influence one another is _____.
> 4. The strongest source of self-efficacy is _____.
> 5. For Rotter, behaviour potential is a function of expectancy and _____.
> 6. Mischel's stable if-then patterns are called _____.
> 7. Acting out a new character for about two weeks is Kelly's _____.
> 8. Witkin's rod-and-frame test measures _____.
>
> **Answers:** (1) operant, (2) reinforcement, (3) reciprocal determinism, (4) performance (enactive) mastery, (5) reinforcement value, (6) behavioural signatures, (7) fixed-role therapy, (8) field dependence-independence.

4.10 The Self, Positive Traits and Indian Approaches to Personality

LOWER YIELD · 16 QS

Who am I, how do I feel about that person, and what makes a life go well? This last section turns from personality seen from outside to the self as people experience it, then to narrow

traits and the strengths that positive psychology studies, and ends with India's oldest personality theory, the three gunas, and its bodily cousin, the Ayurvedic doshas.

The self: James, Cooley and Mead ●●

The modern study of the self begins with **William James** (1842 to 1910), whose *Principles of Psychology* (1890) split it in two. The **I** is the self as knower, the subject doing the thinking; the **me** is the self as known, everything a person can call his own. James divided the me into three parts. The **material self** is the body, clothes, family, home and possessions, which is why losing a phone can feel like losing a piece of yourself. The **social self** is the recognition one gets from others; a person has as many social selves as there are groups whose opinion he cares about. The **spiritual self** is the inner world of dispositions, conscience and faculties, the most intimate and enduring part.

James also gave a formula: **self-esteem = success / pretensions**. A student who aimed only to pass is content with 55 percent; one who pretended to the first rank is crushed by the second (James's example was a man ashamed to be only the second-best boxer in the world). So self-esteem rises in two ways: succeed more, or lower the pretensions.

For **Charles Horton Cooley** (1864 to 1929), writing in *Human Nature and the Social Order* (1902), the self is social from the start. His **looking-glass self** is the self-concept we build by seeing ourselves in other people as in a mirror, in three steps (Figure 4.17): we imagine how we appear to the other person, we imagine that person's judgement of that appearance, and we develop a self-feeling, such as pride or shame, from that imagined judgement. Every step is imagined, so the mirror can distort.

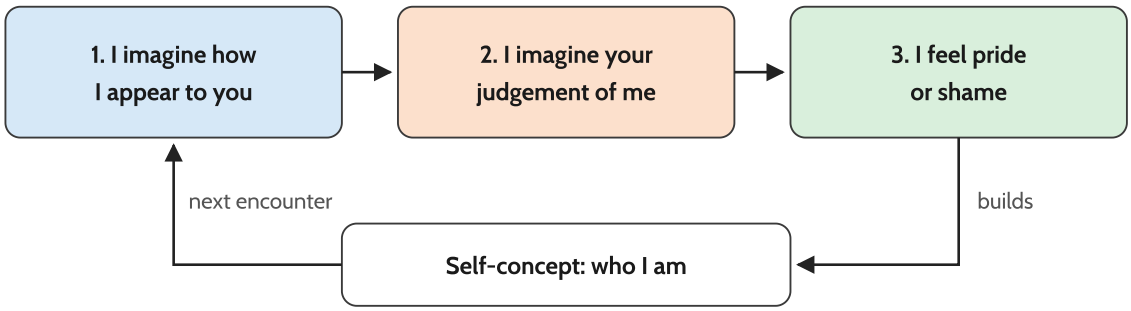


Figure 4.17 Cooley's looking-glass self. Notice that all three steps happen in the imagination: the self is built from what we think others think of us.

George Herbert Mead (1863 to 1931), whose lectures appeared after his death as *Mind, Self and Society* (1934), explained how the mirror gets inside us. With Cooley he founded the tradition later called **symbolic interactionism**: the self emerges through social interaction and shared symbols, above all language. Mead's "I" is the spontaneous, creative, unpredictable response of the person; his "me" is the organised set of others' attitudes that the person has taken in, the socialised self. The child builds the me by taking roles. In the **play stage** the child takes the role of one particular other at a time, playing at being mother or policeman. In the **game stage**, organised games demand holding everyone's role at once: a fielder in cricket must anticipate bowler, batter and captain within shared rules. The organised attitudes of

the whole group, and finally of the community, form the **generalised other**, the standpoint from which the person judges himself.

> **EXAM TRAP**
>
> Looking-glass self = imagined appearance, imagined judgement, self-feeling. The distractors are other theories: innate, unconscious drives (Freud, Section 4.2); comparing ourselves with similar others (Leon Festinger's social comparison theory, 1954, Chapter 3); resolving psychosocial crises (Erik Erikson, Section 6.10); operant reinforcement of self-statements (Section 4.9); solitary introspection or genetic inheritance (Cooley's self is neither private nor inborn). James's I is the knower, Mead's the spontaneous actor; the generalised other is Mead's.

Self-concept, self-esteem and the self-system

Two terms that are often blurred need separating. The **self-concept** is the cognitive description of who one is, the organised set of beliefs about one's own attributes: "I am a Kannadiga, a fast bowler, shy, good at maths." **Self-esteem** is the evaluative and affective component, how one feels about that self: "and I am worthwhile", or "and I am a failure." Two students may share the self-concept "I am shy", one at peace with it and one ashamed of it. The standard measure of global self-esteem is the ten-item Self-Esteem Scale (1965) of **Morris Rosenberg** (1922 to 1992), with statements such as "On the whole, I am satisfied with myself" rated from strongly agree to strongly disagree.

Nor is self-esteem the same as **self-efficacy** (Albert Bandura), the belief that one can carry out a specific task such as a statistics problem (Section 4.9), or **self-actualisation** (Kurt Goldstein's term, made famous by Abraham Maslow), the striving to fulfil one's potential (Section 4.8). Later researchers treated the self as a system that stores and uses information about itself (Table 4.30).

Table 4.30 The self as a cognitive and motivational system.

Concept	Proposed by	What it means
Self-schema	**Hazel Markus**, 1977	A generalisation about the self, drawn from past experience, that organises and guides how self-related information is processed
Possible selves	Markus and **Paula Nurius**, 1986	Selves one hopes to become (a clinical psychologist) or fears becoming (an unemployed graduate); they motivate present action
Self-discrepancy theory	**E. Tory Higgins**, 1987	Gaps between the actual self and the ideal self (hopes) or the ought self (duties) produce distinct emotions
Self-monitoring	**Mark Snyder**, 1974	High self-monitors tailor their self-presentation to each situation; low self-monitors act from inner attitudes and stay consistent

Higgins's theory has a sharp clinical edge. An actual-ideal gap ("I am not who I hoped to be") brings **dejection-related** emotions such as disappointment, sadness and depression; an actual-ought gap ("I am not who I should be") brings **agitation-related** emotions such as anxiety, guilt and fear. Elsewhere: Carl Rogers's self and ideal self (Section 4.8), Gordon Allport's proprium (4.6), Heinz Kohut's self psychology (4.5), the self-reference effect (1.7), social identity and self-presentation (Chapter 3), and the Indian jiva, ahamkara and atman (1.2).

EXAM TRAP

Self-concept is cognitive ("who am I?"); self-esteem is evaluative and affective ("how do I feel about me?"). The reversed pairing is wrong, the two are not identical, and the individual versus collective self (a self defined by personal traits versus one defined by group memberships) is a different distinction altogether.

Limited-domain traits: sensation seeking and explanatory style

A **limited-domain theory** studies one trait in depth rather than the whole person. **Marvin Zuckerman** (1928 to 2018) defined **sensation seeking** as the need for varied, novel, complex and intense sensations and experiences, and the willingness to take physical, social, legal and financial risks for them. His Sensation Seeking Scale (SSS; first form 1964, later the 40-item Form V) offers forced choices such as "I would like to try parachute jumping". Factor analysis yielded four components:

- **Thrill and adventure seeking:** a desire for speed, danger and defying gravity, as in bungee jumping or scuba diving.
- **Experience seeking:** novel experience through travel, music, art or a nonconformist lifestyle.
- **Disinhibition:** release through wild parties, drinking and risky sex.
- **Boredom susceptibility:** an aversion to routine, repetition and predictable people.

Sensation seeking is strongly heritable and belongs within Eysenck's extraversion, with its impulsive form also tied to psychoticism (Section 4.7). It rises in adolescence, falls with age and is higher in men. High sensation seekers have low levels of **monoamine oxidase** (MAO), the enzyme that breaks down dopamine and the other monoamines, which Zuckerman linked to a strong dopamine-driven approach system.

Martin E. P. Seligman (born 1942) reached optimism by way of learned helplessness (Section 1.6), in which dogs given inescapable shock later failed to escape avoidable shock. People, he found, differ in how far helplessness spreads, depending on how they explain bad events to themselves. This habitual way of explaining is the **explanatory style**, which *Learned Optimism* (1990) describes by three Ps. **Permanence**: is the cause lasting ("I will always fail") or temporary ("I was tired that day")? **Pervasiveness**: does it spoil everything ("I am useless") or one area ("I am weak in statistics")? **Personalisation**: is it me ("I am stupid") or outside me ("the paper was unusually hard")? The pessimist explains bad events as permanent, pervasive and personal, and good events as temporary, specific and lucky; the optimist does the reverse, and is less prone to depression, illness and giving up. The **Attributional Style Questionnaire** (ASQ; Christopher Peterson, Seligman and colleagues, 1982) measures it: the respondent names the main cause of imagined good and bad events and rates it as internal, stable and global. Being learned, the style can be retrained. Dispositional optimism and hardiness belong to stress (Section 1.11).

Well-being: hedonic and eudaimonic

Positive psychology (its founding, flow and PERMA are in Sections 1.2 and 1.10) asks what makes a life go well, and two traditions answer differently. **Hedonic well-being** equates the good life with pleasure: positive feelings, comfort, enjoyment and satisfaction with life. **Ed Diener** (1946 to 2021) measured it as **subjective well-being** (SWB): high life satisfaction, fre-

quent positive affect and infrequent negative affect, with life satisfaction assessed by his Satisfaction With Life Scale (1985).

Eudaimonic well-being goes back to **Aristotle** (384 to 322 BCE), for whom *eudaimonia*, flourishing, came from living by virtue and realising one's best potential. Here well-being means cultivating personal strengths, meaning and growth and contributing to the greater good, whether or not it feels pleasant at the moment. **Carol Ryff** (1989) gave it six dimensions of **psychological well-being**: self-acceptance, personal growth, purpose in life, environmental mastery (managing one's surroundings and demands), autonomy (self-determination and independence from social pressure) and positive relations with others. A holiday of films and good food yields hedonic well-being; one spent teaching village children to read may leave you tired but richer in eudaimonic well-being. The two correlate but are distinct.

> EXAM TRAP
>
> Eudaimonic = cultivating strengths, growth and contribution to the greater good. "Pleasure and relaxation" and "pleasure and happiness" describe hedonic well-being. "Income and material possessions" is a material yardstick that neither tradition accepts as well-being itself.

Character strengths, grit and sportsmanship

Christopher Peterson (1950 to 2012) and Seligman wrote *Character Strengths and Virtues* (2004) as a "manual of the sanities", a positive counterpart to the DSM. Surveying moral and religious traditions across the world, they found six virtues valued almost everywhere, each reached through specific **character strengths**, 24 in all (Table 4.31). These are measured by the VIA Inventory of Strengths (VIA stands for Values in Action); a person's top few are his **signature strengths**.

Table 4.31 The six virtues and 24 character strengths (Peterson and Seligman, 2004).

Virtue	Character strengths
Wisdom and knowledge	Creativity, curiosity, judgement (open-mindedness), love of learning, perspective
Courage	Bravery, perseverance (persistence), honesty (integrity), zest
Humanity	Love, kindness, social intelligence
Justice	Teamwork (citizenship), fairness, leadership
Temperance	Forgiveness, humility, prudence, self-regulation (self-control)
Transcendence	Appreciation of beauty and excellence, gratitude, hope (optimism), humour, spirituality

Angela Duckworth (born 1970) added **grit**: perseverance and passion for long-term goals, kept up for years despite failure and plateaus.

KEY STUDY: GRIT (DUCKWORTH AND COLLEAGUES, 2007)

Why do some equally able people achieve more? Duckworth's team gave a Grit Scale to adults, to cadets entering West Point's gruelling first summer and to Scripps National Spelling Bee finalists. Grit predicted educational attainment, which cadets completed the summer and how far spellers advanced, over and above measures of ability. Criticism: a later meta-analysis found grit so close to conscientiousness (Section 4.7) that it may be an old trait under a new name.

The same strengths define character in sport. **Sportsmanship** is playing hard within the rules and with respect for opponents: the true sportsman is persistent (perseverance), keeps self-control under provocation (self-regulation), stays optimistic after a defeat (hope) and plays fair (fairness). Focusing on winning at all costs, by cheating, injuring or gamesmanship, is its opposite: the batter who walks after edging the ball shows character; the bowler who tampers with the ball does not.

The triguna theory of personality ●●●

The **triguna** theory comes mainly from the Samkhya school, attributed to the sage **Kapila**, together with the Bhagavad Gita (chapters 14, 17 and 18), with earlier hints in the Upanishads. It does not come from Nyaya (the school of logic), Mimamsa (the school of Vedic ritual) or the materialist Charvaka. Samkhya divides reality into purusha, pure consciousness, the passive witness, and prakriti, primordial nature, from which body and mind evolve (Section 1.2 gives the full scheme, the inner instrument and the koshas). The three gunas (strands or qualities) belong to prakriti, not purusha; the Gita calls them "born of prakriti" (14.5). Because the mind evolves from prakriti, the gunas are the stuff of personality:

- **Sattva** is clarity, goodness, light, calm, balance and knowledge; the sattvic person is truthful, self-controlled, content and wise.
- **Rajas** is activity, passion, desire and restlessness; the rajasic person is ambitious, driven and attached to results, and tips into over-activity, greed, anger and anxiety.
- **Tamas** is inertia, dullness, darkness, lethargy and ignorance; the tamasic person is heavy, careless and confused, sleeps too much and cannot bring himself to act.

Every **jiva**, every embodied individual self, has all three in shifting proportions. None is ever absent; the dominant guna colours the personality (Figure 4.18), and the Gita traces that colouring through daily conduct (Table 4.32).

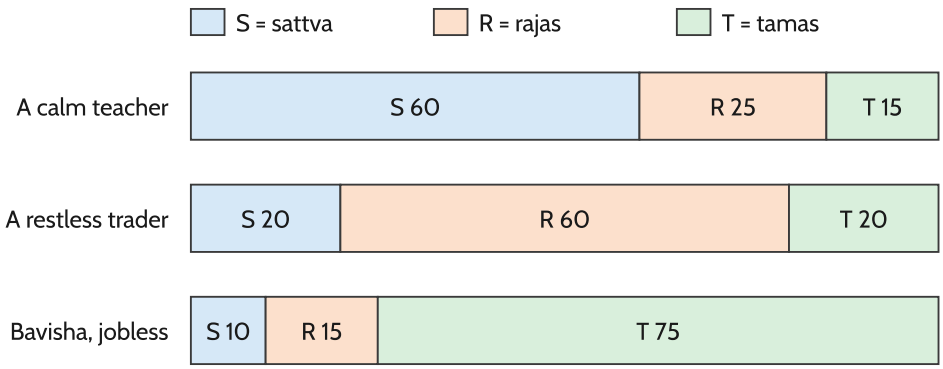


Figure 4.18 Everyone has all three gunas; personality follows the dominant one. The percentages are illustrative, not measured: what matters is that no bar lacks any guna.

Table 4.32 The sattvic, rajasic and tamasic person in the Bhagavad Gita (chapters 17 and 18).

Domain	Sattvic	Rajasic	Tamasic
Work	Done as duty, without craving for its fruit	Driven by desire and ego, with strain	Begun in delusion, careless of consequences
Food	Fresh, wholesome, nourishing	Very hot, sour, salty, pungent; brings pain and disease	Stale, tasteless, left over, impure
Knowledge	Sees one unity in all beings	Sees only separate, different things	Clings to one thing as if it were all
Giving	As duty, to the worthy, expecting nothing	For a return, or grudgingly	At the wrong time, to the unworthy, with contempt

A worked example. Bavisha has lost her job. Since then she is stressed, has stopped eating properly, has withdrawn and cannot get herself to do anything. Run her through the gunas. No calm clarity: not sattva. No restless over-activity, anger or frantic job-hunting: not rajas, which is how the same stress looks in someone who cannot sit still. Heaviness, withdrawal and inability to act: **tamas alone**. Yama cannot be her problem, alone or with tamas, because it is not a mental state but the first limb of Patanjali's yoga, the ethical restraints; pratyahara (withdrawal of the senses, the fifth limb) and jnana (the path of knowledge) are practices, not afflictions.

> **EXAM TRAP**
>
> The only gunas are sattva, rajas and tamas. Not gunas: ahimsa (non-violence, the first of the yamas); yama, pratyahara and samadhi (absorption, the eighth limb), all limbs of yoga; jnana (knowledge); samskara (the latent impression a past act leaves on the mind); ahamkara (the I-maker or ego, a product of prakriti); prakriti itself; vata (an Ayurvedic dosha). The gunas belong to prakriti, not to purusha, atman (the true self), Brahman (ultimate reality), ahamkara, samadhi or samskara. Distractor triads: prakriti, purusha and kshetra (the "field", the body that the self knows); atman, Brahman and maya (appearance), the Vedanta triad; karma, dharma and moksha (action, duty and liberation).

The gunas have been turned into questionnaires: **David Wolf**'s Vedic Personality Inventory (VPI, 1998) and Indian instruments such as the Mysore Triguna Scale score sattva, rajas and

tamas separately. Such studies tend to link sattva with mental health and life satisfaction, and tamas with depression and poor adjustment.

Ayurvedic constitution and Indian typologies

Ayurveda, India's classical medicine, adds a bodily typology. One of its foundational texts, the *Charaka Samhita* of **Charaka**, describes three doshas (the **tridosha**), bodily forces each built from two of the five elements (Table 4.33). Their proportions at conception fix a person's **prakriti** in the Ayurvedic sense: one's inborn bodily and mental constitution, a second meaning quite different from Samkhya's primordial nature. Ayurveda also describes a mental constitution in terms of the gunas, and calls rajas and tamas the two doshas of the mind.

Table 4.33 The three doshas and their Western parallels.

Dosha	Elements	Body	Temperament	Rough parallel
Vata	Air and space	Thin, light, dry	Quick, restless, anxious, creative	Ectomorph
Pitta	Fire and water	Medium, muscular, warm	Sharp, ambitious, irritable	Choleric; mesomorph
Kapha	Earth and water	Heavy, solid	Calm, steady, slow	Phlegmatic; endomorph

The parallels with the four humours of Greek medicine and with William Sheldon's somatotypes (Section 4.1) are real: all three systems tie a bodily fluid or build to a temperament. What are Indian typologies worth as personality science? They are holistic, joining body, diet, routine and mind; the guna model is dimensional, since everyone has all three in varying amounts, which is closer to modern trait thinking than to rigid boxes; and they point to change, since diet, routine, yoga and right action are meant to raise sattva. Their limits are those of all typologies: people get forced into types, the scales rest mostly on self-report with limited validation, what they add to the Big Five is unsettled, and body-temperament links, like Sheldon's, have little experimental support.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. For William James, self-esteem equals success divided by _____.
2. In Cooley's looking-glass self, the third step is a self-feeling such as pride or _____.
3. Mead's term for the organised attitudes of the whole community is the _____.
4. In Higgins's theory, an actual-ought discrepancy produces _____-related emotions.
5. Seligman's three Ps of explanatory style are permanence, pervasiveness and _____.
6. Ryff's six dimensions include self-acceptance, purpose in life and environmental _____.
7. The three gunas are strands of _____, not of purusha.
8. The dosha made of fire and water is _____.

Answers: (1) pretensions, (2) shame, (3) generalised other, (4) agitation, (5) personalisation, (6) mastery, (7) prakriti, (8) pitta.

The Boy on the Vienna Tram, Revisited

Return to Allport's tram story with the whole chapter behind you. Freud's question was psychoanalysis in one line: an innocent anecdote read as the surface of hidden conflict (Sections 4.2 and 4.3). Allport's reply became the trait approach: take conscious motives seriously and describe the stable dispositions a person actually shows (Sections 4.6 and 4.7). Adler would have seized on the "generalized inferiority feeling" Allport admitted to (Section 4.4); Rogers would have asked how the young man saw himself (Section 4.8); Bandura and Mischel would have pointed to the situation, an awkward silence in a famous man's study (Section 4.9). The modern answer is interactionism: a neat, orderly young man, in a situation that made him anxious, telling a story that revealed something of both.

Chapter summary

4.1 What Personality Is: Definitions, Approaches and Early Type Theories. Personality, from the Latin *persona* (the actor's mask), is the relatively enduring, distinctive pattern of thought, feeling and behaviour; **Allport** (1937) called it the dynamic organisation of psychophysical systems. **Temperament** is its biologically based, early-appearing emotional core; character is personality evaluated; traits are dimensions, types are categories. Theories describe, explain, predict and generate research, and differ on Schultz's six questions about human nature. Personality reflects heredity, environment and the situation in interaction, and is studied by clinical, experimental and correlational methods, nomothetically or idiographically (Windelband). The early type theories, the four humours, physiognomy, Kretschmer's body types and **Sheldon's somatotypes**, did not survive testing.

4.2 Freud's Psychoanalysis: Instincts, Structure and Psychosexual Stages. Sigmund Freud (1856 to 1939), a Viennese neurologist, built psychoanalysis from hysteria: Charcot's hypnosis, Bernheim's suggestion and Breuer's Anna O. (catharsis; *Studies on Hysteria*, 1895). Free association, dream analysis and parapraxes open the unconscious. Instincts (Eros and its libido; the death instinct, 1920) supply energy invested by cathexis. The topographic model divides conscious, preconscious and unconscious; the structural model (1923) sets the id (pleasure principle), ego (reality principle, the executive) and superego (conscience and ego-ideal) in conflict, and anxiety (reality, neurotic, moral) signals danger. Fixation at the oral, anal or phallic (Oedipus) stage shapes adult character; latency and the genital stage follow. Research supports unconscious processes, not the stages.

4.3 Defence Mechanisms: How the Ego Protects Itself. Defence mechanisms are the ego's unconscious, reality-distorting strategies against anxiety, described by Sigmund Freud and catalogued by Anna Freud in *The Ego and the Mechanisms of Defence* (1936); they are adaptive in moderation, pathological when rigid, and never rational. **Repression**, the involuntary blocking of threatening material from awareness, is the primary defence; suppression is its conscious, mature counterpart. Displacement changes the target, sublimation the activity, projection the owner, reaction formation the direction and rationalisation the stated reason. **George Vaillant** (1977) ranked defences as psychotic, immature, neurotic and mature. Obsessive-compulsive disorder favours isolation, undoing and reaction formation; phobias, displacement; hysteria, repression; borderline personality disorder, splitting.

4.4 Jung's Analytical Psychology and Adler's Individual Psychology. Jung and Adler were the first neo-Freudians to break with Freud (Adler in 1911, Jung in 1913). Jung's **analytical psychology** widened libido into general psychic energy and placed, beneath the personal unconscious and its complexes (detected with the word association test), an inherited **collective unconscious** of archetypes: persona, shadow, anima, animus and self. His two attitudes and four functions give eight types, the basis of the MBTI, and individuation belongs to midlife. Adler's **individual psychology** sees behaviour as goal-directed: universal inferiority feelings drive compensation and striving for superiority, shaped by fictional finalism, the style of life, social interest and birth order.

4.5 Horney, Fromm, Sullivan and the Object-Relations Theorists. **Karen Horney** traced neurosis to **basic anxiety**, the feeling of being lonely and helpless in a hostile world, managed by ten neurotic needs grouped into moving toward, against and away from people; she added the idealised self, the **tyranny of the shoulds** and **womb envy**. **Erich Fromm** described mechanisms of escape from freedom, five existential needs and character orientations; **Harry Stack Sullivan** saw personality as recurrent interpersonal situations. Ego psychology (Anna Freud, Hartmann, Erikson) strengthened the ego. **Object relations** are internalised images of self and others: Klein's splitting and two positions, Winnicott's good-enough mother. Kohut's **selfobjects** and Lacan's mirror stage complete the section.

4.6 Traits I: Allport, Cattell and the Lexical Search. The **trait approach** describes personality through stable, consistent dimensions of individual difference, unlike types, states or situationism. The **lexical hypothesis** led Allport and Odbert (1936) to cull 17,953 trait words from a dictionary. **Gordon Allport** treated traits as real neuropsychic structures, separated common traits from **personal dispositions** (cardinal, central, secondary), preferred the **idiographic** study of single cases (*Letters from Jenny*), and proposed **functional autonomy**, the proprium and six marks of maturity. **Raymond Cattell** factor-analysed the lexicon into **surface** and **source traits**, described ergs and sentiments, required L-, Q- and T-data, and built the **16PF** (1949).

4.7 Traits II: Eysenck, the Big Five and the Biology of Personality. **Hans Eysenck** joined factor analysis to experiment and biology: a hierarchy from specific responses to types, and three superfactors, extraversion (cortical arousal set by the ARAS), neuroticism (the visceral brain) and psychoticism; crossing E and N gives Galen's four temperaments. Gray, Cloninger, Kagan, and Buss and Plomin extended the biology of temperament. The **Big Five** (OCEAN) came from lexical and questionnaire research; Costa and McCrae's NEO-PI-R has 240 items and 30 facets. High conscientiousness and low neuroticism predict health. HEXACO adds honesty-humility. Heritability is about 40 to 50 percent, with the non-shared environment mattering most. **Mischel** (1968) argued that situations often override traits; aggregation and interactionism replied.

4.8 Humanistic and Existential Theories: Rogers, Maslow and May. The **humanistic approach** stresses free will, an innate tendency to grow toward one's maximum potential, the whole person and conscious experience; disorder is blocked growth, and critics call it vague and blind to culture. Rogers's **actualising tendency**, **organismic valuing process**, **conditions of worth** and **incongruence** explain how conditional love splits self from experience; the **fully functioning person** is the ideal, and the **Q-sort** measures self-ideal congruence. Maslow contrasted D-motivation with **metamotivation** and described self-actualisers, peak experiences and the **Jonah complex**. Existentialists

(Heidegger, Sartre, Binswanger, May, Frankl) describe Dasein in the Umwelt, Mitwelt and Eigenwelt, normal and neurotic anxiety, the will to meaning and noogenic neurosis.

4.9 Behavioural, Social-Cognitive and Cognitive Theories. Skinner's radical behaviourism treats personality as learned operant behaviour (emitted, acting on the environment) with no inner causes. Dollard and Miller (1950) recast Freud in Hull's drive, cue, response and reinforcement: early social learning. Bandura (later social learning, with Walters) proposed reciprocal determinism among person, behaviour and environment, self-regulation and self-efficacy (1977), fed by mastery, vicarious experience, persuasion and arousal. Rotter: behaviour potential = f(expectancy, reinforcement value); internal versus external locus of control (I-E Scale, 1966). Mischel's CAPS gives if-then signatures. Kelly (1955): the person as scientist, bipolar constructs, the Rep test and fixed-role therapy. Witkin: field dependence; Kagan: reflection-impulsivity.

4.10 The Self, Positive Traits and Indian Approaches to Personality. William James (1890) split the self into the knowing I and the known me (material, social and spiritual selves) and defined self-esteem as success divided by pretensions. Cooley's looking-glass self (1902) builds the self-concept from imagined appearance, imagined judgement and self-feeling; Mead (1934) added the play and game stages and the generalised other. Self-concept is cognitive, self-esteem evaluative (Rosenberg, 1965). Zuckerman's sensation seeking and Seligman's explanatory style are limited-domain traits. Hedonic well-being is pleasure; eudaimonic well-being is strengths, growth and contribution (Ryff's six dimensions). The triguna theory (Samkhya and the Gita) describes sattva, rajas and tamas, strands of prakriti; Ayurveda adds vata, pitta and kapha.

Who's who

Table 4.34 The people in this chapter.

Name	When	Examined for
Gordon Willard Allport	1897 to 1967	Definition of personality (1937, revised 1961); character is personality evaluated; idiographic study; met Freud in 1920
Alexander Thomas	New York Longitudinal Study, 1956 onwards	With Chess: easy, difficult and slow-to-warm-up temperaments (pointer)
Stella Chess	New York Longitudinal Study, 1956 onwards	With Thomas: easy, difficult and slow-to-warm-up temperaments (pointer)
Duane Schultz	Theories of Personality, 2017 edition	With Sydney Ellen Schultz: six questions about human nature
Sydney Ellen Schultz	Theories of Personality, 2017 edition	With Duane Schultz: six questions about human nature
Sigmund Freud	1856 to 1939	Psychoanalysis; the darkest image of human nature; built theory from case studies
Carl Gustav Jung	1875 to 1961	Analytical psychology; introvert and extravert types (Section 4.4)
Alfred Adler	1870 to 1937	Individual psychology; birth order (Section 4.4)
Karen Horney	1885 to 1952	Neo-psychoanalytic theory (Section 4.5)
Erich Fromm	1900 to 1980	Neo-psychoanalytic theory (Section 4.5)
Harry Stack Sullivan	1892 to 1949	Neo-psychoanalytic interpersonal theory (Section 4.5)
Anna Freud	1895 to 1982	Ego psychology; the ego's defences (Section 4.3)

Name	When	Examined for
Melanie Klein	1882 to 1960	Object relations theory (Section 4.5)
Erik Erikson	1902 to 1994	Life-span psychosocial stages (Chapter 6)
Raymond Bernard Cattell	1905 to 1998	Factor-analytic trait theory; 16PF (Section 4.6)
Hans Jurgen Eysenck	1916 to 1997	Factor analysis; extraversion and neuroticism; mapped the four temperaments (Section 4.7)
Paul T. Costa Jr.	NEO-PI, 1985	With McCrae: the Five-Factor Model (Section 4.7)
Robert R. McCrae	NEO-PI, 1985	With Costa: the Five-Factor Model (Section 4.7)
Abraham Maslow	1908 to 1970	Humanistic theory (Section 4.8)
Carl Ransom Rogers	1902 to 1987	Humanistic theory; free will, growth and optimism (Section 4.8)
Rollo May	1909 to 1994	Existential theory (Section 4.8)
Burrhus Frederic Skinner	1904 to 1990	Behavioural approach; experimental method (Section 4.9)
Albert Bandura	1925 to 2021	Social-cognitive theory; experimental method (Section 4.9)
Julian Rotter	1916 to 2014	Social-cognitive theory; locus of control (Section 4.9)
Walter Mischel	1930 to 2018	Person-situation critique (1968); social-cognitive theory (Section 4.9)
George Kelly	1905 to 1967	Personal construct theory, the cognitive approach (Section 4.9)
Marvin Zuckerman	1928 to 2018	Sensation seeking, a limited-domain theory
Martin Seligman	born 1942	Learned helplessness and optimism, a limited-domain theory
Wilhelm Windelband	1848 to 1915	Coined nomothetic and idiographic (1894)
Hippocrates	about 460 to 377 BC	The four bodily humours
Galen	AD 129 to about 216	Four temperaments from the dominant humour
Aristotle	384 to 322 BC	The Physiognomonica is attributed to him
Johann Kaspar Lavater	1741 to 1801	Popularised physiognomy with essays of the 1770s
Cesare Lombroso	1835 to 1909	Criminal Man (1876); criminal stigmata
Franz Joseph Gall	1758 to 1828	Founder of phrenology (skull bumps, localised faculties)
Johann Gaspar Spurzheim	1776 to 1832	Popularised phrenology under that name
Ernst Kretschmer	1888 to 1964	Physique and Character (1921): pyknic, asthenic, athletic, dysplastic
William Herbert Sheldon	1898 to 1977	Somatotypes (1940) and temperaments (1942): endomorph, mesomorph, ectomorph
Eduard Spranger	1882 to 1963	Six value types behind the Study of Values
Philip Ewart Vernon	1905 to 1987	Allport-Vernon-Lindzey Study of Values
Gardner Lindzey	1920 to 2008	Allport-Vernon-Lindzey Study of Values
Meyer Friedman	1910 to 2001	With Rosenman: Type A behaviour pattern (1959)
Ray Rosenman	1920 to 2013	With Friedman: Type A behaviour pattern (1959)
Johan Denollet	1990s	Type D (distressed) personality
Jean-Martin Charcot	1825 to 1893	Paris neurologist; Freud studied hysteria and hypnosis with him (1885 to 1886)
Hippolyte Bernheim	1840 to 1919	Nancy; showed that hypnosis works through suggestion
Josef Breuer	1842 to 1925	Treated Anna O.; catharsis; Preliminary Communication (1893) and Studies on Hysteria (1895) with Freud
Bertha Pappenheim	1859 to 1936	Anna O., Breuer's hysteria patient; named the talking cure
Granville Stanley Hall	1844 to 1924	Invited Freud to lecture at Clark University (1909)
Daniel Paul Schreber	1842 to 1911	Freud's case of paranoia (1911), analysed only from his published memoir
Sophocles	5th century BCE	Oedipus Rex, source of the name Oedipus complex
John Broadus Watson	1878 to 1958	Behaviourism, contrasted with psychoanalytic theory
Abraham Harold Maslow	1908 to 1970	Humanistic theory; needs, contrasted with Freud's instincts
Erik Homburger Erikson	1902 to 1994	Psychosocial stages; adolescent identity formation
Henry Alexander Murray	1893 to 1988	Psychogenic needs, contrasted with Freud's instincts
Clark Leonard Hull	1884 to 1952	Drives, contrasted with Freud's instincts
Aaron Temkin Beck	1921 to 2021	Cognitive distortions, contrasted with Freudian slips
Howard Shevrin	1977	Subliminal study: an unseen picture of a bee shaped free associations

Name	When	Examined for
Michael Motley	1987	Laboratory Freudian slips under anxiety (damn shock for sham dock)
Seymour Fisher	1977, 1996	With Greenberg, reviewed about 2,500 studies testing Freud's ideas
Roger Greenberg	1977, 1996	With Fisher, reviewed about 2,500 studies testing Freud's ideas
Girindrasekhar Bose	1887 to 1953	Founded the Indian Psychoanalytical Society (1922); corresponded with Freud; theory of opposite wishes
Elizabeth Loftus	born 1944	False memory research; critic of recovered repressed memories (2006 review)
Ernest Jones	1879 to 1958	Introduced the term rationalisation (1908)
Lawrence K. Frank	1939	Named projective methods; the projective hypothesis rests on the broad sense of projection
George Eman Vaillant	born 1934	Four-level hierarchy of defences (psychotic, immature, neurotic, mature); Adaptation to Life (1977); Grant Study
Otto Kernberg	born 1928	Borderline personality organisation; splitting and projective identification
Herbert Graf (Little Hans)	1909 case	Freud's case of horse phobia: fear of the father displaced onto horses
Joseph Wolpe	1915 to 1997	With Rachman (1960), a conditioning account of Little Hans's phobia
Stanley Rachman	1960	With Wolpe (1960), a conditioning account of Little Hans's phobia
Phebe Cramer	1987	Developmental sequence of defences: denial, then projection, then identification
Sam Glucksberg and L. J. King	1967	Laboratory analogue of repression: words linked with shock were forgotten
David S. Holmes	1990	Review of sixty years of studies: no convincing controlled evidence for repression
Hans Eysenck	1916 to 1997	Trait theorist (Section 4.7); not a neo-Freudian
Carl Rogers	1902 to 1987	Humanist (Section 4.8); empathic understanding; not a neo-Freudian
Eugen Bleuler	1857 to 1939	Directed the Burghölzli hospital, Zurich, where Jung worked from 1900
Franz Riklin	1878 to 1938	Jung's partner in the word association studies (1904)
Francis Galton	1822 to 1911	Studied word association before Jung
Wilhelm Wundt	1832 to 1920	Studied word association before Jung
Henry Murray	1893 to 1988	Thematic Apperception Test (Chapter 2); not the word association test
Donald Winnicott	1896 to 1971	Object relations: transitional object, good enough mother (Section 4.5); not the word association test
Katharine Cook Briggs	1875 to 1968	Co-creator of the Myers-Briggs Type Indicator from Jung's types (read Psychological Types in 1923)
Isabel Briggs Myers	1897 to 1980	Co-creator of the Myers-Briggs Type Indicator (first form 1943)
Raymond Cattell	1905 to 1998	16PF (Section 4.6); not based on Jung's types
Paul Costa	born 1942	NEO-PI-R and the Big Five, with McCrae (Section 4.7)
Robert McCrae	born 1949	NEO-PI-R and the Big Five, with Costa (Section 4.7)
Demosthenes	384 to 322 BC	Greek orator who overcame a stutter: Adler's example of compensation
Viktor Frankl	1905 to 1997	Logotherapy (Chapter 7); not the inferiority complex
Franz Alexander	1891 to 1964	Psychosomatic medicine; did not coin the inferiority complex
Gordon Allport	1897 to 1967	Trait theorist (Section 4.6); did not coin the inferiority complex
Hans Vaihinger	1852 to 1933	The Philosophy of As If (1911), source of Adler's fictional finalism
Albert Ellis	1913 to 2007	Rational emotive therapy (Chapter 7); influenced by Adler
Jean Piaget	1896 to 1980	Formal operations stage (Chapter 6)
Lawrence Kohlberg	1927 to 1987	Moral reasoning (Chapter 6); wrongly paired with formal operations
Sudhir Kakar	Born 1938	Psychoanalyst of Indian childhood and society (The Inner World, 1978); keyed for basic anxiety in the Indian context
Anand C. Paranjpe	2008 (Handbook)	Co-editor of the Handbook of Indian Psychology; distractor for basic anxiety in India
Durganand Sinha	1922 to 1998	Indigenisation of psychology; distractor for basic anxiety in India
Philip Wylie	1942	Coined momism in Generation of Vipers
Edward Strecker	1946	Psychiatrist who took up momism in Their Mothers' Sons; keyed when Wylie is absent

Name	When	Examined for
Edward Deci	1985	Self-determination theory (competence, autonomy, relatedness) with Ryan; distractor for Fromm's needs
Richard Ryan	1985	Self-determination theory with Deci
Heinz Hartmann	1894 to 1970	Father of ego psychology; conflict-free ego sphere and ego autonomy (1939)
Ronald Fairbairn	1889 to 1964	Libido is object-seeking, not pleasure-seeking
Margaret Mahler	1897 to 1985	Separation-individuation
John Bowlby	1907 to 1990	Attachment theory, an offshoot of object relations
Heinz Kohut	1913 to 1981	Self psychology; selfobjects; mirroring, idealising and twinship (alter ego); grandiose self; transmuting internalisation; fragmented self; failed mirroring behind narcissistic pathology
Jacques Lacan	1901 to 1981	The French Freud; return to Freud; mirror stage; unconscious structured like a language; Imaginary, Symbolic and Real
Jean-Étienne Esquirol	1772 to 1840	Modern meaning of hallucination; monomania; distractor for the French Freud
Jules Falret	1877	Folie à deux with Lasègue; distractor for the French Freud
Charles Lasègue	1877	Folie à deux with Falret
Paul Briquet	1859	Briquet's syndrome (now somatisation); distractor for the French Freud
Charles Spielberger	1927 to 2013	State versus trait anxiety; State-Trait Anxiety Inventory (1970)
William Sheldon	1940	Somatotypes, an example of a type approach (Section 4.1)
Henry Odbert	1936	With Allport, culled 17,953 trait words from a dictionary, about 4,500 of them stable traits
Jenny (pseudonym)	1965	Writer of the 300-plus letters in Allport's Letters from Jenny, the classic personal-document study
Philip Vernon	1931	Study of Values with Allport; consistency of expressive behaviour (1933)
Charles Spearman	1863 to 1945	Pioneer of factor analysis; supervised Cattell's PhD (1929)
Christiana Morgan	1935	Co-developer of the Thematic Apperception Test with Murray
Joy Paul Guilford	1897 to 1987	Factor-analytic temperament traits; Guilford-Zimmerman Temperament Survey (1949)
Sybil B. G. Eysenck	EPI 1964, EPQ 1975	Co-author of the Eysenck personality questionnaires
Carl Jung	1875 to 1961	Psychological types behind the MBTI dichotomies (Section 4.4), a distractor for Eysenck and the Big Five
Jeffrey Alan Gray	1934 to 2004	Reinforcement sensitivity theory: behavioural inhibition system (anxiety) and behavioural approach system (impulsivity)
C. Robert Cloninger	1987, extended 1993	Temperament and character model: novelty seeking (dopamine), harm avoidance (serotonin), reward dependence (noradrenaline), persistence
Jerome Kagan	1929 to 2021	Behavioural inhibition: inhibited and uninhibited children
Arnold Buss	1984	EAS temperaments (emotionality, activity, sociability), with Robert Plomin
Robert Plomin	1984; 1987	EAS temperaments with Arnold Buss; the non-shared environment with Denise Daniels
Donald Fiske	1949	First found five factors in ratings on Cattell's variables
Ernest Tupes	1961	Five recurrent factors in US Air Force ratings, with Raymond Christal
Raymond Christal	1961	Five recurrent factors in US Air Force ratings, with Ernest Tupes
Warren Norman	1963	Replicated the five factors
Lewis Goldberg	1981	Coined the label Big Five
William James	1842 to 1910	Character 'set like plaster' by about 30 (1890), a phrase borrowed for adult trait stability
Lewis Terman	1877 to 1956	Study of gifted children from 1921, the data in which conscientious children lived longer
Howard Friedman	1993	Showed that childhood conscientiousness predicted longevity in Terman's sample
Murray Barrick	1991	Meta-analysis with Michael Mount: conscientiousness predicts job performance in every occupational group

Name	When	Examined for
Michael Mount	1991	Meta-analysis with Murray Barrick: conscientiousness predicts job performance
Michael Ashton	2004	HEXACO six-factor model, with Kibeom Lee
Kibeom Lee	2004	HEXACO six-factor model, with Michael Ashton
Delroy Paulhus	2002; 2013	Dark Triad with Kevin Williams; everyday sadism, the fourth trait of the Dark Tetrad
Kevin Williams	2002	Dark Triad (narcissism, Machiavellianism, psychopathy), with Delroy Paulhus
Thomas Bouchard	1990	Minnesota Study of Twins Reared Apart: heritability of personality about 0.40 to 0.50
Denise Daniels	1987	Non-shared environment, with Robert Plomin
Sandra Scarr	1983	Passive, evocative and active gene-environment correlation, with Kathleen McCartney
Kathleen McCartney	1983	Passive, evocative and active gene-environment correlation, with Sandra Scarr
David Buss	born 1953	Evolutionary personality psychology: traits as adaptive individual differences
Hugh Hartshorne	1928	Honesty studies with Mark May: honesty looked situation-specific
Mark May	1928	Honesty studies with Hugh Hartshorne
Seymour Epstein	1979	Aggregation: behaviour averaged over many occasions is stable
Norman Endler	1976	Interactionism, with David Magnusson
David Magnusson	1976	Interactionism, with Norman Endler
Charlotte Buhler	1893 to 1974	Co-founder of humanistic psychology; goals across the life course
Kurt Goldstein	1878 to 1965	Coined the term self-actualisation (The Organism, German 1934, English 1939)
William Stephenson	1953	Devised the Q-sort (The Study of Behavior, 1953), later used by Rogers
J. M. Butler	1954	With Haigh, Q-sort study of 25 clients: average self-ideal correlation .01 before therapy, +.31 after
G. V. Haigh	1954	With Butler, Q-sort study of self-ideal congruence before and after client-centred therapy
Everett Shostrom	1964	Personal Orientation Inventory (POI), a measure of self-actualisation
Abraham Lincoln	1809 to 1865	One of Maslow's historical self-actualisers
Thomas Jefferson	1743 to 1826	One of Maslow's historical self-actualisers
Albert Einstein	1879 to 1955	One of Maslow's self-actualisers
Eleanor Roosevelt	1884 to 1962	One of Maslow's self-actualisers
Soren Kierkegaard	1813 to 1855	Danish philosopher at the root of existentialism; the anxiety of choice
Martin Heidegger	1889 to 1976	Being and Time (1927); Dasein, being-in-the-world
Jean-Paul Sartre	1905 to 1980	Existence precedes essence
Ludwig Binswanger	1881 to 1966	Daseinsanalysis; Umwelt, Mitwelt and Eigenwelt
Medard Boss	1903 to 1990	Daseinsanalysis
Viktor Emil Frankl	1905 to 1997	Man's Search for Meaning (German original 1946); will to meaning, existential vacuum, noogenic neurosis; logotherapy
Irvin Yalom	born 1931	Four ultimate concerns: death, freedom, isolation, meaninglessness
Fritz Perls	1893 to 1970	Gestalt therapy, a wrong option for noogenic anxiety
Ivan Petrovich Pavlov	1849 to 1936	Respondent (classical) conditioning, cited as the example of elicited behaviour
John Dollard	1900 to 1980	Early social learning theory with Miller (1941, 1950); frustration-aggression hypothesis (1939)
Neal Elgar Miller	1909 to 2002	Early social learning with Dollard; fear as a learned drive; approach-avoidance conflict
Edward B. Blanchard	1969	Co-author of the snake-phobia participant modelling study with Bandura and Ritter
B. Ritter	1969	Co-author of the snake-phobia participant modelling study with Bandura and Blanchard

Name	When	Examined for
Richard Walters	1963	Co-author with Bandura of Social Learning and Personality Development: later social learning
Robert Trivers	1971	Reciprocal altruism, a distractor for reciprocal determinism
Aaron Beck	1921 to 2021	Cognitive therapy and schemas; not the author of self-efficacy or fixed-role therapy
Julian Bernard Rotter	1916 to 2014	Social learning theory (1954): behaviour potential, expectancy, reinforcement value; locus of control and the I-E Scale (1966)
Hanna Levenson	1973	Three-part locus of control (IPC) scale: internality, powerful others, chance
Yuichi Shoda	1995	Co-author of the cognitive-affective personality system (CAPS) with Mischel
George Alexander Kelly	1905 to 1967	Personal construct theory (1955); Role Construct Repertory Test; fixed-role therapy
Harrison Gough	1921 to 2016	California Psychological Inventory (CPI)
Herman Witkin	1916 to 1979	Field dependence-independence; rod-and-frame test; Embedded Figures Test
Charles Horton Cooley	1864 to 1929	Looking-glass self (Human Nature and the Social Order, 1902)
George Herbert Mead	1863 to 1931	The I and the me, play and game stages, generalised other (Mind, Self and Society, 1934)
Leon Festinger	1919 to 1989	Social comparison theory (1954), a looking-glass distractor (Chapter 3)
Morris Rosenberg	1922 to 1992	Rosenberg Self-Esteem Scale (1965)
Hazel Markus	Key year 1977	Self-schemas; possible selves with Paula Nurius (1986)
Paula Nurius	Key year 1986	Possible selves, with Hazel Markus
E. Tory Higgins	Key year 1987	Self-discrepancy theory: actual, ideal and ought selves; dejection versus agitation
Mark Snyder	Key year 1974	Self-monitoring
Martin E. P. Seligman	Born 1942	Explanatory style and the three Ps (Learned Optimism, 1990); character strengths and virtues (2004)
Christopher Peterson	1950 to 2012	Attributional Style Questionnaire (1982); Character Strengths and Virtues with Seligman (2004)
Ed Diener	1946 to 2021	Subjective well-being; Satisfaction With Life Scale (1985)
Carol Ryff	Key year 1989	Six dimensions of psychological well-being
Angela Duckworth	Born 1970	Grit, perseverance and passion for long-term goals (2007)
Kapila	Ancient, date uncertain	Traditional founder of Samkhya, the main source of the triguna theory
Patanjali	Ancient, date uncertain	Yoga Sutras; the eight limbs, from yama to samadhi
David Wolf	Key year 1998	Vedic Personality Inventory (1998), measuring sattva, rajas and tamas
Charaka	Ancient, date uncertain	Charaka Samhita; the tridosha theory of vata, pitta and kapha

Key terms

personality, Temperament, Character, trait, type, Disposition, Habit, Attitude, personality theory, historical determinism, interactionism, clinical method, experimental method, correlational method, factor analysis, Nomothetic, idiographic, humours, Sanguine, Choleric, Melancholic, Phlegmatic, Physiognomy, criminal stigmata, Phrenology, Graphology, pyknic, asthenic, endomorphy, mesomorphy, ectomorphy, somatotype, Viscerotonia, Somatotonia, Cerebrotonia, psychoanalysis, hysteria, seduction theory, psychodynamic perspective, free association, manifest content, latent content, parapraxis, instincts, life instincts, libido, death instinct, Cathexis, anticathexis, psychic determinism, topographic model, preconscious, unconscious, id, pleasure principle, primary process thought, ego, reality principle, secondary process thought, superego, conscience, ego-ideal, Reality anxiety, Neurotic anxiety, Moral anxiety, psychosexual stages, Oedipus

complex, castration anxiety, penis envy, Electra complex, defence mechanisms, coping, repression, primal repression, repression proper, displacement, projection, sublimation, rationalisation, reaction formation, isolation of affect, intellectualisation, undoing, identification, identification with the aggressor, introjection, regression, dissociation, conversion, splitting, projective identification, projective hypothesis, empathy, denial, delusional projection, psychotic denial, distortion, schizoid fantasy, hypochondriasis, altruism, humour, anticipation, suppression, neo-Freudians, analytical psychology, principle of opposites, principle of equivalence, principle of entropy, personal unconscious, complex, word association test, collective unconscious, archetypes, persona, shadow, anima, animus, self, mandala, extraversion, introversion, rational functions, irrational functions, inferior function, individuation, Synchronicity, amplification, individual psychology, organ inferiority, compensation, inferiority complex, superiority complex, striving for superiority, fictional finalism, style of life, creative power of the self, social interest, early recollections, basic anxiety, neurotic needs, neurotic trends, idealised self-image, tyranny of the shoulds, womb envy, Momism, mechanisms of escape, existential needs, character orientations, interpersonal theory, self-system, security operations, personifications, prototaxic, parataxic, syntaxic, Ego psychology, conflict-free ego sphere, epigenetic principle, Object relations, part objects, paranoid-schizoid position, depressive position, good-enough mother, transitional object, true self, borderline personality organisation, self psychology, selfobjects, grandiose self, transmuting internalisation, mirror stage, trait approach, Situationism, lexical hypothesis, nomothetic approach, idiographic approach, common traits, personal dispositions, cardinal trait, Central traits, Secondary traits, morphogenic, personal-document technique, Study of Values, functional autonomy, proprium, open system, Surface traits, Source traits, constitutional traits, environmental-mold traits, dynamic traits, Ergs, Sentiments, subsidiation, dynamic lattice, L-data, Q-data, T-data, P-technique, Sixteen Personality Factor Questionnaire, global factors, specification equation, hierarchical model, PEN model, neuroticism, psychoticism, ascending reticular activating system, cortical arousal, visceral brain, Eysenck Personality Inventory, Eysenck Personality Questionnaire, reinforcement sensitivity theory, behavioural inhibition, behavioural inhibition system, behavioural approach system, Big Five, Five-Factor Model, NEO-PI-R, five-factor theory, basic tendencies, characteristic adaptations, maturity principle, HEXACO, Dark Triad, Behaviour genetics, heritability, nonshared environment, gene-environment correlation, gene-environment interaction, personality coefficient, person-situation debate, aggregation, strong situation, humanistic approach, actualising tendency, organismic valuing process, phenomenal field, self-concept, ideal self, need for positive regard, Unconditional positive regard, conditions of worth, incongruence, fully functioning person, Q-sort, deficiency needs, metamotivation, metaneeds, metapathology, Self-actualisation, peak experience, plateau experience, Jonah complex, Personal Orientation Inventory, existence precedes essence, Dasein, Daseinsanalysis, Umwelt, Mitwelt, Eigenwelt, Normal anxiety, Intentionality, daimonic, will to meaning, existential vacuum, noogenic neurosis, radical behaviourism, Respondent behaviour, Operant behaviour, Shaping, Superstitious behaviour, Self-control, drive, cue, learned drive, approach-avoidance conflict, frustration-aggression hypothesis, social cognitive theory, reciprocal determinism, self-regulation, self-reinforcement, moral disengagement, self-efficacy, outcome expectancy, efficacy expectancy, Collective efficacy, Behaviour potential, Expectancy, Reinforcement value, Psychological situation, generalised expectancies, Locus of control, cognitive-affective personality system, behavioural signature, personal constructs, constructive alternativism, fundamental postulate, Role Construct Repertory Test, fixed-role therapy, cognitive style, field dependence-independence, reflection-impulsivity, material self, social self, spiritual self, looking-glass self, symbolic interactionism, play stage, game stage, generalised other, Self-esteem, Self-schema, Possible selves, Self-discrepancy theory, Self-monitoring, limited-domain theory, sensation seeking, explanatory style, Hedonic well-being, subjective well-

being, Eudaimonic well-being, psychological well-being, character strengths, signature strengths, grit, Sportsmanship, triguna, Sattva, Rajas, Tamas, jiva, tridosha, prakriti, Vata, Pitta, Kapha

From the past papers

1. Biologically based characteristic way of reacting is known as (JEMAS 2025)

   (a) Disposition (b) Character (c) Trait (d) Temperament

   Answer: (d) Temperament. Temperament is the biologically based, early-appearing emotional style. Character is personality evaluated morally and learned, a trait is a descriptive dimension not defined by its biological origin, and disposition is too loose a term.

2. Sheldon's theory that people with different body types have different personalities has been: (RML 2020)

   (a) supported by research (b) discredited (c) shown to be accurate for thin people but not overweight people (d) shown to be accurate for women but not for men

   Answer: (b) discredited. Sheldon rated physique and temperament himself, so his expectations inflated the correlations, and independent studies found only weak links. It holds for no subgroup, and his subjects were men.

3. Physiognomy, an early approach to personality study, was based on : (IHBAS 2025)

   (a) Brain localization of functions (b) Dreams and unconscious motives (c) Handwriting style analysis (d) Facial features as indicators of character

   Answer: (d) Facial features as indicators of character. Brain localisation read from skull bumps is phrenology (Gall), handwriting analysis is graphology, and dreams as clues to unconscious motives belong to psychoanalysis.

4. According to Freud, fixation at which stage of development is associated with sarcasm, criticalness, overeating, and nail-biting in adults? (RML 2020)

   (a) Oral (b) Anal (c) Phallic (d) Latency

   Answer: (a) Oral. Overeating and nail-biting are oral-incorporative habits, and sarcasm and criticalness are oral-aggressive (biting) traits; both come from problems of feeding and weaning. Anal fixation gives orderliness, stinginess and obstinacy, or messiness and cruelty; latency is not a fixation point.

5. According to Freud, which term refers to the part of the personality that mediates between desires and social rules? (IHBAS 2024)

   (a) Id (b) Superego (c) Ego (d) Self

   Answer: (c) Ego. Working on the reality principle, the ego is the executive that mediates between the id's desires, the superego and reality or social rules. The id is pure desire (pleasure principle), the superego is internalised morality, and the self is not one of Freud's structures.

6. In psychoanalytic theory, what term describes the mental representation of psychic energy being invested or attached to a particular object, person, or idea? (Real PYQ 2025)

   (a) Libido (b) Cathexis (c) Fixation (d) Sublimation

   Answer: (b) Cathexis. Cathexis is the investment of psychic energy in an object, person or idea. Libido is the energy itself, fixation is libido left stuck at a psychosexual stage, and sublimation channels it into socially valued activity.

7. A woman harassed by her boss at work, initiates an argument with her husband. She is using which of the following ego's defense mechanisms: (RML 2024)

   (a) Projection (b) Identification (c) Sublimation (d) Displacement

   Answer: (d) Displacement. Anger at the boss, who could punish her, is redirected onto a safer target, her husband. Projection would attribute her anger to someone else, identification would take on another's qualities, and sublimation would channel the anger into a socially valued activity.

8. Which type of defence mechanism is dissociation? (NIMHANS 2026 recall)

   (a) Mature (b) Immature (c) Neurotic (d) Conscious

   Answer: (c) Neurotic. Vaillant (1977) placed dissociation at Level III with intellectualisation, isolation, repression, displacement and reaction formation. It is not immature (projection, acting out and the like), and defences are not conscious as a class; only suppression is deliberate.

9. Which of the following defense mechanisms is not involved in obsessive-compulsive disorder? (IHBAS 2020)

   (a) Reaction formation (b) Repression (c) Isolation (d) Undoing

   Answer: (b) Repression. In obsessional neurosis the disturbing idea stays conscious: its feeling is isolated, rituals undo it and reaction formation turns it into its opposite. Repression, which pushes the idea itself out of awareness, is the hallmark of hysteria.

10. Who believed that the universe of the collective unconscious could be accounted for by evolutionary theory, through the similarity of brain structure evident in all human races? (Amity 2023)

    (a) Freud (b) Jung (c) Horney (d) None of these

    Answer: (b) Jung. The collective unconscious is Jung's inherited, universal layer of the psyche, and he explained it by the same brain structure in all human races. Freud's unconscious was personal, and Horney stressed social and cultural forces.

11. The basic assumption of Adler's individual psychology is (JEMAS 2025)

    (a) human behaviour is purposeful and goal directed. (b) an optimistic view of human life. (c) human beings are social beings who could be understood in terms of their relationship with others. (d) development is a lifelong process and ego identity is granted a central place in the process.

    Answer: (a) Human behaviour is purposeful and goal directed. Adler explained behaviour by future goals that pull us forward. Optimism and the social emphasis are features of his view, not its basic assumption, and lifelong development with ego identity at the centre is Erikson's.

12. According to Adler, middle children tend to be: (RML 2022)

    (a) Overachieving (b) Competitive (c) Resentful of the freedom of the older child (d) Filled with feelings of inferiority

    Answer: (b) Competitive. With a pacesetter always ahead, the middle (second) child behaves as if in a race: ambitious and competitive. Such a child may also give up and underachieve, and inferiority feelings are universal, not the mark of the middle child.

13. Karen Horney emphasized personality development as influenced primarily by (JEMAS 2025)

    (a) Social and cultural factors. (b) Biological instincts exclusively. (c) Unconscious sexual conflicts exclusively. (d) Classical conditioning only.

    Answer: (a) Social and cultural factors. Horney broke with Freud's instinct theory: basic anxiety grows from parental indifference and the culture around the child. Instincts and sexual conflicts are Freud's emphasis, and conditioning belongs to behaviourism.

14. In paranoia, the ego's defense mechanism as given by Melaine Klein is seen as: (RML 2024)

    (a) Reaction formation (b) Splitting (c) Suppression (d) Projection

    Answer: (b) Splitting. Klein's paranoid-schizoid position rests on splitting good from bad objects; projection then puts the bad part outside. Projection is Freud's classic mechanism of paranoia, while reaction formation and suppression are unrelated defences.

15. A client tells you that they are "no one" and feels "empty inside." They report chronic fears of abandonment and have a history of intense, unstable relationships. According to psychoanalytic object relations theory, this pathology of the self is most likely a result of: (Real PYQ 2024)

    (a) Fixation at the oral stage of psychosexual development. (b) A failure in parental mirroring and idealizing, leading to a fragmented self. (c) Unresolved Oedipal complex and castration anxiety. (d) Harsh superego demands leading to constant guilt.

    Answer: (b) A failure in parental mirroring and idealizing, leading to a fragmented self. This is Kohut's self psychology: unmet selfobject needs leave an empty, fragmented self. Oral fixation, the Oedipus complex and superego guilt are classical Freudian conflict explanations, not deficits of the self.

16. Cattlell (sic) distinguished between which two types of traits? (JEMAS 2025)

    (a) Openness and Conscientiousness (b) Cardinal and central traits (c) Surface traits and Source traits (d) Id and Ego

    Answer: (c) Surface traits and source traits. Surface traits are clusters of correlated behaviours; source traits are the stable underlying factors that factor analysis reveals. Cardinal and central traits are Allport's, openness and conscientiousness are Big Five factors, and id and ego are Freud's.

17. Allport believed that traits could be organised into three levels: (RML 2020)

    (a) primary, secondary, tertiary (b) cognitive, emotional, physiological (c) id, ego, superego (d) cardinal, central, secondary

    Answer: (d) cardinal, central, secondary. A cardinal trait is a ruling passion found in few people, central traits are the five to ten that describe a person, and secondary traits are minor and inconsistent. Primary-secondary-tertiary is invented, and id-ego-superego is Freud's structure.

18. According to Raymond Cattell's classification of personality data sources, which of the following is NOT recognized as a legitimate data type? (Real PYQ 2025)

    (a) Q-data (Questionnaire data) (b) L-data (Life record data) (c) T-data (Objective test data) (d) P-data (Projective data)

    Answer: (d) P-data. Cattell used exactly three sources: L-data (life records), Q-data (questionnaires) and T-data (objective tests). He classed hard-to-fake tasks such as inkblots as T-data, and his P-technique is a research design, not a data type.

19. Hans Eysenck proposed which three dimension of personality? (JEMAS 2025)

    (a) Extraversion-Introversion, Neuroticism-Stability, Psychoticism (b) Extraversion-Introversion, Sensing-Intuition, Thinking-Feeling (c) Openness, Extraversion, Judging, Perceiving (d) Extraversion-Introversion, Neuroticism-Stability, Psychoticism-Conscientiousness

    Answer: (a) Extraversion-Introversion, Neuroticism-Stability, Psychoticism. These are Eysenck's PEN superfactors. Sensing-intuition, thinking-feeling, judging and perceiving are Myers-Briggs dichotomies from Jung's types, and conscientiousness is a Big Five factor, not part of Eysenck's model.

20. Which personality trait is linked to better health outcomes? (JEMAS 2025)

    (a) Low Agreeableness and Low Neuroticism (b) Neuroticism exclusively (c) Introversion exclusively (d) High conscientiousness and Low neuroticism

    Answer: (d) High conscientiousness and low neuroticism. Conscientious people look after their health and lived longer in Terman's sample, and low neuroticism means less distress. Low agreeableness (hostility) predicts heart disease, high neuroticism harms health, and introversion alone has no health advantage.

21. Mischel's critique of personality traits mainly emphasized that : (IHBAS 2025)

    (a) Situational factors often override traits in determining behavior (b) Personality is entirely biologically based (c) Traits are more important than situations in predicting behavior (d) Traits cannot be measured reliably

    Answer: (a) Situational factors often override traits in determining behaviour. In Personality and Assessment (1968) Mischel found trait-behaviour correlations rarely above about 0.30. He did not claim personality is biological, (c) is the opposite of his point, and trait questionnaires were reliable; they simply predicted single acts poorly.

22. In existential psychology, Dasein encompasses three fundamental psychological dimensions of human existence. Which of the following correctly identifies these three modes of being-in-the-world? (Real PYQ 2025)

    (a) Id, Ego, and Superego (b) Umwelt, Mitwelt, and Eigenwelt (c) Conscious, Preconscious, and Unconscious (d) Sensorimotor, Preoperational, and Concrete Operational

    Answer: (b) Umwelt, Mitwelt, and Eigenwelt. Binswanger and May describe the biological and physical world (Umwelt), the world of relationships (Mitwelt) and one's relation to oneself (Eigenwelt). Id, ego and superego is Freud's structural model, conscious to unconscious his topographic model, and the last option Piaget's stages.

23. Noogenic anxiety is described under: (IHBAS 2020)

    (a) Humanistic School (b) Gestalt school (c) Existential school (d) Psychodynamic school

    Answer: (c) Existential school. Viktor Frankl's noogenic neurosis and anxiety arise from a frustrated will to meaning, an existential problem, not from drive conflict (psychodynamic), blocked growth (humanistic) or Gestalt ideas.

24. A split between the organismic valuing process and a self-concept burdened by conditions of worth which results in feelings of tension and confusion: (Amity 2023)

    (a) Conflict (b) Incongruence (c) Introjection (d) Actualizing tendency

    Answer: (b) Incongruence. Rogers's term for the gap between experience and a self-concept shaped by conditions of worth. Introjection is how those conditions are taken in, conflict is a general term, and the actualising tendency is the growth force that incongruence blocks.

25. Which concept is central to Bandura's Personality Theory? (JEMAS 2025)

    (a) Self-efficacy (b) Inferiority complex (c) Collective unconscious (d) Congruence

    Answer: (a) Self-efficacy. Self-efficacy, the belief that one can organise and execute the actions needed for a specific goal, is the core of Bandura's social cognitive theory. The inferiority complex is Adler's, the collective unconscious Jung's and congruence Rogers's.

26. Which of the following best describes the concept of 'reciprocal determinism' as proposed by Albert Bandura? (RML 2025)

    (a) Individual behavior is solely controlled by environmental factors. (b) Cognitive processes influence both behavior and the environment in a bidirectional manner. (c) Personality develops strictly from genetic inheritance and biological predispositions. (d) Social reinforcement has no impact on self-concept formation.

    Answer: (b). In reciprocal determinism person (cognitive) factors, behaviour and environment influence one another in both directions. Environment alone is Skinner's position, genes alone a biological one, and Bandura never denied the role of social reinforcement.

27. Julian Rotter's concept of locus of control describes whether individuals believe that their outcomes are controlled by : (IHBAS 2024)

    (a) External forces or internal efforts (b) Their cognitive schemas (c) Genetic predisposition (d) Social status

    Answer: (a) External forces or internal efforts. Locus of control is a generalised expectancy that outcomes depend either on one's own effort (internal) or on luck, fate, chance and powerful others (external). Cognitive schemas belong to Beck; genes and social status are not part of the construct.

28. The "looking-glass self," a concept introduced by Charles Horton Cooley, posits that our self-concept is primarily formed by: (Real PYQ 2024)

    (a) Our innate, unconscious drives. (b) Comparing ourselves to others who are similar. (c) Imagining how we appear to others and how they judge us. (d) Successfully resolving psychosocial crises at each life stage.

    Answer: (c) Imagining how we appear to others and how they judge us. Cooley's three steps are imagined appearance, imagined judgement and a self-feeling such as pride or shame. Unconscious drives are Freud's, comparison with similar others is Festinger's social comparison, and psychosocial crises are Erikson's.

29. Which of the following best describes the concept of "self-esteem" in contrast to "self-concept"? (Real PYQ 2024)

    (a) Self-esteem is the cognitive appraisal of one's traits, while self-concept is the evaluative and emotional component. (b) Self-esteem is the evaluative and affective component of the self (how one feels about oneself), while self-concept is the cognitive description of who one is. (c) Self-esteem and self-concept are identical and can be used interchangeably. (d) Self-esteem is a collective term for the self in relation to others, while self-concept is the individual self.

    Answer: (b) Self-esteem is evaluative and affective; self-concept is the cognitive description of who one is. Option (a) reverses the two, they are not identical, and the individual versus collective self is a different distinction altogether.

30. Bavisha lost her job. Since then she is stressed, has stopped eating properly and cannot get herself to do anything. According to yogic philosophy, the main issue she is dealing with is: (NIMHANS 2026 recall)

    (a) Yama alone (b) Tamas alone (c) Both yama and tamas (d) Rajas alone

    Answer: (b) Tamas alone. Not eating, withdrawal and the inability to act are the inertia and lethargy of tamas. Rajas would show as restless over-activity and agitation, and yama is not a mental state at all but the first limb of Patanjali's yoga, the ethical restraints.

High-yield sheet

The facts this chapter's questions ask most, most-asked first. Last read before the exam.

1 The **id** works on the **pleasure principle** and the **ego** on the **reality principle**; the ego is the **executive** that mediates between the id's desires, the superego and reality or social rules. ASKED 7 TIMES

2 The **Big Five** are Openness, Conscientiousness, Extraversion, Agreeableness and Neuroticism (OCEAN); psychoticism, intelligence, intuition, sensing, anxiety (a facet of N) and sense of humour are not Big Five factors. ASKED 7 TIMES

3 The three gunas are **sattva** (clarity, calm, balance, knowledge), **rajas** (activity, passion, desire, restlessness) and **tamas** (inertia, darkness, lethargy, ignorance); a person sunk in withdrawn inertia who cannot act shows tamas alone. ASKED 7 TIMES

4 **Free association** (the fundamental rule: on the couch, say whatever comes to mind) is the central technique of psychoanalysis, and Freudian slips (**parapraxes**), mistakes that reveal the unconscious, are analysed through it. ASKED 6 TIMES

5 **Repression** is the primary defence: the involuntary, unconscious blocking of threatening memories, wishes and feelings from awareness. **Suppression** is its conscious, deliberate counterpart and is mature in Vaillant's scheme. ASKED 6 TIMES

6 **Vaillant (1977)** ranked defences in four levels: psychotic (delusional projection, psychotic denial, distortion), immature (projection, schizoid fantasy, hypochondriasis, passive aggression, acting out), neurotic (intellectualisation, isolation, repression, displacement, reaction formation, dissociation) and mature (altruism, anticipation, humour, sublimation, suppression). ASKED 6 TIMES

7 **Displacement** redirects feelings from a threatening target to a safer one (the woman harassed by her boss who picks a fight with her husband); it is the classic defence of phobias. ASKED 6 TIMES

8 **Alfred Adler** coined the **inferiority complex** (helplessness, hopelessness, lack of self-worth; from organ inferiority, pampering or neglect) and paired it with the striving for superiority; **organ inferiority** (1907) is also his. ASKED 6 TIMES

9 Incestuous fantasies, the **Oedipus complex** and the emergence of the superego belong to the **phallic stage**; the girls' counterpart was named the **Electra complex** by Jung in 1913. ASKED 5 TIMES

10 **Projection** attributes one's own unacceptable thoughts and impulses to others; in its broad sense it is any distorting effect of a person's inner world on his perception of the outer world. ASKED 5 TIMES

11 **Carl Jung**'s **collective unconscious** is the deepest, inherited, universal layer of the psyche, containing the archetypes; with the personal unconscious it forms his two levels of the unconscious. ASKED 5 TIMES

12 The **Sixteen Personality Factor Questionnaire (16PF)** was developed by **Raymond B. Cattell** (1949), who used factor analysis to derive 16 primary source traits; not Eysenck, Allport, Sheldon, Murray or Costa and McCrae. ASKED 5 TIMES

13 Smoking, drinking (alcohol dependence), overeating, nail-biting, sarcasm and criticalness indicate **oral** fixation, which arises from problems of feeding and weaning. ASKED 4 TIMES

14 The **id** is the wholly unconscious reservoir of unorganised instinctual drives; impulsive and irrational, it alone seeks immediate gratification of primitive needs ("Just do it"), whereas the superego says "Don't do it". ASKED 4 TIMES

15 **Sublimation** rechannels sexual or aggressive energy into a socially valued activity such as kickboxing, competitive sport or art; Freud's healthiest defence and mature in Vaillant's scheme. ASKED 4 TIMES

16 **Rationalisation** (Ernest Jones, 1908) is a logical, socially acceptable reason invented after the act for behaviour driven by unconscious or unacceptable motives. ASKED 4 TIMES

17 In Jung's theory the **anima** is the feminine side of a man and the **animus** the unconscious masculine side of a woman; the persona is the social mask and the shadow the dark, disowned side. ASKED 4 TIMES

18 Cattell's three sources of personality data are **L-data** (life records and observers' ratings, such as school performance and criminal records), **Q-data** (self-report questionnaires) and **T-data** (objective tests whose purpose is hidden); P-data and Z-data do not exist. ASKED 4 TIMES

19 The **NEO-PI-R** (Costa and McCrae, 1992) is a trait-approach inventory of the Five-Factor Model: 240 items, five domains with six facets each, 30 facets in all. ASKED 4 TIMES

20 The **humanistic** perspective (Maslow and Rogers) emphasises free will, personal growth and an innate tendency in everyone to grow toward their maximum potential, not unconscious conflict, conditioned responses or statistical norms. ASKED 4 TIMES

21 **Self-efficacy** (Bandura, 1977), a person's belief in his capability to organise and execute the actions needed to achieve a specific goal, is the central concept of **Albert Bandura**'s social cognitive theory and its key factor in motivation. ASKED 4 TIMES

22 The basic assumption of Adler's **individual psychology** is that human behaviour is purposeful and goal-directed; people are motivated primarily by social urges. ASKED 3 TIMES

23 Jung's two attitudes are **extraversion and introversion** (thinking, feeling, sensing and intuiting are his functions), and the **Myers-Briggs Type Indicator** is built directly on his psychological types. ASKED 3 TIMES

24 **Karen Horney** held that personality is shaped primarily by social and cultural factors; her **basic anxiety** is the all-pervading feeling of being lonely and helpless in a hostile world, the foundation of neurosis. ASKED 3 TIMES

25 **Dasein** (Heidegger's being-in-the-world, the basic unity of person and environment) is existential; its three modes are the **Umwelt** (biological and physical world), **Mitwelt** (relationships with others) and **Eigenwelt** (relation to oneself). ASKED 3 TIMES

26 Existential psychology holds that existence precedes essence, so meaning is never fixed; Frankl's **noogenic neurosis and anxiety** belong to the existential school, and uncovering unconscious fears and desires is psychoanalytic, not existential. ASKED 3 TIMES

27 **Reciprocal determinism** (Bandura): person (cognitive) factors, behaviour and environment influence one another in both directions; it is central to social cognitive theory and is not reciprocal inhibition (Wolpe) or reciprocal altruism (Trivers). ASKED 3 TIMES

28 **Julian Rotter**'s **locus of control** (I-E Scale, 1966) is either internal (outcomes depend on one's own effort and preparation) or external (luck, fate, chance, powerful others); there is no neutral or situational locus. ASKED 3 TIMES

29 Cooley's **looking-glass self** (1902): we imagine how we appear to others, imagine their judgement, and develop a self-feeling such as pride or shame. ASKED 3 TIMES

30 The gunas belong to **prakriti**, not purusha, and come mainly from the **Samkhya** school (Kapila) with the Bhagavad Gita (chapters 14, 17 and 18). ASKED 3 TIMES

CHAPTER 5

Physiological Psychology

IN THIS CHAPTER

5.1 ● Biopsychology and the Organisation of the Nervous System: Explain behaviour from the biological perspective, name the six divisions of biopsychology, and map the nervous system: central and peripheral, somatic and autonomic (sympathetic and parasympathetic), the reflex arc, spinal and cranial nerves, and the meninges, cerebrospinal fluid and blood-brain barrier that protect the brain.

5.2 ●● The Brain: Brainstem, Diencephalon and Cerebral Cortex: Name the brain's five developmental divisions and state where the medulla, pons, cerebellum, midbrain, reticular formation, thalamus, hypothalamus, limbic system, basal ganglia, cortical lobes, Brodmann areas and fibre systems lie and what each does.

5.3 ●● Neurons, Neural Conduction, Synapses and Neurotransmitters: Describe the neuron and the glial cells, explain the resting potential, the all-or-none action potential, refractory periods and saltatory conduction, follow synaptic transmission from Ca2+-triggered release to receptor binding and deactivation, and name the main neurotransmitters with their receptors, source nuclei and functions.

5.4 ●● Drugs, Reward and Psychopharmacology: Explain how psychoactive drugs reach and act on the brain, how tolerance, dependence and addiction develop, how the mesolimbic reward pathway and the four dopamine pathways work, and how antipsychotics, antidepressants, mood stabilisers, anxiolytics and drugs for alcohol dependence act and what side effects they cause.

5.5 ●● Research Methods, Behaviour Genetics and the Developing Brain: Choose the right method for studying brain structure, activity or timing, reason through Mendelian and behaviour-genetic designs, identify the main single-gene and chromosomal disorders, and trace how the nervous system develops, prunes itself and stays plastic.

5.6 ● Sensory Pathways and the Motor System: Trace each sense from receptor to cortex, locate a lesion from the visual field it destroys, and follow a movement down from association cortex through the motor cortex, descending tracts, basal ganglia, cerebellum and spinal circuits to the muscles.

5.7 ●●● The Endocrine System, Stress and the Emotional Brain: Explain how the hypothalamus, pituitary and other endocrine glands work, trace the two stress systems with their feedback and allostasis, and identify the limbic, amygdala and Klüver-Bucy findings that underlie emotion.

5.8 ●● Sleep, Biological Rhythms, Hunger, Thirst and Sex: Explain how the suprachiasmatic clock, the sleep and arousal switches of the brainstem and hypothalamus, and the hypothalamic circuits of hunger, thirst and sex work, and answer the classic lesion, EEG and hormone questions on them.

5.9 ●●● The Biology of Learning, Memory and Amnesia: Distinguish anterograde from retrograde amnesia and explain which brain structures (hippocampus, amygdala, diencephalon) and synaptic mechanisms (LTP, LTD, NMDA receptors, dopamine prediction errors) form and store memories, from H.M. to Korsakoff's syndrome.

5.10 ●● Lateralisation, the Split Brain and Language: Explain how the two hemispheres were found to differ (Dax, Broca, the Wada and dichotic listening tests, split-brain studies), list the functions each hemisphere favours, locate Broca's area, Wernicke's area, the arcuate fasciculus and the angular gyrus, trace the Wernicke-Geschwind model and its dual-stream successor, and tell the aphasias and related reading, writing and speech disorders apart by fluency, comprehension, repetition and lesion site.

5.11 ●● Brain Damage, Neurodegeneration and Neuropsychological Syndromes: Explain how the brain is damaged (tumours, stroke, head injury, infection, toxins, genes), identify the lesion, transmitter and signs of epilepsy, Parkinson's, Huntington's and Alzheimer's diseases, the other dementias and multiple sclerosis, and recognise agnosia, apraxia, hemispatial neglect and the prefrontal syndromes by their lesion sites.

Weightage. 364 of the 3180 questions mapped in this book (11.4%) test this chapter, 151 of them from real past papers. 2 of its 11 sections are High yield.

Graz, Austria, 1921. In the night before Easter Sunday the pharmacologist **Otto Loewi** woke with an idea, switched on the light and jotted a few notes on a slip of thin paper. At six in the morning he knew he had written down something important, but he could not read his own scrawl. The question behind it was an old one. Physiologists knew that stimulating the **vagus nerve** slows the heart, but how does a nerve tell a muscle what to do? Most assumed the message simply crossed as electricity. Loewi had long wondered whether the nerve might instead release a chemical, but he had never found a way to test it. The next night, at three o'clock, the idea came back. This time he got up, went straight to his laboratory and ran the experiment.

The design was simple enough to fit on that slip of paper. Loewi kept two frog hearts beating in small chambers of salt solution, the first with its vagus nerve attached. He stimulated the vagus, and the first heart slowed. Then he drew off the fluid that had bathed it and gave it to the second heart, which had been isolated without its nerves. The second heart slowed too. Something had passed from the nerve endings into the fluid, and that something, not an electric current, had carried the message. Loewi called it *Vagusstoff*, 'vagus substance'. It was later identified as **acetylcholine**, and in 1936 Loewi shared the Nobel Prize in Physiology or Medicine with the British scientist **Henry Dale** for discoveries about the chemical transmission of nerve impulses.

Loewi's two frog hearts hold this chapter in miniature. A change in what an organ does had a physical cause that could be isolated, carried from one heart to another and named. Physiological psychology studies behaviour in that spirit: through the nerves that carry messages, the cells and chemicals that pass them on, the drugs that interfere with them, and the brain systems behind sensation, movement, emotion, sleep, memory and language.

5.1 Biopsychology and the Organisation of the Nervous System

LOWER YIELD · 21 QS

Loewi's second frog heart slowed because a nerve had released a chemical. To see how such simple events add up to fear, memory and speech, you need a point of view that explains behaviour through the body, and a map of the machinery. This section supplies both; the structures inside the brain follow in section 5.2.

The biological perspective

The **biological perspective**, also called the neurobiological approach, holds that behaviour and mental processes are produced by the body: the brain and nervous system, hormones and genes. Every psychological event, from a memory to a mood, is assumed to have a physical basis in neural activity, so changing the body with a drug, a lesion or a hormone changes the mind.

Three neighbouring approaches (Chapter 1) sharpen the contrast. The *psychophysical* approach relates the physical intensity of a stimulus to the sensation it produces; the *cognitive* approach explains behaviour by mental processes such as perceiving, remembering and thinking; the *neo-behaviouristic* approach explains it by learned stimulus-response connec-

tions, with intervening variables such as habit strength inside the organism. The science built on this view, biopsychology, owes its emergence largely to **Donald Hebb**'s *The Organization of Behavior* (1949), the first comprehensive theory of how thought and memory could arise from brain activity (section 5.9).

> **EXAM TRAP**
>
> Asked which system the biological perspective links behaviour to, answer the **body**: the mind is the cognitive answer, the soul a dualist one and the unconscious the psychodynamic one. Behaviour "governed by the brain and nervous system" names the **neurobiological** approach.

Biopsychology as a neuroscience

Biopsychology is the scientific study of the biology of behaviour. For **John P. J. Pinel** and **Steven J. Barnes** (*Biopsychology*, 11th edition, 2021) it is a neuroscience: biopsychologists bring expertise in behaviour to what neuroanatomy, neurochemistry and neurophysiology discover. Their research falls into six divisions (Table 5.1).

Table 5.1 The six divisions of biopsychology (Pinel).

Division	What it studies	Typical subjects and methods
Physiological psychology	Neural mechanisms of behaviour, by manipulating and recording the brain	Laboratory animals; lesions, stimulation; mostly pure research
Psychopharmacology	Drug effects on brain and behaviour	Animals and humans; often applied
Neuropsychology	Psychological effects of brain damage	Patients; case studies; the most applied
Psychophysiology	Physiology of psychological processes	Humans; non-invasive EEG, heart rate, skin conductance
Cognitive neuroscience	Neural bases of cognition	Humans; brain imaging; the youngest division
Comparative psychology	Behaviour across species: evolution, genetics, adaptiveness	Many species, laboratory and wild

Biopsychologists study both humans, who can report what they experience, and nonhuman animals, whose simpler brains allow comparison across species and experiments that would be unethical in people. In an *experiment* the researcher assigns subjects to conditions and varies one independent variable, so only an experiment shows cause. A **quasi-experimental study** compares groups already exposed to a condition in real life, such as heavy drinkers and abstainers, so confounding variables remain. A *case study* examines one person in depth (Chapter 9).

KEY STUDY: CONVERGING OPERATIONS AND KORSAKOFF'S SYNDROME

Because each method has weaknesses, progress comes from **converging operations**: aiming several approaches at one problem so that each covers the others' weaknesses. Korsakoff's amnesia (section 5.9) was first blamed on alcohol from quasi-experiments with heavy drinkers; case studies of malnourished people who drank little, and experiments in which thiamine-deficient rats developed similar memory loss, then pointed to thiamine (vitamin B1) deficiency.

Behind all this lies the mind-brain problem. **René Descartes** (1596 to 1650) proposed **dualism**: the body, brain included, is a machine obeying natural laws, while a nonphysical mind controls it, the two meeting, he suggested, at the pineal gland. **Monism** holds that only one kind of substance exists; its materialist form, the working view of biopsychology, treats mental events as brain activity.

The divisions of the nervous system ●●

The nervous system is a system of twos (Figure 5.1). The **central nervous system** (CNS), the brain and spinal cord, lies inside the skull and spine; the **peripheral nervous system** (PNS) is all the nerves outside them. The PNS has two divisions. The **somatic nervous system** deals with the outside world: its **afferent** nerves carry sensory signals from skin, eyes and ears to the CNS, and its **efferent** nerves carry commands to the skeletal muscles, which are under voluntary control (afferent arrives, efferent exits). The **autonomic nervous system** (ANS) regulates the internal environment, controlling smooth muscle, cardiac muscle and glands largely involuntarily.

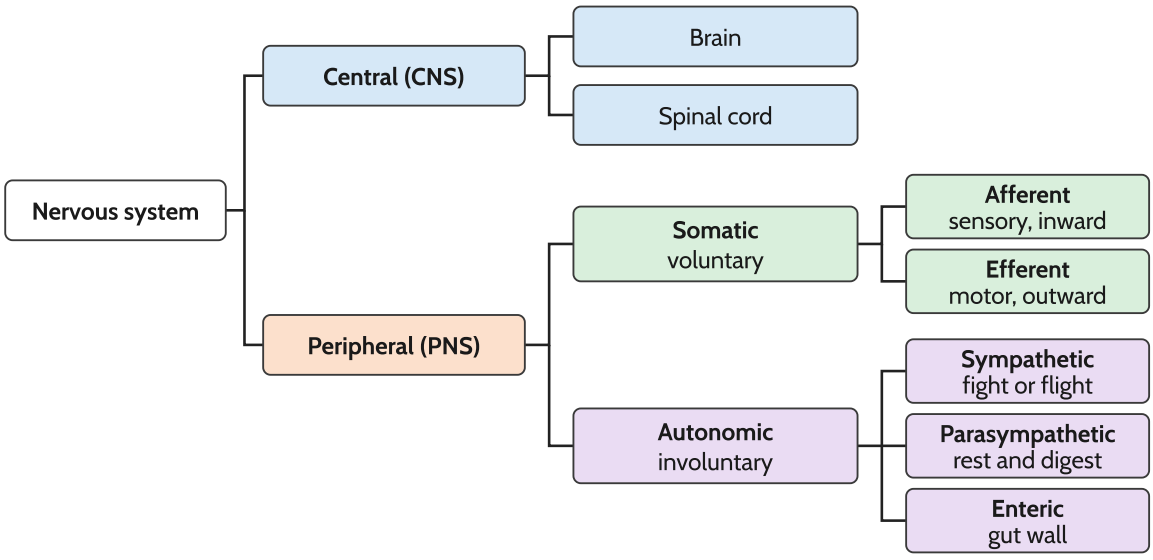


Figure 5.1 Somatic and autonomic are both peripheral; many books count the enteric network of the gut wall as a third autonomic division.

> **EXAM TRAP**
>
> The parts of the ANS are the **sympathetic and parasympathetic** divisions. "Central and peripheral" divide the whole nervous system, "voluntary and involuntary" describe somatic versus autonomic, and "cerebrospinal and cortical" is no division at all. The brain and spinal cord form the **CNS**.

The autonomic nervous system ●●

Each division of the ANS reaches its targets through two neurons: a preganglionic neuron from the CNS synapses in an autonomic ganglion on a postganglionic neuron that runs to the organ.

The **sympathetic division** mobilises and spends energy. **Walter Bradford Cannon** (1871 to 1945) described its emergency role in 1915 as the **fight-or-flight response**: the heart races, pupils and airways widen, blood shifts to the muscles and the liver releases glucose. Its preganglionic neurons leave spinal segments T1 to L2, the **thoracolumbar outflow**; their short fibres end mostly in the sympathetic chain, a string of ganglia beside the spine, from which long postganglionic fibres run to the organs. Acetylcholine is the transmitter at the ganglia, noradrenaline (norepinephrine) at most targets. Some preganglionic fibres go straight to the adrenal medulla, the core of the adrenal gland, which acts like a sympathetic ganglion and releases adrenaline (epinephrine) and some noradrenaline into the blood. This **sympathetic adrenal-medullary system** is the fast arm of the stress response.

The **parasympathetic division** serves rest and digest: it slows the heart, narrows the pupils and promotes digestion, conserving and storing energy. Its preganglionic neurons leave the brainstem in cranial nerves III, VII, IX and X and the spinal cord at S2 to S4, the **craniosacral outflow**. Their long fibres end in ganglia near or inside the organ, and both synapses use acetylcholine. The **vagus nerve** (X) carries most of these fibres, supplying the heart, lungs and most of the gut.

Most organs receive both divisions (*dual antagonistic innervation*), so your heart rate reflects the balance of sympathetic acceleration and vagal braking; sweat glands and most blood vessels are sympathetic only, and in sexual response the two cooperate (Table 5.2). The **enteric nervous system**, a network of neurons in the gut wall, runs local reflexes of gut movement and secretion on its own.

Table 5.2 The sympathetic and parasympathetic divisions compared.

Feature	Sympathetic	Parasympathetic
Outflow	Thoracolumbar (T1 to L2)	Craniosacral (III, VII, IX, X; S2 to S4)
Fibres	Short preganglionic, long postganglionic	Long preganglionic, short postganglionic
Transmitter at target	Noradrenaline (sweat glands: acetylcholine)	Acetylcholine
Heart	Faster, stronger	Slower
Pupils	Dilated	Constricted
Saliva	Reduced (dry mouth)	Increased
Bronchi	Dilated	Constricted
Digestion	Inhibited	Stimulated

Feature	Sympathetic	Parasympathetic
Bladder	Wall relaxed; urine held	Wall contracted; emptied
Sweating, skin conductance	Increased	No effect
Liver glucose	Released into blood	Stored as glycogen
Sexual response	Ejaculation, orgasm	Erection, lubrication

EXAM TRAP

Fight or flight is **sympathetic**; the fullest answer is "sympathetic nervous system plus adrenal medulla". Rest and digest and conserving energy are **parasympathetic**; allocating energy for expenditure is sympathetic. A stem naming the vagus and acetylcholine means parasympathetic, not the sympathetic adrenal-medullary system. The enteric system cannot slow the heart or narrow the pupils, and the pituitary-thyroid axis is endocrine, setting metabolic rate rather than the emergency response (section 5.7).

Grey matter, white matter and the spinal cord ●

Slice fresh brain or spinal cord and you see two tissues. **Grey matter** is mainly neuron cell bodies, dendrites, unmyelinated axons and synapses, where information is processed. **White matter** is mainly myelinated axons, the wiring between areas; myelin is a fatty insulating sheath (section 5.3), and its fat makes the tissue white. A cluster of cell bodies is a nucleus in the CNS but a **ganglion** in the PNS; a bundle of axons is a tract in the CNS but a nerve in the PNS. (The basal ganglia, despite the name, are CNS nuclei.)

In the spinal cord (Figure 5.2) a butterfly-shaped core of grey matter, with dorsal and ventral horns, is surrounded by white columns of ascending and descending tracts: the reverse of the cerebral cortex, which is grey outside. Thirty-one pairs of spinal nerves attach to the cord: 8 cervical, 12 thoracic, 5 lumbar, 5 sacral and 1 coccygeal. Each joins it by two roots. The dorsal root is purely sensory, its cell bodies sitting just outside the cord in the **dorsal root ganglion**; the ventral root is purely motor, its cell bodies lying in the ventral horn. This is the **Bell-Magendie law**, named after **Charles Bell** (1774 to 1842), who reported the motor role of the ventral roots in 1811, and **François Magendie** (1783 to 1855), who demonstrated both halves of the law in 1822.

EXAM TRAP

White matter is **myelinated axons**; cell bodies and unmyelinated axons are grey, and dendrites are never myelinated. Cell bodies in the PNS form **ganglia** (nuclei in the CNS); tracts and nerves are bundles of axons.

Neurons by function and the reflex arc ●●

Neurons fall into three functional classes. Sensory (afferent) neurons carry signals from receptors to the CNS. A **motor neuron** (efferent) carries commands from the brain and spinal cord to muscles and glands. An **interneuron** (connector or association neuron), by far the commonest kind, lies wholly within the CNS and links the two. Muscles and glands are effectors: targets, not neurons.

Together they form the **reflex arc** of an automatic response. Touch a hot pan and the signal runs: receptor in the skin, afferent neuron through the dorsal root, interneuron in the spinal grey matter, efferent neuron through the ventral root, effector muscle that pulls your hand away (Figure 5.2). The arc closes in the spinal cord, so you withdraw before you feel the pain. This withdrawal reflex is *polysynaptic*. The knee-jerk, a stretch reflex, is *monosynaptic*: the sensory neuron synapses directly on the motor neuron, with no interneuron.

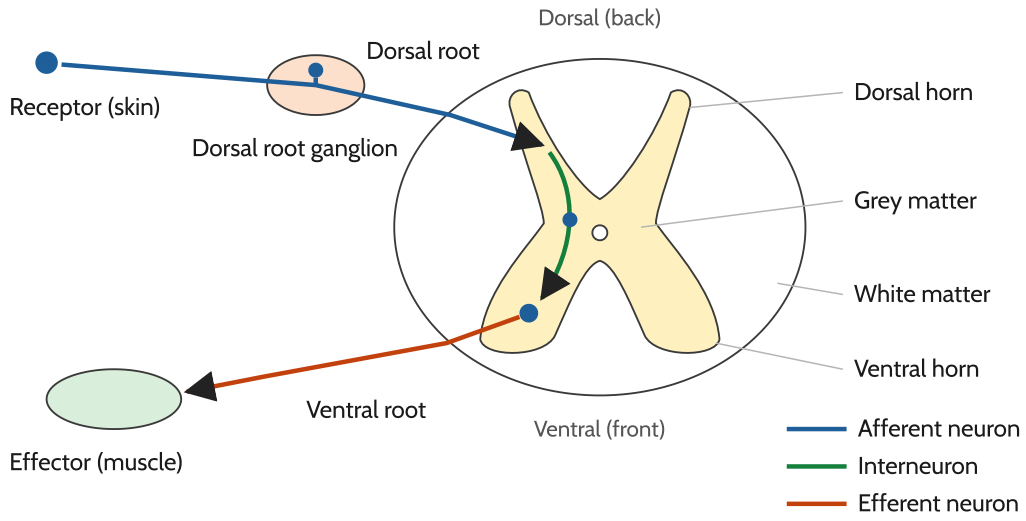


Figure 5.2 The spinal cord in cross section with a three-neuron withdrawal reflex: in by the dorsal root, across by an interneuron in the grey matter, out by the ventral root. The knee-jerk skips the interneuron.

> **EXAM TRAP**
>
> The order is **afferent neuron, interneuron, efferent neuron**; no arc runs afferent to efferent or starts with an efferent. A cell carrying information from the brain to muscles and glands is a **motor neuron**: "effector cell" names the target and "connector neuron" the interneuron.

The twelve cranial nerves

Twelve pairs of **cranial nerves** leave the brain directly, numbered from front to back. They serve mainly the head and neck, except the "wandering" vagus, which reaches the chest and abdomen (Table 5.3). Three are sensory (I, II, VIII), five are motor (III, IV, VI, XI, XII) and four are mixed (V, VII, IX, X). Four carry parasympathetic fibres: III (pupil), VII (tears, saliva), IX (saliva) and X (heart, lungs, gut).

Table 5.3 The twelve cranial nerves.

No.	Name	Type	Main functions
I	Olfactory	Sensory	Smell
II	Optic	Sensory	Vision
III	Oculomotor	Motor	Four of the six eye muscles; lifts eyelid; constricts pupil
IV	Trochlear	Motor	Superior oblique eye muscle

No.	Name	Type	Main functions
V	Trigeminal	Mixed	Facial sensation; chewing
VI	Abducens	Motor	Lateral rectus (turns eye outward)
VII	Facial	Mixed	Facial expression; taste, front of tongue; tears, saliva
VIII	Vestibulocochlear	Sensory	Hearing and balance
IX	Glossopharyngeal	Mixed	Taste, back of tongue; swallowing; saliva
X	Vagus	Mixed	Parasympathetic supply to heart, lungs, gut; throat
XI	Accessory	Motor	Neck and shoulder muscles
XII	Hypoglossal	Motor	Tongue movement

Nerve I is special: smell is the only sense whose main pathway reaches the cortex (the piriform cortex of the medial temporal lobe) without first relaying in the thalamus, as vision, hearing, taste and touch all do (section 5.6).

> REMEMBER
>
> Names: "On Old Olympus' Towering Tops, A Finn And German Viewed Some Hops" (A for acoustic, an old name for VIII). Types: "Some Say Marry Money, But My Brother Says Big Brains Matter More" (S sensory, M motor, B both).

Protecting the CNS ●●

The CNS is encased in bone and wrapped in three **meninges**: from outside in, the tough dura mater, the web-like arachnoid mater and the thin pia mater, which clings to the brain's surface. Between arachnoid and pia lies the subarachnoid space.

Cerebrospinal fluid (CSF), made by the **choroid plexus** (networks of capillaries in the ventricles), fills the brain's four ventricles, the central canal of the spinal cord and the subarachnoid space. It flows from the two lateral ventricles, the largest, to the third ventricle, through the cerebral aqueduct to the fourth, and out into the subarachnoid space, and is absorbed through the arachnoid granulations into the venous sinuses of the dura. CSF cushions the brain, which floats in it. If its flow is blocked, for example at the aqueduct, the ventricles swell and compress the brain: *hydrocephalus*, treated with a draining shunt.

The **blood-brain barrier** exists because the endothelial cells of brain capillaries are sealed together by tight junctions, supported by the end-feet of astrocytes (star-shaped glial cells). Small fat-soluble molecules such as oxygen, alcohol and nicotine pass; glucose is carried across by active transport; large or water-soluble molecules, many toxins and many drugs are kept out (section 5.4). Weak spots include the **area postrema** in the floor of the fourth ventricle, which detects poisons in the blood and triggers vomiting.

> EXAM TRAP
>
> The barrier is **tight junctions plus astrocyte end-feet**: not the pia mater (a meninx), microglia (immune scavengers) or oligodendrocyte myelin (around axons). CSF is made by the **choroid plexus** and absorbed by the arachnoid granulations, and the ventricles hold CSF, not blood, myelin or "synaptic fluid".

Directions and planes

Anterior (rostral) means toward the nose, posterior (caudal) toward the tail. Dorsal means toward the back or the top of the head, ventral toward the belly or the bottom of the head; because humans stand upright, this axis bends at the brainstem, so dorsal is up in the brain but back in the spinal cord. Medial means toward the midline, lateral away from it. Ipsilateral means on the same side, **contralateral** on the opposite side: each hemisphere controls the contralateral hand. A horizontal section is parallel to the ground, a frontal (coronal) section parallel to the face and a sagittal section parallel to the midline; the midsagittal section runs exactly between the hemispheres.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Explaining behaviour through the brain and nervous system is the _____ perspective.
2. Aiming several research approaches at one problem is called _____.
3. Sympathetic preganglionic neurons leave spinal segments _____.
4. Parasympathetic fibres reach most thoracic and abdominal organs through the _____ nerve.
5. A cluster of cell bodies in the PNS is a _____.
6. By the Bell-Magendie law, _____ roots are motor.
7. The knee-jerk reflex, with no interneuron, is _____.
8. Cerebrospinal fluid is made by the _____.

Answers: (1) biological (neurobiological), (2) converging operations, (3) T1 to L2, (4) vagus (X), (5) ganglion, (6) ventral, (7) monosynaptic, (8) choroid plexus.

5.2 The Brain: Brainstem, Diencephalon and Cerebral Cortex

MEDIUM YIELD · 27 QS

A lesion the size of a pea can stop your breathing in one place and merely blur a corner of your vision in another. Location is everything. This section tours the brain from the bottom up: the plan that names its five divisions, the hindbrain and midbrain that keep you alive and awake, the diencephalon that relays and regulates, and the cortex that perceives, plans and thinks.

Five divisions from three swellings

The brain's tongue-twisting names come from its development. The embryonic nervous system begins as a fluid-filled neural tube, and three swellings appear at its front end: the **forebrain** (prosencephalon), **midbrain** (mesencephalon) and **hindbrain** (rhombencephalon). Before birth the forebrain divides into the **telencephalon** and **diencephalon**, and the hindbrain into the **metencephalon** and **myelencephalon**, giving the five divisions of Table 5.4. A memory aid: the telencephalon is on top, and the other four lie below it in alphabetical order.

Table 5.4 The divisions of the brain.

Embryonic swelling	Division	Main adult structures
Forebrain (prosencephalon)	Telencephalon	Cerebral cortex, limbic system, basal ganglia, corpus callosum
Forebrain (prosencephalon)	Diencephalon	Thalamus, hypothalamus (and pineal gland)
Midbrain (mesencephalon)	Mesencephalon	Tectum, tegmentum, cerebral peduncles
Hindbrain (rhombencephalon)	Metencephalon	Pons, cerebellum
Hindbrain (rhombencephalon)	Myelencephalon	Medulla oblongata

The **brainstem** is the stalk on which the cerebral hemispheres sit: usually the midbrain, pons and medulla, though some texts use it for all four divisions below the telencephalon. Either way the cerebellum is treated separately (Figure 5.3).

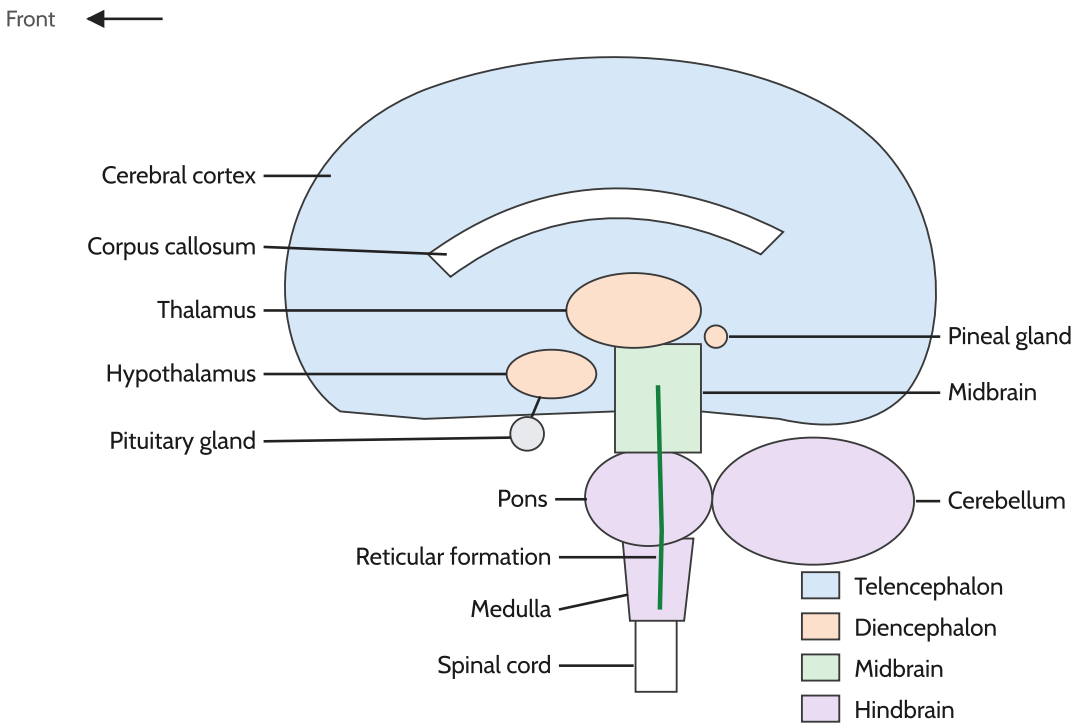


Figure 5.3 The divisions in a schematic midline section. The diencephalon lies beneath the corpus callosum; the reticular formation (dotted) runs through the brainstem core.

EXAM TRAP

The corpus callosum lies just above the thalamus, so it is often filed under the diencephalon. It is the white matter of the cerebral hemispheres, so it belongs to the **telencephalon**. The diencephalon is the thalamus and hypothalamus, the mesencephalon the midbrain, the metencephalon the pons and cerebellum.

The hindbrain: medulla, pons and cerebellum ●●●

The **medulla oblongata** is where the spinal cord widens into the brain. Much of it is tracts: on its front surface two ridges, the **pyramids**, carry the corticospinal motor tract, which crosses sides at the lower medulla (Section 5.6). It also holds the **vital centres**: the cardiovascular centre for heart rate and blood pressure, the respiratory centre that generates the

rhythm of breathing, and reflex centres for swallowing, coughing, sneezing and vomiting. So medullary damage is the brain injury most likely to kill, and an overdose of alcohol or opioids kills by silencing its respiratory centre.

Above it, the **pons** (Latin "bridge") bulges from the front of the brainstem; its nuclei relay signals from the cortex across to the cerebellum. It also helps set the rhythm of breathing, and its reticular nuclei contribute to arousal, sleep and REM sleep (Section 5.8).

Behind the pons sits the **cerebellum** ("little brain"): about 10 percent of the brain's mass but more than half its neurons. It does not start movements; the primary motor cortex does. It makes them accurate, comparing the intended movement with the actual one and correcting the error. So it coordinates and smooths movement, keeps balance and posture, times rapid sequences and supports motor learning, such as the practised timing of a cricket stroke. Damage produces **ataxia**: clumsy, uncoordinated movement and a wide, staggering gait, like drunkenness. It does not produce language (Broca's area), store declarative memories (hippocampus and cortex) or regulate emotion (amygdala); its circuitry is in Section 5.6.

> EXAM TRAP
>
> Heartbeat and respiration are modulated by the **hindbrain** (medulla and pons). The hypothalamus adjusts them for homeostasis but does not pace them; the amygdala is for fear; the cerebellum, though in the hindbrain, does not keep you breathing. Damage most likely to cause **death**: the medulla, not the pons, cerebellum or reticular formation (whose damage causes coma).

The midbrain and the reticular activating system ●●

Two words organise the midbrain. The **tectum** ("roof") is its back surface, with two pairs of bumps: the **superior colliculi** turn your eyes and head towards a sudden flash (visual orienting), and the **inferior colliculi** relay hearing (superior for sight, inferior for sound). The **tegmentum** beneath holds the **substantia nigra**, whose dopamine neurons feed the basal ganglia and die in Parkinson's disease; the **red nucleus**, a motor nucleus; the **periaqueductal grey**, which suppresses pain and mediates opioid analgesia; and the **ventral tegmental area**, source of the dopamine reward pathway (Section 5.4). On the front, the **cerebral peduncles** are thick stalks of axons descending from the cortex.

Through the core of the whole brainstem, from medulla to midbrain, runs the **reticular formation**, a net-like web of about 100 small nuclei (*reticulum*, "little net"). Its ascending part, the **reticular activating system** (RAS), projects through the thalamus and directly to the cortex. It regulates **arousal, wakefulness, attention and consciousness**, filtering stimulation so that a parent sleeps through traffic yet wakes at a baby's cry. Destroy it and the result is **coma**. Its descending parts help control muscle tone and cardiac and respiratory reflexes.

KEY STUDY: MORUZZI AND MAGOUN (1949)

Giuseppe Moruzzi and **Horace Magoun** stimulated the reticular formation of sleeping cats electrically. The cats woke, and their EEG changed from the slow, synchronised waves of sleep to the fast, desynchronised pattern of alertness. With brainstem transection and lesion findings, this led them to propose that low reticular activity produces sleep and high activity wakefulness. Later work showed that sleep is also actively generated elsewhere (Section 5.8).

EXAM TRAP

RAS questions offer its neighbours. Fine motor coordination, balance and posture belong to the cerebellum, emotional memory to the amygdala, episodic memory to the hippocampus, motor planning to premotor and prefrontal cortex, reading comprehension to Wernicke's area and the angular gyrus. The RAS means arousal, wakefulness, attention and consciousness.

The diencephalon: thalamus and hypothalamus ●●

The **thalamus** is the large two-lobed structure at the top of the brainstem, one lobe on each side of the third ventricle. It is the brain's **relay station**: every sense except smell synapses in a thalamic nucleus before reaching cortex. The **lateral geniculate nucleus** relays vision, the **medial geniculate nucleus** hearing (lateral for light, medial for music), and the **ventral posterior nucleus** touch, pain, temperature, body position and taste. Smell goes from the olfactory bulb directly to olfactory cortex. Nor is the thalamus a passive switchboard: each relay nucleus receives feedback from its cortex, and other nuclei relay motor loops (basal ganglia and cerebellum to motor cortex) and association loops (to prefrontal cortex).

Just below the front of the thalamus (*hypo*, "below") lies the almond-sized **hypothalamus**, master regulator of **homeostasis**, the maintenance of a stable internal environment. It controls hunger, thirst, body temperature, the sleep-wake cycle, sexual behaviour, and emotion and aggression, acting through the autonomic nervous system, through motivated behaviour and through the endocrine system. The **pituitary gland** hangs from its underside, and the hypothalamus directs it, with releasing hormones carried in local blood vessels to the anterior pituitary and axons to the posterior pituitary (Section 5.7). The pituitary is the "master gland", but the hypothalamus is its master: the control centre integrating the endocrine, autonomic and visceral systems.

Its nuclei are named here and taught later: the **suprachiasmatic nucleus**, the circadian clock above the optic chiasm; the lateral and ventromedial areas and arcuate nucleus in feeding (both Section 5.8); the preoptic area in temperature and sexual behaviour; and the supraoptic and paraventricular nuclei, which make oxytocin and vasopressin (Section 5.7). On its underside sit the two **mammillary bodies**, part of the limbic memory circuit and damaged in Korsakoff's syndrome (Section 5.9). Behind the thalamus, the epithalamus holds the **pineal gland**, which secretes melatonin at night.

EXAM TRAP

"Relays every sense except smell" is the **thalamus**. "Homeostasis", "master regulator", "controls the pituitary" and "integrates the endocrine, autonomic and visceral systems" are the **hypothalamus**. The brainstem runs autonomic reflexes without integrating them with hormones; the spinal cord carries messages and simple reflexes; the pineal makes melatonin; the corpus callosum, basal ganglia and hippocampus relay no senses.

The limbic system and basal ganglia

Beneath the cortex lie two loose groupings. The **limbic system** is a ring of midline structures circling the thalamus (*limbus*, "border"): the **hippocampus** of the medial temporal lobe (forming new declarative memories and spatial maps), the almond-shaped **amygdala** in front of it (emotion, especially fear, and emotional memory), the **cingulate cortex** above the corpus callosum, the **fornix** (the main limbic tract, arching from hippocampus to mammillary bodies and septum), the **septum** and the mammillary bodies. Together they serve emotion, motivation and memory (Sections 5.7 and 5.9).

The **basal ganglia** are large nuclei deep in each hemisphere. The tail-like **caudate nucleus** and the **putamen** receive cortical input; both look striped, so together they are the **striatum**. Its output goes to the **globus pallidus** ("pale globe"). The **subthalamic nucleus** and the substantia nigra, whose dopamine input to the striatum fails in Parkinson's disease, are functional partners; older texts also counted the amygdala. The basal ganglia select and initiate voluntary movements, build habits and support procedural learning (pathways in Section 5.6).

EXAM TRAP

"Which is NOT a basal ganglia structure?" The **red nucleus**, a motor nucleus of the midbrain tegmentum (source of the rubrospinal tract). The caudate and globus pallidus are core members; the subthalamic nucleus belongs to the circuit.

The cerebral cortex: layers and landmarks

The **cerebral cortex** ("bark") is a sheet of grey matter a few millimetres thick over the hemispheres, with white matter (myelinated axons) beneath. It is crumpled into ridges, **gyri**, and grooves, **sulci**; the deepest grooves are **fissures**. The **longitudinal fissure** separates the hemispheres. The **central sulcus** separates frontal from parietal lobe, with the **precentral gyrus** (primary motor cortex) in front and the **postcentral gyrus** (primary somatosensory cortex) behind. The **lateral fissure** (Sylvian fissure) separates the temporal lobe from the frontal and parietal lobes. The **calcarine sulcus**, on the medial surface of the occipital lobe, is lined by primary visual cortex (Figure 5.4).

About 90 percent of human cortex is **neocortex** ("new cortex", or isocortex), the most recently evolved part, with six layers numbered I to VI from the surface. Large **pyramidal cells** send long axons out of the area; small **stellate cells** are interneurons. **Layer IV** receives thalamic input, so it is thick in sensory cortex; **layer V** holds the large pyramidal output cells projecting to brainstem and spinal cord, so it is thick in motor cortex, where it contains the giant **Betz cells**. The neocortex is the seat of higher cognitive functions: perception, language, rea-

soning, planning and voluntary movement. The older **allocortex** has fewer layers: the hippocampus (three) and olfactory cortex.

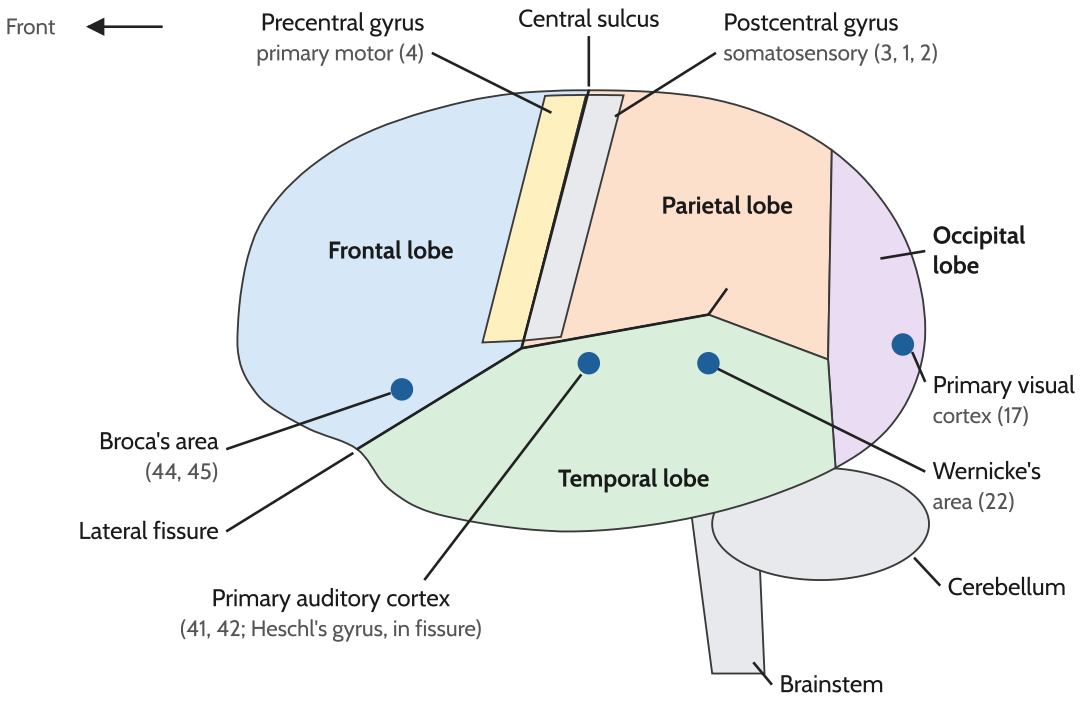


Figure 5.4 The left hemisphere from the side, with Brodmann numbers in brackets. Primary visual cortex lies mostly on the medial surface, along the calcarine sulcus.

EXAM TRAP

The neocortex means **higher cognitive functions**. Basic reflexes belong to the spinal cord and brainstem, olfactory processing to the old olfactory cortex, autonomic regulation to the hypothalamus and medulla.

The lobes, association cortex and Brodmann's map ●●●

The central sulcus and lateral fissure divide each hemisphere into four lobes, with the **insula** hidden inside the lateral fissure (Table 5.5). Lobes are folds of one continuous sheet, not functional units: each holds several different areas.

Table 5.5 The lobes of the cerebral cortex.

Lobe	Landmark and primary area	Main functions
Frontal	In front of the central sulcus; primary motor cortex in the precentral gyrus	Voluntary movement; prefrontal: planning, problem solving, reasoning, intellectual functioning, personality, social behaviour; Broca's area: speech production
Parietal	Behind the central sulcus; primary somatosensory cortex in the postcentral gyrus	Touch and body sense; interpreting and integrating sensory information; spatial attention
Temporal	Below the lateral fissure; primary auditory cortex in Heschl's gyrus	Hearing; Wernicke's area: language comprehension; object and face recognition; medial part: memory

Lobe	Landmark and primary area	Main functions
Occipital	Back of the brain; primary visual cortex (V1, striate cortex) along the calcarine sulcus	Vision
Insula	Inside the lateral fissure	Taste; interoception (sensing the body's internal state)

The frontal lobe, the largest, does two jobs. Its back strip, the **primary motor cortex**, commands voluntary movement of the opposite side of the body, and premotor cortex and the supplementary motor area plan it. Everything in front is **prefrontal cortex**: planning, problem solving, reasoning, working memory, intellectual functioning, personality and social behaviour. In the parietal lobe, posterior parietal cortex integrates touch, vision and body position into a sense of where objects and limbs are. Primary visual cortex is called **striate cortex** for the white stripe in its layer IV.

Cortex is also graded by distance from the senses. **Primary sensory cortex** receives most of its input directly from a thalamic relay nucleus; **secondary sensory cortex** mostly from primary cortex of the same sense. **Association cortex** receives input from more than one sensory system and integrates it; it makes up most of human cortex (prefrontal, posterior parietal, temporal).

In 1909 the German neuroanatomist **Korbinian Brodmann** (1868 to 1918) divided the cortex into numbered areas by **cytoarchitecture**, the size, density and layering of its cells. His numbers remain the common language of cortical maps (Table 5.6).

Table 5.6 Brodmann areas worth knowing.

Area	Region	Lobe
1, 2, 3	Primary somatosensory cortex	Parietal
4	Primary motor cortex	Frontal
6	Premotor cortex and supplementary motor area	Frontal
8	Frontal eye fields	Frontal
9 to 12	Prefrontal cortex	Frontal
17	Primary visual cortex (V1, striate)	Occipital
18, 19	Visual association cortex	Occipital
22	Wernicke's area (posterior part)	Temporal
39	Angular gyrus (reading)	Parietal
40	Supramarginal gyrus	Parietal
41, 42	Primary auditory cortex	Temporal
44, 45	Broca's area	Frontal

Four prefrontal regions return in later sections: **dorsolateral prefrontal cortex** (working memory, planning, mental flexibility), **orbitofrontal cortex** above the eye sockets (impulse control, social behaviour, reward value), **ventromedial prefrontal cortex** (emotion in decision making) and **anterior cingulate cortex** on the medial surface (detecting errors and conflict; the unpleasantness of pain). Their syndromes are in Section 5.11.

EXAM TRAP

"Interpreting sensory information" is **parietal**: every lobe processes some sense, but the phrase means integrating bodily sensation. "Intellectual functioning", "problem solving" and "cognitive functions and voluntary movement" are **frontal**. Vision and V1 are **occipital**; auditory processing is **temporal**; motor commands and language production are frontal.

Fibre systems: wiring the parts together

The white matter comes in three kinds. **Commissural fibres** cross the midline to connect the hemispheres. The largest commissure is the **corpus callosum**, an estimated 200 million axons linking the left and right hemispheres; the smaller **anterior commissure** links the temporal lobes and olfactory regions. **Association fibres** stay within one hemisphere: short U-fibres join neighbouring gyri, and long bundles such as the **arcuate fasciculus** link the frontal and temporal lobes (Broca's and Wernicke's areas). **Projection fibres** connect cortex with lower centres in both directions, from thalamus to cortex and from cortex to brainstem and spinal cord. They funnel through the **internal capsule** between the thalamus and basal ganglia, so a small stroke there can paralyse the opposite side of the body. Three pairs of **cerebellar peduncles** link the cerebellum to the brainstem. Cutting the corpus callosum produces the split brain (Section 5.10).

EXAM TRAP

The corpus callosum connects the **left and right hemispheres**. The wrong options name other fibre systems: brainstem and cerebellum (cerebellar peduncles), frontal and temporal lobes (arcuate fasciculus), thalamus and cortex (projection fibres).

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The corpus callosum belongs to the _____.
2. The vital centres for breathing and heart rate lie in the _____.
3. Cerebellar damage produces uncoordinated movement called _____.
4. Moruzzi and Magoun woke sleeping cats by stimulating the _____.
5. The thalamic relay for hearing is the _____ nucleus.
6. The hypothalamic circadian clock is the _____ nucleus.
7. The _____ nucleus is a midbrain motor nucleus, not a basal ganglion.
8. Primary visual cortex is Brodmann area _____.

Answers: (1) telencephalon, (2) medulla oblongata, (3) ataxia, (4) reticular formation, (5) medial geniculate, (6) suprachiasmatic, (7) red, (8) 17

5.3 Neurons, Neural Conduction, Synapses and Neurotransmitters

MEDIUM YIELD · 45 QS

Every word you read here is carried by cells that work like tiny batteries and talk in chemical whispers. This section follows one message from start to finish: how a neuron is built, how it

holds a charge at rest, how that charge flips into an action potential that races down the axon, and how chemicals carry the signal across the synapse.

The neuron: the basic unit ●●

In the early 1870s the Italian physician **Camillo Golgi** (1843 to 1926) found that silver chromate stains a few neurons in a slice completely black, showing single cells in silhouette for the first time: the **Golgi stain**. Golgi held that the nervous system is one continuous net of fused fibres, the **reticular theory**. The Spanish histologist **Santiago Ramón y Cajal** (1852 to 1934), using Golgi's own stain, showed that it is built of separate cells that touch but do not fuse: the **neuron doctrine**. The rivals shared the Nobel Prize in 1906, and Cajal was right. The **neuron** is the basic structural and functional unit of the nervous system; a dendrite is only part of one, the brain is an organ made of billions of them, and glial cells support neurons rather than carry their messages.

Figure 5.5 shows the parts. The **cell body** (soma) holds the nucleus and keeps the cell alive. Branching **dendrites**, studded with tiny **dendritic spines**, receive most incoming synapses. The cone-shaped **axon hillock** joins the soma to the **axon**, the long fibre that carries signals away from the soma; action potentials begin here (strictly, Pinel notes, in the axon initial segment just beyond it). A fatty **myelin sheath** wraps many axons, broken at intervals by bare gaps, the **nodes of Ranvier**. The axon ends in fine branches, the **telodendria**, each tipped by an **axon terminal** (terminal button) whose **synaptic vesicles** store and release the transmitter.

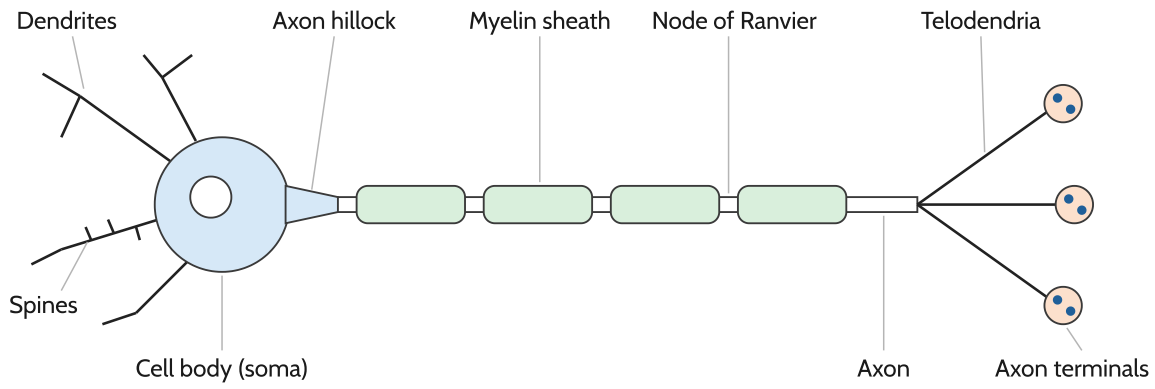


Figure 5.5 A multipolar neuron. Signals flow from the dendrites and soma, through the hillock and down the myelinated axon, to the terminals, whose vesicles (dots) hold the transmitter.

A **multipolar neuron** (one axon, many dendrites) is by far the commonest type and includes motor neurons. A **bipolar neuron** has two processes, one at each end, as in the retina. A **unipolar neuron** has a single process that splits in two, as in most somatosensory neurons.

> EXAM TRAP
>
> Transmitter is released from the **axon terminals**. The telodendria are the branches leading to them, the axon hillock is where the spike starts, and myelin only insulates. The basic unit is the neuron, not the dendrite, the brain or the glial cell.

Glial cells ●●

Neurons share the nervous system with **glial cells** (glia, "glue"). They are not ten times as numerous as neurons, as is often claimed: Pinel puts it at about two glia for every three neurons. Glia do not fire action potentials, but they do far more than hold neurons in place (Table 5.7).

Table 5.7 The glial cells.

Glial cell	Where	Main jobs
Astrocytes (star-shaped; largest)	CNS	End-feet on capillaries help form the blood-brain barrier; buffer extracellular K^+; recycle glutamate to neurons as glutamine (glutamate-glutamine cycle); supply fuel
Oligodendrocytes	CNS	Myelin; one cell wraps segments of many axons
Schwann cells	PNS: nerves outside the skull and spine	Myelin; one cell per segment; guide regrowth of cut nerves
Microglia (smallest)	CNS	Resident macrophages: immune surveillance, phagocytosis, synaptic pruning
Ependymal cells	Lining of the ventricles	Help make and circulate cerebrospinal fluid

EXAM TRAP

The **myelin sheath** is made by glia but is not a glial cell. Oligodendrocytes myelinate the CNS (brain, spinal cord, basal ganglia) and Schwann cells the PNS; options that swap them are wrong. Astrocytes never conduct action potentials.

The resting membrane potential ●●

Push a fine electrode into a resting neuron and its inside reads about 70 millivolts more negative than the outside: the **resting membrane potential**, about **-70 mV** (range about -60 to -80 mV). The neuron is **polarised** because ions are unevenly spread: sodium (Na^+) and chloride (Cl^-) ions are concentrated outside, potassium (K^+) ions and negatively charged proteins inside.

Two forces act on each ion: the **concentration gradient**, from crowded to scarce, and **electrostatic pressure**, as like charges repel and opposite charges attract. Both push Na^+ inwards, yet little enters, because at rest the membrane is far more permeable to K^+ than to Na^+: K^+ channels are open, Na^+ channels closed. K^+ leaks out until the negative charge inside holds the rest back. The **sodium-potassium pump**, powered by ATP, exchanges **three Na^+ out for every two K^+ in**, preserving the gradients against the leak. The Nernst equation gives one ion's equilibrium potential; the Goldman equation combines several ions.

KEY STUDY: HODGKIN AND HUXLEY'S SQUID GIANT AXON (1952)

Alan Hodgkin (1914 to 1998) and **Andrew Huxley** (1917 to 2012) recorded inside the giant axon of the squid, up to 1 mm across. In 1952 they described the action potential as a timed rise in Na^+ and then K^+ permeability, in equations that won a Nobel Prize in 1963. Criticism: the large, simple squid axon fits many mammalian neurons only roughly.

> **EXAM TRAP**
>
> The resting potential is about -70 mV, inside negative: not -40 mV and never positive. It rests on high K^+ permeability (not equal Na^+ and K^+ permeability) plus the pump. Open Na^+ channels, Ca^{2+} influx and Cl^- channels are distractors.

Postsynaptic potentials and the action potential ●●

Transmitter binding nudges a neuron's membrane potential. A depolarisation (say from -70 to -67 mV) makes firing more likely and is an **excitatory postsynaptic potential** (EPSP); a hyperpolarisation (to -72 mV) makes it less likely and is an **inhibitory postsynaptic potential** (IPSP). Both are **graded potentials**, sized in proportion to the input, and they spread fast but decrementally, fading like ripples on a pond.

Whether a neuron fires depends on the sum of thousands of PSPs reaching the axon initial segment, a process called **integration**. In **spatial summation**, PSPs arriving together from different synapses add up (an EPSP and an IPSP cancel); in **temporal summation**, PSPs arriving in quick succession at one synapse add up. If the total reaches the **threshold of excitation**, about -65 mV in Pinel (many texts say -55 mV), an action potential follows.

The **action potential** (AP, spike) is a massive reversal of the membrane potential lasting about 1 millisecond (Figure 5.6). At threshold, voltage-gated Na^+ channels open and Na^+ rushes in, driving the membrane from -70 to about **+50 mV**. The Na^+ channels then close and inactivate, while voltage-gated K^+ channels (the delayed rectifiers) let K^+ out, repolarising the membrane. They close slowly, so the membrane is briefly hyperpolarised before it returns to rest.

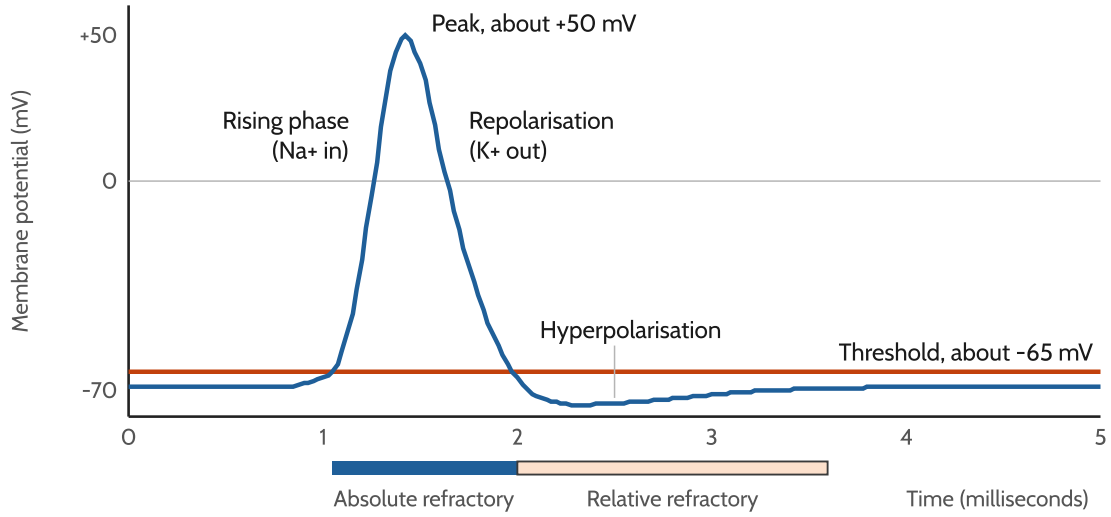


Figure 5.6 An action potential. A depolarisation that reaches the threshold line fires a full spike; the bars show when no stimulus can fire another (absolute) and when only a stronger one can (relative).

Unlike PSPs, action potentials obey the **all-or-none law**: once threshold is reached, the spike fires at full, constant size whatever the stimulus strength; below threshold, nothing fires. Squeeze a gun's trigger hard or gently and the bullet leaves at the same speed. A strong stimulus is signalled instead by firing more often: intensity is coded by frequency, the **rate law**.

EXAM TRAP

Only **action potentials** obey the all-or-none law. EPSPs and IPSPs are graded, and the resting potential is a steady baseline that does not vary with the stimulus. Stronger stimuli give more spikes, not bigger ones; sub-threshold stimuli give no "miniature" spikes; and the law says nothing about neighbouring neurons firing together or one spike per second.

Refractory periods and conduction ●●

For about 1 to 2 milliseconds after a spike begins, no stimulus, however strong, can fire another: the **absolute refractory period**, during which the voltage-gated Na^+ channels are inactivated and cannot reopen until the membrane repolarises. In the **relative refractory period** that follows, K^+ channels are still open and the membrane is hyperpolarised, so only a stronger than normal stimulus works. Two consequences follow. A spike cannot turn back onto the axon it has just left, so it travels one way, soma to terminals: **orthodromic conduction** (backward, antidromic travel is seen mainly under artificial stimulation). And firing is capped at about 1,000 spikes per second, intense stimulation refiring the cell as soon as the absolute period ends.

In an unmyelinated axon the spike is regenerated at each Na^+ channel in turn: **continuous conduction**, which never weakens but is slow, about 1 m per second in thin axons. In a myelinated axon ions can cross only at the nodes of Ranvier, where voltage-gated Na^+ channels are concentrated, so the spike is regenerated node by node and seems to jump: **saltatory conduction** (Latin *saltare*, to leap). The function of myelin is thus to increase conduction speed, up to about 100 m per second in the largest axons (Pinel cites up to 60 m per second for human motor neurons); wider axons are faster too. Multiple sclerosis destroys CNS myelin and slows or blocks conduction (Section 5.11).

EXAM TRAP

The absolute refractory period reflects **inactivated Na^+ channels**, not pump failure, inactivated K^+ channels or Cl^- influx. Myelin speeds conduction; it does not support the neuron, store transmitter or generate spikes. Saltatory conduction needs Na^+ channels clustered at the nodes, not spread evenly along the axon.

How a synapse works ●

A **synapse** is the junction where one neuron passes its message to another across a narrow gap, the **synaptic cleft**, in four steps.

1. **Synthesis and packaging.** Small-molecule transmitters are made and packed into vesicles in the terminal; neuropeptides are made in the soma and carried down the axon on microtubules.
2. **Release.** The arriving action potential opens **voltage-gated Ca^{2+} channels**; Ca^{2+} entry makes vesicles fuse with the membrane and spill their contents into the cleft: **exocytosis**.
3. **Binding.** Transmitter diffuses across the cleft and binds postsynaptic receptors.
4. **Deactivation.** Reuptake or an enzyme ends the message (below).

Most synapses are **axodendritic** (often onto a spine) or **axosomatic**. **Axoaxonic** synapses facilitate or inhibit a single terminal button, and **dendrodendritic** synapses can often transmit

both ways. Most are **directed synapses**, with release site and receptors side by side. At **nondirected synapses**, Pinel's string-of-beads synapses, transmitter is released from **varicosities**, swellings along the axon, and spreads to distant targets; most monoamine neurons work this way. Some neurons are linked by **gap junctions**, protein channels (connexins) joining their cytoplasm to form electrical synapses, faster than chemical ones. And one button often holds a small-molecule transmitter and a neuropeptide together: **coexistence**.

> EXAM TRAP
>
> Release is triggered by **Ca^{2+} influx**, not by K^+ efflux (which repolarises), closure of the pump, or Cl^- influx (which makes IPSPs). Varicosities are **non-directed** synapses that release **monoamines**.

Receptors and deactivation ●

A transmitter acts by binding a **receptor**, a membrane protein shaped for it, and most transmitters have several **receptor subtypes**, so one chemical can send different messages to different places. An **ionotropic receptor** is a ligand-gated ion channel that opens at once (Na^+ in gives an EPSP; Cl^- in or K^+ out an IPSP), so its effect is fast and brief. A **metabotropic receptor** is coupled to a G protein, which opens a nearby channel or triggers a **second messenger** such as cyclic AMP that can alter the cell's metabolism and even its genes; these receptors are more common, and their effects slower, longer lasting and more varied. Small-molecule transmitters bind both kinds, but virtually all neuropeptides bind only metabotropic receptors. Metabotropic receptors on the releasing neuron itself, **autoreceptors**, sense its own transmitter and cut release when levels are high.

Transmitter left in the cleft would jam the line. **Reuptake** by transporters, the commoner route, returns monoamines and amino acids to the button. In **enzymatic degradation** an enzyme breaks the transmitter apart: acetylcholine is deactivated solely by **acetylcholinesterase** in the cleft, and its choline is recycled. Monoamines are degraded by **monoamine oxidase** (MAO-A: serotonin, noradrenaline and dopamine; MAO-B: dopamine and phenylethylamine) and by catechol-O-methyltransferase (COMT).

> EXAM TRAP
>
> Ionotropic receptors bind **small transmitters only**. The transmitter cleared exclusively by an enzyme is **acetylcholine**; dopamine, GABA and glutamate are taken back up. MAO never acts on GABA, glutamate, glycine or acetylcholine.

Amino acids and acetylcholine ●●●

More than 100 transmitters are known, in five families (Table 5.8); only the neuropeptides are large molecules.

Table 5.8 The neurotransmitter families (after Pinel).

Family	Members	Notes
Amino acids	Glutamate, aspartate, glycine, GABA	Fast, directed synapses; GABA is made from glutamate

Family	Members	Notes
Monoamines: catecholamines	Dopamine, noradrenaline (norepinephrine), adrenaline (epinephrine)	Made from tyrosine, via L-dopa
Monoamines: indolamine	Serotonin (5-hydroxytryptamine, 5-HT)	Made from tryptophan
Monoamines: other	Histamine	Made from histidine
Acetylcholine	Acetylcholine (ACh)	A class by itself: choline plus an acetyl group
Unconventional	Nitric oxide, carbon monoxide; endocannabinoids such as anandamide (Sanskrit *ananda*, bliss)	Made on demand; often signal backwards to the presynaptic cell
Neuropeptides	Endorphins, enkephalins, substance P, neuropeptide Y, orexin (hypocretin), oxytocin, vasopressin	Made in the soma; metabotropic receptors only

The main excitatory transmitter of the central nervous system is **glutamate**, acting on three ionotropic receptors, AMPA, kainate and NMDA, and on metabotropic receptors. The **NMDA receptor** is plugged by a magnesium ion (Mg^{2+}): it opens only when glutamate is bound (glycine is a co-agonist) *and* the membrane is depolarised enough, usually by AMPA receptors, to expel the Mg^{2+}. It then admits Ca^{2+}. This coincidence detection underlies long-term potentiation in learning (Section 5.9). After a stroke, glutamate floods out, Ca^{2+} pours in through NMDA receptors and neurons die: **excitotoxicity**.

GABA (gamma-aminobutyric acid) is the main inhibitory transmitter of the brain: an amino acid, not a hormone, neuropeptide or catecholamine. The **GABA-A receptor** is an ionotropic Cl^- channel giving fast IPSPs, with binding sites for benzodiazepines, barbiturates, alcohol and neurosteroids, all of which enhance it (Section 5.4). The **GABA-B receptor** is metabotropic: through a G protein it opens K^+ channels and closes Ca^{2+} channels. **Glycine** is the main inhibitory transmitter of the spinal cord.

At the **neuromuscular junction** the transmitter that makes skeletal muscle contract is **acetylcholine** (ACh), which also serves the autonomic ganglia and parasympathetic endings and, in the brain, learning, memory, attention and REM sleep. Its **nicotinic receptors** (muscle, ganglia) are ionotropic, its **muscarinic receptors** metabotropic. The **nucleus basalis of Meynert** in the basal forebrain supplies the cortex with ACh and is depleted in Alzheimer's disease, hence treatment with cholinesterase inhibitors. Curare blocks nicotinic receptors and botulinum toxin blocks ACh release, both paralysing; in myasthenia gravis, antibodies attack muscle nicotinic receptors, causing weakness that worsens with use.

> EXAM TRAP
>
> Inhibitory: **GABA** (brain) and glycine (spinal cord). Excitatory: **glutamate**. GABA-A is the **ionotropic, ligand-gated Cl^- channel** (fast IPSPs), not a voltage-gated or excitatory channel; GABA-B is metabotropic. NMDA is ionotropic, needs depolarisation and admits Ca^{2+}, not Cl^-.

Monoamine systems and endogenous opioids ●●●

Monoamine neurons are few, clustered mostly in the brainstem, but their branching axons and varicosities reach almost the whole brain, tuning broad states such as arousal and mood.

Table 5.9 lists them with the other widely projecting systems.

Table 5.9 Where the modulating transmitters come from.

Transmitter	Main cell bodies	Main roles
Dopamine	Substantia nigra (nigrostriatal pathway); ventral tegmental area	Movement, reward, attention (Section 5.4)
Noradrenaline	Locus coeruleus (pons)	Arousal, vigilance; also the sympathetic postganglionic transmitter (heart, blood vessels)
Adrenaline	Adrenal medulla (as a hormone); a few medullary neurons	Fight or flight
Serotonin	Raphe nuclei, along the brainstem midline	Mood, appetite, sleep, impulse control
Histamine	Tuberomammillary nucleus (hypothalamus)	Wakefulness; older antihistamines sedate
Acetylcholine	Nucleus basalis of Meynert (basal forebrain); pons	Attention, memory; REM sleep
Orexin (hypocretin)	Lateral hypothalamus	Wakefulness (Section 5.8)
Oxytocin, vasopressin	Hypothalamus (supraoptic, paraventricular nuclei)	Released from the posterior pituitary (Section 5.7)

Low serotonin activity, shown by low cerebrospinal fluid levels of its breakdown product 5-hydroxyindoleacetic acid (**5-HIAA**), is linked to impulsive aggression and violent suicide.

> **EXAM TRAP: IS ADRENALINE A NEUROTRANSMITTER?**
>
> The IHBAS 2020 key marks **adrenaline** as "not a neurotransmitter" beside dopamine, serotonin and noradrenaline. Strictly it is one: Pinel lists it among the catecholamine transmitters, released by adrenergic neurons. It is the odd one out only because it acts mainly as a hormone of the adrenal medulla. Mark adrenaline, knowing why.

Why should the brain have receptors for morphine, a drug from the opium poppy? In 1973 **Candace Pert** (1946 to 2013) and **Solomon Snyder** (born 1938) demonstrated opiate receptors in the brain, plentiful in the periaqueductal grey, a pain-control area. A lock implies a natural key: in 1975 **John Hughes** and **Hans Kosterlitz** (1903 to 1996) isolated the **enkephalins** ("in the head"), and the **endorphins** ("endogenous morphine") followed. These **endogenous opioids**, the body's own morphine, are neuropeptides acting on metabotropic receptors; they control pain and give the "feel-good" glow often credited for the runner's high. Among other neuropeptides, substance P carries pain signals and neuropeptide Y stimulates eating.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Cajal's _____ doctrine replaced Golgi's reticular theory.
2. In the central nervous system myelin is made by _____; in the peripheral nervous system, by Schwann cells.
3. The sodium-potassium pump moves _____ Na^+ ions out for every two K^+ ions in.
4. At its peak the action potential reaches about _____ mV.
5. No stimulus can fire a neuron during the _____ refractory period, when its Na^+ channels are inactivated.
6. Transmitter release is triggered by the influx of _____ ions.
7. Acetylcholine is broken down in the cleft by _____.
8. Serotonin neurons lie in the _____ nuclei; noradrenaline neurons in the locus coeruleus.

Answers: (1) neuron, (2) oligodendrocytes, (3) three, (4) +50, (5) absolute, (6) calcium (Ca^{2+}), (7) acetylcholinesterase, (8) raphe.

5.4 Drugs, Reward and Psychopharmacology

MEDIUM YIELD · 40 QS

Every drug in this section works through the synaptic machinery of Section 5.3: filter coffee, a peg of whisky and a lithium tablet differ only in which synapses they touch. The section runs from the principles of drug action and addiction, through recreational drugs and the reward circuit they hijack, to the drugs prescribed for schizophrenia, depression, mania and anxiety.

Psychoactive drugs and their route to the brain

A **psychoactive drug** is a substance that alters mood, cognition, perception or behaviour by acting on the nervous system; the term does not mean a drug-induced hallucination, a particularly active psychopath or the psychology of physical activity. Drugs are taken in four ways. **Oral ingestion** is easy and safe but unpredictable, because food in the stomach alters absorption. **Injection** (subcutaneous, intramuscular or intravenous) is fast, strong and predictable, but an intravenous overdose cannot be recalled. **Inhalation** through the lung capillaries (tobacco, cannabis, anaesthetics) makes the dose hard to regulate. **Absorption through mucous membranes** of the nose, mouth or rectum is how snorted cocaine enters.

From the blood a drug must cross the blood-brain barrier (Section 5.3), which small, fat-soluble molecules pass most easily. Most drugs are then deactivated by liver enzymes, a process called **drug metabolism**, whose products usually can no longer penetrate the barrier.

Agonists, antagonists and synaptic mechanisms

Drugs that facilitate the effects of a particular neurotransmitter are its **agonists**; drugs that reduce them are its **antagonists**. A drug can act at any step in a transmitter's life. Agonists may increase synthesis (L-dopa, the precursor of dopamine), increase release (amphetamine), block reuptake (cocaine, SSRIs), block degradation (MAO inhibitors) or acti-

vate receptors (nicotine, morphine). Antagonists may block synthesis, empty the vesicles so the transmitter leaks out and is destroyed (reserpine), block release, or occupy receptors without activating them, as receptor blockers do (chlorpromazine, naloxone). Presynaptic autoreceptors normally slow release, so a drug that blocks them increases release and one that activates them reduces it.

An **allosteric modulator** binds a site on the receptor other than the transmitter's own and changes the receptor's response to the transmitter; benzodiazepines are positive allosteric modulators of the GABA-A receptor. A **partial agonist** activates a receptor only partly, raising activity when the transmitter is scarce but lowering it when the transmitter is plentiful; buspirone at the serotonin 5-HT1A receptor is an example.

Tolerance, withdrawal and addiction

Drug tolerance is decreased sensitivity to a drug that develops through exposure to it: a given dose has less effect, or more drug is needed for the same effect. The dose-response curve shifts to the right (Figure 5.7). **Metabolic tolerance** comes from changes that reduce the amount of drug reaching its sites of action (a heavy drinker's liver clears alcohol faster); **functional tolerance** comes from changes that reduce the reactivity of those sites, such as fewer receptors, and accounts for most tolerance to psychoactive drugs. Exposure to one drug also brings tolerance to others that act by the same mechanism, called **cross tolerance**: a heavy drinker needs more of a benzodiazepine or barbiturate, because all three enhance the GABA-A receptor. Tolerance may develop to some effects but not others, and some effects may instead grow with repeated use: **drug sensitisation** (amphetamine's motor effects in rats).

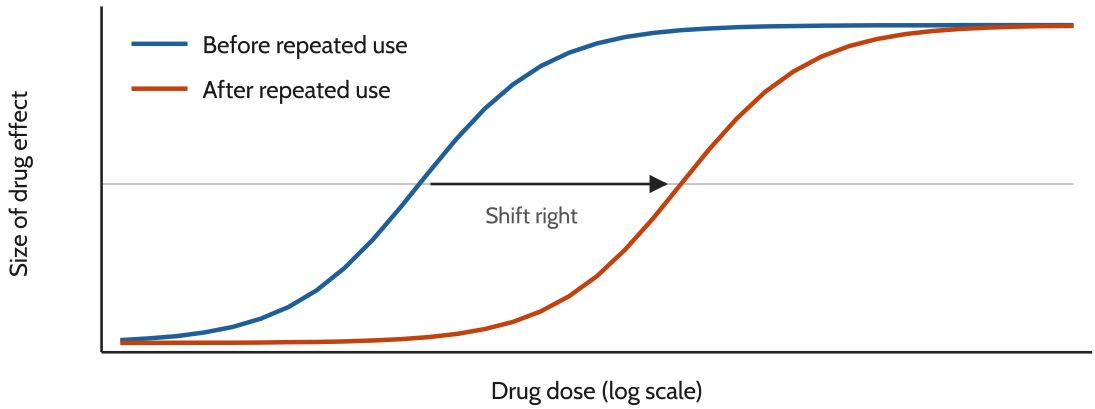


Figure 5.7 Tolerance as a rightward shift of the dose-response curve. After repeated use the same dose has less effect, and a larger dose is needed for the same effect.

When a drug that has been in the body for days is suddenly removed, a **withdrawal syndrome** follows whose effects are almost always opposite to the drug's: sleeping pills leave insomnia, alcohol leaves tremor, seizures and delirium tremens. The compensatory neural changes that produced tolerance are now unopposed. Suffering withdrawal on stopping defines **physical dependence**; **psychological dependence** is a compulsive craving for the drug's effects without bodily withdrawal. Neither defines **drug addiction**: habitual use that continues despite harm to health and social life and despite repeated efforts to stop.

KEY STUDY: SIEGEL'S HEROIN OVERDOSE RATS (1982)

Shepard Siegel treated each dose as a Pavlovian conditioning trial: the usual room and paraphernalia become conditioned stimuli that elicit compensatory responses opposite to the drug's effect, so tolerance is strongest where the drug is usually taken (**conditioned drug tolerance**). Siegel and colleagues (1982) made rats tolerant to heroin in one environment, then gave a large dose there or in a novel one. Many more died in the novel environment, where no compensatory responses opposed the drug, which may explain human overdoses in unfamiliar places. Criticism: some drug cues elicit drug-like rather than opposite responses.

EXAM TRAP

Cross tolerance means exposure to one drug reduces the effect of another acting by the same mechanism; it is not "no addiction to any drug". Withdrawal is usually the *opposite* of the drug's effect, and physical dependence is not addiction: detoxified users still relapse.

Sedative-hypnotics: alcohol, benzodiazepines, barbiturates ●●

Sedatives calm and **hypnotics** induce sleep; both are broad classes covering alcohol, barbiturates, benzodiazepines and newer sleeping pills, not one chemical family. These drugs depress the central nervous system mainly by enhancing GABA, the brain's main inhibitory transmitter, at the **GABA-A receptor**, a chloride (Cl-) channel whose opening hyperpolarises the neuron.

Alcohol is a positive modulator, functionally an agonist, at the GABA-A receptor, and also blocks the NMDA glutamate receptor, so it raises inhibition and lowers excitation. Drinking in pregnancy causes fetal alcohol syndrome (Chapter 6), and the thiamine deficiency of chronic drinkers causes Wernicke-Korsakoff syndrome (Section 5.9).

The **benzodiazepines** form one chemical class: diazepam (Valium), alprazolam (Xanax), flunitrazepam (Rohypnol), triazolam (Halcion) and chlordiazepoxide. They are used as anxiolytics, hypnotics and anticonvulsants. The older **barbiturates** (phenobarbital) bind another site on the same receptor. As Figure 5.8 shows, a benzodiazepine increases the *frequency* of Cl- channel opening and works only when GABA is present, so its effect has a ceiling; a barbiturate increases the *duration* of each opening and at high doses opens the channel by itself. Hence barbiturates, and benzodiazepines taken with alcohol, are dangerous in overdose, whereas a benzodiazepine overdose alone is rarely fatal and is reversed by the antagonist **flumazenil**. Withdrawal from any of these drugs is rebound excitation (anxiety, insomnia, tremor, seizures) and can kill.

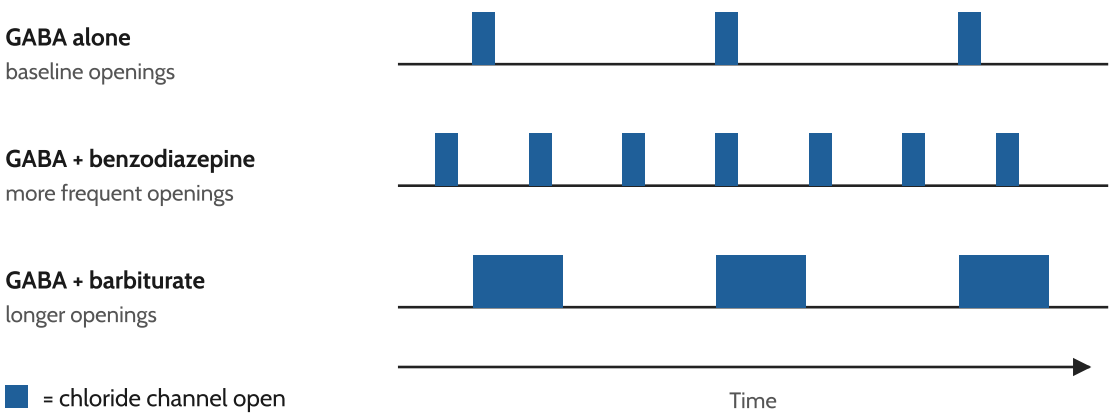


Figure 5.8 How the two drug families enhance the GABA-A receptor. Benzodiazepines make the channel open more often; barbiturates keep each opening longer.

EXAM TRAP

Frequency goes with benzodiazepines and **duration** with barbiturates, never the reverse. Both act on GABA-A, not GABA-B or NMDA receptors, and the receptor benzodiazepines modulate binds GABA, not glutamate, acetylcholine or glycine (the spinal cord's inhibitory transmitter). Valium, Xanax, Rohypnol and Halcion are all benzodiazepines; "sedatives" and "hypnotics" are too broad.

Opioids, stimulants, cannabis and hallucinogens ●●

Opium is the dried sap of the opium poppy's seed pods; its natural alkaloids are morphine, codeine, thebaine and papaverine. **Heroin** (diacetylmorphine) is not one of them: it is made by adding two acetyl groups to morphine, a *semisynthetic* opioid that enters the brain faster. Methadone and fentanyl are fully synthetic. Opioids act mainly at **mu opioid receptors**, the receptors for the body's endorphins (Section 5.3): they relieve pain, cough and diarrhoea, bring euphoria, and in overdose suppress breathing. **Naloxone** (short acting, the emergency reversal of overdose) and **naltrexone** (long acting, oral) are opioid antagonists. Opioid withdrawal mirrors the drug: dilated pupils, runny nose, sweating, gooseflesh and diarrhoea; miserable but rarely fatal. Table 5.10 covers the other classes.

Table 5.10 How stimulants, cannabis and hallucinogens act.

Drug	Main action	Note
Cocaine	Blocks reuptake of dopamine and noradrenaline	Binges; cocaine psychosis
Amphetamine, methamphetamine	Increase release of dopamine and noradrenaline	Psychosis like paranoid schizophrenia
Nicotine	Agonist at nicotinic acetylcholine receptors	Addictive agent of tobacco
Caffeine	Antagonist at adenosine receptors	Blocks adenosine's sleep signal
Cannabis (THC)	Agonist at CB1 receptors	Little risk of fatal overdose
LSD, psilocybin, mescaline	Agonists at serotonin 5-HT2A receptors	Classic hallucinogens
PCP, ketamine	NMDA glutamate receptor antagonists	Dissociative; mimic schizophrenia

The active ingredient of cannabis, delta-9-tetrahydrocannabinol (THC), acts at **CB1 receptors**, among the most common metabotropic receptors in the brain. They are densest in the hippocampus (memory impairment), basal ganglia and cerebellum (motor effects) and cortex, and present in reward circuits, but sparse in the brainstem centres that control breathing, so fatal overdose is unlikely. **CB2 receptors** lie mainly on immune cells. The brain makes its own ligands, the **endocannabinoids**; the first found, **anandamide**, is named from the Sanskrit *ananda*, bliss.

> EXAM TRAP
>
> Asked which is **not** a natural opium alkaloid, choose **heroin**: it is semisynthetic, made from morphine. Morphine, codeine, thebaine and papaverine all occur in the poppy. Cocaine blocks reuptake, whereas amphetamine increases release.

Reward circuits and theories of addiction ●●

> KEY STUDY: OLDS AND MILNER'S 'PLEASURE CENTRE' (1954)
>
> In 1954, at McGill University, **James Olds** and **Peter Milner** let rats press a lever that delivered brief electrical stimulation through an electrode in their own brains. With electrodes in the **septal area** and along the **medial forebrain bundle**, rats pressed thousands of times an hour, ignoring food, until exhausted. This **intracranial self-stimulation** seemed to reveal sites that mediate the pleasure of food, water and sex: 'pleasure centres'. Later work found no single centre; self-stimulation depends largely on dopamine axons running in the medial forebrain bundle.

Those axons form the **mesolimbic pathway**, from the ventral tegmental area (VTA) of the midbrain to the **nucleus accumbens** in the ventral striatum and other limbic sites. It is the core reward pathway of addiction: natural rewards and almost every addictive drug raise dopamine in the nucleus accumbens, directly (cocaine, amphetamine) or indirectly (opioids, alcohol, nicotine), and dopamine antagonists reduce self-stimulation. The reward transmitter is dopamine, not serotonin, noradrenaline or acetylcholine. With chronic use, control of drug taking shifts to the dorsal striatum (habit), and weakened **prefrontal control** makes stopping harder.

Physical-dependence theories said addicts take drugs to escape withdrawal, but detoxified users relapse months later, and cocaine causes little withdrawal distress. **Positive-incentive theories** replaced them: people take drugs for their anticipated pleasure. Yet addicts often crave a drug they no longer enjoy. The **incentive-sensitisation theory** of **Terry Robinson** and **Kent Berridge** (1993) explains this: in addiction-prone people repeated use sensitises the drug's incentive value ('wanting') while tolerance develops to its pleasure ('liking'). Mesolimbic dopamine signals wanting more than liking.

EXAM TRAP

The reward pathway is **mesolimbic** (VTA to nucleus accumbens), not nigrostriatal (movement), tuberoinfundibular (prolactin) or the raphe serotonergic pathway. Olds and Milner's sites were the **septal area and medial forebrain bundle**, not the cerebellar tonsils, the lateral geniculate nucleus (a visual relay) or the globus pallidus (a motor nucleus). The cerebellar vermis and vestibular nuclei serve balance, not addiction.

Dopamine pathways, schizophrenia and antipsychotics ●●●

Four dopamine pathways explain both what antipsychotic drugs fix and what they spoil (Table 5.11).

Table 5.11 The four dopamine pathways.

Pathway	Origin to target	Normal role	Disease or drug effect
Mesolimbic	VTA to nucleus accumbens and limbic structures	Reward, motivation	Overactive in positive symptoms; D2 blockade treats them
Mesocortical	VTA to prefrontal cortex	Cognition, motivation	Underactive in negative and cognitive symptoms
Nigrostriatal	Substantia nigra to caudate and putamen	Movement	Degenerates in Parkinson's disease; blockade gives extrapyramidal effects
Tuberoinfundibular	Hypothalamic arcuate nucleus to median eminence and pituitary	Inhibits prolactin	Blockade gives hyperprolactinaemia

The first antipsychotic, **chlorpromazine**, was developed as an antihistamine; **Jean Delay** and **Pierre Deniker** reported its calming effect in psychotic patients in 1952. **Arvid Carlsson** (Nobel Prize 2000) showed that dopamine is a transmitter, and in 1963 he and **Margit Lindqvist** found that chlorpromazine raises the levels of dopamine's metabolites and argued that it blocks dopamine receptors. Since amphetamine and L-dopa, which raise dopamine activity, can produce psychotic symptoms, the **dopamine hypothesis of schizophrenia** followed. Its key early support came in the mid-1970s, when **Philip Seeman**, **Solomon Snyder** and colleagues showed that the clinical potency of typical antipsychotics correlates with their affinity for the **D2 receptor**. In the revised form, **mesolimbic hyperactivity** produces positive symptoms (delusions, hallucinations) and **mesocortical hypoactivity** negative and cognitive symptoms, which L-dopa does not improve. The **glutamate hypothesis** adds NMDA receptor hypofunction (PCP and ketamine mimic both symptom types); a serotonin view rests on 5-HT2A hallucinogens and atypical drugs. Identical twin concordance is only about 50 percent, so genes are not everything.

EXAM TRAP

The classic hypothesis blames **excess mesolimbic dopamine (D2)**, not decreased serotonin, decreased cerebellar GABA, excess acetylcholine at the neuromuscular junction or glutamate excess (the glutamate view proposes NMDA *hypo*function). Its original support was the potency-D2 correlation, not antipsychotics raising serotonin, 100 percent twin concordance or L-dopa helping negative symptoms.

Antipsychotics (neuroleptics) are the front-line treatment for schizophrenia; CBT, stress management and social skills training are adjuncts (Chapter 7), and neuroleptics are not mainly for depression or anxiety. **Typical antipsychotics** (first generation: chlorpromazine, haloperidol) block D2 receptors; chlorpromazine also blocks D1, and blockade of muscarinic M1 and histamine receptors adds dry mouth and sedation. Because they block D2 in all four pathways, they also cause movement disorders, raise prolactin and may worsen negative symptoms. **Atypical antipsychotics** (second generation: clozapine, risperidone, olanzapine, quetiapine) block 5-HT2A as well as D2: fewer extrapyramidal effects, more weight gain and metabolic syndrome. **Clozapine**, kept for treatment-resistant cases, can cause agranulocytosis (loss of white blood cells), so blood counts are monitored.

The movement effects of nigrostriatal blockade are the **extrapyramidal side effects**, which appear in a fixed order (Table 5.12). **Akathisia** is subjective inner restlessness with an urge to pace; **tardive dyskinesia** comes last and may be irreversible; **neuroleptic malignant syndrome** is a rare emergency. Drug-induced conditions such as akathisia are **secondary movement disorders**, with an identifiable external cause; Huntington's chorea (inherited) and Parkinson's disease are primary.

Table 5.12 Antipsychotic movement side effects in order of onset.

Side effect	Onset	Features
Acute dystonia	Hours to days	Sustained spasms: torticollis, oculogyric crisis
Akathisia	Days to weeks	Inner restlessness, compelling urge to pace
Drug-induced parkinsonism	Weeks	Tremor, rigidity, bradykinesia
Tardive dyskinesia	Months to years	Involuntary lip-smacking, tongue and facial movements, grimacing
Neuroleptic malignant syndrome	Any time, often early	Fever, rigidity, autonomic instability, clouded consciousness

EXAM TRAP

Extrapyramidal effects come from **nigrostriatal** D2 blockade, not mesolimbic (the therapeutic action), mesocortical or tuberoinfundibular blockade. Restlessness soon after starting is **akathisia**; lip-smacking years later is **tardive dyskinesia**, caused by neuroleptics, not antidepressants, anxiolytics or barbiturates.

Antidepressants, mood stabilisers and anxiolytics ●●

The **monoamine hypothesis of depression** holds that depression reflects underactivity at serotonergic and noradrenergic synapses, because the antidepressants in Table 5.13 are all monoamine agonists.

Table 5.13 Antidepressant classes.

Class	Examples	Action	Note
MAO inhibitors	Iproniazid, phenelzine	Block monoamine oxidase	Tyramine 'cheese reaction': hypertensive crisis
Tricyclics	Imipramine, amitriptyline	Block noradrenaline and serotonin reuptake	Anticholinergic effects; dangerous in overdose

Class	Examples	Action	Note
SSRIs (selective serotonin reuptake inhibitors)	Fluoxetine, sertraline	Block serotonin reuptake	Fewer side effects
SNRIs (serotonin-noradrenaline reuptake inhibitors)	Venlafaxine	Block serotonin and noradrenaline reuptake	Some texts use 'SNRI' for selective noradrenaline reuptake inhibitors (reboxetine)
Atypical	Bupropion	Blocks noradrenaline and dopamine reuptake	Also aids quitting smoking

Reuptake blockade raises synaptic serotonin within hours, yet mood lifts only after 2 to 4 weeks. At first the extra serotonin stimulates **5-HT1A autoreceptors**, which damp the firing of serotonin neurons; over weeks these desensitise, firing recovers, and slower changes build up: more brain-derived neurotrophic factor (BDNF) and more hippocampal neurogenesis. The delay is not due to liver enzyme induction, slow entry into the brain or metabolite build-up. An antidepressant such as fluoxetine is no treatment for mania and can trigger it in bipolar patients.

For bipolar disorder the gold standard for treatment and prophylaxis is **lithium**, introduced by the Australian psychiatrist **John Cade** in 1949; it also lowers suicide risk. It has a **narrow therapeutic index**: the useful blood level is about 0.6 to 1.2 mEq/L, and toxicity (coarse tremor, ataxia, confusion) begins just above it and can overlap it. Lithium is cleared by the kidneys, not the liver, so dehydration and renal impairment raise levels, and blood levels, thyroid and kidney function are monitored. Long-term effects include hypothyroidism and nephrogenic diabetes insipidus (thirst, copious urine). Alternatives are the anticonvulsants valproate (divalproex), carbamazepine and lamotrigine. In acute mania, lithium, valproate and atypical antipsychotics such as olanzapine and risperidone are all first line, with haloperidol and the benzodiazepine lorazepam as adjuncts for agitation.

Among anxiolytics, benzodiazepines act within minutes but bring dependence and cross tolerance with alcohol. **Buspirone**, a 5-HT1A partial agonist, is neither sedating nor habit forming but takes weeks to act. **Beta blockers** such as propranolol quiet the bodily signs of anxiety (tremor, palpitations), as in stage fright.

Three drugs help in alcohol dependence. **Disulfiram** (Antabuse) inhibits **aldehyde dehydrogenase**, so acetaldehyde builds up and a drink brings flushing, nausea, headache and palpitations. **Naltrexone**, an opioid antagonist, dampens alcohol's reward and craving, which run partly through endorphin release. **Acamprosate** modulates glutamate (NMDA) transmission and reduces relapse.

> EXAM TRAP
>
> Disulfiram works by **aldehyde dehydrogenase inhibition**, not GABA-A partial agonism, NMDA blockade (closer to acamprosate) or reduced accumbens dopamine (closer to naltrexone). **Fluoxetine** is the odd one out among acute-mania treatments. Lithium toxicity (coarse tremor, ataxia, confusion) overlaps the therapeutic range; constipation and acute liver failure are not its problems, and renal impairment demands monitoring. Cade's lithium report dates from 1949; a 1954 date in some textbooks is an error.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. A heavy drinker who needs more diazepam for the same effect shows _____ tolerance.
2. Benzodiazepines increase the _____ of Cl- channel opening; barbiturates increase its _____.
3. Heroin is a _____ opioid made from morphine.
4. THC acts at _____ receptors, densest in the hippocampus.
5. The reward pathway runs from the _____ to the nucleus accumbens.
6. Extrapyramidal side effects follow D2 blockade in the _____ pathway.
7. The last extrapyramidal effect to appear, often irreversible, is _____.
8. Disulfiram inhibits the enzyme _____.

Answers: (1) cross, (2) frequency; duration, (3) semisynthetic, (4) CB1, (5) ventral tegmental area, (6) nigrostriatal, (7) tardive dyskinesia, (8) aldehyde dehydrogenase.

5.5 Research Methods, Behaviour Genetics and the Developing Brain

MEDIUM YIELD · 25 QS

Every fact in this chapter came from a method, and each method sees the brain differently: its shape, its activity, or the timing of that activity. This section surveys those methods, then asks how genes are inherited and studied, which genetic errors derail development, and how the brain builds and keeps rewiring itself.

Imaging the living brain ●●

Ordinary X-rays show little of the soft brain, so **contrast X-ray** techniques inject a substance that absorbs X-rays differently: in **cerebral angiography** a radio-opaque dye in a cerebral artery outlines the blood vessels, revealing vascular damage and, by displaced vessels, tumours. **Computed tomography** (CT), introduced in the early 1970s by **Godfrey Hounsfield**, passes X-rays through the head from many angles and computes horizontal slices. It is fast and shows fresh blood, so it is the first scan after acute head injury. **Magnetic resonance imaging** (MRI) uses a strong magnetic field and radio waves: hydrogen atoms then emit signals that differ between structures. With the highest **spatial resolution** (precision about where), MRI gives the most detailed view of brain structure. **Diffusion tensor imaging** (DTI), an MRI method, follows water diffusing along axons to map the white matter tracts.

CT and plain MRI are structural; **functional imaging** shows activity. **Positron emission tomography** (PET) injects a radioactive tracer such as 2-deoxyglucose (FDG in human PET), which active neurons absorb like glucose but cannot metabolise, so the scan maps metabolic activity. **Single-photon emission computed tomography** (SPECT) maps blood flow with a gamma-emitting tracer, more cheaply but less sharply. **Functional MRI** (fMRI) needs no injection: it detects the **BOLD signal** (blood-oxygen-level-dependent; **Seiji Ogawa**, 1990), which rises as active regions draw oxygenated blood. Its spatial resolution is good but its **temporal resolution** (precision about when) only moderate, because blood flow takes seconds to respond. **Functional near-infrared spectroscopy** (fNIRS) measures cortical oxygenation with

light shone through the scalp; it is portable and suits infants. Table 5.14 compares the methods.

Table 5.14 Where and when: the main methods compared (figures approximate).

Method	Spatial resolution	Temporal resolution
CT (structural)	Good	None (still image)
MRI (structural)	Best (under 1 mm)	None (still image)
PET (functional)	Moderate (4 to 5 mm)	Poor (about a minute or more)
SPECT (functional)	Poorer (about 1 cm)	Poor (minutes)
fMRI (functional)	Good (1 to 3 mm)	Moderate (seconds)
EEG	Poor (centimetres)	Best (milliseconds)
MEG	Better than EEG	Best (milliseconds)

A working brain does many things at once, so the **paired-image subtraction technique** images two tasks that differ in a single process (reading nouns aloud versus saying a related verb for each) and subtracts one image from the other. Subtraction revealed regions more active in the resting baseline than during tasks: the **default mode network** (DMN), after the "default mode" of brain function described by **Marcus Raichle** and colleagues in 2001. Its hubs are the medial prefrontal cortex, the posterior cingulate cortex and precuneus, and the lateral parietal cortex. It is most active at rest, during mind-wandering and self-referential thought, deactivates during externally focused, goal-directed tasks (moving, seeing, calculating), and is altered in depression, Alzheimer's disease and schizophrenia.

> EXAM TRAP
>
> The functional techniques are **PET and SPECT** (and fMRI), not CT or plain MRI, so "all of the above" is wrong. For brain structure after head injury choose **MRI**; CT is the quick emergency scan for bleeding. EEG, PET and fMRI record activity, not structure.

Recording electrical activity ●●

Imaging shows where; electrical recording shows when. The **electroencephalogram** (EEG), first reported in humans by **Hans Berger** in 1929, records through scalp electrodes the summed electrical activity of cortical neurons. The brain's counterpart of the electrocardiogram (ECG), it has the highest temporal resolution, in milliseconds, but poor spatial resolution, and it is the standard test for epilepsy and sleep (wave bands in Section 5.8). **Magnetoencephalography** (MEG) records the magnetic fields of the same currents: as fast, better at localising, far costlier. In animals, a microelectrode records one neuron's action potentials (**single-unit recording**) or a larger electrode the firing of many neighbours (**multiple-unit recording**).

The response to a single click is lost in the background EEG, so the EEG after hundreds of clicks is averaged (**signal averaging**): random activity cancels and the time-locked **event-related potential** (ERP) remains, named by polarity (P or N) and latency in milliseconds. In the **oddball paradigm**, a rare target among frequent standard stimuli evokes a large positive wave about 300 ms later, the **P300**, first described by **Samuel Sutton** and colleagues in 1965. It

reflects attention allocation and stimulus categorisation, and it is smaller and later in schizophrenia and Alzheimer's disease (Table 5.15).

Table 5.15 Event-related potentials to know.

Component	Timing	Reflects	Note
P50	50 ms, positive	Sensory gating: the response to a second click is normally suppressed	Gating deficient in schizophrenia
N100	100 ms, negative	Early sensory response	Larger when attended
P300	About 300 ms, positive	Attention allocation, categorisation (oddball)	Smaller, delayed in schizophrenia and Alzheimer's
Bereitschaftspotential (readiness potential)	Slow negative rise about 1 s before a voluntary movement	Preparation to move (**Hans Kornhuber** and **Lüder Deecke**, 1965)	Precedes the act; not the motor command

EXAM TRAP

Direct electrical recording "like the ECG" is **EEG**; MEG records magnetic fields, fMRI and PET blood and metabolism. No ERP marks REM sleep onset (Section 5.8) or issues motor commands (Section 5.6); the first visual cortical response is an early component, not the P300.

The **electromyogram** (EMG) records muscle tension and the **electrooculogram** (EOG) eye movements. **Skin conductance** (electrodermal activity, the **galvanic skin response**) rises as sympathetic arousal activates the sweat glands. Heart rate and blood pressure are other autonomic indices; the **polygraph** records several at once and, as a "lie detector", detects arousal, not lies.

Lesions, stimulation and genetic tools

Invasive methods, used mostly in animals, begin with **stereotaxic surgery**: a **stereotaxic atlas** gives each structure's coordinates, and a **stereotaxic instrument** holds the head and guides an electrode or cannula to them. A **lesion** destroys tissue to reveal its function by what is lost: **aspiration** sucks away cortex, **radio-frequency** current heats small subcortical areas, **knife cuts** sever tracts, and **cryogenic blockade** (cooling) makes a reversible lesion. Neurotoxins are selective: **kainic acid** kills cell bodies but spares passing axons, and **6-hydroxydopamine** destroys dopamine and noradrenaline neurons. **Bilateral** lesions usually have far larger effects than **unilateral** ones.

Conversely, **electrical stimulation** or **chemical stimulation** (a microinjected transmitter or agonist) can evoke behaviour. **Deep brain stimulation** (DBS) applies continuous current through implanted electrodes, chiefly in Parkinson's disease. **Transcranial magnetic stimulation** (TMS) induces current in the cortex from a coil on the scalp, a non-invasive "virtual lesion"; repeated TMS treats depression. **Optogenetics** inserts genes for light-sensitive ion channels into one type of neuron, so light switches those neurons on or off. **Microdialysis** samples extracellular fluid in behaving animals to measure transmitter release. In humans, neuropsychological tests (Chapter 2) also map damage.

Genetic methods alter genes. A **gene knockout** mouse lacks a working gene; a **knock-in** replaces or adds one; a **transgenic** animal carries another species' gene, such as a human disease gene. **CRISPR-Cas9**, developed by **Jennifer Doudna** and **Emmanuelle Charpentier** in 2012 (Nobel Prize in Chemistry, 2020), uses a guide RNA to lead the Cas9 protein to a chosen DNA sequence, where it edits the gene or switches it on or off. Other tools do not edit: the **polymerase chain reaction** (PCR) of **Kary Mullis** copies DNA millions of times, **gel electrophoresis** separates DNA or proteins by size, and **Western blotting** detects a particular protein.

Mendel, chromosomes and epigenetics ●

Gregor Mendel (1822 to 1884), a monk who crossed pea plants and published in 1866, found that each trait depends on a pair of inherited factors, now called **genes**, one from each parent. The alternative forms of a gene are **alleles**: two alike make an organism **homozygous** (DD or dd), two different **heterozygous** (Dd). The **genotype** is the alleles carried, the **phenotype** the trait shown. A **dominant** allele is expressed even in a heterozygote; a **recessive** trait appears only in the homozygote, as the **Punnett square** in Figure 5.9 shows. Blending occurs only in **incomplete dominance** (red and white snapdragons give pink) and **codominance** (blood group AB shows both), so with a stated dominant allele "blue-green" is wrong.

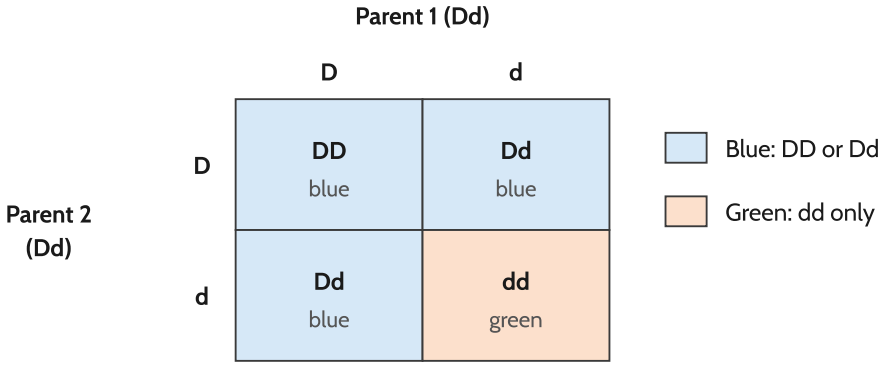


Figure 5.9 Two Dd parents, with blue (D) dominant over green (d): genotypes 1 DD : 2 Dd : 1 dd, phenotypes 3 blue : 1 green. Every Dd child has blue eyes. (The alleles are hypothetical.)

Genes lie on **chromosomes**, strands of **DNA** (deoxyribonucleic acid) whose base sequence codes for proteins. Human cells have 23 pairs: 22 pairs of **autosomes** and the **sex chromosomes**, XX in females and XY in males. **Meiosis** makes eggs and sperm with 23 single chromosomes, and **crossing over** (paired chromosomes swapping segments) makes each one unique. A **mutation** is an accidental change in a gene. X-linked recessive **sex-linked traits**, such as red-green colour blindness, are commoner in males, who have no second X to mask them. **Epigenetics** studies changes in gene expression without any change in the DNA sequence: **DNA methylation** usually silences genes, and **histone modification** loosens or tightens the packaging of DNA, making genes more or less readable. Experience, such as early stress or maternal care, alters these marks.

Separating heredity and environment ●●

Family studies ask whether risk rises with closeness of kinship, but relatives share homes as well as genes. **Twin studies** compare **monozygotic twins** (MZ), from one fertilised egg that

splits, who have the same heredity and share virtually all their genes, with **dizygotic twins** (DZ), who share about 50 percent like other siblings. The **concordance rate** is the percentage of pairs in which, when one twin has a disorder, the other has it too: for schizophrenia, about 48 percent in MZ and 17 percent in DZ twins in the pooled data of **Irving Gottesman** (1991). **Adoption studies** ask whether adoptees resemble their biological parents (genes) or adoptive parents (environment).

KEY STUDY: THE MINNESOTA STUDY OF TWINS REARED APART

From 1979, **Thomas Bouchard** and colleagues (main report in *Science*, 1990) gave adult MZ and DZ twins separated in infancy about 50 hours of intelligence and personality tests. MZ twins reared apart were about as alike as those reared together, and more alike than DZ twins, on almost every measure. Critics note that adoptive homes vary less than homes in general.

EXAM TRAP

Twins reared apart combine twin and adoption logic (Pinel even calls the Minnesota study an adoption study), but the keyed answer is **twin studies**, because comparing MZ with DZ pairs defines the design. Choose adoption studies only if twin studies is not offered.

Linkage studies trace a DNA marker inherited along with a disorder through affected families, locating the gene nearby. **Genome-wide association studies** (GWAS) compare hundreds of thousands of variants across the genome in thousands of patients and controls, usually finding many genes of tiny effect. **Heritability** is the proportion of variation in a trait, in a given population and environment, due to genetic differences; it describes populations, not individuals, and changes when environments change.

Diagnoses are too broad to map onto genes, so Gottesman and **Todd Gould** (2003) popularised the **endophenotype**: a measurable, heritable intermediate trait between genes and symptoms, through which genetic risk produces a disorder. P50 gating, P300 and smooth-pursuit eye-tracking deficits in schizophrenia are examples, also found in unaffected relatives. By contrast, the **phenotype** is the observable symptoms, the **genotype** the genes, and **molecular genetics** the study of DNA itself. The diathesis-stress model is taught in Chapter 8.

Single-gene and chromosomal disorders ●●●

Phenylketonuria (PKU), discovered in 1934 by the Norwegian **Ivar Asbjørn Følling**, is an autosomal recessive lack of **phenylalanine hydroxylase**, the enzyme that converts phenylalanine into **tyrosine**. Phenylalanine accumulates and damages the developing brain, but newborn screening and a low-phenylalanine diet prevent the intellectual disability. In a **trinucleotide repeat** disorder a three-base sequence is repeated too often; in **genomic imprinting** a gene is expressed only from one parent's copy, so one deletion causes different syndromes depending on its parent of origin (Table 5.16).

Table 5.16 Single-gene and chromosomal disorders.

Disorder	Cause	Inheritance	Key features
Phenylketonuria	No phenylalanine hydroxylase, so no tyrosine from phenylalanine	Autosomal recessive	Intellectual disability unless diet from birth

Disorder	Cause	Inheritance	Key features
Huntington's disease	CAG repeat in the HTT gene, chromosome 4 (marker 1983, gene 1993)	Autosomal dominant	Chorea, dementia, onset about 40 (Section 5.11)
Fragile X syndrome	CGG repeat on the X silences the FMR1 gene	X-linked	Commonest inherited intellectual disability
Down syndrome	Trisomy 21	Usually new; risk rises with maternal age	Intellectual disability, early Alzheimer's
Turner syndrome	45,X: one X missing	Sex chromosome error	Short, infertile female
Klinefelter syndrome	47,XXY: extra X	Sex chromosome error	Tall male, small testes, infertility, learning difficulties
Prader-Willi syndrome	Deletion at 15q11 to q13, paternal chromosome	Imprinting	Hyperphagia, obesity, short stature, hypotonia
Angelman syndrome	Same deletion, maternal chromosome	Imprinting	Severe intellectual disability, absent speech, seizures, laughter

EXAM TRAP

Asperger's syndrome (autism spectrum, Chapter 8) and transsexualism (a gender identity term) are not chromosomal syndromes. PKU blocks tyrosine: urea is a waste product of protein breakdown, tryptophan the precursor of serotonin, histidine of histamine. CGG and FMR1 mean fragile X; CAG on chromosome 4 means Huntington's.

Building the nervous system

After fertilisation the **zygote** divides into a solid ball (the **morula**) and then a hollow ball that implants (the **blastocyst**); these stages (Chapter 6) contain no nervous tissue. The nervous system begins at **gastrulation**, in the third week, when the embryo forms three germ layers: ectoderm, mesoderm and endoderm. Induced by the mesoderm beneath, a patch of ectoderm becomes the **neural plate**, which folds into the neural groove, whose lips fuse into the **neural tube**, forerunner of the brain and spinal cord. Failures of this **neurulation** cause **spina bifida** (tail end) and **anencephaly** (head end).

Development then proceeds in phases: **proliferation** in the **ventricular zone** lining the tube, **migration** of young neurons along **radial glial cells**, **aggregation** into structures, and axon growth led by **growth cones**. **Roger Sperry**, who rotated frogs' eyes and found that regenerating optic axons regrew to their original targets, proposed the **chemoaffinity hypothesis** (1963): each target releases a chemical label that attracts its axon. After synapses form, far more neurons exist than survive; in **apoptosis** (programmed cell death) those that fail to obtain **neurotrophins** from their targets die. **Rita Levi-Montalcini** discovered the first, **nerve growth factor** (Nobel Prize 1986). In **synapse rearrangement**, surviving axons take over the vacated sites.

EXAM TRAP

The nervous system begins forming at the **gastrula** stage. Zygote, morula and blastocyst come earlier.

Pruning, myelination and lifelong plasticity ●●

The young brain overproduces synapses; childhood synaptic density far exceeds the adult level. **Synaptic pruning** (neural pruning) then eliminates unused connections and keeps those in use: "use it or lose it". It is heaviest in childhood and continues in the prefrontal cortex through adolescence, partly carried out by **microglia**, which engulf unwanted synapses. **Myelination** also continues into early adulthood, the prefrontal cortex last, one reason judgement and impulse control mature late.

> EXAM TRAP
>
> Pruning is the **elimination of unused connections**, a process of neuronal development and plasticity. It is not memory consolidation, not long-term potentiation or any strengthening of pathways (both Section 5.9), not neurotransmitter release (Section 5.3), not the growth of new neurons and not an increase in brain size.

Neuroplasticity is the brain's ability to change its structure and function throughout life with experience, learning and injury; it is not a protective coating (skull, meninges or myelin), memory loss or cell death. In the 1960s **Mark Rosenzweig** and colleagues found that rats reared in **enriched environments** developed a thicker, heavier cortex than rats in barren cages. Plasticity peaks in **critical periods**: when **David Hubel** and **Torsten Wiesel** sewed one eye of a young kitten shut (**monocular deprivation**), the visual cortex permanently lost its responses to that eye, whereas adult cats were barely affected. Adults still make new neurons (**adult neurogenesis**) in the hippocampal dentate gyrus and, in rodents, the olfactory bulb; **Peter Eriksson** and colleagues confirmed it in the human hippocampus in 1998. **Brain-derived neurotrophic factor** (BDNF) supports synaptic plasticity, neuronal survival (it prevents apoptosis) and hippocampal neurogenesis, not action potential firing (Section 5.3) or myelin synthesis. Exercise and antidepressants raise it; chronic stress lowers it (Section 5.7).

> SCAN YOUR BRAIN
>
> Fill in the blanks. The answers are below; cover them first.
>
> 1. The brain's counterpart of the ECG, with the highest temporal resolution, is the _____.
> 2. CT and plain MRI are structural; PET and _____ are functional.
> 3. A positive wave about 300 ms after a rare oddball stimulus is the _____.
> 4. The tool that edits or switches genes at chosen DNA sequences is _____.
> 5. A heterozygous Dd child, with D dominant, shows the _____ trait.
> 6. The karyotype 47,XXY gives _____ syndrome.
> 7. In PKU, phenylalanine is not converted into _____.
> 8. The nervous system begins forming at the _____ stage.
>
> **Answers:** (1) EEG, (2) SPECT, (3) P300, (4) CRISPR-Cas9, (5) dominant, (6) Klinefelter, (7) tyrosine, (8) gastrula.

5.6 Sensory Pathways and the Motor System LOWER YIELD · 22 QS

Section 1.3 followed light, sound and touch to the sense organs. This section follows their signals into the brain, then traces motor commands back out to the muscles.

Principles of sensory systems

Every sensory system begins with **transduction**, the conversion of physical energy into neural signals by receptors. A neuron's **receptive field** is the area of the sensory surface (a patch of retina or skin) in which a stimulus changes its firing. The systems are hierarchical: receptors, thalamic relay nuclei, **primary sensory cortex** (input mainly from the thalamus), **secondary sensory cortex** (input mainly from primary cortex of the same sense) and **association cortex** (input from several senses). Each level contains areas with different jobs (**functional segregation**), and information flows along several routes at once (**parallel processing**).

Refractive errors •

The cornea and lens must focus the image on the retina; their failures are **refractive errors**, optical rather than neural faults. In **myopia** (near-sightedness) distant objects blur because the eyeball is too long, so their image focuses in front of the retina; a concave lens corrects it. In **hypermetropia** (far-sightedness) a short eyeball focuses near objects behind the retina; a convex lens corrects it. **Astigmatism** is an irregularly curved cornea that blurs some orientations, and **presbyopia** the stiffening of the ageing lens that brings reading glasses after about 40. Visual field defects, by contrast, are blind areas left by pathway damage (5.6.4).

Retinal circuits and lateral inhibition •

The retina is built inside out (the **inverted retina**): light passes the ganglion, amacrine, bipolar and horizontal cells, in that order, before reaching the rods and cones at the back. Signals return along the **through pathway**, receptors to bipolar cells to ganglion cells, whose axons form the optic nerve, while **horizontal** and **amacrine cells** carry signals sideways. Where the axons leave, at the optic disc, there are no receptors: the blind spot, filled in by **completion**.

Stephen Kuffler (1953) found that cat ganglion cells have circular receptive fields with an opposing centre and surround: an **on-centre cell** fires to light on its centre and is inhibited by light on its surround, and an off-centre cell does the reverse. Such cells signal contrast, not even illumination.

The mechanism is **lateral inhibition**, the suppression of a neuron by its active neighbours, analysed by **Haldan Keffer Hartline** in the eye of the horseshoe crab *Limulus* (Nobel Prize 1967). At a light-dark border, a receptor just on the bright side has dark neighbours, receives less inhibition and signals "brighter still"; one just on the dark side receives extra inhibition and signals "darker still". The result is the illusory bright and dark strips called **Mach bands** (**Ernst Mach**, 1865).

> EXAM TRAP
>
> Mach bands come from lateral inhibition, not the rod-cone dichotomy (dim-light versus colour vision), the opponent process (colour afterimages) or temporal summation (adding weak signals over time).

From retina to striate cortex

At the **optic chiasm**, fibres from the nasal half of each retina cross; temporal fibres do not. Because nasal hemiretinas see the outer visual fields, the **left visual field reaches the right hemisphere**. Each optic tract ends in the **lateral geniculate nucleus** (LGN) of the thalamus, the visual relay (hearing uses the medial geniculate nucleus, touch the ventral posterior nucleus, and smell no thalamic relay).

The LGN has six layers. Layers 1 and 2 are **magnocellular**: large, fast, colour-blind cells with large receptive fields, for motion, contrast and depth. Layers 3 to 6 are **parvocellular**: small, slower cells for colour and fine detail. Koniocellular cells lie between the layers. Optic radiations carry the output to the **primary visual cortex** (V1, striate cortex, area 17) around the calcarine fissure. V1 is **retinotopic**, and the tiny fovea claims about a quarter of it.

Field loss therefore locates damage (Figure 5.10). An optic nerve lesion blinds one eye. A pituitary tumour pressing on the chiasm cuts the crossing fibres: **bitemporal hemianopia**. Damage behind the chiasm (tract, radiations or V1) causes **contralateral homonymous hemianopia**, the same opposite half-field lost in both eyes. A posterior cerebral artery stroke of V1 often spares central vision (**macular sparing**), because the occipital pole also draws blood from the middle cerebral artery. A small V1 lesion leaves a **scotoma**, an island of blindness often hidden by completion; in **blindsight**, studied by **Lawrence Weiskrantz** in the patient D.B. (1974), people point accurately at stimuli in the scotoma that they deny seeing.

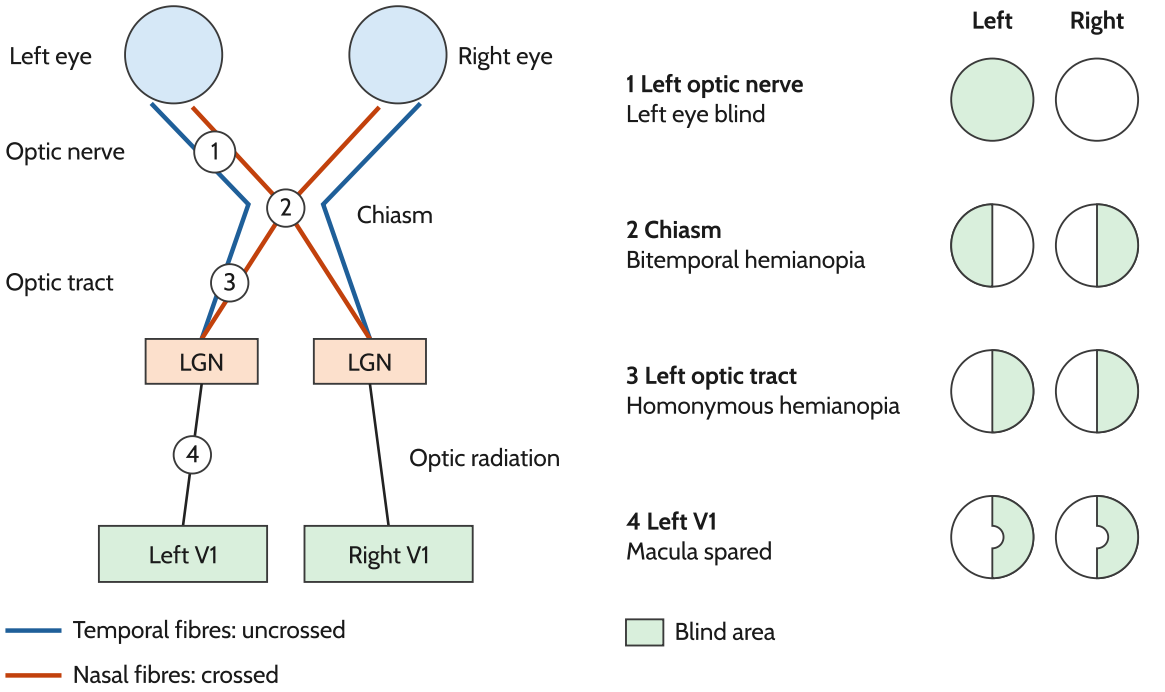


Figure 5.10 The visual pathway seen from above, and the field loss from damage at four sites. Circles show each eye's view as the patient sees it; shading marks blindness.

EXAM TRAP

The LGN relays vision only. Magnocellular is motion, parvocellular colour and detail; the two differ, and neither projects only to V4. Hemianopia with macular sparing means V1, not the chiasm, optic tract or optic nerve.

Visual cortex and the two streams

In V1, **David Hubel** and **Torsten Wiesel** (Nobel Prize 1981; Section 1.4) found **simple cells**, responding to a line of one orientation in one exact position, and **complex cells**, responding to that orientation anywhere in a larger field, often only when it moves. V2 (area 18) elaborates contours and feeds specialised areas: V4 for colour (damage causes **cerebral achromatopsia**, a grey world) and V5, or MT, for motion (damage causes **akinetopsia**, movement seen as still frames).

Leslie Ungerleider and **Mortimer Mishkin** (1982) grouped these areas into two streams leaving V1. The **dorsal stream** runs to the posterior parietal cortex: "where" things are. The **ventral stream** runs to the inferotemporal cortex: "what" they are, faces included (the fusiform face area, Section 5.11). **Melvyn Goodale** and **David Milner** (1992) recast the dorsal stream as "how", vision for guiding action such as grasping, and the ventral stream as vision for conscious perception.

EXAM TRAP

The streams are not colour against motion (motion is dorsal) or hearing against vision, and they diverge from V1 rather than converging on it.

Hearing, touch, pain, smell and taste ●●

The **hair cells** of the organ of Corti transduce mechanically: basilar membrane movement bends their stereocilia, opening cation channels through which potassium from the endolymph flows in and depolarises the cell. Inner hair cells supply almost all auditory nerve signals; outer hair cells amplify the vibration. The pathway runs from the cochlear nuclei to the **superior olives** on both sides, where the two ears' signals first meet and are compared to locate sounds, then to the inferior colliculi, the medial geniculate nucleus and the primary auditory cortex in Heschl's gyrus (areas 41 and 42), which is **tonotopic** (each frequency has its own place). **Conductive deafness** arises in the outer or middle ear, **sensorineural deafness** in the hair cells or auditory nerve.

Table 5.17 lists the skin's receptors; **proprioceptors**, the muscle spindles and Golgi tendon organs (5.6.8), signal body position.

Table 5.17 Skin receptors.

Receptor	Detects	Adapts
Free nerve endings (nociceptors)	Pain and temperature	Slowly
Pacinian corpuscles (deep)	Vibration, deep pressure	Fast
Meissner's corpuscles (fingertips)	Light touch	Fast
Merkel's discs	Sustained touch and pressure	Slowly
Ruffini endings	Skin stretch	Slowly

The **dorsal column-medial lemniscus system** carries touch, vibration and proprioception; it ascends uncrossed and crosses in the medulla. The **anterolateral system** (spinothalamic tract and others) carries nociceptive pain and temperature; it crosses in the spinal cord near the level of entry. Both relay in the ventral posterior nucleus of the thalamus to **primary somatosensory cortex** (SI; postcentral gyrus, areas 3, 1 and 2). **Wilder Penfield** and **Theodore Rasmussen** (1950) mapped its **somatosensory homunculus** by stimulating patients' cortex during surgery: the hands, lips and tongue, used for fine discrimination, have the largest areas, the trunk, hips and feet small ones. Secondary somatosensory cortex (SII), below SI, receives input from both sides of the body.

Pain has no single cortical centre; the anterior cingulate cortex tracks its unpleasantness. It can be blocked from above: **David Reynolds** (1969) operated on rats with no pain relief but electrical stimulation of the **periaqueductal grey** (PAG) (**stimulation-produced analgesia**). Opioid receptors were then found in the PAG, endorphins were isolated, and a **descending analgesia** circuit was traced from the PAG through the serotonergic raphe nuclei to the dorsal horn, where interneurons block incoming pain (compare the gate-control theory of Ronald Melzack and Patrick Wall, 1965, Section 1.3). **Phantom limbs** reflect cortical reorganisation: **Vilayanur Ramachandran** showed the face area of SI invading the lost hand's territory, and his mirror box eases phantom pain.

Smell alone reaches the cortex without a thalamic relay: the olfactory bulb projects to the **piriform cortex** and amygdala, and on to the orbitofrontal cortex. Taste travels in cranial nerves VII, IX and X to the solitary nucleus of the medulla, the ventral posterior thalamus and the gustatory cortex in the insula.

EXAM TRAP

Pain uses free nerve endings (Merkel, Meissner and Ruffini receptors serve touch) and the anterolateral system (not the route for proprioception, vibration and touch). Hair cells work by mechanical bending and potassium entry, not sodium-channel closure (the photoreceptor's response to light), GABA or chemical signalling. Phantoms are cortical remapping, not nerve regrowth, denial or cerebellar damage.

The sensorimotor hierarchy and mirror neurons ●●

The motor system is a hierarchy run in reverse, from goals down to muscles. At the top, posterior parietal cortex integrates where the body and nearby objects are (damage causes apraxia and neglect, Section 5.11), and dorsolateral prefrontal cortex decides to initiate voluntary responses. **Secondary motor cortex**, the premotor cortex and supplementary motor area (area 6), programmes patterns of movement and fires just before movement starts. **Primary motor cortex** (area 4, precentral gyrus), the main cortical exit, is laid out as a **motor homunculus** with outsized hands and mouth. Brainstem nuclei and spinal circuits execute. Primary, secondary and supplementary areas thus all control movement. The levels are functionally segregated and parallel, output is guided by sensory feedback, and practice hands control down to lower levels.

KEY STUDY: RIZZOLATTI'S MIRROR NEURONS (1990S)

Giacomo Rizzolatti and colleagues found neurons in area F5 of macaque ventral premotor cortex that fired both when the monkey grasped an object and when it watched an experimenter grasp it: **mirror neurons**, active for performing and for observing the same action. Humans show a mirror system in premotor cortex, Broca's area (inferior frontal gyrus) and the inferior parietal lobule, proposed to support imitation, action understanding and empathy. Criticism: human single-cell evidence is scarce, and the "broken mirror" theory of autism is debated.

EXAM TRAP

Mirror neurons fire for doing and watching, not one alone, and not in REM sleep. The system is not in the hippocampus, amygdala or cerebellum.

Descending pathways and spinal circuits ●●

Four tracts descend from the motor cortex (Table 5.18). The corticospinal, or pyramidal, tract passes through the pyramids of the medulla, and at the **pyramidal decussation** in the lower medulla about 85 to 90 percent of its fibres cross, so each hemisphere moves the opposite side of the body.

Table 5.18 The descending motor tracts.

Tract	Crossing	Controls
Dorsolateral corticospinal	Medullary pyramids	Limbs, fine finger movement
Dorsolateral corticorubrospinal (via red nucleus)	Brainstem	Distal limb muscles
Ventromedial corticospinal	Uncrossed	Trunk, proximal muscles

Tract	Crossing	Controls
Ventromedial cortico-brainstem-spinal	Bilateral	Posture, walking

Upper motor neurons run from the motor cortex down the corticospinal tract; lower motor neurons run from the anterior horn of the spinal cord to the muscle (Table 5.19). An upper lesion frees spinal reflexes from descending inhibition, so tone and reflexes rise and stroking the sole makes the big toe extend (an extensor plantar response), the **Babinski sign** (**Joseph Babinski**, 1896). Amyotrophic lateral sclerosis destroys both kinds of neuron and shows both patterns.

Table 5.19 Upper against lower motor neuron lesions.

Feature	Upper motor neuron	Lower motor neuron
Weakness	Spastic, tone raised	Flaccid, hypotonia
Reflexes	Brisk; Babinski sign	Reduced or absent
Muscle	Little wasting	Atrophy, fasciculations

Spinal circuits also work alone. A tendon tap stretches a muscle, and its muscle spindles, which detect length, excite its motor neurons through a single synapse: the **stretch reflex**, which holds limbs steady against loads. Golgi tendon organs detect tension and relax a dangerously contracting muscle. **Reciprocal innervation** relaxes a muscle's antagonist as it contracts. **Central pattern generators** produce rhythms: **Sten Grillner** (1985) showed that cats whose spinal cords were cut off from the brain could walk on a treadmill.

> EXAM TRAP
>
> The corticospinal tract crosses at the medullary pyramids, not at the optic chiasm (optic fibres), the spinal level of entry (pain fibres), the pons, the cerebral peduncles or the third ventricle, and "it does not cross" is false.

Basal ganglia and cerebellum

Neither the basal ganglia nor the cerebellum sends commands to the muscles; both modulate the cortex through the thalamus. In the basal ganglia, the striatum (caudate and putamen) receives cortical input, and the output nuclei, the internal globus pallidus (GPi) and substantia nigra pars reticulata (SNr), constantly inhibit the thalamus (Figure 5.11).

In the **direct pathway**, cortex excites striatal D1 neurons, which inhibit GPi and SNr; the thalamus, released, excites the cortex and movement is facilitated. In the **indirect pathway**, striatal D2 neurons inhibit the external globus pallidus (GPe), freeing the subthalamic nucleus (STN) to excite GPi; the thalamus is inhibited and movement suppressed. Dopamine from the substantia nigra pars compacta excites the direct and inhibits the indirect pathway. Its loss causes the slow, scarce movement of Parkinson's disease, a **hypokinetic** disorder; loss of indirect-pathway neurons causes the chorea of Huntington's disease, a **hyperkinetic** one (Section 5.11). The basal ganglia thus perform **action selection**, initiate and gate voluntary movement, and learn habits, motor sequences and rewarded actions. Depth perception, memory consolidation and cardiorespiratory control belong to the visual cortex, hippocampus and medulla.

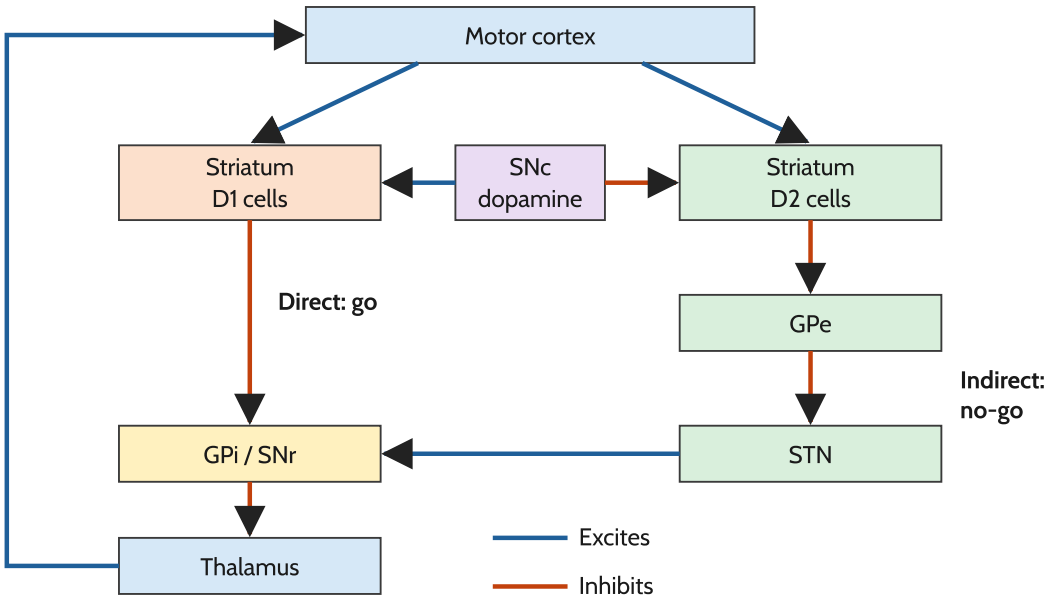


Figure 5.11 The basal ganglia loops. GPi and SNr brake the thalamus; the direct pathway lifts the brake, and the indirect pathway, through GPe and STN, tightens it. Dopamine favours movement through both.

The cerebellum compares intended with actual movement and corrects the error: **climbing fibres** from the inferior olive carry error signals to its Purkinje cells. It coordinates and times movement, maintains balance, and supports motor adaptation and eyeblink conditioning (Section 5.9). Damage causes **ataxia** (clumsy, unsteady movement), intention tremor (worse as the hand nears its target) and dysmetria (over- or undershooting).

Table 5.20 Cerebellum against basal ganglia.

Feature	Cerebellum	Basal ganglia
Core job	Error correction, coordination	Action selection, gating
Learning	Motor adaptation, eyeblink	Habits, procedural, reward-based
Damage	Ataxia, intention tremor	Parkinson's, Huntington's

EXAM TRAP

Climbing-fibre error correction is cerebellar; habit and reward-based selection are basal ganglia jobs. The route through GPe and STN is the indirect pathway, hippocampus-fornix-thalamus is a memory circuit, and cortex to cerebellum to spinal cord is no basal ganglia route.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Mach bands are explained by _____ in the retina.
2. A pituitary tumour pressing on the optic chiasm causes _____ hemianopia.
3. Motion and contrast are carried by the _____ layers of the LGN.
4. The ventral "what" stream runs from V1 to the _____ cortex.
5. Pain and temperature ascend in the _____ system.
6. Most corticospinal fibres cross at the _____ in the lower medulla.
7. Climbing fibres carry error signals to the cerebellum's _____ cells.

Answers: (1) lateral inhibition, (2) bitemporal, (3) magnocellular, (4) inferotemporal, (5) anterolateral, (6) pyramidal decussation, (7) Purkinje.

5.7 The Endocrine System, Stress and the Emotional Brain

HIGH YIELD · 46 QS

A bus swerves towards you on a Mysuru road. Before you have thought a word, your heart pounds and your palms are wet, and an hour later you are still shaky: nerves reacted in milliseconds, but hormones in your blood kept you on edge for hours. This section follows the hormones from the hypothalamus to the glands and the stress systems they build, then the brain circuits that turn a threat into fear.

Glands, ducts and hormones

Glands come in two kinds. **Exocrine glands** are **duct glands**: they deliver their secretions through ducts onto a surface or into a cavity, and those secretions are not hormones. Sweat, salivary, tear and digestive glands are exocrine. **Endocrine glands** are **ductless**: they release **hormones**, chemical messengers carried by the blood to distant cells that have receptors for them. The **pancreas** is both: most of it sends digestive enzymes down a duct, while islands of cells within it release insulin into the blood.

Hormones belong to three chemical classes. **Amino acid derivatives** are made in a few steps from one amino acid (adrenaline, thyroxine, melatonin). **Peptide and protein hormones** are short and long chains of amino acids (insulin, oxytocin, vasopressin, growth hormone, ACTH). **Steroid hormones** are made from cholesterol: the gonadal hormones (testosterone, oestrogens, progesterone) and the adrenal cortex hormones (cortisol, aldosterone). Most hormones act on receptors in the cell membrane, but steroids, small and fat-soluble, also enter the cell, bind receptors in the cytoplasm or nucleus and change **gene expression** directly, so their effects are the most diverse and long-lasting.

EXAM TRAP

Exocrine glands are **duct glands**; endocrine glands are **ductless**. "Primary" is no class of gland, and sweat, saliva and tears are never hormones.

The hypothalamus and the pituitary ●●●

The **pituitary gland**, a pea-sized gland hanging from the base of the brain, is called the **master gland** because most of its hormones are **tropic hormones**, which make other glands release theirs; it also releases growth hormone, which regulates growth. Yet the master gland has a master: the **hypothalamus** above it (Section 5.2), which integrates the endocrine, autonomic and visceral systems. In an emergency the hypothalamus supervises the fight-or-flight response, driving the sympathetic nervous system and the pituitary at once.

The pituitary is really two glands, controlled in two different ways (Table 5.21). The **anterior pituitary** (**adenohypophysis**) is glandular tissue that releases growth hormone, adrenocorticotropic hormone (ACTH), thyroid-stimulating hormone (TSH), the gonadotropins (follicle-stimulating hormone, FSH, and luteinising hormone, LH) and prolactin, yet no nerve fibres reach it. How, then, does the hypothalamus command it? **Geoffrey Harris** (1913 to 1971) proposed in 1955 that hypothalamic neurons release **releasing** and **release-inhibiting hormones** into a network of blood vessels, the **hypothalamo-hypophyseal portal system**, which carries them down the pituitary stalk; cutting the portal veins halted anterior pituitary release until they regrew. In the late 1960s **Roger Guillemin** and **Andrew Schally** separately isolated one, thyrotropin-releasing hormone (TRH), sharing the 1977 Nobel Prize.

The **posterior pituitary** (**neurohypophysis**) is neural tissue, an outgrowth of the hypothalamus, and makes nothing itself. Oxytocin and vasopressin are synthesised in neurons of the **supraoptic** and **paraventricular nuclei** of the hypothalamus, carried down their axons and released from the posterior pituitary into the blood. **Vasopressin**, or **antidiuretic hormone** (ADH), makes the kidneys reabsorb water; its failure causes **diabetes insipidus**, with copious dilute urine and constant thirst. **Oxytocin** contracts the uterus in labour, ejects milk during suckling and promotes bonding.

Table 5.21 How the hypothalamus runs the pituitary. The first five rows are anterior, the last two posterior.

Hypothalamic signal	Pituitary hormone	Target	Main effect
CRH	ACTH	Adrenal cortex	Cortisol release
TRH	TSH	Thyroid	Thyroxine release
GnRH	FSH and LH	Ovaries, testes	Sex hormones, eggs, sperm
GHRH; somatostatin inhibits	Growth hormone	Most tissues	Growth; excess gigantism, deficit dwarfism
Dopamine inhibits	Prolactin	Breasts	Milk production
Made in supraoptic and paraventricular nuclei; sent down axons	Oxytocin	Uterus, breasts, brain	Labour, milk ejection, bonding
Same nuclei, same route	Vasopressin (ADH)	Kidneys	Water reabsorption

EXAM TRAP

Master gland: the pituitary, controlled by the hypothalamus; not the hypothalamus, thalamus, pineal, thyroid or adrenal. **Supervisor** of fight or flight: the hypothalamus, not the pituitary, the adrenal gland or the parasympathetic system. The anterior pituitary is reached by **portal blood**, not by axons, cerebrospinal fluid or the vagus nerve. **Not anterior pituitary:** oxytocin (posterior) and melatonin (pineal). The posterior pituitary only stores and releases oxytocin and vasopressin; orexin (a hypothalamic wakefulness peptide, Section 5.8), noradrenaline and "vasodilators" are impostors. Oxytocin and vasopressin are made in the supraoptic and paraventricular nuclei, not the suprachiasmatic nucleus (body clock), the arcuate nucleus (feeding; the dopamine that inhibits prolactin) or the mammillary bodies. Prolactin makes milk; oxytocin ejects it.

The other endocrine glands ●●●

Three facts about the remaining glands (Table 5.22) recur: the **adrenal glands** run the stress response, the **thyroid** sets the metabolic rate and the **pancreas** controls blood glucose.

Each adrenal gland sits on a kidney and is two glands in one. The outer **adrenal cortex** makes steroids: **cortisol**, the chief glucocorticoid and the stress hormone; aldosterone, for salt and water balance; and adrenal androgens. The inner **adrenal medulla**, a modified sympathetic ganglion, releases **adrenaline** and noradrenaline when sympathetic nerves fire. Adrenaline and **epinephrine** are two names for one hormone (Latin and Greek for "on the kidney"), as noradrenaline is norepinephrine.

The **thyroid gland** releases thyroxine (T4) and triiodothyronine (T3), which set the body's metabolic rate and energy expenditure; the thyroid itself is controlled by TSH. In the islets of Langerhans of the pancreas, beta cells release **insulin**, which lowers blood glucose by letting cells take it up, and alpha cells release glucagon, which raises it. **Type 1 diabetes** is autoimmune destruction of the beta cells, usually in childhood. **Type 2 diabetes**, far commoner and widespread in India, is a disorder of **insulin resistance**: it runs in families, but it is most strongly associated with poor diet, obesity and a sedentary lifestyle.

Table 5.22 The other endocrine glands.

Gland	Hormones	Main functions	Disorders
Pineal	Melatonin, at night	Circadian rhythm (Section 5.8)	Used for jet lag
Thyroid	Thyroxine (T4), T3	Metabolic rate, energy use	Hypothyroidism: lethargy, depression; cretinism in infancy. Hyperthyroidism (Graves' disease): anxiety, agitation, weight loss
Parathyroids	Parathyroid hormone	Raises blood calcium	Low calcium: tetany
Adrenal cortex	Cortisol, aldosterone, adrenal androgens	Stress, glucose, immunity; salt and water	Cushing's (excess), Addison's (deficiency)
Adrenal medulla	Adrenaline, noradrenaline	Fast fight-or-flight arousal	Phaeochromocytoma: panic-like attacks
Pancreas (islets)	Insulin (beta cells), glucagon (alpha cells)	Lower and raise blood glucose	Type 1 and type 2 diabetes

Gland	Hormones	Main functions	Disorders
Gonads	Testosterone; oestrogens, progesterone	Sexual development and behaviour (Section 5.8)	Hypogonadism
Thymus	Thymosin	Matures T lymphocytes	Immune deficiency

Endocrine disorders often present as psychiatric ones. In **Cushing's syndrome**, excess cortisol (from a tumour or steroid medicines) brings a round "moon face" and central obesity and, most commonly, depression, anxiety, irritability and poor memory and concentration; a minority develop mania or psychosis (not catatonia or loss of smell). **Addison's disease**, failure of the adrenal cortex with too little cortisol, brings fatigue, apathy and depression. Thyroid disorders are in Table 5.22.

> EXAM TRAP
>
> Stress and stress-related metabolic change: **adrenal glands**. Overall metabolic rate: **thyroid**. Insulin: the **pancreas** only, not the kidney, thyroid or hypothalamus. Amity 2023's "Which controls the thyroid gland? Metabolism" means what the thyroid controls; the thyroid itself obeys TSH, glucose uptake is insulin's job and emotion the limbic system's. The adrenal secretes **both** cortisol and epinephrine; melatonin, thyroxine and insulin are distractors, and noradrenaline and dopamine are adrenaline's relatives, not its other name. For type 2 diabetes genetics matters, and stress and toxins contribute less, but the keyed answer is poor diet and sedentary lifestyle.

Two stress systems, feedback and allostasis ●●

The hypothalamus answers a threat through two systems (Figure 5.12). In the fast **sympathetic adrenal-medullary system** (SAM), sympathetic nerves fire and the adrenal medulla pours adrenaline and noradrenaline into the blood within seconds: the fight-or-flight response named by **Walter Bradford Cannon** (1871 to 1945; the autonomic wiring is in Section 5.1).

The slower **hypothalamic-pituitary-adrenal axis** (HPA axis) works by hormones. The paraventricular nucleus of the hypothalamus releases corticotropin-releasing hormone (CRH) into the portal blood, the anterior pituitary answers with ACTH, and ACTH in the general circulation makes the adrenal cortex release cortisol over minutes. Cortisol mobilises glucose and dampens inflammation and immunity. It also switches the axis off by **negative feedback** on the hypothalamus (less CRH) and the anterior pituitary (less ACTH), helped by the hippocampus, whose dense glucocorticoid receptors let it restrain the axis. The thyroid and gonadal axes work the same way. **Hans Selye**'s general adaptation syndrome, what follows when the axis stays switched on, is taught in Section 1.11.

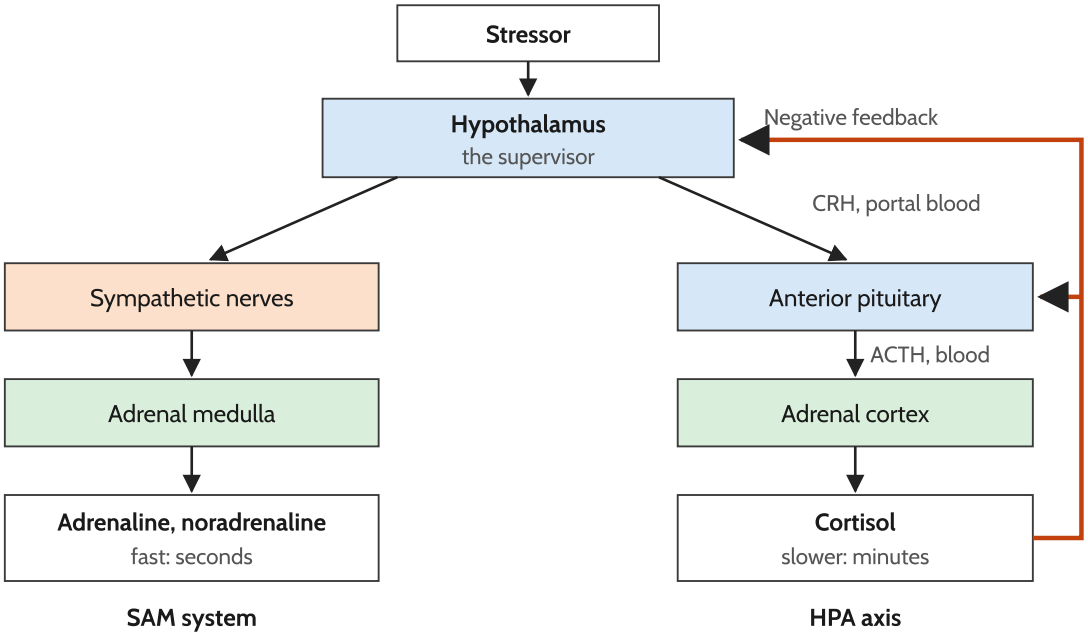


Figure 5.12 The two stress systems. Solid arrows drive; the dashed line is cortisol's negative feedback on the hypothalamus and anterior pituitary. The posterior pituitary and pineal gland play no part.

Brief cortisol surges help; chronic ones harm. Prolonged high cortisol shrinks hippocampal dendrites, and a damaged hippocampus brakes the axis less well, so cortisol climbs further. Chronic stress also suppresses immunity and slows wound healing. The study of such links between mind, nervous system and immune system is **psychoneuroimmunology**.

Cannon also named the goal of these systems: **homeostasis** (popularised in *The Wisdom of the Body*, 1932), keeping the internal state stable around fixed set points, as a thermostat holds a room's temperature. **Peter Sterling** and **Joseph Eyer** (1988) argued that the brain also predicts demands and shifts the set points in advance, raising blood pressure as you stand and cortisol before an examination. They called this **allostasis**, "stability through change": the brain regulating its own stress responses by controlling hormone release. **Bruce McEwen** and **Eliot Stellar** (1993) added **allostatic load**, the cumulative wear and tear when these responses are overused or fail to switch off: high blood pressure, abdominal fat, insulin resistance and hippocampal damage.

EXAM TRAP

The HPA chain is **hypothalamus (CRH), anterior pituitary (ACTH), adrenal cortex (cortisol)**. ACTH comes from the anterior pituitary, not from the adrenal gland it targets; the posterior pituitary and adrenal medulla are not in the axis, and "pineal, pituitary, thyroid" is no axis; the cerebellum plays no part. Cortisol feeds back on the **hypothalamus and anterior pituitary**, not the pineal, posterior pituitary, adrenal medulla or vagal motor nucleus (parasympathetic output). "The brain regulating its own stress responses by controlling hormone release" is **allostasis**, not homeostasis (fixed set points), neuroplasticity (rewiring with experience, Section 5.5) or circadian rhythm (Section 5.8).

CLINICAL LINK

In melancholic depression the HPA axis is often overactive: in the **dexamethasone suppression test**, a night dose of synthetic glucocorticoid fails to suppress next morning's cortisol (Chapter 8). Because endocrine disease mimics depression and anxiety, hormones are checked before a psychiatric diagnosis.

Emotion and the autonomic nervous system •

Fear and anger bring one broad sympathetic pattern: heart rate, blood pressure and breathing rise, the pupils dilate, blood shifts to the muscles, glucose enters the blood and sweating increases. Sweat conducts electricity, so skin conductance rises and skin resistance falls (the galvanic skin response). Pupil constriction is parasympathetic. The **polygraph** records heart rate, blood pressure, breathing and skin conductance together; it detects arousal, not lies. Whether each emotion has its own autonomic pattern divides the James-Lange, Cannon-Bard and Schachter-Singer theories (Section 1.11).

EXAM TRAP

Odd one out among signs of sympathetic activation: **pupil constriction**. "Lowered skin resistance" is sympathetic: it is the same event as raised skin conductance.

From sham rage to the limbic system •••

Which part of the brain makes emotion? In 1929 **Philip Bard** found that **decorticate** cats, their cortex removed, raged at the lightest touch, arching their backs, hissing and baring their teeth, but excessively and at no target. He called this **sham rage**. It survived removal of everything down to the hypothalamus but vanished once the hypothalamus went too. The hypothalamus organises rage; the cortex inhibits and directs it.

In 1937 **James Papez** (1883 to 1958) proposed a circuit for emotion (Figure 5.13). The **Papez circuit** runs from the hippocampus through the fornix to the mammillary bodies, by the mammillothalamic tract to the anterior thalamic nucleus, on to the cingulate cortex, through the cingulum to the entorhinal (parahippocampal) cortex and back to the hippocampus. Emotion, he argued, is expressed through the hypothalamus and experienced through the cortex. The loop is now thought to serve memory more than emotion (Section 5.9), but it began the search for an emotional brain.

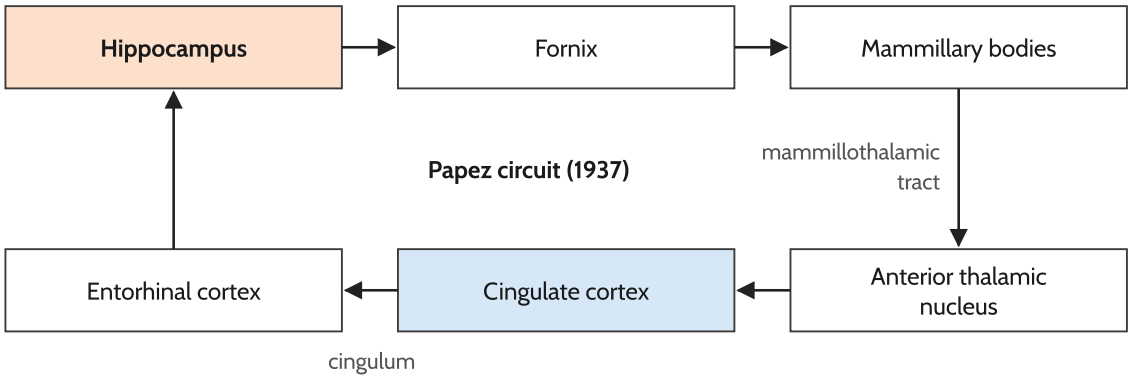


Figure 5.13 The Papez circuit, a closed loop. Loops through the caudate, putamen, globus pallidus or substantia nigra (basal ganglia), from retina to visual cortex, or through the cerebellum are other circuits.

Paul MacLean (1913 to 2007) enlarged the circuit in 1952, adding the amygdala, septum and other structures, and named it the **limbic system** (Latin *limbus*, border, because it rings the thalamus at the inner edge of the cortex); later it became the middle layer of his triune brain. Its members are the hippocampus, amygdala, cingulate gyrus, fornix, septum and mammillary bodies, with the hypothalamus and anterior thalamus closely tied in. It is the classic "seat of emotion", regulating emotion, motivation and memory.

> EXAM TRAP
>
> "Seat of emotion", "emotion and motivation", "emotional behaviour": **limbic system**; offered single parts (hippocampus, thalamus, hypothalamus) beside it, choose the system. Not the reticular formation (arousal), hindbrain (breathing, heart rate), whole forebrain (too broad), pituitary (hormones), cerebellum (motor coordination), corpus callosum (joins the hemispheres), frontal lobe (voluntary movement, decisions), thalamus and sensory cortex (sensory processing), visual cortex (acuity) or Broca's area (syntax). Hunger, thirst and hormone regulation are hypothalamic, so Amity 2023 keys "regulation of emotions", not "both".

The amygdala and fear ●●●

The **amygdala** (Greek for almond), a cluster of nuclei in each anterior medial temporal lobe, is the key emotional structure: it detects threat, gives events emotional salience, triggers fear and defensive aggression, and learns what predicts danger. Its role was mapped by fear conditioning: after a tone has preceded mild shocks, the tone alone makes a rat freeze, heart racing. Signals enter through the **lateral nucleus**, where tone and shock are associated, and leave through the **central nucleus** for the hypothalamus (autonomic and hormonal responses) and the **periaqueductal grey** of the midbrain (freezing). Amygdala lesions abolish conditioned fear.

Joseph LeDoux (born 1949) traced two routes in (Figure 5.14). The **low road**, from the sensory thalamus straight to the amygdala, is fast and crude: you jump back from a coiled shape before you know what it is. The **high road**, from the thalamus through the sensory cortex to the amygdala, is slower but accurate: it tells you the shape is a rope.

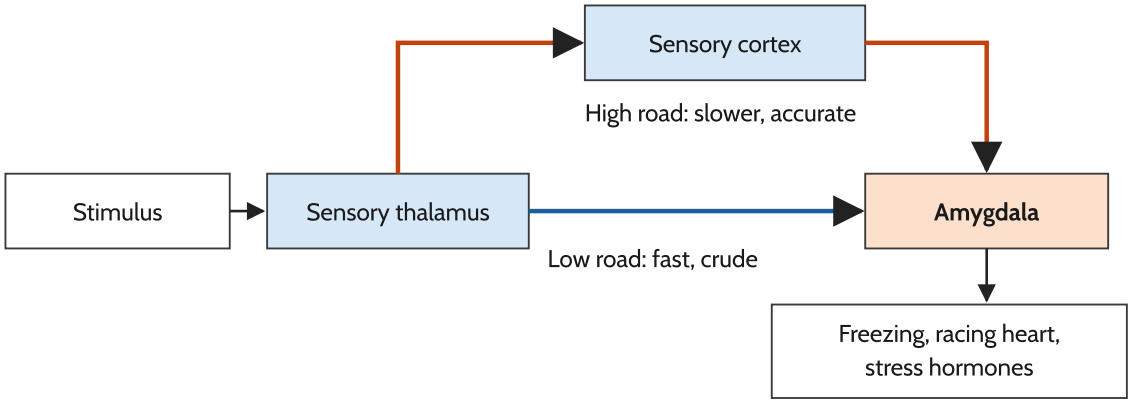


Figure 5.14 LeDoux's two roads to fear. The solid low road reacts before you consciously recognise the stimulus; the dashed high road identifies it.

KEY STUDY: PATIENT S.M. (ADOLPHS AND COLLEAGUES, 1994)

Does the human amygdala matter for emotion? **Ralph Adolphs** and colleagues tested S.M., whose amygdalae had been destroyed on both sides by Urbach-Wiethe disease, a rare genetic disorder that calcifies the amygdala and nearby medial temporal lobe. She recognised faces and most expressions but not fear, and could not produce a fearful expression herself. Later work found she barely feels fear, reaching for snakes in a pet shop (Justin Feinstein and colleagues, 2011). Criticism: one rare patient.

In **affective blindsight**, described by **Beatrice de Gelder** and colleagues in 1999, patients blinded in part of the visual field by damage to the primary visual cortex guess well above chance which emotion a face shown there expresses, while insisting they see nothing. The signal is thought to travel subcortically, from the superior colliculus through the pulvinar of the thalamus to the amygdala. "Affect neglect", "emotional neglect" and "motivational blindsight" are not established terms.

EXAM TRAP

Fear and aggression, emotion processing, threat detection, fear conditioning: **amygdala**. Not the hippocampus (conscious recall, declarative memory, Section 5.9), thalamus (relay), hypothalamus (executes the amygdala's orders), frontal lobe (verbal working memory), brainstem (vital functions), cerebellum and basal ganglia (starting movement, motor sequence learning) or suprachiasmatic nucleus (circadian rhythm). "Slower, more accurate identification" is the high road; conscious vision and voluntary motor output are cortical.

Klüver-Bucy syndrome and other influences ●●

What happens to an animal without its amygdalae? In 1937 and 1939 **Heinrich Klüver** and **Paul Bucy** removed both anterior temporal lobes, amygdalae included, in rhesus monkeys. Unmanageable monkeys became tame and placid, approaching snakes without fear. They showed **psychic blindness**, a visual agnosia (seeing objects without recognising their meaning); examined everything by mouth (**hyperorality**); ate almost anything, even meat, which rhesus monkeys normally refuse (**hyperphagia** and dietary change); directed sexual activity at inappropriate partners and objects (**hypersexuality**); and attended compulsively to every

sight (**hypermetamorphosis**). This **Klüver-Bucy syndrome** follows mainly from loss of the amygdala. Human cases, after herpes simplex encephalitis or head injury, are partial and often add aphasia and amnesia from wider damage (Table 5.23).

Table 5.23 Klüver-Bucy features and their look-alikes.

Option	Where it belongs
Tameness, placidity, loss of fear; psychic blindness, hyperorality, hyperphagia, hypersexuality	Klüver-Bucy (amygdala and temporal lobe)
Aphasia, amnesia	Human Klüver-Bucy with wider damage
Intense fear, attack posture, piloerection, hyperalertness	Amygdala stimulation; sham rage
Hyperphagia with obesity	Ventromedial hypothalamus (Section 5.8)
Pure anterograde amnesia	Hippocampus, as in H.M. (Section 5.9)
Avolition, alogia	Negative symptoms of schizophrenia (Chapter 8)
Spastic paralysis; loss of language comprehension	Motor pathways (Section 5.6); Wernicke's area (Section 5.10)

EXAM TRAP

JEMAS 2025 keys "agnosia, aphasia, hyperorality, hyperphagia, hypersexuality" (aphasia from human cases). Each rival cluster hides an impostor: hyperalertness, avolition and alogia, asexuality, hyposensitivity and delayed reaction. The structure is the **amygdala with adjacent temporal lobe**, not the cerebellum, hippocampus or caudate nucleus (Huntington's disease, Section 5.11).

Three other influences need a line each. The **ventromedial prefrontal cortex** regulates emotion, and its damage leaves people impulsive and socially inept (Section 5.11). **Testosterone** raises social aggression in the males of many species, less clearly in humans. Low **serotonin** activity is linked to impulsive aggression (Section 5.3).

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Exocrine glands release their secretions through _____; endocrine glands are ductless.
2. The hypothalamus reaches the anterior pituitary through the hypothalamo-hypophyseal _____ system.
3. Oxytocin and vasopressin are made in the supraoptic and _____ nuclei.
4. In the HPA axis, CRH makes the anterior pituitary release _____.
5. Sterling and Eyer's "stability through change" is _____.
6. Bard's decorticate cats showed _____ rage.
7. LeDoux's _____ road runs from the thalamus straight to the amygdala.
8. Tameness, hyperorality and hypersexuality after bilateral temporal lobectomy make up _____ syndrome.

Answers: (1) ducts, (2) portal, (3) paraventricular, (4) ACTH, (5) allostasis, (6) sham, (7) low, (8) Klüver-Bucy.

5.8 Sleep, Biological Rhythms, Hunger, Thirst and Sex

MEDIUM YIELD · 31 QS

Why does a student who revised all night still feel alert in the evening, hungry at lunchtime however large breakfast was, and thirsty after salty chaat? The answers lie in small clusters of neurons, mostly in the hypothalamus.

Biological rhythms and the master clock ●●●

A **circadian rhythm** (Latin *circa diem*, about a day) repeats about every 24 hours, like the sleep-wake cycle; **ultradian rhythms** are shorter (the 90-minute sleep cycle) and **infradian rhythms** longer (the menstrual cycle). Circadian rhythms are entrained, locked to the world, by **zeitgebers** ("time givers"), above all light. Without cues, in a cave or windowless laboratory, they persist as **free-running rhythms** slightly longer than 24 hours (about 24.2 in humans). Jet lag and shift work put the clock out of step with the day; eastbound flights, which demand a phase advance, are the harder.

Curt Richter found in 1967 that large medial hypothalamic lesions disrupted the daily cycles of eating, drinking and activity in rats; smaller lesions located the clock in the **suprachiasmatic nucleus** (SCN), just above the optic chiasm. SCN lesions abolish the circadian timing of sleep but not its total amount.

KEY STUDY: THE SCN TRANSPLANT (RALPH AND COLLEAGUES, 1990)

Martin Ralph and colleagues transplanted SCN tissue from fetal mutant hamsters with a 20-hour free-running cycle into normal adults whose 25-hour rhythms SCN lesions had abolished. Rhythms returned with the donor's 20-hour period (and the reverse transplant gave 25 hours): the SCN sets the period. It is the master clock, though most body cells carry clock machinery that it keeps in step.

Light reaches the SCN by the **retinohypothalamic tract**, from retinal ganglion cells containing the photopigment **melanopsin**, not from rods or cones. The clock itself is a molecular loop: clock genes such as *period*, *timeless* and *clock* are transcribed, and their proteins build up and switch off their own transcription over about 24 hours, as **Jeffrey Hall**, **Michael Rosbash** and **Michael Young** showed in fruit flies (Nobel Prize, 2017).

The SCN drives the **pineal gland**, a small endocrine gland of the epithalamus, to secrete **melatonin** at night, the "hormone of darkness"; light suppresses it. Melatonin promotes sleepiness, and taken at bedtime in the new time zone it eases jet lag.

> EXAM TRAP
>
> The master circadian pacemaker is the **SCN of the hypothalamus**, not the paraventricular or ventromedial nucleus or mammillary bodies, and no centre for hunger, temperature, anterior pituitary release, memory, language or pain (Table 5.25). Yet RML 2022 ("which gland regulates circadian rhythm") and RML 2023 ("which structure regulates the body's internal clock") keyed the **pineal gland**, which turns the clock's signal into melatonin. Choose the SCN whenever it is offered; for a gland, or "hypothalamus" against "pineal", these keys want the pineal. The pituitary (Section 5.7), cerebellum and thalamus keep no clock.

The EEG of sleep and REM sleep ●●●

When cortical neurons fire independently, the electroencephalogram (EEG) is desynchronised (small, fast waves); when they fire together it is synchronised (large, slow waves). Stages 3 and 4 together form **slow-wave sleep**, merged as N3 in the 2007 scoring rules of the American Academy of Sleep Medicine (Table 5.24). The NREM stages and a REM period make a cycle of about 90 minutes, four to six a night (Section 1.5).

Table 5.24 The EEG of waking and sleep.

State	EEG	Frequency	Marks
Alert waking	Beta: low amplitude, desynchronised	13 to 30 Hz	Eyes open, active thought
Relaxed waking	Alpha, occipital	8 to 12 (13) Hz	Eyes closed
N1 (stage 1)	Theta	4 to 7 Hz	Drowsy transition
N2 (stage 2)	Sleep spindles, K-complexes	Spindles 12 to 14 Hz	Light sleep
N3 (stages 3, 4)	Delta: high amplitude, synchronised	0.5 to 4 Hz	Slow-wave sleep; growth hormone surge
REM	Low voltage, fast, mixed; sawtooth waves	Waking-like	Eye movements, atonia, dreams

In 1953 **Eugene Aserinsky** and **Nathaniel Kleitman** found recurring bursts of rapid eye movements in sleepers, and with **William Dement** linked them to dreaming: about 80 per cent of REM awakenings yield a dream report. REM is **paradoxical sleep**: the EEG looks awake while the body is deeply asleep. Its signs are rapid eye movements; **atonia**, loss of tone in the core skeletal muscles; a low-voltage, fast, desynchronised EEG; penile and clitoral tumescence; irregular heart rate and breathing; and the most vivid, story-like dreams.

> **EXAM TRAP**
>
> REM is **not** marked by EEG synchrony, delta waves or the growth hormone surge: all three belong to N3. Spindles and K-complexes mark N2, theta N1, and occipital alpha relaxed waking with eyes closed. Slow-wave sleep is stages 3 and 4 only, not stages 1 to 4 or NREM plus REM.

The switches for sleep, waking and dreams

In 1936 **Frederic Bremer** cut cats' brainstems between the colliculi (the cerveau isolé, "isolated forebrain"), and the cortex showed almost constant slow-wave sleep; a cut at the bottom of the brainstem (the encéphale isolé, 1937) left the sleep-wake cycle intact. Something between the cuts keeps the forebrain awake. **Giuseppe Moruzzi** and **Horace Magoun** (1949) found that stimulating the brainstem reticular formation woke sleeping cats: it is the **reticular activating system**.

It is now seen as several ascending arousal systems named by transmitter: noradrenaline (locus coeruleus), serotonin (raphe nuclei), histamine (tuberomammillary nucleus), acetylcholine (basal forebrain and pons) and **orexin**, or hypocretin (lateral hypothalamus, in its posterior, wake-promoting part). Slow-wave sleep begins when GABA neurons of the **ventrolateral preoptic area** (VLPO) become active and inhibit these systems. As the two sides inhibit each other, **Clifford Saper** (2001) likened them to a flip-flop switch: one side wins, so we change state quickly (Figure 5.15).

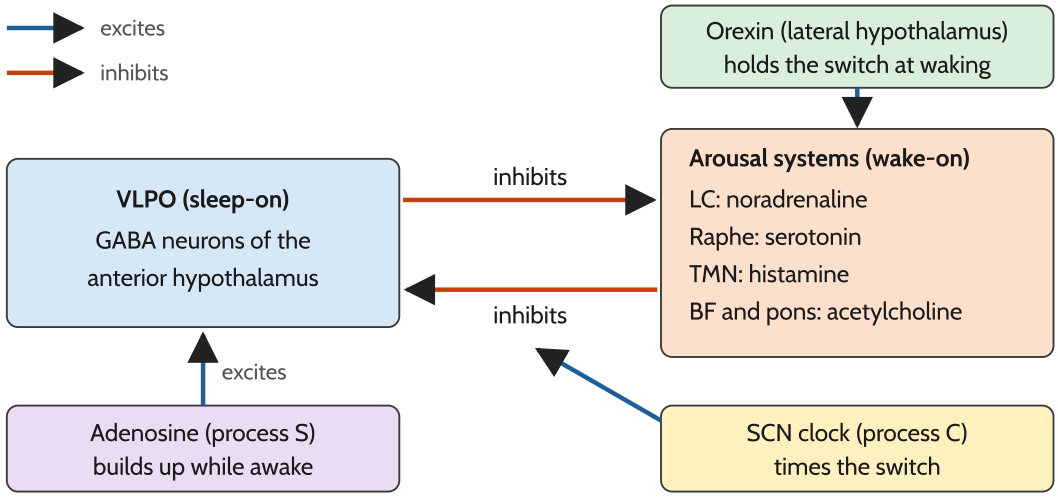


Figure 5.15 The sleep-wake flip-flop. The VLPO and the arousal systems inhibit each other, so one side wins; orexin holds it at waking, adenosine pushes it toward sleep, the SCN times it. LC, locus coeruleus; TMN, tuberomammillary nucleus; BF, basal forebrain.

The chemical of drowsiness is **adenosine**, a by-product of the brain's energy use that builds up during waking and falls during sleep: it is the homeostatic sleep pressure, and caffeine keeps you awake by blocking its receptors. Orexin, acetylcholine and serotonin promote arousal instead. **Alexander Borbély**'s **two-process model** (1982) joins this process S, rising with time awake, to the circadian process C from the SCN.

REM has its own switch: REM-on neurons of the sublaterodorsal nucleus in the pons and REM-off neurons of the ventrolateral periaqueductal grey inhibit each other. In REM the

sublaterodorsal nucleus drives medullary neurons that inhibit motor neurons, producing atonia.

In the activation-synthesis hypothesis of **J. Allan Hobson** and **Robert McCarley** (1977; Section 1.5), the cortex weaves a dream from random pontine activity. In REM the **dorsolateral prefrontal cortex** is relatively inactive, so dreams lack organisation, planning, logic and a sense of time; the amygdala is active, so they are emotional; and the **extrastriate** visual association cortex is active while the **striate** (primary visual) cortex is quiet, so we see vivid images without input from the eyes. The somatosensory cortex plays no special part.

Sleep disorders and why we sleep

In **narcolepsy** the orexin neurons of the lateral (posterior) hypothalamus are lost. Its signs are excessive daytime sleepiness with sleep attacks; cataplexy, a sudden loss of muscle tone triggered by emotion, in full consciousness; sleep paralysis; hypnagogic hallucinations while falling asleep; and sleep-onset REM. All but the sleepiness are REM breaking into waking.

In **REM sleep behaviour disorder** (RBD), REM atonia is lost and patients act out their dreams; it often precedes Parkinson's disease and Lewy body dementia by years (Section 5.11). Sleepwalking (somnambulism), night terrors and bedwetting are NREM parasomnias, arising from partial arousal out of N3 early in the night; nightmares are frightening REM dreams, mostly late in the night. In sleep apnoea, breathing stops repeatedly during sleep. Insomnia and its treatment are in Chapters 7 and 8.

> EXAM TRAP
>
> Sleepwalking and night terrors: **N3**, not REM, N1 or N2 (a body in REM is paralysed). RBD is loss of REM atonia, not an inability to enter REM, not bedwetting in N3 and not daytime sleepiness without night-time symptoms.

Why sleep at all? Recuperation theories say sleep restores what waking depletes; adaptation theories say it evolved to keep animals inactive and safe when they need not forage. Sleep also aids memory consolidation (Section 5.9).

Hunger: set points and positive incentives

Most people assume hunger works like a thermostat. On this **set-point assumption**, eating starts when energy falls below a set point and stops when it is restored: **Jean Mayer**'s glucostatic theory (1953) made blood glucose the regulated quantity, and the lipostatic theory made it body fat. **John Pinel** and colleagues argue instead for **positive-incentive theory**: we eat mainly for the anticipated pleasure of food, shaped by taste, learning and company, and body weight drifts to a settling point.

The circulating hormone that triggers hunger is **ghrelin**, from the empty stomach. Satiety signals come from the gut and from fat. In 1973 **James Gibbs**, **R. C. Young** and **Gerard Smith** injected **cholecystokinin** (CCK), released by the duodenum as food arrives, into hungry rats, which then ate smaller meals: the classic satiety peptide. Insulin, released by the pancreas as blood glucose rises, is both a satiety and a body-fat signal.

KEY STUDY: THE OB/OB MOUSE AND LEPTIN (1994)

Mice of the ob/ob strain overeat and grow up to three times normal weight. In 1994 **Jeffrey Friedman**'s group cloned the mutated gene: expressed only in fat cells, it codes a hormone they named **leptin**, which ob/ob mice lack, and leptin injections cut their eating and fat. Released in proportion to fat stores, leptin is the long-term regulator of body weight and fat. Yet obese humans usually have high leptin and do not respond to injections (leptin resistance). The long-term fat signal is leptin, not the hypothalamus (a brain area, not a hormone), oestrogen or cortisol.

The hypothalamus and the arcuate nucleus ●●●

Section 1.10 told how two lesion studies built the **dual-centre hypothesis**. In 1940 **Albert Hetherington** and **Stephen Ranson** destroyed the ventromedial hypothalamus (VMH) of rats, which overate (hyperphagia) and grew grossly obese, first in a **dynamic phase** of rapid gain, then in a **static phase** holding the new weight. In 1951 the Indian physiologist **Bal Krishan Anand** and **John Brobeck** found that lateral hypothalamus (LH) lesions caused aphagia, a fatal refusal to eat; **Philip Teitelbaum** and **Alan Epstein** (1962) showed that it comes with adipsia, a refusal to drink, and that tube-fed rats partly recover. **Eliot Stellar** (1954) made the LH the feeding centre and the VMH the satiety centre.

The model has since been revised: VMH lesions raise blood insulin, so food is turned into fat and the rats overeat because they are getting fat, not the reverse; cut fibres from the paraventricular nucleus explain much of the syndrome; and LH lesions cause broad motor and sensory deficits. The nuclei are now parts of a network (Table 5.25) whose hub is the arcuate nucleus.

Table 5.25 Hypothalamic nuclei met in this section.

Nucleus	Main role	Damage or note
Suprachiasmatic (SCN)	Circadian clock	Rhythms lost, total sleep kept
Ventrolateral preoptic	Switches on sleep	Insomnia
Medial preoptic area	Male sex; temperature	Sexually dimorphic nucleus
Ventromedial (VMH)	Satiety; female sex	Hyperphagia, obesity
Lateral (LH)	Feeding; orexin	Aphagia, adipsia
Arcuate	Appetite, energy balance	NPY and POMC neurons
Paraventricular	Stress (CRH); satiety	Hyperphagia
Supraoptic	ADH, oxytocin	Diabetes insipidus
Mammillary bodies	Memory	Amnesia (Section 5.9)

The **arcuate nucleus** is where hunger and fat signals meet, in two opposing groups of neurons (Figure 5.16). NPY/AgRP neurons release **neuropeptide Y** and agouti-related peptide; they are orexigenic, and NPY injected into the hypothalamus is among the strongest triggers of eating known. POMC/CART neurons release alpha-melanocyte-stimulating hormone (alpha-MSH) and cocaine-and-amphetamine-regulated transcript (CART); they are anorexigenic, and alpha-MSH acting on melanocortin receptors brings satiety. Leptin and insulin in-

hibit the NPY neurons and excite the POMC neurons; ghrelin excites the NPY neurons. Serotonin also promotes satiety.

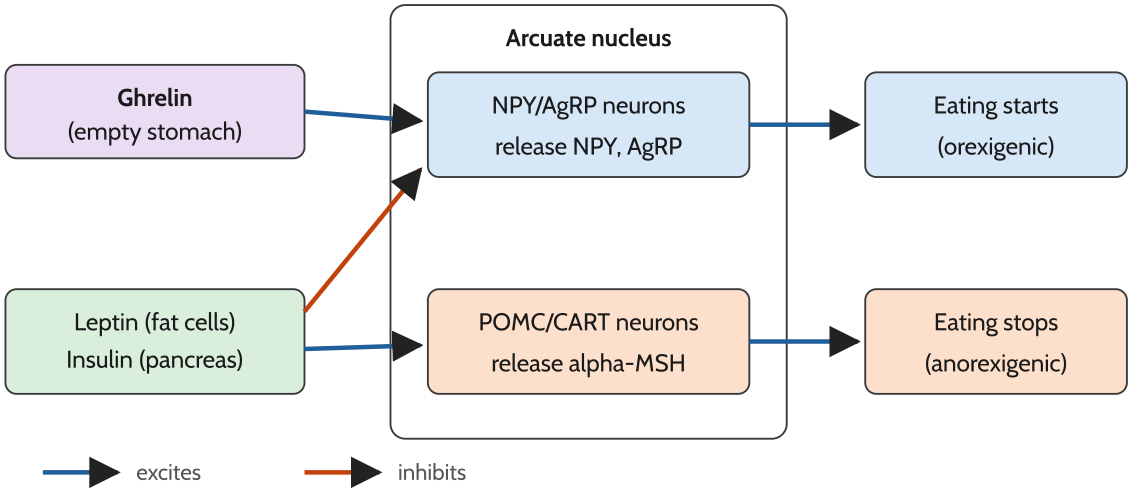


Figure 5.16 The arcuate nucleus. Ghrelin excites the NPY/AgRP (eating) neurons; leptin and insulin inhibit them and excite the POMC/CART (satiety) neurons.

EXAM TRAP

Satiety centre: **VMH**; its lesion gives hyperphagia and obesity, not excessive drinking, insomnia, diabetes insipidus or lost thermoregulation or circadian rhythm. Feeding centre: **LH**; its lesion gives aphagia, adipsia and weight loss (Lateral, Lose). "Ventromedial hippocampus" and "lateral thalamus" are invented; the cerebrum and basal ganglia are not hunger centres. The arcuate nucleus is not for motor coordination, memory or circadian rhythm. NPY triggers eating; leptin, CART and CCK are satiety signals.

Learned influences on eating, and thirst

What is on the plate, and who is at the table, matter too. In the **portion size effect**, people eat more when served larger portions: **Barbara Rolls** and colleagues (2002) served lunches of macaroni and cheese in different sizes, and the bigger the portion, the more people ate, without feeling fuller. In **sensory-specific satiety**, a food loses its appeal as you eat it while other foods stay attractive, so a new dish restarts eating. A small first course increases hunger (the appetiser effect), and a cafeteria diet of varied, palatable foods made rats eat 84 per cent more calories. Through social facilitation we eat more with others than alone, though we also eat less when companions do. Prader-Willi syndrome is in Section 5.5, eating disorders in Chapter 8.

Thirst has two triggers. In **osmotic thirst** the salt concentration of body fluid rises, as after a salty meal: water leaves the cells, and hypothalamic osmoreceptors sense this cellular dehydration, start thirst and trigger the release of antidiuretic hormone (ADH) so the kidneys keep water. In **hypovolaemic thirst** blood volume falls, as after bleeding: the kidneys release renin, which forms angiotensin II, and angiotensin II acts on the subfornical organ to start drinking. When ADH fails, in diabetes insipidus, the kidneys pass large volumes of dilute urine and the person is constantly thirsty.

Hormones, sexual development and sexual response

The **organisational effects** of hormones act early in development and permanently shape body and brain; activational effects act in adulthood, only while the hormone is present. **Charles Phoenix** and colleagues (1959) drew this distinction after female guinea pigs exposed to testosterone before birth mounted like males when given testosterone as adults.

The default course of a fetus is female. The **SRY gene** on the Y chromosome turns the primitive gonads into testes, which secrete testosterone, masculinising the body, and Müllerian-inhibiting substance, which makes the female ducts degenerate; **Alfred Jost** found that fetuses deprived of their gonads develop female ducts whatever their chromosomes. From puberty the pituitary gonadotropins, follicle-stimulating hormone (FSH) and luteinising hormone, make the gonads secrete androgens (chiefly testosterone), oestrogens (oestradiol) and progestins (progesterone). Both sexes make all three, in different amounts; hence Pinel's attack on the **mamawawa**, the men-are-men-and-women-are-women assumption that the sexes are separate, opposite categories.

In congenital adrenal hyperplasia, the adrenal glands of XX fetuses make excess androgens, partly masculinising girls' genitals; in androgen insensitivity syndrome, XY individuals with testes but unresponsive androgen receptors develop a female body without a uterus. The medial preoptic area, whose sexually dimorphic nucleus is several times larger in male rats (**Roger Gorski**, 1978), is critical for male sexual behaviour, the VMH for female. In 1991 **Simon LeVay** reported that INAH3, a nucleus of the anterior hypothalamus, was smaller in gay than in heterosexual men, nearer the size in women; the finding has not been consistently replicated, and reading it as a "female-like" brain is mamawawa thinking.

William Masters and **Virginia Johnson** (1966) observed volunteers in the laboratory and described a four-phase **sexual response cycle**: excitement, plateau (arousal levels off near its peak), orgasm and resolution, with a male refractory period after orgasm. **Helen Singer Kaplan** (1979) put desire first, in a three-phase model of desire, excitement and orgasm.

> **EXAM TRAP**
>
> Masters and Johnson: **excitement, plateau, orgasm, resolution**. A sequence that starts with plateau, or inserts desire anywhere, is not theirs: desire first is Kaplan, and the refractory period lies within the male resolution phase rather than being a phase of its own.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The master circadian pacemaker is the _____ nucleus of the hypothalamus.
2. Stages 3 and 4 (N3), with high-amplitude delta waves, form _____ sleep.
3. Slow-wave sleep begins when neurons of the _____ become active.
4. The chemical that builds up during waking and makes us drowsy is _____.
5. Narcolepsy follows the loss of _____ neurons in the lateral hypothalamus.
6. Lesions of the _____ hypothalamus produce hyperphagia and obesity.
7. In the arcuate nucleus, neurons releasing _____ directly trigger eating.

Answers: (1) suprachiasmatic, (2) slow-wave, (3) ventrolateral preoptic area, (4) adenosine, (5) orexin (hypocretin), (6) ventromedial, (7) neuropeptide Y.

5.9 The Biology of Learning, Memory and Amnesia HIGH YIELD · 47 QS

Every time you learn something, your brain physically changes. Where, and how? The answers come from patients whose memory broke in revealing ways, from animal lesion studies, and from the synapse itself. Section 1.7 treated memory as a psychological process; here the subject is its biology.

Amnesia: forward and backward in time ●●

Clinically, **amnesia** is a pathological loss of memory, far beyond ordinary forgetting. It usually follows brain damage or disease (head injury, stroke, surgery, encephalitis, thiamine deficiency); more rarely its cause is psychological, as in dissociative amnesia (Chapter 8). It is a disorder of memory, not a behavioural disturbance, a depressed mood or impaired perception.

Amnesias come in two directions in time from the onset (Figure 5.17). The first, **anterograde amnesia**, is the inability to form new long-term memories after the injury or onset: the patient chats normally, yet ten minutes later the conversation is gone. The second, **retrograde amnesia**, is the loss of memories formed before it. It usually has a **temporal gradient**: recent memories are lost most completely while remote ones survive, a pattern **Théodule Ribot** (1839 to 1916) described in *Les maladies de la mémoire* (1881), hence **Ribot's law**. The gradient shows that memories grow sturdier with age, through consolidation (5.9.4). In **source amnesia** you remember a fact but not where or when you learned it, a lapse common in old age and after frontal lobe damage.

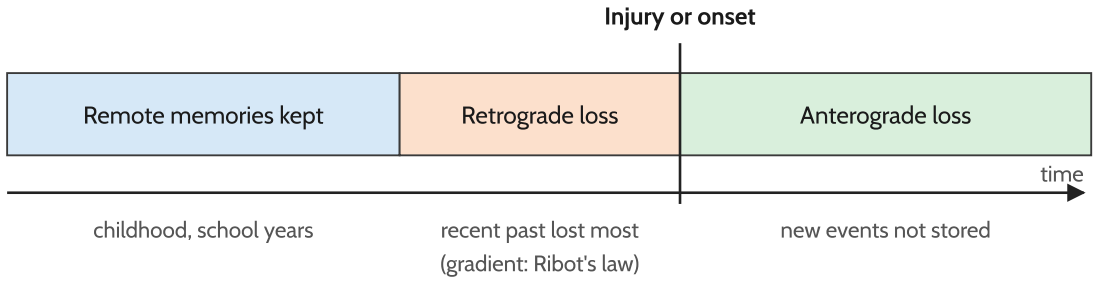


Figure 5.17 Amnesia is named by its direction from the onset. Retrograde loss is worst for the most recent past; anterograde loss blocks everything that follows.

> **EXAM TRAP**
>
> Anterograde means forward from the injury: **no new memories**. It is not "temporary memory loss" (transient global amnesia, Table 5.27), a visual deficit, or an inability to speak or recognise faces. Losing memories from **before** a concussion or stroke, or failing to retrieve old memories, is retrograde amnesia; dissociative amnesia is the psychological kind.

The search for the engram ●●

If learning changes the brain, the change must be somewhere. In 1904 the German zoologist **Richard Semon** (1859 to 1918) named it the **engram**: the physical trace an experience leaves in the brain. It is not a process of forgetting, a retrieval failure or a type of explicit memory.

Its most determined hunter, **Karl Spencer Lashley** (1890 to 1958), trained rats on mazes and then removed different parts of their cortex, expecting one lesion to erase the habit. None did. In *Brain Mechanisms and Intelligence* (1929) he stated two principles: **mass action** (the deficit depended on how much cortex was removed, not where) and **equipotentiality** (within a functional area, any part could do the work of any other). His review "In search of the engram" (1950) concluded that memories are not held in one place but distributed throughout the cortex.

His former student **Donald Olding Hebb** (1904 to 1985) explained why in *The Organization of Behavior* (1949). A memory is held by a **cell assembly**, a network of neurons linked because they fired together during an experience; spread across the cortex, it survives any single lesion in part. The links form at a **Hebbian synapse**: when neuron A repeatedly helps to fire neuron B, the connection from A to B strengthens ("cells that fire together wire together", a later summary). Long-term potentiation, found in 1973, bore him out (5.9.7). The Montreal neurosurgeon **Wilder Penfield** (1891 to 1976), stimulating the temporal lobes of awake epileptic patients, sometimes evoked vivid recollections, now regarded as reconstructions rather than stored replays.

> EXAM TRAP
>
> Lashley placed memories **throughout the cortex**: not in specific localised regions, the hippocampus alone or only the frontal and temporal lobes. The engram is the **physical trace** of a memory; the term is Semon's, the search Lashley's.

H.M. and the medial temporal lobes ●●●

Section 1.7 told H.M.'s story; here is the neural detail. **Henry Gustav Molaison** (1926 to 2008) had severe epilepsy that drugs could not control, apparently arising in both medial temporal lobes. In 1953, when he was 27, the neurosurgeon **William Beecher Scoville** (1906 to 1984) performed a **bilateral medial temporal lobectomy**, removing the inner parts of both temporal lobes: most of the hippocampus, the amygdala and adjacent cortex. The seizures eased. Scoville and the psychologist **Brenda Milner** (born 1918) reported the cost in 1957.

H.M. was left with severe anterograde amnesia for **declarative memory** (explicit memory: consciously recalled facts and events). He never learned his new address or recognised doctors he met after surgery, and he reread magazines with fresh interest. He had some retrograde amnesia for the few years before surgery. Spared were his intelligence (above average), language, personality, working memory (a normal digit span of about six) and remote memories, such as people and events from his childhood.

Most surprising, he could learn skills. Tracing a star while watching his hand only in a mirror, ten times a day on three days, he made fewer errors each day, yet each day denied ever doing the task. He also improved at rotary pursuit (keeping a stylus on a moving target), identified fragmented drawings sooner on the incomplete-pictures test (priming) and acquired a conditioned eyeblink: all implicit (nondeclarative) memory.

Three lessons followed. The medial temporal lobes consolidate new declarative memories but do not store remote ones, since his old memories survived. Short-term and long-term memory are separable. Explicit and implicit memory dissociate.

KEY STUDY: SCOVILLE AND MILNER (1957)

Their report on H.M. and other patients established medial temporal amnesia as a distinct syndrome. **Criticism:** the lesion also took the amygdala and nearby cortex, so the case alone could not show which structure mattered.

EXAM TRAP

H.M.'s most profound deficit was **anterograde amnesia** for declarative memory, not retrograde amnesia or a loss of procedural, semantic or working memory. Improving mirror drawing shows preserved **procedural** memory, not episodic, declarative or semantic memory. The surgery removed the **medial temporal lobes**, not prefrontal, parietal (angular gyrus) or occipital (lingual gyrus) cortex. It did not abolish motor coordination, selectively erase semantic memory, or cause "global retrograde amnesia with intact new learning", the reverse of his case.

The hippocampus: consolidation and space ●●●

Which structure mattered? Patient R.B., described by **Stuart Zola-Morgan**, **Larry Squire** and **David Amaral** (1986), suffered ischaemia (interrupted blood supply to the brain) at 52, when a pump failed during heart bypass surgery. His anterograde amnesia was milder than H.M.'s, and after his death the only obvious damage was to pyramidal cells in field CA1 of the hippocampus. Hippocampal damage alone, then, causes anterograde amnesia: a man who since a road accident remembers his past but not what he said ten minutes ago has probably injured his hippocampus.

The hippocampus is needed for **consolidation**, the conversion of fragile new memories into stable long-term ones, especially episodic and spatial declarative memories. In **standard consolidation theory**, it binds the scattered cortical pieces of an experience and replays them, partly during sleep (5.8), until cortical connections can hold the memory alone; remote memories then no longer need it. This explains both H.M.'s intact childhood and Ribot's gradient.

It is also the brain's map. In 1971 **John O'Keefe** (born 1939) and **Jonathan Dostrovsky** found **place cells** in the rat hippocampus, each firing only when the rat is in one particular spot; O'Keefe and **Lynn Nadel** argued in *The Hippocampus as a Cognitive Map* (1978) that the hippocampus builds a **cognitive map** of the environment. **May-Britt Moser** (born 1963) and **Edvard Moser** (born 1962) in 2005 found **grid cells** in the entorhinal cortex, the hippocampus's main input, each firing in a regular triangular grid of locations, a coordinate system for space. The three shared the 2014 Nobel Prize in Physiology or Medicine. **Eleanor Maguire** and colleagues (2000) found enlarged posterior hippocampi in London taxi drivers, larger the longer they had driven.

K.C., **Kent Cochrane** (1951 to 2014), studied by **Endel Tulving** (1927 to 2023), suffered diffuse brain damage including the medial temporal lobes in a motorcycle accident in 1981. He lost episodic, autobiographical memory for his whole life and could not imagine his future, yet his semantic knowledge, skills (chess, card games) and working memory survived: episodic and semantic memory are separable. **Clive Wearing** (born 1938), a British musician, lost both hippocampi and more to herpes simplex encephalitis in 1985; left with a memory span of seconds, he could still play the piano and conduct.

EXAM TRAP

The hippocampus **forms and consolidates new declarative memories** (especially episodic and spatial); remote memories end up in the wider cortex. It does not control temperature (hypothalamus), balance (cerebellum), speech (Broca's area, 5.10), procedural memory (basal ganglia, cerebellum), smell transduction (olfactory receptors, 5.6) or voluntary movement (motor cortex). Failure to form new memories after head injury points to the **hippocampus**, not the amygdala, prefrontal cortex, cerebellum, thalamus, hypothalamus, pituitary, brainstem, reticular formation (arousal) or spinal cord (reflexes).

Memory systems and the amygdala ●●

Memory, then, is several systems, each with its own structures (Table 5.26). Skills and habits depend on the cerebellum, basal ganglia and motor cortex, spared in H.M. but damaged in Parkinson's and Huntington's disease, which impair skill learning. **Mortimer Mishkin** (1926 to 2021) and **Jean Delacour** (1975) devised the **delayed nonmatching-to-sample test**: a monkey sees a sample object, then after a delay must choose the new object rather than the sample to find food. Monkeys with medial temporal lesions failed at longer delays, an animal model of H.M.; later lesions traced this object-recognition memory chiefly to the perirhinal cortex.

Table 5.26 Memory systems and the structures they depend on.

System	What it holds	Key structures	Classic evidence
Declarative (explicit)	Facts and events, consciously recalled	Hippocampus, medial temporal lobe; medial diencephalon	H.M., R.B., Korsakoff patients
Spatial	Places and routes	Hippocampus (place cells); entorhinal cortex (grid cells)	Water maze; taxi drivers
Object recognition	"Seen this before?"	Perirhinal cortex	Delayed nonmatching-to-sample
Semantic knowledge	Word meanings, general facts	Anterior and inferior temporal cortex	Semantic dementia (5.11)
Working memory	Items held and used now	Dorsolateral prefrontal cortex	H.M.'s normal digit span
Procedural	Motor and cognitive skills, habits	Cerebellum, basal ganglia (striatum), motor cortex	Mirror drawing; Parkinson's, Huntington's
Priming	Easier processing of a repeated stimulus	Neocortex	Incomplete-pictures test
Classical conditioning	Learned reflexes and fears	Cerebellum (eyeblink); amygdala (fear)	H.M.'s eyeblink conditioning
Emotional enhancement	Stronger memory for arousing events	Amygdala, acting on the hippocampus	Propranolol study (1994)

Why do you remember where you were when you heard shocking news, but not last Tuesday's lunch? **James McGaugh** (born 1931) showed that arousal hormones (adrenaline and cortisol) activate the amygdala, which then strengthens consolidation in the hippocampus and cortex: it boosts memories rather than storing them. **Larry Cahill**, McGaugh and colleagues (1994) told participants an emotional story; those given propranolol, a beta-adren-

ergic blocker, lost the usual memory advantage for its emotional part, while memory for a neutral version was unaffected. Flashbulb memories (1.7) are the everyday result.

> EXAM TRAP
>
> Procedural learning: **cerebellum, basal ganglia and motor cortex**, not the hippocampus with amygdala (declarative and emotional), the frontal association cortex (working memory, planning) or the hindbrain with spinal cord (vital functions, reflexes). Enhanced memory for emotional events: **amygdala**, not the hippocampus (which stores the event), prefrontal cortex, thalamus (sensory relay), hypothalamus (homeostasis) or suprachiasmatic nucleus (circadian clock, 5.8).

Korsakoff's syndrome and its look-alikes ●●●

The second classic site of amnesia is the medial diencephalon. **Korsakoff's syndrome** (Korsakoff's psychosis), described in the 1880s by the Russian neuropsychiatrist **Sergei Korsakoff** (1854 to 1900), is an amnesic disorder caused by deficiency of **thiamine (vitamin B1)**, classically in chronic alcoholism with a poor diet: alcohol supplies calories without vitamins and impairs thiamine absorption. It also occurs in malnourished non-drinkers, so the vitamin, not the alcohol, is the main cause. Thiamine is needed to metabolise glucose, so energy-hungry neurons die first.

It usually follows **Wernicke's encephalopathy**, described by **Carl Wernicke** (1848 to 1905) in 1881: an acute emergency with a triad of confusion (disorientation, lapses of attention), ataxia (loss of muscle coordination) and ophthalmoplegia (paralysis of eye movements), often with nystagmus (jerky eye movements). Attention and coordination are both impaired. Untreated survivors settle into Korsakoff's amnesia, and the continuum is called **Wernicke-Korsakoff syndrome**.

Damage centres on the mammillary bodies and the mediodorsal (dorsomedial) thalamus, a **diencephalic amnesia**, though it is diffuse. The result is dense anterograde amnesia, severe retrograde amnesia with a temporal gradient reaching back decades, confusion, apathy, poor insight and **confabulation**: detailed, plausible but false stories that fill memory gaps, told sincerely without intent to deceive. Malingering, by contrast, is deliberate faking for gain; delirium is an acute, fluctuating disturbance of consciousness; confusion is muddled thinking without invented stories. Thiamine supplements for heavy drinkers help prevent it. Treatment is high-dose thiamine given before any glucose (a glucose load can precipitate Wernicke's encephalopathy); it halts damage, but established amnesia is largely permanent.

> EXAM TRAP
>
> Korsakoff's: **vitamin B1 (thiamine)**, not B12, niacin, B6, folate or vitamin D, hypoxia or lead. Wernicke-Korsakoff syndrome impairs **both** attention and muscle coordination. Memory loss with confabulation and nystagmus in a heavy drinker is Wernicke-Korsakoff syndrome, not Alzheimer's, vascular or frontotemporal dementia. Sincere invented memories are **confabulation**, not malingering, delirium or confusion.

Table 5.27 Look-alike options: what each actually is or causes.

Option	What it is or causes

Option	What it is or causes
Vitamin B12 (cobalamin) deficiency	Anaemia (as in pernicious anaemia); subacute combined degeneration of the spinal cord
Niacin (B3) deficiency	Pellagra: dermatitis, diarrhoea, dementia
Pyridoxine (B6) deficiency	Peripheral neuropathy; seizures in infants
Folate (B9) deficiency	Neural tube defects in pregnancy; anaemia
Vitamin D deficiency	Rickets (soft bones)
Hypoxia	Loss of CA1 neurons: hippocampal amnesia, as in R.B.
Lead poisoning	Lowered IQ, inattention and anaemia in children
Alzheimer's disease	Progressive dementia; recent episodic memory fails first (5.11)
Parkinson's and Huntington's disease	Basal ganglia diseases (Huntington's is inherited, with chorea: jerky involuntary movements); procedural learning impaired
Pick's disease, frontotemporal dementia	Frontal and temporal atrophy; personality and language change before memory
Creutzfeldt-Jakob disease	Prion disease; rapidly progressive dementia with myoclonus (muscle jerks)
Vascular dementia	Repeated strokes; stepwise decline
Multiple sclerosis	Demyelination in brain and spinal cord; not a classic amnesic syndrome
Schizophrenia	Psychotic disorder (Chapter 8); not an amnesic disorder
Transient global amnesia	Sudden dense anterograde amnesia lasting hours, then full recovery
Concussion; electroconvulsive therapy	Short retrograde gap before the blow or treatment, some anterograde loss; mostly recovers

Memory at the synapse: Aplysia and LTP ●●

Learning was first traced to particular synapses in a sea slug. **Eric Kandel** (born 1929) chose *Aplysia* because its large neurons can be identified one by one. A touch to its siphon makes it withdraw its gill. Repeated harmless touches weaken the reflex (**habituation**) because sensory neurons release less transmitter onto motor neurons; a tail shock strengthens it (**sensitisation**) because interneurons release serotonin onto the sensory terminals, raising cyclic AMP and transmitter release. Long-term forms need the transcription factor CREB, protein synthesis and new synapses. Kandel shared the 2000 Nobel Prize.

In mammals the breakthrough came in 1973: **Timothy Bliss** (born 1940) and **Terje Lømo** (born 1935) stimulated the perforant path into the dentate gyrus of the hippocampus in anaesthetised rabbits. After a brief burst of high-frequency (tetanic) pulses, single test pulses produced larger responses for hours, and in later work for weeks or months. This **long-term potentiation** (LTP) is a lasting strengthening of synaptic transmission after high-frequency stimulation. Four properties fit a memory mechanism: **longevity** (it lasts); **input specificity** (only the stimulated synapses strengthen); **cooperativity** (several inputs must be active together); and **associativity** (a weak input active with a strong one is potentiated too, a cellular version of pairing a conditioned with an unconditioned stimulus).

The secret lies in the receptors for glutamate, the brain's main excitatory transmitter (Figure 5.18). **AMPA receptors** carry ordinary fast excitation. The **NMDA receptor** is special: at resting potential its channel is plugged by a magnesium ion (Mg^{2+}), so glutamate alone lets al-

most nothing through. Only when glutamate is bound *and* the postsynaptic membrane is already depolarised is the magnesium expelled, letting calcium (Ca^{2+}) flood in. The NMDA receptor is thus a **coincidence detector** of presynaptic and postsynaptic activity: Hebb's rule in molecular form. (Glycine must also bind as a co-agonist but cannot open it alone; the receptor is excitatory and postsynaptic, so it does not trigger vesicle release.)

In the dendritic spine, calcium activates CaMKII (calcium/calmodulin-dependent protein kinase II), which strengthens AMPA receptors and inserts extra ones, so the same glutamate release produces a larger response. Late LTP, lasting days or more, needs CREB-driven gene expression, new proteins and growth of more and larger dendritic spines; LTP does not involve fewer spines, glutamate depletion or neuron death (apoptosis). **Richard Morris** and colleagues (1986) tied it to behaviour: rats given AP5, an NMDA receptor blocker, showed no hippocampal LTP and were badly impaired at learning where a platform lay hidden in the Morris water maze, a pool of opaque water.

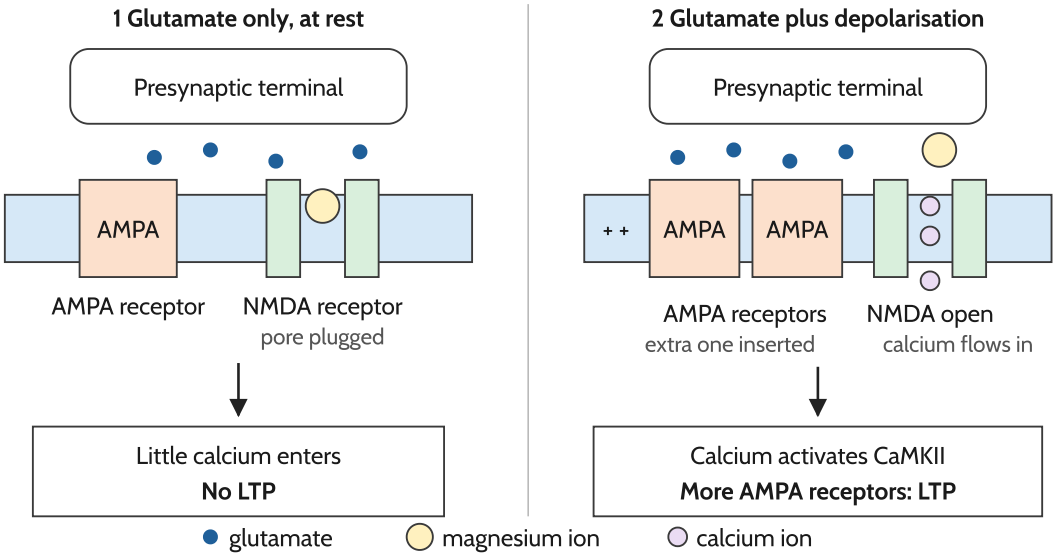


Figure 5.18 The NMDA receptor as a coincidence detector: magnesium plugs it when glutamate arrives alone (left) and leaves when the membrane is also depolarised (+ +), letting calcium in (right).

> **EXAM TRAP**
>
> LTP is a lasting **strengthening** after **high-frequency** stimulation, not a weakening (that is LTD), neuron death or myelination, and it is linked to **memory**, not thermoregulation, circadian rhythm or sleep. Its induction depends on **NMDA receptors**, not AMPA receptors alone, muscarinic M1 or GABA-A receptors. Magnesium blocks the NMDA channel, not the AMPA channel.

LTD and the dopamine teaching signal ●●

Synapses that could only strengthen would saturate. LTP's counterpart, **long-term depression** (LTD), is a lasting weakening of synaptic transmission. **Serena Dudek** and **Mark Bear** (1992) produced it in hippocampal field CA1 with prolonged low-frequency stimulation, about 1 Hz for 15 minutes. LTD there also needs NMDA receptors, but the modest, sustained calcium rise activates phosphatases (enzymes that remove phosphate groups) rather than kinases,

and AMPA receptors are pulled out of the membrane. A large, fast calcium surge gives LTP; a small, slow one gives LTD. LTD is not cell death, presynaptic GABA release, magnesium block of AMPA receptors or glutamate release without postsynaptic effect; high-frequency stimulation with NMDA receptors blocked produces neither LTP nor LTD.

Learning also needs a signal saying what is worth learning. **Wolfram Schultz** recorded from dopamine neurons of the ventral tegmental area and substantia nigra in monkeys learning that a light predicted juice. Before learning, the neurons burst when juice arrived unexpectedly. After learning, they burst at the light and ignored the expected juice; if the juice was withheld, firing paused when it should have come. Schultz, **Peter Dayan** and **Read Montague** (1997) identified this as a **reward prediction error**: reward received minus reward expected, positive for a pleasant surprise, zero for a predicted reward, negative for a disappointment. It is the surprise term of the Rescorla-Wagner model (Chapter 1), and the dopamine released strengthens the striatal synapses that were just active. Drug reward is covered in 5.4.

> **EXAM TRAP**
>
> The prediction-error signal comes from **dopamine neurons of the ventral tegmental area**, not serotonin release, GABA inhibition or acetylcholine release. LTD needs **prolonged low-frequency** stimulation; LTP needs brief high-frequency bursts.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Loss of memories from before an injury, worst for the most recent, is _____ amnesia.
2. Lashley concluded that memories are _____ throughout the cortex.
3. H.M.'s 1953 operation removed both _____ lobes.
4. R.B.'s amnesia followed damage to hippocampal field _____.
5. The beta blocker _____ abolished the memory advantage of emotional events.
6. Korsakoff's syndrome is caused by deficiency of _____ (vitamin B1).
7. At rest the NMDA channel is blocked by _____ ions.
8. Dopamine neurons that pause when an expected reward is omitted signal a reward _____ error.

Answers: (1) retrograde, (2) distributed, (3) medial temporal, (4) CA1, (5) propranolol, (6) thiamine, (7) magnesium, (8) prediction.

5.10 Lateralisation, the Split Brain and Language

MEDIUM YIELD · 28 QS

In 1861 a Paris surgeon found that a man who could say only one syllable had damage on the left side of his brain. That finding raised the two questions of this section: why do the hemispheres do different jobs, and where is language produced and understood?

Discovering the dominant hemisphere

Cerebral lateralisation is the tendency of a function to depend more on one cerebral hemisphere than on the other. In 1836 **Marc Dax**, a French country doctor, reported that none of his 40 or so patients with speech loss had damage confined to the right hemisphere, but nobody took notice. In 1861 **Paul Broca** (1824 to 1880) described Louis Victor Leborgne, nicknamed 'Tan' because it was almost the only syllable he could utter. Leborgne understood speech; at autopsy his lesion lay in the left inferior frontal cortex, and by 1864 Broca had seen the same in more patients. Because language, and later skilled movement, proved to depend on the left hemisphere, it was called the **dominant hemisphere** or major hemisphere, and the right the **minor hemisphere**. The second label is outdated.

Two tests reveal the language hemisphere of a living person. In the **Wada test** (sodium amytal test) of **Juhn Atsushi Wada** (1949), the barbiturate sodium amytal is injected into one carotid artery, anaesthetising that hemisphere for a few minutes; if it holds language, the patient falls mute. It is used before brain surgery. The harmless **dichotic listening** test plays different words or digits to the two ears at once. **Doreen Kimura** (1961) found that most people report more right-ear items, a right-ear advantage for words, because each ear's stronger pathway runs to the opposite hemisphere and the left hemisphere handles language; for melodies the advantage passes to the left ear.

About 90 percent of people are right-handed, and handedness is loosely tied to language. Language is in the left hemisphere in about 95 percent of right-handers and in most left-handers (roughly 70 percent); the rest have it on the right or in both hemispheres.

> EXAM TRAP
>
> The left hemisphere is the **major** (dominant) hemisphere. 'Spatial and pattern sense' and 'centre of musical ability' describe the right hemisphere, and 'minor hemisphere' is the old name for the right. Most left-handers are still left-dominant for language.

What each hemisphere does best

Table 5.28 summarises the asymmetries. Phonological processing means handling the sound structure of words; right-hemisphere control of spatial attention explains why neglect follows right parietal damage (5.11).

Table 5.28 Functional asymmetries in most right-handers. Each is a relative bias, not a monopoly.

Left hemisphere	Right hemisphere
Speech, phonology, grammar and syntax	Emotional prosody; understands simple words
Reading, writing, arithmetic	Visuospatial processing; spatial and pattern sense
Analytic, sequential processing	Holistic processing of whole patterns
Verbal memory	Non-verbal (visual, spatial) memory
Speech sounds	Music, melody, non-speech sounds
Skilled, complex movement	Faces; emotional expression; spatial attention

One right-hemisphere skill lies inside language itself. **Prosody** is the melody of speech: the intonation and emotional tone that let 'Oh, great' mean delight or sarcasm. Emotional

prosody depends on right-hemisphere regions homotopic to the language areas, that is, in their mirror-image positions. **Elliott Ross** (1981) described the **aprosodias**: damage opposite Broca's area leaves speech flat and monotonous, and damage opposite Wernicke's area leaves the patient deaf to the emotion in other voices, while words and grammar are spared.

The hemispheres differ in structure too. **Norman Geschwind** (1926 to 1984) and **Walter Levitsky** (1968) found the **planum temporale**, auditory cortex behind Heschl's gyrus that includes part of Wernicke's area, larger on the left in about two thirds of 100 brains. Popular talk of logical 'left-brained' and creative 'right-brained' people is a myth. Lateralisation is a matter of degree (the right-ear advantage is slight), every complex skill, creativity included, uses both hemispheres together, and imaging finds no 'left-brained' individuals.

> EXAM TRAP
>
> The function most reliably lateralised to the right is **visuospatial processing and prosody**; phonological processing, grammatical syntax and verbal memory are left-hemisphere. Speech intonation depends on **right-hemisphere regions homotopic to the language areas**, not Broca's area, the primary motor cortex or the occipital lobe.

The split brain ●●

The most striking evidence came from cutting the corpus callosum, the largest commissure linking the hemispheres (5.2). In 1953 **Ronald Myers** and **Roger Wolcott Sperry** (1913 to 1994) cut both the corpus callosum and the optic chiasm of cats, so each eye fed one hemisphere. A cat trained on a visual task with one eye patched had to relearn it from scratch with the other eye, as if it had two brains: the corpus callosum transfers learning between hemispheres.

From the 1960s surgeons cut the corpus callosum in people with intractable epilepsy, to stop seizures spreading between hemispheres. This **corpus callosotomy** (often called commissurotomy) reduced the seizures and left patients almost unchanged in daily life. Sperry and his student **Michael Gazzaniga** (born 1939) then tested them, work that won Sperry the Nobel Prize in 1981.

> KEY STUDY: SPERRY AND GAZZANIGA'S SPLIT-BRAIN PATIENTS (1960S)
>
> **Method:** the left visual field projects to the right hemisphere and the right field to the left (5.6). In **tachistoscopic presentation** the patient fixates a central dot while a picture is flashed to one side for about a tenth of a second, too fast for an eye movement. Each hand, hidden from view, reports to and is controlled by the opposite hemisphere. **Result:** objects flashed to the right visual field were named at once. Objects flashed to the left field reached only the non-speaking right hemisphere: the patient said he saw nothing, yet his left hand picked out the object by touch (Figure 5.19). Neither eye nor field was blind. **Criticism:** the patients were few, and long-standing epilepsy may have reorganised their brains.

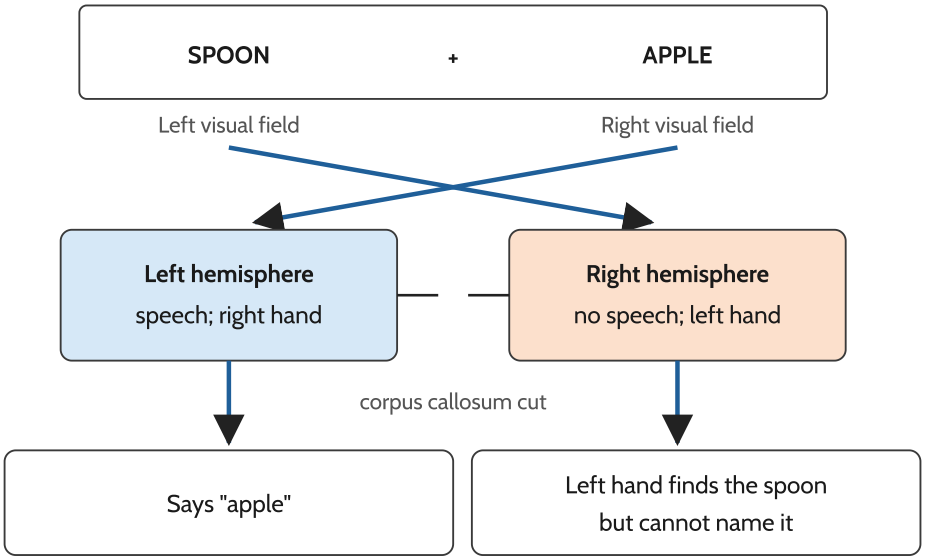


Figure 5.19 The split-brain test. Each visual field reaches only the opposite hemisphere, so the object seen by the right hemisphere can be found by the left hand but not named.

Four conclusions followed. Each hemisphere perceives, learns and remembers independently: shown two pictures at once, each hand fetches the object its own hemisphere saw. Language is not bilateral: in right-handers speech comes from the left hemisphere, and the right cannot speak, though it understands simple words. The right hemisphere is better at spatial tasks. And each hemisphere completes what it sees: shown **chimeric figures**, photographs joining halves of two different faces, patients of **Jerre Levy**, **Colwyn Trevarthen** and Sperry (1972) reported one whole face, naming the right-field half but pointing to the left-field half.

In Gazzaniga's best-known demonstration, one patient's left hemisphere saw a chicken claw and his right hemisphere a snow scene; his right hand chose a chicken, his left a shovel. Asked why, he said at once that the shovel was for cleaning out the chicken shed. Gazzaniga called this the **left-hemisphere interpreter**, which invents plausible reasons for behaviour whose real cause it does not know. A few patients show **intermanual conflict**: one man's left hand would switch off a television programme he was enjoying.

Outside the laboratory both hemispheres share the same eyes, ears and body, and some information crosses through the brainstem or by cross-cueing (one hemisphere signalling to the other with a frown). Cutting the corpus callosum therefore does not kill, blind or abolish language; its cost is difficulty coordinating activities that need both hemispheres.

> EXAM TRAP
>
> An object in the left visual field cannot be named but can be selected with the left hand; the patient is not blind in that eye or field and has not forgotten the object. The Nobel Prize went to Roger Sperry, not to **Karl Spencer Lashley** (1890 to 1958; the engram, 5.9), **Wilder Graves Penfield** (1891 to 1976; cortical stimulation mapping and the homunculus, 5.6) or **Donald Olding Hebb** (1904 to 1985; cell assemblies, 1949).

Broca's area and Wernicke's area ●●●

Lateralisation tells you which side; localisation tells you where within it. **Broca's area** lies in the left inferior frontal gyrus, in the pars opercularis and pars triangularis (areas 44 and 45, 5.2), just in front of the face area of the primary motor cortex. This frontal-lobe area serves speech production: programming articulation and assembling grammatical sentences. It is not the seat of comprehension, memory consolidation (the hippocampus, 5.9) or motor coordination (the cerebellum, 5.6).

In 1874 **Carl Wernicke** (1848 to 1905) described patients who spoke fluently but understood little, with damage further back. **Wernicke's area** is the posterior superior temporal gyrus of the dominant hemisphere (area 22), the classic centre for comprehending spoken and written language. In front of it, in Heschl's gyrus, the primary auditory cortex registers sounds without understanding them. The **arcuate fasciculus**, a bundle of association fibres, joins the two language areas. Behind Wernicke's area, where the temporal, parietal and occipital lobes meet, the **angular gyrus** (area 39) links a word's seen form to its sound and meaning, so it is vital for reading. The usual decoys have other jobs: the occipital pole holds primary visual cortex, the postcentral gyrus primary somatosensory cortex, the right parietal lobe spatial attention, and the anterior cingulate attention and error monitoring.

> EXAM TRAP
>
> Ria speaks slowly and laboriously in keywords that make sense. 'Frontal lobe' is true but too broad, and a temporal-lobe (Wernicke's area) lesion leaves speech fluent: the answer is **Broca's area**, also the site after a left frontal stroke with non-fluent speech. For comprehension the answer is **Wernicke's area** (posterior superior temporal gyrus), never the inferior frontal gyrus, Heschl's gyrus or the angular gyrus.

The Wernicke-Geschwind model and the dual streams

In 1965 Geschwind revived the ideas of Broca, Wernicke and Dejerine as the **Wernicke-Geschwind model**, a left-hemisphere circuit of seven parts (Figure 5.20). To repeat a heard word, the primary auditory cortex passes the sound to Wernicke's area, where it is understood; the arcuate fasciculus carries it to Broca's area, which selects an articulation programme; and the face area of the motor cortex drives the speech muscles. To read aloud, the primary visual cortex sends the written word to the angular gyrus, which converts it into its auditory code for Wernicke's area, and the same output path follows. Damage to each box or link predicts a particular aphasia.

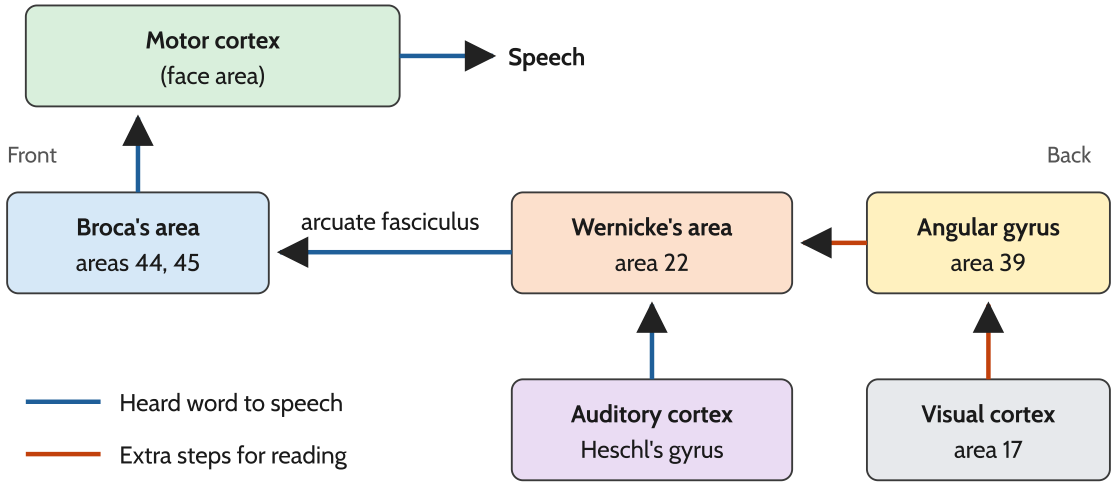


Figure 5.20 The Wernicke-Geschwind model, left hemisphere, front at the left. The solid route repeats a heard word; reading aloud enters by the dashed route and then follows the same output path.

The model's limits are serious. Removing Broca's area alone often leaves no lasting aphasia, real lesions are large and include white matter, and almost every aphasic patient has both expressive and receptive problems. Modern research favours the **dual-stream model** of **Gregory Hickok** and **David Poeppel** (2007). A ventral stream along the temporal lobe maps speech sounds onto meaning in both hemispheres; a dorsal stream from the temporoparietal junction to the frontal lobe, including the arcuate fasciculus, maps sounds onto articulation and is strongly left-lateralised. That is why damage to the dorsal route can spare understanding but ruin repetition.

Aphasia and its syndromes ●●●

Aphasia is an acquired disturbance of language function due to organic brain disease, most often a left-hemisphere stroke, and is the common term for all types of language loss. Dysphasia is often used for partial loss, dyslexia is a disorder of reading, and 'lexphasia' is not a term. Aphasia is a disorder of language, not of intellect, abstract thinking or memory, which fail in dementia and amnesia (5.9, 5.11). Clinicians classify aphasias by fluency, comprehension, repetition and naming (Table 5.29).

Broca's aphasia (expressive or non-fluent aphasia) is primarily a deficit of speech production. Speech is slow, effortful, poorly articulated and telegraphic, made of content words with the small grammatical words missing (**agrammatism**): 'Yes … Monday … Dad and Dick … Wednesday nine o'clock … doctors … and … teeth.' The keywords make sense, comprehension is relatively preserved, repetition is impaired, and the patient knows it and is frustrated. Right hemiparesis is common because the motor cortex lies next door. **Wernicke's aphasia** (receptive or fluent aphasia) is the mirror image: fluent, grammatical speech that is empty of meaning, full of **paraphasias** (wrong words or sounds) and neologisms (invented words), sometimes reaching word salad, with severely impaired comprehension, poor repetition and little awareness.

Table 5.29 The aphasia syndromes. Naming is impaired in all of them.

Aphasia	Speech	Comprehension	Repetition	Usual lesion

Aphasia	Speech	Comprehension	Repetition	Usual lesion
Broca's	Non-fluent, telegraphic	Relatively good	Poor	Broca's area
Wernicke's	Fluent, empty	Poor	Poor	Posterior superior temporal gyrus
Conduction	Fluent	Good	Very poor	Arcuate fasciculus, supramarginal gyrus
Global	Non-fluent	Poor	Poor	Whole perisylvian region
Transcortical motor	Non-fluent	Good	Good	Anterior or superior to Broca's area
Transcortical sensory	Fluent, echolalic	Poor	Good	Posterior watershed zone
Mixed transcortical	Non-fluent, echolalic	Poor	Good	Both watershed zones
Anomic	Fluent, word-finding pauses	Good	Good	Variable

A lesion of the arcuate fasciculus or the supramarginal region produces **conduction aphasia**: fluent speech and good comprehension but very poor repetition. A large lesion around the whole lateral fissure (the perisylvian region) causes **global aphasia**, with everything impaired. The transcortical aphasias follow lesions in the border (watershed) zones that spare the perisylvian loop, so repetition survives. In **transcortical motor aphasia** the lesion lies anterior or superior to Broca's area, sparing it: speech is non-fluent but repetition is preserved. **Transcortical sensory aphasia** resembles Wernicke's aphasia with intact repetition, often as echolalia (parroting what is heard); mixed transcortical aphasia combines both. In anomic aphasia the main problem is word finding, but **anomia** occurs in every aphasia.

EXAM TRAP

Damage to Broca's area gives non-fluent, effortful speech with relatively intact comprehension. Fluent aphasia with poor comprehension is Wernicke's, anomia occurs in every aphasia, failure to recognise faces is prosopagnosia (5.11), and hearing, reading and colour naming depend on other regions.

EXAM TRAP

Amity 2023 asked what '_____ is closely related to grammar and pronunciation' and keyed **aphasia**, the disorder in which grammar and pronunciation break down. The key is doubtful: the brain region concerned is **Broca's area**, which assembles grammar and programmes articulation, so expect the key 'aphasia' but know that the region is Broca's area. Wernicke's area serves comprehension, and agnosia is failure to recognise despite intact senses (5.11).

Reading, writing and speech beyond aphasia

Alexia is an acquired loss of reading and agraphia of writing. **Joseph Jules Dejerine** (1849 to 1917) linked alexia with agraphia to the left angular gyrus in 1891, and in 1892 described **alexia without agraphia** (pure alexia) after damage to the left occipital cortex and the splenium, the rear of the corpus callosum. His patient could write but not read what he had written: words seen by the intact right visual cortex could not cross to the left-hemisphere language areas. Such patients often cannot name colours either. Acalculia is an acquired loss of calcu-

lation. **Gerstmann syndrome** (**Josef Gerstmann**, 1920s) follows left angular gyrus damage and combines finger agnosia, left-right confusion, agraphia and acalculia.

Not every speech problem is aphasia. **Dysarthria** is faulty articulation from weak or uncoordinated speech muscles, and **apraxia of speech** is faulty programming of articulatory movements; in both, language itself is intact. **Developmental dyslexia** and other language-based learning difficulties are present from childhood despite adequate intelligence and teaching. Their core weakness is phonological, linked to the left perisylvian language network including Broca's area rather than to the occipital lobe (vision), cerebellum (coordination) or amygdala (emotion); the disorders are in Chapter 8. Finally, deaf signers with left-hemisphere damage develop sign-language aphasias of the Broca and Wernicke types, while right-hemisphere damage spares signing: the left hemisphere is specialised for language, not for sound.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The Wada test injects _____ into one carotid artery.
2. Dichotic listening shows a _____-ear advantage for words.
3. An object flashed to the left visual field reaches the _____ hemisphere and can be picked out with the _____ hand.
4. Emotional prosody depends on _____-hemisphere regions homotopic to the language areas.
5. Wernicke's area lies in the posterior _____ gyrus (area 22).
6. Fluent speech and good comprehension but very poor repetition is _____ aphasia.
7. Non-fluent speech with preserved repetition is _____ aphasia.

Answers: (1) sodium amytal, (2) right, (3) right; left, (4) right, (5) superior temporal, (6) conduction, (7) transcortical motor.

5.11 Brain Damage, Neurodegeneration and Neuropsychological Syndromes

MEDIUM YIELD · 32 QS

Much of what we know about the healthy brain comes from damaged ones: what a patient loses, and what survives, shows what the damaged part did. This section covers how brains are damaged, the major neurological diseases and the classic neuropsychological syndromes.

Causes of brain damage

Brain damage has six main causes. A **brain tumour** (neoplasm) is a mass of cells growing independently of the rest of the body. About 20 percent are **meningiomas**, encapsulated in their own membrane between the meninges, harmful only by pressure and almost always benign. Most others are **infiltrating**, growing diffusely into brain tissue and usually malignant; **gliomas**, from glial cells, are the commonest. About 10 percent are **metastatic**, seeded through the blood from cancers elsewhere, often the lungs.

A **stroke** is a sudden-onset cerebrovascular disorder. Around the dead tissue (the infarct) lies the **penumbra**, tissue that may recover or die over the next few days and that treatment tries

to save. **Cerebral haemorrhage** is bleeding from a ruptured vessel, often an aneurysm. **Cerebral ischaemia** is a blocked blood supply, from *thrombosis* (a plug formed on the spot), *embolism* (a plug carried from elsewhere) or *arteriosclerosis* (walls narrowed by fat deposits). Starved neurons release excess glutamate, which overactivates NMDA receptors and floods neighbouring neurons with Na^+ and Ca^{2+}, killing them over a day or two: **excitotoxicity** (Section 5.3).

A **closed-head traumatic brain injury** leaves the skull intact. A **contusion** (bruised brain) forms where the brain strikes the skull, at the site of the blow (coup) and often on the opposite side (**contrecoup injury**). **Concussion**, now called mild traumatic brain injury, disturbs consciousness without structural damage, but repeated ones produce **chronic traumatic encephalopathy**, the dementia of boxers and American football players. The confused period after a head injury in which new memories are not stored is **post-traumatic amnesia**; its length is the best single guide to severity.

Infection causes **encephalitis** (brain inflammation): herpes simplex encephalitis attacks the medial temporal lobes (Clive Wearing, Section 5.9), bacteria cause meningitis, neurosyphilis causes **general paresis** (psychosis and dementia) and HIV can cause dementia. **Neurotoxins** include lead, mercury, alcohol (Korsakoff's syndrome, Section 5.9) and antipsychotic drugs (tardive dyskinesia, Section 5.4). **Genetic** causes include recessive genes, the dominant gene of Huntington's disease and the extra chromosome 21 of Down syndrome. Whatever the cause, neurons die by **necrosis**, passive swelling and bursting within hours that inflames nearby tissue, or by **apoptosis**, programmed shrinkage over a day or two without inflammation.

Epilepsy

Epilepsy is a disorder of recurrent, spontaneous seizures arising from chronic brain dysfunction; a single seizure from a high fever does not count. Neurons discharge in synchronous bursts, often because GABA inhibition fails, and the EEG shows high-amplitude spikes. Many patients first sense an **aura** (a strange smell, déjà vu), which betrays where the seizure starts. **Generalised** seizures involve the whole brain, **focal** (partial) seizures only part of it (Table 5.30). Simple focal seizures are sometimes called Jacksonian after **John Hughlings Jackson** (1835 to 1911), whose **Jacksonian march** is jerking that spreads from fingers to arm to face as the discharge crosses the motor homunculus. Complex focal seizures usually begin in the temporal lobe (**temporal lobe epilepsy**) and bring **automatisms** such as fumbling with buttons. The International League Against Epilepsy (ILAE) classification of 2017 grades focal seizures by awareness instead: **focal aware** (simple) and **focal impaired awareness** (complex).

Table 5.30 The main seizure types.

Seizure	Consciousness	What you see	EEG
Tonic-clonic (grand mal)	Lost	Tonic stiffening, then clonic rhythmic jerking; tongue biting, cyanosis; postictal confusion	Spikes across the whole cortex
Absence (petit mal)	Lost for seconds	Blank stare, eyelid flutter; tone kept; mainly children	3-per-second spike-and-wave
Simple focal	Kept	Signs in one limb; Jacksonian march	Focal spikes
Complex focal	Impaired	Aura, automatisms, no memory of the episode	Temporal spikes

A seizure lasting more than five minutes, or repeated seizures without recovery of consciousness between them, is **status epilepticus** (older definitions said 30 minutes), a medical emergency. Anticonvulsants control seizures but cure none; resistant cases may have the focus removed (H.M., Section 5.9) or the corpus callosum cut (split brain, Section 5.10).

> EXAM TRAP
>
> Epilepsy is a neurological disorder but **not a neurodegenerative one**. Parkinson's, Alzheimer's and Huntington's diseases are progressive losses of neurons; epilepsy is episodic, often with normal function between seizures.

Parkinson's and Huntington's diseases ●●

Parkinson's disease, described in 1817 by **James Parkinson** (1755 to 1824) as the "shaking palsy", affects 1 to 2 percent of people over 65. Its signs are resting tremor, rigidity, **bradykinesia** (slowness) or akinesia (difficulty starting movement), postural instability, a masked face and a shuffling gait; loss of smell and depression may come years earlier, and dementia often follows. Dopamine neurons of the **substantia nigra pars compacta** die, so the **nigrostriatal pathway** no longer delivers dopamine to the striatum (Figure 5.21); in 1960 **Oleh Hornykiewicz** found the patients' striatum almost empty of dopamine. Surviving nigral neurons hold **Lewy bodies**, clumps of alpha-synuclein. **L-dopa**, the dopamine precursor, which unlike dopamine crosses the blood-brain barrier, and dopamine agonists relieve symptoms for some years; **deep brain stimulation** of the subthalamic nucleus helps when drugs fail. In 1982 young drug users in California who injected a "synthetic heroin" contaminated with **MPTP** developed permanent parkinsonism (reported by **J. William Langston** in 1983); MPTP destroys nigral dopamine neurons in monkeys too, giving the best animal model of the disease.

> EXAM TRAP
>
> Parkinson's disease is **dopamine** loss from the **substantia nigra pars compacta**. It is not degeneration of the locus coeruleus (noradrenaline), raphe nuclei (serotonin), spinal motor neurons (motor neuron disease, polio), hippocampal pyramidal cells (memory) or cholinergic basal forebrain (Alzheimer's), nor a serotonin, acetylcholine or GABA deficiency. The subthalamic nucleus is the stimulation target, not the lesion.

Huntington's disease, described by **George Huntington** in 1872, is rarer (about 1 in 7,500): **chorea** (jerky, dance-like involuntary movements), psychiatric symptoms and cognitive decline ending in dementia, beginning at about 40. An autosomal dominant expanded CAG repeat in the *huntingtin* gene on chromosome 4 (Section 5.5) kills the GABA neurons of the striatum, caudate first. Parkinson's is hypokinetic, Huntington's hyperkinetic.

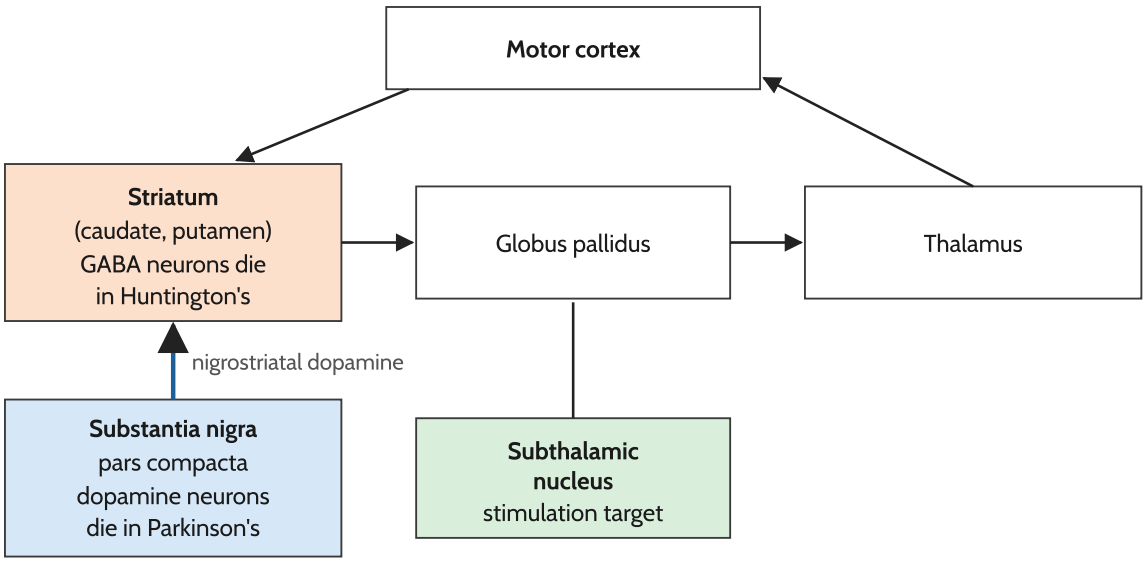


Figure 5.21 The basal ganglia motor loop. Parkinson's disease removes the nigrostriatal dopamine input (thick arrow); Huntington's disease destroys the striatum's own GABA neurons.

Alzheimer's disease and other dementias ●●

Dementia is an acquired, usually progressive decline of global intellect (memory, abstraction, judgement) that disrupts daily life; its criteria belong to Chapter 8, and "senile dementia" and "atrophy" are vague older labels. The commonest cause is **Alzheimer's disease**, described in 1906 by **Alois Alzheimer** (1864 to 1915) in his patient **Auguste Deter**. Onset is insidious and progression gradual: recent episodic memory fails first, then language (anomia), visuospatial and executive functions. The brain shows **amyloid plaques** (beta-amyloid, outside cells), **neurofibrillary tangles** (tau, inside cells) and neuron loss that begins in the entorhinal cortex and hippocampus. Death of cholinergic neurons in the **nucleus basalis of Meynert** leaves the cortex short of acetylcholine, so treatment uses **cholinesterase inhibitors** (donepezil, rivastigmine, galantamine) and **memantine**, an NMDA receptor antagonist. The **APOE e4** allele raises risk; rare early-onset families carry APP or presenilin mutations, and because APP lies on chromosome 21, almost everyone with Down syndrome has plaques and tangles by 40.

> EXAM TRAP
>
> Acetylcholine deficiency means **Alzheimer's disease**, not "senile dementia", "atrophy" or Korsakoff's syndrome (thiamine deficiency, Section 5.9). Alzheimer's is not a loss of GABA (Huntington's), noradrenaline or histamine (Section 5.3).

Table 5.31 compares the other dementias. Smell loss is an early sign of both Parkinson's and Alzheimer's disease.

Table 5.31 The dementias compared.

Dementia	Pathology	Course and distinctive signs
Alzheimer's disease	Plaques, tangles; cholinergic loss	Insidious, gradual; memory first, then language and visuospatial skills

Dementia	Pathology	Course and distinctive signs
Dementia with Lewy bodies	Cortical Lewy bodies	Fluctuating cognition; recurrent well-formed visual hallucinations; parkinsonism; REM sleep behaviour disorder; severe antipsychotic sensitivity
Vascular dementia	Multiple infarcts	Stepwise decline; patchy deficits
Frontotemporal (Pick's disease)	Frontal and temporal atrophy; Pick bodies	Early personality change, disinhibition, or language variants; memory spared early
Creutzfeldt-Jakob disease	Prions (infectious mis-folded protein)	Very rapid decline; myoclonic jerks

EXAM TRAP

Lewy body dementia means **visual** hallucinations, not auditory ones (typical of schizophrenia). Aphasia points to frontotemporal or late Alzheimer's disease, and smell loss is not specific.

Multiple sclerosis ●

Multiple sclerosis is an autoimmune attack on the myelin that oligodendrocytes make in the central nervous system. Hard scars, sclerotic **plaques**, form in the white matter, **disseminated in time and space**. It usually starts in early adulthood with a **relapsing-remitting** course of visual disturbance, weakness, numbness and fatigue, and is commoner in women and far from the equator. Contrast alpha-synuclein deposits (Parkinson's), beta-amyloid plaques (Alzheimer's) and degeneration of lower motor neurons (polio, and amyotrophic lateral sclerosis, which kills upper motor neurons too).

Agnosia, apraxia and other higher-function disorders ●●●

Damage to association cortex disturbs higher functions rather than simple sensation or movement. Most of their names begin with Greek "a-" (without) and are best learned together (Table 5.32).

Table 5.32 Disorders of higher function.

Disorder	What is lost	Note
Dementia	Global intellect and abstraction	Acquired, progressive
Amnesia	Memory	Section 5.9
Aphasia, alexia, acalculia	Language, reading, calculation	Section 5.10
Anomia	Naming	Word-finding failure
Agnosia	Recognition	Sensation intact
Apraxia	Learned skilled movement (motor skills)	Strength, coordination intact
Ataxia	Coordination	Cerebellum (Section 5.6)
Dyspraxia	Coordination, from childhood	Developmental coordination disorder
Anosmia	Smell	Olfactory damage

Agnosia, a term coined by **Sigmund Freud** in 1891, is failure to recognise despite intact sensation and intellect. **Heinrich Lissauer** (1890) split visual object agnosia into **apperceptive** (no whole percept forms, so the patient cannot copy a drawing) and **associative** (the drawing is copied but not recognised). In **prosopagnosia** (face blindness), faces are seen as faces but not recognised, even those of close family, who are known at once by voice. It follows bilateral or right-sided damage to the fusiform gyrus at the occipitotemporal junction, the **fusiform face area** identified by **Nancy Kanwisher** and colleagues in 1997. **Simultanagnosia**, seeing only one object at a time, is part of Bálint's syndrome after bilateral posterior parietal damage. **Anosognosia** (**Joseph Babinski**, 1914) is unawareness of one's own deficit. **Astereognosis** is failure to recognise objects by touch, and **auditory agnosia** failure to recognise sounds.

> EXAM TRAP
>
> Prosopagnosia is inability to **recognise faces**. The IHBAS 2024 key gives "distinguish tastes", which is wrong (loss of taste is ageusia); it is also not loss of facial expression or of object recognition. Face-recognition loss after a stroke is an **agnosia**, not aphasia, apraxia or ataxia, and prosopagnosia beats the broader "visual agnosia". The lesion is fusiform, not in the primary motor cortex (movement), dorsolateral prefrontal cortex (executive function) or cerebellar vermis (balance).

Apraxia (**Hugo Liepmann**, 1900) is inability to perform learned purposeful movements despite intact strength, sensation, coordination and comprehension. In **ideomotor apraxia** the patient cannot mime an act on command (show how you use a comb) though he does it naturally; in **ideational apraxia** he cannot sequence the steps of an act such as brushing his hair. **Constructional apraxia** spoils copying and block building, **dressing apraxia** putting on clothes. The lesion is usually left parietal or frontal.

> EXAM TRAP
>
> "Impairment in motor performance and coordination" is **dyspraxia** (developmental coordination disorder; Amity 2023 key). Apraxia is acquired loss of skilled movement with coordination intact; ataxia is cerebellar incoordination; anomia is failure to name, alexia to read.

Hemispatial neglect •

After right posterior parietal damage, patients with **hemispatial neglect** ignore the left side of space and of their own body. They eat from the right of the plate, shave only the right side of the face, bisect lines to the right and miss left targets on cancellation tests (Figure 5.22). In **extinction**, a left stimulus is noticed alone but missed when paired with a right one, and anosognosia is common. Left neglect is far commoner and more severe than right neglect, because the right hemisphere attends to both sides of space. Neglect is inattention, not blindness: in **homonymous hemianopia** the patient knows half the visual field is lost and compensates. Bilateral damage to primary visual cortex causes **cortical blindness** (bilateral homonymous hemianopia), sometimes denied in **Anton's syndrome**; **hemiplegia** (one-sided paralysis) follows motor cortex or internal capsule damage.

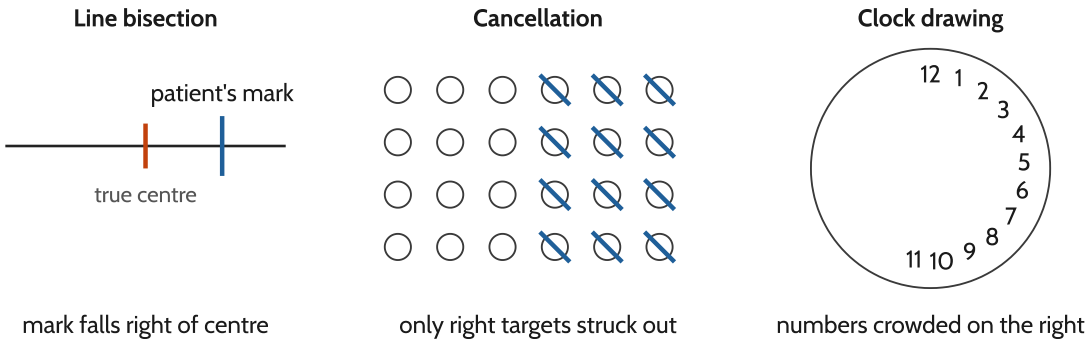


Figure 5.22 Left neglect after right parietal damage on three bedside tests (dashed line: true centre). Everything on the left is ignored.

EXAM TRAP

Right parietal lesion: **left** neglect. Anterograde amnesia (Section 5.9), expressive (Broca's) aphasia (Section 5.10), anosmia, prosopagnosia and bilateral homonymous hemianopia are distractors.

Phineas Gage and the prefrontal syndromes ●●●

KEY STUDY: PHINEAS GAGE (1848)

In 1848 an explosion drove a tamping iron through the left cheek and frontal lobe of **Phineas Gage** (1823 to 1860), a 25-year-old railway construction foreman in Vermont. His speech, memory and intelligence survived, but the reliable foreman became impulsive, profane and poor in judgement, and soon lost his job. His physician **John Martyn Harlow** reported the case in 1848 and, in 1868, that friends found him "no longer Gage". In 1994 **Hanna Damasio** and colleagues reconstructed the injury from his skull and placed it in the **ventromedial prefrontal cortex**. Criticism: the record is thin, and Gage later drove stagecoaches in Chile for several years before dying in San Francisco, so he may have partly recovered.

Gage is the classic evidence that the frontal lobes govern personality and social behaviour. Damage to different prefrontal regions produces different frontal lobe syndromes (Table 5.33). Dorsolateral damage causes the **dysexecutive syndrome**: **executive functions** (planning, sequencing, set-shifting and inhibition, using working memory) break down, and the patient shows **perseveration**, repeating a response after it has stopped working. In the **Wisconsin Card Sorting Test** of **David Grant** and **Esta Berg** (1948) cards are sorted by colour, shape or number, the rule being learned only from feedback and changed without warning; in 1963 **Brenda Milner** showed that frontal patients keep sorting by the old rule. Planning is tested by the Tower of London of **Tim Shallice** (1982), inhibition by the Stroop test, and verbal fluency by naming words that begin with a given letter.

Table 5.33 The prefrontal syndromes.

Region	Main deficits	Tests and cases
Dorsolateral	Working memory, planning, set-shifting, abstraction; perseveration	Wisconsin Card Sorting, Tower of London, Stroop, fluency

Region	Main deficits	Tests and cases
Orbitofrontal and ventromedial	Disinhibition, impulsivity, poor social judgement (pseudopsychopathy)	Iowa gambling task; Gage, E.V.R.
Medial frontal and anterior cingulate	Apathy, abulia (loss of will), akinetic mutism	Bilateral medial frontal damage

Orbitofrontal and ventromedial damage spares intelligence but wrecks conduct, as in patient E.V.R. (**Paul Eslinger** and **Antonio Damasio**, 1985), who kept a superior IQ after removal of an orbitofrontal tumour yet made ruinous decisions. Antonio Damasio's **somatic marker hypothesis** holds that this cortex tags choices with bodily emotional signals ("gut feelings"); without them, as the **Iowa gambling task** of **Antoine Bechara** and colleagues (1994) shows, patients keep choosing risky card decks. Prefrontal leucotomy or lobotomy (**António Egas Moniz**, Nobel Prize 1949; popularised by **Walter Freeman**) cut these connections and left patients apathetic.

> EXAM TRAP
>
> Gage means **ventromedial prefrontal cortex**, not the occipital lobe (vision), temporal lobe (hearing, memory), posterior parietal cortex (spatial attention), hippocampus and amygdala (memory, fear) or cerebellum and vermis (coordination). The Wisconsin test detects **dorsolateral** dysfunction, not hippocampal (long-term declarative memory), motor cortex or cerebellar damage, colour perception (visual cortex) or reflexes (spinal cord). Orbitofrontal damage gives disinhibition, not pure aphasia, cortical blindness or hemiplegia. Impaired executive function is the hallmark of the frontal lobe syndrome, not of obsessive-compulsive disorder, catatonic schizophrenia (Chapter 8) or temporal lobe epilepsy.

Recovery and plasticity after damage

When an axon is cut, its distal segment breaks up within days (**anterograde degeneration**, or Wallerian degeneration, after **Augustus Waller**, 1850), while **retrograde degeneration** creeps back to the cell body. Peripheral axons often **regenerate**, guided by Schwann cells; central axons rarely do, blocked by oligodendrocyte inhibitors and glial scars. Instead, healthy neighbouring axons grow branches into vacated synapses (**collateral sprouting**), and neighbouring cortex takes over lost territory (**cortical reorganisation**). Recovery is best with small lesions and young patients, and people with high **cognitive reserve** (roughly, education and intelligence) compensate better. In **constraint-induced movement therapy** (**Edward Taub**, 1993) a stroke patient's good arm is restrained while the weak arm is trained intensively, enlarging its cortical representation. Batteries such as the Halstead-Reitan, Luria-Nebraska and NIMHANS battery chart the losses (Chapter 2).

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Tissue around a stroke's infarct that treatment tries to save is the _____.
2. A 3-per-second spike-and-wave EEG marks the _____ seizure.
3. Parkinson's disease kills dopamine neurons of the _____ pars compacta.
4. Alzheimer's disease kills cholinergic neurons of the nucleus basalis of _____.
5. Recurrent well-formed visual hallucinations suggest dementia with _____.
6. Knowing one's wife by voice but not by face is _____.
7. Right posterior parietal damage causes neglect of the _____ side.
8. The Wisconsin Card Sorting Test is most sensitive to _____ prefrontal damage.

Answers: (1) penumbra, (2) absence, (3) substantia nigra, (4) Meynert, (5) Lewy bodies, (6) prosopagnosia, (7) left, (8) dorsolateral.

Loewi's Two Frog Hearts, Revisited

Every section of this chapter has passed through Loewi's laboratory. The nerve he stimulated was the vagus, the main parasympathetic nerve, whose rest-and-digest outflow slows the heart (5.1). The slowing was a chemical message crossing a synapse: acetylcholine binding to receptors and then broken down by acetylcholinesterase (5.3). Drugs act by imitating, boosting or blocking such messages (5.4), and Loewi's logic of changing one thing and measuring another runs through the methods of 5.5. The opposite state, the pounding heart of fear, belongs to the sympathetic division and the adrenal medulla (5.7). His idea surfaced at night, out of sleep (5.8). And when Alzheimer's disease strips the cortex of acetylcholine, the drugs prescribed are cholinesterase inhibitors, which keep Vagusstoff working longer (5.11).

Chapter summary

5.1 Biopsychology and the Organisation of the Nervous System. The biological perspective explains behaviour through the body, brain and nervous system. Biopsychology (Pinel) has six divisions and progresses by converging operations; Descartes's dualism opposes monism. The CNS is the brain and spinal cord; the PNS has somatic (voluntary) and autonomic divisions. The sympathetic division (fight or flight, Cannon 1915; T1 to L2; adrenal medulla) spends energy; the parasympathetic (rest and digest; craniosacral; vagus; acetylcholine) conserves it. A reflex arc runs afferent neuron, interneuron, efferent neuron. White matter is myelinated axons; PNS cell-body clusters are ganglia; dorsal roots are sensory, ventral roots motor. Twelve cranial nerves, the meninges, CSF and the blood-brain barrier complete the map.

5.2 The Brain: Brainstem, Diencephalon and Cerebral Cortex. The embryonic forebrain, midbrain and hindbrain become five divisions: telencephalon (cortex, limbic system, basal ganglia, corpus callosum), diencephalon (thalamus, hypothalamus), mesencephalon, metencephalon (pons, cerebellum) and myelencephalon (medulla). The medulla holds the vital centres for breathing and heart rate; the cerebellum coordinates movement and balance; the reticular activating system maintains arousal (Moruzzi and Magoun, 1949). The thalamus relays every sense except smell,

and the hypothalamus governs homeostasis through the pituitary. Six-layered neocortex, about 90 percent of cortex, serves higher cognition across frontal, parietal, temporal and occipital lobes, mapped by Brodmann (1909); commissural, association and projection fibres connect it.

5.3 Neurons, Neural Conduction, Synapses and Neurotransmitters. The neuron is the basic unit of the nervous system (Cajal's neuron doctrine; Nobel 1906, shared with Golgi); glia include astrocytes, microglia and the myelin makers, oligodendrocytes (CNS) and Schwann cells (PNS). The resting potential, about -70 mV, rests on K+ permeability and the pump (3 Na+ out, 2 K+ in). Summed EPSPs reaching threshold fire an all-or-none action potential (Na+ in, K+ out, peak about +50 mV), then refractory periods. Myelin gives fast saltatory conduction. Ca2+ influx triggers release; ionotropic and metabotropic receptors respond; reuptake or enzymes clear the transmitter. Glutamate excites, GABA inhibits, acetylcholine drives muscle, monoamines arise in brainstem nuclei, and endorphins control pain.

5.4 Drugs, Reward and Psychopharmacology. Psychoactive drugs act as agonists or antagonists at synapses. Tolerance (metabolic, functional, cross and conditioned, as in Siegel's 1982 heroin study) and withdrawal are distinct from addiction. Benzodiazepines raise the frequency and barbiturates the duration of GABA-A channel opening; heroin is a semisynthetic opioid; THC acts at CB1 receptors. Olds and Milner (1954) found self-stimulation sites in the septal area, and the mesolimbic pathway (VTA to nucleus accumbens) is the core reward circuit. Antipsychotics block D2, and nigrostriatal blockade causes extrapyramidal effects ending in tardive dyskinesia. SSRIs work after 2 to 4 weeks; lithium (Cade 1949) is the gold-standard mood stabiliser; disulfiram blocks aldehyde dehydrogenase.

5.5 Research Methods, Behaviour Genetics and the Developing Brain. Structural imaging (contrast X-ray, CT, MRI, DTI) shows anatomy; functional imaging (PET, SPECT, the BOLD signal of fMRI, fNIRS) shows activity, and paired-image subtraction revealed the default mode network, active at rest. EEG has the best temporal resolution, and averaging it yields ERPs such as the P300 to oddball targets. Lesions, stimulation (DBS, TMS, optogenetics), microdialysis, knockouts and CRISPR-Cas9 manipulate the brain. Mendelian inheritance and epigenetics underlie twin, adoption, linkage and GWAS designs, heritability and endophenotypes. PKU, Huntington's, fragile X, Down, Turner, Klinefelter and Prader-Willi syndromes show genetic errors. The neural tube forms at gastrulation; pruning, myelination and neuroplasticity continue for years.

5.6 Sensory Pathways and the Motor System. Sensory systems transduce energy, build receptive fields and climb from receptors through thalamic relays to primary, secondary and association cortex. In the inverted retina, lateral inhibition sharpens edges and creates Mach bands. Nasal fibres cross at the optic chiasm; the LGN's magnocellular layers carry motion, its parvocellular layers colour and detail, to V1, whose damage gives contralateral homonymous hemianopia, often with macular sparing. Dorsal 'where/how' and ventral 'what' streams leave V1. Hair cells, two somatosensory pathways and PAG analgesia complete the senses. Corticospinal fibres cross at the pyramidal decussation; mirror neurons fire for doing and watching; the basal ganglia select actions and the cerebellum corrects errors.

5.7 The Endocrine System, Stress and the Emotional Brain. Endocrine glands are ductless and release hormones into the blood; exocrine glands use ducts. Hormones are amino acid derivatives, peptides or steroids. The hypothalamus controls the pituitary, the master gland: releasing hormones reach the anterior lobe (GH, ACTH, TSH, FSH, LH, prolactin) through portal blood,

while oxytocin and vasopressin, made in the supraoptic and paraventricular nuclei, travel down axons to the posterior lobe. The adrenals run stress: the fast SAM system releases adrenaline, and the slower HPA axis (CRH, ACTH, cortisol) is braked by negative feedback. Allostasis is stability through change. Sham rage, the Papez circuit and the limbic system led to the amygdala, the hub of fear.

5.8 Sleep, Biological Rhythms, Hunger, Thirst and Sex. Circadian rhythms are timed by the suprachiasmatic nucleus (SCN), entrained by light through the retinohypothalamic tract, which drives night-time melatonin from the pineal gland. Sleep EEG runs from beta and alpha waking through theta (N1) and spindles (N2) to delta slow-wave sleep (N3); REM is paradoxical, with a waking-like EEG and atonia. Sleep begins when the VLPO inhibits the arousal systems; adenosine builds sleep pressure, and orexin, lost in narcolepsy, holds waking. Ghrelin starts hunger, CCK and leptin bring satiety, VMH lesions cause obesity and LH lesions aphagia, and arcuate NPY neurons trigger eating. Masters and Johnson described excitement, plateau, orgasm and resolution.

5.9 The Biology of Learning, Memory and Amnesia. Amnesia is a pathological loss of memory: anterograde (no new memories after the onset) or retrograde (loss of earlier memories, worst for recent ones, Ribot's law). Lashley's search for Semon's engram found memories distributed across the cortex, which Hebb explained with cell assemblies. H.M.'s bilateral medial temporal lobectomy (1953) showed that the hippocampus consolidates new declarative memories but not skills; R.B., place and grid cells, K.C. and Clive Wearing refined the picture, and the amygdala strengthens emotional memories. Korsakoff's syndrome follows thiamine deficiency and damages the mammillary bodies and mediodorsal thalamus. At the synapse, NMDA-dependent LTP strengthens and LTD weakens connections, while dopamine signals reward prediction errors.

5.10 Lateralisation, the Split Brain and Language. Dax (1836) and Broca (1861, patient Leborgne, 'Tan') tied language to the left, dominant hemisphere; the Wada test (1949) and Kimura's dichotic listening right-ear advantage (1961) confirm it in about 95 percent of right-handers and most left-handers. The right hemisphere leads in visuospatial processing, faces, music and emotional prosody. Split-brain studies (Myers and Sperry 1953; Sperry and Gazzaniga, Nobel 1981) showed each hemisphere working independently. Broca's area (inferior frontal gyrus, areas 44 and 45) serves speech production; Wernicke's area (posterior superior temporal gyrus, area 22) comprehension. Aphasia is acquired language loss from organic brain disease: Broca's, Wernicke's, conduction, global, transcortical and anomic types.

5.11 Brain Damage, Neurodegeneration and Neuropsychological Syndromes. Brain damage comes from tumours (encapsulated meningiomas, infiltrating gliomas, metastases), strokes (haemorrhage or ischaemia, with glutamate excitotoxicity in the penumbra), closed-head injury (contusions, contrecoup, chronic traumatic encephalopathy), infections, neurotoxins and genes; neurons die by necrosis or apoptosis. Epilepsy (tonic-clonic, absence, focal seizures) is neurological but not neurodegenerative. Parkinson's disease is dopamine loss from the substantia nigra pars compacta; Huntington's disease kills striatal GABA neurons; Alzheimer's disease brings plaques, tangles and acetylcholine loss. Multiple sclerosis is autoimmune demyelination. Agnosia (prosopagnosia), apraxia, right parietal left neglect and the prefrontal syndromes (Gage, Wisconsin Card Sorting Test) complete the section.

Who's who

Table 5.34 The people in this chapter.

Name	When	Examined for
Otto Loewi	1873 to 1961	Two frog hearts experiment (1921): vagus stimulation releases a chemical, Vagusstoff (acetylcholine); Nobel Prize 1936 with Henry Dale
Henry Hallett Dale	1875 to 1968	Acetylcholine and chemical transmission of nerve impulses; Nobel Prize 1936 shared with Loewi
Donald Olding Hebb	1904 to 1985	The Organization of Behavior (1949), which played a key role in the emergence of biopsychology
John P. J. Pinel	2021 (Biopsychology, 11th edition)	Biopsychology as a neuroscience; its six divisions; converging operations
Steven J. Barnes	2021	Co-author with Pinel of Biopsychology, 11th edition
René Descartes	1596 to 1650	Mind-body dualism; mind and body meeting at the pineal gland
Walter Bradford Cannon	1871 to 1945	Fight-or-flight response of the sympathetic division (1915)
Charles Bell	1774 to 1842	Bell-Magendie law: motor role of the ventral roots (1811)
François Magendie	1783 to 1855	Bell-Magendie law: dorsal roots sensory, ventral roots motor (1822)
Giuseppe Moruzzi	1949	With Magoun, showed that stimulating the brainstem reticular formation wakes sleeping cats and desynchronises the EEG: the reticular activating system
Horace Magoun	1949	With Moruzzi, proposed that reticular activity controls sleep and wakefulness (the reticular activating system)
Korbinian Brodmann	1868 to 1918	Numbered map of the cortex by cytoarchitecture (1909): areas 4 motor, 17 visual, 41 and 42 auditory, 44 and 45 Broca's area
Camillo Golgi	1843 to 1926	Golgi silver stain (early 1870s); reticular theory; shared the 1906 Nobel Prize with Cajal
Santiago Ramón y Cajal	1852 to 1934	Neuron doctrine: the nervous system is made of separate cells; shared the 1906 Nobel Prize
Alan Hodgkin	1914 to 1998	Squid giant axon; ionic model of the action potential (1952, with Huxley); Nobel Prize 1963
Andrew Huxley	1917 to 2012	Squid giant axon; ionic model of the action potential (1952, with Hodgkin); Nobel Prize 1963
Candace Pert	1946 to 2013	Demonstrated opiate receptors in the brain, with Snyder (1973)
Solomon Snyder	born 1938	Demonstrated opiate receptors in the brain, with Pert (1973)
John Hughes	1975 (key year)	Isolated the enkephalins, the first endogenous opioids, with Kosterlitz (1975)
Hans Kosterlitz	1903 to 1996	Isolated the enkephalins, with Hughes (1975)
Shepard Siegel	1982	Conditioned drug tolerance; heroin-tolerant rats died more often when dosed in a novel environment
James Olds	1954	Intracranial self-stimulation; the 'pleasure centre' in the septal area and medial forebrain bundle (with Peter Milner)
Peter Milner	1954	Co-discoverer of intracranial self-stimulation with James Olds
Terry Robinson	1993	Incentive-sensitisation theory of addiction with Kent Berridge: sensitised wanting, not liking
Kent Berridge	1993	Incentive-sensitisation theory; mesolimbic dopamine signals wanting more than liking
Jean Delay	1952	Reported chlorpromazine's antipsychotic effect with Pierre Deniker
Pierre Deniker	1952	Reported chlorpromazine's antipsychotic effect with Jean Delay
Arvid Carlsson	1963; Nobel Prize 2000	Dopamine as a transmitter; with Margit Lindqvist (1963) argued that chlorpromazine blocks dopamine receptors
Margit Lindqvist	1963	With Arvid Carlsson, found chlorpromazine raises dopamine metabolites and argued it is a dopamine receptor blocker
Philip Seeman	mid-1970s	Antipsychotic clinical potency correlates with D2 receptor affinity
John Cade	1949	Introduced lithium for mania

Name	When	Examined for
Godfrey Hounsfield	1919 to 2004	Introduced computed tomography (CT) in the early 1970s; Nobel Prize 1979
Seiji Ogawa	born 1934	Described the BOLD signal on which functional MRI depends (1990)
Marcus Raichle	born 1937	Described the brain's default mode (2001), which gave the default mode network its name
Hans Berger	1873 to 1941	First human electroencephalogram (EEG), reported in 1929
Samuel Sutton	1965	First described, with colleagues, the P300 event-related potential
Hans Kornhuber	1965	With Lüder Deecke, discovered the Bereitschaftspotential (readiness potential)
Lüder Deecke	1965	With Hans Kornhuber, discovered the Bereitschaftspotential (readiness potential)
Jennifer Doudna	born 1964	CRISPR-Cas9 gene editing with Emmanuelle Charpentier (2012); Nobel Prize in Chemistry 2020
Emmanuelle Charpentier	born 1968	CRISPR-Cas9 gene editing with Jennifer Doudna (2012); Nobel Prize in Chemistry 2020
Kary Mullis	1944 to 2019	Invented the polymerase chain reaction (PCR), which copies DNA
Gregor Mendel	1822 to 1884	Laws of inheritance from pea plants (published 1866): dominant and recessive traits
Irving Gottesman	1930 to 2016	Pooled schizophrenia twin concordance (about 48 percent MZ, 17 percent DZ; 1991); endophenotypes with Todd Gould (2003)
Todd Gould	2003	With Irving Gottesman, popularised the endophenotype concept in psychiatry
Thomas Bouchard	born 1937	Minnesota Study of Twins Reared Apart (begun 1979; main report in Science, 1990)
Ivar Asbjørn Følling	1888 to 1973	Discovered phenylketonuria (PKU) in 1934
Roger Sperry	1913 to 1994	Chemoaffinity hypothesis of axon growth (1963), from frog eye-rotation studies
Rita Levi-Montalcini	1909 to 2012	Discovered nerve growth factor, the first neurotrophin; Nobel Prize 1986
Mark Rosenzweig	1922 to 2009	Enriched environments produce a thicker, heavier rat cortex (1960s)
David Hubel	1926 to 2013	With Torsten Wiesel, monocular deprivation and the critical period in kitten visual cortex
Torsten Wiesel	born 1924	With David Hubel, monocular deprivation and the critical period in kitten visual cortex
Peter Eriksson	1998	Confirmed, with colleagues, adult neurogenesis in the human hippocampus
Stephen Kuffler	1913 to 1980	On-centre and off-centre receptive fields of retinal ganglion cells (1953)
Haldan Keffer Hartline	1903 to 1983	Lateral inhibition in the eye of Limulus, the horseshoe crab; Nobel Prize 1967
Ernst Mach	1838 to 1916	Mach bands, the illusory strips at light-dark edges (1865)
Lawrence Weiskrantz	1926 to 2018	Blindsight in the patient D.B. (1974)
Leslie Ungerleider	1946 to 2020	Dorsal 'where' and ventral 'what' visual streams (1982, with Mishkin)
Mortimer Mishkin	1926 to 2021	Dorsal 'where' and ventral 'what' visual streams (1982, with Ungerleider)
Melvyn Goodale	1992	Recast the dorsal stream as 'how', vision for action (with Milner)
David Milner	1992	Recast the dorsal stream as 'how', vision for action (with Goodale)
Wilder Penfield	1891 to 1976	Somatosensory and motor homunculi mapped by stimulating the cortex in surgery (1950, with Rasmussen)
Theodore Rasmussen	1910 to 2002	Somatosensory homunculus (1950, with Penfield)
Ronald Melzack	1929 to 2019	Gate-control theory of pain (1965, with Wall); taught in Section 1.3
Patrick Wall	1925 to 2001	Gate-control theory of pain (1965, with Melzack); taught in Section 1.3
David Reynolds	1969	Stimulation-produced analgesia: surgery on rats with only periaqueductal grey stimulation for pain relief
Vilayanur Ramachandran	born 1951	Phantom limbs explained by somatosensory cortical remapping; the mirror box
Giacomo Rizzolatti	born 1937	Mirror neurons in macaque ventral premotor area F5 (1990s)
Joseph Babinski	1857 to 1932	Babinski sign, the extensor plantar response of upper motor neuron lesions (1896)

Name	When	Examined for
Sten Grillner	1985	Spinal central pattern generators: cats with the spinal cord cut off from the brain walked on a treadmill
Geoffrey Harris	1913 to 1971	Proposed (1955) that the hypothalamus controls the anterior pituitary through hormones carried in portal blood vessels
Roger Guillemin	Nobel Prize 1977	Isolated thyrotropin-releasing hormone (TRH) from sheep hypothalami in the late 1960s
Andrew Schally	Nobel Prize 1977	Isolated TRH from pig hypothalami in the late 1960s, independently of Guillemin
Hans Selye	1907 to 1982	General adaptation syndrome (taught in Section 1.11)
Peter Sterling	1988	Allostasis, stability through change (with Joseph Eyer)
Joseph Eyer	1988	Allostasis, stability through change (with Peter Sterling)
Bruce McEwen	1993	Allostatic load (with Eliot Stellar)
Eliot Stellar	1993	Allostatic load (with Bruce McEwen)
Philip Bard	1929	Sham rage in decorticate cats: the hypothalamus organises rage
James Papez	1883 to 1958	Papez circuit of emotion (1937)
Paul MacLean	1913 to 2007	Named the limbic system (1952); triune brain
Joseph LeDoux	born 1949	Amygdala and fear conditioning; the low road and high road to the amygdala
Ralph Adolphs	1994	Patient S.M.: bilateral amygdala damage (Urbach-Wiethe disease) abolishes recognition of fear in faces
Justin Feinstein	2011	Showed that S.M. barely experiences fear
Heinrich Klüver	1937, 1939	Klüver-Bucy syndrome after bilateral anterior temporal lobectomy in rhesus monkeys (with Paul Bucy)
Paul Bucy	1937, 1939	Klüver-Bucy syndrome (with Heinrich Klüver)
Beatrice de Gelder	1999	Affective blindsight: recognising emotional faces in the blind field without awareness
Curt Richter	1967	Search for the circadian clock; medial hypothalamic lesions disrupt daily rhythms
Martin Ralph	1990	SCN transplants impose the donor's free-running period
Jeffrey Hall	Nobel 2017	Molecular clock genes (period) in fruit flies
Michael Rosbash	Nobel 2017	Molecular clock genes (period) in fruit flies
Michael Young	Nobel 2017	Molecular clock genes (timeless) in fruit flies
Eugene Aserinsky	1921 to 1998	Discovered REM sleep with Kleitman (1953)
Nathaniel Kleitman	1895 to 1999	Discovered REM sleep with Aserinsky (1953)
William Dement	1928 to 2020	Linked REM sleep to dreaming; the sleep cycle
Frederic Bremer	1936, 1937	Cerveau isolé and encéphale isolé preparations
Clifford Saper	2001	VLPO flip-flop switch for sleep and waking
Alexander Borbély	1982	Two-process model of sleep (process S and process C)
J. Allan Hobson	1977	Activation-synthesis hypothesis of dreams, with McCarley
Robert McCarley	1977	Activation-synthesis hypothesis of dreams, with Hobson
Jean Mayer	1953	Glucostatic theory of hunger
John Pinel	Biopsychology	Positive-incentive theory of eating; critique of the mamawawa
James Gibbs	1973	CCK reduces meal size (with Young and Smith)
R. C. Young	1973	CCK reduces meal size (with Gibbs and Smith)
Gerard Smith	1973	CCK reduces meal size (with Gibbs and Young)
Jeffrey Friedman	1994	Cloned the ob gene and discovered leptin
Albert Hetherington	1940	VMH lesions cause hyperphagia and obesity (with Ranson)
Stephen Ranson	1940	VMH lesions cause hyperphagia and obesity (with Hetherington)
Bal Krishan Anand	1951	LH lesions cause aphagia (with Brobeck); Indian physiologist
John Brobeck	1951	LH lesions cause aphagia (with Anand)
Philip Teitelbaum	1962	LH lesions cause adipsia as well as aphagia, with partial recovery (with Epstein)

Name	When	Examined for
Alan Epstein	1962	LH lesions cause adipsia as well as aphagia, with partial recovery (with Teitelbaum)
Barbara Rolls	2002	Portion size effect
Charles Phoenix	1959	Organisational and activational effects of hormones
Alfred Jost	1916 to 1991	Fetuses without gonads develop female reproductive ducts
Roger Gorski	1978	Sexually dimorphic nucleus of the medial preoptic area
Simon LeVay	1991	INAH3 and sexual orientation
William Masters	1915 to 2001	Four-phase sexual response cycle (1966), with Johnson
Virginia Johnson	1925 to 2013	Four-phase sexual response cycle (1966), with Masters
Helen Singer Kaplan	1929 to 1995	Desire, excitement and orgasm model (1979)
Théodule Ribot	1839 to 1916	Ribot's law (1881): in retrograde amnesia recent memories are lost before remote ones
Richard Semon	1859 to 1918	Coined the term engram, the physical trace of a memory (1904)
Karl Spencer Lashley	1890 to 1958	Search for the engram; mass action and equipotentiality (1929); 'In search of the engram' (1950): memories distributed throughout the cortex
Henry Gustav Molaison (H.M.)	1926 to 2008	Bilateral medial temporal lobectomy (1953); anterograde amnesia for declarative memory with preserved skill learning
William Beecher Scoville	1906 to 1984	Neurosurgeon who operated on H.M. in 1953; Scoville and Milner (1957)
Brenda Milner	born 1918	Studied H.M. (Scoville and Milner, 1957); mirror-drawing learning without recollection
Stuart Zola-Morgan	1986	Patient R.B.: damage to hippocampal field CA1 alone causes anterograde amnesia (with Squire and Amaral)
Larry Squire	1986	Patient R.B. (with Zola-Morgan and Amaral); memory systems of the medial temporal lobe
David Amaral	1986	Patient R.B. (with Zola-Morgan and Squire)
John O'Keefe	born 1939	Hippocampal place cells (1971, with Dostrovsky); The Hippocampus as a Cognitive Map (1978, with Nadel); Nobel Prize 2014
Jonathan Dostrovsky	1971	Discovered place cells with O'Keefe
Lynn Nadel	1978	The Hippocampus as a Cognitive Map, with O'Keefe
May-Britt Moser	born 1963	Grid cells of the entorhinal cortex (2005); Nobel Prize 2014
Edvard Moser	born 1962	Grid cells of the entorhinal cortex (2005); Nobel Prize 2014
Eleanor Maguire	2000	Enlarged posterior hippocampi in London taxi drivers
Kent Cochrane (K.C.)	1951 to 2014	Lost episodic, autobiographical memory after a 1981 accident; semantic, procedural and working memory spared
Endel Tulving	1927 to 2023	Episodic versus semantic memory; studied patient K.C.
Clive Wearing	born 1938	Musician left with profound amnesia by herpes simplex encephalitis (1985); musical skill preserved
Jean Delacour	1975	Delayed nonmatching-to-sample test (with Mishkin)
James McGaugh	born 1931	The amygdala modulates consolidation, so emotional events are remembered better
Larry Cahill	1994	Propranolol blocks the enhanced memory for an emotional story
Sergei Korsakoff	1854 to 1900	Described Korsakoff's syndrome (1880s): thiamine-deficiency amnesia with confabulation
Carl Wernicke	1848 to 1905	Described Wernicke's encephalopathy (1881): confusion, ataxia, ophthalmoplegia
Eric Kandel	born 1929	Habituation and sensitisation of the Aplysia gill-withdrawal reflex; Nobel Prize 2000
Timothy Bliss	born 1940	Discovered long-term potentiation with Lømo in the rabbit hippocampus (1973)
Terje Lømo	born 1935	Discovered long-term potentiation with Bliss (1973)
Richard Morris	1986	Morris water maze; the NMDA blocker AP5 blocks LTP and spatial learning
Serena Dudek	1992	Long-term depression by prolonged low-frequency stimulation (with Bear)

Name	When	Examined for
Mark Bear	1992	Long-term depression by prolonged low-frequency stimulation (with Dudek)
Wolfram Schultz	1997	Dopamine neurons of the ventral tegmental area signal reward prediction error
Peter Dayan	1997	Reward prediction error account of dopamine (with Schultz and Montague)
Read Montague	1997	Reward prediction error account of dopamine (with Schultz and Dayan)
Marc Dax	1836	First report linking speech loss to left-hemisphere damage; ignored in his lifetime
Paul Broca	1824 to 1880	Patient Leborgne ('Tan', 1861); left inferior frontal speech area (Broca's area)
Louis Victor Leborgne	1861	Broca's patient 'Tan': understood speech but could hardly speak; left frontal lesion
Juhn Atsushi Wada	1949	Wada (sodium amytal) test of language lateralisation
Doreen Kimura	1961	Dichotic listening; right-ear advantage for words, left-ear advantage for melodies
Elliott Ross	1981	Aprosodias after right-hemisphere damage homotopic to the language areas
Norman Geschwind	1926 to 1984	Larger left planum temporale (with Levitsky, 1968); Wernicke-Geschwind model (1965)
Walter Levitsky	1968	With Geschwind, left-right asymmetry of the planum temporale
Ronald Myers	1953	With Sperry, split-brain cats: the corpus callosum transfers learning
Roger Wolcott Sperry	1913 to 1994	Split-brain research in cats and humans; Nobel Prize 1981
Michael Gazzaniga	born 1939	Split-brain patients with Sperry; the left-hemisphere interpreter
Jerre Levy	1972	Chimeric figures test (with Trevarthen and Sperry)
Colwyn Trevarthen	1972	Chimeric figures test (with Levy and Sperry)
Wilder Graves Penfield	1891 to 1976	Cortical stimulation mapping and the homunculus; a distractor for split-brain research
Gregory Hickok	2007	Dual-stream (dorsal and ventral) model of speech processing, with Poeppel
David Poeppel	2007	Dual-stream (dorsal and ventral) model of speech processing, with Hickok
Joseph Jules Dejerine	1849 to 1917	Alexia with agraphia (angular gyrus, 1891); alexia without agraphia (1892)
Josef Gerstmann	1920s	Gerstmann syndrome: finger agnosia, left-right confusion, agraphia, acalculia
John Hughlings Jackson	1835 to 1911	Jacksonian (simple focal) seizures and the Jacksonian march
James Parkinson	1755 to 1824	Described the shaking palsy, Parkinson's disease (1817)
Oleh Hornykiewicz	1926 to 2020	Showed dopamine depletion in the striatum in Parkinson's disease (1960)
J. William Langston	1983	Reported MPTP parkinsonism in young drug users (1982 cases, 1983 report)
George Huntington	1850 to 1916	Described Huntington's disease (1872)
Alois Alzheimer	1864 to 1915	Described Alzheimer's disease in Auguste Deter (1906): plaques and tangles
Auguste Deter	1850 to 1906	Alois Alzheimer's first described patient with Alzheimer's disease
Sigmund Freud	1856 to 1939	Coined the term agnosia (1891)
Heinrich Lissauer	1861 to 1891	Apperceptive versus associative visual agnosia (1890)
Nancy Kanwisher	1997	Identified the fusiform face area with fMRI (1997)
Hugo Liepmann	1863 to 1925	Analysed apraxia and its link to left-hemisphere damage (1900)
Phineas Gage	1823 to 1860	Tamping-iron injury of the frontal lobe (1848): personality change, classic case for the ventromedial prefrontal cortex; later drove stagecoaches in Chile
John Martyn Harlow	1819 to 1907	Gage's physician; reported the case in 1848 and in 1868 (friends found him 'no longer Gage')
Hanna Damasio	1994	Reconstructed Gage's injury from his skull (1994), placing it in the ventromedial prefrontal cortex
David Grant	1948	Wisconsin Card Sorting Test (1948, with Esta Berg)

Name	When	Examined for
Esta Berg	1948	Wisconsin Card Sorting Test (1948, with David Grant)
Tim Shallice	1982	Tower of London test of planning (1982)
Antonio Damasio	1994	Somatic marker hypothesis of ventromedial prefrontal function
Paul Eslinger	1985	Described patient E.V.R. with Antonio Damasio (1985)
Antoine Bechara	1994	Iowa gambling task (1994): risky choices after ventromedial damage
E.V.R.	1985	Patient of Paul Eslinger and Antonio Damasio (1985): orbitofrontal damage, superior IQ but ruinous decisions
António Egas Moniz	1874 to 1955	Prefrontal leucotomy; Nobel Prize 1949
Walter Freeman	1895 to 1972	Popularised lobotomy in the United States
Augustus Waller	1816 to 1870	Wallerian (anterograde) degeneration of cut axons (1850)
Edward Taub	1993	Constraint-induced movement therapy for stroke
H.M. (Henry Molaison)	1926 to 2008	Amnesia after temporal lobe surgery for epilepsy (Section 5.9)

Key terms

biological perspective, Biopsychology, quasi-experimental study, converging operations, dualism, Monism, central nervous system, peripheral nervous system, somatic nervous system, afferent, efferent, autonomic nervous system, sympathetic division, fight-or-flight response, thoracolumbar outflow, sympathetic adrenal-medullary system, parasympathetic division, craniosacral outflow, vagus nerve, enteric nervous system, Grey matter, White matter, ganglion, dorsal root ganglion, Bell-Magendie law, motor neuron, interneuron, reflex arc, cranial nerves, meninges, Cerebrospinal fluid, choroid plexus, blood-brain barrier, area postrema, contralateral, telencephalon, diencephalon, brainstem, medulla oblongata, pons, cerebellum, ataxia, tectum, tegmentum, reticular formation, reticular activating system, thalamus, lateral geniculate nucleus, medial geniculate nucleus, hypothalamus, homeostasis, suprachiasmatic nucleus, mammillary bodies, limbic system, basal ganglia, striatum, cerebral cortex, central sulcus, lateral fissure, neocortex, allocortex, primary motor cortex, prefrontal cortex, Association cortex, cytoarchitecture, Commissural fibres, corpus callosum, Association fibres, Projection fibres, internal capsule, Golgi stain, neuron doctrine, neuron, dendrites, axon hillock, myelin sheath, nodes of Ranvier, telodendria, axon terminal, glial cells, resting membrane potential, sodium-potassium pump, excitatory postsynaptic potential, inhibitory postsynaptic potential, spatial summation, threshold of excitation, action potential, all-or-none law, absolute refractory period, relative refractory period, saltatory conduction, exocytosis, varicosities, gap junctions, coexistence, ionotropic receptor, metabotropic receptor, autoreceptors, acetylcholinesterase, monoamine oxidase, glutamate, NMDA receptor, GABA, acetylcholine, endorphins, psychoactive drug, drug metabolism, agonists, antagonists, allosteric modulator, partial agonist, Drug tolerance, Metabolic tolerance, functional tolerance, cross tolerance, drug sensitisation, withdrawal syndrome, physical dependence, psychological dependence, drug addiction, conditioned drug tolerance, benzodiazepines, barbiturates, CB1 receptors, endocannabinoids, intracranial self-stimulation, mesolimbic pathway, nucleus accumbens, incentive-sensitisation theory, dopamine hypothesis of schizophrenia, Antipsychotics, Typical antipsychotics, Atypical antipsychotics, extrapyramidal side effects, Akathisia, tardive dyskinesia, neuroleptic malignant syndrome, secondary movement disorders, monoamine hypothesis of depression, narrow therapeutic index, Computed tomography, Magnetic resonance imaging, spatial resolution, Diffusion tensor imaging, Positron emission tomography, Single-photon emission computed tomography, BOLD signal, temporal resolution, paired-image subtraction technique, default mode network, electroencephalogram, Magnetoencephalography, event-related potential, P300, galvanic skin response, stereotaxic surgery, Transcranial magnetic stimulation,

Optogenetics, gene knockout, CRISPR-Cas9, alleles, heterozygous, phenotype, Epigenetics, Twin studies, concordance rate, Heritability, endophenotype, Phenylketonuria, genomic imprinting, neural tube, chemoaffinity hypothesis, apoptosis, Synaptic pruning, Neuroplasticity, transduction, receptive field, functional segregation, myopia, on-centre cell, lateral inhibition, Mach bands, optic chiasm, magnocellular, parvocellular, primary visual cortex, bitemporal hemianopia, contralateral homonymous hemianopia, blindsight, simple cells, complex cells, dorsal stream, ventral stream, hair cells, superior olives, dorsal column-medial lemniscus system, anterolateral system, somatosensory homunculus, periaqueductal grey, mirror neurons, pyramidal decussation, Upper motor neurons, Babinski sign, stretch reflex, direct pathway, indirect pathway, action selection, climbing fibres, Exocrine glands, Endocrine glands, hormones, Steroid hormones, pituitary gland, tropic hormones, anterior pituitary, hypothalamo-hypophyseal portal system, posterior pituitary, Vasopressin, Oxytocin, adrenal cortex, cortisol, adrenal medulla, adrenaline, thyroid gland, insulin, Cushing's syndrome, Addison's disease, hypothalamic-pituitary-adrenal axis, negative feedback, psychoneuroimmunology, allostasis, allostatic load, sham rage, Papez circuit, amygdala, low road, high road, affective blindsight, Klüver-Bucy syndrome, circadian rhythm, zeitgebers, free-running rhythms, retinohypothalamic tract, melanopsin, pineal gland, melatonin, slow-wave sleep, paradoxical sleep, atonia, orexin, ventrolateral preoptic area, adenosine, two-process model, narcolepsy, REM sleep behaviour disorder, set-point assumption, positive-incentive theory, ghrelin, cholecystokinin, leptin, dual-centre hypothesis, arcuate nucleus, neuropeptide Y, portion size effect, sensory-specific satiety, osmotic thirst, hypovolaemic thirst, organisational effects, SRY gene, mamawawa, sexual response cycle, amnesia, anterograde amnesia, retrograde amnesia, temporal gradient, Ribot's law, source amnesia, engram, mass action, equipotentiality, cell assembly, Hebbian synapse, bilateral medial temporal lobectomy, declarative memory, consolidation, standard consolidation theory, place cells, cognitive map, grid cells, delayed non-matching-to-sample test, Korsakoff's syndrome, Wernicke's encephalopathy, Wernicke-Korsakoff syndrome, diencephalic amnesia, confabulation, habituation, sensitisation, long-term potentiation, AMPA receptors, coincidence detector, long-term depression, reward prediction error, Cerebral lateralisation, dominant hemisphere, minor hemisphere, Wada test, dichotic listening, Prosody, aprosodias, planum temporale, corpus callosotomy, tachistoscopic presentation, chimeric figures, left-hemisphere interpreter, intermanual conflict, Broca's area, Wernicke's area, arcuate fasciculus, angular gyrus, Wernicke-Geschwind model, dual-stream model, Aphasia, Broca's aphasia, agrammatism, Wernicke's aphasia, paraphasias, conduction aphasia, global aphasia, transcortical motor aphasia, Transcortical sensory aphasia, anomia, Alexia, alexia without agraphia, Gerstmann syndrome, Dysarthria, apraxia of speech, Developmental dyslexia, brain tumour, meningiomas, penumbra, excitotoxicity, contusion, contrecoup injury, chronic traumatic encephalopathy, post-traumatic amnesia, general paresis, necrosis, Epilepsy, Jacksonian march, status epilepticus, Parkinson's disease, Lewy bodies, Huntington's disease, Dementia, Alzheimer's disease, amyloid plaques, neurofibrillary tangles, Multiple sclerosis, Agnosia, prosopagnosia, Anosognosia, Apraxia, hemispatial neglect, extinction, dysexecutive syndrome, perseveration, somatic marker hypothesis, anterograde degeneration, collateral sprouting, cognitive reserve

From the past papers

1. Parts of the autonomic nervous system are: (Amity 2023)

   (a) Sympathetic and parasympathetic (b) Central and peripheral (c) Voluntary and involuntary (d) Cerebrospinal and cortical

   Answer: (a) Sympathetic and parasympathetic. These are the two main divisions of the ANS (many books add the enteric system of the gut wall). Central and peripheral divide the whole nervous system, voluntary and involuntary describe somatic versus autonomic control, and cerebrospinal and cortical is not a division at all.

2. Which of the following is the correct path of a reflex arc? (Amity 2023)

   (a) Afferent neuron to interneuron to afferent neuron (b) Afferent neuron to afferent neuron to interneuron (c) Afferent neuron to interneuron to efferent neuron (d) Efferent neuron to interneuron to the brain

   Answer: (c) Afferent neuron to interneuron to efferent neuron. The signal enters the spinal cord by a sensory (afferent) neuron, crosses an interneuron in the grey matter and leaves by a motor (efferent) neuron to the effector. An arc must end in an efferent neuron and closes in the spinal cord, not the brain; the knee-jerk even skips the interneuron.

3. White matter is primarily composed of (JEMAS 2025)

   (a) Myelinated axons. (b) Myelinated dendrites. (c) Cell bodies. (d) Unmyelinated axons.

   Answer: (a) Myelinated axons. The fat of the myelin sheath makes the tissue look white. Cell bodies and unmyelinated axons make up grey matter, and dendrites are not myelinated.

4. The reticular activating system (RAS), located in the brainstem, is crucial for the regulation of: (Real PYQ 2024)

   (a) Fine motor coordination (b) Arousal, wakefulness, and attention (c) Emotional memory (d) Balance and posture

   Answer: (b) Arousal, wakefulness, and attention. The RAS, the ascending part of the reticular formation in the brainstem core, keeps the cortex awake and alert; its destruction causes coma. Fine motor coordination and balance and posture belong to the cerebellum, emotional memory to the amygdala.

5. Which brain structure would most likely result in death if damaged? (Amity 2023)

   (a) Cerebellum (b) Medulla (c) Reticular formation (d) Pons

   Answer: (b) Medulla. The medulla oblongata holds the vital centres for breathing, heart rate and blood pressure, so its damage is the most likely to kill. Cerebellar damage causes ataxia and reticular damage coma; the pons helps set the breathing rhythm but is not the main vital centre.

6. Which brain structure serves as the primary control center for integrating the endocrine, visceral, and autonomic nervous systems? (Real PYQ 2025)

   (a) Cerebellum (b) Hypothalamus (c) Thalamus (d) Brainstem

   Answer: (b) Hypothalamus. It controls the autonomic nervous system and directs the pituitary, so it integrates neural and hormonal control of the body for homeostasis. The thalamus is the sensory relay, the cerebellum coordinates movement, and the brainstem carries out autonomic reflexes without integrating them with the endocrine system.

7. Which of the following is an inhibitory neurotransmitter? (JEMAS 2025)
   (a) Norepinephrine (b) GABA (c) Dopamine (d) Epinephrine

   Answer: (b) GABA. GABA is the main inhibitory transmitter of the brain: at GABA-A receptors it opens Cl- channels and produces IPSPs. Noradrenaline, dopamine and adrenaline are catecholamines that modulate arousal, movement and reward; none is the standard inhibitory transmitter.

8. Which of the following is NOT a type of glial cell? (Amity 2023)
   (a) Microglia (b) Astrocytes (c) Oligodendrocytes (d) Myelin sheath

   Answer: (d) Myelin sheath. Myelin is the fatty insulation that glial cells make (oligodendrocytes in the CNS, Schwann cells in the PNS); it is not a cell. Microglia (immune cells), astrocytes and oligodendrocytes are all glia.

9. Which of the following is not a neurotransmitter? (IHBAS 2020)
   (a) Dopamine (b) Serotonin (c) Nor-epinephrine (d) Adrenalin

   Answer: (d) Adrenalin (as keyed). Adrenaline is the odd one out because it acts mainly as a hormone of the adrenal medulla. Strictly it is also a catecholamine neurotransmitter, released by a few adrenergic brainstem neurons, so treat this as an odd-one-out question; dopamine, serotonin and noradrenaline are classic brain monoamine transmitters.

10. Valium, Xanax, Rohypnol, and Halcion are: (Amity 2023)
    (a) Benzodiazepines (b) Barbiturates (c) Sedatives (d) Hypnotics

    Answer: (a) Benzodiazepines. Diazepam, alprazolam, flunitrazepam and triazolam all belong to this one chemical class, which raises the frequency of GABA-A chloride channel opening. Barbiturates are a different class (they raise the duration of opening), and 'sedatives' and 'hypnotics' are broad functional labels that also cover alcohol and barbiturates.

11. Which of the following substances is NOT a natural alkaloid compound? (Amity 2023)
    (a) Morphine (b) Heroin (c) Codeine (d) Papaverine

    Answer: (b) Heroin. Heroin (diacetylmorphine) is semisynthetic, made by adding two acetyl groups to morphine. Morphine, codeine, papaverine and thebaine occur naturally in opium, the dried sap of the poppy.

12. A 45-year-old man with schizophrenia has been on antipsychotic medication for years. He presents with repetitive, involuntary, and purposeless movements of his mouth, tongue, and face, including lip-smacking and grimacing. This condition is most likely: (Real PYQ 2024)
    (a) Akathisia (b) Acute Dystonia (c) Tardive Dyskinesia (d) Parkinsonism

    Answer: (c) Tardive Dyskinesia. Involuntary orofacial movements after months or years of neuroleptic treatment are the late, often irreversible extrapyramidal effect. Acute dystonia (spasms) comes within hours to days, akathisia (inner restlessness, pacing) within days to weeks, and parkinsonism (tremor, rigidity, slowness) within weeks.

13. Which neuroscientific technique provides the highest temporal resolution for studying neural activity? (RML 2025)
    (a) fMRI (b) PET scan (c) EEG (d) CT scan

    Answer: (c) EEG. EEG records the brain's electrical activity directly, so it follows changes millisecond by millisecond. fMRI depends on a blood-flow response that takes seconds, PET on tracer uptake over a minute or more, and CT is a still structural image with no temporal resolution at all.

14. In Phenylketonuria (PKU), the body fails to convert phenylalanine into which substance due to a deficiency of phenylalanine hydroxylase? (Real PYQ 2025)

    (a) Urea (b) Tyrosine (c) Tryptophan (d) Histidine

    Answer: (b) Tyrosine. Phenylalanine hydroxylase converts phenylalanine into tyrosine; in autosomal recessive PKU the enzyme is missing, so phenylalanine accumulates and damages the brain unless a low-phenylalanine diet starts at birth. Urea is a waste product of protein breakdown, tryptophan the precursor of serotonin and histidine of histamine.

15. A male child presents with tall stature, small testes, and learning difficulties. Karyotype analysis shows an extra X chromosome. This condition is : (IHBAS 2025)

    (a) Klinefelter syndrome (b) Turner syndrome (c) Prader-Willi syndrome (d) Down syndrome

    Answer: (a) Klinefelter syndrome. An extra X in a male (47,XXY) gives tall stature, small testes, infertility and learning difficulties. Turner syndrome (45,X) gives a short, infertile female; Prader-Willi syndrome is a paternal deletion at 15q11 to q13; Down syndrome is trisomy 21.

16. The Mach bands phenomenon is best explained by : (IHBAS 2025)

    (a) Rod-cone dichotomy in the fovea (b) Opponent-process theory of color vision (c) Lateral inhibition in the retina (d) Temporal summation in photoreceptors

    Answer: (c) Lateral inhibition in the retina. At a light-dark border, cells on the bright side receive less inhibition from their dark neighbours and cells on the dark side receive more, so the edge is exaggerated into a brighter and a darker strip. The rod-cone dichotomy concerns dim-light versus colour vision, the opponent process colour coding and afterimages, and temporal summation the adding of weak signals over time.

17. Difficulty in seeing objects at a distance is called : (IHBAS 2024)

    (a) Myopia (b) Hypermetropia (c) Astigmatism (d) Visual Field Defects

    Answer: (a) Myopia. In myopia (near-sightedness) the eyeball is too long, so distant objects focus in front of the retina and blur; a concave lens corrects it. Hypermetropia blurs near objects, astigmatism comes from an irregular cornea, and visual field defects are blind areas caused by damage to the visual pathway, not focusing errors.

18. Which of the following areas of the brain is involved in controlling movements of the body? (Amity 2023)

    (a) Primary (b) Secondary (c) Supplementary (d) All of the above

    Answer: (d) All of the above. Movement is controlled by a hierarchy: primary motor cortex (area 4, precentral gyrus) is the main cortical output, and secondary motor cortex, the premotor cortex and supplementary motor area, programmes the patterns of movement it executes. No single area does the job alone.

19. In carrying out the 'fight or flight' response, the role of supervisor is assigned to the: (RML 2024)

    (a) Adrenal gland (b) Pituitary gland (c) Hypothalamus (d) Parasympathetic nervous system

    Answer: (c) Hypothalamus. It integrates the endocrine and autonomic systems and drives both the sympathetic nervous system and the pituitary. The pituitary and the adrenal gland carry out its orders, and the parasympathetic system calms the body instead.

20. Which cluster of symptoms is seen in Kluver-Bucy syndrome? (JEMAS 2025)

    (a) Agnosia, aphasia, hyper-orality, hyperphagia, hypersexuality (b) Agnosia, avolition, alogia, asexuality (c) Hypersexuality, altered emotional response, hyperorality, hyperalertness (d) Hypersexuality, hypo-sensitivity, delayed reaction

    Answer: (a). Bilateral anterior temporal lobe and amygdala damage gives psychic blindness (agnosia), hyperorality, hyperphagia and hypersexuality, with aphasia in human cases. Avolition and alogia are negative symptoms of schizophrenia, and the animals become placid, not hyperalert.

21. What is the term for the brain's process of regulating its own responses to stress by controlling hormone release? (IHBAS 2024)

    (a) Neuroplasticity (b) Homeostasis (c) Allostasis (d) Circadian rhythm

    Answer: (c) Allostasis. Sterling and Eyer (1988) defined it as stability through change: the brain anticipates demands and adjusts hormone release. Homeostasis is Cannon's stability around fixed set points, neuroplasticity is rewiring with experience, and circadian rhythm is the 24-hour cycle.

22. A lesion of the ventromedial hypothalamus (VMH) produces: (Recall set)

    (a) Loss of circadian rhythms (b) Aphagia and adipsia (c) Diabetes insipidus (d) Hyperphagia and obesity

    Answer: (d) Hyperphagia and obesity. The VMH was the classic satiety centre (Hetherington and Ranson, 1940). Aphagia and adipsia follow lateral hypothalamic lesions (Anand and Brobeck, 1951), circadian rhythms depend on the suprachiasmatic nucleus, and diabetes insipidus follows failure of ADH from the supraoptic and paraventricular nuclei.

23. Which gland regulates the circadian rhythm in the human beings? (RML 2022)

    (a) Hypothalamus (b) Cerebellum (c) Pituitary gland (d) Pineal gland

    Answer: (d) Pineal gland. The clock itself is the suprachiasmatic nucleus of the hypothalamus, but the question asks for a gland, and the pineal gland turns the clock's signal into night-time melatonin. The cerebellum coordinates movement and the pituitary is the master gland of other hormones; neither keeps a clock.

24. Masters and Johnson's sexual response cycle is distinct from Kaplan's model. Which of the following correctly reflects Masters and Johnson's sequence? (IHBAS 2025)

    (a) Excitement, Plateau, Orgasm, Resolution (b) Plateau, Excitement, Orgasm, Resolution (c) Excitement, Desire, Plateau, Orgasm, Refractory (d) Desire, Arousal, Plateau, Orgasm, Resolution

    Answer: (a) Excitement, Plateau, Orgasm, Resolution. Masters and Johnson (1966) described these four phases, with a male refractory period after orgasm. Putting desire first is Helen Singer Kaplan's (1979) change, and no model starts with plateau.

25. A person suffered a head injury in a road accident several years ago. Since the accident, he is unable to form new memories. He can remember many things which he did before the accident, but he cannot remember what he said ten minutes ago. The part of the brain which is most probably injured is: (IHBAS 2020)

    (a) Brain stem (b) Hippocampus (c) Spinal cord (d) Reticular formation

    Answer: (b) Hippocampus. Failing to form new memories while old ones survive is anterograde amnesia, the signature of hippocampal damage (H.M., R.B.). The brainstem and reticular formation serve vital functions and arousal, and the spinal cord carries reflexes and sensorimotor signals; none consolidates memory.

26. According to Karl Lashley's research on memory storage, where did he conclude that memories are primarily stored in the brain? (Real PYQ 2025)

    (a) In specific localized brain regions dedicated to particular memories (b) Distributed throughout the entire brain cortex (c) Exclusively in the hippocampus (d) Only in the frontal and temporal lobes

    Answer: (b) Distributed throughout the entire brain cortex. No cortical lesion erased a maze habit in Lashley's rats; the deficit depended on how much cortex was removed (mass action), not where (equipotentiality). He found no localised engram, and the hippocampus's role in forming memories emerged only later, with H.M.

27. Wernicke-Korsakoff syndrome results in: (Amity 2023)

    (a) Lapse of attention (b) Loss of muscle coordination (c) Both a & b (d) None

    Answer: (c) Both a and b. Thiamine deficiency first causes Wernicke's encephalopathy: confusion with lapses of attention, ataxia (loss of muscle coordination) and ophthalmoplegia with nystagmus. Survivors develop the amnesia of Korsakoff's syndrome, so the combined syndrome affects attention and coordination as well as memory.

28. The left hemisphere is: (RML 2024)

    (a) Concerned to a large extent with spatial and pattern sense (b) The minor hemisphere (c) The major hemisphere (d) Centre of musical ability

    Answer: (c) The major hemisphere. The left hemisphere is dominant for language in about 95 percent of right-handers and most left-handers, so it was called the major or dominant hemisphere. Spatial and pattern sense and musical ability are right-hemisphere strengths, and 'minor hemisphere' is the old label for the right.

29. Cutting through a person's corpus callosum causes the person to: (RML 2024)

    (a) Die (b) Become blind (c) Be unable to speak or understand language (d) Have difficulty coordinating certain kind of activities

    Answer: (d) Have difficulty coordinating certain kinds of activities. Corpus callosotomy for intractable epilepsy leaves patients alive, sighted and able to speak (the left hemisphere still speaks); the cost is that the hemispheres cannot share information, which shows up in tasks needing both.

30. Ria has difficulty enunciating words correctly and speaks in a slow and laboured way; her speech often makes sense, but it includes only keywords. Which part of her brain is damaged? (RML 2022)

    (a) Wernicke's area (b) Broca's area (c) Temporal lobe (d) Frontal lobe

    Answer: (b) Broca's area. Slow, effortful, telegraphic speech that still makes sense is Broca's (non-fluent) aphasia. 'Frontal lobe' is true but too broad; Wernicke's area and the temporal lobe produce fluent, empty speech with poor comprehension.

31. Prosopagnosia results in an inability to : (IHBAS 2024)

    (a) Recognize faces (b) Distinguish tastes (c) Express feelings through facial expressions (d) Recognize objects

    Answer: (a) Recognize faces. The source key gives (b), which is wrong: prosopagnosia is a visual agnosia for faces after fusiform gyrus damage, and loss of taste is ageusia. Loss of facial expression is a motor sign (the masked face of Parkinson's disease), and failure to recognise objects is object agnosia.

32. A patient presents with an inability to perform coordinated, purposeful movements, despite having intact motor function, sensation, and comprehension of the task. When asked to brush their hair, they cannot sequence the movements. This deficit is best described as: (Real PYQ 2024)

    (a) Agnosia (b) Ataxia (c) Apraxia (d) Alexia

    Answer: (c) Apraxia. Loss of learned purposeful movement with strength, sensation, coordination and comprehension intact is apraxia (here ideational: the steps cannot be sequenced). Agnosia is a failure of recognition, ataxia is cerebellar incoordination and alexia is loss of reading.

33. Which of the following neurological conditions is NOT classified as a neurodegenerative disorder? (Real PYQ 2025)

    (a) Parkinson's disease (b) Alzheimer's disease (c) Epilepsy (d) Huntington's disease

    Answer: (c) Epilepsy. Epilepsy is recurrent seizures from synchronous neural discharge, an episodic disorder with normal function between attacks. Parkinson's (substantia nigra dopamine neurons), Alzheimer's (cortical and hippocampal neurons) and Huntington's (striatal GABA neurons) are progressive losses of neurons.

High-yield sheet

The facts this chapter's questions ask most, most-asked first. Last read before the exam.

1 **Korsakoff's syndrome** is caused by thiamine (vitamin B1) deficiency, classically in chronic alcoholism; it damages the mammillary bodies and mediodorsal thalamus and causes dense anterograde amnesia, severe retrograde amnesia and confabulation. ASKED 13 TIMES

2 Extrapyramidal side effects come from D2 blockade in the **nigrostriatal** pathway and appear in order: acute dystonia (hours to days), akathisia (days to weeks, inner restlessness), parkinsonism (weeks) and **tardive dyskinesia** (months to years, orofacial movements, may be irreversible); akathisia is a secondary movement disorder. ASKED 10 TIMES

3 Damage to **Broca's area** (left inferior frontal gyrus, pars opercularis and triangularis, areas 44 and 45) causes Broca's aphasia: non-fluent, effortful, telegraphic speech with relatively preserved comprehension and impaired repetition. ASKED 9 TIMES

4 The **hippocampus** consolidates new declarative (especially episodic and spatial) memories; its damage, as after a head injury, causes anterograde amnesia while remote memories survive. ASKED 8 TIMES

5 The ANS has two divisions: the **sympathetic** mediates fight or flight (Walter Cannon, 1915) together with the adrenal medulla, which releases adrenaline, and spends energy; the **parasympathetic** (rest and digest) conserves energy, slowing the heart and constricting the pupils mainly through the vagus nerve and acetylcholine. ASKED 7 TIMES

6 The **limbic system**, named by Paul MacLean in 1952, is the classic seat of emotion, regulating emotion, motivation and memory; hunger and thirst belong to the hypothalamus. ASKED 7 TIMES

7 The **adrenal glands** run the stress response: the cortex releases cortisol at the end of the HPA axis (hypothalamus CRH, anterior pituitary ACTH, adrenal cortex cortisol, with negative feedback on the hypothalamus and anterior pituitary), and the medulla releases adrenaline, also called epinephrine. ASKED 7 TIMES

8 The **ventromedial hypothalamus** was the satiety centre: its lesions cause hyperphagia and obesity (Hetherington and Ranson, 1940), whereas **lateral hypothalamus** lesions cause aphagia and adipsia (Anand and Brobeck, 1951). ASKED 7 TIMES

9 **H.M.** (Henry Molaison) had both medial temporal lobes (hippocampus, amygdala, adjacent cortex) removed by William Scoville in 1953; he was left with severe anterograde amnesia for declarative memory but improved at mirror drawing (preserved procedural memory). ASKED 7 TIMES

10 The **dopamine hypothesis** of schizophrenia: mesolimbic dopamine (D2) hyperactivity produces positive symptoms and mesocortical hypoactivity negative symptoms; its original support was that typical antipsychotics' clinical potency correlates with their D2 affinity. ASKED 6 TIMES

11 The **amygdala** is the key structure for fear, aggression, threat detection and fear conditioning; in LeDoux's account the low road (thalamus straight to amygdala) is fast and crude, the high road through the cortex slower and accurate. ASKED 6 TIMES

12 **Parkinson's disease** is a **dopamine** deficiency caused by degeneration of the **substantia nigra pars compacta** (nigrostriatal pathway), with alpha-synuclein Lewy bodies; not the locus coeruleus, raphe nuclei or cholinergic basal forebrain. ASKED 6 TIMES

13 The **mesolimbic pathway**, from the ventral tegmental area to the nucleus accumbens (ventral striatum), is the core reward pathway of addiction, and dopamine is the reward transmitter. ASKED 5 TIMES

14 **Oxytocin and vasopressin** (ADH) are made in the supraoptic and paraventricular nuclei of the hypothalamus and released from the posterior pituitary; neither oxytocin nor melatonin is an anterior pituitary hormone. ASKED 5 TIMES

15 **Long-term potentiation** (Bliss and Lømo, 1973) is a lasting strengthening of synaptic transmission after high-frequency stimulation; its induction needs NMDA receptors, which pass calcium only when glutamate binds and the depolarised membrane expels their magnesium block. ASKED 5 TIMES

16 **Wernicke's area** is the posterior superior temporal gyrus of the dominant hemisphere (area 22), the classic region for comprehending spoken and written language; damage gives fluent, empty speech with poor comprehension. ASKED 5 TIMES

17 The **medulla oblongata** (hindbrain) holds the vital centres for breathing, heart rate and blood pressure; its damage is the brain injury most likely to cause death. ASKED 4 TIMES

18 **GABA** is the main inhibitory transmitter of the brain (glycine in the spinal cord); **glutamate** is the main excitatory transmitter of the CNS. ASKED 4 TIMES

19 **All-or-none law**: once threshold is reached, an action potential fires at full, constant amplitude whatever the stimulus strength, and intensity is coded by firing frequency; EPSPs, IPSPs and the resting potential do not obey it. ASKED 4 TIMES

20 **Saltatory conduction**: in myelinated axons the action potential jumps from one node of Ranvier to the next, where voltage-gated Na+ channels cluster; the function of myelin is to increase conduction speed. ASKED 4 TIMES

21 **Benzodiazepines** (Valium, Xanax, Rohypnol, Halcion) and alcohol enhance GABA at the GABA-A receptor; benzodiazepines increase the frequency of chloride channel opening, barbiturates its duration. ASKED 4 TIMES

22 The **pituitary** is the master gland, regulating most other glands and growth, and is itself controlled by the hypothalamus. ASKED 4 TIMES

23 **Klüver-Bucy syndrome** (Klüver and Bucy, 1937 and 1939), after bilateral removal of the anterior temporal lobes and amygdala, brings tameness and loss of fear, psychic blindness, hyperorality, hyperphagia and hypersexuality. ASKED 4 TIMES

24 The **suprachiasmatic nucleus** of the hypothalamus is the master circadian pacemaker, entrained by light through the retinohypothalamic tract; its lesions abolish circadian rhythms. ASKED 4 TIMES

25 **REM sleep** shows rapid eye movements, skeletal muscle atonia and a low-voltage, fast, desynchronised EEG; EEG synchrony, delta waves and the growth hormone surge belong to slow-wave sleep. ASKED 4 TIMES

26 Beta waves (13 to 30 Hz) mark alert waking and **alpha** (8 to 12 Hz) relaxed waking with eyes closed; **slow-wave sleep** is stages 3 and 4 (N3), with high-amplitude delta waves of 0.5 to 4 Hz. ASKED 4 TIMES

27 **Anterograde amnesia** is the inability to form new memories after the onset; **retrograde amnesia** is the loss of memories formed before it, usually worst for the most recent (Ribot's law). ASKED 4 TIMES

28 In split-brain patients (corpus callosotomy) an object flashed to the left visual field reaches only the right hemisphere, so it cannot be named but can be picked out with the left hand; each hemisphere works independently, and Roger Sperry won the 1981 Nobel Prize. ASKED 4 TIMES

29 **Prosopagnosia** is a visual **agnosia** for faces (recognised at once by voice), after bilateral or right-sided damage to the **fusiform gyrus** (fusiform face area); it is not loss of taste, despite the IHBAS 2024 key. ASKED 4 TIMES

30 **Cerebrospinal fluid** is made by the choroid plexus, fills the ventricles (the two lateral ventricles are the largest) and the subarachnoid space, and is absorbed through the arachnoid granulations; the **blood-brain barrier** is formed by tight junctions between capillary endothelial cells, supported by astrocyte end-feet. ASKED 3 TIMES

CHAPTER 6

Developmental Psychology

IN THIS CHAPTER

6.1 ● Foundations: The Lifespan View, Key Issues and Ecological Contexts: Define development, growth and maturation, explain the lifespan perspective, the principles and enduring issues of development and the difference between critical and sensitive periods, and place a child's settings in Bronfenbrenner's ecological systems.

6.2 ● Heredity, Prenatal Development, Birth and the Newborn: Explain how genes are inherited and expressed, how heredity and environment are separated and combined, how prenatal development unfolds and what threatens it, and how the newborn is assessed, including the Apgar scale and the newborn reflexes.

6.3 ● Infancy: Brain, Body, Motor Skills and Perception: Describe growth and brain development in infancy, apply the principles, milestones and theories of motor development, explain how infant perception is studied and what it reveals (senses, visual cliff, affordances), and order the early learning, imitation and milestone sequence.

6.4 ●●● Piaget's Theory of Cognitive Development: Explain Piaget's constructivist theory, from schemes, assimilation, accommodation and equilibration through the four stages and their landmark tasks (object permanence, conservation, the pendulum), and judge how later research and postformal theorists have revised it.

6.5 ●● Vygotsky, Bruner, Information Processing and Theory of Mind: Explain Vygotsky's sociocultural theory (the zone of proximal development, scaffolding, private speech) and how it differs from Piaget's, Bruner's modes of representation, the information-processing view of development, and how theory of mind and social perspective taking develop.

6.6 ● Emotional Development, Temperament and Self-Regulation: Trace emotional development from Bridges' first emotions through the self-conscious emotions, stranger and separation anxiety and the still face, and classify temperament (Thomas and Chess, Rothbart and Bates, Kagan) and self-regulation (Kopp, Mischel's marshmallow test).

6.7 ●● Attachment: Imprinting, Bowlby, Ainsworth and Beyond: Explain why infants form attachments, trace the phases of attachment over the first three years, classify Strange Situation patterns by reunion behaviour, and weigh the evidence on deprivation, culture and adult attachment.

6.8 ● Parenting, Family, Play and Gender in Childhood: Compare Baumrind's parenting styles on demandingness and responsiveness, recognise the forms of maltreatment and neglect, trace how play develops from Piaget to Parten, and explain how children acquire gender through socialisation, Kohlberg's stages and gender schemas.

6.9 • Moral Development: Piaget, Kohlberg, Gilligan and Beyond: Trace moral development from Piaget's heteronomous and autonomous morality through Kohlberg's three levels and six stages, place any moral reason at its level, and weigh the critiques from Gilligan, cross-cultural research and later theories.

6.10 •• Erikson's Psychosocial Stages, Identity and Adolescence: Name all eight of Erikson's psychosocial stages with their ages, crises, virtues and maldevelopments, and explain adolescent identity (Marcia's statuses), Elkind's adolescent egocentrism, puberty and the adolescent brain, and modern views of adolescence.

6.11 • Adulthood, Ageing, Death and Dying: Trace adult development from emerging adulthood through midlife and old age to death and grief, naming the theorist behind each stage model, type of marriage and theory of ageing, and telling the climacteric apart from general physical ageing.

Weightage. 230 of the 3180 questions mapped in this book (7.2%) test this chapter, 101 of them from real past papers. 1 of its 11 sections is High yield.

Southern France, January 1800. On a winter morning a boy walked out of the woods near the village of Saint-Sernin-sur-Rance, in the district of Aveyron, looking for food. He was naked but for the rags of a shirt, scarred, about twelve years old, and he could not speak. Local people had glimpsed a wild child in the forests for years, and more than once he had been caught and had escaped. This time he stayed. Within months he was in Paris, the most talked-about child in France, and learned men came to see what a human being is like when society has had no hand in shaping him.

The verdict of the greatest authority was harsh. **Philippe Pinel**, reformer of the Paris asylums, examined the boy and judged him an incurable idiot, perhaps abandoned because he was defective from birth. A 26-year-old physician at the Paris institution for deaf-mutes, **Jean-Marc-Gaspard Itard**, disagreed. The boy, he argued, was backward because he had grown up alone; give him the experiences he had missed and he would become fully human. Itard named him Victor and, with his carer Madame Guérin, worked with him daily for about five years, training his senses, his attention and his feelings, and trying above all to teach him language.

The results, reported in 1801 and 1806, were mixed. Victor came to recognise written words, follow instructions, show affection for Madame Guérin and protest when he was punished unjustly. But he never learned to speak more than a word or two, and Itard ended the lessons disappointed. Victor lived quietly with Madame Guérin in Paris until his death in 1828, aged about forty. Was his failure born or made, and had the teaching simply come too late? This chapter gives you the tools to answer.

6.1 Foundations: The Lifespan View, Key Issues and Ecological Contexts

LOWER YIELD · 13 QS

Before the story of how a fertilised cell becomes an ageing adult, you need its frame: what developmental change means, what drives it, the questions still argued over, and the contexts in which every child grows.

The science of change and stability

Developmental psychology is the scientific study of change and stability from conception to death. It takes the **developmental perspective**, the approach concerned with the changes that occur as people mature. The other perspectives of Section 1.2 ask other questions: the psychoanalytic about unconscious conflicts rooted in childhood, the biological about genes, brain and hormones, and the humanistic about free choice and self-actualisation.

Keep two kinds of change apart. **Growth** is quantitative change, an increase in size or amount: height, weight, vocabulary. **Development** is qualitative change in structure and organisation, an orderly, progressive series of changes towards maturity: the child does not just know more words but starts combining them into sentences. Change runs through three interacting domains: physical (body, brain, motor skills), cognitive (memory, language, thinking) and psychosocial, or socioemotional (emotions, personality, relationships).

The lifespan is divided into periods, but the divisions are a social construction, an idea a society shares rather than a fact of nature: adolescence became a recognised stage only with industrialisation. Indian courses follow **Elizabeth Hurlock** (1898 to 1988), whose *Developmental Psychology: A Life-Span Approach* marks ten periods (Table 6.1).

Table 6.1 Hurlock's ten periods of the lifespan.

Period	Age span	Period	Age span
1 Prenatal	Conception to birth	6 Puberty	10 or 12 to 13 or 14
2 Infancy	Birth to 2 weeks	7 Adolescence	13 or 14 to 18
3 Babyhood	2 weeks to 2 years	8 Early adulthood	18 to 40
4 Early childhood	2 to 6 years	9 Middle age	40 to 60
5 Late childhood	6 to 10 or 12	10 Old age	60 onward

Older texts treated development as ending with adolescence. **Paul Baltes** (1939 to 2006) countered in 1987 with the **lifespan perspective**: development is *lifelong* (no age holds supremacy), *multidimensional* (physical, cognitive and social strands), *multidirectional* (some abilities rise while others fall), *plastic* (open to change at every age), *contextual* (embedded in history and culture) and *multidisciplinary*, and every age brings both gains and losses (an older adult loses speed but may gain wisdom). Baltes named three kinds of influence. **Normative age-graded influences** are tied to age and shared by most people (puberty, retirement). **Normative history-graded influences** are shared by a cohort, a generation, because of when it lived (Partition, the COVID-19 pandemic). **Non-normative influences** are unusual events in one life (a parent's early death). Longitudinal, cross-sectional and cross-sequential designs, which separate age changes from cohort effects, are in Chapter 9; infant research methods are in Section 6.3.

From philosophers to baby biographies

Two philosophers framed the debate. **John Locke** (1632 to 1704) held that the newborn mind is a **tabula rasa**, a blank slate written on by experience: the root of the nurture view. **Jean-Jacques Rousseau** (1712 to 1778), in *Émile* (1762), saw the child as a noble savage, born good, whose natural plan of growth unfolds in stages: the root of the nature, maturation and stage views.

Scientific observation began with the **baby biography**, a detailed diary of one child. **Charles Darwin** (1809 to 1882) published one on his first son in 1877, and **Wilhelm Preyer** (1841 to 1897) followed with *The Mind of the Child* (1882), often taken as the start of scientific child psychology. **G. Stanley Hall** (1844 to 1924) led the child study movement with questionnaires given to large numbers of children; his *Adolescence* (1904) called the teens a time of storm and stress (Section 6.10) and proposed **recapitulation**, the idea that each child's development replays the evolutionary history of the species. Hall's student **Arnold Gesell** (1880 to 1961) took the **normative approach**, observing many children to find the typical age, or norm, of each behaviour. Against him stood the behaviourist **John B. Watson** (1878 to 1958), who boasted that he could train any healthy infant into any kind of specialist.

Maturation and learning ●●

Gesell's lasting idea is **maturation**, the biological unfolding of genetic potential: changes that appear on a genetic timetable with little dependence on practice. **Learning** is change through experience and practice. Maturation supplies readiness; learning builds on it. Three terms that depend on experience are not maturation: **socialisation**, acquiring the values, norms and behaviour of one's society; accommodation, Jean Piaget's term for changing a mental scheme to fit new information (Section 6.4); and conditioning, learning by association (Section 1.6).

KEY STUDY: GESELL AND THOMPSON'S CO-TWIN CONTROL (1929)

Can practice hurry an unready body? Gesell and **Helen Thompson** studied identical twin girls, T and C, holding heredity constant: the method of **co-twin control**. Twin T practised stair climbing daily for six weeks from 46 weeks of age. Twin C's training began only at 53 weeks, yet after two weeks she had caught up. Maturation, not early training, set the pace. Criticism: one pair of twins and a narrow set of skills.

Myrtle McGraw (1899 to 1988) refined the point in *Growth: A Study of Johnny and Jimmy* (1935). Intensive training from infancy did not hurry Johnny's *phylogenetic* skills, those common to the species (crawling, walking), but it advanced *ontogenetic* skills particular to the individual (swimming, roller skating) once he was ready; his twin Jimmy was left untrained. So a child's increasing skill at using his muscles is due mainly to maturation, not to the opportunity to exercise, pushing by parents or learning. Gesell turned such observations into age norms and developmental schedules for motor, adaptive, language and personal-social behaviour (the scales are in Chapter 2).

EXAM TRAP

"The biological unfolding of genetic potential" is **maturation** (Gesell), never socialisation, accommodation or conditioning. A child's growing muscular skill is credited to **maturation**, not exercise, parental pushing or learning. Growth is **quantitative** (size, amount); development is **qualitative** (structure, organisation).

Principles of development and developmental tasks

Development follows regularities that hold across cultures (Table 6.2). Two give the direction of physical growth: **cephalocaudal**, from head to tail (head control comes before walking), and **proximodistal**, from the centre outwards (arms before fingers). Motor milestones are in Section 6.3.

Table 6.2 Principles of development (Hurlock and standard texts).

Principle	Meaning and example
Orderly, predictable pattern	Same sequence in all children: sit, stand, walk
Cephalocaudal	Head to foot
Proximodistal	Centre to extremities
General to specific (mass to specific)	Whole-body wriggling comes before one hand reaching
Continuous but uneven in rate	Never stops; spurts in infancy and puberty

Principle	Meaning and example
Individual differences	Same sequence, different rates: walking at 10 or at 15 months
Maturation and learning interact	Reading needs both readiness and practice
Early foundations are critical	Early patterns tend to persist
Each stage has its own pattern	Characteristic behaviour, hazards and social expectations

Robert J. Havighurst (1900 to 1991), in *Developmental Tasks and Education* (1948), defined a **developmental task** as one that arises at about a certain period of life; success brings happiness and success with later tasks, while failure brings unhappiness, social disapproval and later difficulty. Tasks arise from physical maturation (walking), cultural pressure (reading) and personal values (choosing a vocation). Early childhood tasks include walking, talking, taking solid food and controlling elimination; middle childhood brings the skills of ordinary games, getting along with age-mates, and reading, writing and arithmetic. Adolescent and adult tasks are in Sections 6.10 and 6.11.

The enduring issues

Every theory takes a stand on the issues of Table 6.3. On **nature versus nurture** (heredity or environment?) today's answer is interaction: a genetic tendency shows itself only in some environments, and inherited traits help select the environments a child meets (Section 6.2). **Continuity versus discontinuity** asks whether development is gradual, quantitative accumulation, a ramp, or a series of qualitatively different stages, a staircase.

Table 6.3 The core issues of developmental psychology.

Issue	The question	Typical answers
Nature versus nurture	Heredity or environment?	Rousseau, Gesell (nature); Locke, Watson (nurture); today, interaction
Continuity versus discontinuity	Gradual change or stages?	Learning theory (continuous); Freud, Piaget, Erikson (stages)
Stability versus change	Do early traits persist?	Freud (personality set by 5 or 6); Baltes (plastic for life)
Early versus later experience	Do the first years decide?	Freud, Bowlby (early); Clarke and Clarke (later can repair)
Active versus passive child	Does the child shape development?	Piaget (active builder); Watson (moulded by environment)
Universal versus context-specific	One path everywhere?	Piaget (universal stages); Vygotsky, Bronfenbrenner (culture)

Try a worked example. In one Native American tribe, mothers answer an infant's distress at once and hold the baby almost constantly for three years; for generations the children have grown up quiet, even-tempered, obedient and respectful. Is the calm inborn? No: it follows a child-rearing practice handed down in the culture, so it is **learnt behaviour**, acquired through experience. **Innate behaviour** appears without experience, as reflexes do. The calm is also **overt behaviour** (observable action), not **covert behaviour** (private thoughts and feelings), and "complex behaviour" describes how many parts an act has, not where it comes from.

EXAM TRAP

Behaviour that follows a child-rearing practice handed down in a culture is **learnt**; stability across generations is not proof of heredity. Stage theorists (Freud, Piaget, Erikson) take the **discontinuity** side; learning and information-processing theorists take the **continuity** side.

Critical and sensitive periods ●●

A **critical period** is a fixed window in which a particular experience must occur for normal development; if it passes without that experience, the effects are largely irreversible. The model case comes from ethology, the study of behaviour in its evolutionary setting: **Konrad Lorenz** (1903 to 1989) reported in 1935 that greylag goslings imprint on the first moving object they see after hatching, even Lorenz himself (Section 6.7). In human development the clearest critical periods are in the embryo: each organ forms within a brief window, so a teratogen such as thalidomide or rubella seriously damages an organ only if it strikes while that organ is forming, mostly in weeks 3 to 8 (Section 6.2).

A **sensitive period** is an optimal but more flexible window: the experience matters most then, but some development remains possible later (Figure 6.1). First language is the standard case. **Eric Lenneberg** (1967) proposed a critical period for language from about age two to puberty (Section 1.8); the evidence now favours a sensitive period. Genie, found in Los Angeles in 1970 at 13 after years of isolation, learned many words but never mastered grammar. Such lasting limits after early language deprivation support sensitive periods. They say nothing about semantics (word meaning) or syntax (rules of word order), which are levels of language, or about nativism, the view of **Noam Chomsky** (born 1928) that language ability is innate.

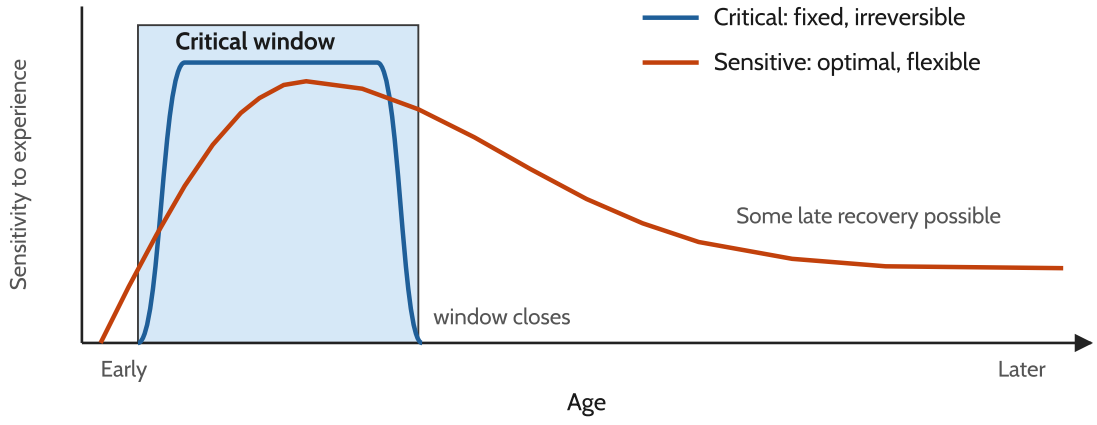


Figure 6.1 A critical period (solid line) opens and closes sharply; a sensitive period (dashed line) peaks early but never falls to zero.

Can later experience repair a poor start? **Ann Clarke** and **Alan Clarke** argued that it often can in *Early Experience: Myth and Evidence* (1976). The strongest test is the English and Romanian Adoptees study led by **Michael Rutter** (1933 to 2021): children adopted into British homes from Romanian orphanages after 1989 caught up almost fully if adopted before about six months, and greatly, though some kept lasting difficulties, if adopted later (Section 6.7).

Later experience compensating for earlier deficits contradicts the strict critical-period view. Sensitive periods allow late recovery by definition; continuity concerns gradual versus stepwise change, and stability whether traits persist.

> **EXAM TRAP**
>
> RML 2024 asked for the strongest evidence for critical periods "in the case of human development" and keyed **imprinting**. Imprinting is the classic critical period, but it is animal evidence. Within human development the clearest critical periods are in **embryonic development** (teratogen timing); language shows a sensitive period, and vocational interests show none. Expect imprinting as the key; know why embryonic development is the better human answer.

The grand theories at a glance

Six families of theory answer these issues differently (Table 6.4), each taught in full elsewhere. Psychoanalytic and cognitive theories propose stages; learning theories see continuous, environment-driven change; ethology stresses evolved behaviour and critical periods; ecological and dynamic systems theories stress context and many interacting causes. In the dynamic systems theory of **Esther Thelen** (1941 to 2004), a new skill self-organises out of body, task and environment.

Table 6.4 The grand theories of development.

Family	Key theorists	Where taught
Psychoanalytic	Sigmund Freud (psychosexual stages); Erik Erikson (psychosocial stages)	Chapter 4; Section 6.10
Learning	John B. Watson, B. F. Skinner, Albert Bandura	Section 1.6
Cognitive	Jean Piaget; Lev Vygotsky; information processing	Sections 6.4, 6.5
Ethological	Konrad Lorenz, Nikolaas Tinbergen, John Bowlby	Section 6.7
Ecological	Urie Bronfenbrenner	Below
Dynamic systems	Esther Thelen	Section 6.3

Bronfenbrenner's ecological systems ●●

Most theories put the child under a microscope; **Urie Bronfenbrenner** (1917 to 2005) widened the lens. His **ecological systems theory**, set out in *The Ecology of Human Development* (1979), places the child inside environmental systems nested like Russian dolls (Figure 6.2), each affecting the child and affected in turn.

- The **microsystem** is the innermost layer: the settings in which the child takes part directly, face to face, such as family, school, peer group and neighbourhood. Influence is bidirectional: parents shape the child, and the child shapes the parents.
- The **mesosystem** is the set of links between microsystems: home and school (a parent-teacher meeting), family and peers. It is a relationship between settings, not a setting.
- The **exosystem** holds settings the child is not in that still affect the child: a parent's workplace (a night shift), the school board, the mass media, extended family.
- The **macrosystem** is the overarching culture: values, laws, ideology and the economic system.

- The **chronosystem**, added later, is time: transitions such as parental divorce and historical events such as a pandemic.

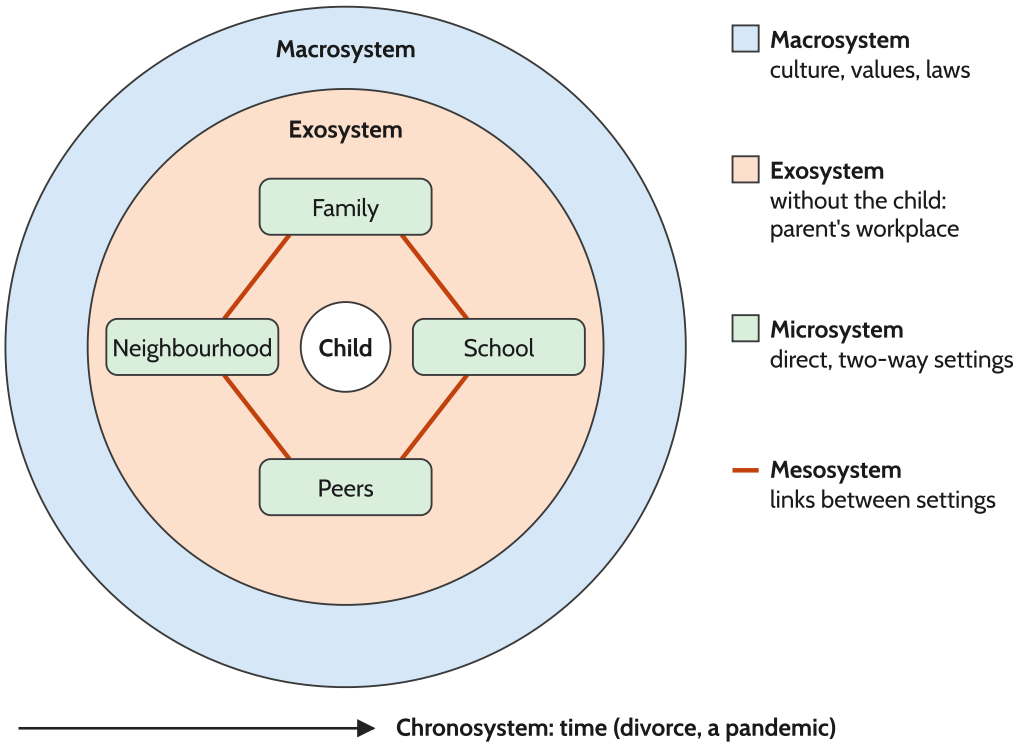


Figure 6.2 Bronfenbrenner's ecological systems. The dashed mesosystem lines are links between microsystem settings, not a setting; the exosystem holds settings without the child.

Later Bronfenbrenner gave biology more weight. In the **bioecological model**, developed with **Stephen Ceci** (1994) and later with **Pamela Morris** (1998), development is the product of Process-Person-Context-Time (PPCT). Its engine is the **proximal process**: regular, increasingly complex, two-way interaction between the child and nearby people, objects and symbols, such as a parent reading with a child every night. Person characteristics (temperament, abilities) shape these processes, context is the five systems, and time runs from moments to history. Bronfenbrenner also helped found Head Start (1965), the American preschool programme for poor children.

EXAM TRAP

Family, school and peer group are **all microsystem**; the immediate family is never meso-, exo- or macrosystem. The **mesosystem is a link** (school-family, parent-teacher), not a setting. The **exosystem does not contain the child** (the parent's workplace). Culture is macrosystem; divorce is chronosystem. Nested systems are **Bronfenbrenner's**, not Skinner's (operant learning), Gesell's (maturation) or Hall's (child study).

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Increase in size or amount is growth; qualitative change in structure and organisation is _____.
2. The biological unfolding of genetic potential is called _____.
3. Gesell and Thompson held heredity constant with the method of _____ control.
4. Head control before walking illustrates the _____ principle.
5. Events shared by a whole generation, such as a pandemic, are normative _____-graded influences.
6. A fixed window with irreversible effects is a _____ period.
7. Links between a child's home and school belong to the _____.
8. A parent's workplace belongs to the _____.

Answers: (1) development, (2) maturation, (3) co-twin, (4) cephalocaudal, (5) history, (6) critical, (7) mesosystem, (8) exosystem.

6.2 Heredity, Prenatal Development, Birth and the Newborn

LOWER YIELD · 7 QS

Every life begins as one cell; nine months later a newborn arrives able to suck, startle and grip. This section follows that journey, from genes to the newborn's first reflexes.

Genes, chromosomes and conception

The nucleus of each body cell holds 46 **chromosomes** in 23 pairs, one of each pair from each parent. A chromosome is a long strand of **DNA** (deoxyribonucleic acid); a **gene** is a segment of DNA that codes for a protein. Twenty-two pairs are **autosomes**; the twenty-third, the **sex chromosomes**, is XX in females and XY in males. Body cells multiply by **mitosis** into identical copies; sperm and ova form by **meiosis**, which halves the number to 23 and reshuffles genes, so siblings differ. At conception, usually in a fallopian tube, sperm and ovum unite into one cell, the **zygote**; since every ovum carries an X, the father's X or Y sperm decides the child's sex. A zygote that splits yields **monozygotic twins** (identical, sharing all their genes); two ova fertilised by two sperm yield **dizygotic twins** (fraternal, sharing about half, like any siblings).

Patterns of inheritance •

Each gene comes in alternative forms called **alleles**. The alleles a person carries are the **genotype**; the observable traits, shaped also by the environment, are the **phenotype**. Two identical alleles make a person **homozygous** for that gene; two different ones, **heterozygous**.

In **dominant-recessive inheritance**, the pattern **Gregor Mendel** (1822 to 1884) reported from pea plants in 1866, a heterozygote shows only the dominant allele's trait; the recessive allele is masked but can be passed on, so the person is a **carrier**. Let blue eyes (D) be dominant and green (d) recessive: DD and Dd children have blue eyes, dd children green. Figure 6.3 shows why two blue-eyed Dd parents can have a green-eyed child.

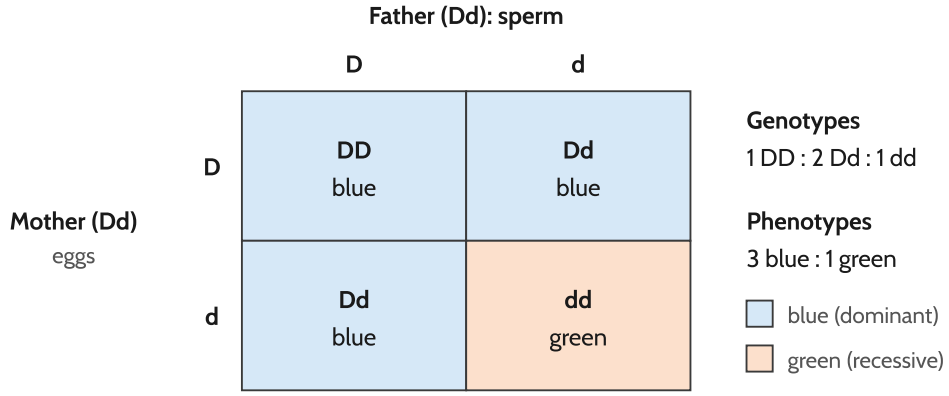


Figure 6.3 A Punnett square for two Dd parents, blue (D) dominant. Every Dd child is blue-eyed; only the dd child, one chance in four, has green eyes.

> **EXAM TRAP**
>
> Heterozygous does not mean blended. A Dd child shows the dominant phenotype (blue). Blue-green would need incomplete dominance, and "cannot be determined" is wrong whenever dominance is stated.

Two carriers thus face a one-in-four risk, in each pregnancy, of a child with a recessive disorder: **phenylketonuria** (PKU), cystic fibrosis (thick mucus clogs lungs and pancreas), sickle-cell anaemia or Tay-Sachs disease (fatal nervous system degeneration in early childhood). In PKU the enzyme phenylalanine hydroxylase is missing, so phenylalanine builds up and damages the brain; newborn screening and a low-phenylalanine diet prevent the intellectual disability. A dominant disorder needs one copy: **Huntington's disease** (chorea and dementia) appears in midlife, often after it has been passed on. **Genetic counselling** estimates such risks from family history and tests.

In **incomplete dominance** the heterozygote is intermediate (sickle-cell carriers sickle mildly when oxygen is low); in **codominance** both alleles are fully expressed (A and B alleles give blood group AB). In **X-linked inheritance** a recessive allele on the X is usually masked by a female's second X, but a male has none, so haemophilia, red-green colour blindness and Duchenne muscular dystrophy strike mainly males, through carrier mothers. **Polygenic inheritance**, many genes of small effect, underlies continuous traits such as height and intelligence. In **genomic imprinting** one parent's allele is chemically silenced, so the effect depends on which parent it came from. A **mutation** is a sudden, permanent change in DNA, spontaneous or caused by radiation or chemicals.

Chromosomal abnormalities ●●

Whole chromosomes can be extra or missing, usually because a pair fails to separate in meiosis (nondisjunction), and parts can be lost (deletion). **Down syndrome**, described by **John Langdon Down** (1828 to 1896) in 1866, is usually **trisomy 21**, three copies of chromosome 21; its risk rises with maternal age. It is the commonest genetic cause of intellectual disability; fragile X is the commonest inherited one (Table 6.5).

Table 6.5 Chromosomal and related syndromes.

Syndrome	Karyotype or cause	Key features
Down	Trisomy 21	Intellectual disability, flat face, heart defects
Klinefelter	47,XXY male	Tall, small testes, infertility, language and learning difficulties
XYY	47,XYY male	Tall, usually typical; no proven link with crime
Turner	45,X female	Short, webbed neck, ovarian failure, spatial difficulties
Triple X	47,XXX female	Tall; mild verbal and learning difficulties
Fragile X	FMR1 gene on the X	Intellectual disability, long face, large ears; autism
Cri du chat	Deletion of 5p	Cat-like cry, small head, severe intellectual disability
Prader-Willi	Paternal 15q11-q13 deletion	Hypotonia, hyperphagia, obesity, short stature, hypogonadism
Angelman	Maternal 15q11-q13 deletion	Severe intellectual disability, no speech, seizures, frequent laughter
Williams	Deletion of 7q11.23	Very sociable, fluent speech, weak spatial skills

EXAM TRAP

Klinefelter (47,XXY): tall male, small testes. **Turner** (45,X): short female, webbed neck. **Prader-Willi** and **Angelman** share one region of chromosome 15: lost from the father's copy it gives Prader-Willi, the always-hungry child; from the mother's, Angelman. This is genomic imprinting.

Heredity and environment

How far is a trait inherited? (The nature-nurture debate itself is in Section 6.1.) **Family studies** are weak evidence, since relatives share homes as well as genes. **Adoption studies** ask whether adopted children resemble their biological or adoptive parents; **twin studies**, whether identical (MZ) twins are more alike than fraternal (DZ) twins. Most informative are **MZ twins adopted by different parents**: identical genes, different homes, so their similarities reflect heredity and their differences environment. MZ twins reared together share genes and home, confounding the two; DZ twins or siblings adopted apart share only about half their genes. **Albert Stunkard** and colleagues (1990) found identical twins reared apart nearly as alike in body-mass index as those reared together. **Heritability**, the proportion of variation in a trait within a population due to genetic differences, describes groups, not individuals (IQ figures: Section 1.9; behaviour genetics: Chapter 5).

Genes and environment also work together. **Sandra Scarr** and **Kathleen McCartney** (1983) described three kinds of **gene-environment correlation**: **passive** (parents supply both genes and a matching home: musical parents, a musical house), strongest in infancy; **evocative** (a child's traits evoke responses: a cheerful baby draws more play); and **active**, or **niche-picking** (children seek settings that suit them), which grows with age. In **gene-environment interaction** one environment affects genotypes differently: an ordinary diet harms only the child with PKU. **Irving Gottesman**'s (1963) **reaction range** is the band of outcomes a genotype permits, the environment deciding where in it a person falls. **Conrad Hal Waddington**'s (1905 to 1975) **canalisation** is heredity's channelling of some traits, such as early motor milestones, to the same outcome in almost any environment. **Epigenetics** studies how experience switches genes on or off (for example by DNA methylation) without changing the DNA sequence.

The three prenatal periods

Prenatal development lasts about 38 weeks from conception, in three periods (Figure 6.4). In the **germinal period**, about the first two weeks, the zygote divides (cleavage) as it travels down the fallopian tube, becoming a solid ball, the **morula** (about day 4), then a hollow **blastocyst** (about day 5), whose inner cell mass becomes the baby and outer layer the placenta. **Implantation** in the uterine wall ends the period.

The **embryonic period** (weeks 3 to 8) opens with **gastrulation**, when the inner cell mass forms three germ layers: ectoderm (nervous system, skin), mesoderm (muscle, bone, heart, blood) and endoderm (gut, lungs, glands). The nervous system begins here: the neural plate arises from ectoderm and folds into the **neural tube**, the future brain and spinal cord, by about week 4. Because every major organ forms now (**organogenesis**), this period is the most vulnerable to teratogens. The **fetal period** (week 9 to birth) brings growth and refinement: **quickening**, the first movements the mother feels, at about 16 to 20 weeks, and the **age of viability**, when a baby born early may survive with intensive care, at about 22 to 24 weeks.

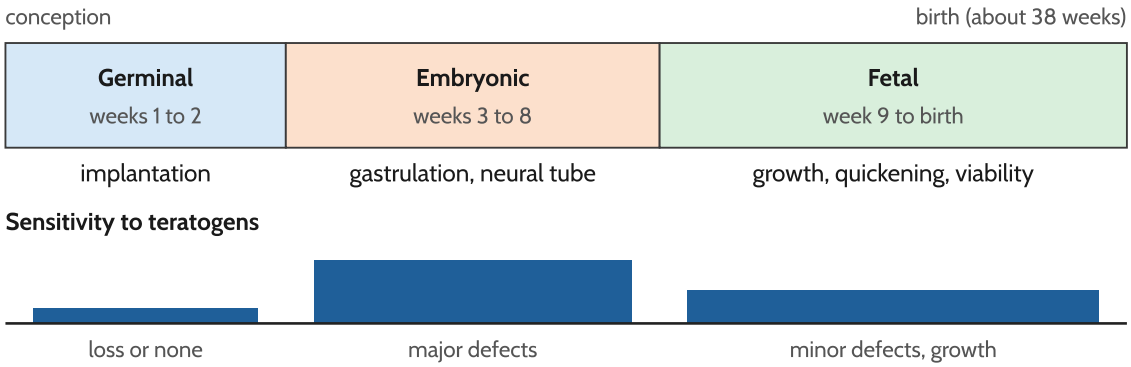


Figure 6.4 The three prenatal periods (not to scale). Bar height shows sensitivity to teratogens, greatest while the organs form.

> EXAM TRAP
>
> The nervous system begins at the **gastrula** stage, not at the zygote, morula or blastocyst stage, whose cells have not yet formed germ layers. The **embryonic**, not the fetal, period is the most vulnerable.

Teratogens, maternal factors and prenatal tests

A **teratogen** is any environmental agent that harms the developing child. Its effect depends on **timing** (each organ has its own critical period, as Figure 6.4 shows), **dose**, the **genetic susceptibility** of mother and child, and **delayed effects** that surface years later. Alcohol (first in Table 6.6) is the commonest preventable cause of intellectual disability: **fetal alcohol spectrum disorder** (FASD) ranges up to full fetal alcohol syndrome, with growth delay, a small head, learning problems and facial signs (short eye openings, a thin upper lip, a smooth philtrum).

Table 6.6 The main teratogens.

Agent	Main effects on the child

Agent	Main effects on the child
Alcohol	Fetal alcohol spectrum disorder
Nicotine	Low birth weight, prematurity, miscarriage
Cocaine, opioids	Prematurity, low birth weight, newborn withdrawal
Thalidomide (late 1950s)	Missing or shortened limbs
Valproate (antiepileptic)	Neural tube defects, lower IQ
Radiation	Small head, intellectual disability
Rubella	Deafness, cataracts, heart defects
Toxoplasmosis	Eye and brain damage
Syphilis	Stillbirth; eye, bone and brain damage
HIV	Infection before, during or after birth
Zika virus	Microcephaly (small head and brain)

Maternal factors matter too. Teenage mothers risk low birth weight; mothers over 35 face more chromosomal abnormalities and miscarriage. **Folic acid** before conception and in early pregnancy prevents most **neural tube defects**: **spina bifida** (an unclosed spinal column) and **anencephaly** (most of the brain missing). Poorly controlled diabetes raises birth defects and birth size; severe stress raises prematurity and low birth weight. In **Rh incompatibility** an Rh-negative mother's antibodies attack the red cells of an Rh-positive fetus, usually in a later pregnancy; an anti-D injection prevents it. Prenatal tests are compared in Table 6.7.

Table 6.7 Prenatal tests.

Test	How and when	Detects
Ultrasound	Sound-wave image; non-invasive	Age, growth, twins, structural defects
Amniocentesis	Amniotic fluid by needle; 15 to 20 weeks	Chromosomal and genetic disorders; small miscarriage risk
Chorionic villus sampling	Placental tissue; 10 to 13 weeks	The same, earlier; slightly higher risk
Maternal blood screening	Alpha-fetoprotein (AFP); 15 to 20 weeks	High: neural tube defect; low: Down syndrome risk

> **INDIAN CONTEXT**
>
> The **PCPNDT Act** (Pre-Conception and Pre-Natal Diagnostic Techniques Act; 1994, amended 2003) bans prenatal sex determination and sex selection; ultrasound centres must register and may not reveal the sex of the fetus.

Birth and assessing the newborn

Labour has three stages: **dilation** of the cervix (the longest), **delivery** of the baby, and delivery of the **placenta**. Its main danger is **anoxia**, too little oxygen (as when the cord is squeezed), which can damage the brain and cause cerebral palsy. A **preterm** baby is born before 37 weeks, perhaps at a weight right for its age; a baby **small for gestational age** weighs less than expected for its age, even at term, and usually fares worse. **Low birth weight** means

under 2,500 g. **Kangaroo care**, skin-to-skin holding on a parent's chest, steadies temperature, breathing and heart rate and improves survival.

The **Apgar scale**, devised in 1952 (published 1953) by the anaesthesiologist **Virginia Apgar** (1909 to 1974), rates five signs 0, 1 or 2 (total 10) at 1 and 5 minutes: **a**ppearance (colour), **p**ulse, **g**rimace (reflex irritability), **a**ctivity (muscle tone) and **r**espiration. A total of 7 or more is good; 3 or less is an emergency. The **Neonatal Behavioral Assessment Scale** (1973) of the paediatrician **T. Berry Brazelton** (1918 to 2018) assesses reflexes, muscle tone, state changes and responses to people.

Peter Wolff (1966) distinguished the newborn's **states of arousal**: regular (non-REM) sleep, irregular (REM) sleep, drowsiness, quiet alertness, and waking activity and crying. Newborns sleep about 16 to 18 hours a day, half of it REM sleep. **Sudden infant death syndrome** (unexplained death of a sleeping infant under one year) fell sharply after back-to-sleep campaigns taught parents to lay babies on their backs.

Newborn reflexes

A **reflex** is an inborn, automatic response to a specific stimulus (Table 6.8). **Survival reflexes** (breathing, blinking, sucking, swallowing, rooting) have clear value, and some last for life. **Primitive reflexes** (Moro, grasp, Babinski, stepping, tonic neck) are controlled by the brainstem and spinal cord and fade over the first year as the cortex matures. A reflex absent at birth, unequal on the two sides or persisting well past its age signals damage to the nervous system.

Table 6.8 Newborn reflexes.

Reflex	Stimulus	Response	Disappears
Rooting	Cheek stroked	Turns toward touch, opens mouth	3 to 4 months
Sucking	Object in mouth	Rhythmic sucking	About 4 months
Moro (startle)	Loss of support, loud noise	Arms fling out, then in as if embracing	4 to 6 months
Palmar grasp	Palm pressed	Fingers grip	3 to 6 months
Plantar grasp	Ball of foot pressed	Toes curl under	9 to 12 months
Babinski	Sole stroked	Big toe up, toes fan out	About 12 months
Stepping	Held upright, feet down	Walking movements	About 2 months
Swimming	Face down in water	Paddles, kicks, holds breath	4 to 6 months
Tonic neck	Head turned, lying on back	Fencing posture	4 to 6 months
Galant	Back stroked beside spine	Trunk curves to that side	4 to 6 months

EXAM TRAP

Texts differ, most of all for the Babinski (8 to 12 months, or up to 2 years). Answer **about 12 months**; an upgoing toe after about 2 years, or in an adult, signals an upper motor neuron lesion. Loss of support triggers the Moro; the sole, the Babinski; the cheek, rooting; the palm, the palmar grasp.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. If D is dominant, a Dd child shows the _____ phenotype.
2. A tall male with small testes and karyotype 47,XXY has _____ syndrome.
3. Prader-Willi syndrome is a 15q11-q13 deletion on the chromosome from the _____.
4. Heredity and environment are best separated in _____ twins adopted by different parents.
5. The nervous system begins forming at the _____ stage.
6. The period most vulnerable to teratogens is the _____ period.
7. The Babinski reflex is normal until about _____ months.

Answers: (1) dominant, (2) Klinefelter, (3) father, (4) monozygotic, (5) gastrula, (6) embryonic, (7) 12.

6.3 Infancy: Brain, Body, Motor Skills and Perception

LOWER YIELD · 12 QS

A newborn cannot hold up her own head. A year later she walks across the room and picks a grain of rice off the floor between thumb and finger. This section follows that change through brain, body, movement and the senses, and asks how we can know what a baby who cannot speak perceives and learns.

Physical growth in the first two years

Birth weight roughly doubles by about 5 months and triples by the first birthday. Growth has two directions. In the **cephalocaudal trend** (Latin, "head to tail") the head grows first and fastest; it is a quarter of body length at birth but an eighth in adulthood. In the **proximodistal trend** growth runs from the centre outward, trunk before arms and arms before hands. Breast milk is the ideal food. Severe malnutrition causes **marasmus**, wasting of the whole body from too little food of any kind, usually in the first year, or **kwashiorkor**, a protein deficiency after weaning in which belly and limbs swell with fluid.

The developing brain ●●

Neurogenesis, the birth of new neurons, is largely prenatal. Why, then, does the brain grow so much after birth? Because existing neurons form connections and gain insulation. In **synaptogenesis** the infant brain forms far more synapses than it will keep, and **synaptic pruning** then removes the rarely used ones and keeps those used often: "use it or lose it". Pruning is a process of neuronal development and plasticity that fits circuits to the child's own experience, continuing in the prefrontal cortex through adolescence (Section 5.5). **Myelination**, the wrapping of axons in fatty myelin that speeds conduction, reaches the prefrontal cortex last, continuing into early adulthood. Together these take the brain from about 25 percent of adult weight at birth to about half by 6 months, 75 percent by age 2 and 90 percent by age 5 to 6 (Figure 6.5).

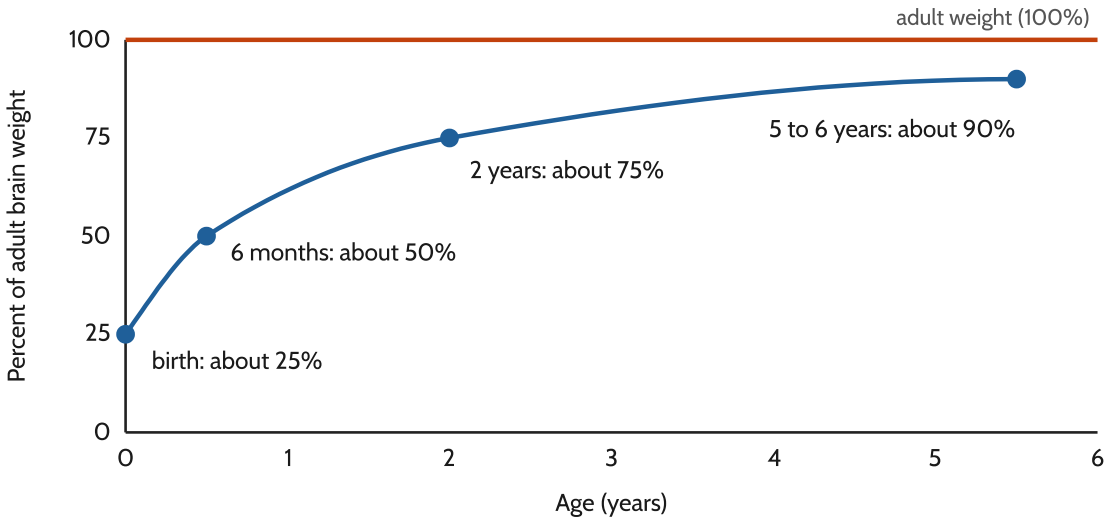


Figure 6.5 Brain weight as a percentage of adult weight (approximate values joined by a smooth line). The curve climbs most steeply in the first two years.

William Greenough and colleagues (1987) distinguished two kinds of plasticity. **Experience-expectant plasticity** depends on experiences every member of the species can count on, such as patterned light and speech; the brain overproduces synapses in readiness and keeps those the expected input uses. Hence sensitive periods: a cataract left untreated in an infant's eye leaves lasting poor vision. **Experience-dependent plasticity** builds connections from experiences unique to the individual and continues for life.

EXAM TRAP

Synaptic pruning belongs to **neuronal development and plasticity**, not memory consolidation (stabilising a new memory, Section 1.7), neurotransmitter release (the moment-to-moment signal at a synapse) or long-term potentiation (lasting strengthening of a synapse by repeated stimulation, Chapter 5). Half of adult brain size is keyed at **about 6 months**, not birth, 1 year or 3 years; some texts imply nearer the first year, so treat these percentages as approximate.

Principles and milestones of motor development

Motor control follows the same two directions as growth (Figure 6.6). By the **cephalocaudal** principle it spreads from head to foot: the infant holds up her head before she sits, and sits before she stands. By the **proximodistal** principle it spreads from the centre of the body to the extremities: she swings her whole arm at a toy months before her fingers can pick it up. Development also runs from **gross motor skills** (large movements of trunk and limbs, such as walking) to **fine motor skills** (precise movements of hands and fingers), and from **mass to specific**: shown a toy, a young infant waves, kicks and wriggles all at once, and only later reaches with one hand. Finally, motor development shapes social and behavioural expectations: a child who walks and feeds herself is treated differently and expected to do more.

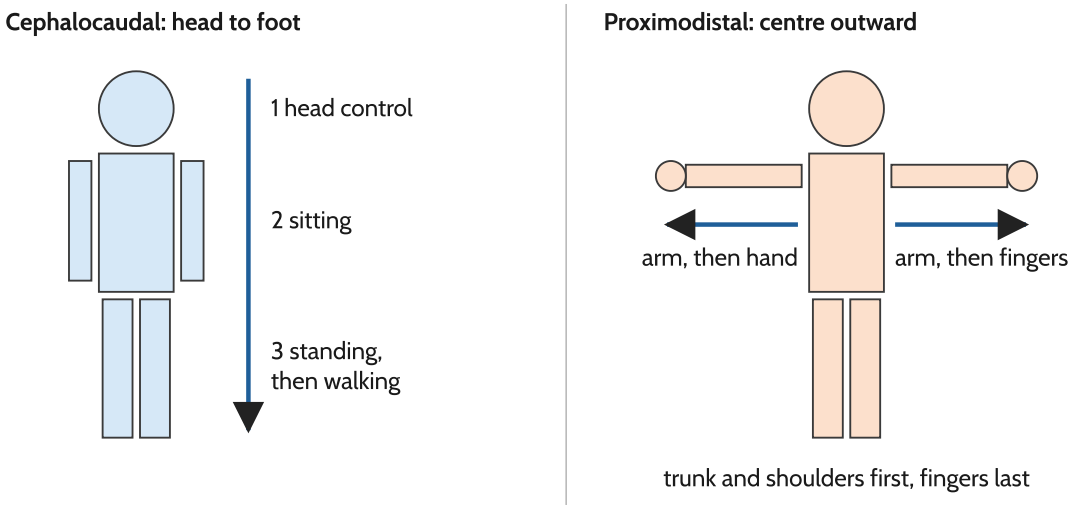


Figure 6.6 The two directions of motor development: head to foot (left) and centre to fingers (right). "Distal to proximal" runs the wrong way.

Table 6.9 gives average ages. Screening norms such as the Denver Developmental Screening Test (Chapter 2) show the ages by which 25, 50, 75 and 90 percent of babies pass each item: healthy babies vary by months.

Table 6.9 Gross motor milestones (approximate average ages).

Milestone	Approximate age
Holds head up steadily when held upright	6 weeks to 3 months
Rolls over	3 to 5 months
Sits without support	about 6 months
Crawls	7 to 9 months
Pulls to stand	8 to 9 months
Walks alone	about 12 months (9 to 17)

EXAM TRAP

Proximodistal means **centre to extremities**. "Distal to proximal" reverses it and is the false option; cephalocaudal, gross to fine, mass to specific, and shaping social and behavioural expectations are all true principles.

Fine motor skills: from reflex to pincer grasp

The hand tells the proximodistal story in miniature. The newborn's **palmar grasp reflex** closes her fingers round anything pressed into her palm. Her **prereaching** is poorly aimed swiping at objects in view, fading after about 2 months. At 3 to 4 months voluntary reaching brings the first voluntary grasp, the **ulnar grasp**: the fingers close clumsily against the palm, holding the object on the little-finger (ulnar) side. By 4 to 5 months she transfers objects from hand to hand. Control then shifts to the thumb side in **radial grasps** (the radius is the forearm bone on the thumb side), and by about 9 to 12 months the **pincer grasp**, thumb against forefinger, lifts that grain of rice.

EXAM TRAP

The keyed first grasp is the **ulnar grasp**, the first voluntary one; the palmar grasp at birth is a reflex. Texts order the middle steps differently (one even puts a clumsy palmar grip before the ulnar grasp), but the pincer grasp always comes last.

Explaining motor development

Why do babies sit, crawl and walk in this order? **Arnold Gesell** (1880 to 1961) gave the **maturational view**: motor development unfolds a genetically programmed sequence as nerves and muscles mature, so practice adds little (his co-twin study is in Section 6.1). Experience matters more than he thought. **Wayne Dennis** and Marsena Dennis (1940) found that Hopi infants bound to cradleboards walked at about the same age as unbound Hopi infants, which seemed to favour maturation; but in Iranian orphanages, where babies lay on their backs with almost no chance to move, Wayne Dennis (1960) found many 3- and 4-year-olds who could not walk alone. Cultures that train sitting and walking, such as the Kipsigis of Kenya, see these skills earlier. Maturation is necessary but not sufficient.

The modern account is **dynamic systems theory**, applied to motor skills by **Esther Thelen** (1941 to 2004): a skill is assembled from the body's current size and strength, the task and the support the environment offers. Take the newborn stepping reflex, which seems to vanish at about 2 months. It vanishes, Thelen showed, because the legs grow heavier faster than the muscles grow stronger: with her legs held in water, the infant steps again.

Methods for studying infant perception

How can you know what a baby perceives? Watch where she looks, how hard she sucks and how her brain responds. **Robert Fantz** (1961) used **preferential looking**: two stimuli are shown side by side and an observer times the looking at each. Infants from a few days to 6 months old looked longest at a face, then at patterns such as newsprint and a bull's-eye, and least at plain discs: they prefer patterns, and faces most of all.

The **habituation-dishabituation** method needs no favourite. Show a picture again and again and looking falls (**habituation**, the simplest form of learning); show a new one, and if looking recovers (**dishabituation**) the infant has noticed the difference. It is the standard test of discrimination and categorisation before language: habituate a baby to many cats, then show a new cat and a dog, and extra looking at the dog shows a category "cat". In **high-amplitude sucking**, strong sucks on a recording dummy switch on a sound, and sucking revives when a new sound is noticed. In the **violation-of-expectation** method of **Renée Baillargeon** (1985), infants look longer at an impossible event than a possible one, revealing what they expect of objects (Section 6.4). **Evoked potentials**, electrical brain responses recorded from the scalp, show whether the brain registers a difference when behaviour shows nothing.

EXAM TRAP

Habituation paradigms assess **discrimination and categorisation before language**, not long-term declarative memory (recognition here lasts minutes within one session), object permanence specifically (tested by violation of expectation) or moral reasoning (Section 6.9).

The infant's senses

Every sense works at birth, but vision is the least mature. Newborn **visual acuity**, the sharpness of sight, is poor (about 20/200 to 20/600, against the adult 20/20) but improves rapidly over the first year. Colour vision is present by about 2 to 3 months, and newborns already prefer faces.

Hearing works before birth. **Anthony DeCasper** and **William Fifer** (1980) found that newborns under 3 days old sucked in whichever pattern played their own mother's voice rather than another woman's. DeCasper and **Melanie Spence** (1986) had mothers read *The Cat in the Hat* aloud in late pregnancy; after birth the babies sucked to hear that story rather than a new one. Infants at first hear speech-sound contrasts their own language does not use, then lose them: **perceptual narrowing**. **Janet Werker** and **Richard Tees** (1984) found that most English-learning infants of 6 to 8 months heard the Hindi contrast between a dental and a retroflex "ta", but few did by 10 to 12 months.

Taste and smell are keen at birth: newborns relax at sweet tastes and grimace at bitter, and breastfed babies about a week old turn towards a pad their own mother has worn. Touch is well developed, and newborns feel pain. **Intermodal perception**, combining several senses into one event, appears early: **Elizabeth Spelke** (1976) showed 4-month-olds two films side by side with one soundtrack, and they looked longer at the film that matched the sound.

Depth, locomotion and affordances ●●

Eleanor Gibson and **Richard Walk** (1960) built the **visual cliff**, a glass-topped table with a checkerboard just under the glass on the shallow side and far below it on the deep side (full study in Section 1.4). Infants of crawling age, about 6 months onward, crossed the shallow side to their mothers but refused the deep side. Depth perception is therefore present by the time infants crawl: not fully formed at birth, not delayed until age 2 and not taught in words. Chicks and young goats tested within a day of birth also avoided the drop, which Gibson and Walk read as evidence that depth perception is largely innate.

Is the crawler's fear inborn? **Joseph Campos** and colleagues measured heart rate on the cliff. Two-month-olds on the deep side showed a slowing heart rate, a sign of interest; crawlers showed a quickening one, a sign of fear, and the more weeks an infant had crawled, the warier she was. Locomotor experience brings wariness of heights.

The **ecological view** of **James J. Gibson** (1904 to 1979) and Eleanor Gibson explains why: we perceive the world in terms of what we can do in it. An **affordance** is an action possibility the environment offers a particular perceiver: a slope affords crawling down to an experienced crawler but a fall to a novice. Because affordances depend on the perceiver's abilities, the change that most transforms them is **independent locomotion**: once the infant crawls and walks, distant toys become reachable and drops dangerous. Starting to speak, the attachment relationship and new caretaking reshape her social world, but none changes what her surroundings afford her actions nearly as much.

EXAM TRAP

The visual cliff tested **depth perception**, present **by crawling age**, not colour perception, object permanence (Section 6.4), language comprehension, imprinting (Section 6.7) or audio-visual integration (intermodal perception). Toddlers' affordances change most with **independent locomotion**.

Learning, imitation and the milestone sequence ●●

Infants learn from birth. Newborns can be classically conditioned: stroke a newborn's forehead just before giving sweet water and soon the stroke alone sets off sucking. In **Carolyn Rovee-Collier**'s operant studies a ribbon tied a 2- or 3-month-old's ankle to a mobile, so kicking made it move; days later infants kicked again at the sight of the same mobile, showing memory long before language.

KEY STUDY: MELTZOFF AND MOORE'S IMITATING NEWBORNS (1977)

Question. Can newborns imitate gestures they cannot see themselves make? **Method.** **Andrew Meltzoff** and **M. Keith Moore** showed infants 12 to 21 days old an adult protruding the tongue, opening the mouth and protruding the lips; coders blind to the gesture scored videotapes. **Result.** Infants made more of whichever gesture they had seen; imitation was later reported within hours of birth. **Conclusion.** Imitation is an important early feature of infant learning, appearing long before 1 year (and before the 8 to 12 months **Jean Piaget** expected), and not only for tongue protrusion. **Criticism.** Later studies often found only tongue protrusion matched, so the finding is debated. Deferred imitation is in Section 6.4.

Table 6.10 gathers the milestones of the first two years. Of the social smile, stranger anxiety, the pincer grasp and symbolic play, **symbolic play** (letting one object stand for another, a block for a phone) comes last, because it needs mental representation.

Table 6.10 The milestone sequence of the first two years (approximate ages).

Approximate age	Milestone
Birth	Reflexive smile, often in sleep
6 to 8 weeks	Social smile, at faces (Section 6.6)
About 2 months	Cooing (vowel-like sounds)
3 to 4 months	Laughter
About 6 months	Babbling ("ba-ba-ba")
About 8 months	Stranger anxiety (Section 6.6)
9 to 12 months	Pincer grasp
About 12 months	First word; first steps
18 to 24 months	Symbolic play; two-word speech

The language rows hide a gap: children understand about 50 words months before they can say 50. This **receptive-expressive gap** is the gap between what infants understand (receptive language) and what they can produce (expressive language); it concerns language, not cogni-

tive, motor or social skills. Language stages and theories (holophrases, telegraphic speech, overregularisation, the language acquisition device, motherese) are in Section 1.8.

> **EXAM TRAP**
>
> Imitation **may be seen as early as 12 to 21 days**, not only after 1 year and not only for tongue protrusion. Of the four milestones, **symbolic play** appears last.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The brain reaches about half its adult weight by about _____ months.
2. Removing rarely used synapses is synaptic _____.
3. Motor control spreading from the centre of the body outward is the _____ principle.
4. The first voluntary grasp is the _____ grasp.
5. Thelen's _____ theory explains the vanishing stepping reflex by heavier legs.
6. Looking less at a repeated picture and more at a new one underlies the _____ method.
7. English-learning infants losing a Hindi contrast by 10 to 12 months shows perceptual _____.
8. An action possibility the environment offers a perceiver is an _____.

Answers: (1) 6, (2) pruning, (3) proximodistal, (4) ulnar, (5) dynamic systems, (6) habituation-dishabituation, (7) narrowing, (8) affordance.

6.4 Piaget's Theory of Cognitive Development HIGH YIELD · 43 QS

Pour a five-year-old's juice from a short, wide glass into a tall, thin one and she will say she now has more, although she watched every drop. She is not careless; she is reasoning with the tools she has. A Swiss biologist built the most influential theory of children's thinking on mistakes like hers. This section follows it from method and core concepts through four stages and beyond, then weighs the evidence.

Piaget and his method ●●

Jean Piaget (1896 to 1980) was a biologist first, with a 1918 doctorate on molluscs. Around 1920, in the Paris laboratory of **Théodore Simon**, co-author with **Alfred Binet** (1857 to 1911) of the first intelligence scale (1905; Section 1.9), he scored children's answers to reasoning items. The wrong answers fascinated him: children of the same age made the same kinds of error, so younger children seemed not duller than older ones but to think differently.

In Geneva from 1921 he pursued one question for sixty years: how does knowledge grow? He called the field **genetic epistemology**, the study of the origins (genesis) of knowledge. His tool was the **clinical method** (*méthode clinique*), a flexible interview that follows each answer with further questions to uncover the reasoning behind it; for infancy he observed his own three children closely. His name-year books are *The Language and Thought of the Child* (1923),

The Moral Judgment of the Child (1932; Section 6.9), *The Origins of Intelligence in Children* (1936) and *The Construction of Reality in the Child* (1937); his stages of play are in Section 6.8.

Piaget treated **intelligence as biological adaptation**: as a species adapts through bodily structures, a child adapts through ever better mental structures. In his **constructivism** the child is an active "little scientist" who builds knowledge by acting on objects, first physically and later mentally.

> EXAM TRAP
>
> Piaget = intelligence as a biological adaptation, and the stages of cognitive development. Free association is **Sigmund Freud**'s (1856 to 1939), operant conditioning **Burrhus Frederic Skinner**'s (1904 to 1990) and aggression as learned social behaviour **Albert Bandura**'s (1925 to 2021). Psychosocial stages belong to **Erik Erikson** (1902 to 1994) and moral stages to **Lawrence Kohlberg** (1927 to 1987).

Schemes, adaptation and equilibration ●●

The unit of knowledge is the **scheme** (often called a schema), an organised pattern of action or thought for dealing with a class of experiences: a newborn's sucking and grasping, a ten-year-old's adding and classifying. Two **functional invariants** shape schemes at every age. **Organisation** combines simple schemes into larger systems, as looking and grasping merge into reaching for what one sees. **Adaptation** adjusts schemes to the environment through two complementary processes.

In **assimilation** the child fits new experience into an existing scheme: a two-year-old whose "doggie" scheme means "four-legged animal" calls a cow "doggie". In **accommodation** the child changes the scheme, that is, his way of thinking, to fit experience that will not fit: the cow moos, an adult says "cow", and the scheme splits in two. Assimilation keeps knowledge stable; accommodation makes it grow.

When schemes handle events, the child is in **equilibrium**. A misfit produces **disequilibrium**, an uncomfortable cognitive conflict, resolved by assimilating and accommodating until a new, more advanced balance is reached. This self-regulating process, **equilibration** (Figure 6.7), drives the child from stage to stage. Mature schemes become **operations**: internalised mental actions that are reversible (they can be undone in thought) and organised into logical systems; the stage names describe the child's relation to them.

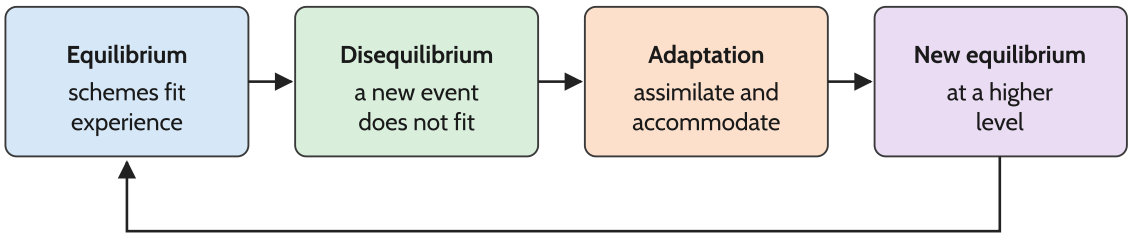


Figure 6.7 Equilibration. A misfit between schemes and experience creates disequilibrium; assimilation and accommodation restore balance at a more advanced level.

Piaget named four factors in development: **maturation** of the nervous system; **experience**, both physical (properties of objects, such as weight) and logico-mathematical (relations one's own actions create, such as ten pebbles staying ten in any order of counting); **social transmission** (being taught); and equilibration, which coordinates the other three. Development thus comes from maturation and learning together, never from either alone, and not from social and cultural practice alone, which is nearer Vygotsky's emphasis (Section 6.5).

> EXAM TRAP
>
> Piaget's two basic processes of learning and adaptation are **assimilation and accommodation**; encoding and retrieval are memory processes (Section 1.7), and attention and perception, or sensory input and motor output, are not his pair. Changing one's way of thinking to fit new experience is **accommodation**, not assimilation, conservation (below) or **habituation**, the decline in responding to a repeated stimulus (Section 6.3).

The four stages at a glance ●●

Piaget's stages are qualitatively different ways of thinking, not simply more knowledge. Every child passes through them in the same **invariant sequence**, none skipped or reversed, each built on the last. The ages are approximate; the order is what the theory fixes. Table 6.11 maps the rest of this section.

Table 6.11 Piaget's four stages of cognitive development.

Stage	Age (years)	Main achievements	Main limits
Sensorimotor	Birth to 2	Object permanence; intentional action; deferred imitation and first symbols by 18 to 24 months	Thinks through action; no mental symbols until the end
Preoperational	2 to 7	Symbolic function: language, pretend play, drawing	Egocentrism, animism, centration, irreversibility, no conservation
Concrete operational	7 to 11	Conservation, decentration, reversibility, seriation, transitivity, classification	Logic only about concrete, real things
Formal operational	11 or 12 onward	Abstract, hypothetical, propositional and hypothetico-deductive reasoning	Idealism, adolescent egocentrism; not reached by everyone

EXAM TRAP

The stages are sensorimotor, preoperational, concrete operational and formal operational. **Hypothetico-deductive reasoning is not a stage**; it is the hallmark of the formal operational stage. Preoperational = 2 to 7 years; concrete operational begins at about 7.

Sensorimotor stage and object permanence ●●●

For the first two years, Piaget argued, infants think with senses and actions rather than mental symbols: the **sensorimotor stage**. Its motor is the **circular reaction**, an action the infant hits on by chance and repeats because the result is interesting; its target moves outward from her own body (primary) to objects (secondary) to deliberate experiment (tertiary), as Table 6.12 shows. By 8 to 12 months acts are intentional, one scheme serving as the means to another (pushing a cushion aside to reach a toy). By 18 to 24 months she represents the world mentally: she solves problems by sudden insight, pretends, and shows **deferred imitation**, copying an action seen hours or days earlier.

Table 6.12 The six sensorimotor substages.

Substage	Months	What the infant does	Object permanence
1 Reflexes	0 to 1	Exercises inborn reflexes: sucking, grasping	None
2 Primary circular reactions	1 to 4	Repeats chance actions on own body (thumb sucking); coordinates schemes across senses (looks towards a sound)	Looks briefly where it vanished
3 Secondary circular reactions	4 to 8	Repeats interesting effects on objects (shakes a rattle)	Retrieves a partly hidden object
4 Coordination of secondary schemes	8 to 12	Intentional, goal-directed, means-end acts	Searches for a fully hidden object; A-not-B error
5 Tertiary circular reactions	12 to 18	Varies actions to see what happens	Follows visible moves; fails invisible ones
6 Mental representation (invention of new means)	18 to 24	Insight, deferred imitation, symbolic thought, pretend play	Complete: solves invisible displacements

The stage's best-known achievement is **object permanence**, the understanding that objects go on existing when they can no longer be seen, heard or touched. To a 4-month-old, Piaget claimed, a toy covered by a cloth has ceased to be. Search for a fully hidden object begins in substage 4, about 8 to 12 months, with a revealing mistake. Hide a toy under cloth A several times and let the infant find it; then, as she watches, hide it under cloth B. She searches under A. This **A-not-B error** shows that object permanence is still incomplete: the object belongs where her action last succeeded. At 12 to 18 months she follows visible moves but not invisible ones (the toy, hidden in a hand, is slipped under a cushion); only at 18 to 24 months does she infer where it must be, and object permanence is complete.

Adele Diamond (1985) showed that the A-not-B error grows with the delay between hiding and search, and later tied its disappearance to maturation of the dorsolateral prefrontal cortex, which supports working memory and inhibition of a *prepotent* (habitual, previously rewarded) reach.

KEY STUDY: BAILLARGEON'S DRAWBRIDGE (1985, 1987)

Renée Baillargeon used the **violation-of-expectation** method: infants look longer at events that surprise them. A screen rotated like a drawbridge towards a box placed behind it. In the possible event it stopped at the hidden box; in the impossible event it swung through the box's space. Five-month-olds (1985), and later some infants of 3.5 to 4.5 months (1987), looked longer at the impossible event, as if they knew the hidden box still existed. Criticism: longer looking may reflect novelty or perceptual preference rather than knowledge.

EXAM TRAP

Object permanence is a **sensorimotor** achievement, never preoperational, concrete or formal, and not to be confused with conservation. Keyed ages: it **begins about 8 to 12 months** (substage 4, A-not-B error) and is **fully established by 18 to 24 months**; 2 months, 4 to 6 months, 8 to 10 months (onset only) or 3 years are wrong for full mastery. The A-not-B error reflects incomplete object permanence with a prepotent reach, not centration or failed reversibility.

Preoperational thought ●●

Around the second birthday the **symbolic function** appears: the ability to make one thing (a word, image or gesture) stand for another, seen in language, pretend play (a banana becomes a telephone) and drawing. The **preoperational stage** (2 to 7) is named for what the child cannot yet do: perform operations. It has a **symbolic function substage** (2 to 4) and an **intuitive thought substage** (4 to 7), when children ask endless "why" questions and judge by intuition rather than logic. **Judy DeLoache** (1987) found that children under about 3 cannot use a scale model of a room to find a toy hidden in the real room: they lack **dual representation**, treating an object both as a thing and as a symbol for something else.

In the Piagetian view, then, preschoolers are unable to reason properly, and their thought shows a family of related limits:

- **Egocentrism**: the inability to distinguish one's own visual or conceptual perspective from another person's; a cognitive limit, not selfishness.
- **Animism**: attributing life and feelings to inanimate things ("the sun is following me").
- **Artificialism**: believing natural things were made by people ("someone painted the sky").
- **Centration**: focusing on one salient feature while ignoring the others. A 5-year-old who sorts a deck of cards by colour but cannot then re-sort the same deck by suit is centring on colour.
- **Irreversibility**: the inability to run an action backwards in the mind, with a focus on **states, not transformations**: the child compares before and after and ignores the pouring between.
- **Transductive reasoning**: reasoning from particular to particular with no general rule. Piaget's daughter, having missed her nap, announced, "It isn't afternoon."
- Failure of **class inclusion**: shown wooden beads, mostly brown, and asked "More brown beads or more wooden beads?", the child says "more brown", comparing subclasses instead of a subclass with the whole.
- Lack of conservation, the subject of the next subsection.

KEY STUDY: THE THREE MOUNTAINS TASK

Piaget and **Bärbel Inhelder** (1913 to 1997) seated children before a model of three mountains (one snow-capped, one with a house, one with a red cross) and asked them to pick the photograph showing the view of a doll sitting elsewhere. Four-year-olds usually picked their own view; correct choices became common only around 7 to 8. Piaget read this as egocentrism. Criticism: the task is unfamiliar and spatially hard; meaningful versions show perspective taking much earlier (Evaluating Piaget).

EXAM TRAP

Centration = focusing on one salient feature while ignoring others (the card sorter; judging liquid by height), not egocentrism, animism, transductive reasoning, irreversibility or the reversible operation behind conservation. Coordinating schemes across senses is a sensorimotor achievement, not centration. "Preschoolers cannot reason properly" is the **Piagetian** view, not that of **Arthur Robert Jensen** (1923 to 2012, Level I and Level II abilities), **Joy Paul Guilford** (1897 to 1987, structure of intellect) or a "holistic" view (Section 1.9).

Conservation ●●●

Conservation is the understanding that a quantity stays the same despite changes in shape or arrangement, so long as nothing is added or taken away. Every task follows one script (Figure 6.8). The child agrees that two identical short, wide glasses hold equal water. One is then poured, as she watches, into a tall, thin glass, where the water stands higher (it does not stay at the same height; that is the point). The preoperational child now says the tall glass has more: her judgement is swayed by one dimension, height, and she ignores the narrower width.

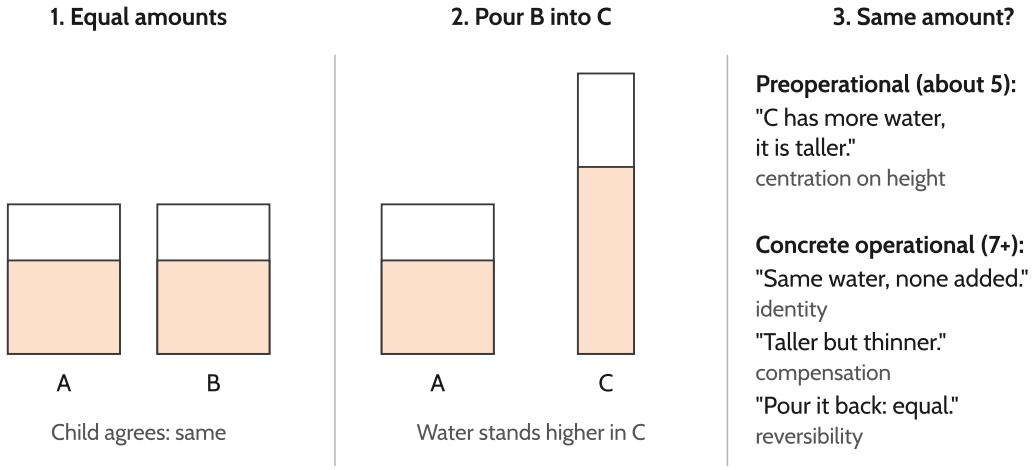


Figure 6.8 The liquid conservation task. Beaker C is half as wide as A, so the same water stands twice as high; only the conserving child sees that the amount is unchanged, and justifies it in one of three ways.

The conserving child gives one of three justifications, exactly the operations the preoperational child lacks:

- **Identity**: "It's the same water; you didn't add any or take any away." The principle of identity is the recognition that certain characteristics of an object remain fixed even when its appearance changes.
- **Compensation**: "It's taller, but it's thinner." A change in one dimension is offset by a change in another.
- **Reversibility** (negation): "If you pour it back, it will be the same as before."

Conservation is several achievements, mastered in a typical order (Table 6.13). A child who conserves substance at 7 or 8 may fail with weight until 9 or 10, though both need the same operation. Piaget called this **horizontal décalage** (French for "gap"): one operation, applied to different contents, mastered at different ages within a stage. **Vertical décalage** is the same achievement recurring at a higher level in a later stage: a toddler finds her way round the house by action but maps it only years later. Conservation of liquid, loosely called liquid volume, differs from volume by displacement, which comes last; all count as concrete operational achievements.

Table 6.13 Conservation tasks in their usual order of mastery (ages approximate; sources differ by a year).

Conservation of	The change the child sees	Usual age (years)
Number	One of two equal rows of counters spread out	6 to 7 (first)
Liquid quantity	Water poured into a taller, thinner glass	7 to 8
Mass (substance)	One of two equal clay balls rolled into a sausage	7 to 8
Length	One of two equal sticks moved ahead	7 to 8
Area	Equal blocks clustered on one field, scattered on another	8 to 9
Weight	Reshaped clay: will it still balance the scale?	9 to 10
Volume (displacement)	Reshaped clay: will it still raise the water as much?	11 to 12 (last)

EXAM TRAP

The **principle of identity** (certain characteristics of objects remain fixed when their appearance changes) is not recognising positive characteristics in oneself (self-esteem), identifying with one's gender (gender identity, Section 6.8) or bringing behaviour into harmony with the world. In the liquid task the liquid does **not** stay at the same height, and the test is not only for under-5s.

Concrete operational stage ●●

At about 7 children begin to perform true operations on real objects and events: the **concrete operational stage** (7 to 11). Shown the clay ball rolled into a sausage, the child now says it is still the same amount, and can say why. The gains behind this are:

- **Decentration**: attending to several features at once, such as height and width.
- Reversibility: mentally undoing an action, which first appears in this stage.
- **Seriation**: arranging items along a dimension (sticks from shortest to longest) systematically, not by trial and error.
- **Transitive inference** (transitivity): if A is longer than B and B is longer than C, then A is longer than C, without comparing A and C directly.

- Classification by more than one attribute, and success on class inclusion ("more wooden beads").
- Reduced egocentrism in perception and conversation.

The limit is in the name: the concrete thinker reasons logically about real, picturable things but balks at the purely hypothetical. Told "If all dogs were pink and Bruno is a dog, what colour is Bruno?", she objects that dogs are not pink.

> EXAM TRAP
>
> In the concrete stage reduced egocentrism, transitive inference and conservation of mass are present, object permanence is long established and deferred imitation dates from infancy. What the concrete operational child struggles with, and does **not** attain, is **hypothetico-deductive reasoning**.

Formal operations and postformal thought ●●

From about 11 or 12 the **formal operational stage** frees thought from the concrete. Adolescents reason about abstractions (justice, infinity), about the possible as well as the real, and about **propositions**, statements whose conclusions follow from their form ("If A then B; A; therefore B"). The signature is **hypothetico-deductive reasoning**: deducing hypotheses from a general theory and testing them systematically, one variable at a time. With it come **counterfactual reasoning** (from premises contrary to fact: "If the moon were made of cheese…"), **combinatorial reasoning** (generating every possible combination of factors) and metacognitive reflection, thinking about one's own thinking.

Inhelder and Piaget's tasks appear in *The Growth of Logical Thinking from Childhood to Adolescence* (1958). In the **pendulum problem** the child must find what controls how fast a pendulum swings: string length, weight, height of release or force of push. The concrete thinker changes several things at once; the formal thinker varies one at a time and finds that only length matters. In the chemical combination task the adolescent mixes colourless liquids in every combination to find which turns yellow. The new power brings idealism (measuring the real world against ideal ones) and adolescent egocentrism (Section 6.10). Nor is the stage universal: many adults do not reason formally on unfamiliar content.

Is formal logic the end point? Several theorists proposed **postformal thought**, adult reasoning that accepts contradiction, uncertainty and context (Table 6.14). Its core is **dialectical reasoning**: seeing that opposing views each hold part of the truth and integrating them in a synthesis.

Table 6.14 Postformal thought.

Theorist	Year	Idea
Klaus Riegel	1973	Dialectical operations, "the final period of cognitive development": integrating contradiction
William Perry	1970	College students move from dualism (right versus wrong) through relativism to commitment
Patricia Arlin	1975	Problem finding: a fifth stage beyond formal problem solving
Michael Basseches	1984	Dialectical thinking: seeing systems in constant change and interaction

Theorist	Year	Idea
Gisela Labouvie-Vief	1980	Pragmatic thought: logic adapted to real-life limits and emotion
Jan Sinnott	1998	Relativistic thought: choosing among logics, each valid in its context

EXAM TRAP

The stage of abstract thought is the **formal operational** stage. It includes hypothetical, hypothetico-deductive, propositional, counterfactual, combinatorial and metacognitive reasoning. **Dialectical reasoning is postformal** (Riegel, Basseches), not formal operational.

Evaluating Piaget

Piaget's broad picture survives: children actively construct knowledge, thinking changes in kind as well as amount, and his sequence recurs across cultures. His timing and stages have fared worse.

First, he **underestimated infants and preschoolers**. Baillargeon credited infants with object knowledge months early, and many preschool failures came from the task rather than the mind. **Margaret Donaldson** (1926 to 2020) drew the lesson in *Children's Minds* (1978): children reason well when a task makes "human sense" to them.

KEY STUDY: POLICEMEN AND A NAUGHTY TEDDY

Martin Hughes (1975) asked children of 3.5 to 5 to hide a boy doll from two policeman dolls among crossing walls; about 90 percent succeeded, though the task needs two perspectives at once, because hiding makes sense to them. **James McGarrigle** and Donaldson (1974) let a "naughty teddy" disarrange one row of the number-conservation task as if by accident; far more 4- to 6-year-olds conserved than when the adult changed the row on purpose. Criticism: some may have ignored the accidental change rather than truly conserved.

Second, development is less stage-like than claimed: décalage is the rule, and abilities grow gradually, domain by domain. Third, culture and schooling matter more than he allowed, the core of Lev Vygotsky's critique (Section 6.5). Fourth, he did not separate **competence**, the understanding a child has, from **performance**, what she shows on a task with its own language, memory and attention demands.

Neo-Piagetian theories keep the stages but explain them by growing processing capacity. **Juan Pascual-Leone** (1970) proposed **M-space**, the number of schemes a child can hold in mind at once, which grows with age. **Robbie Case** (1944 to 2000) traced stage progress to more efficient use of limited working memory as skills become automatic. **Kurt Fischer** (1943 to 2020) proposed skill theory (1980): skills develop through levels, separately in each domain, and run higher with support than alone.

EXAM TRAP

Spot the false statement: "Cognitive structures of children under 7 reflect abstract mental operations" is **false**; their thought is preoperational. True Piagetian statements: children construct knowledge by acting on objects physically or mentally; infant schemes are organised patterns of action; older children's operations let them see that quantities are conserved despite changes in appearance.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Changing one's schemes to fit a new experience is _____; fitting it into existing schemes is _____.
2. Piaget saw intelligence as a form of biological _____.
3. Object permanence begins at about _____ months and is complete by _____ months.
4. An infant who searches where a toy was last found, not where she saw it hidden, shows the _____ error.
5. A 5-year-old who sorts cards by colour but cannot re-sort them by suit shows _____.
6. Saying "it's the same clay; nothing was added" is the conservation justification called _____.
7. Mastering conservation of mass years before conservation of volume is _____ décalage.
8. Dialectical reasoning belongs to _____ thought, not formal operations.

Answers: (1) accommodation, assimilation; (2) adaptation; (3) 8 to 12, 18 to 24; (4) A-not-B; (5) centration; (6) identity; (7) horizontal; (8) postformal.

6.5 Vygotsky, Bruner, Information Processing and Theory of Mind

MEDIUM YIELD · 25 QS

Piaget's child (Section 6.4) works alone, a small scientist testing the world. Yet a real child learning to tie her shoelaces usually has an older sister kneeling beside her. Vygotsky and Bruner put that helper back into the picture; the information-processing approach treats the mind as a computer that gets faster and smarter; and research on theory of mind asks when children discover that other people have minds of their own.

Vygotsky's sociocultural theory

Lev Semyonovich Vygotsky (1896 to 1934), a Russian psychologist who died of tuberculosis at 37, saw cognitive growth as a socially mediated activity: children acquire new ways of thinking through cooperative dialogue with more skilled members of their culture. This is his **sociocultural theory**, set out in *Thinking and Speech* (1934) and known to English readers mainly through *Mind in Society* (1978), an edited collection of his essays.

Vygotsky separated elementary mental functions, natural capacities we share with animals (simple attention, perception, involuntary memory), from **higher mental functions** such as

voluntary attention, deliberate memory and verbal reasoning, which are human and cultural. What turns the first into the second is **mediation**: we act on the world, and on our own minds, through tools. Each culture supplies **tools of intellectual adaptation**, signs such as number systems, writing, maps and memory aids, that let children use their basic functions more adaptively; a knot tied in a handkerchief to remember an errand is a sign steering one's own memory. Language is the master tool.

How does a tool that exists between people get inside one head? The **general genetic law of cultural development** states that every higher function appears twice: first between people (interpsychological), then within the child (intrapsychological). This passage from outer to inner is **internalisation**. A mother who asks "Where did you last see your bag?" remembers together with her son; years later he asks himself silently. Joint work rests on **intersubjectivity**, the process by which two people who begin a task with different understandings arrive at a shared one. Inter-subjectivity therefore precedes intra-subjectivity.

Development, Vygotsky added, must be studied at four interrelated levels: microgenetic (change over minutes or days), ontogenetic (over a lifetime), phylogenetic (over evolution) and sociohistorical (change in the culture's values and technologies).

> EXAM TRAP
>
> For Vygotsky, **inter-subjectivity precedes intra-subjectivity**. "Intra precedes inter" reverses the law, and "children do not engage in inter-subjectivity" contradicts it. "The two are related" is true but empty: the law is about their order.

The zone of proximal development and scaffolding ●●●

Vygotsky described two children who, working alone, both solve test problems set for eight-year-olds. Given hints, one solves problems set for twelve-year-olds, the other only those for nine-year-olds. The distance from eight to twelve, or from eight to nine, is what he called the **zone of proximal development** (ZPD): the gap between the child's actual developmental level, shown by independent problem solving, and her level of potential development, shown by problem solving under adult guidance or with more capable peers. That helper is now called the **more knowledgeable other**. As a range of tasks, the ZPD holds those too difficult to master alone but achievable with help (Figure 6.9). Teaching should aim inside it, at functions still ripening.

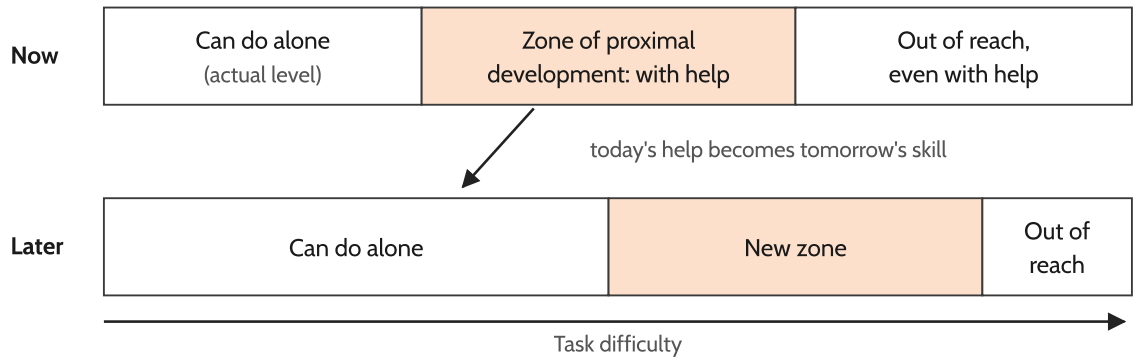


Figure 6.9 The zone of proximal development. As the child internalises what she first did with help, the zone moves towards harder tasks.

Teaching within the ZPD is now usually called **scaffolding**: support fitted to the learner's current level and withdrawn as competence grows. The term is not Vygotsky's: it comes from **David Wood**, Jerome Bruner and **Gail Ross** (1976).

KEY STUDY: WOOD, BRUNER AND ROSS (1976)

A tutor helped 3- to 5-year-olds build a pyramid of interlocking wooden blocks, a task too hard for them alone. Good tutoring recruited interest, simplified the task, kept the goal in view, marked critical features, controlled frustration and demonstrated, stepping in only as far as the child needed. **Criticism:** the metaphor implies a fixed structure built by the adult, whereas real tutoring is a two-way exchange.

Barbara Rogoff (*Apprenticeship in Thinking*, 1990) described development as an apprenticeship built on **guided participation**: children learn by taking part, beside adults and peers, in their community's valued activities. In **reciprocal teaching** (**Annemarie Palincsar** and **Ann Brown**, 1984), teacher and pupils take turns leading discussion of a text through questioning, summarising, clarifying and predicting, the teacher fading as pupils take over. **Dynamic assessment** (test, teach, retest) measures how much a child gains from help, as in **Reuven Feuerstein**'s Learning Potential Assessment Device (1979).

EXAM TRAP

The ZPD is the **gap between independent and assisted performance**, a term coined by **Vygotsky** (not Piaget, Bruner, Kohlberg, Bronfenbrenner, Erikson, Skinner or Freud). Wrong options name something else: tasks done alone (the actual level), too simple, too hard even with help, or learned without instruction; innate abilities, structures or the language acquisition device (Section 1.8); a critical, optimal or language-learning period (Section 6.1) or the usual age for a milestone; a stage of logical operations or object permanence (Section 6.4); a fixed capacity or genetic ceiling; mental age minus chronological age (Section 1.9); a brain or adolescent growth spurt; a neural region; a peer group.

Thought, speech and concepts ●●

On language the two theorists disagreed most clearly. For Piaget thought comes first, and language expresses concepts already built through action; the preschooler's talk to no one, his **egocentric speech** (*The Language and Thought of the Child*, 1923), mirrors egocentric thought and simply fades. For Vygotsky thought and language have separate roots: prelinguistic thought (a toddler using a stick to reach a toy) and preintellectual speech (babbling). At about age 2 they merge; thought becomes verbal, and from then on language shapes thought. So thought depends on language for Vygotsky, and language depends on thought for Piaget.

Speech then turns inward. Social speech, used to communicate, comes first. From about 3 children talk aloud to themselves as they work ("the red one goes… no, here"). Vygotsky saw this talk, now called **private speech**, as social speech turned to guiding one's own actions; at school age it goes underground as **inner speech**, silent verbal thought. **Laura E. Berk**'s studies in the 1980s and 1990s supported him: children use more private speech on hard tasks and after errors, those who use it more tend to improve, and it fades to whispers before going inward.

> EXAM TRAP
>
> **Vygotsky: thought depends on language; Piaget: language depends on thought.** The reversed option swaps the names, and "one studied early, the other later language" is false. Private speech guides and regulates behaviour and becomes inner speech. It is not a regression to "autistic" (undirected, wishful) thinking, a word Piaget used, nor mere imitation, a language disorder or egocentrism that fades with schooling.

Vygotsky traced concept formation with block-sorting tasks. Young children form heaps (syncretic groupings), objects lumped together by chance impression. Next come complexes, groups built on concrete, shifting links (a red block, then a red triangle, then any triangle). The most advanced complex, the **pseudoconcept**, looks like a true concept (every triangle gathered) but rests on visible likeness, not an abstract rule. True concepts mature in adolescence. Vygotsky linked them to the distinction between the acting "I" and the reflecting "I" (the self that does and the self that thinks about its doing), to the problems and developmental tasks society sets adolescents, and to participation in its institutions. Gender differences play no part.

Piaget and Vygotsky compared ●

Both men were born in 1896, both saw the child as active, and neither held that development is solely environmental or predetermined. Table 6.15 sets out where they differed.

Table 6.15 Piaget versus Vygotsky.

Issue	Piaget	Vygotsky
The child	Lone scientist	Social apprentice
Universality	Universal structures, invariant stages	Skills often specific to the culture, learned from more competent people
Learning and development	Development leads learning	Learning leads development

Issue	Piaget	Vygotsky
Engine	Equilibration through acting on objects	Social interaction and cultural tools
Language and thought	Thought first; language reflects it	Separate roots, merge at about 2; language shapes thought
Self-talk	Egocentric speech, fades	Private speech, becomes inner speech
Partners	Peers, whose differing views create conflict	Adults and more capable peers
Classroom	Discovery, readiness	Scaffolding in the ZPD

> EXAM TRAP
>
> Options often swap the theorists: it was **Piaget**, not Vygotsky, who held that all children pass through the same stages. "Solely environmental" and "predetermined" fit neither man.

Jerome Bruner: representation and discovery

Jerome Seymour Bruner (1915 to 2016), an American leader of the cognitive revolution, shared Vygotsky's interest in culture and teaching. In 1966 he proposed three **modes of representation**. The **enactive** mode represents knowledge through action (a baby knows a rattle as a shaking movement); the **iconic** mode, through images (picturing the route to school); the **symbolic** mode, through language, numbers and other arbitrary symbols. They appear roughly in that order from infancy to the school years, but unlike Piaget's stages none is left behind.

In *The Process of Education* (1960) Bruner argued that any subject can be taught in some intellectually honest form to any child at any stage. Hence the **spiral curriculum**, which returns to the same big ideas at rising levels of complexity, and **discovery learning**, in which learners find principles for themselves by working on problems. His language acquisition support system (LASS) is covered in Section 1.8.

Information processing: the mind as a computer

The **information-processing approach** likens the mind to a computer: information comes in, is held and transformed, and leaves as an answer. Development means better hardware (the brain and nervous system) and better software (attention, memory strategies and problem-solving routines), and it is continuous: many small quantitative gains, not qualitative stages.

The hardware speeds up. **Robert V. Kail** (1991) showed that processing speed rises steeply in childhood and more slowly in adolescence, at a similar rate across very different tasks, which suggests a general biological cause such as myelination. Attention grows more selective, working memory holds more, and the **executive functions** of the prefrontal cortex mature: inhibitory control (stopping a dominant response), working memory (holding and updating information) and cognitive flexibility (switching rules or viewpoints).

The software improves too. Memory strategies appear in order: rehearsal (repeating items) from about 7, organisation (grouping items by meaning) in middle childhood and elaboration (linking items through an image or sentence) mostly in adolescence. A child with a **produc-**

tion deficiency, **John H. Flavell**'s term, can use a strategy when told but does not produce it spontaneously; one with a **utilisation deficiency** (**Patricia H. Miller**) produces it but at first gains little from it.

Flavell (1979) also made **metacognition**, knowledge and control of one's own thinking, central; its memory part is **metamemory**, such as knowing that a long list needs more study. Knowledge matters as much as capacity: **Michelene Chi** (1978) found that child chess experts recalled board positions better than adult novices, who still recalled digit lists better. **Robert S. Siegler**'s **overlapping waves model** (1996) holds that children always have several strategies (for 3 + 5: counting on fingers, counting on from the larger number, simply retrieving 8) whose frequencies shift gradually, like overlapping waves, rather than in stages. In **fuzzy-trace theory** (**Charles Brainerd** and **Valerie Reyna**, 1990) we store separate verbatim and gist traces and rely more on gist with age.

Theory of mind ●●

A **theory of mind** is the understanding that people have mental states (beliefs, desires, intentions) that guide their actions and can differ from one's own and from reality. The term comes from **David Premack** and **Guy Woodruff** (1978), who asked whether a chimpanzee has one. Its precursors appear late in the first year: **joint attention** (following another's gaze or point to a shared object), social referencing (Section 6.6) and intention reading (treating a reach as aimed at a goal).

Gyorgy Gergely and **Gergely Csibra** (from 1995) call this early framework the **teleological stance**: around the end of the first year infants read an action as efficient and goal-directed, expecting an agent to take the most direct route its situation allows. In the philosopher **John Searle**'s (1983) terms, toddlers understand **intentions in action**, the goal visible in an ongoing act. They have moved beyond mere **contingencies** (this event follows that one) but not yet to **prior intentions** (plans formed before acting) or mental states such as belief and desire.

Henry Wellman (1990) traced the next steps: a desire psychology at about 2, explaining actions by wants, then a **belief-desire psychology** at 3 to 4, in which actions follow from what people want and what they believe, even falsely. The test is the **false-belief task**: the child must predict the action of someone whose belief the child knows is wrong. In **Heinz Wimmer** and **Josef Perner**'s original (1983), Maxi puts his chocolate in one cupboard and goes out; his mother moves it. Where will Maxi look? In the Smarties (unexpected-contents) task of Perner, **Susan Leekam** and Wimmer (1987), a child who expected sweets in a Smarties tube is shown pencils and asked what a friend will think is inside. Three-year-olds answer from reality; most children pass at about 4, in the preoperational years. Flavell's **appearance-reality distinction** comes at a similar age: shown a sponge painted like a rock, a 3-year-old says it looks like a sponge, while a 5-year-old says it looks like a rock but is really a sponge.

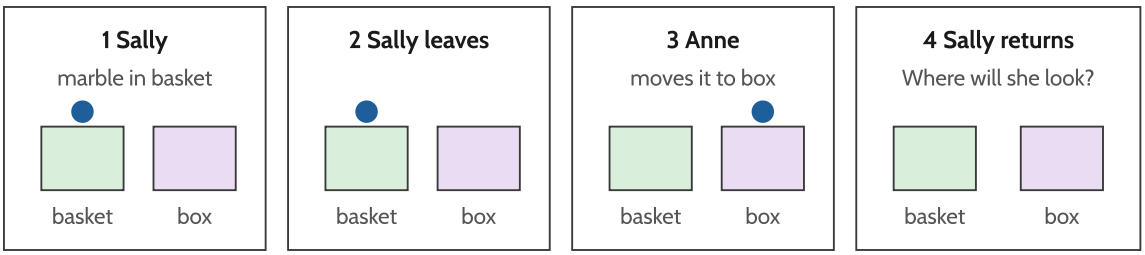


Figure 6.10 The Sally-Anne task. To pass, the child must answer from Sally's mistaken belief, not from what she herself knows.

KEY STUDY: BARON-COHEN, LESLIE AND FRITH (1985)

Simon Baron-Cohen, **Alan Leslie** and **Uta Frith** acted out the **Sally-Anne task** with dolls (Figure 6.10). About 85 percent of typically developing preschoolers and 86 percent of children with Down syndrome answered "the basket", but 80 percent of autistic children, though of higher verbal mental age than the Down syndrome group, pointed to the box. The authors proposed a specific theory-of-mind deficit in autism, later called mindblindness (Chapter 8). **Criticism:** the task also loads on language and memory, and some autistic children pass it.

EXAM TRAP

False-belief tasks (Maxi, Sally-Anne, Smarties) assess **theory of mind** and are passed at **about age 4**, in the preoperational period: not at 12 months (only the teleological stance), age 8 (concrete operations) or adolescence. They are not the A-not-B task (object permanence), a conservation task (both Section 6.4), the Strange Situation (attachment, Section 6.7) or a working memory test.

Recursive thought and social perspective taking

Passing a false-belief task is first-order reasoning: "Sally thinks the marble is in the basket." A **second-order theory of mind** nests one belief inside another ("John thinks that Mary thinks the ice-cream van is in the park"); Perner and Wimmer (1985) found it at about 6 to 7. The ability to nest perspectives is **recursive thought**, a form of perspective taking that views a situation from at least two perspectives, reasoning simultaneously about what two or more people are thinking ("I think that you think that I think..."). Patricia Miller, **Frank Kessel** and Flavell (1970) showed it growing through the school years. It is not **referential style**, **Katherine Nelson**'s term for toddlers whose first words mostly name objects; not **reversibility**, Piaget's mental undoing of an operation (Section 6.4); and not **reconstruction**, the rebuilding of memories (Section 1.7).

Robert L. Selman (1976, 1980) studied **social perspective taking**, understanding another person's thoughts and feelings, through dilemmas. In the best known, Holly has promised her father not to climb trees, then finds a friend's kitten stuck in one. From the answers

Selman described five levels (Table 6.16), a progression from egocentric to multiple, mutual perspective taking.

Table 6.16 Selman's levels of social perspective taking (ages approximate and overlapping).

Level	Age	What the child understands
0 Undifferentiated (egocentric)	3 to 6	Confuses others' views with her own: "He'll be happy; he likes kittens."
1 Social-informational (differentiated, subjective)	4 to 9	Views differ because information differs: "If he knows why, he won't be cross."
2 Self-reflective (reciprocal)	7 to 12	Sees her own thoughts from another's viewpoint, and knows others can do the same
3 Third-party (mutual)	10 to 15	Sees both views as a neutral third person would
4 Societal (in-depth)	Adolescence onward	Sees views as shaped by society's values and norms

EXAM TRAP

Selman's levels run from **egocentric to multiple, mutual perspective taking**, not from concrete to abstract number, inductive to deductive logic or procedural to declarative memory. Age bands differ across sources (level 1 is given as 4 to 9, 5 to 9 or 6 to 8), and one key places the move from level 1 to level 2 at **5 to 9 years**: choose it over 3 to 5 (level 0), 10 to 12 (level 3) or late adolescence (level 4).

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The gap between what a learner can do alone and with guidance is the _____.
2. Every higher function appears first between people, then within the child: the general _____ law of cultural development.
3. Self-guiding talk aloud that later becomes inner speech is _____ speech.
4. Bruner's modes of representation are enactive, iconic and _____.
5. Using a strategy when told but never spontaneously is a _____ deficiency.
6. Toddlers taking the teleological stance understand intentions in _____.
7. Reasoning simultaneously about what two or more people are thinking is _____ thought.

Answers: (1) zone of proximal development, (2) genetic, (3) private, (4) symbolic, (5) production, (6) action, (7) recursive

6.6 Emotional Development, Temperament and Self-Regulation

LOWER YIELD · 19 QS

Follow one baby through her first three years: at two months she smiles at your face, at eight months she cries when a stranger reaches for her, and at two years she hides her face in embarrassment. Alongside this shared timetable runs an individual style, temperament, and the growing capacity to manage it, self-regulation.

The first emotions

What can a newborn feel? In the **differentiation theory** of **Katharine Banham Bridges** (1932), built from observing infants of different ages, the newborn has only a diffuse *general excitement*. Within about three months this splits into two poles, **distress** and **delight**, which are therefore the first emotions infants express. Each pole then divides again, distress into anger, disgust and fear, delight into elation and affection, until by two years most adult emotions can be seen (Figure 6.11).

Two rival views differ. **John Broadus Watson** (1878 to 1958) held that only **fear, rage and love** are innate and all other emotions are conditioned onto them. **Carroll Izard** (1923 to 2017), in his **differential emotions theory**, held that each basic emotion is a separate innate system with its own facial expression, switched on by maturation (Section 1.11). Pairs such as "pain and anger" or "fear and euphoria" belong to no one's scheme.

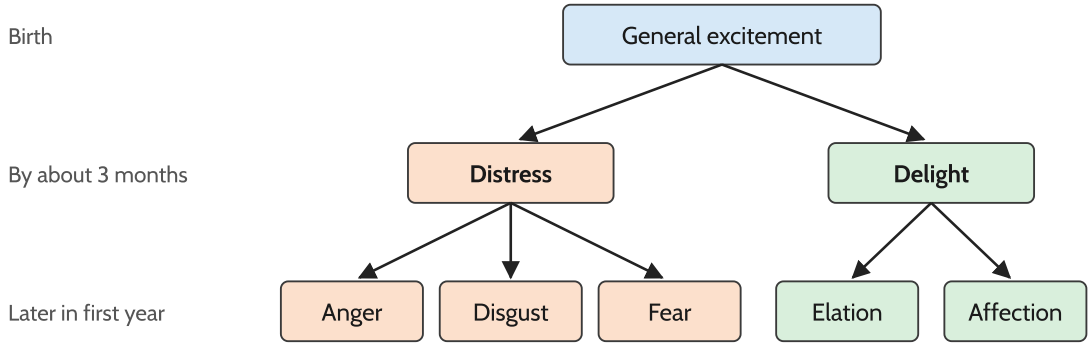


Figure 6.11 Bridges' differentiation of emotion (1932): general excitement divides first into distress and delight, then into specific emotions.

Modern research, much of it by **Michael Lewis**, places the **primary emotions** (joy, interest, surprise, sadness, disgust, anger and fear) within the first six months; none needs a sense of self. Is the first smile joy? Not yet. The **reflexive smile** (endogenous smile) of the first weeks comes from internal states, mostly in sleep. The **social smile**, a smile in response to a human face or voice, appears at about 6 to 10 weeks, and **laughter** at about 3 to 4 months (Figure 6.12).

Self-awareness and the self-conscious emotions

A second family of emotions must wait for a self. Embarrassment, pride, shame, guilt, envy and empathy are **self-conscious emotions** (secondary or self-evaluative emotions): each involves judging yourself through others' eyes or against a standard, so each needs *self-awareness* and appears only after 12 months, from the second half of the second year. Lewis places embarrassment, envy and empathy at about 18 to 24 months, once the child recognises himself, and pride, shame and guilt at about two and a half to three years, once he has rules to judge himself by. **Joseph Campos** (2009) and **Jerome Kagan** (1929 to 2021; 2010) agree on this timing.

KEY STUDY: THE ROUGE TEST (LEWIS AND BROOKS-GUNN, 1979)

Michael Lewis and **Jeanne Brooks-Gunn** secretly dabbed rouge on infants' noses and placed them before a mirror. A baby who touches her own nose, not the reflection's, knows the face is hers. Infants under a year treated the image as another baby; nose-touching was typical by 18 to 24 months. This **mirror self-recognition** marks the arrival of the self, just as embarrassment appears.

EXAM TRAP

Self-conscious emotions appear **after 12 months** (from the second half of the second year), never at 3 months, 6 to 9 months or "before 9 months". Those early ages belong to the primary emotions.

Strangers, separations and the still face

In the second half of the first year emotion turns sharply social. **Stranger anxiety**, wariness or crying when an unfamiliar adult approaches, emerges at about 6 to 8 months; **René Spitz** (1887 to 1974) called it eight-month anxiety (Section 6.7). **Separation protest** (separation anxiety), crying when the caregiver leaves, follows; Kagan found the same curve in several very different cultures, rising from about 8 months, peaking at about 13 to 15 months and then declining. Both fears therefore peak within roughly 8 to 15 months, because the baby now has a specific attachment figure and enough memory to notice an unfamiliar face or an absent caregiver. Excessive distress far beyond this age is separation anxiety disorder (Chapter 8).

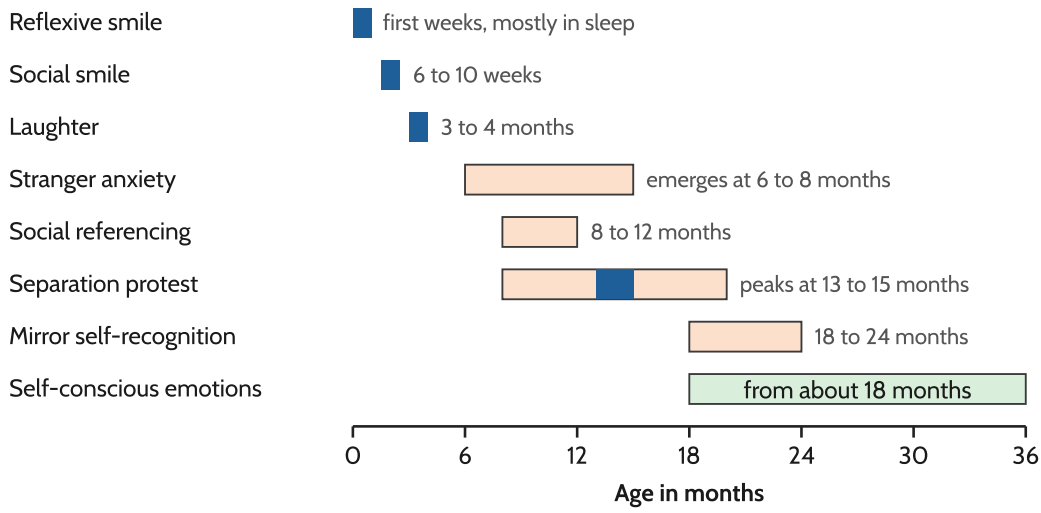


Figure 6.12 Emotional milestones of the first three years. The dark block marks the peak of separation protest; both fears overlap at 8 to 15 months.

EXAM TRAP

Stranger anxiety and separation anxiety peak at about **8 to 15 months**, not at 1 to 2 months (the age of the social smile), 3 years or 6 years.

Infants also learn to use others' emotions as information. In **social referencing**, from about 8 to 12 months, an infant facing an uncertain situation reads a trusted adult's face and lets it guide her. **James Sorce**, Campos and colleagues (1985) placed 12-month-olds on a visual cliff (Section 6.3) with a drop just deep enough to make them hesitate: when the mother looked happy, about three in four crossed; when she looked afraid, none did.

KEY STUDY: THE STILL-FACE PARADIGM (TRONICK AND COLLEAGUES, 1978)

Are young infants simply soaked in their caregivers' mood, or partners in the exchange? **Edward Tronick** and colleagues filmed mothers with infants a few months old in three episodes: normal face-to-face play; a still face, in which the mother keeps looking at the baby with a blank, unresponsive face; and reunion. Faced with the still face, babies first tried to restart the exchange with smiles, coos and reaching, then turned their gaze away, slumped and became distressed and disorganised. The **still-face paradigm** showed that even very young infants actively expect and seek affective reciprocity (emotional give-and-take); it is now used to study the effects of maternal depression.

EXAM TRAP

The still face shows infants as **active partners who expect reciprocity**, distressed and disorganised when it stops. They are not passive recipients of caregiver mood and do not habituate at once; the paradigm says nothing about fathers versus mothers, preferring strangers, or face recognition before 12 months.

From soothing to self-regulation

Caregivers first manage a baby's feelings by rocking and feeding. **Emotion regulation**, the processes by which we monitor and adjust the intensity and timing of our emotional reactions, develops as a handover to the child: by a few months babies avert their gaze from overstimulation or suck a thumb. **Claire Kopp** (1982) described three steps: *control* (about 12 to 18 months: complying with caregivers' demands), *self-control* (from about 2 years: complying without being watched) and *self-regulation* (from about 3 years: flexibly guiding one's own behaviour). Children also learn **display rules**, a culture's norms for when, where and how much emotion may be shown (Section 1.11): in many Indian homes a child learns early to hide anger before elders.

What temperament is

Some babies take a new face in their stride; others scream. **Temperament** names these differences. In **Mary Rothbart**'s definition, it is constitutionally based individual differences in *reactivity* (how quickly and strongly a child is aroused) and *self-regulation* (how she modulates that arousal). It appears early, within the first months. It is moderately stable: a fearful infant often becomes a cautious child, but many change. And it is generally considered biologically and genetically based: identical twins are more alike in temperament than fraternal twins. The environment does not create temperament but shapes its expression, the idea of goodness of fit (below). Temperament is the raw material of adult personality (Chapter 4).

> **EXAM TRAP**
>
> Asked what temperament is "generally considered to be", choose **genetically (biologically) based**. "Learned behaviour", "a result of parenting" and "environmentally influenced" are the distractors: the environment modulates temperament but is not its origin.

Thomas and Chess: the New York Longitudinal Study ●●

In 1956 the psychiatrists **Alexander Thomas** (1914 to 2003) and **Stella Chess** (1914 to 2007), husband and wife, launched the **New York Longitudinal Study** (NYLS), following children from infancy into adulthood. At a time when psychiatry blamed children's problems on parents, they asked whether babies arrive with behavioural styles of their own. They **interviewed parents**, asking not for impressions ("a good baby") but for concrete descriptions of feeding, bathing, a new food or a stranger's visit, and they **observed the babies** to check the reports. The nine dimensions in Table 6.17 were drawn from these records, not from clinical experience or earlier studies.

Table 6.17 Thomas and Chess's nine dimensions of temperament and the three types.

Dimension	What it describes	Easy	Difficult	Slow-to-warm-up
Activity level	Proportion of active to quiet periods	Not defining	Not defining	Low
Rhythmicity	Regularity of sleep, hunger, elimination	Regular	Irregular	Fairly regular
Approach-withdrawal	First response to a new food, toy or person	Approaches	Withdraws	Withdraws at first
Adaptability	Speed of adjusting to change	Quick	Slow	Slow but gradual
Threshold of responsiveness	Stimulus strength needed to evoke a response	Not defining	Not defining	Not defining
Intensity of reaction	Energy of a response, pleasant or not	Mild to moderate	Intense	Mild
Quality of mood	Pleasant versus unpleasant behaviour	Positive	Negative	Mildly negative
Distractibility	How easily outside stimuli interrupt an activity	Not defining	Not defining	Not defining
Attention span and persistence	Keeping at an activity despite obstacles	Not defining	Not defining	Not defining
Share of the sample	About 35 percent fitted no single type	About 40 percent	About 10 percent	About 15 percent

Easy, difficult and slow-to-warm-up children ●●●

The ratings clustered into three patterns, defined mainly by rhythmicity, approach-withdrawal, adaptability, intensity and mood.

- The **easy child** (about 40 percent) is regular in eating and sleeping, cheerful, approaches new things and adapts quickly.

- The **difficult child** (about 10 percent) has irregular biological rhythms, a generally negative mood and intense reactions, withdraws from new stimuli and is slow to adapt.
- The **slow-to-warm-up child** (about 15 percent) is low in activity, withdraws from new stimuli at first with mild rather than intense negativity, and adapts slowly but gradually with repeated exposure.

The remaining 35 percent or so showed mixed patterns: they are unclassified, not a fourth named type. Picture the first week at a Mysuru playschool. Ananya plays at once and naps on schedule: easy. Kabir screams, flings his snack and still protests a fortnight later: difficult. Meera clings quietly to her mother's dupatta, watches from the edge and joins in by the second week: slow-to-warm-up, because her withdrawal is mild and gives way.

The **difficult child syndrome** means that some infants are difficult even in the first few weeks after birth, before parenting could have produced it. It is a temperament, not a later behaviour problem (though such children run a higher risk of one), and has nothing to do with auditory discrimination (turning the head to one sound and not another) or a delay in voluntary control.

Why do some difficult children do well? Thomas and Chess's answer was **goodness of fit**: the match between a child's temperament and the demands and expectations of caregivers and the wider environment. A difficult child whose parents stay patient and consistent often thrives; the same child with impatient, punitive parents tends to develop problems.

> EXAM TRAP
>
> Thomas and Chess are known for **temperament**, not human sexuality, needs and drives or marital therapy. Their three types are **easy, difficult and slow-to-warm-up**. Distractors: sanguine, choleric, melancholic and phlegmatic (Hippocrates' humours, Galen's temperaments; Chapter 4); Type A, B and C (behaviour patterns linked to heart disease and cancer; Section 1.11); "inconsistent", "sensitive, resilient, withdrawn" and "inhibited only" (inhibition is Kagan's construct, below); and "insecurely attached", an attachment category (Section 6.7). Goodness of fit is never IQ differences between twins, the number of siblings or parent-child genetic concordance.

Rothbart and Bates, Kagan, Buss and Plomin

Later researchers condensed the overlapping nine dimensions. Mary Rothbart, who built the parent-report **Infant Behavior Questionnaire** (1981), and **John Bates** proposed a **structural model of temperament** (1998) with exactly three broad dimensions. **Surgency**, or extraversion, covers approach, activity, positive anticipation, smiling and sociability. **Negative affectivity** covers fear, frustration, sadness, discomfort and difficulty being soothed. **Effortful control** is the capacity to focus and shift attention voluntarily and to inhibit a dominant response; it is the self-regulatory side of temperament and matures in the preschool years.

Kagan studied reaction to the unfamiliar. Shown novel mobiles, voices and smells, about one 4-month-old in five thrashed and cried. These **high-reactive infants** tended to become *inhibited*, shy and wary children, while low-reactive infants tended to become *uninhibited*, bold and sociable. **Behavioural inhibition**, the tendency to withdraw from and be distressed by the unfamiliar, is Kagan's construct; he traced it to an easily excited amygdala. Its stability is moderate: many inhibited toddlers grow less shy. **Arnold Buss** and **Robert Plomin** (1984)

proposed the **EAS model** of three heritable temperaments: emotionality (how easily a child becomes distressed), activity (tempo and vigour) and sociability (preference for company). Table 6.18 matches each scheme to its authors.

Table 6.18 Who proposed which scheme of temperament.

Theorists	Units	Basis or method
Thomas and Chess (NYLS, 1956)	Nine dimensions; easy, difficult, slow-to-warm-up	Parent interviews, observation; goodness of fit
Rothbart and Bates	Surgency (extraversion), negative affectivity, effortful control	Infant Behavior Questionnaire; reactivity and self-regulation
Kagan	Inhibited versus uninhibited (behavioural inhibition)	Reactions to the unfamiliar; amygdala
Buss and Plomin (1984)	Emotionality, activity, sociability (EAS)	Heritable traits; twin studies
Hippocrates and Galen (ancient)	Sanguine, choleric, melancholic, phlegmatic	Bodily humours (Chapter 4)

EXAM TRAP

Asked which is **not** in Rothbart and Bates's structure, choose **inhibition to the unfamiliar**: it is Kagan's behavioural inhibition. Extraversion is on the list, as surgency.

CLINICAL LINK

Behavioural inhibition raises the later risk of social anxiety (Chapter 8), and difficult temperament with poor fit predicts behaviour problems: a vulnerability, not a sentence.

Effortful control and the marshmallow test

Effortful control is the core of **self-regulation**, the ability to manage attention, emotion and action in pursuit of a goal; its most famous test asks a child to wait.

KEY STUDY: THE MARSHMALLOW TEST (MISCHEL, FROM THE LATE 1960S)

Walter Mischel (1930 to 2018) tested preschoolers at the Bing Nursery School at Stanford University. A child could eat one treat now or have two if she waited, up to 15 minutes, for the experimenter to return; ringing a bell brought him back at once, but then she got only one. The **marshmallow test** measures **delay of gratification**: giving up a smaller immediate reward for a larger later one. Follow-ups found that longer waiters were rated more competent as adolescents and scored higher on the SAT, the American college admission test (**Yuichi Shoda**, Mischel and **Philip Peake**, 1990). A larger, more diverse replication by **Tyler Watts**, **Greg Duncan** and **Haonan Quan** (2018) found the link about half as strong, and weaker still once family background was controlled.

Waiting turned out to be a skill of attention. Children waited longer when the treats were covered or when they distracted themselves by singing. Dwelling on the treat's *hot* features (sweet, chewy) made waiting almost impossible; thinking of its *cool*, abstract features ("puffy, like a cloud") made it easier. Mischel later described a hot, emotional "go" system and a cool, cognitive "know" system that matures with age.

Keep delay of gratification apart from three neighbours. **Sustained attention** is keeping focus on one task over time. **Resistance to temptation** is obeying a prohibition when no one is watching, studied with the **forbidden-toy paradigm**: the child is told not to touch an attractive toy and left alone with it, with no bigger reward for complying. **Impulse control** in social settings is holding back an unacceptable act such as grabbing or interrupting. The marshmallow task draws on all three, but its defining feature is the choice between a smaller-sooner and a larger-later reward. Mischel's personality theory is in Chapter 4.

> EXAM TRAP
>
> The marshmallow test primarily measures **delay of gratification**, not sustained attention, resistance to temptation (the forbidden toy) or impulse control in social settings.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Bridges held that the newborn's general excitement first differentiates into _____ and _____.
2. Embarrassment, pride and guilt need _____, which the rouge test shows at about 18 to 24 months.
3. Separation protest peaks at about _____ months.
4. The still-face paradigm was devised by _____ and colleagues (1978).
5. A child who withdraws mildly from novelty but adapts gradually is _____.
6. The match between temperament and environmental demands is _____.
7. Rothbart and Bates's three dimensions are surgency, negative affectivity and _____.
8. Mischel's marshmallow test measures _____.

Answers: (1) distress; delight, (2) self-awareness, (3) 13 to 15, (4) Edward Tronick, (5) slow-to-warm-up, (6) goodness of fit, (7) effortful control, (8) delay of gratification.

6.7 Attachment: Imprinting, Bowlby, Ainsworth and Beyond

MEDIUM YIELD · 37 QS

Leave a one-year-old in a strange room and watch the door. She cries when her mother walks out, crawls to her the moment she returns and, within a minute, is back among the toys. That small drama is attachment, the first close relationship of life. This section asks why the bond forms, how it unfolds over the first three years, how its quality is measured, what its absence does and how it echoes into adulthood.

What attachment is: four theories

An **attachment** is a strong, enduring emotional bond to a specific person who cannot be swapped for anyone else. Four behaviours reveal it: the infant keeps close to the person (proximity seeking), explores from her as a **secure base**, runs back to her as a **safe haven** when frightened, and shows **separation distress** when she leaves. A baby who smiles at everyone is sociable; one who protests when one particular person leaves is attached.

Why does the bond form? Four answers compete (Table 6.19). For **Sigmund Freud** (1856 to 1939) it was feeding: the mother gratifies oral needs, so the infant loves her; **Erik Erikson** (1902 to 1994) added that responsive care builds basic trust (Section 6.10). The learning theorists **John Dollard** and **Neal Miller** (1950) and **Robert Sears** said the same in behavioural terms: the mother, paired again and again with hunger reduction, becomes a secondary reinforcer, so attachment is a learned **secondary drive**, nicknamed 'cupboard love'. The cognitive-developmental view adds a precondition: a baby must tell familiar people from strangers and know that people go on existing when out of sight, an achievement called **person permanence** (Section 6.4). The **ethological theory** of John Bowlby holds that attachment is an evolved behavioural system that kept helpless infants near a protector and so alive. Geese and monkeys, as the next two subsections show, undermined the feeding theories, and the ethological theory is the predominantly accepted account today.

Table 6.19 Four theories of why infants attach.

Theory	Key figures	Why the infant attaches	Status
Psychoanalytic	Freud; Erikson	Feeding gratifies oral needs; care builds trust	Feeding overrated
Learning (secondary drive)	Dollard and Miller; Sears	Mother becomes a secondary reinforcer ('cupboard love')	Refuted by Harlow
Cognitive-developmental	Builds on Piaget	Needs person permanence	Explains timing only
Ethological	Bowlby, after Lorenz	Evolved system for survival	Predominantly accepted

EXAM TRAP

The **ethological** theory is today's predominantly accepted account, not the psychoanalytic, learning or cognitive-developmental one. 'Cupboard love' is the learning view.

Lorenz and imprinting ●●

KEY STUDY: LORENZ'S GOSLINGS (1935)

Konrad Lorenz (1903 to 1989), an Austrian zoologist, split a clutch of greylag goose eggs. Half hatched under their mother; half hatched in an incubator, where the first moving thing the goslings saw was Lorenz. The incubator goslings followed him everywhere, and when the two broods were mixed and released, each gosling ran to its own 'mother', goose or man.

Lorenz called this **imprinting**: a rapid, largely irreversible following response to the first moving object, formed only within a narrow **critical period** after hatching (Section 6.1). It needs no practice and no food reward. It also lasts: birds imprinted on humans later courted humans, a long-term effect called sexual imprinting. **Eckhard Hess** (1916 to 1986) timed the window in mallard ducklings: imprinting peaked at about 13 to 16 hours after hatching and then fell away sharply. Later work found the window more flexible than Lorenz thought, so many now say sensitive period. Lorenz shared the 1973 Nobel Prize in Physiology or Medicine with **Nikolaas Tinbergen** and **Karl von Frisch**. Human babies do not imprint, but Lorenz gave Bowlby two ideas: early bonds are biologically prepared, and they have a critical period.

EXAM TRAP

Imprinting is **rapid, largely irreversible and limited to a critical period**: not slow, reversible lifelong learning, operant conditioning with food or a purely visual system unrelated to attachment. Nor is it habituation (Section 6.3), newborn reflexive smiling (Section 6.6) or same-sex playmate preference at 5 (Section 6.8). **Lorenz**, not Ainsworth, Bowlby or Erikson, argued that newborn creatures imprint during a critical period.

Harlow's surrogate mothers ●●

If feeding builds the bond, an infant should love whoever feeds it. **Harry Harlow** (1905 to 1981) tested this at the University of Wisconsin with his wife and colleague **Margaret Harlow**.

KEY STUDY: 'THE NATURE OF LOVE' (HARLOW, 1958)

Infant rhesus monkeys, taken from their mothers at birth, were raised with two surrogate mothers: a bare wire-mesh figure and one covered in terry cloth. For some infants only the wire mother held a feeding bottle. Whichever fed them, they spent most of the day clinging to the cloth mother and visited the wire one only to feed. Frightened by a mechanical toy, they fled to the cloth mother, calmed, then turned to explore; in a strange room they explored only when she was present. Conclusion: **contact comfort**, not feeding, is the primary basis of attachment, and the cloth mother served as safe haven and secure base. Criticism: the work is ethically troubling, and monkeys are not children.

Monkeys reared in total isolation for six months to a year grew up rocking, clasping themselves and unable to play or mate normally; isolate females who later had infants became 'motherless mothers', neglecting or abusing them. Even a cloth mother could not produce normal social development. Yet **Stephen Suomi** and Harlow (1972) found that housing isolates with younger 'therapist' monkeys, which clung and played without threat, restored much social behaviour. Early deprivation is serious but partly reversible, a theme the human studies below repeat.

EXAM TRAP

Harlow showed **contact comfort, not feeding**, as the basis of attachment: not conditioned reinforcement, operant shaping of clinging, classical conditioning of hunger or innate fear of strangers. He did not show imprinting in primates, critical periods for language or genetic determinism, and his isolates show that maternal absence harms even when nutrition is met.

Bowlby's attachment theory ●●

John Bowlby (1907 to 1990), a British psychiatrist and psychoanalyst, wove these threads into attachment theory. In *Forty-Four Juvenile Thieves* (1944) he found that 14 of the young thieves at his London clinic showed **affectionless psychopathy** (no guilt, no affection for others), and 12 of those 14 had suffered prolonged early separation from their mothers. His 1951 World Health Organization report, *Maternal Care and Mental Health*, set out the **maternal deprivation hypothesis**: an infant needs a warm, continuous relationship with the mother or a per-

manent substitute, and its prolonged disruption in the early years does lasting harm. The trilogy *Attachment* (1969), *Separation* (1973) and *Loss* (1980) completed the theory.

Its claims are these. Attachment is an evolved behavioural system: crying, smiling, clinging and following bring a caregiver close; the system switches on under threat or separation, and when the infant feels safe it explores from its secure base. **Monotropy** means that one bond, usually with the mother, is special and qualitatively different from the rest. The bond must form within a critical period, roughly the first two and a half years. And the infant builds an **internal working model**: a mental representation of the self (am I lovable?), of attachment figures (will they come?) and of relationships, built from early experience, that serves as a prototype, a schema, for later relationships yet can be revised by new experience.

What does a broken bond look like? Bowlby's colleague **James Robertson** filmed *A Two-Year-Old Goes to Hospital* (1952) and later, with **Joyce Robertson**, other brief separations. Young children passed through a **protest-despair-detachment** sequence: protest (crying, searching), despair (listless misery) and detachment (seeming recovery, but coolness to the mother on her return). Bowlby applied a similar sequence to grief (Section 6.11).

> EXAM TRAP
>
> Attachment theory is **Bowlby's**, not Freud's, Rogers's or Bandura's. The **internal working model** is a revisable representation of self, others and relationships. It is not the proximity-seeking mechanism or the attachment system itself (which the model guides), a 'bio-social relationship model' (not Bowlby's term), Tolman's cognitive map (Chapter 1), a hippocampus-amygdala circuit, a genetic fear template, a reflex repertoire, a language substrate or object permanence.

Phases of attachment: Bowlby, Schaffer and Emerson, Mahler ●●

Attachment assembles itself over the first two years. Bowlby described four phases (Figure 6.13):

- **Pre-attachment** (birth to 6 weeks): signals bring adults close; the baby knows her mother's smell and voice but does not mind strangers.
- **Attachment-in-the-making** (6 weeks to 6 to 8 months): she prefers familiar people but does not yet protest at separation.
- **Clear-cut attachment** (6 to 8 months to 18 months to 2 years): clear preference for the primary caregiver, separation protest, stranger anxiety (Section 6.6) and secure-base behaviour.
- **Formation of a reciprocal relationship**, the **goal-corrected partnership** (from 18 months to 2 years onwards): language lets the toddler understand and predict the parent's comings and goings, so protest gives way to negotiation.

Rudolph Schaffer and **Peggy Emerson** (1964) followed 60 Glasgow infants at home, monthly through the first year and again at 18 months, recording protest at everyday separations and fear of strangers. They found four stages: asocial (0 to 6 weeks), indiscriminate (6 weeks to about 7 months, accepting comfort from anyone), **specific attachment** (from about 7 months, with separation protest and stranger fear) and **multiple attachments** (soon after; most infants had several by 18 months). Infants attached most strongly to whoever responded most sensitively to them, not necessarily the person who fed them.

Margaret Mahler (1897 to 1985), a psychoanalyst, traced how the child separates psychologically within the bond (*The Psychological Birth of the Human Infant*, 1975, with **Fred Pine** and **Anni Bergman**). A normal autistic phase (the first weeks) and a symbiotic phase (to about 5 months, mother and infant felt as one) give way to **separation-individuation** in four subphases: differentiation or hatching (about 5 to 10 months: turning outward, first stranger reactions); practising (about 10 to 16 months: delighted exploration on new legs); **rapprochement** (about 15 to 24 months: aware now of being separate, the toddler alternates independent exploration with returns to the caregiver for **emotional refuelling**, and the clash of autonomy and closeness brings the clinging and tantrums of the rapprochement crisis); and consolidation (about 24 to 36 months), when **object constancy**, a stable inner image of the mother, lets the child bear her absence. Her developmental theory of schizophrenia and childhood psychosis belongs to Chapter 8.

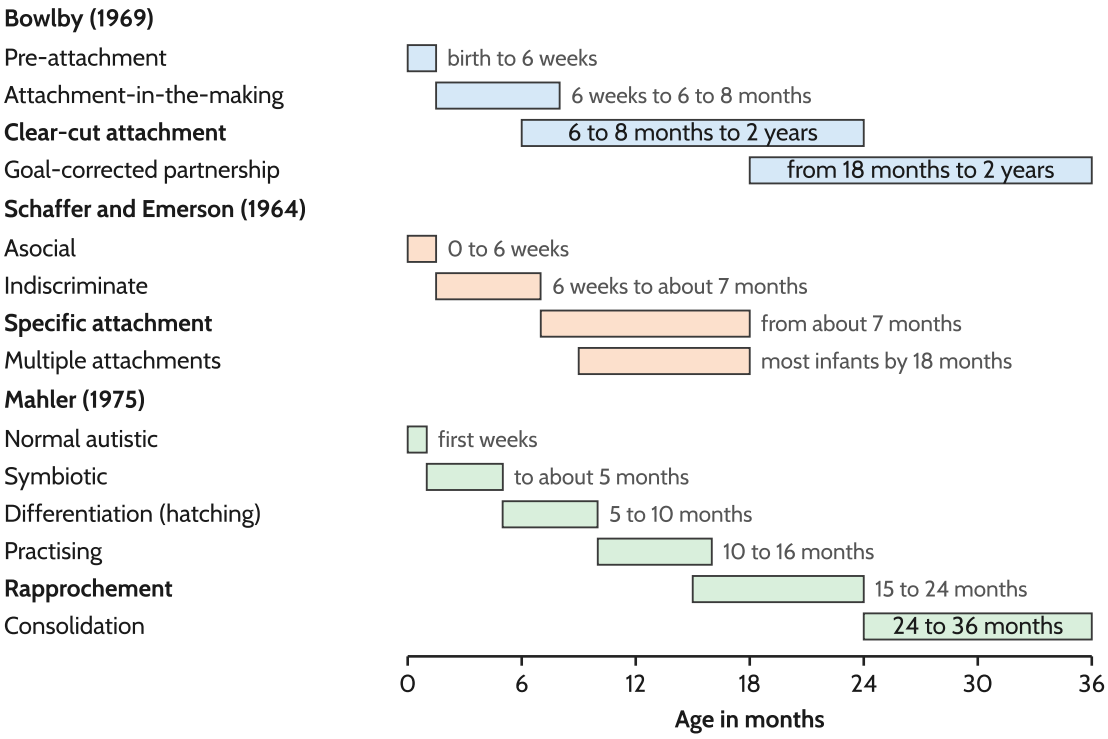


Figure 6.13 Three stage schemes of the first three years on one time line. Bars show usual age ranges; boundaries are approximate, and Schaffer and Emerson's bars stop where their study ended, at 18 months. Bold labels mark the key phases.

EXAM TRAP

Clear preference, separation protest and stranger anxiety mark Bowlby's **clear-cut** phase (6 to 8 months to 2 years; '6 months to 2 years' in keys), not pre-attachment, attachment-in-the-making or the goal-corrected partnership. Schaffer and Emerson's specific attachment starts at about **7 months**. **Rapprochement** (15 to 24 months) is exploration alternating with refuelling: not symbiotic merger, the first stranger reactions (differentiation) or full object constancy.

Ainsworth and the Strange Situation ●●

Bowlby's theory said every infant attaches; it did not say how well. That measure came from **Mary Ainsworth** (1913 to 1999), who worked with Bowlby in London and then watched mothers and babies at home in Uganda and Baltimore. To capture differences in security in one short session she devised the **Strange Situation** (Ainsworth and **Barbara Wittig**, 1969; reported in full in *Patterns of Attachment*, 1978, with **Mary Blehar**, **Everett Waters** and **Sally Wall**). Over eight episodes of about three minutes each, a 12- to 18-month-old explores a playroom with the caregiver, meets a stranger, is left with the stranger, is reunited, is left alone, is joined by the stranger and is reunited again. Observers score exploration, separation distress, reaction to the stranger and, above all, **reunion** behaviour, which best reveals what the infant expects of the caregiver.

Ainsworth's **maternal sensitivity hypothesis** explains the differences: security grows from noticing the baby's signals, reading them correctly and answering them promptly and appropriately. In Baltimore, mothers who answered crying promptly in the early months had securely attached one-year-olds; later studies confirm a real but more modest link.

> EXAM TRAP
>
> **Ainsworth** (not Bowlby, Harlow, Lorenz, Piaget or Erikson) devised the Strange Situation, and she is known for attachment, not cognitive, moral or language development. Bowlby is to attachment as Ainsworth is to the Strange Situation; the Bobo doll is Albert Bandura's, Little Albert is John Watson and Rosalie Rayner's, and the visual cliff is Eleanor Gibson and Richard Walk's (Section 6.3).

Four patterns of attachment, and culture ●●●

Ainsworth's infants fell into three groups, A, B and C; **Mary Main** and **Judith Solomon** (1986, 1990) added a fourth, D (Table 6.20). **Secure attachment** (B) is the commonest, about 60 to 70 percent of samples. **Insecure-avoidant attachment** (A, about 20 percent) follows rejecting care: an infant whose bids for comfort are rebuffed learns to suppress them. **Insecure-resistant attachment** (C), or ambivalent (about 10 to 15 percent), follows inconsistent care: never sure that comfort will come, the infant clings and protests. **Disorganised attachment** (D), or disoriented, abounds among maltreated infants and those with a frightened or frightening caregiver, who is both the source of fear and its only refuge.

Table 6.20 The four attachment patterns. Read the reunion column first.

Pattern	Separation	Reunion	Caregiving
Secure (B)	Explores freely; upset, often mildly	Actively seeks proximity, contact or interaction; readily comforted, returns to play	Sensitive
Insecure-avoidant (A)	Little distress; treats stranger like caregiver	Conspicuous avoidance: ignores or turns away	Rejecting
Insecure-resistant or ambivalent (C)	Explores little; very distressed	Seeks contact yet resists it, clinging then angry; cannot be soothed	Inconsistent
Disorganised or disoriented (D)	No consistent strategy	No one clear pattern; inconsistent, often bizarre: contradictory, freezing, dazed or apprehensive	Maltreating; frightened or frightening

EXAM TRAP

Read the reunion. Distressed, seeks interaction but rejects it when offered = **insecure-resistant (C)**. Little distress, ignores the caregiver = **avoidant (A)**. Contradictory, freezing or apprehensive, often after maltreatment = **disorganised (D)**, Main and Solomon's addition to Ainsworth's original A, B and C. 'Secure-resistant' and 'resistance without distress' are not types.

Is the pattern universal? **Marinus van IJzendoorn** and **Pieter Kroonenberg** (1988) pooled 32 studies from eight countries: secure was the commonest type everywhere, and variation within cultures was about one and a half times that between them. Northern German samples (**Klaus Grossmann** and **Karin Grossmann**, Bielefeld) held many avoidant infants because parents valued early independence, which lowers visible distress and reunion seeking. Japanese infants, rarely apart from their mothers, and Israeli kibbutz infants, who seldom met strangers, showed more resistant attachment.

EXAM TRAP

Northern Germany's excess of type A is **cultural** (an emphasis on early independence), not genetic, a modified protocol or a leap into adult attachment.

Deprivation and privation

René Spitz (1887 to 1974) compared babies in a foundling home, fed and clean but rarely held, with babies cared for by their own mothers in a prison nursery. The foundling babies fell behind, sickened and many died, a syndrome he called **hospitalism** (1945). He also described **anaclitic depression** (1946): infants separated from their mothers in the second half of the first year turned weepy, then withdrawn, lost weight and stopped developing, recovering if the mother returned within a few months.

Michael Rutter (1933 to 2021), in *Maternal Deprivation Reassessed* (1972), distinguished **deprivation** (losing a bond already formed) from **privation** (never forming one), the more harmful of the two. **Barbara Tizard** and **Jill Hodges** followed children raised in London residential nurseries by many changing carers: those later adopted formed close bonds with their new parents, closer than children returned to birth families, yet at 16 still had more difficulty with peers. Rutter's English and Romanian Adoptees study found that Romanian orphans adopted into British homes before about 6 months caught up almost fully, while later adoptees more often kept cognitive deficits and disinhibited attachment. Privation harms, but early rescue repairs much. **Donald Winnicott** (1896 to 1971) added the **good-enough mother**, reliably responsive rather than perfect, and the **transitional object**, a blanket or soft toy that comforts the child on the way to independence.

CLINICAL LINK

Severe early neglect can produce **reactive attachment disorder** (a withdrawn child who rarely seeks comfort) or **disinhibited social engagement disorder** (indiscriminate, overfamiliar approach to strangers); both are taught in Chapter 8.

> **EXAM TRAP**
>
> Hospitalism and anaclitic depression are **Spitz's**, not Piaget's, Skinner's or Eysenck's. Deprivation is losing a bond; privation is never having one.

Attachment beyond infancy

Mary Main's **Adult Attachment Interview** classifies adults by how coherently they discuss their childhood: **secure-autonomous** (balanced), **dismissing** (idealises parents, recalls little), **preoccupied** (still angrily entangled) and **unresolved** (incoherent about loss or abuse). These parallel infant types B, A, C and D, and a parent's classification predicts her infant's well above chance. **Cindy Hazan** and **Phillip Shaver** (1987) found secure, avoidant and anxious-ambivalent styles in adult romantic love, in proportions resembling the infant ones. **Kim Bartholomew** and **Leonard Horowitz** (1991) crossed positive or negative models of self and of others into four adult styles: secure, preoccupied, dismissing and fearful.

Monotropy, finally, has softened. Fathers are attachment figures in their own right, and most infants have several attachments by 18 months, as in any Indian joint family; responsiveness, not feeding, decides to whom. On day care, **Jay Belsky** (1986) warned that over 20 hours a week in the first year raises the risk of insecurity, but the large Study of Early Child Care run by the National Institute of Child Health and Human Development (NICHD) found that day care alone does not; risk rises only when poor or very long care meets insensitive mothering.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Lorenz's goslings followed the first moving object they saw during a critical period, a process called _____.
2. Harlow's monkeys clung to the cloth mother even when only the wire mother gave milk, showing the importance of _____.
3. Bowlby's mental representation of self, attachment figures and relationships, a prototype for later bonds, is the _____.
4. Clear preference, separation protest and stranger anxiety mark Bowlby's _____ phase.
5. The Strange Situation was devised by _____.
6. An infant very distressed at separation who on reunion seeks contact yet angrily resists it is classed as _____.
7. The disorganised pattern (D) was added by _____ and Judith Solomon.
8. In Mahler's _____ subphase the toddler alternates exploration with returns for emotional refuelling.

Answers: (1) imprinting, (2) contact comfort, (3) internal working model, (4) clear-cut attachment, (5) Mary Ainsworth, (6) insecure-resistant or ambivalent (C), (7) Mary Main, (8) rapprochement.

6.8 Parenting, Family, Play and Gender in Childhood LOWER YIELD · 12 QS

Watch a four-year-old for an afternoon. A parent sets her limits, siblings and grandparents crowd round, she plays beside and then with other children, and in almost everything she is

learning what it means to be a girl. This section follows those threads; attachment, which comes first, is Section 6.7.

Baumrind's parenting styles ●●

Why do some children grow up confident and self-controlled while others become anxious or unruly? In the 1960s **Diana Baumrind** (1927 to 2018) observed preschoolers at nursery school and at home and interviewed their parents. In papers from 1966 to 1971 she described three patterns of **parenting style**, the overall emotional climate in which a parent raises a child.

Authoritative parenting combines warmth and emotional support with clear boundaries: rules are explained, discipline is reasoned, children are held accountable for their behaviour and their growing autonomy is encouraged. **Authoritarian parenting** makes high demands with little warmth: strict obedience, rules without reasons ("because I said so") and often harsh punishment. **Permissive parenting**, also called indulgent, is warm and accepting but makes few demands and sets few rules.

In 1983 **Eleanor Maccoby** (1917 to 2018) and **John Martin** recast the styles as combinations of two independent dimensions: **demandingness** (control: expecting mature behaviour, supervising, disciplining) and **responsiveness** (warmth: acceptance and attention to the child's needs). The cell low on both became a fourth style, **neglectful parenting** (uninvolved): detached parents who do little beyond meeting basic needs. Figure 6.14 shows the four styles. Children of authoritative parents are the most self-reliant, competent and achievement-oriented; those of authoritarian parents tend to be anxious, withdrawn and unhappy, and those of permissive parents impulsive, with poor self-control; children of neglectful parents fare worst, with low self-esteem, poor schoolwork and later delinquency.

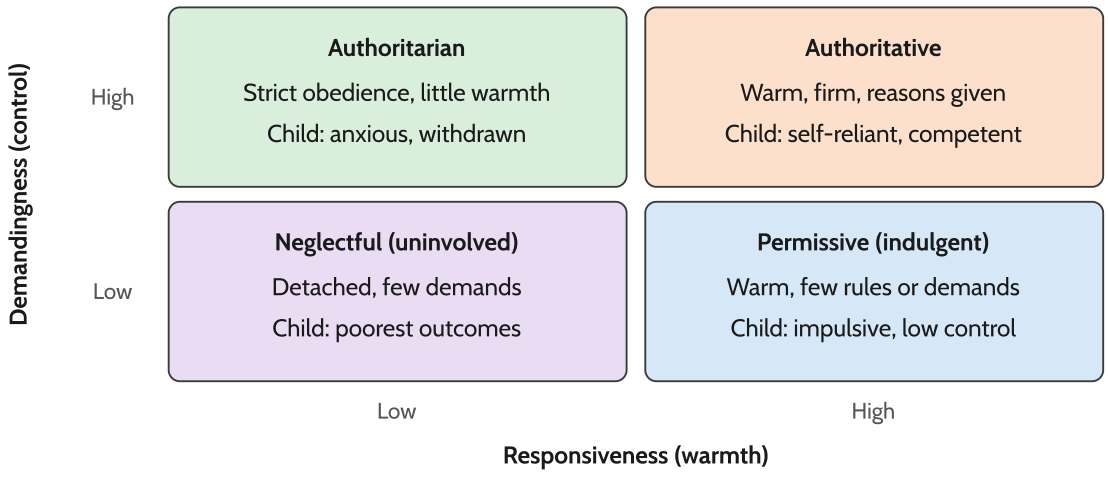


Figure 6.14 Maccoby and Martin's two dimensions give four parenting styles; only authoritative parenting is high on both.

Culture shifts the picture. **Ruth Chao** (1994) showed that Chinese-American parents score as authoritarian on Western scales, yet their children do well at school: their strictness expresses the Chinese idea of *training* (*chiao shun*), devoted involvement that children read as care. Many Indian families show a similar *warm control*, firm expectations of obedience inside close affection. Control harms mainly when it comes without warmth.

EXAM TRAP

Authoritative is the best style; **authoritarian** is strict and cold. "Emotional support, clear boundaries and accountability", "high demandingness and high responsiveness" and "reasoned discipline with autonomy support" all mean authoritative. Permissive is low demandingness with high responsiveness; neglectful is low on both and was added by Maccoby and Martin, not Baumrind.

New labels, discipline and the family

Helicopter parenting is over-involved parenting: parents hover, monitor and step in to solve problems the child could handle alone. The image comes from **Haim Ginott** (1922 to 1973), whose 1969 book *Between Parent and Teenager* quotes a teenager whose mother hovers over him like a helicopter. It has several roots at once: parental anxiety, parents' unfulfilled wishes lived out through the child, and parents' own personality traits such as perfectionism. **Tiger parenting** (after **Amy Chua**'s 2011 memoir) is strict and achievement-driven; **free-range parenting** grants age-appropriate independence with little supervision. **Martin Hoffman**'s (1967) three kinds of discipline, **induction** (explaining how the behaviour affects others), **power assertion** (force, threats) and **love withdrawal** (withholding affection), are compared in Section 6.9.

Parents are one part of a family system in which every member affects every other. Siblings are companions, teachers and rivals; the claim that birth order shapes personality belongs to Alfred Adler (Chapter 4). Only children, despite the "spoilt" stereotype, match others in achievement and adjustment. **E. Mavis Hetherington**, following families for decades (2002), found that most children adjust to divorce within two to three years, though about 25 percent show serious problems against about 10 percent from intact families; remarriage brings a second adjustment. Maternal employment does not in itself harm children, and day care matters through its quality.

INDIAN CONTEXT

In a **joint family** several generations share a home and its childcare, so the child has many attachment figures, more adults to obey and a strong sense of interdependence. Urbanisation has made the **nuclear family** (parents and children only) the commonest household, with day care or grandparents filling the gap.

Child maltreatment and neglect

In 1962 the paediatrician **C. Henry Kempe** (1922 to 1984) and colleagues described the **battered child syndrome**: young children with fractures at different stages of healing that their parents' accounts did not explain. **Child maltreatment** has four types: **physical abuse** (non-accidental injury), **sexual abuse** (any sexual activity with a child), **emotional abuse** (persistent rejecting, belittling or terrorising) and **neglect**, failure to meet a child's basic needs and the most often reported form. Neglect has five categories: **physical** (failure to provide proper nutrition, housing and clothing), **supervisory** (children left unattended), **health** or medical (failure to immunise, seek proper care or follow medical directions), **educational** (not enrolling the child, chronic absence) and **emotional** (withholding affection and attention).

EXAM TRAP

Using **licensed day care** is not neglect: it is approved, supervised care, the odd one out beside nutrition, supervision and immunisation.

Risk factors include parents who were themselves maltreated (about a third repeat it), young, depressed or misusing substances; premature, disabled or difficult children; and poverty, domestic violence and isolation. Effects include disorganised attachment (Section 6.7), aggression, school failure, depression and post-traumatic stress. Management and India's Protection of Children from Sexual Offences (POCSO) Act, 2012 are in Chapter 7.

The development of play ●●

Play is activity chosen freely and done for its own sake. **Jean Piaget** (1896 to 1980) saw it as pure assimilation, the child bending reality to fit existing schemes, so play follows the cognitive stages (Section 6.4). The first form, in infancy during the sensorimotor stage, is **functional play** (practice or sensorimotor play): simple actions repeated for the pleasure of mastering them. With the preoperational symbolic function comes **symbolic play** (pretend play), and in the concrete operational stage **games with rules**. In 1968 the Israeli psychologist **Sara Smilansky** added **constructive play**, making or building something. Table 6.21 gives the standard definitions.

Table 6.21 Types of play, in the order they appear.

Type	Definition	When	Example
Sensorimotor or practice (functional)	Deriving pleasure from exercising sensorimotor schemes	First; infancy	Shaking a rattle
Pretense or symbolic	Transforming the physical environment into a symbol	Preoperational, about 2 to 7	A banana becomes a phone
Constructive	Combining sensorimotor or practice play with symbolic representation of ideas	Preschool onwards	Building a palace from blocks
Social	Interaction with peers	Grows through preschool	Chasing, turn-taking
Games with rules	Play governed by rules agreed in advance	Concrete operational, about 7 to 11	Kho-kho, ludo, gully cricket

Sociodramatic play is shared pretend play in roles (house, shop, doctor), peaking around four to five. **Rough-and-tumble play** is friendly wrestling and chasing with a relaxed "play face"; it peaks in middle childhood and, unlike real aggression, ends with the children still together. For **Lev Vygotsky** (1896 to 1934), every imaginary situation has rules (a child playing "mother" must act as mothers do), so play creates a zone of proximal development in which the child acts "a head taller than himself" (Section 6.5).

EXAM TRAP

The first play of infancy is **functional (practice)** play, not symbolic play, which needs the preoperational symbolic function. "Transforming the physical environment into a symbol" is symbolic play; "combining sensorimotor or practice play with symbolic representation of ideas" is constructive play; "interaction with peers" is social play.

Social play and theories of play

Play also becomes more social. In 1932 **Mildred Parten** observed nursery children aged two to five and ranked their play by social involvement (Table 6.22), from solitary and **parallel play** (beside others, not with them) through **associative play** (together, without a shared goal) to **cooperative play** (organised, with roles and a common aim). Earlier forms persist: solitary and parallel play stay common in older preschoolers.

Table 6.22 Parten's stages of social play (1932).

Stage	What the child does	Rough age
Unoccupied	Not playing; watches whatever catches the eye	Infancy
Solitary	Plays alone, absorbed in own toys	About 2 to 3
Onlooker	Watches others play, may talk, does not join	About 2 to 3
Parallel	Plays beside others with similar toys	About 2 to 3
Associative	Plays with others, sharing, no common goal	About 3 to 4
Cooperative	Organised play, shared goal and roles	About 4 to 5

Why do children play at all? Table 6.23 compares the classical and modern answers; play therapy is in Chapter 7.

Table 6.23 Theories of play.

Theory	Proposed by	Children play to
Surplus energy	**Friedrich Schiller; Herbert Spencer**	Burn off energy left over once survival needs are met
Relaxation or recreation	**Moritz Lazarus**	Restore energy spent in work
Pre-exercise or practice	**Karl Groos**	Rehearse adult skills, as a kitten stalking string practises hunting
Recapitulation	**G. Stanley Hall**	Replay stages of human evolution, from climbing to team games
Psychoanalytic	Sigmund Freud	Fulfil wishes and master anxious experiences by replaying them
Cognitive	Jean Piaget; Lev Vygotsky	Consolidate schemes; create a zone of proximal development

Peers: status, friendship and bullying

Sociometric status is measured by asking classmates whom they like most and least. In 1982 **John Coie**, **Kenneth Dodge** and a colleague identified five groups: **popular** (many "like most"), **rejected** (many "like least"), **neglected** (few nominations of either kind), **controversial** (many of both) and **average**. Rejected children are most at risk of loneliness and later problems; neglected children usually do well once they find a friend. Friendship deepens from shared play to mutual trust and, in adolescence, intimacy. **Bullying**, as defined by the Norwegian psychologist **Dan Olweus** (1931 to 2020), is repeated, intentional aggression against someone less powerful; his whole-school programme roughly halved it in Norway.

Gender concepts and gender socialisation

Gender research needs exact terms. **Sex** is biological: chromosomes, hormones and anatomy. **Gender** is the psychological and social meaning a culture attaches to being male or female. **Gender identity** is one's own sense of being male, female or another gender; a **gender role** is the behaviour a culture expects of each sex. A **gender stereotype** is a rigid, oversimplified belief about what males and females are like ("boys don't cry"); gender stereotyping applies it to individuals. **Gender typing** is the process by which children adopt the preferences and behaviours their culture considers right for their sex.

Gender socialisation is the process by which children learn this "gender map" from the people and institutions around them. Parents come first: pink or blue, dolls or trucks, indoor or outdoor chores, much of it taught unconsciously because their own gender orientations are so deeply embedded. Peers, school and media reinforce the map. *Gender construction*, by contrast, is the sociological view that a society creates gender categories and their meanings, and *gender structuralisation* is not a standard developmental term. From about three, children prefer same-sex playmates; this **gender segregation** peaks in middle childhood, so that, as Maccoby put it, boys (large groups, rough play) and girls (small groups, talk) grow up in two cultures.

> EXAM TRAP
>
> Parents passing on the "gender map" (pink and blue, toys, chores), consciously or unconsciously, is **gender socialisation**. Gender stereotyping is a belief applied to people, not a learning process; gender construction and gender structuralisation are distractors.

Theories of gender development

The **biological** view points to prenatal hormones: girls with **congenital adrenal hyperplasia**, exposed to high androgen levels before birth, prefer boys' toys and rough play more than other girls, although raised as girls. In the **psychoanalytic** view the child of three to six identifies with the same-sex parent (Chapter 4). **Social learning theory**, applied to gender by **Walter Mischel** (1930 to 2018) in 1966, holds that children are reinforced for gender-appropriate behaviour and imitate same-sex models. **Kay Bussey** and **Albert Bandura** (1925 to 2021) extended this in 1999 into a social cognitive theory: reinforcement and modelling build internal gender standards by which children regulate themselves. Either way, shaping from outside comes first.

Lawrence Kohlberg (1927 to 1987) reversed the order in his cognitive-developmental theory of 1966: "I am a boy, therefore I want to do boy things." Understanding comes first, and children then socialise themselves. It grows in three steps (Figure 6.15): **basic gender identity** (about two to three: labelling oneself and others), **gender stability** (about three to five: a boy will become a man) and **gender constancy** or consistency (about five to seven: a boy in a frock is still a boy). These are his only stages; "gender-role achievement" is not one. **Ronald Slaby** and **Karin Frey** (1975) supported him: watching a split-screen film of a man and a woman doing the same things, children high in constancy looked longer at the same-sex adult. Yet children prefer gender-typed toys by about two, long before constancy.

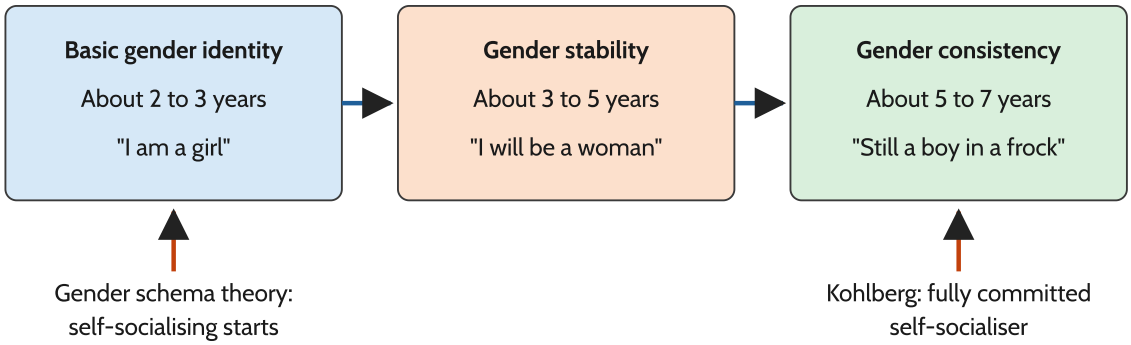


Figure 6.15 Kohlberg's three steps in understanding gender. Gender schema theory starts self-socialisation at the first step; on the stricter reading of Kohlberg, it is complete only at the third.

Gender schema theory, proposed in 1981 by **Sandra Bem** (1944 to 2014) and independently by **Carol Martin** and **Charles Halverson**, explains this. A gender schema is an organised set of beliefs about what goes with each sex. As soon as children have basic gender identity, they sort toys, activities and people into "for boys" and "for girls" and attend to and remember what fits their own group: shown a girl sawing wood, many later recall a boy doing it (Martin and Halverson, 1983). Earlier, Bem (1974) had treated masculinity and femininity as separate dimensions, so a person can be high on both: **androgyny**, measured by the Bem Sex Role Inventory (Chapter 2).

> EXAM TRAP
>
> IHBAS 2020 keyed **basic gender identity** as the starting point of self-socialisation in Kohlberg's theory: it is the first link in his chain. Many textbooks are stricter: for Kohlberg, children become fully committed self-socialisers only at gender consistency (constancy), and it is gender schema theory that starts self-socialisation at basic gender identity. Answer basic gender identity for that stem, but know both.

How large are the resulting differences? In 1974 Eleanor Maccoby and **Carol Jacklin** found only four well established: girls' greater verbal ability and boys' greater visual-spatial ability, mathematical ability and aggression. **Janet Shibley Hyde**'s **gender similarities hypothesis** (2005) went further: across 46 meta-analyses, 78 percent of differences were small or close to zero.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Warmth combined with clear, reasoned control is _____ parenting.
2. Maccoby and Martin's two dimensions are demandingness and _____.
3. In 1962 C. Henry Kempe described the _____ syndrome.
4. For Piaget, the first play of infancy is _____ play.
5. Play combining practice play with symbolic representation of ideas is _____ play.
6. Playing beside others with similar toys but not with them is Parten's _____ play.
7. Kohlberg's steps are basic gender identity, gender stability and gender _____.
8. Being high on both masculine and feminine traits is Bem's _____.

Answers: (1) authoritative, (2) responsiveness, (3) battered child, (4) functional (practice), (5) constructive, (6) parallel, (7) consistency (constancy), (8) androgyny

6.9 Moral Development: Piaget, Kohlberg, Gilligan and Beyond

LOWER YIELD · 20 QS

Read two stories to a six-year-old. John, called to dinner, opens a door and accidentally smashes fifteen cups standing on a tray behind it. Henry, reaching for jam while his mother is out, breaks one cup. Who is naughtier? The six-year-old usually says John, since fifteen cups are worse than one; a ten-year-old says Henry, who was doing wrong. What has changed is not knowledge about cups but reasoning about right and wrong. This section follows that change through Piaget, Kohlberg and Gilligan, then meets the researchers who widened the picture.

Three sides of morality

Moral development is the growth of the standards by which a person judges right and wrong, and of the feelings and actions that go with them. Morality has three sides, and each tradition studied one. The **emotional** side is guilt, shame and empathy (their emergence in toddlers is traced in Section 6.6). **Sigmund Freud** (1856 to 1939) placed it in the **superego**, the conscience formed at about five or six when the child resolves the Oedipus complex by identifying with the same-sex parent; it punishes misdeeds with guilt (Chapter 4). The **behavioural** side is what people do when tempted. Social learning theorists led by **Albert Bandura** (1925 to 2021) held that moral conduct is learned through reinforcement, punishment and above all **modelling**; in 1963 Bandura and **Frederick McDonald** even shifted children's judgments between consequences and intentions simply by letting them hear adults judge the other way. The **cognitive** side, **moral reasoning**, is how a person thinks a moral problem through; it belongs to Piaget and Kohlberg. The study below shows why reasoning and conduct need separate study.

KEY STUDY: THE CHARACTER EDUCATION INQUIRY (1928 TO 1930)

Hugh Hartshorne and **Mark May** asked whether children have a general trait of honesty. They gave more than 10,000 schoolchildren rigged chances to lie, cheat and steal at school, at home, at parties and in games. Most children cheated in some situations and not in others, and cheating in one setting only weakly predicted cheating in another. Honesty, they concluded, is largely **situation-specific**, not a fixed trait, and moral knowledge barely predicted honest conduct.

Piaget: from moral realism to autonomy

Jean Piaget (1896 to 1980) turned from logic (Section 6.4) to morality in *The Moral Judgment of the Child* (1932). He asked Swiss boys playing marbles where the rules came from and whether they could be changed, and he read children **paired stories** like John and Henry's, in which one child does more damage by accident and another less damage while misbehaving. He described a **premoral period**, before about age 5, when children pay little attention to rules, followed by two stages (Table 6.24).

In **heteronomous morality** (Greek for "ruled by another"), also called **moral realism**, from about 5 to 10 (some books say 9), rules come from adults and are sacred and unchangeable: a new marble rule is cheating even if every player agrees. Acts are judged by their consequences, not intentions, which Piaget called **objective responsibility**, so John is naughtier. The child believes in **immanent justice**, that the world itself punishes wrongdoing (a boy who stole apples falls through a rotten bridge "because he stole"), and favours **expiatory punishment**, severe and unrelated to the offence.

From about 10, cooperation among equal peers brings **autonomous morality**, or **moral relativism**. Rules become agreements that players may change by consent; intentions outweigh consequences, so Henry is naughtier; fairness means equality; and **reciprocal punishment** should fit the misdeed and repair it (the child who broke a window pays for it). Piaget underestimated preschoolers, who do use intentions when a story makes them clear, but his idea that moral reasoning grows with cognition and peer experience became Kohlberg's starting point.

Table 6.24 Piaget's two stages of moral reasoning.

Feature	Heteronomous (moral realism), about 5 to 10	Autonomous (moral relativism), from about 10
Rules	Fixed, sacred, made by authority	Social agreements, changeable
Acts judged by	Consequences (objective responsibility)	Intentions (subjective responsibility)
Justice	Immanent justice	Equality among peers
Punishment	Expiatory: severe, arbitrary	Reciprocal: fits the offence
Respect	One-sided, for adults (constraint)	Mutual, among peers (cooperation)

EXAM TRAP

Heteronomous children judge by **rules seen as unchangeable and the consequences of acts**. "Intentions and equality" describe Piaget's **autonomous** stage; "social contracts and individual rights" is Kohlberg's stage 5; "abstract principles of justice" is Kohlberg's stage 6. Piaget did not use the Heinz dilemma.

Kohlberg's method: the Heinz dilemma •

Lawrence Kohlberg (1927 to 1987) carried Piaget's approach into adolescence. For his 1958 doctoral study he interviewed 72 Chicago boys aged 10, 13 and 16, and re-interviewed many of them every three or four years for about 20 years. His method was to pose hypothetical moral dilemmas, stories in which two moral values collide, to children and adolescents of different ages. The most famous is the **Heinz dilemma**:

A woman is dying of cancer. A druggist has a drug that might save her, but he charges $2,000 for a dose that cost him $200 to make. Her husband, Heinz, can raise only about $1,000, and the druggist will not sell it cheaper or let him pay later. Heinz breaks in and steals the drug. Should he have done that? Why?

The decisive word is "why". Kohlberg scored the **reasoning**, not the choice: "steal" and "don't steal" can come from the same stage, and the same answer from different stages. His interview and scoring system became the **Moral Judgment Interview**. The theory describes moral reasoning, not moral behaviour.

> EXAM TRAP
>
> Kohlberg studied morality by **posing moral dilemmas to children of different ages**, not through daily journals, court records of delinquents, or making people perform legal and illegal acts (watching real cheating was Hartshorne and May's method). His stages concern **moral reasoning**; language acquisition (Chapter 1), motor skills (Section 6.3) and psychosexual conflict (Freud, Chapter 4) belong to other theories.

Three levels, six stages •••

Kohlberg grouped the reasons into three levels of two stages each (Table 6.25, Figure 6.16), named by the person's relation to **conventions**, society's rules and expectations.

At the **preconventional level** (Level I), typical of children under about 9, morality is under external control: the child obeys rules to avoid punishment or gain rewards, so rewarded acts are right and punished acts wrong. **Stage 1**, the **punishment-and-obedience orientation**, judges an act by the punishment it brings. **Stage 2**, **instrumental purpose and exchange** (naive hedonism), counts as right whatever serves one's own interest: what is in it for me?

At the **conventional level** (Level II), reached by most adolescents and where most adults stay, the person conforms to social rules to be a good group member and maintain the social order. **Stage 3**, the **good boy-nice girl orientation**, counts as right whatever wins the approval of close others; intentions matter, and the golden rule guides relationships. **Stage 4**, the **law-and-order orientation**, takes the whole society's view: do your duty, respect authority and obey laws, because society would collapse if everyone broke them.

At the **postconventional level** (Level III), reached by a minority of adults and rarely before 20, principles can stand above a particular law. **Stage 5**, the **social-contract orientation**, sees laws as instruments that express the will of the majority and further human welfare, to be changed by democratic agreement when they fail. **Stage 6**, **universal ethical principles**, follows self-chosen principles of justice and human dignity, even against an unjust law.

Table 6.25 Kohlberg's levels and stages, with the alternative labels textbooks use (the older ones come from Kohlberg's 1960s papers).

Level	Stage	Right means	Other labels
I Preconventional	1 Punishment and obedience	Avoiding punishment	Heteronomous morality (Kohlberg 1976)
I Preconventional	2 Instrumental purpose and exchange	Rewards; deals that pay me	Naive (instrumental) hedonism; individualism; self-interest
II Conventional	3 Good boy-nice girl	Approval of close others	Mutual interpersonal expectations; Papalia: maintaining mutual relations, approval of others, the golden rule
II Conventional	4 Law and order	Duty, authority, social order	Social-order-maintaining; social system and conscience; Papalia: social concern and conscience
III Postconventional	5 Social contract	Agreed laws serving rights and welfare	Papalia: morality of contract, of individual rights and of democratically accepted law
III Postconventional	6 Universal ethical principles	Self-chosen principles	Morality of individual principles of conscience; Papalia: morality of universal ethical principles

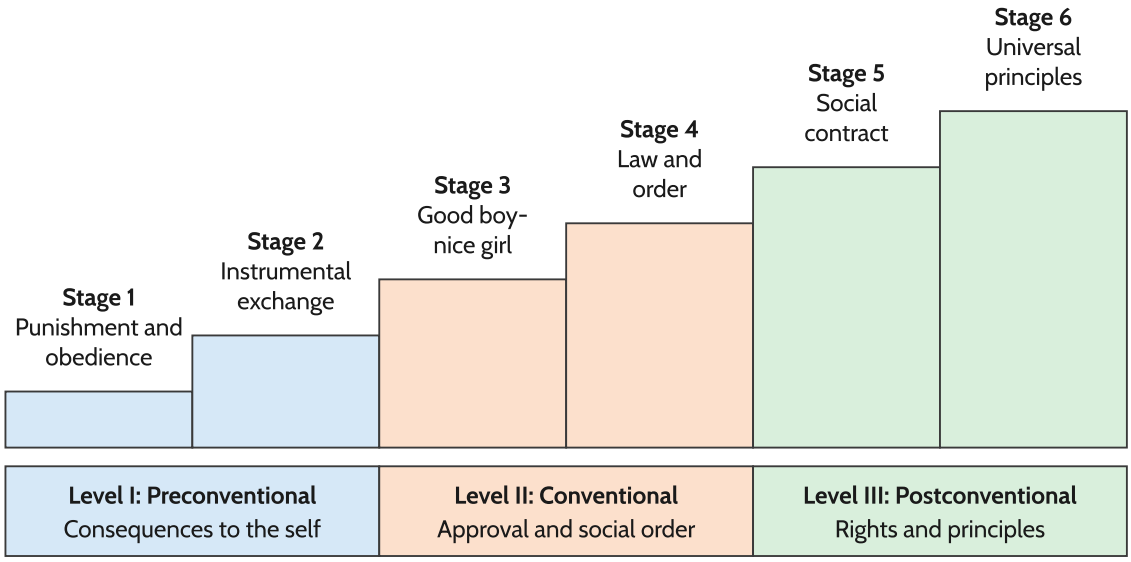


Figure 6.16 Kohlberg's stages as a staircase. Each level holds two stages, everyone climbs in the same order without skipping, and fewer people reach the higher steps.

EXAM TRAP

The levels are **preconventional, conventional and postconventional**; "unconventional", "formal conventional", "premoral" and "amoral" are not among them. "Premoral" is Piaget's period before rules matter (only Kohlberg's earliest papers used it for Level I), so treat it as not a level and never as the answer for reasoning about laws. Papalia calls Level III the **morality of autonomous moral principles**; when both names are offered for the third level, choose **postconventional**. Conformity to peers is stage 3, so conventional. In the four-statement item, "obedience, self-interest and conformity" (stages 1 to 3) and "law and order, human rights and universal ethics" (stages 4 to 6) are true; "law and order, self-interest and purpose" mixes levels, and "conformity, identity and competence" borrows Erikson's terms (Section 6.10). Answer: 1 and 3 only.

Classifying a reason: worked examples

To place a reason, ignore the choice and ask what makes the act right or wrong for the speaker. Consequences to the self mark Level I; approval, duty, the law and keeping society working mark Level II; rights and principles that could justify changing a law mark Level III (Table 6.26).

Table 6.26 Classifying moral reasons.

Reason given	Stage and level	The tell
"Don't steal; you will be punished."	1, preconventional	Punishment
"Help him; he will reward me."	2, preconventional	Reward, exchange
"Steal it; people will think him heartless otherwise."	3, conventional	Approval
"Don't steal; it is against the law, and laws keep society functioning."	4, conventional	Social order
"Keep your promise; society depends on people honouring agreements."	4, conventional	Social order
"Steal it; the right to life outweighs the right to property."	5, postconventional	Rights the law should serve
"Steal it and accept jail; every life has equal worth."	6, postconventional	Universal principle

EXAM TRAP

"Promises", "agreements" and "laws" do not by themselves mean stage 5. Keeping a promise **because society depends on it**, or obeying a law **because laws keep society functioning**, is stage 4: **conventional**. It becomes stage 5 only when the speaker asks whether the agreement serves people's rights and welfare, and would change it if not.

How sound is Kohlberg's theory?

Kohlberg held that the stages form an **invariant sequence**: everyone climbs them in the same order, without skipping, and regression is rare; the 20-year Chicago data largely agreed. Each stage needs a cognitive base that is necessary but not sufficient: postconventional reasoning needs formal operations (Section 6.4), yet many formal-operational adults never reach it. It also needs **role-taking**, seeing a problem from other people's viewpoints (Robert Selman's levels of social perspective taking, Section 6.5), which grows through discussion and education. Stage 6 proved so rare that Kohlberg later dropped it from his scoring manual. Because his interview needs trained scorers, **James Rest** built the **Defining Issues Test** in the 1970s, a questionnaire in which people rate and rank prepared statements about each dilemma.

Three criticisms dominate. The first is **cultural bias**. **John Snarey** (1985) reviewed 45 studies from 27 cultural areas: stages 1 to 4 appeared in order almost everywhere, but stage 5 mainly in urban, educated groups and hardly ever in village societies, whose communal reasoning the scoring missed. **Richard Shweder**, studying Hindu families in Bhubaneswar, Odisha, argued that the scale measures only an **ethic of autonomy** (harm, rights, justice) and misses an **ethic of community** (duty, hierarchy) and an **ethic of divinity** (purity, sanctity). The second is the **gap between reasoning and behaviour**: higher-stage reasoners act morally only somewhat more often. The third is **gender bias**: the founding sample was all male.

INDIAN CONTEXT

Shweder's Oriya Brahmin informants judged it a serious moral wrong for an eldest son to get a haircut and eat chicken the day after his father's death; Chicago informants saw a personal choice. **Joan Miller**, comparing Hindu Indians in Mysore with Americans, found that Indians treated helping family, friends and even strangers as **moral obligations**, where Americans saw many of the same acts as personal decisions. A scale built around individual rights scores such duty-based reasoning as merely conventional.

Gilligan: the ethic of care

Carol Gilligan (born 1936), who had worked with Kohlberg at Harvard, noticed that women often scored at stage 3, apparently below men. In *In a Different Voice* (1982) she argued that his scale rests on a male **ethic of justice**, reasoning from rights, rules and fairness, and misses the **ethic of care** more often heard in women, reasoning from relationships and responsibility, wanting no one hurt. Her 11-year-old Jake solved Heinz like "a math problem with humans": life outweighs property. Amy worried that if Heinz went to jail his wife would have no one, and wanted them to talk it out.

From interviews with 29 women deciding about abortion, Gilligan described three levels of care: an **orientation to individual survival** (caring only for oneself), **goodness as self-sacrifice** (caring for others at one's own expense) and the **morality of nonviolence** (caring for self and others, hurting no one).

Is there a sex difference? **Lawrence Walker**'s 1984 review found little difference in Kohlberg scores once education and occupation were allowed for, and later meta-analyses found only small differences in care and justice orientation: both sexes use both voices.

EXAM TRAP

The pairing is fixed: **females, ethic of care; males, ethic of justice**. Options that reverse it are wrong, and "males have an ethic of justice", though part of her argument, is what Kohlberg already measured; her departure is the female ethic of care. Gilligan added a voice; she did not replace Kohlberg.

Beyond the stage theories

Elliot Turiel's **social domain theory** (1983) holds that children build separate kinds of social knowledge from early on: the **moral domain** (harm, fairness, rights), the **social-conventional domain** (agreed customs such as dress codes or forms of address) and the **personal domain** (choices such as friends or hairstyle). Even preschoolers say hitting would be wrong in a school with no rule against it, but calling the teacher by her first name would then be fine: they already separate moral from conventional rules.

Nancy Eisenberg studied **prosocial moral reasoning**, about helping at a cost to oneself (should Mary, on her way to a party, stop to help a girl who has fallen?). Children move through five levels: hedonistic, needs-oriented, approval-oriented, empathic and finally internalised reasons.

Martin Hoffman saw **empathy** as the main motive of moral action and traced four stages: **global empathy** in the first year (an infant cries when another cries), **egocentric empathy** in

the second (a toddler fetches his own mother to comfort a crying friend), empathy for **another's feelings** from about 2 or 3, and in late childhood empathy for **another's life condition** (for the poor as a group). He also compared three kinds of discipline. Power assertion (force, threats, removing privileges) and love withdrawal (coldness, ignoring the child) produce compliance out of fear or anxiety. **Induction**, explaining how the child's act affects others ("Riya is crying because you snatched her book"), recruits empathy and guilt and so best fosters **internalisation**, behaving well when no one is watching (parenting styles are in Section 6.8).

Finally, **Jonathan Haidt**'s **social intuitionist model** (2001) reverses the cognitive tradition: moral judgments are fast gut intuitions, and reasoning mostly follows to justify them.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Hartshorne and May concluded that honesty is largely _____, not a general trait.
2. In Piaget's _____ morality, rules are unchangeable and acts are judged by their consequences.
3. The belief that the world itself punishes misdeeds is _____ justice.
4. Kohlberg posed moral _____, chiefly the Heinz story, and scored the reasoning, not the choice.
5. At the _____ level, rewarded acts are right and punished acts are wrong.
6. "Don't steal; laws keep society functioning" is stage 4, the _____ orientation.
7. Seeing laws as instruments that express the majority's will and further welfare is the _____ orientation.
8. Gilligan argued that women more often reason with an ethic of _____.

Answers: (1) situation-specific, (2) heteronomous, (3) immanent, (4) dilemmas, (5) preconventional, (6) law-and-order, (7) social-contract, (8) care.

6.10 Erikson's Psychosocial Stages, Identity and Adolescence

MEDIUM YIELD · 40 QS

Ask a nineteen-year-old in a Mysuru hostel who she is, and what she will make of her life, and you may get a long pause. Erik Erikson put that question at the centre of a theory that runs from the first feed to the last years, and he gave it a name now used everywhere: the identity crisis. His eight stages come first; then identity is followed into adolescence itself, with its identity statuses, its egocentrism, its changing body and brain, and the old argument over storm and stress.

Erikson and the psychosocial approach ●●

Erik Homburger Erikson (1902 to 1994) lived his own theory. Born in Frankfurt to a Danish mother who never told him who his father was, he carried his stepfather's surname, Homburger, and wandered Europe as an artist in his late teens. At 25 he began teaching at a small school in Vienna, was analysed by **Anna Freud** (1895 to 1982), and became a psychoanalyst without any university degree. He moved to the United States in 1933 and later treated

Second World War veterans whose trouble seemed to be a confused sense of who they were rather than repressed conflict. His key books are *Childhood and Society* (1950), which set out the eight stages, *Young Man Luther* (1958), *Identity: Youth and Crisis* (1968) and *Gandhi's Truth* (1969).

Erikson belonged to **ego psychology**, which treats the ego as a partly independent, adaptive part of the personality rather than a servant of the id. Development follows the **epigenetic principle**: like an embryo whose organs appear on a fixed timetable (*epi* means "upon"), the personality unfolds through a built-in sequence of stages, while society decides how each turns out. Each stage poses a **psychosocial crisis**, a turning point between two opposed tendencies set off by new social demands; it is a choice of direction, not a catastrophe. A favourable resolution leaves the ego with a basic strength, or **virtue**. Favourable means a balance tilted towards the positive pole, not the absence of the negative: a child who trusts everyone is gullible. An unresolved crisis makes later ones harder but not hopeless, because development continues across the whole lifespan.

Erikson coined the term **identity crisis** for the adolescent struggle to know who one is. His own labels for the eight stages of the life cycle run from the oral-sensory stage through the muscular-anal, locomotor-genital, latency, puberty and adolescence, young adulthood and adulthood stages to the stage of maturity. The first four echo Freud's oral, anal, phallic and latency stages; the adult stages are Erikson's own.

> **EXAM TRAP**
>
> The identity crisis, and the eight stages from the "oral-sensory stage" to the "stage of maturity", are Erikson's. Freud described five psychosexual stages, Harry Stack Sullivan developmental epochs from infancy to adulthood, and Melanie Klein the paranoid-schizoid and depressive positions of infancy (all Chapter 4). John Bowlby is attachment (Section 6.7); Eric Berne is transactional analysis (Chapter 7).

> **INDIAN CONTEXT**
>
> *Gandhi's Truth* (1969) is a psychohistory, a great life read through Erikson's stages and psychoanalytic ideas. Centred on the 1918 Ahmedabad textile mill strike, it asks how one man's identity struggle became a nation's method of nonviolent protest, as *Young Man Luther* had asked of Martin Luther. It won the Pulitzer Prize.

The eight stages ●●●

Table 6.27 is the map to memorise: for each stage an age, a crisis named by its two poles, the virtue a good resolution leaves, and the people who matter most, a "radius of significant relations" that widens from one mother to all of humankind.

Table 6.27 Erikson's eight psychosocial stages. Ages are approximate and vary between texts.

Stage (Erikson's label)	Age	Crisis	Virtue	Key relationships
1 Infancy (oral-sensory)	Birth to about 1 (some: 18 months)	Trust versus mistrust	Hope	Maternal person
2 Early childhood (muscular-anal)	About 1 or 1.5 to 3	Autonomy versus shame and doubt	Will	Parents

Stage (Erikson's label)	Age	Crisis	Virtue	Key relationships
3 Play age (locomotor-genital)	3 to 5 or 6	Initiative versus guilt	Purpose	Basic family
4 School age (latency)	6 to puberty	Industry versus inferiority	Competence	Neighbourhood and school
5 Adolescence (puberty and adolescence)	About 12 to 18	Identity versus role confusion	Fidelity	Peer groups; models of leadership
6 Young adulthood	About 18 to 40	Intimacy versus isolation	Love	Partners in friendship, sex, competition and cooperation
7 Adulthood (middle)	About 40 to 65	Generativity versus stagnation	Care	Shared household and work
8 Maturity (late adulthood)	65 onward	Ego integrity versus despair	Wisdom	Humankind, "my kind"

1. Trust versus mistrust. The helpless infant learns whether the world is dependable. Reliable feeding, comfort and contact build basic trust, an expectation of "consistency, continuity and sameness"; neglect or inconsistency breeds suspicion. The virtue is hope, a lasting confidence that wishes can be met.

2. Autonomy versus shame and doubt. The walking, talking toddler wants to do things alone. Erikson's key acts are holding on and letting go, and the classic battleground is toilet training. Patient parents leave the child with will, the determination to choose and to restrain oneself; harsh or overprotective ones leave shame and self-doubt.

3. Initiative versus guilt. The preschooler plans, explores, asks endless questions and acts out adult roles in play. Encouraged, the child gains purpose, the courage to pursue goals; punished or mocked for each venture, the child learns guilt.

4. Industry versus inferiority. School brings tools, tasks and classmates to compare oneself with. Finishing tasks and earning recognition build industry and competence; repeated failure, or being made to feel behind one's peers, brings inferiority.

5. Identity versus role confusion. With puberty and new choices, the young person must fuse childhood identifications, skills, values and future roles into one answer to "Who am I?" Success brings fidelity, loyalty to freely chosen values and people; failure brings role confusion (see Identity, moratorium and Marcia's statuses).

6. Intimacy versus isolation. With an identity in hand, the young adult can risk merging it with another's, in friendship and love, without losing it. The virtue is love, mutual devotion; failure brings isolation and self-absorption.

7. Generativity versus stagnation. Erikson's **generativity** is concern for establishing and guiding the next generation: raising children, mentoring juniors, teaching, building work that lasts (parenthood is neither necessary nor sufficient). Its failure is **stagnation**, a sense of having achieved little and having little to offer the next generation, with boredom and self-indulgence. The virtue is care.

8. Ego integrity versus despair. Looking back, the older person either accepts the one life lived, failures included, as meaningful (**ego integrity**), or feels regret, bitterness and fear of death because there is no time to start again (despair). The virtue is wisdom, a detached concern with life itself in the face of death.

Some personality texts put the adult boundaries at 35 and 55. The first four outcomes depend mainly on what others do to the child; in the last four the person increasingly chooses.

> **EXAM TRAP**
>
> An 18-month-old has been keyed to autonomy versus shame and doubt, yet another key gives trust versus mistrust as 0 to 18 months. Eighteen months is the boundary, so read the options; when both stages are offered for a child of exactly 18 months, the keyed answer has been autonomy.
>
> For sequence items, write the stage number under each option. In one mock item they run (a) 1, 2, 5, 7; (b) 2, 4, 5, 7; (c) 3, 4, 5, 8; (d) 1, 3, 6, 8. All rise, so the item is flawed; the key is (a), trust, autonomy, identity, generativity: of the two series that start at stage 1, it skips fewer stages. Virtues in order: hope, will, purpose, competence, fidelity, love, care, wisdom.

Applying the stages, and how they go wrong ●●

The stages are easiest to see in people. Mrs Rao, 52, finds her deepest satisfaction in training young engineers: generativity. Her colleague of the same age, who feels he has achieved little and has nothing to pass on, is stagnating. Her father, 78, would live his life the same way again, mistakes and all: integrity. His neighbour broods over wasted chances and dreads death: despair. Her niece in Class 4, measuring herself against classmates while mastering long division, is in industry versus inferiority.

> **EXAM TRAP**
>
> Erikson's term for the mid-life feeling of having achieved little and having little to offer the next generation is **stagnation**; generativity is the healthy pole. "Mid-life crisis" is a popular term, not Erikson's (Section 6.11), and presbyopia is the age-related loss of near focusing in the eye's lens.

A crisis can also be resolved too one-sidedly. In *The Life Cycle Completed* (1982) Erikson split this maldevelopment two ways. Too much of the positive pole, with none of the negative, gives a **maladaptive tendency**; too much of the negative pole gives a **malignant tendency**, the more serious (Table 6.28). He thought maladaptations could lead towards neurosis and malignancies towards psychosis, and that therapy could correct both, later experience the milder maladaptations.

Table 6.28 Erikson's maladaptive and malignant tendencies (1982).

Crisis	Maladaptive (too much positive)	Malignant (too much negative)
Trust versus mistrust	Sensory maladjustment: gullible	Withdrawal
Autonomy versus shame and doubt	Shameless wilfulness	Compulsion
Initiative versus guilt	Ruthlessness: goals at any cost	Inhibition: tries nothing
Industry versus inferiority	Narrow virtuosity: all skill, no childhood	Inertia
Identity versus role confusion	Fanaticism: one view only	Repudiation of adult roles
Intimacy versus isolation	Promiscuity: closeness without depth	Exclusivity: shuts others out
Generativity versus stagnation	Overextension: no time for self	Rejectivity: stops caring

Crisis	Maladaptive (too much positive)	Malignant (too much negative)
Integrity versus despair	Presumption	Disdain for life

Joan Erikson (1903 to 1997), his wife and collaborator, added a ninth stage in the book's extended edition (1997): in the eighties and nineties the negative poles of all eight crises return as the body fails, and a calm move beyond the self, gerotranscendence, becomes possible.

Erikson versus Freud

Erikson called his theory an extension of Freud's, but Table 6.29 shows how far it moved: from id to ego, from body zones to society, and from a personality set in early childhood to one still developing in old age. Freud's psychosexual stages are taught in Chapter 4.

Table 6.29 Freud and Erikson compared.

Point	Freud	Erikson
Kind of stage	Psychosexual: body zones	Psychosocial: crises with society
Driving structure	Id and instinct	Ego (ego psychology)
Span	Five stages; personality set by about 5 or 6	Eight stages, the whole lifespan
Culture and history	Little role; the family triangle	Shape every crisis
Poor outcome	Fixation	Weak virtue; maladaptive or malignant tendency
Adolescence	Genital stage	Identity crisis, the heart of the theory

Identity, moratorium and Marcia's statuses ●●●

For Erikson adolescence is the hinge of the life cycle. Childhood identifications (my father's son, a good student, a cricketer) must be sorted, kept or dropped, and fused with new abilities, sexual maturity and choices of work, beliefs and love into an **ego identity**: a sense of being the same person across time and situations, recognised by the people who count. Society helps by granting a **psychosocial moratorium**, a socially sanctioned delay of adult commitments in which the young can try out courses, friendships and causes before choosing; college years and Erikson's own wandering years are examples.

When the work fails the result is **identity confusion** (role confusion in Erikson's earlier wording; also identity diffusion): inner fragmentation, little or no sense of where one's life is headed, and an inability to gain the support of a social role or vocation. Some young people instead seize a **negative identity**, built from the roles their family or community holds up as most undesirable, such as the delinquent, on the logic that it is better to be somebody bad than nobody. Adolescence, then, is a true developmental stage and a moratorium, the fifth of eight and not the last.

> EXAM TRAP
>
> "Inner fragmentation, no sense of direction, no support from a social role or vocation" is **identity confusion**. Incongruence is Carl Rogers's gap between self-concept and experience (Chapter 4); dissociation is a break in the integration of memory, identity and consciousness (Chapter 8); "conflict" is too general.

KEY STUDY: MARCIA'S IDENTITY STATUS INTERVIEW (1966)

James E. Marcia asked whether Erikson's identity could be measured. He interviewed male college students about occupation, religion and politics and scored two dimensions: **crisis or exploration**, a period of actively weighing alternatives, and **commitment**, a firm investment in a choice. Crossing them gave four **identity statuses** (Figure 6.17) that predicted behaviour: achieved students coped best with a stressful task, foreclosed students endorsed authoritarian values most. Criticism: an all-male first sample, and a person can hold different statuses in different areas of life.

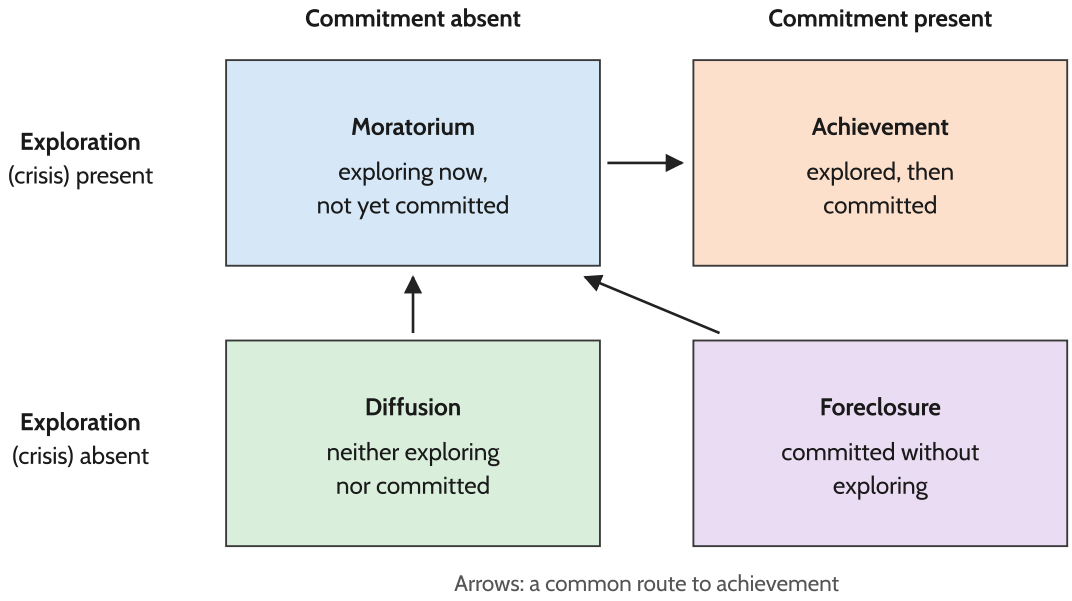


Figure 6.17 Marcia's identity statuses cross exploration (crisis) with commitment. Foreclosure and moratorium are mirror images: commitment without exploration, and exploration without commitment.

- Neither exploration nor commitment is **identity diffusion**. The young person drifts; adjustment is poorest.
- Commitment without exploration is **foreclosure**. Choices are taken over ready-made, often from parents, like the son who becomes a doctor because the family always assumed he would; such youth tend to be rigid and conforming.
- Exploration without commitment is **moratorium**. The person is in the middle of the crisis, anxious and actively trying options.
- Exploration followed by commitment is **identity achievement**, the most mature status, with the highest self-esteem.

Statuses are not fixed stages. Achievement grows through late adolescence and young adulthood, and adults reopen settled questions after a divorce, a job loss or migration and then close them again: the **MAMA cycles** (moratorium, achievement, moratorium, achievement) described by Marcia and colleagues in 1992. **Jean Phinney** applied the same logic to **ethnic identity**, from unexamined through exploration to achieved; a secure ethnic identity goes with higher self-esteem in minority youth.

EXAM TRAP

Foreclosure is commitment WITHOUT exploration; moratorium is exploration WITHOUT commitment; achievement has both; diffusion neither. Marcia's dimensions are crisis (exploration) and commitment, not IQ and emotional intelligence, locus of control (Chapter 4) or approach and avoidance motivation (Chapter 1).

Adolescent egocentrism

Formal operational thought (Section 6.4) lets adolescents think about thinking, including other people's. **David Elkind** argued in 1967 that this power brings a new **adolescent egocentrism**: the teenager fails to separate what others are thinking about from his or her own preoccupations. It shows in two mental constructions. The **imaginary audience** is the belief that everyone is watching and judging: the boy who will not enter class with a pimple on his chin, sure that all forty classmates will notice. The **personal fable** is the belief that one's experiences and feelings are unique, that no one, least of all a parent, could understand them ("You don't know what it's like to be me"), and with it a sense of invulnerability ("It won't happen to me") that feeds reckless riding. Some notes list the invulnerability strand separately as the "mythological fable". Both are strongest in early adolescence and fade by the late teens.

EXAM TRAP

Adolescent egocentrism = imaginary audience ("everyone is looking at me") plus personal fable ("my experiences are unique"). Conservation and reversibility are concrete operational (Section 6.4), inner speech and scaffolding are Vygotsky's (Section 6.5), and object permanence and stranger anxiety belong to infancy (Sections 6.4 and 6.6).

Puberty and the adolescent brain

Adolescence begins with **puberty**, the biological changes that bring sexual maturity; adolescence itself is the wider psychological and social transition. The hormones come in two waves. In **adrenarche**, from about 6 to 8, the adrenal glands raise their output of weak androgens. In **gonadarche**, from about 9 or 10, the hypothalamus and pituitary switch on the ovaries and testes, which pour out oestrogens and testosterone (Chapter 5). The visible result is the growth spurt, a rapid gain in height and weight that starts about two years earlier in girls (around 10) than in boys (around 12), so that for a while many girls are taller than boys of their age. The **primary sex characteristics** are the reproductive organs themselves (ovaries, uterus and vagina; testes, penis and scrotum); **secondary sex characteristics** are the other visible signs (breasts, body and facial hair, a deeper voice, wider hips or shoulders). In girls, **menarche**, the first menstrual period, comes late in the sequence, at about 12 or 13; **spermarche**, the first ejaculation, at about 13 or 14. Better nutrition and health have pulled menarche earlier, from about 16 or 17 in mid-nineteenth-century Europe to about 12 or 13 today, and made adults taller: the **secular trend**.

Timing matters too. Early-maturing boys tend to be confident and popular, though they try alcohol and delinquency earlier; late-maturing boys are often more anxious. Early-maturing girls carry the greatest risk: looking older than they feel, they attract older friends and show more depression, poor body image, eating problems and substance use.

Inside the skull, regions mature at different speeds. **Laurence Steinberg**'s **dual systems model** (2008) pairs a reward system in the limbic system and ventral striatum, which becomes sharply more responsive around puberty, with a cognitive control system centred on the prefrontal cortex, which matures slowly into the mid-twenties (Figure 6.18). In mid-adolescence the accelerator is pressed while the brakes are still being fitted, so risk-taking peaks, above all among peers.

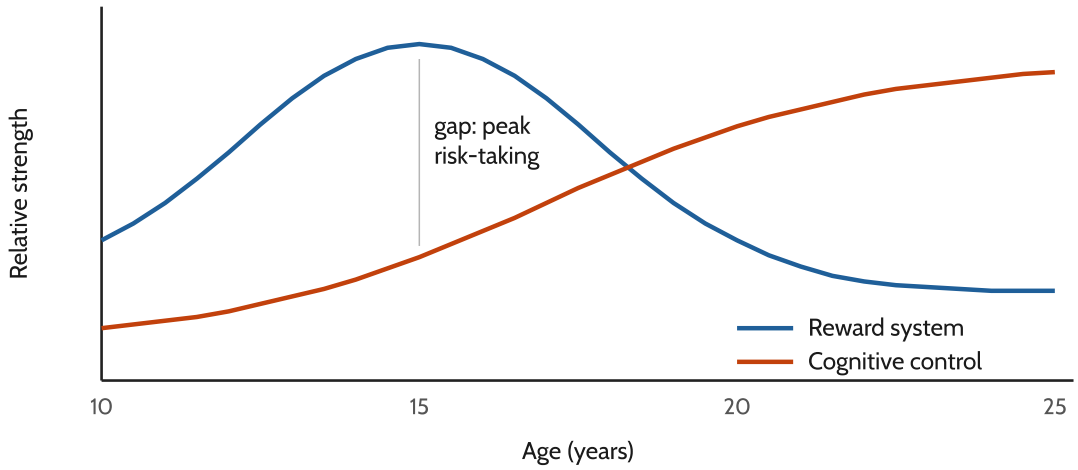


Figure 6.18 Steinberg's dual systems model (schematic). Reward sensitivity (solid line) peaks in mid-adolescence; cognitive control (dashed line) matures gradually into the mid-twenties. The gap marks the years of peak risk-taking.

KEY STUDY: RISK WITH FRIENDS WATCHING (GARDNER AND STEINBERG, 2005)

Margo Gardner and Steinberg had adolescents (13 to 16), college-age youths and adults play a video driving game in which they could brake at an amber light or gamble on getting through, alone or with two friends watching. Friends' presence roughly doubled the adolescents' risk-taking, raised the youths' by about half and left the adults' unchanged. Criticism: a laboratory game, though it matches teen drivers' higher crash rates with teen passengers.

Storm and stress, tasks, parents and peers ●●

Must adolescence be turbulent? **G. Stanley Hall** (1844 to 1924), in *Adolescence* (1904), said yes: the teens were a biologically driven period of **storm and stress**, of mood swings, conflict with parents and risk-taking. **Margaret Mead** (1901 to 1978) answered from the field. In *Coming of Age in Samoa* (1928) she described Samoan girls passing smoothly into adulthood in a relaxed culture with many caring adults, and concluded that turmoil is made by culture, not biology. **Derek Freeman** (1983) later attacked her fieldwork, but modern surveys support a middle view: most adolescents cope well and share their parents' basic values, while a minority run into serious trouble. Conflict and moodiness do rise in early adolescence; storm and stress is real but not universal.

Robert J. Havighurst (1900 to 1991) listed the **developmental tasks of adolescence** (the idea itself is in Section 6.1): new and more mature relations with age-mates of both sexes; a masculine or feminine social role; accepting one's body; emotional independence from parents and

other adults; preparing for marriage and family life; preparing for a career; a set of values and an ethical system to guide behaviour; and socially responsible behaviour. Emotional separation from parents, strong peer identification and heterosexual peer interest are therefore all tasks of adolescence.

Relations with parents are renegotiated, not cut. The older psychoanalytic view of Anna Freud and **Peter Blos** held that healthy adolescents must detach from their parents. Research now shows that autonomy grows best inside continued connection: teenagers who stay securely attached explore more confidently and adjust better, so it can be healthy for interdependence to continue. Detachment does not typically happen earlier, in middle childhood; peers do not simply replace parents, who keep their influence on values, education and career while peers sway dress, music and leisure; and parents' goals differ across families and cultures, many Indian families valuing lasting interdependence.

Dexter Dunphy (1963), observing teenagers in Sydney, distinguished **cliques**, small groups of close friends, from crowds, larger groups formed when several cliques join for parties and outings. He traced five stages: isolated same-sex cliques; same-sex cliques beginning to interact; their leaders forming the first mixed-sex clique; a full crowd of mixed-sex cliques; and the crowd loosening into groups of couples. Conformity to peers rises in early adolescence, peaks at about 14 and then declines as autonomy matures.

EXAM TRAP

Today's view of adolescents and parents: continued interdependence can be healthy. Not that detachment comes earlier in middle childhood, that the peer group matters more than parents, or that all parents share the same goals.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The virtue that emerges from initiative versus guilt is _____.
2. A toddler of 18 months who insists on feeding herself faces _____ versus shame and doubt.
3. A middle-aged man who feels he has achieved little and has nothing to offer the next generation shows Erikson's _____.
4. Erikson called the socially sanctioned delay of adult commitments a psychosocial _____.
5. Commitment without exploration is Marcia's status of _____.
6. The teenager convinced that no one has ever felt what she feels shows Elkind's _____.
7. Too much of the positive pole of the identity crisis gives the maladaptive tendency of _____.
8. In Steinberg's dual systems model the _____ cortex matures later than the reward system.

Answers: (1) purpose, (2) autonomy, (3) stagnation, (4) moratorium, (5) foreclosure, (6) personal fable, (7) fanaticism, (8) prefrontal.

6.11 Adulthood, Ageing, Death and Dying LOWER YIELD · 2 QS

Development does not stop at twenty. A Mysuru engineer of 24 unsure of his career, his mother of 50 nursing an ailing father, and his grandfather of 80 trimming his circle of friends are all still developing.

Emerging adulthood and adult development

In 2000 **Jeffrey Jensen Arnett** named the years from about 18 to 25 **emerging adulthood**, a period created in industrialised societies by later marriage, parenthood and settled work. Its five features are identity exploration (in love and work), instability, self-focus, feeling in-between (neither adolescent nor adult) and possibilities (life can still go many ways). Its Eriksonian task, intimacy versus isolation, is in Section 6.10.

Daniel Levinson (1920 to 1994) drew the best-known map of what follows. In *The Seasons of a Man's Life* (1978), built on interviews with 40 men aged 35 to 45, each man has a **life structure**, the underlying pattern of his life at a given time, built mainly around work and family. Overlapping eras alternate stable periods, when a structure is built, with transitions of about five years, when it is questioned and remade (Figure 6.19): the early adult transition (17 to 22) leaves adolescence behind, and the age-30 transition (28 to 33) revises the first adult structure. The young man pursues **the Dream**, a vision of the adult he hopes to become, helped by a mentor, an older guide at work. In the **midlife transition** (about 40 to 45) he reappraises the Dream against what he has achieved. Table 6.30 adds four other theorists.

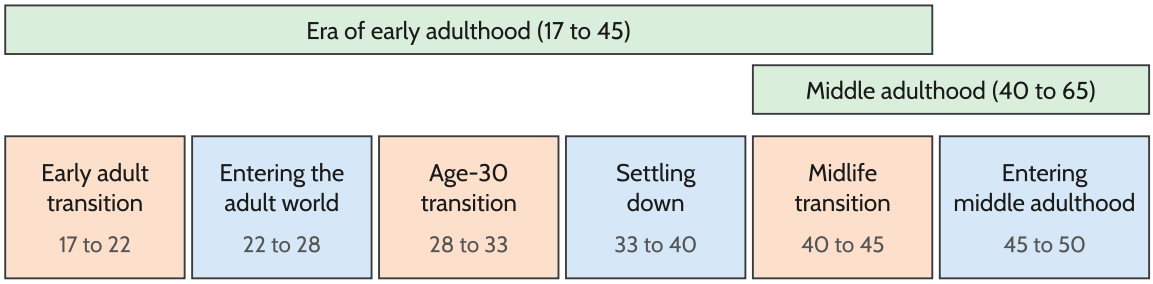


Figure 6.19 Levinson's seasons. Transitions alternate with stable periods; the midlife transition belongs to both eras.

Table 6.30 Other theories of adult development.

Theorist	Key idea
George Vaillant (born 1934), Grant Study	Followed 268 Harvard men from 1938; added career consolidation (after intimacy) and keeper of the meaning (after generativity) to Erikson's stages
Roger Gould, *Transformations* (1978)	Growth means shedding false childhood assumptions, such as "life is simple and controllable"
Bernice Neugarten (1916 to 2001)	Social clock: a cultural timetable for life events; off-time events (widowhood at 35) are harder than on-time ones
Robert Peck	Tasks of old age: ego differentiation, body transcendence and ego transcendence, versus work-role, body and ego preoccupation

Robert J. Havighurst (1900 to 1991) set adult developmental tasks (Section 6.1). Early adulthood: selecting a mate, learning to live with a partner, starting a family, rearing children, managing a home, starting an occupation, civic responsibility, a congenial social group. Middle age: civic and social responsibility, an economic standard of living, helping teenagers become adults, adult leisure, relating to the spouse as a person, accepting physical change, adjusting to ageing parents. Later maturity: adjusting to declining strength and health, to retirement and reduced income and to a spouse's death, affiliating with one's age group, civic obligations, satisfactory living arrangements.

Love, marriage and work

Why we love, including **Robert Sternberg**'s triangular theory, is social psychology (Chapter 3). Developmental psychology asks how a marriage divides roles. **Elizabeth Hurlock** (1898 to 1988), in *Developmental Psychology*, described three types (Table 6.31).

Table 6.31 Hurlock's types of marriage.

Type	Roles	What is emphasised
Traditional marriage	Sharply separate: husband provider, wife homemaker	Division of labour; husband heads the family
Companionship marriage	Shared and interchangeable	Romantic love, emotional intimacy, equality
Collegial marriage	Different but complementary; each spouse responsible for his or her own	Comradeship and sharing, as between colleagues

> **EXAM TRAP**
>
> Comradeship and sharing with different roles is a *collegial* marriage (JEMAS 2025). Companionship marriage shares the same roles; traditional marriage separates roles without stressing comradeship. "Satisfactory marriage" is not a type.

> **INDIAN CONTEXT**
>
> Most Indian marriages are still arranged by families, now usually with the couple's consent; love is expected to grow after the wedding, and the marriage joins two families.

Elsewhere more adults cohabit, divorce is commoner, and a first child usually lowers marital satisfaction for a while. Work is the other pillar. **Eli Ginzberg** (1911 to 2002; *Occupational Choice*, 1951) described three periods of career choice: fantasy (to about 11), wishing without regard to ability; tentative (about 11 to 17), weighing interests, capacities and values; and realistic (17 to the early twenties), exploring, crystallising and specifying one choice. **Donald Super**'s career stages are in Chapter 7.

Middle adulthood: the climacteric and midlife

From about 40 the body changes slowly. Near vision blurs as the lens stiffens (**presbyopia**), hearing for high pitches fades (presbycusis), bone mass falls (especially in women after menopause), hair recedes and greys, muscle strength and agility decline, and reaction time and manual dexterity slow. All of this is general physical ageing.

The **climacteric** is narrower: the end of ovulation and termination of reproductive capacity. In women it culminates in **menopause**, the last menstrual period (confirmed after 12 months without one), as ovarian oestrogen falls. The average age is about 51 in Western samples and about 46 to 47 in Indian surveys. The male climacteric, or **andropause**, is a slow decline in testosterone; most men stay fertile into old age.

> EXAM TRAP
>
> Climacteric means the end of ovulation and termination of reproductive capacity (JEMAS 2025). Lost strength, receding or greying hair, and slower reactions and dexterity are general ageing. Menopause is the event within the climacteric, not a synonym.

Elliott Jaques (1917 to 2003) coined **midlife crisis** in 1965, from creative artists confronting mortality in their mid-thirties. Large surveys find that only a minority report one, usually after illness or job loss. Midlife is more often a peak of competence and generativity (Section 6.10). The **empty nest**, when the last child leaves, more often brings relief than depression. The **sandwich generation** are middle-aged adults, usually women, caring for children and ageing parents at once. **K. Warner Schaie**'s Seattle Longitudinal Study (begun 1956) found that most mental abilities peak in middle adulthood, while perceptual speed declines from early adulthood.

Late adulthood and the biology of ageing

Gerontology studies the biological, psychological and social sides of ageing; geriatrics is the medicine of old age. **Robert Butler** (1927 to 2010) coined **ageism**, prejudice against people because they are old, in 1969. Older adults are divided into the young-old (65 to 74), old-old (75 to 84) and oldest-old (85 and over). Life span is the maximum possible length of life, about 120 years (**Jeanne Calment** lived 122); **life expectancy** is the average number of years a person born in a given year can expect to live, about 70 in India.

Primary ageing is gradual, inevitable and universal bodily decline. Secondary ageing comes from disease, disuse and lifestyle, so it is partly preventable. Tertiary ageing is the rapid loss shortly before death; its cognitive form, a sharp fall in test scores, is the **terminal drop** (**Robert Kleemeier**, 1962).

Programmed theories say the genes set ageing's timetable. **Leonard Hayflick** and **Paul Moorhead** (1961) found that human cells in culture divide only about 40 to 60 times (some textbooks say 75 to 80), the **Hayflick limit**, apparently because telomeres, the caps on chromosome ends, shorten with each division. Damage, or variable-rate, theories blame accumulated random injury instead (Table 6.32).

Table 6.32 Biological theories of ageing.

Family	Theory	Claim
Programmed	Cellular clock (Hayflick)	Cells divide a fixed number of times as telomeres shorten
Programmed	Programmed senescence	Genes switch on and off in sequence
Programmed	Immunological	Immune function declines on schedule, so disease rises
Programmed	Hormonal stress	An ageing hormonal system lowers resistance to stress
Damage	Wear and tear	Body parts, like machine parts, wear out with use

Family	Theory	Claim
Damage	Free radicals (**Denham Harman**, 1956)	Unstable oxygen molecules from metabolism damage DNA and cells
Damage	Rate of living	Faster metabolism, shorter life
Damage	Autoimmune	The immune system attacks the body's own cells

Psychosocial theories of ageing

Is a good old age a matter of stepping back or of staying busy? **Elaine Cumming** and **William Henry**'s **disengagement theory** (*Growing Old*, 1961) said old people and society withdraw from each other by natural, mutual consent. Research disagreed: withdrawal is usually forced by illness or loss, and it lowers well-being. Havighurst's **activity theory** replied that the more active older people stay, the more satisfied they are. Table 6.33 sets out the main theories.

Table 6.33 Psychosocial theories of ageing.

Theory	Proposed by	Core idea
Disengagement	Cumming and Henry, 1961	Mutual, natural withdrawal of the person and society
Activity	Havighurst	Stay active and replace lost roles
Continuity theory	**Robert Atchley**, 1989	Keep inner (values) and outer (roles, routines) continuity, using lifelong coping strategies
Socioemotional selectivity	Laura Carstensen	As time feels short, emotional goals beat knowledge goals; smaller, closer networks
Selective optimisation with compensation	Paul and Margret Baltes, 1990	Select fewer goals, optimise them, compensate for losses
Gerotranscendence	**Lars Tornstam**, 1989	From a materialistic, rational outlook to a cosmic, transcendent one
Successful ageing	**John Rowe** and **Robert Kahn**, 1997	Low disease and disability, high mental and physical functioning, active engagement

Laura Carstensen's **socioemotional selectivity theory** explains why the grandfather of 80 drops acquaintances but sees more of his family.**Paul Baltes** (1939 to 2006) and **Margret Baltes** (1939 to 1999) illustrated **selective optimisation with compensation** (SOC) with the pianist **Arthur Rubinstein**, who in old age played fewer pieces (selection), practised them more (optimisation) and slowed before fast passages so that they sounded faster (compensation). Butler's life review, used in therapy, is in Chapter 7.

> **EXAM TRAP**
>
> Disengagement says withdraw; activity, stay busy; continuity, stay yourself; socioemotional selectivity, prune the network for emotional reasons. SOC belongs to the Baltes couple, not Carstensen; the three criteria of successful ageing, to Rowe and Kahn.

Cognition, retirement and elder care

Fluid intelligence declines from early adulthood while crystallised intelligence holds or rises into old age (Section 1.9). **Timothy Salthouse**'s **processing-speed theory** (1996) traces much of the loss to a general slowing of processing. Episodic and working memory decline; semantic and procedural memory hold. Paul Baltes's **Berlin wisdom paradigm** defines wisdom as expert knowledge about the fundamental pragmatics of life, judged by factual and procedural knowledge, lifespan contextualism, value relativism and managing uncertainty; it reflects experience more than age. Dementia is disease, not normal ageing (Chapters 5 and 8).

Retirement is a process: Atchley described a honeymoon, then possible disenchantment, reorientation and stability; it goes best when chosen, healthy and funded. Widowhood falls mostly on women, who live longer and usually marry older men.

> **INDIAN CONTEXT**
>
> As joint families shrink, the law has stepped in. The Maintenance and Welfare of Parents and Senior Citizens Act 2007 obliges children (or, for the childless, heirs) to maintain parents and senior citizens (60 and over). Tribunals can order up to Rs 10,000 a month, and abandoning a senior citizen is punishable by up to three months in prison, a Rs 5,000 fine, or both.

Death, dying and grief

Death was once the stopping of heart and breathing. Life support broke that link, so India (Transplantation of Human Organs Act 1994) and most countries also recognise **brain death**: irreversible loss of all brain function or, in Indian law, of all brainstem function. In a persistent vegetative state the brainstem still works.

Maria Nagy (1948) questioned 378 Budapest children aged 3 to 10. Under about 5, death was a departure or sleep, and reversible; from 5 to 9 it was final but personified, a "death-man" who could be escaped; after 9 it was universal and inevitable. **Mark Speece** and **Sandor Brent** (1984) named three parts of a mature concept, irreversibility, universality and nonfunctionality (all functions stop), grasped by most children at about 5 to 7.

Elisabeth Kübler-Ross's five stages of dying (*On Death and Dying*, 1969) are denial, anger, bargaining, depression and acceptance (Section 7.9). **John Bowlby** (1907 to 1990) saw mourning as a broken attachment (Section 6.7) with four phases: numbness, yearning and searching, disorganisation and despair, and reorganisation. **Margaret Stroebe** and **Henk Schut**'s **dual process model** (1999) replaces fixed stages with oscillation between loss-oriented coping (grief work, yearning) and restoration-oriented coping (new tasks and roles, time off from grief).

Cicely Saunders (1918 to 2005) opened St Christopher's Hospice, London, in 1967; hospices give **palliative care**, aiming at comfort and dignity rather than cure (Section 7.9). Euthanasia is active when a lethal act ends life and **passive euthanasia** when life support is withheld or withdrawn; India permits only the passive kind. In *Aruna Shanbaug v Union of India* (2011), about a Mumbai nurse left vegetative by a 1973 assault, the Supreme Court refused the plea but allowed passive euthanasia with High Court approval. *Common Cause v Union of India* (2018) held that dying with dignity is part of the right to life (Article 21) and upheld the **living will**, an advance directive refusing life support.

EXAM TRAP

Kübler-Ross's stages describe the dying; Bowlby's phases, the bereaved; Stroebe and Schut's model is oscillation, not stages. Shanbaug (2011) allowed passive euthanasia; Common Cause (2018) added living wills.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Arnett called ages 18 to 25 _____ adulthood.
2. Levinson's vision of the adult a young man hopes to become is the _____.
3. Comradeship with different, complementary roles marks a _____ marriage.
4. The end of ovulation and of reproductive capacity is the _____.
5. Cumming and Henry proposed _____ theory.
6. Selection, optimisation and compensation is the model of Paul and Margret _____.
7. A sharp cognitive decline shortly before death is the _____.
8. The dual process model contrasts loss-oriented and _____-oriented coping.

Answers: (1) emerging, (2) Dream, (3) collegial, (4) climacteric, (5) disengagement, (6) Baltes, (7) terminal drop, (8) restoration

The Wild Boy of Aveyron, Revisited

Look at Victor again with this chapter behind you. Pinel and Itard were arguing nature versus nurture (Section 6.1), and the modern answer is that each may have been partly right: Victor may have had a developmental disability, perhaps autism, before he was abandoned (Sections 6.2 and 6.3). His failure to speak after years of teaching fits a **sensitive period** for language that had largely closed (Section 6.1), while his growing problem solving and his bond with Madame Guérin show cognition and attachment still developing (Sections 6.4 to 6.7). His protest at unjust punishment touches moral development (Section 6.9), the turmoil puberty brought him touches adolescence (Section 6.10), and his quiet life to about forty is a reminder that development runs across the whole lifespan (Section 6.11).

Chapter summary

6.1 Foundations: The Lifespan View, Key Issues and Ecological Contexts. Developmental psychology studies change and stability from conception to death. Growth is quantitative, development qualitative, and maturation is the biological unfolding of genetic potential (Gesell; co-twin control, 1929). Hurlock divides life into ten periods; Baltes's lifespan perspective (1987) sees development as lifelong, multidirectional and plastic, shaped by age-graded, history-graded and non-normative influences. Development runs cephalocaudally and proximodistally, with Havighurst's developmental tasks at each period. Core issues include nature versus nurture and continuity versus discontinuity. A critical period is fixed and irreversible (imprinting, embryonic organ formation); a sensitive period is flexible (language). Bronfenbrenner (1979) nests the child in micro-, meso-, exo- and macrosystems, with the chronosystem added later.

6.2 Heredity, Prenatal Development, Birth and the Newborn. Each body cell carries 46 chromosomes in 23 pairs; meiosis halves them in sperm and ovum, which unite as a **zygote**. In **dominant-recessive inheritance** a heterozygote shows the dominant trait, while incomplete dominance, codominance, X-linked, polygenic inheritance and genomic imprinting vary the pattern. Chromosomal errors include Down (trisomy 21), Klinefelter (47,XXY), Turner (45,X) and Prader-Willi (paternal 15q11-q13 deletion). **MZ twins reared apart** best separate heredity from environment, which also correlate and interact. Prenatal life runs through germinal, embryonic (gastrulation, neural tube, greatest vulnerability to **teratogens**) and fetal periods. Newborns are scored on the **Apgar scale** and show primitive reflexes such as the Moro and Babinski.

6.3 Infancy: Brain, Body, Motor Skills and Perception. Birth weight doubles by about 5 months and triples by 1 year, and growth and motor control run **cephalocaudal** and **proximodistal** (never distal to proximal). Neurons are mostly made before birth; synaptogenesis, **synaptic pruning** and myelination take the brain from about 25 percent of adult weight to half by about 6 months. The first voluntary grasp is ulnar; the pincer grasp follows by 9 to 12 months. Gesell stressed maturation, Dennis experience and Thelen dynamic systems. Looking methods reveal face preference and perceptual narrowing; the visual cliff shows depth perception by crawling age, when **independent locomotion** transforms affordances. Newborns imitate (Meltzoff and Moore, 1977), and comprehension leads production.

6.4 Piaget's Theory of Cognitive Development. Jean Piaget (1896 to 1980) saw intelligence as biological adaptation: the child, an active little scientist, builds **schemes** and adapts them by **assimilation** (fitting experience into existing schemes) and **accommodation** (changing schemes), driven by **equilibration**. Four stages follow in invariant order: **sensorimotor** (birth to 2; object permanence begins at 8 to 12 months with the A-not-B error and is complete by 18 to 24), **preoperational** (2 to 7; egocentrism, centration, no conservation), **concrete operational** (7 to 11; conservation, reversibility, transitivity) and **formal operational** (11 or 12 onward; hypothetico-deductive reasoning). Dialectical reasoning is postformal. Baillargeon, Donaldson and the neo-Piagetians showed he underestimated young children.

6.5 Vygotsky, Bruner, Information Processing and Theory of Mind. Lev Vygotsky's sociocultural theory holds that higher mental functions arise in social interaction, mediated by cultural tools (language the master tool): every function appears first between people, then within the child. The **zone of proximal development** is the gap between what a learner can do alone and with a more knowledgeable other; support there is **scaffolding** (Wood, Bruner and Ross, 1976). For Vygotsky language shapes thought and private speech becomes inner speech. Bruner described enactive, iconic and symbolic representation. Information-processing research tracks speed, strategies and metacognition. False-belief tasks are passed at about 4, and Selman traced perspective taking from egocentric to societal.

6.6 Emotional Development, Temperament and Self-Regulation. Bridges (1932) held that the newborn's general excitement differentiates first into **distress and delight**; primary emotions are present within six months, the social smile at 6 to 10 weeks, and the **self-conscious emotions** only after 12 months. Stranger and separation anxiety peak at about 8 to 15 months, and Tronick's still face shows that infants expect affective reciprocity. Temperament is early, moderately stable and biologically based: Thomas and Chess gave nine dimensions, the **easy, difficult and slow-to-warm-up** types and goodness of fit; Rothbart and Bates reduced it to surgency, negative affec-

tivity and effortful control; Kagan studied behavioural inhibition. Mischel's marshmallow test measures **delay of gratification**.

6.7 Attachment: Imprinting, Bowlby, Ainsworth and Beyond. Attachment is an enduring bond shown by proximity seeking, secure-base and safe-haven behaviour and separation distress. Lorenz's **imprinting** and Harlow's **contact comfort** undermined the feeding-based psychoanalytic and secondary-drive theories, leaving Bowlby's **ethological theory** (an evolved system, monotropy, a critical period, the **internal working model**) predominant. Bowlby's four phases, Schaffer and Emerson's stages and Mahler's separation-individuation chart the first three years. Ainsworth's **Strange Situation** sorts infants by reunion behaviour into secure (B), avoidant (A) and resistant (C), plus Main and Solomon's disorganised (D). Spitz, Rutter and the adoption studies show privation harms but is partly reversible.

6.8 Parenting, Family, Play and Gender in Childhood. Diana Baumrind described authoritative, authoritarian and permissive parenting; **Maccoby and Martin** (1983) crossed demandingness with responsiveness and added the neglectful style. Authoritative parenting, high on both, gives the best outcomes, though warm control also works in Chinese and Indian families. Helicopter parenting has several roots, and the family acts as a system. Neglect is physical, supervisory, medical, educational or emotional; licensed day care is not neglect. Piaget's play runs from functional to symbolic play to games with rules, Smilansky added constructive play, and Parten traced solitary to cooperative play. Children learn the gender map through socialisation; **Kohlberg** and gender schema theory explain how they then socialise themselves.

6.9 Moral Development: Piaget, Kohlberg, Gilligan and Beyond. Morality has emotional (guilt, empathy, the superego), behavioural (social learning; Hartshorne and May found honesty situation-specific) and cognitive sides. **Piaget** (1932) traced a shift from **heteronomous morality** (fixed rules, judging by consequences, immanent justice, expiatory punishment) to **autonomous morality** (rules as agreements, intentions, reciprocal punishment). **Kohlberg** posed dilemmas such as Heinz's and scored the reasoning: **preconventional** (punishment, exchange), **conventional** (approval, law and order) and **postconventional** (social contract, universal principles). Critics cite cultural bias (Snarey, Shweder, Miller), the reasoning and behaviour gap and gender bias; **Gilligan** added the female **ethic of care**. Turiel, Eisenberg, Hoffman and Haidt widened the field.

6.10 Erikson's Psychosocial Stages, Identity and Adolescence. Erik Erikson (1902 to 1994), an ego psychologist, proposed eight psychosocial stages governed by the epigenetic principle, each a crisis whose favourable resolution yields a virtue: trust (hope), autonomy (will), initiative (purpose), industry (competence), identity (fidelity), intimacy (love), generativity (care) and integrity (wisdom). One-sided resolution gives maladaptive or malignant tendencies. Adolescence is a psychosocial moratorium whose failure is identity confusion. **James Marcia** (1966) crossed exploration with commitment to give diffusion, foreclosure, moratorium and achievement. **David Elkind** described the imaginary audience and personal fable. Puberty, Steinberg's dual systems model and modern research temper Hall's storm and stress: connection with parents stays healthy.

6.11 Adulthood, Ageing, Death and Dying. Arnett's **emerging adulthood** (about 18 to 25) leads into Levinson's life structures, eras, transitions and Dream, joined by Havighurst's developmental tasks, Vaillant, Gould, Neugarten's social clock and Peck. Hurlock's marriages are traditional, companionship or **collegial** (comradeship with different roles). The **climacteric** is the end of ovulation and reproductive capacity, culminating in menopause; the midlife crisis (Jaques, 1965) is

not universal. Ageing is explained by programmed theories (Hayflick limit, telomeres) and damage theories (free radicals), and adapted to through disengagement, activity, continuity, socioemotional selectivity or SOC. Speed and fluid abilities fall; crystallised knowledge holds. Grief follows Bowlby's phases or Stroebe and Schut's dual process; India allows passive euthanasia and living wills.

Who's who

Table 6.34 The people in this chapter.

Name	When	Examined for
Victor of Aveyron	found 1800; died 1828	The wild boy of Aveyron, educated by Itard; nature versus nurture and language deprivation (chapter opening case)
Jean-Marc-Gaspard Itard	1774 to 1838	Physician who trained Victor of Aveyron (reports of 1801 and 1806); pioneer of special education
Philippe Pinel	1745 to 1826	Paris asylum reformer who judged Victor of Aveyron an incurable idiot (1800)
Elizabeth Hurlock	1898 to 1988	Developmental Psychology: A Life-Span Approach; ten periods of the lifespan; principles of development
Paul Baltes	1939 to 2006	Lifespan perspective (1987); normative age-graded, history-graded and non-normative influences
John Locke	1632 to 1704	Tabula rasa, the blank slate (nurture)
Jean-Jacques Rousseau	1712 to 1778	Emile (1762); the child as noble savage, natural unfolding in stages (nature)
Charles Darwin	1809 to 1882	Baby biography of his son (1877)
Wilhelm Preyer	1841 to 1897	The Mind of the Child (1882), start of scientific child psychology
G. Stanley Hall	1844 to 1924	Child study movement; Adolescence (1904); storm and stress; recapitulation
Arnold Gesell	1880 to 1961	Maturation; normative approach; co-twin control with Thompson (1929); developmental schedules
John B. Watson	1878 to 1958	Behaviourism; development moulded by environment (nurture)
Jean Piaget	1896 to 1980	Accommodation; active child; universal cognitive stages
Helen Thompson	1929	Co-twin control study of stair climbing with Gesell
Myrtle McGraw	1899 to 1988	Growth: A Study of Johnny and Jimmy (1935); training speeds ontogenetic, not phylogenetic, skills
Robert J. Havighurst	1900 to 1991	Developmental tasks; Developmental Tasks and Education (1948)
Sigmund Freud	1856 to 1939	Psychosexual stages; early experience decisive
Erik Erikson	1902 to 1994	Psychosocial stages
John Bowlby	1907 to 1990	Ethological attachment theory; importance of early experience
Ann Clarke	1976	Early Experience: Myth and Evidence, with Alan Clarke; later experience can repair early deficits
Alan Clarke	1976	Early Experience: Myth and Evidence, with Ann Clarke
Lev Vygotsky	1896 to 1934	Sociocultural theory; context-specific development
Urie Bronfenbrenner	1917 to 2005	Ecological systems theory, The Ecology of Human Development (1979); bioecological model and PPCT; Head Start (1965)
Konrad Lorenz	1903 to 1989	Imprinting in greylag goslings (1935); critical periods
Eric Lenneberg	1921 to 1975	Critical period hypothesis for language (1967)
Genie	found 1970	Isolated child who learned words but never mastered grammar; sensitive period for language
Noam Chomsky	born 1928	Nativism: language ability is innate
Michael Rutter	1933 to 2021	English and Romanian Adoptees study; recovery after early deprivation
Esther Thelen	1941 to 2004	Dynamic systems theory of motor development
B. F. Skinner	1904 to 1990	Operant conditioning; learning theory of development
Albert Bandura	1925 to 2021	Observational learning; learning theory of development

Name	When	Examined for
Nikolaas Tinbergen	1907 to 1988	Ethology
Stephen Ceci	1994	Bioecological model with Bronfenbrenner
Pamela Morris	1998	Bioecological model and PPCT with Bronfenbrenner
Gregor Mendel	1822 to 1884	Dominant-recessive inheritance, from pea plants (1866)
John Langdon Down	1828 to 1896	Described Down syndrome (1866)
Albert Stunkard	1990	Twins reared apart: body-mass index of identical twins nearly as alike apart as together
Sandra Scarr	1983	Passive, evocative and active (niche-picking) gene-environment correlation, with Kathleen McCartney
Kathleen McCartney	1983	Gene-environment correlation, with Sandra Scarr
Irving Gottesman	1963	Reaction range
Conrad Hal Waddington	1905 to 1975	Canalisation
Virginia Apgar	1909 to 1974	Apgar scale for the newborn (1952, published 1953)
T. Berry Brazelton	1918 to 2018	Neonatal Behavioral Assessment Scale (1973)
Peter Wolff	1966	Newborn states of arousal
William Greenough	1987	Experience-expectant versus experience-dependent plasticity
Wayne Dennis	1940; 1960	Hopi cradleboard study (1940, with Marsena Dennis); severe motor delays in Iranian orphanages (1960)
Marsena Dennis	1940	Co-author of the Hopi cradleboard study
Robert Fantz	1961	Preferential looking; infants prefer patterns, and faces most of all
Renée Baillargeon	1985	Violation-of-expectation method
Anthony DeCasper	1980; 1986	Newborns prefer the mother's voice (with Fifer, 1980) and a story heard in the womb (with Spence, 1986)
William Fifer	1980	With DeCasper, newborn preference for the mother's voice
Melanie Spence	1986	With DeCasper, The Cat in the Hat study of prenatal learning
Janet Werker	1984	With Tees, perceptual narrowing: English-learning infants lose a Hindi contrast by 10 to 12 months
Richard Tees	1984	Co-author with Werker of the perceptual narrowing study
Elizabeth Spelke	1976	Intermodal perception: infants look at the film that matches the soundtrack
Eleanor Gibson	1910 to 2002	Visual cliff with Walk (1960); ecological view of perception and affordances
Richard Walk	1960	Visual cliff with Eleanor Gibson
Joseph Campos	1970 onwards	Heart-rate studies on the visual cliff; locomotor experience brings wariness of heights
James J. Gibson	1904 to 1979	Ecological view of perception; affordances
Carolyn Rovee-Collier	1942 to 2014	Mobile-kicking (operant conditioning) studies of infant memory
Andrew Meltzoff	1977	With Moore, neonatal imitation at 12 to 21 days
M. Keith Moore	1977	Co-author with Meltzoff of the neonatal imitation study
Théodore Simon	1873 to 1961	Co-author of the Binet-Simon scale; Piaget scored test items in his Paris laboratory around 1920
Alfred Binet	1857 to 1911	First intelligence scale with Simon (1905)
Burrhus Frederic Skinner	1904 to 1990	Operant conditioning (a distractor in Piaget items)
Lawrence Kohlberg	1927 to 1987	Moral stages (a distractor for who proposed the cognitive stages)
Adele Diamond	1985	A-not-B error grows with delay; linked to prefrontal maturation and inhibition of a prepotent reach
Judy DeLoache	1987	Scale-model task; dual representation
Bärbel Inhelder	1913 to 1997	Three mountains task with Piaget; The Growth of Logical Thinking (1958)
Arthur Robert Jensen	1923 to 2012	Level I and Level II abilities (a distractor in the 'preschoolers cannot reason' item)
Joy Paul Guilford	1897 to 1987	Structure of intellect (a distractor in the 'preschoolers cannot reason' item)
Klaus Riegel	1973	Dialectical operations as the final period of cognitive development (postformal)

Name	When	Examined for
William Perry	1970	College students move from dualism through relativism to commitment
Patricia Arlin	1975	Problem finding as a fifth, postformal stage
Michael Basseches	1984	Dialectical thinking in adulthood
Gisela Labouvie-Vief	1980	Pragmatic postformal thought
Jan Sinnott	1998	Relativistic postformal thought
Margaret Donaldson	1926 to 2020	Children's Minds (1978): children reason well when tasks make human sense
Martin Hughes	1975	Policeman task: preschoolers take two perspectives when hiding makes sense
James McGarrigle	1974	Naughty teddy conservation study with Donaldson
Juan Pascual-Leone	1970	Neo-Piagetian M-space (mental capacity that grows with age)
Robbie Case	1944 to 2000	Neo-Piagetian: stage progress through more efficient use of working memory
Kurt Fischer	1943 to 2020	Neo-Piagetian skill theory (1980)
Lev Semyonovich Vygotsky	1896 to 1934	Sociocultural theory; zone of proximal development; general genetic law; private and inner speech; Thinking and Speech (1934), Mind in Society (1978)
David Wood	1976	Coined scaffolding with Bruner and Ross (the block-pyramid tutoring study)
Gail Ross	1976	Co-author of the scaffolding study with Wood and Bruner
Jerome Seymour Bruner	1915 to 2016	Enactive, iconic and symbolic modes of representation (1966); spiral curriculum and discovery learning (The Process of Education, 1960); scaffolding (1976); LASS
Barbara Rogoff	1990	Guided participation; development as apprenticeship (Apprenticeship in Thinking)
Annemarie Palincsar	1984	Reciprocal teaching, with Ann Brown
Ann Brown	1984	Reciprocal teaching, with Annemarie Palincsar
Reuven Feuerstein	1921 to 2014	Dynamic assessment; Learning Potential Assessment Device (1979)
Laura E. Berk	1980s to 1990s	Research supporting Vygotsky on private speech
Robert V. Kail	1991	Processing speed rises with age at a similar rate across tasks
John H. Flavell	1970, 1979	Production deficiency; metacognition and metamemory (1979); appearance-reality distinction; recursive thought study (1970)
Patricia H. Miller	1970, 1990	Utilisation deficiency; recursive thought study with Kessel and Flavell (1970)
Frank Kessel	1970	Recursive thought study with Miller and Flavell
Michelene Chi	1978	Child chess experts out-recall adult novices: the knowledge base
Robert S. Siegler	1996	Overlapping waves model of strategy development
Charles Brainerd	1990	Fuzzy-trace theory (verbatim and gist traces), with Valerie Reyna
Valerie Reyna	1990	Fuzzy-trace theory, with Charles Brainerd
David Premack	1925 to 2015	Coined theory of mind with Guy Woodruff (1978), asking whether the chimpanzee has one
Guy Woodruff	1978	Coined theory of mind with David Premack
Gyorgy Gergely	1995	Teleological stance (with Gergely Csibra): toddlers understand intentions in action
Gergely Csibra	1995	Teleological stance, with Gyorgy Gergely
John Searle	1983	Philosopher who distinguished prior intentions from intentions in action
Henry Wellman	1990	Desire psychology at about 2, belief-desire psychology at 3 to 4
Heinz Wimmer	1983	First false-belief task (Maxi and the chocolate) with Josef Perner; second-order belief (1985)
Josef Perner	1983	False-belief Maxi task with Wimmer (1983); second-order theory of mind (1985); Smarties task (1987)
Susan Leekam	1987	Smarties unexpected-contents task, with Perner and Wimmer
Simon Baron-Cohen	1985	Sally-Anne task with Leslie and Frith; theory-of-mind deficit (mindblindness) in autism
Alan Leslie	1985	Sally-Anne task, with Baron-Cohen and Frith

Name	When	Examined for
Uta Frith	1985	Sally-Anne task, with Baron-Cohen and Leslie
Katherine Nelson	1973	Referential style of early vocabulary (a wrong option for recursive thought)
Robert L. Selman	1976, 1980	Levels of social perspective taking, from egocentric to societal; the Holly dilemma
Katharine Banham Bridges	1932	Differentiation theory: general excitement splits first into distress and delight
John Broadus Watson	1878 to 1958	Three innate emotions: fear, rage and love
Carroll Izard	1923 to 2017	Differential emotions theory: basic emotions as separate innate systems
Michael Lewis	1979	Timetable of primary and self-conscious emotions; rouge test with Brooks-Gunn
Jeanne Brooks-Gunn	1979	Rouge (mirror self-recognition) test with Lewis, typical by 18 to 24 months
Jerome Kagan	1929 to 2021	Behavioural inhibition and high-reactive infants; cross-cultural separation protest peaking at 13 to 15 months
René Spitz	1887 to 1974	Eight-month anxiety (stranger anxiety)
James Sorce	1985	Social referencing on the visual cliff: no infant crossed when the mother looked afraid
Edward Tronick	1978	Still-face paradigm: infants expect affective reciprocity
Claire Kopp	1982	Control, then self-control, then self-regulation
Mary Rothbart	1981	Temperament as reactivity and self-regulation; Infant Behavior Questionnaire; structural model with Bates
John Bates	1998	Structural model of temperament with Rothbart: surgency, negative affectivity, effortful control
Alexander Thomas	1914 to 2003	New York Longitudinal Study (1956); nine dimensions; easy, difficult, slow-to-warm-up; goodness of fit
Stella Chess	1914 to 2007	New York Longitudinal Study with Thomas; difficult child syndrome; goodness of fit
Arnold Buss	1984	EAS model of temperament with Plomin: emotionality, activity, sociability
Robert Plomin	1984	EAS model of temperament with Buss
Walter Mischel	1930 to 2018	Marshmallow test of delay of gratification at Stanford's Bing Nursery School; hot and cool systems
Yuichi Shoda	1990	Follow-up linking preschool waiting to later competence and SAT scores
Philip Peake	1990	Co-author of the 1990 marshmallow follow-up
Tyler Watts	2018	Marshmallow replication: about half the original effect
Greg Duncan	2018	Co-author of the 2018 marshmallow replication
Haonan Quan	2018	Co-author of the 2018 marshmallow replication
Hippocrates	about 460 to 370 BC	Four humours, the ancient distractor to Thomas and Chess's types
Galen	AD 129 to about 216	Turned the humours into four temperaments: sanguine, choleric, melancholic, phlegmatic
John Dollard	1900 to 1980	With Neal Miller, secondary drive ('cupboard love') theory of attachment (1950)
Neal Miller	1909 to 2002	With John Dollard, secondary drive theory of attachment (1950)
Robert Sears	1908 to 1989	Learning (secondary drive) account of dependency and attachment
Eckhard Hess	1916 to 1986	Imprinting in mallard ducklings peaks about 13 to 16 hours after hatching
Karl von Frisch	1886 to 1982	Ethologist; shared the 1973 Nobel Prize with Lorenz and Tinbergen
Harry Harlow	1905 to 1981	Wire and cloth surrogate mothers: contact comfort, 'The Nature of Love' (1958); isolation and motherless mothers
Margaret Harlow	1918 to 1971	Harry Harlow's wife and colleague in the contact comfort research
Stephen Suomi	1972	With Harlow, rehabilitation of isolate monkeys by younger 'therapist' monkeys
James Robertson	1952	Film A Two-Year-Old Goes to Hospital; protest, despair and detachment
Joyce Robertson	1967 to 1973	Films of young children in brief separations, with James Robertson

Name	When	Examined for
Rudolph Schaffer	1964	With Peggy Emerson, 60 Glasgow infants; specific attachment from about 7 months; attachment to the most responsive person
Peggy Emerson	1964	With Rudolph Schaffer, stages of attachment in Glasgow infants
Margaret Mahler	1897 to 1985	Separation-individuation; rapprochement at about 15 to 24 months; object constancy
Fred Pine	1975	Co-author with Mahler of The Psychological Birth of the Human Infant
Anni Bergman	1975	Co-author with Mahler of The Psychological Birth of the Human Infant
Mary Ainsworth	1913 to 1999	Strange Situation; secure, avoidant and resistant types; maternal sensitivity hypothesis; Uganda and Baltimore studies
Barbara Wittig	1969	With Ainsworth, first report of the Strange Situation
Mary Blehar	1978	Co-author of Patterns of Attachment
Everett Waters	1978	Co-author of Patterns of Attachment
Sally Wall	1978	Co-author of Patterns of Attachment
Mary Main	1986, 1990	With Judith Solomon, the disorganised or disoriented (D) type; Adult Attachment Interview
Judith Solomon	1986, 1990	With Mary Main, the disorganised or disoriented (D) type
Marinus van IJzendoorn	1988	With Kroonenberg, cross-cultural meta-analysis: secure most common everywhere; more variation within than between cultures
Pieter Kroonenberg	1988	With van IJzendoorn, cross-cultural meta-analysis of the Strange Situation
Klaus Grossmann	1985	Many avoidant (A) infants in northern Germany, from a cultural emphasis on early independence
Karin Grossmann	1985	With Klaus Grossmann, northern German attachment samples
Barbara Tizard	1926 to 2015	With Jill Hodges, follow-up of children from London residential nurseries
Jill Hodges	1989	With Barbara Tizard, institution-reared children followed to age 16
Donald Winnicott	1896 to 1971	Good-enough mother; transitional object
Cindy Hazan	1987	With Phillip Shaver, attachment styles in adult romantic love
Phillip Shaver	1987	With Cindy Hazan, adult romantic attachment
Kim Bartholomew	1991	With Leonard Horowitz, four adult styles: secure, preoccupied, dismissing, fearful
Leonard Horowitz	1991	With Kim Bartholomew, four-category model of adult attachment
Jay Belsky	1986	Day care debate: more than 20 hours a week in the first year as a risk to security
John Watson	1878 to 1958	With Rosalie Rayner, Little Albert (conditioned fear); a distractor here
Rosalie Rayner	1920	With John Watson, Little Albert
Edward Tolman	1886 to 1959	Cognitive map (not the internal working model)
Hans Eysenck	1916 to 1997	Personality; a distractor for anaclitic depression
Carl Rogers	1902 to 1987	Person-centred theory; a distractor for attachment theory
Diana Baumrind	1927 to 2018	Parenting styles: authoritative, authoritarian, permissive (papers from 1966 to 1971)
Eleanor Maccoby	1917 to 2018	With John Martin (1983): demandingness and responsiveness, neglectful style; gender segregation; with Jacklin, sex differences (1974)
John Martin	1983	With Maccoby: two dimensions of parenting; added the neglectful (uninvolved) style
Ruth Chao	1994	Chinese parenting as training (chiao shun), not authoritarian control
Haim Ginott	1922 to 1973	Helicopter image in Between Parent and Teenager (1969)
Amy Chua	2011	Tiger parenting (her memoir of strict, achievement-driven parenting)
Martin Hoffman	1967	Discipline types: induction, power assertion, love withdrawal
Alfred Adler	1870 to 1937	Birth order and personality (Chapter 4)
E. Mavis Hetherington	2002	Long-term studies of divorce and remarriage
C. Henry Kempe	1922 to 1984	Battered child syndrome (1962)
Sara Smilansky	1968	Added constructive play; sociodramatic play
Mildred Parten	1932	Six stages of social play: unoccupied to cooperative
Friedrich Schiller	1759 to 1805	Surplus energy theory of play

Name	When	Examined for
Herbert Spencer	1820 to 1903	Surplus energy theory of play
Moritz Lazarus	1824 to 1903	Relaxation or recreation theory of play
Karl Groos	1861 to 1946	Pre-exercise or practice theory of play
John Coie	1982	Sociometric status groups (with Dodge)
Kenneth Dodge	1982	Sociometric status groups (with Coie)
Dan Olweus	1931 to 2020	Definition of bullying; whole-school prevention programme
Kay Bussey	1999	Social cognitive theory of gender (with Bandura)
Ronald Slaby	1975	With Frey: gender constancy and attention to same-sex models
Karin Frey	1975	With Slaby: gender constancy and attention to same-sex models
Sandra Bem	1944 to 2014	Gender schema theory (1981); androgyny and the Bem Sex Role Inventory (1974)
Carol Martin	1981	Gender schema theory (with Halverson); memory for gender-inconsistent pictures (1983)
Charles Halverson	1981	Gender schema theory (with Martin)
Carol Jacklin	1974	With Maccoby: review of sex differences
Janet Shibley Hyde	2005	Gender similarities hypothesis
Carol Gilligan	Born 1936	In a Different Voice (1982): ethic of care (women) versus ethic of justice (men); levels of care
Hugh Hartshorne	Character Education Inquiry, 1928 to 1930	With Mark May: honesty is largely situation-specific
Mark May	Character Education Inquiry, 1928 to 1930	With Hugh Hartshorne: honesty is largely situation-specific
Frederick McDonald	1963	With Bandura: modelling shifted children's judgments between consequences and intentions
James Rest	1970s	Defining Issues Test, a questionnaire measure of moral reasoning
John Snarey	1985	Cross-cultural review of Kohlberg research: 45 studies, 27 cultural areas; stage 5 mainly in urban, educated groups
Richard Shweder	1980s Bhubaneswar studies	Indian research: ethics of autonomy, community and divinity
Joan Miller	1980s to 1990s Mysore studies	Indian and American comparisons: Indians treat interpersonal duties as moral obligations
Lawrence Walker	1984	Review finding little sex difference in Kohlberg scores
Elliot Turiel	1983	Social domain theory: moral, social-conventional and personal domains
Nancy Eisenberg	1980s	Levels of prosocial moral reasoning: hedonistic, needs-oriented, approval-oriented, empathic, internalised
Jonathan Haidt	2001	Social intuitionist model: intuition first, reasoning follows
Diane Papalia	Human Development textbook	Alternative stage labels: social concern and conscience (stage 4); morality of autonomous moral principles (Level III)
Robert Selman	1976, 1980	Levels of social perspective taking (Section 6.5); role-taking underlies Kohlberg's stages
Erik Homburger Erikson	1902 to 1994	Eight psychosocial stages; coined identity crisis; Childhood and Society (1950), Young Man Luther (1958), Identity: Youth and Crisis (1968), Gandhi's Truth (1969)
Joan Erikson	1903 to 1997	Erikson's collaborator; added a ninth stage for the very old (1997)
Anna Freud	1895 to 1982	Analysed Erikson; older psychoanalytic view that healthy adolescents detach from parents
Harry Stack Sullivan	1892 to 1949	Developmental epochs from infancy to adulthood; not the oral-sensory to maturity scheme
Melanie Klein	1882 to 1960	Paranoid-schizoid and depressive positions; not the identity crisis
Eric Berne	1910 to 1970	Transactional analysis; a distractor for the identity crisis
Martin Luther	1483 to 1546	Subject of Erikson's psychohistory Young Man Luther (1958)

Name	When	Examined for
Mohandas Karamchand Gandhi	1869 to 1948	Subject of Erikson's psychohistory Gandhi's Truth (1969), centred on the 1918 Ahmedabad mill strike
James E. Marcia	1966	Four identity statuses from crisis (exploration) and commitment; MAMA cycles (1992)
Jean Phinney	1989	Ethnic identity: unexamined, exploration, achieved
David Elkind	1967	Adolescent egocentrism: imaginary audience and personal fable
Laurence Steinberg	2008	Dual systems model of adolescent risk-taking
Margo Gardner	2005	With Steinberg, peers watching doubled adolescents' risk-taking in a driving game
Margaret Mead	1901 to 1978	Coming of Age in Samoa (1928): adolescent turmoil is cultural
Derek Freeman	1983	Criticised Mead's Samoan fieldwork
Peter Blos	1967	Psychoanalytic view of adolescence as detachment (second individuation) from parents
Dexter Dunphy	1963	Cliques and crowds; five stages of adolescent peer group structure
Jeffrey Jensen Arnett	2000	Emerging adulthood (about 18 to 25): identity exploration, instability, self-focus, feeling in-between, possibilities
Daniel Levinson	1920 to 1994	The Seasons of a Man's Life (1978): life structure, eras and transitions, the Dream and the mentor, midlife transition at about 40 to 45
George Vaillant	born 1934	Grant Study of Harvard men; added career consolidation and keeper of the meaning to Erikson's stages
Roger Gould	1978	Transformations: adult growth as shedding false childhood assumptions
Bernice Neugarten	1916 to 2001	Social clock; on-time and off-time life events
Robert Peck	1950s	Tasks of old age: ego differentiation, body transcendence, ego transcendence
Robert Sternberg	born 1949	Triangular theory of love (Chapter 3)
Eli Ginzberg	1911 to 2002	Occupational Choice (1951): fantasy, tentative and realistic periods of career choice
Donald Super	1910 to 1994	Life-span career stages (Chapter 7)
Elliott Jaques	1917 to 2003	Coined the term midlife crisis (1965)
K. Warner Schaie	1956	Seattle Longitudinal Study: most mental abilities peak in middle adulthood
Robert Butler	1927 to 2010	Coined ageism (1969); life review
Jeanne Calment	1875 to 1997	Longest confirmed human life, 122 years
Robert Kleemeier	1962	Terminal drop in cognitive scores before death
Leonard Hayflick	1961	Hayflick limit on cell divisions; cellular clock theory of ageing
Paul Moorhead	1961	With Hayflick, showed that cultured human cells divide a limited number of times
Denham Harman	1956	Free-radical theory of ageing
Elaine Cumming	1961	Disengagement theory (Growing Old, with William Henry)
William Henry	1961	Disengagement theory (Growing Old, with Elaine Cumming)
Robert Atchley	1989	Continuity theory of ageing; phases of retirement
Lars Tornstam	1989	Gerotranscendence
John Rowe	1997	Successful ageing (with Robert Kahn)
Robert Kahn	1997	Successful ageing (with John Rowe)
Laura Carstensen	born 1953	Socioemotional selectivity theory
Margret Baltes	1939 to 1999	Selective optimisation with compensation (1990, with Paul Baltes)
Arthur Rubinstein	1887 to 1982	Pianist used by the Baltes couple to illustrate selective optimisation with compensation
Timothy Salthouse	1996	Processing-speed theory of cognitive ageing
Aruna Shanbaug	1948 to 2015	Nurse in a vegetative state; her 2011 Supreme Court case allowed passive euthanasia in India
Maria Nagy	1948	Children's understanding of death: three stages in 378 Budapest children
Mark Speece	1984	With Sandor Brent: irreversibility, universality and nonfunctionality in children's death concept

Name	When	Examined for
Sandor Brent	1984	With Mark Speece: irreversibility, universality and nonfunctionality in children's death concept
Elisabeth Kubler-Ross	1926 to 2004	On Death and Dying (1969): denial, anger, bargaining, depression, acceptance
Margaret Stroebe	1999	Dual process model of grief (with Henk Schut)
Henk Schut	1999	Dual process model of grief (with Margaret Stroebe)
Cicely Saunders	1918 to 2005	Founded St Christopher's Hospice, London (1967); modern hospice and palliative care

Key terms

Developmental psychology, developmental perspective, Growth, Development, lifespan perspective, Normative age-graded influences, Normative history-graded influences, Non-normative influences, tabula rasa, baby biography, recapitulation, normative approach, maturation, Learning, socialisation, co-twin control, cephalocaudal, proximodistal, developmental task, nature versus nurture, Continuity versus discontinuity, learnt behaviour, Innate behaviour, overt behaviour, covert behaviour, critical period, sensitive period, ecological systems theory, microsystem, mesosystem, exosystem, macrosystem, chronosystem, bioecological model, proximal process, chromosomes, gene, mitosis, meiosis, zygote, monozygotic twins, dizygotic twins, genotype, phenotype, dominant-recessive inheritance, phenylketonuria, incomplete dominance, codominance, X-linked inheritance, Polygenic inheritance, genomic imprinting, Down syndrome, Heritability, gene-environment correlation, gene-environment interaction, reaction range, canalisation, Epigenetics, germinal period, embryonic period, gastrulation, fetal period, teratogen, fetal alcohol spectrum disorder, preterm, small for gestational age, Kangaroo care, Apgar scale, Neonatal Behavioral Assessment Scale, Primitive reflexes, cephalocaudal trend, proximodistal trend, marasmus, kwashiorkor, Neurogenesis, synaptogenesis, synaptic pruning, Myelination, Experience-expectant plasticity, Experience-dependent plasticity, gross motor skills, fine motor skills, mass to specific, palmar grasp reflex, prereaching, ulnar grasp, radial grasps, pincer grasp, maturational view, dynamic systems theory, preferential looking, habituation-dishabituation, habituation, dishabituation, high-amplitude sucking, violation-of-expectation, Evoked potentials, visual acuity, perceptual narrowing, Intermodal perception, visual cliff, affordance, independent locomotion, receptive-expressive gap, genetic epistemology, clinical method, constructivism, scheme, functional invariants, Adaptation, assimilation, accommodation, equilibration, operations, sensorimotor stage, deferred imitation, object permanence, A-not-B error, symbolic function, preoperational stage, Egocentrism, Animism, Centration, Irreversibility, Transductive reasoning, class inclusion, Conservation, Identity, Reversibility, horizontal décalage, concrete operational stage, Decentration, Transitive inference, formal operational stage, hypothetico-deductive reasoning, counterfactual reasoning, postformal thought, dialectical reasoning, Neo-Piagetian theories, sociocultural theory, higher mental functions, mediation, tools of intellectual adaptation, general genetic law of cultural development, internalisation, intersubjectivity, zone of proximal development, more knowledgeable other, scaffolding, guided participation, reciprocal teaching, private speech, pseudoconcept, modes of representation, spiral curriculum, discovery learning, information-processing approach, executive functions, production deficiency, utilisation deficiency, metacognition, metamemory, overlapping waves model, fuzzy-trace theory, theory of mind, joint attention, teleological stance, belief-desire psychology, false-belief task, appearance-reality distinction, Sally-Anne task, second-order theory of mind, recursive thought, social perspective taking, differentiation theory, differential emotions theory, primary emotions, reflexive smile, social

smile, self-conscious emotions, mirror self-recognition, Stranger anxiety, Separation protest, social referencing, still-face paradigm, Emotion regulation, display rules, Temperament, New York Longitudinal Study, easy child, difficult child, slow-to-warm-up child, difficult child syndrome, goodness of fit, Infant Behavior Questionnaire, structural model of temperament, Surgency, Negative affectivity, Effortful control, high-reactive infants, Behavioural inhibition, EAS model, self-regulation, marshmallow test, delay of gratification, Sustained attention, Resistance to temptation, forbidden-toy paradigm, attachment, secure base, safe haven, separation distress, secondary drive, person permanence, ethological theory, imprinting, contact comfort, affectionless psychopathy, maternal deprivation hypothesis, Monotropy, internal working model, protest-despair-detachment, Clear-cut attachment, goal-corrected partnership, multiple attachments, separation-individuation, rapprochement, emotional refuelling, object constancy, Strange Situation, maternal sensitivity hypothesis, Secure attachment, Insecure-avoidant attachment, Insecure-resistant attachment, Disorganised attachment, hospitalism, anaclitic depression, deprivation, privation, good-enough mother, transitional object, Adult Attachment Interview, parenting style, Authoritative parenting, Authoritarian parenting, Permissive parenting, demandingness, responsiveness, neglectful parenting, Helicopter parenting, battered child syndrome, Child maltreatment, neglect, functional play, symbolic play, games with rules, constructive play, Sociodramatic play, Rough-and-tumble play, parallel play, associative play, cooperative play, Sociometric status, Bullying, Gender, Gender identity, gender role, gender stereotype, Gender typing, Gender socialisation, gender segregation, congenital adrenal hyperplasia, gender constancy, Gender schema theory, androgyny, Moral development, superego, moral reasoning, situation-specific, premoral period, heteronomous morality, moral realism, objective responsibility, immanent justice, expiatory punishment, autonomous morality, moral relativism, reciprocal punishment, Heinz dilemma, Moral Judgment Interview, preconventional level, punishment-and-obedience orientation, instrumental purpose and exchange, conventional level, good boy-nice girl orientation, law-and-order orientation, postconventional level, social-contract orientation, invariant sequence, role-taking, Defining Issues Test, ethic of justice, ethic of care, social domain theory, prosocial moral reasoning, empathy, Induction, social intuitionist model, ego psychology, epigenetic principle, psychosocial crisis, virtue, identity crisis, generativity, stagnation, ego integrity, maladaptive tendency, malignant tendency, ego identity, psychosocial moratorium, identity confusion, negative identity, identity statuses, identity diffusion, foreclosure, identity achievement, MAMA cycles, ethnic identity, adolescent egocentrism, imaginary audience, personal fable, puberty, adrenarche, gonadarche, primary sex characteristics, secondary sex characteristics, menarche, spermarche, secular trend, dual systems model, storm and stress, developmental tasks of adolescence, cliques, emerging adulthood, life structure, the Dream, midlife transition, keeper of the meaning, Social clock, Traditional marriage, Companionship marriage, Collegial marriage, presbyopia, climacteric, menopause, andropause, midlife crisis, empty nest, sandwich generation, Gerontology, ageism, life expectancy, Primary ageing, terminal drop, Hayflick limit, disengagement theory, activity theory, Continuity theory, Gerotranscendence, socioemotional selectivity theory, selective optimisation with compensation, processing-speed theory, Berlin wisdom paradigm, brain death, dual process model, palliative care, passive euthanasia, living will

From the past papers

1. Which term refers to the biological unfolding of genetic potential? (IHBAS 2025)

   (a) Maturation (b) Socialization (c) Accommodation (d) Conditioning

   Answer: (a) Maturation. Maturation (Arnold Gesell) is change on a genetic timetable, largely independent of practice. Socialisation, accommodation (Piaget's adjustment of a scheme) and conditioning all depend on experience.

2. The fact that people can use later experiences to compensate for earlier deficits is a contradiction of which element of earlier developmental psychology? (IHBAS 2020)

   (a) Critical periods (b) Continuity (c) Sensitive periods (d) Stability

   Answer: (a) Critical periods. A strict critical period makes a missed experience irreversible, so recovery through later experience (Clarke and Clarke, 1976; the English and Romanian Adoptees study) contradicts it. Sensitive periods allow late recovery by definition; continuity and stability concern other questions.

3. In the case of human development, the strongest evidence for the critical period hypothesis is found in: (RML 2024)

   (a) Imprinting (b) Language acquisition (c) Embryonic development (d) Development of vocational interests

   Answer keyed: (a) Imprinting. Imprinting (Lorenz) is the textbook critical period, but it is animal evidence; within human development the clearest critical periods are in embryonic development, where teratogen timing decides which organ is damaged. Language shows a sensitive period; vocational interests show none.

4. If a blue eye allele (D) is dominant and a green eye allele (d) is recessive, what is the phenotype of a heterozygous (Dd) child? (Real PYQ 2025)

   (a) Green eyes (b) Blue eyes (c) Blue-green eyes (d) Cannot be determined

   Answer: (b) Blue eyes. In dominant-recessive inheritance a heterozygote shows the dominant allele's trait. Green eyes need the dd genotype, a blue-green blend would need incomplete dominance, and the phenotype can be determined because dominance is stated.

5. During embryonic development, the nervous system begins forming at the : (IHBAS 2025)

   (a) Zygote stage (b) Morula stage (c) Gastrula stage (d) Blastocyst stage

   Answer: (c) Gastrula stage. At gastrulation, in week 3, the three germ layers form and the neural plate arises from ectoderm, folding into the neural tube by about week 4. The zygote, morula and blastocyst are earlier stages with no germ layers yet.

6. A child with hyperphagia, obesity, short stature, and hypotonia is found to have a deletion on chromosome 15 (15q11-q13). This syndrome is : (IHBAS 2025)

   (a) Fragile X syndrome (b) Turner syndrome (c) Prader-Willi syndrome (d) Klinefelter syndrome

   Answer: (c) Prader-Willi syndrome. Loss of the paternal 15q11-q13 region gives hypotonia, hyperphagia, obesity and short stature; loss of the maternal copy gives Angelman syndrome. Fragile X is an FMR1 disorder on the X chromosome, Turner is 45,X and Klinefelter 47,XXY.

7. Human motor development is governed by certain basic principles. Which of the following is Not a basic principle of motor development? (RML 2021)

   (a) Maturation of the motor system proceeds from gross motor to fine motor behaviours. (b) Motor development is from cephalo-to-caudal or from head to feet. (c) Motor development proceeds from the distal to the proximal or from the extremities to the central part of the body. (d) Motor development plays a major role in social and behavioural expectations.

   Answer: (c) Distal to proximal. The proximodistal principle runs from the centre of the body to the extremities, so (c) reverses it. Gross to fine, cephalocaudal and the social role of motor development are all genuine principles.

8. What is the principal factor that changes the affordances available to the toddler, compared with the infant? (RML 2022)

   (a) Independent locomotion (b) Starting to speak (c) The attachment relationship (d) New kinds of caretaking

   Answer: (a) Independent locomotion. Affordances are the action possibilities the environment offers a perceiver, so they change most when the child's own action abilities change: crawling and walking make toys reachable and drops dangerous. Speech, attachment and caretaking change the social world, not what the surroundings afford action.

9. The process of synaptic pruning is most closely associated with : (RML 2023)

   (a) Memory consolidation (b) Neuronal development and plasticity (c) Neurotransmitter release (d) Long-term potentiation

   Answer: (b) Neuronal development and plasticity. The young brain overproduces synapses and prunes the little-used ones, fitting circuits to experience. Memory consolidation, neurotransmitter release and long-term potentiation describe how memories and signals work at existing synapses, not the developmental sculpting of circuits.

10. Which is not attained in Piaget's concrete operational stage of cognitive development? (JEMAS 2025)

    (a) Reduced egocentrism (b) Deferred imitation (c) Hypothetico deductive reasoning (d) Transitive inference

    Answer: (c) Hypothetico deductive reasoning. It is the hallmark of the formal operational stage (11 or 12 onward). Reduced egocentrism and transitive inference are concrete operational gains, and deferred imitation was attained earlier, at 18 to 24 months in the sensorimotor stage, so it is present rather than missing.

11. A 5-year-old child is asked to sort a deck of cards. The cards can be sorted by color (red/black) or by suit (hearts, spades, etc.). The child successfully sorts by color but cannot then re-sort the same deck by suit. This inflexibility is an example of: (Real PYQ 2024)

    (a) Egocentrism (b) Animism (c) Centration (d) Transductive reasoning

    Answer: (c) Centration. The preoperational child focuses on one salient feature (colour) and ignores others (suit). Egocentrism is failure to take another's perspective, animism is giving life to objects, and transductive reasoning runs from particular to particular.

12. Formal operational reasoning involves all the following EXCEPT: (RML 2022)

    (a) hypothetical reasoning (b) counterfactual reasoning (c) dialectical reasoning (d) metacognitive reasoning

    Answer: (c) Dialectical reasoning. Integrating opposing views is postformal thought (Riegel 1973, Basseches 1984). Hypothetical, counterfactual and metacognitive reasoning all belong to Piaget's formal operational stage.

13. Vygotsky's concept of the 'zone of proximal development' (ZPD) refers to: (RML 2025)

    (a) The difference between what a learner can do independently and what they can achieve with guidance. (b) A measure of a person's fixed learning capacity across cognitive domains. (c) A developmental stage characterised by logical reasoning capability. (d) The optimal period during which specific cognitive skills must develop.

    Answer: (a) The difference between what a learner can do independently and what they can achieve with guidance. The ZPD runs from the actual developmental level (independent problem solving) to the level reached with a more knowledgeable other, and it moves as skills are internalised. It is not a fixed capacity, not a stage of logical operations and not a critical or optimal period.

14. What is the primary difference between Piaget and Vygotsky's theories of language development? (RML 2021)

    (a) Piaget argued thought depends on language while Vygotsky argued language depends on thought (b) Vygotsky argued thought depends on language while Piaget argued language depends on thought (c) Vygotsky was concerned with early language development while Piaget was concerned with later language development. (d) None of the above

    Answer: (b). For Piaget language expresses concepts already built through action, so it depends on thought; for Vygotsky thought and speech have separate roots that merge at about 2, after which language shapes thought. Option (a) swaps the names, and (c) is false: both studied the speech of young children (egocentric speech for Piaget, private speech for Vygotsky).

15. In the 'teleological stance', toddlers can understand: (RML 2022)

    (a) Contingencies (b) Intentions in action (c) Prior Intentions (d) Mental states such as belief and desire

    Answer: (b) Intentions in action. In Gergely and Csibra's teleological stance, infants near the end of the first year read an action as efficient and goal-directed, so they grasp the goal visible in an ongoing act. They have moved beyond mere contingencies but do not yet represent prior intentions (plans formed before acting) or beliefs and desires, which come with desire psychology at about 2 and belief-desire psychology at 3 to 4.

16. Which of the following is a key characteristic of the "slow-to-warm-up" temperament category in the Thomas and Chess model? (Real PYQ 2024)

    (a) Positive approach to new stimuli and high regularity of biological functions. (b) Negative withdrawal from new stimuli, slow adaptability, and mild intensity of reactions. (c) Intense and frequent negative emotional reactions with irregular routines. (d) A mix of easy and difficult characteristics with no clear pattern.

    Answer: (b) Negative withdrawal from new stimuli, slow adaptability, and mild intensity of reactions. The slow-to-warm-up child withdraws at first but mildly, and adapts gradually. Option (a) describes the easy child, (c) the difficult child, and (d) the roughly 35 percent who fitted no single type.

17. Which among the following is not included within the classification of structure of temperament given by Rothbart and Bates (RML 2022)

    (a) Extraversion (b) Inhibition to the unfamiliar (c) Negative affectivity (d) Effortful control

    Answer: (b) Inhibition to the unfamiliar. Rothbart and Bates have exactly three dimensions: surgency (extraversion), negative affectivity and effortful control. Inhibition to the unfamiliar is Jerome Kagan's behavioural inhibition.

18. According to experts on socioemotional development like Campos (2009) and Kagan (2010), self-conscious emotions appear at which age? (RML 2022)

    (a) 6-9 months (b) Before 9 months (c) After 12 months (d) 3 months

    Answer: (c) After 12 months. Embarrassment, pride, shame, guilt, envy and empathy need self-awareness, which arrives in the second half of the second year (mirror self-recognition at 18 to 24 months). The earlier ages belong to the primary emotions.

19. Children, who are distressed by their parent's absence, seek interaction but also reject it when offered. How can the reaction be categorised? (RML 2021)

    (a) Insecure-avoidant (b) Insecure-resistant (c) Insecure-disorganized (d) Secure-resistant

    Answer: (b) Insecure-resistant. Ainsworth's type C (ambivalent) infants are very distressed at separation and on reunion seek contact yet resist it. Avoidant (A) infants show little distress and ignore the caregiver, disorganised (D) infants show contradictory or freezing behaviour, and 'secure-resistant' is not a type.

20. According to the schema theory, which hypothetical construct provides a prototype for subsequent attachments and can be adapted through more experience? (RML 2021)

    (a) Proximity-seeking mechanism (b) Attachment system (c) Bio-social relationship model (d) Internal working model

    Answer: (d) Internal working model. Bowlby's internal working model is a schema of self, attachment figures and relationships, built from early experience, that acts as a prototype for later relationships and can be revised. Proximity seeking is a behaviour the attachment system produces, the attachment system is the evolved control system itself, and 'bio-social relationship model' is not Bowlby's term.

21. Which theory of attachment is currently considered the predominantly accepted theory by developmental psychologists? (IHBAS 2020)

    (a) Psychoanalytic (b) Learning (c) Cognitive-developmental (d) Ethological

    Answer: (d) Ethological. Bowlby's ethological theory treats attachment as an evolved behavioural system that keeps the infant near a protector. The psychoanalytic and learning (secondary drive, 'cupboard love') theories rest on feeding, which Harlow's contact comfort findings undermined, and the cognitive-developmental view explains only the timing of attachment.

22. Which parenting style is characterized by the combination of giving emotional support, setting clear boundaries, and maintaining accountability for children's behavior? (Real PYQ 2025)

    (a) Authoritarian parenting (b) Authoritative parenting (c) Permissive parenting (d) Neglectful parenting

    Answer: (b) Authoritative parenting. Warmth (emotional support) plus firm, reasoned control (boundaries, accountability) is Baumrind's authoritative style, high on both demandingness and responsiveness. Authoritarian is high control without warmth, permissive is warmth without control, and neglectful lacks both.

23. According to Jean Piaget's theory of play development, what is the first stage of child play that emerges during infancy? (Real PYQ 2025)

    (a) Symbolic play (b) Functional play (c) Games with rules (d) Constructive play

    Answer: (b) Functional play. Functional (practice or sensorimotor) play, repeating actions for the pleasure of exercising sensorimotor schemes, comes first in infancy. Symbolic play needs the preoperational symbolic function, games with rules belong to the concrete operational years, and constructive play was added later by Smilansky.

24. The starting point for self-socialization, according to Kohlberg's cognitive-developmental theory of gender-role development, is: (IHBAS 2020)

    (a) Gender consistency (b) Basic gender identity (c) Gender stability (d) Gender-role achievement

    Answer: (b) Basic gender identity (as keyed). Labelling oneself a boy or girl is the first link in Kohlberg's chain. Many textbooks say Kohlberg's children become full self-socialisers only at gender consistency, and that it is gender schema theory that starts self-socialisation at basic gender identity. Gender-role achievement is not one of his stages.

25. What is the focus of Kohlberg's preconventional level of moral development? (IHBAS 2025)

    (a) Social contract (b) Avoiding punishment and gaining rewards (c) Obeying laws (d) Ethical principles

    Answer: (b) Avoiding punishment and gaining rewards. At Level I morality is under external control: stage 1 obeys to avoid punishment and stage 2 seeks rewards and fair deals. Obeying laws is stage 4 (conventional); social contract and ethical principles are stages 5 and 6 (postconventional).

26. Jean Piaget's theory of moral development suggests that children in the stage of 'heteronomous morality' (typically ages 5-10) base their moral judgments on: (Real PYQ 2024)

    (a) Intentions and equality (b) Rules as being unchangeable and the consequences of actions (c) Social contracts and individual rights (d) Abstract principles of justice

    Answer: (b) Rules as being unchangeable and the consequences of actions. Heteronomous children (moral realism) treat adult rules as sacred and judge acts by the damage done (objective responsibility). Intentions and equality belong to Piaget's autonomous stage; social contracts and abstract justice are Kohlberg's stages 5 and 6.

27. Carol Gilligan departs from Kohlberg's theory on moral development by highlighting which of these differences? (RML 2022)

    (a) Males have ethics of care (b) Males have ethics of justice (c) Females have ethics care (d) Females have ethics of justice

    Answer: (c) Females have an ethic of care. In *In a Different Voice* (1982) Gilligan argued that Kohlberg's scale captures a male ethic of justice and misses the ethic of care, reasoning from relationships and responsibility, more often heard in women. Options (a) and (d) reverse the pairing; (b) is what Kohlberg already measured.

28. According to Erik Erikson's psychosocial development theory, a child who is 1.5 years old (18 months) would be in which developmental stage? (Real PYQ 2025)

    (a) Trust vs. Mistrust (b) Autonomy vs. Shame and Doubt (c) Initiative vs. Guilt (d) Industry vs. Inferiority

    Answer: (b) Autonomy vs. Shame and Doubt. Eighteen months is the boundary between the first two stages, and the keyed answer places the toddler, now walking and facing toilet training, in autonomy versus shame and doubt. Initiative versus guilt begins at about 3 and industry versus inferiority at school age.

29. According to Erikson, the state of _____ involves feelings of inner fragmentation, little or no sense of where one's life is headed, and an inability to gain the support of a social role or vocation. (Amity 2023)

    (a) Conflict (b) Identity confusion (c) Incongruence (d) Dissociation

    Answer: (b) Identity confusion. It is the negative pole of Erikson's fifth stage, identity versus role confusion. Incongruence is Carl Rogers's gap between self-concept and experience, dissociation is a break in the integration of memory and identity, and conflict is too general.

30. Which term was used by Erikson to refer to the feeling experienced by some individuals in mid-life that they have achieved very little or nothing and have little to offer to the next generation? (RML 2020)

    (a) Stagnation (b) Generativity (c) Presbyopia (d) Mid-life crisis

    Answer: (a) Stagnation. It is the negative pole of generativity versus stagnation in middle adulthood. Generativity is the healthy pole (guiding the next generation), presbyopia is age-related loss of near focusing in the lens, and mid-life crisis is a popular term, not Erikson's.

31. In adulthood climacteric refers to (JEMAS 2025)

    (a) loss of muscular strength and agility. (b) receding hairlines and greying of hair in adults. (c) the end of ovulation and termination of reproductive capacity. (d) gradual decline in reaction time and manual dexterity.

    Answer: (c) the end of ovulation and termination of reproductive capacity. The climacteric is the midlife winding down of reproductive capacity, culminating in menopause in women. Lost strength and agility, receding and greying hair, and slower reaction time and dexterity are real midlife changes, but they are general physical ageing, not the climacteric.

32. The type of marriage in which comradeship and sharing are emphasized, but husband and wife assume responsibility for different roles is known as (JEMAS 2025)

    (a) Traditional marriage. (b) Collegial marriage. (c) Companionship marriage. (d) Satisfactory marriage.

    Answer: (b) Collegial marriage. In Hurlock's typology the collegial marriage stresses comradeship and sharing while each spouse takes responsibility for different, complementary roles. A companionship marriage shares the same roles, a traditional marriage separates roles without stressing comradeship, and satisfactory marriage is not a type.

High-yield sheet

The facts this chapter's questions ask most, most-asked first. Last read before the exam.

1 **Object permanence** is a **sensorimotor** achievement: search for a fully hidden object begins at about 8 to 12 months (substage 4, with the A-not-B error) and object permanence is complete by 18 to 24 months. ASKED 10 TIMES

2 Erikson's crisis of adolescence (about 12 to 18) is **identity versus role confusion**, and its virtue is fidelity. ASKED 10 TIMES

3 The **zone of proximal development** is the gap between what a learner can do independently (the actual developmental level) and what the learner can achieve with guidance from a more knowledgeable other: tasks too hard to master alone but achievable with help. ASKED 8 TIMES

4 **Conservation** (quantity stays the same despite changes in shape or arrangement, as with the clay sausage or the tall glass) is mastered in the **concrete operational** stage (7 to 11), number first and volume last. ASKED 7 TIMES

5 **Hypothetico-deductive reasoning** and abstract thought define the **formal operational** stage (11 or 12 onward); it is not a stage itself, and the concrete operational child does not attain it. ASKED 7 TIMES

6 **Thomas and Chess** (New York Longitudinal Study, 1956) found three types: **easy** (about 40 percent: regular, positive, adaptable, approaches), **difficult** (about 10 percent: irregular rhythms, negative mood, intense reactions, withdrawal, slow adaptation) and **slow-to-warm-up** (about 15 percent: low activity, mild intensity, initial withdrawal, slow gradual adaptation); about 35 percent fitted no single type. ASKED 6 TIMES

7 **Harry Harlow** ('The Nature of Love', 1958): infant rhesus monkeys clung to a terry-cloth mother that gave no milk rather than a wire mother that fed them, so **contact comfort, not feeding**, is the primary basis of attachment. ASKED 6 TIMES

8 In the Strange Situation, reunion behaviour decides the type: **secure (B)** infants seek contact, are readily comforted and return to play; **insecure-avoidant (A)** infants show little distress and ignore the caregiver; **insecure-resistant or ambivalent (C)** infants are very distressed, seek contact yet resist it and cannot be soothed. ASKED 6 TIMES

9 **Authoritative** parenting (Baumrind) is high in demandingness and high in responsiveness: emotional support, clear boundaries, reasoned discipline and accountability, with the best child outcomes; authoritarian is high demand and low warmth, permissive low demand and high warmth, and neglectful (added by Maccoby and Martin, 1983) low on both. ASKED 6 TIMES

10 Kohlberg's **preconventional level** (stage 1 punishment-and-obedience, stage 2 instrumental purpose and exchange, or self-interest) is morality under external control: rules are obeyed to avoid punishment or gain rewards, so rewarded acts are right and punished acts wrong. ASKED 6 TIMES

11 Kohlberg's **conventional level** is conformity to social rules and maintenance of the social order: stage 3 good boy-nice girl (approval, interpersonal expectations, the golden rule) and stage 4 law and order (duty, respect for authority; Papalia's social concern and conscience). Obeying a law or keeping a promise because society depends on it is conventional. ASKED 6 TIMES

12 In **Bronfenbrenner**'s ecological systems theory (1979), the **microsystem** holds the child's direct settings (family, school, peers), the **mesosystem** is the links between them (home-school), the **exosystem** holds settings without the child (parent's workplace), the **macrosystem** is culture and the **chronosystem** (added later) is time. ASKED 5 TIMES

13 **Mary Ainsworth** devised the **Strange Situation** (with Barbara Wittig, 1969; Patterns of Attachment, 1978): Bowlby is to attachment theory as Ainsworth is to the Strange Situation. ASKED 5 TIMES

14 **Centration** is the preoperational focus on one salient feature while ignoring others, as in the 5-year-old who sorts cards by colour but cannot re-sort them by suit. ASKED 4 TIMES

15 The term **zone of proximal development** was coined by **Lev Vygotsky** (not Piaget, Bruner, Kohlberg, Bronfenbrenner, Erikson, Skinner or Freud); **scaffolding**, the support given within it, is a term from Wood, Bruner and Ross (1976). ASKED 4 TIMES

16 The fourth pattern, **disorganised or disoriented (D)**, was added by **Mary Main and Judith Solomon** (1986, 1990): no one clear pattern, with contradictory, freezing, dazed or apprehensive behaviour on reunion, often after maltreatment. ASKED 4 TIMES

17 Kohlberg's three levels are **preconventional, conventional and postconventional**; Level III (Papalia: morality of autonomous moral principles) holds stage 5, the social-contract orientation (laws as instruments expressing the majority's will and furthering welfare), and stage 6, universal ethical principles. Unconventional, formal conventional and amoral are not Kohlberg levels, and premoral is Piaget's period before rules matter. ASKED 4 TIMES

18 Erikson's virtues in order: trust gives **hope**, autonomy **will**, initiative **purpose**, industry **competence**, identity **fidelity**, intimacy **love**, generativity **care**, integrity **wisdom**. ASKED 4 TIMES

19 Marcia's identity statuses cross **crisis (exploration)** with **commitment**: diffusion (neither), foreclosure (commitment without exploration), moratorium (exploration without commitment), achievement (both). ASKED 4 TIMES

20 Middle adulthood is **generativity versus stagnation**: mentoring and guiding the next generation is generativity; feeling one has achieved little and has little to offer is stagnation. ASKED 4 TIMES

21 Erikson's first stage is **trust versus mistrust** in infancy (birth to about 1, some texts 18 months); a child of 18 months is keyed to autonomy versus shame and doubt. ASKED 4 TIMES

22 A **critical period** is a fixed window in which an experience must occur for normal development, with largely irreversible effects (Lorenz's imprinting; embryonic organ formation); later experience compensating for early deficits contradicts it. ASKED 3 TIMES

23 **Eleanor Gibson and Richard Walk** (1960) used the **visual cliff** to test depth perception: infants of crawling age (about 6 months onward) refused the deep side, so depth perception is present by the time infants crawl. ASKED 3 TIMES

24 Piaget's stage ages: **sensorimotor** birth to 2, **preoperational** 2 to 7, **concrete operational** 7 to 11, **formal operational** 11 or 12 onward. ASKED 3 TIMES

25 **False-belief tasks** (Wimmer and Perner's Maxi task, 1983; Baron-Cohen, Leslie and Frith's Sally-Anne task, 1985) assess **theory of mind** and are typically passed at **about age 4**, in the preoperational years. ASKED 3 TIMES

26 Bowlby's **internal working model** is a mental representation of self, attachment figures and relationships, built from early experience, that serves as a prototype for later relationships and can be revised by new experience. ASKED 3 TIMES

27 **Konrad Lorenz** (1935): greylag goslings **imprint** on the first moving object after hatching, a rapid, largely irreversible following response formed within a narrow critical period. ASKED 3 TIMES

28 Late adulthood (65 onward) is **ego integrity versus despair**: accepting the life one has lived versus regret and fear of death; the virtue is wisdom. ASKED 3 TIMES

29 **Maturation** (Arnold Gesell) is the biological unfolding of genetic potential; a child's increasing skill at using his muscles is due mainly to maturation, not exercise, parental pushing or learning.
ASKED 2 TIMES

30 **Klinefelter syndrome** (47,XXY) is a tall male with small testes, infertility and learning difficulties; **Turner syndrome** (45,X) is a short female with a webbed neck and ovarian failure. ASKED 2 TIMES